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strengthening mechanisms in crystal plasticity

A. S. ARGON



OXFORD MATERIALS

OXFORD SERIES ON MATERIALS MODELLING

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OXFORD SERIES ON MATERIALS MODELLING

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Strengthening Mechanisms in Crystal Plasticity

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PREFACE

This book, among a number of others on materials modeling published by Oxford University Press, is primarily addressed to graduate students in materials science and mechanical engineering and to researchers in materials modeling. As a perusal of the roughly 20 titles in the series indicates, the modeling methodology used in such wide-ranging subjects and phenomena will necessarily also be wide-ranging. In the present book on *Strengthening Mechanisms in Crystal Plasticity*, the principal tool is dislocation mechanics, applied to well-defined mechanical phenomena of both industrial and research interest. The principal goal of the book is to develop a basic understanding of the variety of mechanisms that have been developed to control the resistance to dislocation motion in a crystal lattice by some intrinsic, but mostly extrinsic means. The coverage strives to inform rather than to impress. The central tool involves the use of the line properties of crystal dislocations in the treatment of the ways in which dislocation lines sample obstacles in their path under an applied stress and in the presence of possible thermal assistance. The coverage is limited to a low-homologous-temperature range where diffusional processes play little or no role in overcoming the obstacles and where the obstacles are stable in the time window of the deformation. To make the subject negotiable to the reader, a minimal coverage of crystal symmetry and the line properties of dislocations is introduced in Chapters 1 and 2. The main subjects, namely the lattice resistance, solid-solution strengthening, precipitation strengthening, and strain hardening in homogeneous crystals and polycrystals, are then tackled, followed by further applications of hardening in the presence of heterogeneities, and some other special topics.

The various phenomena and mechanisms are always introduced first in overviews to develop a qualitative appreciation of what is to be modeled. The models that follow are presented in some necessary detail to demonstrate the general simplicity and direct nature of the approach and to develop confidence in dealing with similar applications by variations of what is presented. The development of the models refers in most instances to the original publications in the literature, which are always cited, but it is usually significantly different from the published models, because of the attempt to use uniformity in methodology. In all cases, the models are compared with the best experimental results available. Much of the work cited and the associated experimental results are of earlier vintage. The goal is to encourage the reader to familiarize himself/herself through these citations with the richness in the foundations of the subject. At the ends of the chapters, in addition to the full references, other references for further reading in depth are also provided.

Many of the subjects and phenomena presented in this book have been explored by computer simulation using various methods. The best of these available simulations are quite specific and relate to simple unit processes, which include the energies of formation and motion of point defects, the structures of the cores of dislocations, specific interactions of dislocations with obstacles such as solute atoms and precipitate particles, and intersection of dislocations in the process of plastic flow. Some of these simulation results are referred to in the text when the context of the presentation warrants this and results in better understanding of the phenomena. Other simulations, dealing with the collective behavior of dislocations such as the stages of strain hardening, including simulations referred to as “dislocation dynamics” that are presently too inadequate to go much beyond the earliest stages of hardening, have deliberately not been referred to. The basic value of these latter simulations has been to demonstrate that the dislocation interaction processes that had been considered earlier are correct and are internally consistent. However, readers are encouraged to follow modern developments in crystal plasticity and compare these with the topics covered in this book to assess whether further clarification has been achieved.

In the preparation of the book I benefited from many discussions with colleagues in the field, but particularly with A. Ardell, J. Bassani, V. Bulatov, W. Cai, H. Mughrabi, R. Schwarz, S. Takeuchi, and S. Yip. Other colleagues supplied original micrographs or digital data files of micrographs, or helped locate them. They include H. Gleiter, P. Haehner, N. Hansen, H. Hattendorf, D. Laughlin, H. Mughrabi, J. Spence, and H. Teichler. Several colleagues kindly read parts of the book and supplied detailed comments and criticisms. For this I am grateful to H. Mughrabi, L. Kubin, F. Nabarro, and, particularly, D. Parks, who gave detailed comments on the entire book. In addition, the two series editors, A. Sutton and R. Rudd, have also made important suggestions about the entire book. The text was prepared and underwent numerous modifications, all cheerfully performed, by Doris Elsemiller. The illustrations were ably produced by Andrew Standeven. All this would not have been possible without the financial help of Dean T. Magranti and S. Suresh from special funds, and my Department Head R. Abeyaratne, who patiently provided secretarial assistance for the duration and also arranged additional support from the special departmental Pappalardo–Oxford Press funds for the production of books. I also acknowledge the many, not always successful, “needlings” from Sönke Adlung of OUP to speed up the writing process.

My introduction to the subject of the book dates back to my (often stormy) dealings with Egon Orowan, who more than anybody else must get credit for giving me a new direction into materials science. For this reason, this book is dedicated to his memory. Here I also acknowledge another stimulus for dealing with the subject dating back to the 1970s, a collaboration with U. F. Kocks and M. F. Ashby that resulted in a book in the Progress in Materials Science series.

Finally, I acknowledge the support and understanding of my wife Xenia, who told me earlier that “all this will have little effect in a hundred years”. I would personally settle for an effect that lasts fifty years.

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LIST OF SYMBOLS

B	($= kT / \Omega v_D$) drag coefficient
D	distance of separation between braids in Stage II hardening
D_k	kink diffusivity along dislocation
E	Young's modulus
E_b	binding energy of solute to screw dislocation core
$\mathcal{E}$	line tension of dislocation
$\mathcal{E}_e$	line tension of edge dislocation
$\mathcal{E}_s$	line tension of screw dislocation
F	Helmholtz free energy
ΔF	Helmholtz free-energy change
ΔF^*	Helmholtz free energy of activation
F_c	cosine potential of lattice resistance (Helmholtz)
F_k	Helmholtz free energy of a kink
F_0	($= \mathfrak{F}_0$) Helmholtz line energy of a dislocation in potential valley
F_p	parabolic potential of lattice resistance (Helmholtz)
$\mathfrak{F}$	line energy of dislocation
$\mathfrak{F}_e$	line energy of edge dislocation
$\mathfrak{F}_c$	line energy of Cottrell dislocation
$\mathfrak{F}_s$	line energy of screw dislocation
G	Gibbs free energy
ΔG	Gibbs free-energy change
$\Delta G^*(\sigma, \hat{\tau})$	Gibbs free energy of activation, dependent on shear stress and threshold shear resistance
ΔG_{dk}^*	Gibbs free energy of double kink
H	enthalpy
ΔH^*	activation enthalpy
I	interaction energy
I_μ	modulus misfit interaction energy of a solute atom
I_s	size misfit interaction energy of a solute atom
I_{kk}	kink–kink interaction energy
J	flux of mobile configurations
K	bulk modulus; resistive force of discrete particle to dislocation motion
$\hat{K}$	peak resistive force of discrete obstacle
$K(y)$	force against distance (y) curve
K_e	anisotropic line energy factor for edge dislocation

K_s	anisotropic line energy factor for screw dislocation
L_m	mean distance of binding contacts along dislocation line
L^*	activation spacing between kinks in double-kink formation
Q	experimental activation energy
R	size of crystal; radius of curvature; Cottrell–Stokes ratio
S	entropy
ΔS^*	activation entropy
T	absolute temperature
T_0	absolute cutoff temperature above which a resistance mechanism becomes negligible
U	internal energy
V	volume
W	work
ΔW	work increment
ΔW^*	activation work
a	area
δa	infinitesimal area increment
Δa	area increment swept out by dislocation
$\langle a \rangle$	most probable area of precipitate sheared by dislocation
Δa^*	activation area
b	magnitude of Burgers vector
b_p	magnitude of Burgers vector of partial dislocation
c	defect (solute, precipitate) concentration; volume fraction
c_s	velocity of shear wave of sound
c_{ij}	Voigt elastic constants
c_{dk}	thermal-equilibrium concentration of double kinks
d	grain diameter
f	force exerted by dislocation on contacted discrete obstacle
h	height
h_i	enthalpy of formation of interstitial
h_v	enthalpy of formation of vacancy
k	Boltzmann's constant
l	mean distance between obstacles in slip plane
l_i	mean distance between obstacles touching a dislocation
m	phenomenological exponent; mass of dislocation per unit length; ($= \ln(\dot{\gamma}_0/\dot{\gamma})$)
r	radial coordinate
r_0	actual particle radius
r^*	critical precipitate nucleus at formation
$\langle r \rangle$	most probable radius of precipitate sheared by dislocation
s_{ij}	Voigt elastic compliances
v_d	dislocation velocity

v_k	kink velocity along dislocation
$\Delta v_0^* = \Delta v^*$	shear activation volume (often just an activation volume)
Δv_p^*	pressure activation volume
w	effective range of interaction of obstacle with dislocation; dislocation core width; capture cross section of edge dislocation multipole
w_k	kink width
x, y, z	rectangular coordinates
Δy^*	activation distance
$\Theta_I, \Theta_{II}, \Theta_{III}, \Theta_{IV}$	strain hardening rates in Stages I, II, III, and IV
Λ	coherent length of dislocation sampling random positive or negative solute interaction under applied stress; distance between paired dislocations interacting with precipitates in order strengthening; mean free path length; stacking-fault width in extended dislocation
Ω	atomic volume
Ω_f	volume of faulted cluster
α	constant of proportionality
α_{ij}	element of direction cosine matrix
β	constant of proportionality; $\sigma/\hat{\tau}$; $K/2\mathcal{E}$
β_c	critical normalized obstacle strength, $\hat{K}/2\mathcal{E}$
γ	one-dimensional shear strain in mechanisms to be understood as plastic (deviatoric) shear strain
$\gamma_4, \gamma_5, \gamma_6$	Voigt elastic shear strains ($\gamma_{23}, \gamma_{13}, \gamma_{12}$, respectively)
γ^T	freestanding transformation shear strain in shear transformation, also $\Delta\gamma^T$
$\dot{\gamma}$	plastic shear strain rate
$\dot{\gamma}_0$	(= $\dot{\gamma}_G$) preexponential of volume-averaged shear strain rate in thermally assisted deformation mechanisms
$\Delta\gamma$	plastic shear strain increment
$\Delta\gamma_0$	(= $\Delta\gamma^T\Omega_f/V$) volume-averaged shear strain related to shear transformation
δ	infinitesimal length; grain boundary thickness
δ_{ij}	Kronecker delta
ε	normal strain
$\varepsilon_1, \varepsilon_2, \varepsilon_3$	Voigt elastic normal strains ($\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}$, respectively)
ε_{ij}	elements of strain tensor
$\dot{\varepsilon}$	normal strain rate
ϵ_s	size misfit parameter ($((1/a)(da/dc))$)
ϵ_μ	modulus misfit parameter ($((1/\mu)(d\mu/dc))$)

θ	angular coordinate; various other angles in mechanisms
λ	Lamé elastic constant; various length parameters; Friedel sampling length below flow
λ_c	(λ_{FF}) Friedel–Fleischer sampling length (Friedel length) for point obstacles at flow
λ_{SL}	Schwarz–Labusch sampling length for obstacles with finite width
μ	isotropic shear modulus; Lamé elastic constant
ν	Poisson’s ratio
ν_D	Debye frequency
ν_G	($\ll \nu_D$) “attempt frequency” of dislocation overcoming thermally penetrable obstacle
ξ, η, ζ	normalized coordinates x, y , and z
ρ	total dislocation density
ρ_G	geometrically necessary dislocation density
ρ_m	mobile dislocation density
ρ_S	statistically stored dislocation density
ρ_w	dislocation density in cell walls
σ	stress, generally meant to be the applied shear stress
$\sigma_1, \sigma_2, \sigma_3$	Voigt normal stresses ($\sigma_{11}, \sigma_{22}, \sigma_{33}$, respectively)
σ_i	internal (elastic) back stress
σ_{ij}	elements of stress tensor
σ_m	mean normal stress; component of flow stress acting to achieve kink mobility on screw dislocation
σ_s	effective stress in thermal-assistance mechanisms
τ	shear resistance
$\hat{\tau}$	threshold shear resistance
$\hat{\tau}_c$	composite threshold resistance of dislocation cells
$\hat{\tau}_i$	threshold resistance of impenetrable particle
$\hat{\tau}_{km}$	threshold resistance of solute clusters to kink motion along screw dislocations
$\hat{\tau}_p$	threshold resistance of penetrable particle
$\hat{\tau}_w$	threshold resistance of walls of dislocation cells
ϕ	angle between Burgers vector and line direction vector
χ_{APB}	antiphase boundary energy
χ_I	interface free energy
χ_{SF}	stacking-fault energy
ω_{ij}^p	element of plastic rotation matrix

*This book is dedicated to the memory of Egon Orowan who introduced me to
the field of Crystal Plasticity.*

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STRUCTURE OF CRYSTALLINE SOLIDS AND THE “DEFECT STATE”

1.1 Overview

Technologically important solids can range from completely disordered, through to partially ordered, to fully ordered crystalline types. What governs the state of order of a solid depends primarily on the type of atomic bonding and on the processing path that has led to the final state. Our interest in this book will be almost entirely in solids in the crystalline state and in their inelastic response sufficiently below their melting point, where diffusion-controlled alterations of their structure will be too slow to be of importance when compared with their rate of inelastic response. For that purpose, we shall first consider briefly some of the most important crystal structures, which will permit us to deal with their inelastic response mechanisms.

At temperatures sufficiently below the melting point, under low applied stress, all solids exhibit reversible elastic behavior, which is nearly always anisotropic when referred to the perfect crystal. However, in our coverage we shall mostly idealize the elastic response as isotropic and linear, so as to concentrate primarily on the physical mechanisms of inelastic behavior and on how these can be manipulated to enhance the resistance of materials to permanent deformation.

A perfect crystalline solid in single-crystalline form can respond only in a reversible elastic manner in thermal equilibrium with its surroundings when stressed monotonically well below the critical levels that destabilize the crystal structure. Under an applied homogeneous stress, the elastic response is homogeneous down to the atomic level. In contrast, the inelastic or, more narrowly, plastic response is locally heterogeneous and requires crystal defects for its development. The type and intensity of the plastic response depend on the character of the “defect state”. For this purpose we shall introduce the principal crystal defects in a hierarchy of increasing dimensionality from point, through line, to planar defects, and define their characteristic features. Among these defects, we shall be concerned, overwhelmingly, with dislocations as the principal “carriers” of plastic deformation. We shall introduce the line properties of dislocations, which will permit us to focus on their behavior and their interactions as separate entities rather than constantly referring to the properties and response of the crystals that contain them, thereby translating much of “crystal plasticity” into “dislocation mechanics”.

1.2 Principal Crystal Structures of Interest

When solidified from the melt, at normal cooling rates, pure metals and simple compounds can undergo long-range ordering, which results in the formation of crystal structures of varying complexity and symmetry, producing lattices or structures in which a definite arrangement of atoms is repeated in space at regular intervals as is shown generically in Fig. 1.1. The basic arrangement of the symmetry, containing from one to a larger number of atoms, is called a *unit cell*, in distinction from a *primitive cell*, which contains only one atom. The former is of greater interest, since it reflects better the symmetry that governs the elastic and plastic response. Figure 1.1 shows a triclinic lattice of low symmetry, in which the three principal axes make angles of α , β , and γ between the principal lattice identity vectors **a**, **b**, and **c**. The figure also shows other vectors connecting lattice sites, and two crystallographic planes, which are identified by a special notation referred to as the Miller index notation (for an introduction to Miller indices in crystallography, see Allen and Thomas 1999). Most, but not all, technologically interesting crystal structures of pure metals and simple compounds possess a higher degree of symmetry. The symmetry properties of crystals represented as crystallographic point groups is a precise and complex subject, which we need not develop here. Excellent discussions of these forms of representation can be found in Barrett and Massalski (1980) and Allen and Thomas (1999).

Figure 1.2 shows three of the most common crystal structures found in metals: the face-centered cubic (FCC) lattice, the body-centered cubic (BCC) lattice, and

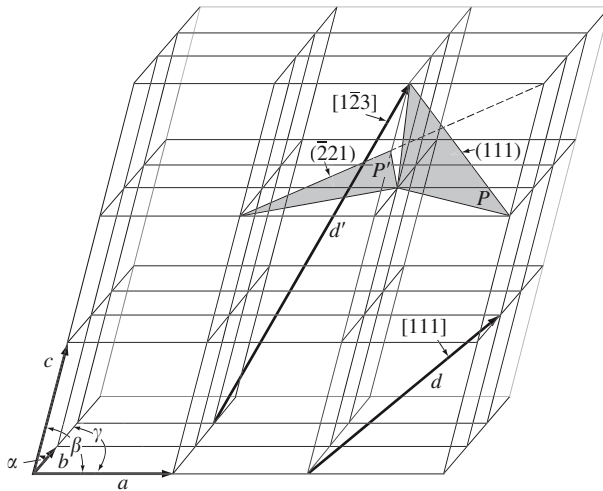


FIG. 1.1. A triclinic lattice showing crystal axes, some directions, and planes identified by Miller indices.

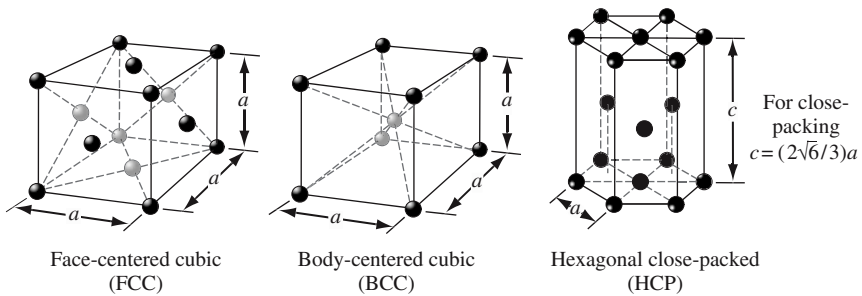


FIG. 1.2. Three prominent crystal structures of metals: (a) face-centered cubic; (b) body-centered cubic; (c) hexagonal close-packed.

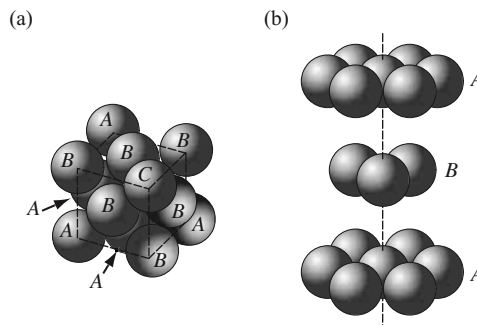


FIG. 1.3. Important differences in the symmetry between the FCC and HCP close-packed structures: (a) in FCC, close-packed layers of hard spheres are packed in the third direction in unbroken close-packed rows of touching spheres, that is, ABCABC, etc.; (b) in HCP, close-packed layers are packed in the third direction as zigzag rows of touching spheres, that is, ABAB, etc.

the hexagonal close-packed (HCP) structure. The latter, with an ideal c/a ratio of 1.633, and the FCC structure can both be idealized as a close packing of hard spheres in a plane, where the planes are packed on top of each other, with the spheres of one layer fitting into specific interstitial sites of the layer below. The HCP and FCC structures, however, have radically different symmetry properties, as is depicted in Figs. 1.3(a) and (b). In the HCP structure, the third close-packed layer is stacked above the atoms of the close-packed first layer (A), while the second layer (B) is stacked in relation to the two (A) layers in only one of the two available interstitial sites, resulting in a stacking order of $\dots ABABA \dots$. In comparison, in the stacking of the FCC planes, the third close-packed layer (C) is stacked on top of the second set of available interstitial sites of the (A) layer, resulting in a stacking order of $\dots ABCABC \dots$, and giving rise to three additional close-packed rows of atoms in addition to the three in the (A) layer. Thus, the FCC lattice possesses six

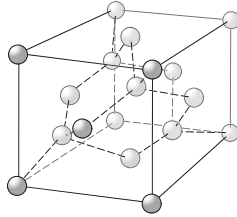


FIG. 1.4. The diamond cubic structure with eight atoms per unit cell, giving a low coordination number of four tetrahedrally bonded atoms.

close-packed rows of atoms in space outlining a tetrahedron, in comparison with only three in the HCP structure, all lying in the basal plane as should be clear from Figs. 1.3(a) and (b). This difference has important consequences for the plasticity of these two structures, which we shall discuss in some detail in later chapters. None of the hexagonal metals possesses the ideal c/a ratio of 1.633, although Co and Mg come close. In both the FCC and the HCP metals, the relationships between atoms in neighboring planes are the same. With energetic differences arising only through the next-nearest-neighbor relations, these differences in energy tend to be small, making it possible to have stacking faults in FCC and HCP structures, which have important consequences for the plasticity of these structures, as we shall also discuss in later chapters. Figure 1.4 shows the diamond cubic lattice, characterized by strong tetrahedral bonding of atoms, which otherwise has very similar crystallographic symmetry characteristics to the FCC lattice.

Table 1.1 lists the most common elemental materials having the FCC, BCC, HCP, and diamond cubic structures, together with their lattice parameters a at room temperature. A large number of other technologically important materials are also of interest. Figure 1.5 shows the structures of some of these: the ordered intermetallic compound Ni_3Al , having FCC symmetry; the ionic compound NaCl , having the structure of two interpenetrating FCC lattices of Na and Cl ions, and representing also the structure of MgO ; the compound CsCl , with BCC symmetry; Al_2O_3 (showing only the cation (Al) lattice and its holes, and not the intervening anion (O) lattice), having hexagonal symmetry; and cubic ZrO_2 . Table 1.2 lists the lattice constants of these and some other compounds of interest.

1.3 Small-Strain Elasticity in Crystals

1.3.1 Hooke's Law

In 1676 Robert Hooke discovered that thin metallic wires when loaded in tension, extended (strained) reversibly in linear proportionality to the applied load (stress), provided the loads were small so as not to result in permanent stretching. This

TABLE 1.1. Crystal structures of some common elements and lattice dimensions at room temperature (20°C)^a

Element	Type of structure	Atoms per unit cell	Lattice constants (Å) ^b		Distance of closest approach (Å) ^c
			<i>a</i>	<i>c</i>	
Aluminum	FCC	4	4.0490		2.862
Beryllium	HCP	2	2.2854	3.5841	2.225
Cadmium	HCP	2	2.9787	5.617	2.979
Carbon (diamond)	Diamond cubic	8	3.568		1.544
Chromium	BCC	2	2.8845		2.498
Cobalt, α	HCP	2	2.507	4.069	2.506
Cobalt, β	FCC	4	3.552		2.511
Copper	FCC	4	3.6153		2.556
Germanium	Diamond cubic	8	5.658		2.450
Gold	FCC	4	4.0783		2.884
Hafnium	HCP	2	3.1946	5.0511	3.127
Iridium	FCC	4	3.8389		2.714
Iron, α	BCC	2	2.8664		2.481
Iron, γ (extrapolated)	FCC	4	3.571		2.525
Lead	FCC	4	4.9495		3.499
Lithium	BCC	2	3.5089		3.039
Magnesium	HCP	2	3.2092	5.2103	3.196
Manganese, α	Cubic	58	8.912		2.24
Manganese, β	Cubic	20	6.300		2.373
Molybdenum	BCC	2	3.1466		2.725
Nickel	FCC	4	3.5238		2.491
Platinum	FCC	4	3.9237		2.775
Potassium	BCC	2	5.344		4.627
Silicon	Diamond cubic	8	5.4282		3.351
Silver	FCC	4	4.0856		2.888
Sodium	BCC	2	4.2906		3.715
Tantalum	BCC	2	3.3026		2.860
Tin, α	Diamond cubic	8	6.47		2.81
Titanium, α	HCP	2	2.9504	4.6833	2.89
Tungsten	BCC	2	3.1648		2.739
Zinc	HCP	2	2.664	4.945	2.664
Zirconium	HCP	2	3.2312	5.1477	3.172

^a Compiled from Barrett and Massalski (1980). Courtesy of Pergamon Press.^b 1 Å = 1 angstrom unit = 10^{-10} m.^c Magnitude of Burgers vector.

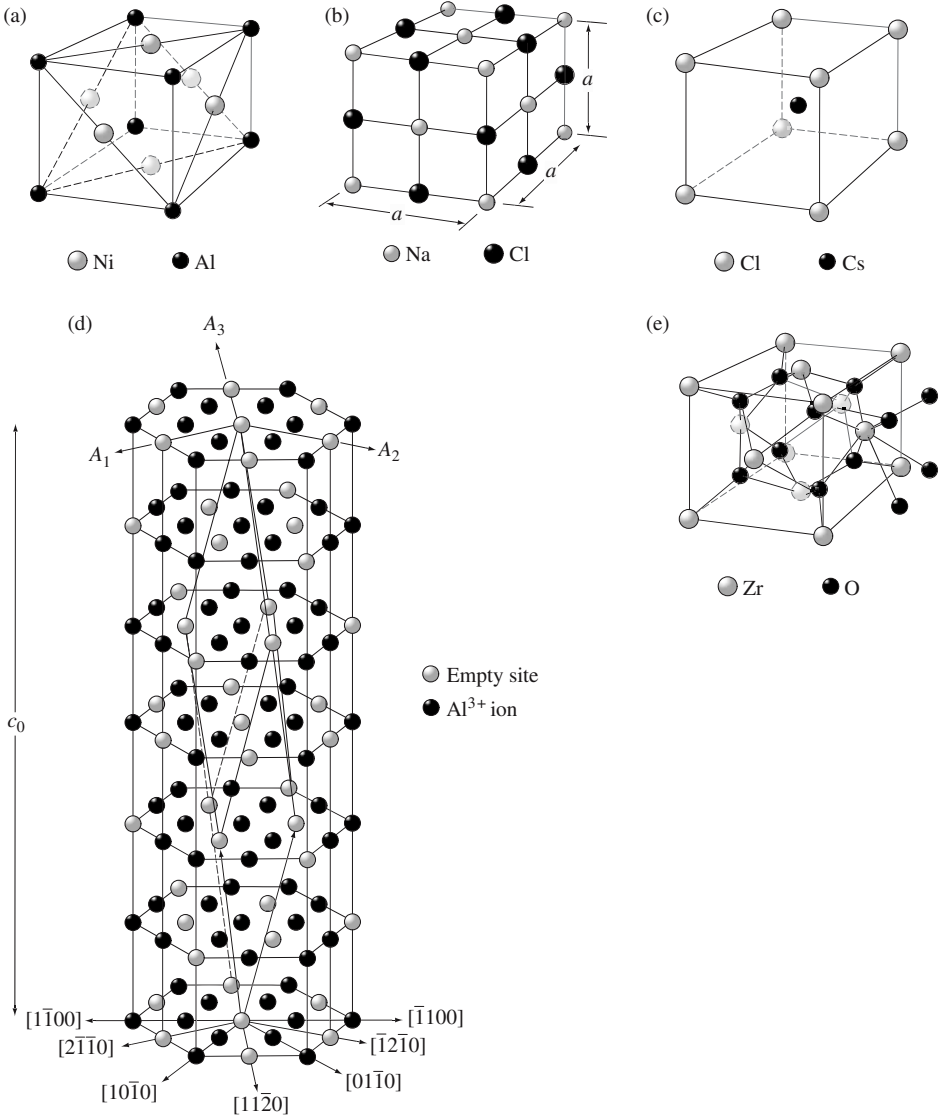


FIG. 1.5. More complex crystal structures: (a) ordered Ni_3Al ; (b) NaCl ; (c) BCC-type CsCl structure; (d) hexagonal structure of Al_2O_3 , where only the cation structure of Al is shown, with the characteristic arrangement of holes; the hexagonal close-packed interspersed layers of O anions fitting between the Al cations are omitted for clarity (after Chiang *et al.* 1997); (e) the cubic-zirconia ZrO_2 structure.

TABLE 1.2. Crystal structures of some common inorganic compounds and their lattice dimensions at room temperature (20 °C)^a

Compound	Structure	Atoms/unit cell	Lattice constants (Å)		
			<i>a</i>	<i>c</i>	α
Al ₂ O ₃ (β)	Hexagonal	50 ^b	5.56	22.55	
CaCO ₃ (calcite)	Rhombohedral	10	6.36		46° 6'
CaF ₂ (fluorite)	Cubic	12	5.45		
CsCl (cesium chloride)	Cubic (BCC)	2	4.11		
GaAs	Cubic (zinc blende)	8	5.635		
GaSb	Cubic (zinc blende)	8	6.118		
MgO	Rock salt (FCC)	8	4.203		
NaCl	Rock salt (FCC)	8	5.627		
UO ₂	Cubic (fluorite)	12	5.47		
ZrO ₂	Cubic (fluorite)	12	5.07		

^a Compiled from *Handbook of Chemistry and Physics*, 42nd edn (Hodgman *et al.* 1961).
(Courtesy of Chemical Rubber Co.)

^b As defined by Kronberg (1957).

linear relationship, which is referred to as Hooke's law, can be generalized to a linear relation between all nine elements of the strain tensor and all nine elements of the stress tensor, implying the presence of 81 constants of proportionality, or *elastic compliances*, s_{ijkl} , relating generically the strain tensor component ε_{ij} to the stress tensor component σ_{kl} in an expression of the type

$$\varepsilon_{ij} = \sum_{l=1}^3 \sum_{k=1}^3 s_{ijkl} \sigma_{kl} \quad (1.1)$$

or, alternatively, as a stress–strain relation

$$\sigma_{ij} = \sum_{l=1}^3 \sum_{k=1}^3 c_{ijkl} \varepsilon_{kl}, \quad (1.2)$$

where the proportionality constants c_{ijkl} are referred to as the *elastic constants*. Clearly, the elastic constants and elastic compliances s_{ijkl} of the generalized Hooke's law are related, and one set can be obtained from the other. This operation, however, is not of interest here (see Aitken 1954).

Since, in our considerations, both the stress and the strain tensors are symmetrical about the diagonal, there are only six independent stress and strain elements, making many of the elastic compliances equal to each other. This makes it possible to simplify the generalized Hooke's law to involve at most 36 elastic compliances or constants, but requires the introduction of a shorthand notation for both stress and strain to obtain a unique representation, which is referred to as the Voigt notation and which we state as follows:

$$\sigma_{11} = \sigma_1, \quad \varepsilon_{11} = \varepsilon_1, \quad (1.3a,b)$$

$$\sigma_{22} = \sigma_2, \quad \varepsilon_{22} = \varepsilon_2, \quad (1.4a,b)$$

$$\sigma_{33} = \sigma_3, \quad \varepsilon_{33} = \varepsilon_3, \quad (1.5a,b)$$

$$\sigma_{23} = \sigma_{32} = \sigma_4, \quad \varepsilon_{23} = \varepsilon_{32} = \frac{\gamma_4}{2}, \quad (1.6a,b)$$

$$\sigma_{13} = \sigma_{31} = \sigma_5, \quad \varepsilon_{13} = \varepsilon_{31} = \frac{\gamma_5}{2}, \quad (1.7a,b)$$

$$\sigma_{12} = \sigma_{21} = \sigma_6, \quad \varepsilon_{12} = \varepsilon_{21} = \frac{\gamma_6}{2}. \quad (1.8a,b)$$

In eqs. (1.6–1.8), the strain elements γ_i are referred to as the tangential shear strain elements; the use of these permits certain convenient economies in representation.

On the basis of the contracted Voigt notation for stresses and strains, the generalized Hooke's law can be written out in the form of a set of linear relations, which are, for the strain–stress relationships,

$$\begin{aligned} \varepsilon_1 &= s_{11}\sigma_1 + s_{12}\sigma_2 + s_{13}\sigma_3 + s_{14}\sigma_4 + s_{15}\sigma_5 + s_{16}\sigma_6, \\ \varepsilon_2 &= s_{21}\sigma_1 + s_{22}\sigma_2 + s_{23}\sigma_3 + s_{24}\sigma_4 + s_{25}\sigma_5 + s_{26}\sigma_6, \\ &\vdots \\ \gamma_6 &= s_{61}\sigma_1 + s_{62}\sigma_2 + s_{63}\sigma_3 + s_{64}\sigma_4 + s_{65}\sigma_5 + s_{66}\sigma_6. \end{aligned} \quad (1.9)$$

They can be abbreviated as matrix products

$$\boldsymbol{\varepsilon} = \mathbf{s}\boldsymbol{\sigma}, \quad (1.10)$$

$$\boldsymbol{\sigma} = \mathbf{c}\boldsymbol{\varepsilon}, \quad (1.11)$$

where

$$\mathbf{c} = \mathbf{s}^{-1}. \quad (1.12)$$

Here, the italic symbols are 6×1 column matrixes, $\mathbf{s}$ and $\mathbf{c}$, the elastic compliances, and the elastic constants are 6×6 matrixes. (For matrix multiplication and other operations, refer to Aitken 1954).

In an elastic solid, the increment of external work done, dW , per unit volume must be equal to the increment of elastic strain energy dU per unit volume, that is,

$$dW = dU = \sigma_1 d\varepsilon_1 + \sigma_2 d\varepsilon_2 + \sigma_3 d\varepsilon_3 + \sigma_4 d\gamma_4 + \sigma_5 d\gamma_5 + \sigma_6 d\gamma_6. \quad (1.13)$$

Since the elastic strain energy is a unique function of the state, independent on how that state was reached, it is possible to demonstrate that the elastic-compliance and elastic-constant matrixes, as defined above, must be symmetrical. This follows directly from the observation, for example, that

$$\frac{\partial^2 U}{\partial \gamma_6 \partial \varepsilon_1} = \frac{\partial \sigma_1}{\partial \gamma_6} = c_{16} = \frac{\partial^2 U}{\partial \varepsilon_1 \partial \gamma_6} = \frac{\partial \sigma_6}{\partial \varepsilon_1} = c_{61}, \quad (1.14)$$

which, of course, also holds for the inverse matrix of the elastic compliances. These arguments indicate that, even in the case of the lowest symmetry, there can only be 21 elastic compliances or constants describing elastic stress–strain relations.

If the principal axes that define the stress and strain are chosen to be parallel to the principal symmetry axes of a material, it can be readily demonstrated that for many crystal systems of interest and even for many technologically important materials possessing special processing symmetry, there will be far fewer elastic coefficients. Below, we consider some of these.

1.3.2 Orthorhombic Crystals

Orthorhombic crystals have three mutually perpendicular principal symmetry axes. Since a 180° rotation about each principal axis gives no change, there can be no linear relations between shear stresses and normal strains or between shear stresses and shear strains with different subscripts. This can be proved immediately by observing that if this were not so, the stated symmetry would not be present. This establishes that in such materials only nine independent elastic compliances (or constants) remain, which are s_{11} , s_{22} , s_{33} , s_{12} , s_{13} , s_{23} , s_{44} , s_{55} , and s_{66} . Many technologically interesting materials, such as rolled metal plates, unidirectionally produced polymer films and paper, composite sheet materials, and even wood, have such a symmetry, which is referred to as *orthotropic* symmetry when it relates to materials rather than crystals.

1.3.3 Hexagonal Crystals

Crystals with hexagonal symmetry possess only five elastic constants by virtue of the fact of having isotropic properties in the basal plane, leaving only the elastic compliances s_{11} ($=s_{22}$), s_{33} , s_{12} , s_{13} ($=s_{23}$), and s_{44} ($=s_{55}$), with s_{66} being equal to $2(s_{11} - s_{12})$ owing to isotropy in the 1–2 plane. Again we note that many technologically important materials, such as drawn fibers and products extruded

through a circular die, which have isotropic properties in the plane across the principal processing direction, have hexagonal symmetry, which in these instances is referred to as *fiber* symmetry.

1.3.4 Cubic Crystals

As Table 1.1 indicates, many crystals of elemental materials possess cubic symmetry with three mutually perpendicular and identical principal axes. For such materials, only three elastic compliances remain, which are s_{11} ($=s_{22} = s_{33}$), s_{12} ($=s_{13} = s_{23}$), and s_{44} ($=s_{55} = s_{66}$).

1.3.5 Isotropic Materials

Polycrystalline solids on a much larger scale than the grain size and possessing no texture, and amorphous solids with no principal processing direction, have uniform properties in all directions and are referred to as *isotropic*. For such materials, only two elastic coefficients are necessary to describe the elastic response. Many choices are possible for these two coefficients. The logical choice is the shear modulus μ and the bulk modulus K , describing two physically uncoupled forms of material response. For operational reasons, however, other choices are often more convenient, such as the shear modulus μ and Poisson's ratio ν . With this latter choice, we represent the general strain–stress and stress–strain relations as

$$\varepsilon_{ij} = \frac{1}{2\mu} \left(\sigma_{ij} - \frac{\nu}{(1 + \nu)} \delta_{ij} \sigma_{kk} \right) \quad (1.15)$$

and

$$\sigma_{ij} = 2\mu \varepsilon_{ij} + \delta_{ij} \lambda \varepsilon_{kk}, \quad (1.16)$$

where δ_{ij} (the Kronecker delta) is unity for $i = j$ and zero for $i \neq j$, and repeated indices such as in σ_{kk} ($=\sigma_{11} + \sigma_{22} + \sigma_{33}$) and ε_{kk} ($=\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}$) imply summation over all indices. Here $\lambda = 2\mu\nu/(1 - 2\nu)$ is one of the two Lamé constants (the other being the shear modulus μ). Finally, the Young's modulus E and the bulk modulus K are

$$E = 2\mu(1 + \nu) \quad (1.17)$$

and

$$K = \frac{1}{3} \left(\frac{\sigma_{kk}}{\varepsilon_{ii}} \right) = \frac{2\mu(1 + \nu)}{3(1 - 2\nu)}, \quad (1.18)$$

where $\sigma_{kk}/3$ is the mean normal stress (negative pressure) and ε_{ii} is the dilatation, the change of volume per unit volume. We note, in passing, that while widely used, the Young's modulus is not a pure measure of material response, as it combines both the shear and the dilatational response.

With few exceptions we shall idealize the elasticity of solids as isotropic, as already stated earlier, so as not to burden the discussion of the physical mechanisms with inessential operational detail. We note here that many cubic crystals are quite anisotropic. Tungsten, W, which is often cited as being isotropic, is only so at room temperature. Thus, we shall make use principally of the elastic relations in eqs. (1.15) and (1.16). The relationship between various combinations of elastic constants of isotropic elasticity are listed in Table 1.3 for ready reference.

1.3.6 Temperature and Strain Dependence of Elastic Response

The elastic constants and compliances of most elemental crystal structures, as well as many common compounds, together with some of their temperature dependences, can be found in Simmons and Wang (1971). The temperature dependence of elastic constants is primarily of vibrational entropic origin and roughly follows a monotonic decline between absolute zero and the melting point by about 50% (Köster, 1948). Figure 1.6 gives the temperature dependence of the Young's modulus for a selection of metals.

The elastic relations introduced above are always defined at zero strain (or stress). As strains increase in magnitude, the relations become progressively

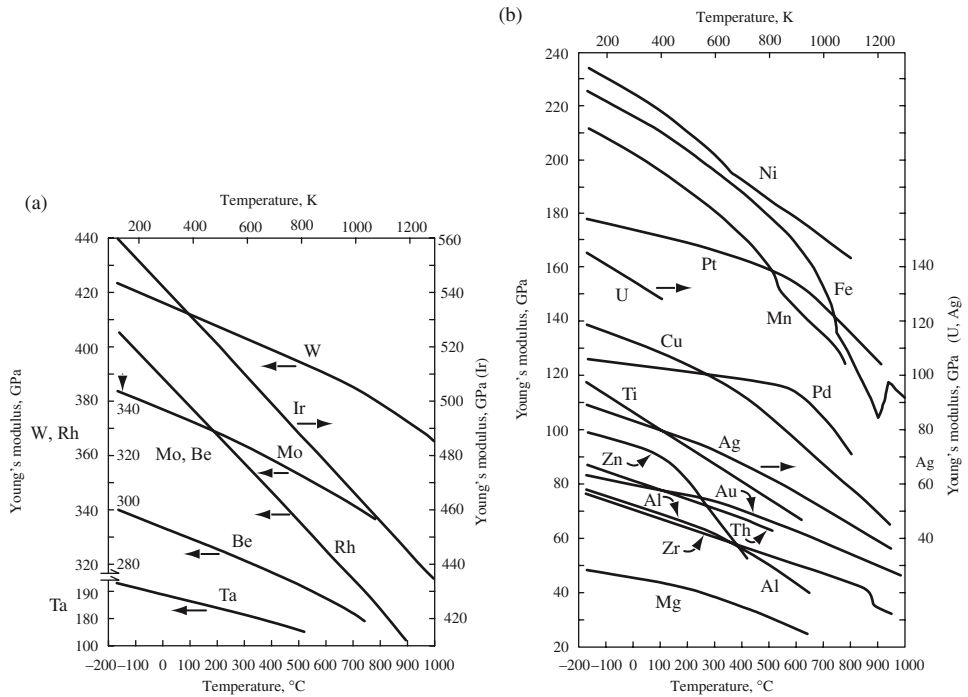


FIG. 1.6. Temperature dependence of the Young's modulus of a number of prominent metals (from Köster, 1948; courtesy of Carl Hanser GmbH). (Note the four different vertical scales in (a) and the two different vertical scales in (b).)

TABLE 1.3. Relations between isotropic elastic constants^a

Elastic constants	In Terms of				
	E, ν	E, μ	K, ν	K, μ	λ, μ
E	$= E$	$= E$	$= 3(1 - 2\nu)K$	$= \frac{9K}{1 + 3K/\mu}$	$= \frac{\mu(3 + 2\mu/\lambda)}{1 + \mu/\lambda}$
ν	$= \nu$	$= -1 + \frac{E}{2\mu}$	$= \nu$	$= \frac{1 - 2\mu/3K}{2 + 2\mu/3K}$	$= \frac{1}{2(1 + \mu/\lambda)}$
μ	$= \frac{E}{2(1 + \nu)}$	$= \mu$	$= \frac{3(1 - 2\nu)K}{2(1 + \nu)}$	$= \mu$	$= \mu$
K	$= \frac{E}{3(1 - 2\nu)}$	$= \frac{E}{9 - 3E/\mu}$	$= K$	$= K$	$= \lambda + \frac{2\mu}{3}$
λ	$= \frac{E\nu}{(1 + \nu)(1 - 2\nu)}$	$= \frac{E(1 - 2\mu/E)}{3 - E/\mu}$	$= \frac{3K\nu}{1 + \nu}$	$= K - \frac{2\mu}{3}$	$= \lambda$

^a From McClintock and Argon (1966) with modifications.

nonlinear and ultimately lead to tensile or volumetric decohesion or ideal shear. The symmetry-preserving bulk decohesion obeys a remarkably simple scaling relationship arising from a universal binding-energy relation demonstrated by Rose *et al.* (1983), which can be given for uniaxial tension simply as

$$\sigma_{11} = E_0 \varepsilon_{11} \exp(-\alpha \varepsilon_{11}), \quad (1.19)$$

where E_0 is the uniaxial “zero strain” Young’s modulus (or $(1 - \nu)E_0/[(1 + \nu)(1 - 2\nu)]$ for the case of only uniaxial strain deformation ($\varepsilon_{22} = \varepsilon_{33} = 0$)) and α is the reciprocal of the ideal uniaxial decohesion strain. From here, it is clear that the strain-dependent decrease of the Young’s modulus with increasing strain can be obtained readily by differentiation:

$$\frac{d\sigma_{11}}{d\varepsilon_{11}} \equiv E(\varepsilon) = E_0(1 - \alpha \varepsilon_{11}) \exp(-\alpha \varepsilon_{11}), \quad (1.20)$$

(this quantity is often called the tangent modulus).

The corresponding ideal shear response at large shear strains is less well understood because of shear-induced breakdown of symmetry. It is a subject under active study.

The fundamental atomic-level basis of the elastic relations and the temperature dependence of the elastic constants has been widely discussed. While first-principles connections are still rare, these are becoming better understood (see, for example, Nastar and Willaime 1995).

While the dilatational and shear responses of an elastic solid are mechanistically distinct, they are not uncoupled. The presence of a pressure results in a stiffening of the shear response and the presence of a shear stress alters the crystal symmetry, and therefore affects the bulk modulus. (For an operationally enlightening and useful treatment of these dependences, see Rice *et al.* 1992.)

1.4 Inelastic Deformation and the Role of Crystal Defects

The elastic deformation of homogeneous solids under stress can be viewed as an affine transformation in which the strains of the macroscopic, smoothly varying deformation field relate uniquely to the strains at any “point” locally, as long as these points contain at least several atoms to make the continuum model appropriate. Thus, elasticity of homogeneous solids is a local field property exhibited by all solids at small strain. Mechanistically, the affine field notion of elastic response is readily acceptable as long as the relative atomic displacements are everywhere small on the scale of interatomic distances and remain in the stable range of response of the solid. Such a solid, when perfect, will undergo a homogeneous shear strain when subjected to a homogeneous shear stress until the *ideal shear strength* is reached

everywhere, at the maximum of the interatomic shear resistance. Beyond this, the perfect solid is no longer stable and can deform to indefinitely large strains, as it undergoes catastrophic shear collapse. Thus, a perfect crystalline solid is a strongly coupled system of atoms, which at low temperatures is required to deform homogeneously and in an affine manner.

Unless extreme precautions are taken to ensure perfection, real crystalline solids usually contain lattice defects such as *vacancies* and *interstitials*, which are of point nature, line defects, consisting of *dislocations* of various types, and planar defects such as *stacking faults*, *antiphase boundaries*, and *grain boundaries*. These crystal defects contain metastable atomic configurations that weaken the coupling between adjacent volume elements and permit the local initiation of inelastic processes involving large local relative atom motions at such sites. Once initiated, some results of inelastic processes such as rearranged point defect dipoles become polarized and lose their kinematic ability to couple with the local stresses as a result of the rearrangement, and remain local. Upon removal of the stress they are driven back by the induced dipole of the local residual stresses and contribute only to the *anelastic* response, which is usually time-dependent.

Other processes, such as shear and dilatational transformations and those that involve dislocations, have the ability to propagate metastable atom configurations in a self-similar manner without any loss of coupling to the local applied stresses. Hence such defects possess mobility, which they often exercise over ranges that are orders of magnitude larger than the sizes of the dislocation cores, the thicknesses of the mobile interfaces, and so on, that constitute the metastable atom configurations. Here, it is the periodic nature of the crystal structure that promotes the mobility; the position-independent, undiminished coupling of these configurations with the local stress is essential for their mobility.

While all defects have a role to play in the inelastic deformation of crystals, it is dislocations which, through their property of self-similar translation, offer a highly efficient means of producing plastic shear strain, and it is that is of primary interest. We introduce their line properties in the required detail in Section 1.6 below, but defer the discussion of the consequences of their mobility to Chapter 2.

1.5 Vacancies and Interstitials

In the low-temperature range, where diffusional processes are too slow to be of importance, point defects play only a supporting role compared with dislocations in plasticity. Nevertheless, they are still of interest in that capacity through their various interactions with dislocations. Point defects both can be produced during glide processes of dislocations and can act as obstacles to their motion, either individually or collectively in various types of clusters. When point defects are produced in abundance by particle irradiation, their interaction with dislocations can

become quite complex. Unlike dislocations, both vacancies and interstitials can be produced by thermal processes and can exist in equilibrium concentrations governed by their energies of formation. They can also play a role in aging phenomena involving their aggregation into clusters, and in pinning dislocations, where the mobility of the latter is of interest. Their energetics, studied both experimentally and through various forms of simulation, can involve many complexities due primarily to local atomic relaxations immediately around them in the case of vacancies and relatively long-range elastic misfit fields in the case of interstitials (see Smallman and Harris 1977; Nordlund and Averback 2005).

In Table 1.4, we list a collection of experimentally measured enthalpies of formation and motion of vacancies in metals. For broad analyses, the rough estimate of Friedel (1964) for the enthalpies of formation h_v^f and h_i^f of vacancies and interstitials, respectively, given in eq. (1.21) below, can be quite useful:

$$2h_v^f \approx \frac{1}{2}h_i^f \approx \frac{1}{2}\mu\Omega, \quad (1.21)$$

where Ω is the atomic volume. The energy of formation of interstitials is much larger than that of vacancies because of the substantial elastic strain energy of the misfit that they create in their surroundings. While there are a number of theoretical

TABLE 1.4. Enthalpies of formation h_v^f and motion h_v^m of Vacancies in some FCC and BCC metals^a

	h_v^f (eV)	h_v^m (eV)	h_v^m/h_v^f
Al (FCC)	0.67 (B)	0.62 (B)	0.93
	0.73 (F)	0.65 (F)	0.89
Ag (FCC)	1.13 (B)	0.66 (B)	0.58
	0.99 (F)	0.86 (F)	0.87
Au (FCC)	0.95 (B)	0.83 (B)	0.87
	0.87 (F)	0.89 (F)	1.02
Cu (FCC)	1.28 (B)	0.71 (B)	0.55
	1.03 (F)	1.06 (F)	1.03
Fc (BCC)	1.50 (F)	1.10 (F)	0.73
Mo (BCC)	3.20 (B)	1.70 (B)	0.53
	2.30 (F)	1.70 (F)	0.74
Nb (BCC)	2.00 (F)	2.10 (F)	1.05
Pb (FCC)	0.50 (F)	0.60 (F)	1.20
Pt (BCC)	1.51 (B)	1.43 (B)	0.95
	1.49 (F)	1.38 (F)	0.93
W (BCC)	3.60 (B)	1.70 (B)	0.47
	3.30 (F)	(3.30) (F)	1.00

^a Compiled from Balluffi (1978) (B) and Franklin (1972) (F).

estimates of the formation enthalpies of interstitials, reliable experimental measurements are rare; these are generally based on the annealing kinetics of metals subjected to particle irradiation at cryogenic temperatures or quenching from high temperatures. Therefore Friedel's estimate given above may be useful. Experiments indicate that the activation enthalpy h_v^m for motion of vacancies in close-packed metals ranges from $0.6h_v^f$ to slightly over unity times h_i^v , as listed in Table 1.4. The activation enthalpies for motion of interstitials, on the other hand, are quite low, owing to their large enthalpies of formation and the local lattice dilatations that they produce around them, which permit them to glide relatively effortlessly in close-packed directions. Estimates gleaned from the annealing kinetics of radiation damage indicate that $h_i^m \approx 0.1h_v^m$ (Friedel 1964).

Vacancies and interstitials, being point defects, can be produced by thermal fluctuations so that definite atomic concentrations of them can exist in thermal equilibrium. In a large crystal, the introduction of a dilute concentration of point defects with given formation enthalpies will increase the overall enthalpy of the crystal in direct proportion to the atomic concentration. However, in compensation, the configurational entropy of the crystal based on the change in the number of separately distinguishable configurations of dispersed point defects in the lattice will initially increase faster than the increase in the overall enthalpy, and will result in a decrease in the Gibbs free energy until a minimum of the free energy is reached. A straightforward analysis of this (see Allen and Thomas 1999 or McClintock and Argon 1966) results in a thermal-equilibrium atomic concentration c of any specific point defect given by a simple Boltzmann expression of the form

$$c = \exp \left(-\frac{h_v^f}{kT} \right), \quad (1.22)$$

where h_v^f is the enthalpy of formation of a vacancy; it could equally well be that of an interstitial or any other point defect.

The formation of point defects, more specifically vacancies or interstitials, will occur by thermal fluctuations only at favorable sites, such as a free surface, incoherent grain boundaries, edge dislocation cores, and the like, where such formation is topologically unhindered. The spontaneous formation of a vacancy and interstitial as a pair inside a perfect crystal is quite unlikely under normal conditions, since the sum of their formation energies would be too high to be supplied by random thermal fluctuations, but it frequently occurs in particle irradiation. Naturally, the establishment of the above-cited equilibrium concentration will not be instantaneous but will require time for it to be established by the dispersal of the point defects formed. We shall note in later chapters that similar arguments for the equilibrium concentrations of other pointlike defects can be made using expressions similar to eq. (1.22).

1.6 Line Properties of Dislocations

1.6.1 Topology and Stress Fields of Dislocations

In crystal plasticity, a dislocation line is best conceived of as the line of termination of an incomplete lattice slip translation that has occurred by a uniform relative displacement across a crystallographic plane. Two limiting forms of dislocation lines serve as useful idealizations. A line terminating an incomplete slip operation normal to the direction vector $\mathbf{d}$ of relative displacement across the plane is called an *edge dislocation*, and one parallel to this relative-displacement direction is called a *screw dislocation*. These are depicted in Fig. 1.7 as the central lines OO' of very long right circular cylinders. The topological nature of a dislocation is characterized with the aid of a right-handed line integration operation around a unit vector $\mathbf{s}$ (parallel to the line, as shown in Fig. 1.7), which accumulates the gradients of the local displacement vector $\mathbf{u}$ of the surrounding linear elastic field along any closed contour around the line:

$$\mathbf{b} = \oint \frac{\partial \mathbf{u}}{\partial \xi} d\xi, \quad (1.23)$$

where $d\xi$ is an increment of the line integration path along the chosen contour sufficiently removed from the center of the dislocation. This line integration operation,

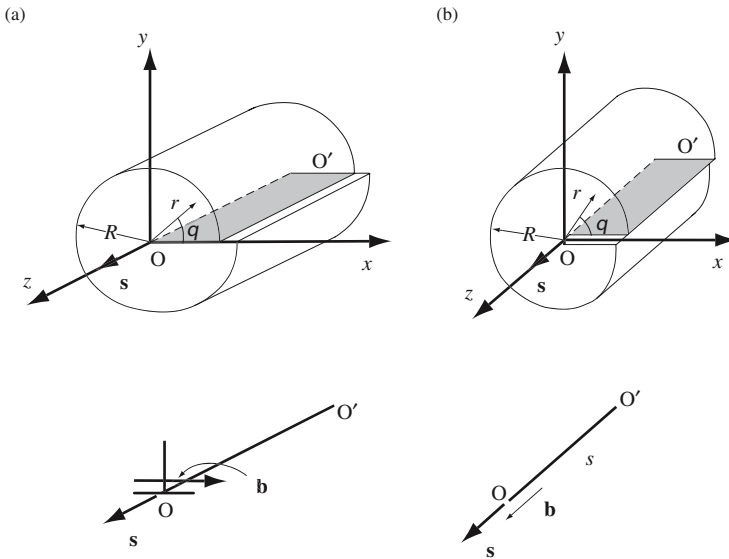


FIG. 1.7. Geometries of (a) an edge dislocation and (b) a screw dislocation, in rectangular cylinders. The vectors $\mathbf{s}$ and $\mathbf{b}$, describing the positive line direction and Burgers displacement, are shown.

which is a quantitative measure of the incompatibility that is enclosed, is a vector quantity and is called the *Burgers vector* $\mathbf{b}$, which, together with the associated line vector $\mathbf{s}$, uniquely represents the topology of the dislocation line. By convention, an edge dislocation of the type shown in Fig 1.7(a) with an extra half-plane of thickness $\mathbf{b}$ pointing upward in the positive y direction is called a positive edge dislocation and is symbolized by an inverted letter T ($\perp$). The screw dislocation in Fig. 1.7(b), having $\mathbf{b}$ parallel to $\mathbf{s}$, is called a right-hand screw dislocation because its displacement field represents a right-handed spiral ramp advancing in the direction parallel to the vector $\mathbf{s}$. Such a dislocation is symbolized by a letter S. Negative edge and left-hand screw dislocations can be represented by inverted versions of these letters.

The stress fields of these dislocations are of interest in both the rectangular x, y, z coordinate system and the polar coordinate system r, θ, z .

A positive edge dislocation in x, y, z coordinates in an isotropic elastic solid of infinite extent ($R \rightarrow \infty$) possesses the following stresses:

$$\sigma_{xx} = -\frac{\mu b}{2\pi(1-\nu)r} \sin \theta (2 + \cos 2\theta) = -\frac{\mu b}{2\pi(1-\nu)} \frac{y(3x^2 + y^2)}{(x^2 + y^2)^2}, \quad (1.24a)$$

$$\sigma_{yy} = \frac{\mu b}{2\pi(1-\nu)r} \sin \theta \cos 2\theta = \frac{\mu b}{2\pi(1-\nu)} \frac{y(x^2 - y^2)}{(x^2 + y^2)^2}, \quad (1.24b)$$

$$\sigma_{xy} = \frac{\mu b}{2\pi(1-\nu)r} \cos \theta \cos 2\theta = \frac{\mu b}{2\pi(1-\nu)} \frac{x(x^2 - y^2)}{(x^2 + y^2)^2}. \quad (1.24c)$$

In cylindrical coordinates, the stresses are

$$\sigma_{rr} = \sigma_{\theta\theta} = \frac{\mu b}{2\pi(1-\nu)r} \sin \theta, \quad (1.24d)$$

$$\sigma_{r\theta} = \frac{\mu b}{2\pi(1-\nu)r} \cos \theta. \quad (1.24e)$$

An edge dislocation represents a plane-strain problem where an out-of-plane stress $\sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$ must always be present.

For a right-hand screw dislocation in the x, y, z coordinate system, the rectangular-axis stresses are

$$\sigma_{xz} = -\frac{\mu b}{2\pi r} \sin \theta = -\frac{\mu b}{2\pi} \frac{y}{(x^2 + y^2)}, \quad (1.25a)$$

$$\sigma_{yz} = \frac{\mu b}{2\pi r} \cos \theta = \frac{\mu b}{2\pi} \frac{x}{(x^2 + y^2)}. \quad (1.25b)$$

In cylindrical coordinates, only one stress element is present,

$$\sigma_{\theta z} = \frac{\mu b}{2\pi r}; \quad (1.25c)$$

all other stress components in the x, y plane vanish. The stress field of a screw dislocation in which all stress components are out of the x, y plane represents a complement to the plane-strain problem and is often referred to as an anti-plane-strain problem. For finite-size cylinders, other so-called *image stresses* are present and must be added. We present these in the appendix to this chapter.

Equations (1.24d) and (1.24e) indicate that the plane-strain stress field of an edge dislocation at any point r, θ is made up of a dilatational field, varying as $\sin \theta$, and an associated shear field, varying as $\cos \theta$. This indicates that while there are local dilatations around an edge dislocation, the dilatation integrated around the dislocation will vanish. The stress field of a screw dislocation in cylindrical coordinates given by eq. (1.25c), on the other hand, reflects directly the spiral-ramp nature of the displacement field, advancing uniformly with equal increments of the angle θ .

In anisotropic crystals, the stress fields of dislocations on specific crystallographic planes with Burgers vectors in specific crystallographic directions require a more complex representation, which, however, can be summarized as

$$\sigma_{ij}^e = \left(\frac{bK_e}{2\pi r} \right) \tilde{\sigma}_{ij}^e(\theta), \quad \sigma_{ij}^s = \left(\frac{bK_s}{2\pi r} \right) \tilde{\sigma}_{ij}^s(\theta), \quad (1.26a,b)$$

where K_e for the edge dislocation and K_s for the screw dislocation are functions of the appropriate elastic constants and the orientation of the dislocation line relative to the lattice, and where the θ -dependent functions $\tilde{\sigma}_{ij}^e(\theta)$ and $\tilde{\sigma}_{ij}^s(\theta)$ for the edge and screw dislocation depend also on the elastic constants and are different from those appearing in eqs. (1.24a–c) and (1.25a,b). Specific forms of them have been calculated for many relevant crystal dislocations in important lattices and can be found in advanced treatises on dislocations such as that by Hirth and Lothe (1982).

When the dislocation line is of mixed nature (part edge and part screw), for which the Burgers vector $\mathbf{b}$ makes an angle ϕ with the line vector $\mathbf{s}$, the stress field is obtained by superposition of the separate stress components, and is given by

$$\sigma_{ij} = \frac{(b \sin \phi) \mu}{2\pi(1-\nu)r} \tilde{\sigma}_{ij}^e + \frac{(b \cos \phi) \mu}{2\pi r} \tilde{\sigma}_{ij}^s, \quad (1.27a)$$

where $\tilde{\sigma}_{ij}^e$ and $\tilde{\sigma}_{ij}^s$ are the θ -dependent functions in eqs. (1.24a–c) for an edge dislocation and (1.25a,b) for a screw dislocation in an isotropic elastic medium. For

anisotropic crystals, the corresponding expression is

$$\sigma_{ij} = \frac{(b \sin \varphi) K_e}{2\pi r} \tilde{\sigma}_{ij}^e(\theta) + \frac{(b \cos \varphi) K_s}{2\pi r} \tilde{\sigma}_{ij}^s(\theta). \quad (1.27b)$$

A dislocation line can “snake around” a crystal along an arbitrary curve in space, assuming all conceivable mixtures between edge and screw. By virtue of the fixed incompatibility that it contains, however, a dislocation line must preserve the magnitude and direction of its Burgers vector along it. Thus a dislocation cannot terminate inside a crystal. It can only terminate on a free surface; on itself, forming a loop; or on other dislocations, at nodes where the sum of the Burgers vectors of the dislocations flowing into and out of the node forms a closed polygon; or it can “diffuse away” on an incoherent grain boundary where the incompatibility can be dispersed by incoherent shear.

Examination shows that all stress components of dislocations diverge as $r \rightarrow 0$. This is an artifact arising from the use of Hooke’s law in the basic solution of the dislocation stress field problem. When strains become unbounded as $r \rightarrow 0$, Hooke’s law breaks down and stresses are no longer linearly proportional to strains. All solids possess an ideal shear strength σ_{is} , which depends on the nature of the interplanar shear coupling and ranges roughly from about 0.03μ for close-packed metals to about 0.1μ for covalent solids. Thus the stresses in eqs. (1.24) and (1.25) could be formally cut off at a radius r_c , the core radius, where the ideal shear strength is reached. An early and still meaningful estimate of this was given by Foreman *et al.* (1951) and is

$$r_c \approx \frac{\mu b}{4\pi \sigma_{is}}. \quad (1.28)$$

Then, for $r < r_c$ as $r \rightarrow 0$, all stresses go to zero by arguments of antisymmetry. In reality, significant nonlinear effects in the stress fields of dislocations reach out considerably beyond r_c (Seeger and Haasen 1958), but our simple cutoff procedure will serve for our considerations. Thus, the stress expressions in eqs. (1.24) and (1.25) are applicable only for $r > r_c$, that is, outside the dislocation core, where most of the important interactions take place. Parenthetically, this also applies to the line integral in eq. (1.23) that defines the Burgers vector. This integral has to be constructed well outside the core region to give unique answers.

1.6.2 Line Energies of Dislocations

The elastic strain energy $\mathfrak{F}$ of a dislocation line is readily calculable from the stress fields given in eqs. (1.24) and (1.25). The result is, for isotropic solids,

$$\mathfrak{F}_e = \frac{\mu b^2}{4\pi(1-\nu)} \ln\left(\frac{\alpha R}{b}\right), \quad \mathfrak{F}_s = \frac{\mu b^2}{4\pi} \ln\left(\frac{\alpha R}{b}\right), \quad (1.29a,b)$$

where $\mathfrak{F}_e$ and $\mathfrak{F}_s$ stand for the elastic line energy per unit length of edge and screw dislocations within a range R of the outer field. These expressions also incorporate the contribution from the energy inside the nonlinear core by means of the factor α , which is obtained from atomistic computations of the core structure and is ~ 1.0 for close-packed crystals and can be as large as 4.0 for covalent crystals with larger ideal shear strains and a higher core energy density (Hirth and Lothe 1982).

In anisotropic solids, these line energies become

$$\mathfrak{F}_e = \left(\frac{b^2 K_e}{4\pi} \right) \ln \left(\frac{\alpha R}{b} \right), \quad (1.30a)$$

$$\mathfrak{F}_s = \left(\frac{b^2 K_s}{4\pi} \right) \ln \left(\frac{\alpha R}{b} \right) \quad (1.30b)$$

for edge and screw dislocations, respectively.

For mixed dislocations, the line energy per unit length is again obtainable by a direct superposition of individual contributions, since there is no interaction between parallel edge and screw dislocations, giving

$$\mathfrak{F}(\phi) = \frac{\mu b^2}{4\pi} \left(\frac{1 - \nu \cos^2 \phi}{(1 - \nu)} \right) \ln \left(\frac{\alpha R}{b} \right) \quad (1.31a)$$

for isotropic crystals, and

$$\mathfrak{F}(\phi) = \frac{b^2}{4\pi} (K_e \sin^2 \phi + K_s \cos^2 \phi) \ln \left(\frac{\alpha R}{b} \right) \quad (1.31b)$$

for the general anisotropic case.

As all the line energy terms indicate, they are logarithmically divergent in R . For real crystals containing many dislocations with a mean spacing l , R is usually taken as l on the assumption that dislocations of opposite sign will pair up to first order as dipoles to shield each other's long-range stress fields.

Further examination of these line energies reveals that, since dislocation lines are relatively inflexible on the atomic scale, the configurational entropy of a crystal containing a mildly wavy dislocation is usually negligible on the scale of the elastic line energy (Cottrell 1953). However, because the shear modulus (and more generally the elastic constants) is temperature-dependent owing to the vibrational entropy of the atoms on the lattice sites, there is substantial vibrational entropy locked into the dislocation strain field. Thus, if the modulus μ or, more generally, the anisotropic factors K_e and K_s are considered with their proper temperature dependence, the line energies given in eqs. (1.29)–(1.31) represent the Helmholtz free energy of the crystal containing the dislocation. The vibrational entropy contribution to the

free energy of the line is then obtainable formally from (Kocks *et al.* 1975)

$$S_v = \frac{\partial \mathfrak{F}}{\partial T} = \mathfrak{F} \left(-\frac{\partial \ln \mu}{\partial T} \right) \quad (1.32)$$

on the assumption that the temperature dependence of the average shear modulus is, to first order, an adequate representation of that of all the elastic constants.

From considerations of the third law of thermodynamics, the logarithmic derivative of the shear modulus must vanish at absolute zero and systematically decrease up to approximately the Debye temperature, where it should reach a roughly constant value. Thus, for most close-packed metals and transition metals, this results in a vibrational entropy contribution in the range of $S_v \approx 0.6\mathfrak{F}_0/T_m$, where $\mathfrak{F}_0$ is the extrapolated line energy at absolute zero and T_m is the melting point.

For order-of-magnitude considerations of the line energy of a mixed dislocation, it is useful to evaluate eqs. (1.29a,b) for a typical case of a moderately deformed crystal, which is usually taken to correspond to a mean dislocation spacing of ~ 100 nm. With this simplification, and not distinguishing between edge and screw dislocations, the numerical terms in eqs. (1.29a,b) become ~ 0.5 , resulting in a line energy per unit length of roughly

$$\mathfrak{F} \approx \mathfrak{F}_s \approx \mathfrak{F}_e \approx \frac{\mu b^2}{2}, \quad (1.33)$$

regardless of the orientation of the line. This represents the “constant-line-energy” model of a dislocation line.

The magnitude of the free energy of a dislocation line per atomic length evaluated from eq. (1.33) is then $\mu b^3/2$ and is typically ~ 3 eV for a close-packed metal such as copper at its Debye temperature. Of this energy, roughly 10–15% resides in the core of the dislocation (Hirth and Lothe 1982). According to eq. (1.32), the free energy per atomic length of a dislocation at the melting point should typically have dropped to $\sim 40\%$ of its value at the Debye temperature, that is, to ~ 1.6 eV. Comparison of this with the average thermal energy per atom of ~ 0.2 eV at the melting point for copper indicates that the free energy of a dislocation line per atomic length is 8–160 times the average thermal energy per atom between the melting point and the Debye temperature. This conclusion has very important consequences for the near-impossibility of generating dislocations by thermal fluctuations. The topic of the forms of generation of dislocations will be discussed in Chapter 2.

1.7 Planar Faults

On a hierarchical level of increasing dimensionality, there are a variety of planar defects of interest, which include *stacking faults*, *coherent twin boundaries*, *antiphase boundaries*, and *grain boundaries*. Each of these could receive expanded treatment in its own right. We shall make contact with all of them in later chapters

to some extent, when their properties influence certain strength-related phenomena. Because of their more widespread effect on dislocation mechanics, however, we discuss here only stacking faults, which were already touched on briefly when discussing the different nature of the atomic close packing of FCC and HCP structures.

Thus, we note again that both FCC lattices and HCP structures with an ideal c/a ratio can be viewed, to a first approximation, as a close packing of hard spheres, differing only in the packing sequence of atomic layers on top of each other; the nearest-neighbor relationships of the atoms, which have a coordination of 12, are indistinguishable between FCC and HCP, and with the differences arising only in the next-nearest-neighbor relationships. If the contribution to the overall free energy of the crystal of the next-nearest-neighbor interactions of atoms is small, *stacking faults* of local hexagonal stacking in FCC crystals and FCC stacking in HCP crystals should be encountered with a finite probability.

Stacking faults in FCC crystals can arise in several ways. If, in the normal stacking order of ... ABCABC ... discussed in Section 1.2 earlier, an atomic layer, say a “C” layer, is removed and the structure is collapsed, the new stacking has the sequence ... AB | ABCABC ..., where the vertical line represents the withdrawn “C” layer, creating an *intrinsic* stacking fault upon the collapse of the right side of the crystal onto the left. This has created, around a plane, an HCP stacking two atomic layers thick. Such a stacking fault could arise in two ways. First, it could occur if a layer of vacancies were to condense in a close-packed plane, followed by collapse of the two parts of the crystal to remove the plane of vacancies. More interestingly, the same result can be achieved if the side of the crystal including the “C” plane is translated from the “C” interstitial site of the “A” layer by a “half step” to the “A” interstitial site of the “B” layer. This would again create the same stacking fault. In the second alternative, the stacking fault is created not by vacancy collapse, but by a shear translation equal to the separation of two neighboring interstitial sites, having a length of $b/\sqrt{3}$, where b is the Burgers identity translation distance between two neighboring atoms in the close-packed plane. As we discuss in Chapter 2, such translations can result from the passage of a *Shockley partial dislocation* (named after Shockley, who considered the possibility first), which produces an intrinsic stacking fault through its passage rather than restoring full order, which a perfect dislocation of *Burgers vector* $\mathbf{b}$ would do.

A stacking fault of a different type could arise when instead of removal of a close-packed plane, an extra plane of close-packed atoms is introduced, associated with a separation of the two sides of the crystal rather than a collapse. This would result in a stacking change given as ... AB | ACABC ..., which is referred to as an *extrinsic* stacking fault and could, for example, form through the aggregation of a platelet of interstitial atoms between the “B” and “C” layers. Such a stacking fault, however, cannot be formed through a “half-step” translation, and as a result is of lesser interest than the intrinsic stacking fault.

Complementary considerations are applicable to HCP structures, where similar faults can arise. Because stacking faults create disturbances only in the next-nearest-neighbor relationships, their energies tend to be quite small in comparison with grain boundary energies or surface free energies. Stacking-fault energies can be measured in a number of ways, some of which we discuss in later chapters in the context of partial dislocations, extended dislocations, and their nodes. There has been considerable recent effort to calculate stacking-fault energies by various first-principles approaches, which on the whole have greater utility for developing a relative scaling of stacking-fault energies of close-packed metals when normalized with experimental measurements, than for giving quantitatively more accurate determinations. In Table 1.5, energies of intrinsic stacking faults for a number of pure FCC and HCP metals, based on a variety of direct experimental measurements are presented. Stacking-fault energies in dilute solid-solution alloys are also of great interest. Some relevant values of these will be presented in later chapters in the context of their usage. Stacking-fault energies in compounds (for a series of II–VI compounds such as CdTe, see Takeuchi *et al.* 1984), in certain oxides and nitrides (Suzuki *et al.* 1994), and in a substantial family of tetrahedrally coordinated layered crystals (Takeuchi and Suzuki 1999) have also been determined experimentally, to name a few more complex cases of other close-packed structures.

TABLE 1.5. Stacking-fault energies in some close-packed materials (in mJ/m^2)^a

Material	Type	χ_{sf}
Al	FCC	200 (HL)
Ag	FCC	17 (HL); 20 (C)
Au	FCC	55 (HL); 35 (C)
Be	HCP	>230 (F)
Cd	HCP	160 (F)
Cu	FCC	73 (HL)
Ge	Diamond cubic	75 (PH)
Mg	HCP	~125 (F)
Ni	FCC	400 (HL); 125 (C)
Pd	FCC	180 (HL)
Pt	FCC	95 (HL)
Rh	FCC	~750 (HL)
Si	Diamond cubic	69 (FC)
Th	FCC	~750 (HL)
Zn	HCP	~230 (F)

^a References: HL, Hirth and Lothe (1982) (and references within); C, Coulomb (1978); F, Fleischer (1986); FC, Föll and Carter (1979); PH, Packeiser and Haasen (1977).

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Appendix: Dislocation Stress Fields in a Finite Cylinder

There are some problems where the stress fields of dislocations in cylinders of finite radius R are of interest. To obtain them, additional image stress solutions need to be added to those represented in eqs. (1.24a–e) for edge dislocations and (1.25a–c) for screw dislocations to render the cylindrical surface stress-free. These solutions are readily obtainable by standard procedures (Cottrell 1953).

Thus, the full solutions for the in-plane stresses of a positive edge dislocation are best represented in the form appearing in eqs. (1.24d,e) and become

$$\sigma_{rr} = \frac{\mu b}{2\pi(1-\nu)r} \left(1 - \left(\frac{r}{R} \right)^2 \right) \sin \theta, \quad (1.34)$$

$$\sigma_{\theta\theta} = \frac{\mu b}{2\pi(1-\nu)r} \left(1 - 3 \left(\frac{r}{R} \right)^2 \right) \sin \theta, \quad (1.35)$$

$$\sigma_{r\theta} = \frac{\mu b}{2\pi(1-\nu)r} \left(1 - \left(\frac{r}{R} \right)^2 \right) \cos \theta. \quad (1.36)$$

The full solution for the stress field of a right-hand screw dislocation, on the other hand, is

$$\sigma_{\theta z} = \frac{\mu b}{2\pi r} \left(1 - 2 \left(\frac{r}{R} \right)^2 \right). \quad (1.37)$$

KINEMATICS AND KINETICS OF CRYSTAL PLASTICITY

2.1 Overview

Crystal plasticity is overwhelmingly a consequence of self-similar translations of dislocations, which we view as a limiting form of more general shear transformations that also include twinning and martensitic transformations. Unlike elastic deformation, which can be homogeneous down to the atomic scale, plastic deformation, involving dislocation translations or other shear transformations, is locally discrete and inhomogeneous. It can be viewed as homogeneous only when the discrete processes are *homogenized* over a representative volume element (RVE), taken large enough to represent *quasi-smooth* behavior. Thus, for the purpose of viewing plasticity as a continuum field theory, its applicability is limited to volume elements no smaller than the RVE over which processes have been homogenized.

In this chapter, we develop the essential kinematics of crystal plasticity, the dislocation line properties that are associated with the production of plastic strain, and the expenditure of plastic work. Moreover, we consider broadly the thermodynamics and kinetics of crystal plasticity for more detailed application in the discussion of the various plastic resistance mechanisms to be considered in later chapters.

2.2 Kinematics of Inelastic Deformation

2.2.1 Plasticity Resulting from Shear Transformations

Inelastic deformation cannot be exhibited by a perfect crystal under any stress less than the ideal shear strength for slip or twinning. It requires for its local development a defective solid state which permits various forms of local strain-producing configurations to take shape. In the broadest sense, all inelastic deformation can be viewed as arising from a set of shear transformations, occurring in small volume elements, stimulated by the prevailing local stress state, and emanating from or involving a variety of crystal or structural defects. Thus, inelastic deformation is always inhomogeneous on the scale of the mechanism and must be defined as an average over a volume element that must contain enough elementary transformations to result in

an acceptably smooth process. The smallest such volume element above which the inelastic response can be considered smooth will be referred to as a representative volume element (RVE). This represents the scale of a so-called “material point” in any continuum field description of the process.

We consider first the kinematic aspects of inelastic deformation, separately from their energetic aspects, or the driving forces that cause them to develop. While such a separation is instructive for clarifying the kinematics, it is not realistic for understanding the overall nature of the development of shear relaxations and their energetics. Therefore, caution is necessary.

Consider, as shown in Fig. 2.1(a), a shear transformation in a small volume element Ω_f occurring inside a large volume V . We denote by $\boldsymbol{\varepsilon}^T$ the unconstrained transformation strain (tensor) that would develop freely outside the body. When constrained by the elastic surroundings, the transformation strain of the interior is only $\boldsymbol{\varepsilon}^c$ when all distant stresses are relaxed. If the volume element Ω_f is an ellipsoid and the exterior volume V is very large (infinite), the inner field is homogeneous (Eshelby 1957). When the outer body is of finite size, the effect of the transformed volume on the outside is to set up a compensating effective homogenous “image strain”, in the same sense as the transformation strain inside Ω_f (Eshelby 1954). Thus, if a total volume fraction c of material has undergone the same transformation,

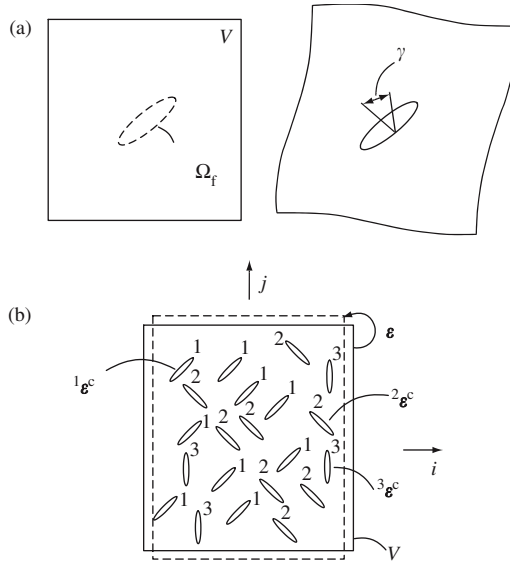


FIG. 2.1. (a) Shear transformation with a transformation strain γ in a volume Ω_f ; (b) three families of transformations with different principal directions, producing constrained tensorial transformations $1\boldsymbol{\varepsilon}^c$, $2\boldsymbol{\varepsilon}^c$, $3\boldsymbol{\varepsilon}^c$ in a representative volume element V , resulting on the whole in an external tensor transformation $\boldsymbol{\varepsilon}$.

then the total volume-average plastic strain in the large body as a whole becomes (Nabarro 1940)¹

$$\boldsymbol{\varepsilon} = c \boldsymbol{\varepsilon}^T, \quad (2.1a)$$

and in incremental form,

$$d\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^T dc. \quad (2.1b)$$

If there are m different types of transformation, as depicted in Fig. 2.1(b), each producing separately unconstrained transformation strains $\boldsymbol{\varepsilon}_m^T$ and having volume fractions c_m , then collectively they produce an overall volume-average plastic strain of

$$\boldsymbol{\varepsilon} = \sum_m c_m \boldsymbol{\varepsilon}_m^T, \quad (2.2a)$$

and in incremental form,

$$d\boldsymbol{\varepsilon} = \sum_m c_m \boldsymbol{\varepsilon}_m^T dc_m, \quad (2.2b)$$

provided that $\sum_m c_m$ remains relatively small so that mutual interactions between volume elements Ω_{fm} can be neglected, and all transformed volume elements have the same mechanical properties.

2.2.2 Plasticity Resulting from Dislocation Glide

When the inelastic deformation is produced by a uniform shear strain $\delta\gamma$ in a set of identical lenticular (penny-shaped) volume elements with principal planes characterized by a unit normal vector $\mathbf{n}$, and a shear direction parallel to $\mathbf{d}$, the average inelastic strain increment in the sample is still the volume average of all inelastic shears and is given by

$$\begin{aligned} \delta \varepsilon_{ij}^p &= \left(\frac{h \delta\gamma}{V} \right) \int_A \frac{1}{2} (d_i n_j + d_j n_i) dA \\ &= c \delta\gamma \frac{1}{2} (d_i n_j + d_j n_i), \end{aligned} \quad (2.3)$$

where h is the average thickness of the lenticular transformations, d_i and n_j are the components of the unit vectors $\mathbf{d}$ and $\mathbf{n}$ in the direction of the macroscopic axes i and j , the area integral is over the principal-plane areas of the entire set of shear transformations, and c is their total volume fraction.

¹ In this and subsequent sections of this chapter, the strains that are being referred to are all inelastic strains. When they are experienced, they occur in an elastically strained background. Thus, they should be conceived of as residual strains that remain after the deformed body is unstressed.

Clearly, if the shear is produced by a set of m separate transformations having distinct plane normals $\mathbf{n}^m$, shear directions $\mathbf{d}^m$, and shear increments $\delta\gamma_m$, the overall increment of plastic strain in the macroscopic sample is

$$\delta\varepsilon_{ij}^p = \sum_m c_m \delta\gamma_m \alpha_{ij}^m, \quad (2.4a)$$

and

$$\alpha_{ij}^m = \frac{1}{2} (d_i^m n_j^m + d_j^m n_i^m) \quad (2.4b)$$

is called the Schmid strain resolution tensor, which serves to resolve the local-mechanism strains onto the external axes i and j . Finally, when the shears are due to individual translations of dislocation loops, or to sweeping of dislocation segments across portions of slip planes having unit normal vectors $\mathbf{n}$, with Burgers vectors in the direction $\mathbf{d}$ and having magnitudes b , the local shears degenerate into discontinuous translations, and the strain increments, hereafter called *plastic*, of a set of m separate slip systems become, as a special case of eqs. (2.4a,b),

$$\delta\varepsilon_{ij}^p = \sum_m b_m \frac{\alpha_{ij}^m \delta a_m}{V}. \quad (2.5)$$

Here α_{ij}^m is of the same form as that given in eq. (2.4b), and the swept-out area δa_m is for the entire length for dislocation lines for the specific set m under consideration. Shear operations of this type, depicted in Fig. 2.2, and resulting from relative translation of parts of the crystal over each other in the direction of the Burgers vector are called *slip*.²

Often, single-slip systems are of interest in mechanistic considerations, for which eq. (2.5) simplifies to

$$\delta\varepsilon_{ij}^p = \left(\frac{b \delta a}{V} \right) \alpha_{ij}, \quad (2.6)$$

where δa is again the entire area increment swept out by mobile segments of dislocations on the slip planes oriented in relation to the external axes by α_{ij} , as depicted in Fig. 2.2. When this motion can be well characterized as an average displacement δx of a total segment length Λ of dislocations, we have

$$\delta\varepsilon_{ij}^p = \left(\frac{b \Lambda \delta x}{V} \right) \alpha_{ij} = b \rho_m \alpha_{ij} \delta x, \quad (2.7)$$

where, in the final form, $\rho_m (= \Lambda/V)$ has been introduced as the density of mobile dislocations (segments), that is, the mobile line length of dislocations per unit volume.

² We shall use *slip* to refer to a completed translation and *glide* to refer to the conservative motion of dislocations.

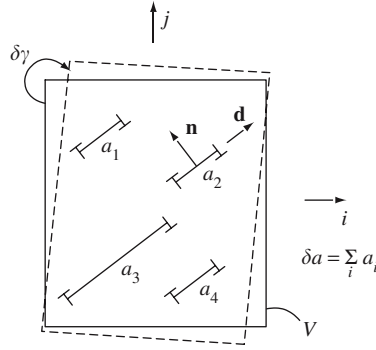


FIG. 2.2. Formation of four dislocation pairs, all on parallel planes with the same unit normal vectors $\mathbf{n}$ and shear directions $\mathbf{d}$, having swept out areas $a_1, \dots, a_4$, producing an overall shear distortion in the representative volume element V .

Finally, whenever time enters into the motion of dislocations to establish a kinetic basis of the process, eq. (2.7) is readily transformed into a rate expression as

$$\dot{\epsilon}_{ij}^p = b \rho_m \alpha_{ij} \dot{x}. \quad (2.8)$$

Equation (2.8) relates the internal glide rate, based on the dislocation velocities $\dot{x}$, to the external strain rate, and constitutes a dislocation-mobility-based kinematic expression.

In certain instances, the kinematics of the strain rate needs to be viewed differently, when a characteristic incubation (activation) time t_a must elapse for a dislocation *embryo* of size δa to be formed. This then produces a variant of eq. (2.6) representing a nucleation-controlled deformation process and gives rise to a plastic strain rate of

$$\dot{\epsilon}_{ij}^p = b \frac{\delta a}{t_a} \frac{1}{V} \alpha_{ij}, \quad (2.9)$$

due to slip.

This limiting form has applications to phenomena such as twinning, martensitic transformations, and, in the case of dislocation glide, to problems of the thermal overcoming of specific slip-plane obstacles.

2.2.3 Lattice Rotations Accompanying Slip

Slip results in lattice rotations that are instrumental in the development of deformation textures, which we shall discuss briefly in chapter 8.

The relevant concepts are illustrated schematically in Fig. 2.3. Figure 2.3(a) illustrates a reference volume element (an RVE or larger), along with the unit

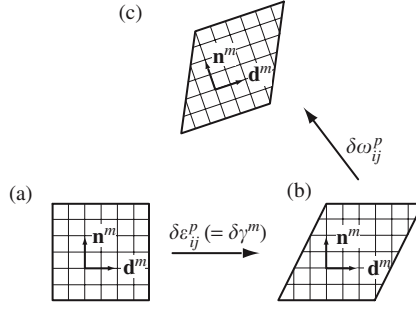


FIG. 2.3. General steps in a deformation process: (a) initial square element; (b) the element in (a) has sheared by an increment $\delta \epsilon_{ij}^p (= \delta \gamma^m)$; (c) the sheared element has undergone a rigid-body rotation $\delta \omega_{ij}^p$.

vectors $\mathbf{n}^m$ and $\mathbf{d}^m$ representing the slip plane normal and the slip direction of the system m . Without loss of generality, the instantaneous configuration can be taken as the external reference for the shape. At fixed lattice orientation, an increment of crystallographic shear $\delta \gamma_m$ in the system m generates an increment of plastic strain $\delta \epsilon_{ij}^p$ and an increment of lattice rotation $\delta \omega_{ij}^p$. These increments provided by the m th system are given by³

$$\delta \epsilon_{ij}^p = \alpha_{ij}^m \delta \gamma_m, \quad \delta \omega_{ij}^p = \beta_{ij}^m \delta \gamma_m, \quad (2.10a,b)$$

where

$$\beta_{ij}^m = \frac{1}{2} (d_i^m n_j^m - d_j^m n_i^m) \quad (2.11)$$

is the skew-symmetric counterpart of the Schmid strain resolution tensor. The resulting intermediate deformed configuration of the body is depicted in Fig. 2.3(b). It must now be noted that the total crystallographic plastic strain increment and plastic-strain-induced rotation must be given by sums over all possible slip systems k , as follows:

$$\delta \epsilon_{ij}^p = \sum_{m=1}^k \alpha_{ij}^m \delta \gamma_m, \quad \delta \omega_{ij}^p = \sum_{m=1}^k \beta_{ij}^m \delta \gamma_m. \quad (2.12a,b)$$

Homogenous (or homogenized on the macroscale) slip processes alone, in general, alter lengths and orientations of fiducial lines and angles drawn on the element,

³ In keeping with the previously declared simplification, we are ignoring the lattice rotation of the associated elastic strain component. This amounts to a so-called rigid plastic idealization.

but leave the lattice unchanged. Lastly, the intermediate configuration could be subject to a strain-free rigid-body rotation increment $\delta\omega_{ij}^p$, which carries both material line elements and lattice vectors to their final orientations. This is shown in Fig. 2.3(c). This final step of rigid-body rotation is important in the operational analysis of deformation-processing problems, but of no great interest to us here in the understanding of mechanisms.

Although all the kinematic relations discussed above have been derived on the basis of shear and translations of parallel planes, they are applicable without change to dilatations as well. Thus, for example, if the areas in eq. (2.5) or (2.6) were to be separated (or contracted) rather than translated across, by a distance of magnitude b , parallel to the unit normal vector $\mathbf{n}$, this would be reflected in the definition of the resolution factor α_{ij} ; in this case of thickening or thinning of the area, $\mathbf{n}$ is parallel to $\mathbf{d}$. The type of dislocation motion which accomplishes this is called *climb* and is associated with material diffusion.

2.3 Flexure and Motion of Dislocations under Stress

2.3.1 Interaction of a Dislocation Line with an External Stress

A stationary dislocation in a solid does not interact with an external stress, and, except for the slight inhomogeneity of the core region due to its extreme nonlinearity, there is no means of detecting the presence of stationary dislocations in a crystal from external stiffness measurements. When a dislocation sweeps out an area δa , however, as eq. (2.7) shows, the solid undergoes plastic strain and an external stress σ_{ij} does an increment of work δW on the solid given by

$$\delta W = \sigma_{ij} \delta \varepsilon_{ij}^p V = b \alpha_{ij} \sigma_{ij} \delta a = b_i \sigma_{ij} \delta a_j = L b_i \sigma_{ij} \epsilon_{jkl} s_k \delta \xi_l, \quad (2.13)$$

where repeated indices imply summation over them. The symbol ϵ_{ijk} is an element of the permutation tensor, which has a value of $+1$ for an even permutation of the indices 1, 2, 3, a value of -1 for an odd permutation of the indices 1, 2, 3, and a value of 0 for any repeated index.⁴

⁴ Equation (2.13) represents the work done on the system consisting of the solid of volume V by an external stressing agency, in response to the displacement of a dislocation line. Alternatively, it is possible to consider a different system, consisting of both the solid volume V and the stressing agency. Then δW , as stated in eq. (2.13), would represent a decrease of the potential energy of this combined system, due to the displacement of the dislocation line, in the presence of no other interaction with anything external to this system. While this would permit us to define “forces” on the dislocation line as negative gradients of the potential energy of this larger system due to the displacement of the dislocation, we choose the more transparent approach outlined in this section.

In the above train of equations, the next to last form results from the translational nature of slip over an area element represented as a vector normal to the area element and having a magnitude equal to that element. The final expression introduces a cross product for the area element, made up of a vector Ls parallel to the dislocation line and equal to the segment length L displaced, multiplied by a vector $\delta\xi$ representing the displacement of that segment of the dislocation in the plane of the sweep. This permits the introduction of a conceptual force $\mathbf{f}$ equal to the gradient of the external work done per unit line length as

$$f_l = \left(\frac{1}{L} \right) \delta W_l = b_i \sigma_{ij} \epsilon_{jkl} s_k, \quad (2.14)$$

where the comma implies differentiation with respect to a quantity, in this case in the direction of l . Thus, by virtue of displacement of distant boundaries due to displacement of the dislocation in any direction l normal to itself, the external stresses do work on the solid, which can be imagined to be affected by a conceptual force f acting along the dislocation line and displacing it by an amount $\delta\xi$. The conceptual force f in eq. (2.14) is known as the Peach–Koehler force (Hirth and Lothe 1982).

For an edge dislocation, two distinct directions of motion are distinguishable: the glide motion parallel to the Burgers vector, which *conserves* the extent of the extra half-plane, and the climb motion perpendicular to the Burgers vector, which is *nonconservative* with regard to the extent of the extra half-plane and requires diffusion of matter. The special forms of the glide and climb components of the force f per unit length are

$$f_{\text{glide}} = b\sigma_s, \quad (2.15a)$$

$$f_{\text{climb}} = b\sigma_n, \quad (2.15b)$$

where σ_s is the shear stress on the glide plane in the glide direction, with a sign that can usually be identified readily by inspection, and σ_n is the normal stress acting across the extra half-plane, which, when positive, produces climb so as to *extend* the extra half-plane. For screw dislocations, all motions in all directions perpendicular to the dislocation line are kinematically conservative and produce glide, giving a force per unit length of the same form as eq. (2.15a), where, however, the sign of the force is best obtained from eq. (2.14).

The stress external to a dislocation can be the result of another dislocation acting at the site of the test dislocation. Thus, in general, dislocations interact, exerting forces on each other, provided that the required stresses σ_s or σ_n are present in the makeup of the interacting dislocation. In the case, for example, of parallel edge and screw dislocations, the stresses of each dislocation that might produce forces on

the other do not exist in their makeup, resulting in no interaction between them—a fact that has already been made use of in obtaining the line energy of a mixed dislocation given by eq. (1.31).

2.3.2 Interaction Energies of Dislocations with Stresses External to Them

An important alternative statement of the above development leading to the introduction of the Peach–Koehler conceptual force is to state the problem in terms of the energy of interaction of the dislocation with the external stress agency, which has wide ranging uses later.

Thus, consider a unit length of a positive edge dislocation as shown in Fig. 2.4, to be displaced in either the x or the y direction by a distance dx or dy in the presence of a uniform stress σ , containing all components of the general stress tensor. The displacement of the dislocation will then interact with the stress field either positively or negatively. If the displacement is in the $y = 0$ plane in the positive x direction by a distance dx , the resulting relative displacement of the dislocation would relax the tractions σ_{xy} across the plane (negative interaction) by an amount

$$dI = -\sigma_{xy}b \, dx, \quad (2.16)$$

resulting in a negative increment of interaction energy, and a driving force in the x direction of the dislocation equal to

$$f_x = -\frac{dI}{dx} = \sigma_{xy}b. \quad (2.17)$$

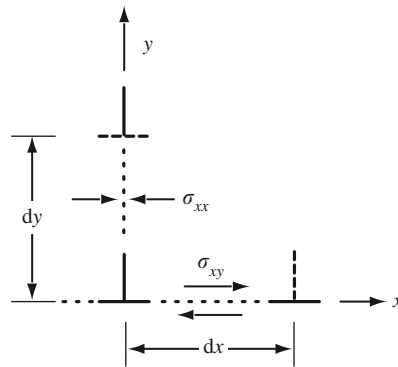


FIG. 2.4. A positive stress σ_{xy} displaces a positive edge dislocation in the positive x direction, and a negative stress σ_{xx} displaces a positive edge dislocation in the positive y direction.

Alternatively, if the dislocation were to be displaced in the $x = 0$ plane in the positive y direction by a distance dy , work would have to be done against the tractions σ_{xx} on this plane, resulting in a positive increment of interaction energy

$$dI = \sigma_{xx}b \, dy, \quad (2.18)$$

and a driving force in the y direction on the dislocation of

$$f_y = -\frac{dI}{dy} = -\sigma_{xx}b. \quad (2.19)$$

We recognize these to be the glide (eq. 2.17) and climb (eq. 2.19) forces of eqs. (2.15a) and (2.15b).

2.3.3 Interaction of a Dislocation with Free Surfaces and Inhomogeneities

In a very large crystal of size R , the energy of a dislocation does not vary with small displacements $\delta\xi$ in its position, that is, those for which $\delta\xi/R \ll 1$. When a dislocation line approaches a free surface closely, however, its energy decays smoothly to zero as it emerges at the surface. The absence of a stress field on the other side of the free surface acts along the dislocation line in the manner of an image dislocation of opposite sign placed symmetrically outside the surface (in an infinite reference crystal), as shown in Fig. 2.5(a) for the case of a screw dislocation. Thus, the force exerted by the free surface on the dislocation is also obtainable from eq. (2.14) by considering the image stress as if it were due to an image dislocation inside a very large reference crystal. The glide and climb forces experienced by an edge dislocation near a free surface are obtainable by considering two image dislocations, as shown in Fig. 2.5(b).

The placement of a negative image screw dislocation relative to the free surface as depicted in Fig. 2.5(a) not only represents the force exerted on the dislocation by the free surface but also represents an exact solution of the stress field of a screw dislocation at a distance a under the surface. The placement of a pair of glide and climb image dislocations as depicted in Fig. 2.5(b) represents the correct force of interaction of the free surface with the edge dislocation but does not make the surface stress-free. This result is part of a more comprehensive elasticity problem of the stress field of an edge dislocation near a free surface that also renders the surface free of stress, giving the convenient result of representing the interaction forces of the dislocation with the free surface by means of these two images (Head 1953). If the dislocation encounters another half-space of different elastic properties instead of a free surface, there will be different interactions that can be either toward or away from the half-space depending on whether or not the half-space is

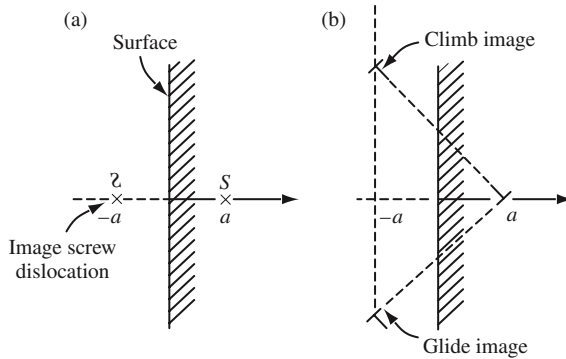


FIG. 2.5. (a) A negative image screw dislocation positioned at a distance $-a$ outside a surface both cancels the stresses of a positive screw dislocation a distance a under the surface and represents properly the force that a free surface exerts on the screw dislocation in the solid; (b) glide and climb images of an opposite edge dislocation, positioned as shown outside the solid, represent properly the forces exerted by the surface on the edge dislocation inside the solid but do not render the surface stress-free.

less or more stiff than the medium that the dislocation is in. Such problems are also discussed in the above reference, to which the reader is referred.

The interaction of a dislocation line with a free surface where no applied stresses are present represents a special case of more general types of occurrence where the interaction of a dislocation with a local inhomogeneity results in a free-energy gradient involving the dislocation. This introduces a conceptual force on the dislocation line that is the negative gradient of the free energy relative to the position of the dislocation segment.

2.3.4 Line Tension of a Dislocation

Because the line energy of a dislocation is predominantly free energy, in an isotropic crystal the lowest-energy configuration of a dislocation line constrained to go through two points is a straight line. When the dislocation line between the two pinning points is displaced by an external agency such as a force $\mathbf{f}$, the dislocation bows out into a line of constant curvature as shown in Fig. 2.6, where the work done by the force sweeping out the cross-hatched area is stored as energy of increased line length between points A and B until a new equilibrium is reached. When the force is removed slowly, the dislocation returns smoothly to its initial straight shape. This property of a dislocation line of equilibrating with a force applied to it, by assuming a curved shape, can be conceived of as a *line tension* and is represented

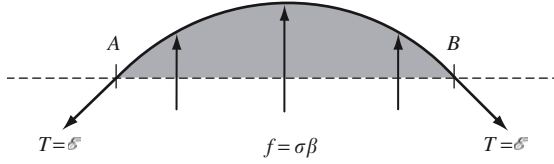


FIG. 2.6. A glide force $f = \sigma b$ normal to a pinned dislocation segment AB that bows it out is equilibrated by line tension forces $T = \mathcal{E}$ acting along the line at points A and B .

by a tangent vector $\mathbf{T}$ acting along the ends of the pinned segment. From simple variational considerations, the magnitude of the line tension must be equal to the line energy per unit length in the constant-line-energy approximation, that is,

$$|\mathbf{T}| = \mathfrak{F}, \quad (2.20)$$

which can be taken, to a first approximation, as being equal to the value given in eq. (1.30). Hereafter, we shall use the symbol $\mathcal{E}$ for the line tension.

The constant-line-energy/line-tension approximation introduced in eq. (2.20) offers a powerful means of solving problems involving complex dislocation interactions, and associated changes of shape.

An immediate application of the line-tension notion is in considering the shape of a dislocation line that is bowed out between two pinning points by an applied force $f = \sigma b$, as is depicted in Fig. 2.6. For the constant-line-energy/line-tension model, this bowed-out shape is a curve of constant curvature, which is immediately obtainable from simple force equilibrium and is given by

$$\frac{1}{R} = \frac{\sigma b}{T} = \frac{\sigma b}{\mathcal{E}}, \quad (2.21)$$

where R is the radius of curvature of the bowed-out shape.

In certain cases, however, where precise answers are necessary and where the constant-line energy/line-tension approximation is too crude, or when crystal anisotropy is important, the full orientation dependence of the line energy, as given by eq. (1.31a), must be taken into account. This results in a more precise definition of the line tension as a virtual variation of the line energy with changes of line length, and is given by (de Wit and Koehler 1959)

$$\mathcal{E}(\varphi) = \mathfrak{F}(\varphi) + \frac{\partial^2 \mathfrak{F}(\varphi)}{\partial \varphi^2}, \quad (2.22)$$

where $\mathfrak{F}(\varphi)$ is the line energy per unit length of a mixed dislocation that has a line vector making an angle φ with its Burgers vector, resulting, for isotropic elasticity, in (eq. 1.31a).

$$\mathfrak{F}(\varphi) = \frac{\mu b^2}{4\pi(1-\nu)}(1 - \nu \cos^2 \varphi) \ln\left(\frac{\alpha R}{b}\right). \quad (2.23)$$

This gives the following for the more general expression for the line tension according to eq. (2.22):

$$\mathcal{E}(\varphi) = \frac{\mu b^2}{4\pi(1-\nu)}(1 + \nu - 3\nu \sin^2 \varphi) \ln\left(\frac{\alpha R}{b}\right) \quad (2.24)$$

for an isotropic solid. With this refinement, the local curvature of a pinned dislocation line under stress is still given by eq. (2.21), where, however, the line tension given by eq. (2.24) needs to be considered.

The expression in eq. (2.22) indicates that even when the crystal can be considered as isotropic but when the constant-line-energy approximation is not appropriate and the line energy is considered to be dependent on the line orientation as given in eq. (2.23), a bowed-out mixed dislocation segment pinned between two points will no longer have a constant curvature. An interesting corollary of this fact is that in many anisotropic crystals the expression in eq. (2.22) may take on negative values in certain ranges of the orientation φ , resulting in the condition that the shape of the dislocation segment of lowest energy will be a kinked line, avoiding the regions where $\mathcal{E}(\varphi)$ is negative. Such cases were considered by Head (1967), particularly in connection with unusual internal damping effects.

2.3.5 Uniformly Moving Dislocations and The Dislocation Mass

When a straight dislocation line moves through a crystal with a constant velocity v , the atoms surrounding the dislocation are accelerated and decelerated as the dislocation moves through the lattice, and the total energy of the crystal increases as a consequence. Additionally, as the velocity nears the various velocities of shear waves or longitudinal waves, the stress field of the moving dislocation is progressively distorted and concentrates the Burgers displacement more and more into a plane normal to the plane of motion of the dislocation line. These distortions were discussed by Hirth and Lothe (1982) but are of less importance to us than the line energy of the moving dislocation. For a screw dislocation, the result is straightforward and can be given by (Hirth and Lothe 1982)

$$\mathfrak{F}(v) = \mathfrak{F}_0 / \sqrt{1 - \left(\frac{v}{c_s}\right)^2}, \quad (2.25)$$

where $c_s = \sqrt{\mu/\rho}$ is the velocity of a shear wave, ρ is the material density of the crystal, and $\mathfrak{F}_0$ is the line energy of a screw dislocation at rest. This expression indicates that the line energy diverges as $v \rightarrow c_s$ and, consequently, barring exceptional considerations, dislocation velocities can never reach the shear wave velocity. Edge dislocations, which have both dilatational and shear energy, with the dilatational field having a terminal velocity different from that of the shear field, shows more complicated behavior when v approaches these terminal velocities. Nevertheless, its energy too diverges when $v \rightarrow c_s$, but not in the same simple form as that given by eq. (2.25).

We note that for velocities $v/c_s \ll 1.0$ the energy in eq. (2.25) can be expanded as

$$\mathfrak{F}(v) = \mathfrak{F}_0 + \frac{1}{2} \left(\frac{\mathfrak{F}_0}{c_s^2} \right) v^2 + \dots, \quad (2.26)$$

leading to the interesting interpretation that the additional kinetic energies of the atoms surrounding the moving dislocation act as if the moving dislocation line, considered as a separate entity, possesses an effective mass m per unit length of

$$m = \frac{\mathfrak{F}_0}{c_s^2} = \frac{1}{2} b^2 \rho. \quad (2.27)$$

for a constant line energy model of $\mu b^2/2$, and, recalling that $c_s^2 = \mu/\rho$, this amounts to the mass of roughly one-half an atom mass per atom length of the dislocation line. While this inertial effect of a dislocation is quite small, it nevertheless plays a significant role in the dynamics of dislocation motion.

2.3.6 The Basic Differential Equation for a Moving Dislocation Line

We consider a segment dx between x and $x + dx$ of a quasi-straight dislocation line, roughly parallel to the x axis, under a variety of forces in the y direction at a time t , and consider its equilibrium. The dislocation segment experiences line tension forces in the y direction; an applied Peach–Koehler force σb in the y direction; a potential resistance arising from any source that results in changes of the line energy, dependent on the y coordinate of the segment in the lattice; and a velocity-dependent drag, arising primarily from the interaction with lattice phonons. If these forces are not in equilibrium, they will produce acceleration in the y direction, or, alternatively, they need to be equilibrated with a d'Alembert force, as depicted in Fig. 2.7. The resulting differential equation is

$$\mathcal{E} \frac{\partial y^2}{\partial x^2} + \sigma b - \frac{\partial \mathfrak{F}(y)}{\partial y} - B \frac{\partial y}{\partial t} = m \frac{\partial y^2}{\partial t^2}, \quad (2.28)$$

where we identify the terms on the left-hand side as the line tension–curvature force, the Peach–Koehler force due to the applied stress, the potential resistance due to the dependence of the line energy on position in the y direction, and the phonon drag,

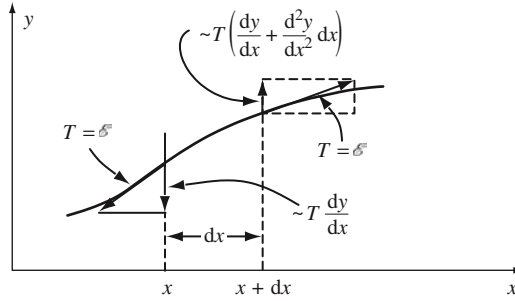


FIG. 2.7. A bowed dislocation segment of length dx under the action of several forces is equilibrated by line tension forces acting at the ends of the line.

while the term on the right-hand side is the d'Alembert force equal to the line mass times the line acceleration. In the following chapters, there will be a number of applications of the solution of this differential equation for both static and dynamic line shape problems. Of these terms, the phonon drag relates to the resistance that a moving dislocation line encounters through its interaction with thermal vibrations of the lattice (lattice phonons). In the classical range, at temperatures above the Debye temperature, $\theta_D = h\nu_D/k$, where $h(= 6.62 \times 10^{-34} \text{ Js})$ is Planck's constant, $k(= 1.38 \times 10^{-23} \text{ J/K})$ is Boltzmann's constant, and ν_D is the fundamental lattice frequency ($\sim 10^{13} \text{ s}^{-1}$), the drag coefficient can be taken as

$$B = \frac{kT}{\Omega\nu_D}, \quad (2.29)$$

where Ω is the atomic volume. The Debye temperatures of most metals tend to be in the range 200–400 K. A selection is given in Table 2.1. At lower temperatures, where the specific heat capacities of solids are subject to quantum restrictions and decrease to zero as $T \rightarrow 0 \text{ K}$, moving dislocations begin to interact with and become retarded by conduction electrons as well (see Kocks *et al.* 1975 for an expanded discussion).

2.3.7 The Multiplication of Dislocation Line Length

As eq. (2.8) indicates, maintenance of a constant plastic strain rate requires maintenance of a constant dislocation flux of $\rho_m \dot{\gamma}$ through the lattice at all times. As mobile dislocations are arrested by various obstacles to glide and are removed from the pool, new dislocation line length needs to be constantly generated. In a perfect crystal, dislocations cannot be generated at any meaningful rate by stresses lower than the ideal shear strength (Xu and Argon 2000). There are, however, many kinematical means of producing new dislocation line length by issuing it from sources, which are generally of the type first conceived of by Frank and Read (1950) and known by those names. The sequences in the production of a dislocation loop

TABLE 2.1. Debye temperatures θ_D of a representative set of solids^a

Substance	θ_D (K)	Substance	θ_D (K)
Ag	215	Mg	290
Al	398	Mo	379
Au	180	Na	159
C (diamond)	1,860	Ni	370
Cd	160	Pb	88
Cu	315	Ta	245
Fe	420	W	310
		NaCl	281

^a Assembled from Kittel (1953).

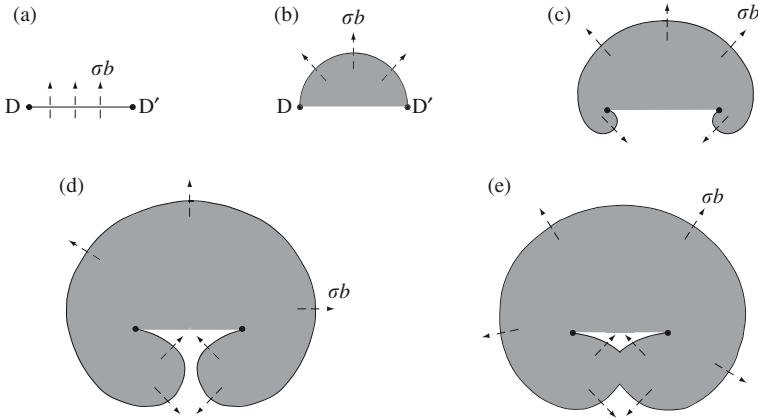


FIG. 2.8. Stages of the action of a Frank-Read source (Read 1953; courtesy of McGraw-Hill).

from a Frank-Read source are shown in Fig. 2.8. A segment DD' of length l of a long dislocation line threading in and out of a slip plane at points D and D' can go through a succession of convolutions and produce a large number of concentric loops. From a starting straight-line configuration DD' , the initial length bows out under a glide force to a critical configuration of a semicircle (Fig. 2.8(b)) of radius $R = l/2$ under a critical stress σ given by eq. (2.17),

$$\sigma = \frac{2\mathcal{E}}{bl} = \frac{\mu b}{l}. \quad (2.30)$$

Beyond the critical configuration, the semicircle expands under decreasing stress (Figs. 2.8(c) and (d)) until the opposing expanding portions of the loop come together, touch, and fuse (Figs. 2.8(d) and (e)), to release a fully formed loop and restore a fresh length between D and D' to repeat the process many times. Figure 2.9 shows such a Frank–Read source in Si (Dash 1957). Sources of this type can be freely produced also when a portion of a screw dislocation cross-slips twice to a parallel plane, where the segment on the parallel plane can act as a Frank–Read source as depicted in Fig. 2.10. Through such purely kinematical possibilities, the dislocation line length can “mushroom out” and spread in three dimensions under stresses as low as that given by eq. (2.30) to make up new dislocation line length with relative ease to compensate for line length that is inactivated elsewhere.

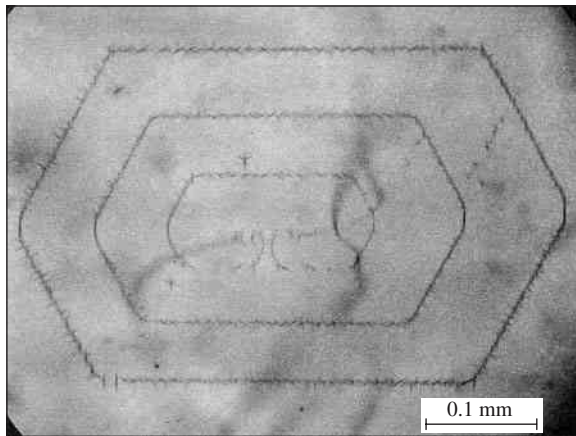


FIG. 2.9. A Frank–Read source in silicon has produced two full dislocation loops and is about to complete a third loop (Dash 1957; courtesy of General Electric Co.)

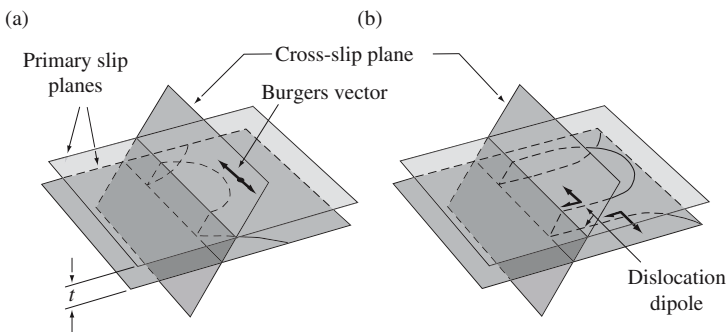


FIG. 2.10. A double cross-slip process resulting in an active Frank–Read source (after McClintock and Argon 1966).

2.4 The Mechanical Threshold of Deformation

It is intuitively clear that large-strain distortions can only be accomplished by shear and that as a result, the driving forces for it must involve shear stresses, or at least a critical deviation from a state of pure mean normal stress. In that sense it is possible to conceive of a fundamental thought experiment carried out at 0 K, in which a nearly perfect crystal containing the necessary complement of microstructural or lattice defects is subjected to increasing levels of shear stress, while the shear strain rate, over and above the ubiquitous rate-independent elastic flexure, is monitored, and plotted against the shear stress. Figure 2.11 shows the likely outcome, of no detectable shear strain rate for shear stresses σ less than a definite value $\hat{\tau}(0)$. When $\sigma = \hat{\tau}(0)$, the shear strain rate should undergo an abrupt and, perhaps, only inertially limited increase through an apparent initiation of inelastic deformation by a specific mechanism. Clearly, in real materials with the usual internal variability of defect structure, the transition from no deformation to a high rate of deformation should be somewhat less abrupt but should still exhibit a threshold behavior. We take this material property $\hat{\tau}(0)$ as the *mechanical threshold* for initiating deformation by a specific mechanism, such as the intrinsic lattice resistance to dislocation glide or resistance to dislocation glide caused by a field of solute atoms, precipitate particles, and the like, associated with a specific level of defect structure. Thus, under this idealization, we expect that, at $T = 0$ K,

$$\dot{\gamma} = 0 \quad \text{for } \sigma < \hat{\tau}(0), \quad (2.31a)$$

$$\dot{\gamma} > 0 \text{ (and large)} \quad \text{for } \sigma = \hat{\tau}(0), \quad (2.31b)$$

and that $\sigma > \hat{\tau}(0)$ should be unattainable under quasi-static conditions with a given initial defect structure.

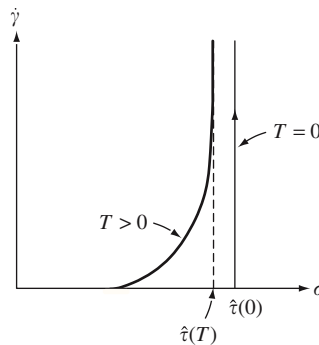


FIG. 2.11. The strain rate $\dot{\gamma}$ becoming unbounded at a threshold $\hat{\tau}(0)$, at $T = 0$ K, or when $\sigma \rightarrow \hat{\tau}(T)$ at $T > 0$.

At $T > 0$, the above behavior should be modified in two important ways. Since the basic process governing $\hat{\tau}$ will usually be an elastic interaction on the atomic level, an increase in temperature should decrease $\hat{\tau}(T) (< \hat{\tau}(0))$ to below the 0 K reference value, in proportion to the temperature dependence of the elastic constants. In this modified situation, we shall refer to $\hat{\tau}(T)$ as the rate-independent (athermal) reference mechanical threshold. Moreover, with increasing temperature, the local energy barriers of the inelastic strain-producing process can be overcome at lower resolved shear stresses with the help of thermal fluctuations. Thus, a finite inelastic strain rate should be observable even below the now reduced $\hat{\tau}(T)$. This should give rise to the typical softened behavior pattern shown in Fig. 2.11.

We note briefly here that the reference mechanical threshold can be increased by strain hardening, through the accumulation of immobile dislocations. Alternatively, at elevated temperatures in the range of half the melting point and above, where self-diffusion becomes rapid, the reference mechanical threshold can be made meaningless by recovery processes resulting in steady-state deformation with a substantially stationary defect structure. In addition, other purely diffusional transport processes can also be present which are capable of producing inelastic strain; these, however, we shall not discuss (see Argon 1996).

2.5 Elements of Thermally Activated Deformation

2.5.1 General Principles

As stated earlier, inelastic deformation requires a “defective” crystal in which local shear relaxations are kinematically possible. Thus, we consider as a specific case a cluster of volume Ω_f in a volume V , able to undergo a freestanding transformation shear strain $\Delta\gamma^T$ which converts the unconstrained volume Ω_f from an unflexed state A to a flexed state B as depicted in Fig. 2.12(a). When the shear transformation is completed inside the body, it results in shear strain increment of

$$\Delta\gamma_0 = \Delta\gamma^T \frac{\Omega_f}{V} \quad (2.32)$$

in the overall volume V .

We now consider a reversible thought experiment in which the overall system is flexed in small increments of strain $d\gamma$ from $\gamma = 0$ to $\gamma = \Delta\gamma_0$ in constrained equilibrium, and the resulting Helmholtz free energy ΔF of the system is monitored continuously. We expect that the resulting change in ΔF with increasing inelastic strain γ will look somewhat like what is shown in Fig. 2.12(b), where we show $\Delta F_B > \Delta F_A$, indicating that, in this case, the final transformed state is less stable than the initial one. We now define the associated deformation resistance τ for this

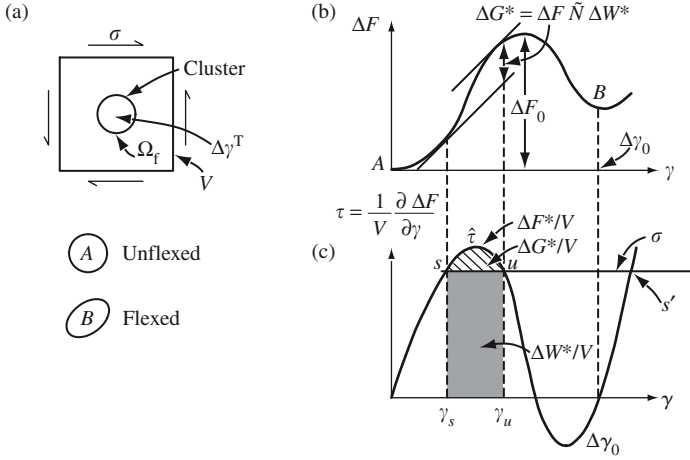


FIG. 2.12. (a) Thermally activated overcoming of an energy barrier for a shear transformation taking place in a volume Ω_f inside a volume V ; (b) the Helmholtz potential energy ΔF as a function of the applied shear strain γ ; (c) graph of the shear resistance τ as a function of shear strain γ .

process as

$$\tau \equiv \frac{1}{V} \frac{\partial \Delta F}{\partial \gamma} \quad (2.33)$$

and plot it in Fig. 2.12(c), in line with ΔF as a function of γ . We take τ as a material property, and note that there is a maximum level of deformation resistance $\hat{\tau}$, reached where the slope of the ΔF vs. γ relation reaches a maximum value. Under an applied shear stress σ , incremented under quasi-static and constrained conditions, it should become possible to flex the system V reversibly up and down the configuration path from $\gamma = 0$ to $\gamma = \Delta\gamma_0$, and beyond.

We note that at any particular level of applied stress $\sigma < \hat{\tau}$ the stress is in equilibrium with the deformation resistance at two points of flexure, s and u , for inelastic system strains γ_s and γ_u . If the system at this level of stress is given a virtual configurational variation $\Delta\gamma$, the external agency applying the stress will do an increment of work on the system given by

$$\Delta W = V \sigma \Delta\gamma. \quad (2.34)$$

On the ΔF diagram, this can be shown by sloping lines tangent to the curve at the points s and u where $\sigma = \tau$. We note that at s and u , the resulting changes in the

Gibbs free energy,⁵

$$\Delta G_s = \Delta F_s - \Delta W_s > 0, \quad (2.35a)$$

$$\Delta G_u = \Delta F_u - \Delta W_u < 0, \quad (2.35b)$$

are of different sign, for both positive and negative flexure of the system. Clearly, the equilibrium at s is stable, while that at u is unstable. The difference ΔF between the sloping lines on the ΔF diagram and the gray area under the curve in the deformation resistance line represents a Gibbs free-energy difference, or an energy barrier, ΔG^* , which we recognize as the required activation free energy that must be supplied by thermal fluctuations to permit the system to reach the unstable (saddle point) configuration u from the stable configuration s under the applied shear stress σ . Once the system has been activated over the energy barrier, it can spontaneously flex into a new stable configuration, shown by s' on the deformation resistance diagram, as it develops a total elemental strain increment $\Delta\gamma_0$. As is clearly delineated in Figs. 2.12(b) and (c), the activation free energy ΔG^* has two components: the Helmholtz free-energy change ΔF^* and the external work increment ΔW^* during the activation:

$$\Delta F^* = V \int_{\gamma_s}^{\gamma_u} \tau(\gamma) d\gamma, \quad (2.36a)$$

$$\Delta W^* = V \int_{\gamma_s}^{\gamma_u} \sigma d\gamma = V\sigma(\gamma_u - \gamma_s). \quad (2.36b)$$

Here we note that, as indicated in Fig. 2.12(b), during the strain-producing activated event, the Helmholtz free energy will often show an increase in the new flexed state. If this is a regularly recurring feature, it is often useful to recognize that a certain component of the applied stress $\sigma = \tau_{\text{STOR}}$ must be reserved to provide this rate of free-energy storage. Thus, the component of the stress that is available to assist the system over the thermally penetrable obstacle is $\sigma - \tau_{\text{STOR}}$, which is often called the effective stress.

The activation rate R of the inelastic process, in the direction of absorbing positive work, can then be given as

$$R = \nu_G \exp\left(\frac{-\Delta G^*(\sigma, \hat{\tau})}{kT}\right) \quad (2.37a)$$

⁵ In classical thermodynamics, the Gibbs free energy has a more restricted definition including only the effects of work done by the negative mean normal component (the pressure) of the stress tensor. In view of this, the broader interpretation that has been used here has often been referred to as the free enthalpy (see Kocks *et al.* 1975). However, in honor of Gibbs, who profoundly influenced this field, we shall still refer to this generalized definition by his name.

for stress levels close to $\hat{\tau}$, where ΔG^* is the activation free energy for the *forward* process and ν_G is a characteristic normal-mode frequency of the cluster or other configuration along the activation path, where the “cluster” could be a dislocation line segment in front of an obstacle, or a twin embryo, etc. The exponential factor, called the Boltzmann factor, gives the probability that a thermal fluctuation of the required energy (or larger) can be supplied by a thermal reservoir at a temperature T . We note that ν_G , being a cluster frequency involving many atoms, will be much smaller than the usual Debye frequency of the atoms, ν_D , to which it is related crudely by

$$\nu_G \approx \nu_D \left(\frac{\Omega}{\Omega_f} \right)^{1/3}, \quad (2.37b)$$

where Ω is the atomic volume, and the factor in parentheses is typically of the order of $10^{-2} - 10^{-1}$. Thus, the activation rate expression related to inelastic processes in its classical form holds valid well below the usual Debye temperature $\theta_D = (h\nu_D/k)$, where h and k are, respectively, Planck’s constant and Boltzmann’s constant.

Here we are interested in the net inelastic strain rate, which is

$$\dot{\gamma}_{\text{net}} = \dot{\gamma}_0 \left[\exp\left(-\frac{\Delta G^*}{kT}\right) - \exp\left(-\frac{\Delta G_r^*}{kT}\right) \right], \quad (2.38)$$

where the second term represents the *reverse* rate of deformation associated with a possible return of configuration B back to A, during which work needs to be done against the applied stress. For this process, ΔG_r^* is the activation free energy for the reverse transformation reaction under stress. The preexponential factor $\dot{\gamma}_0$ has a typical composition

$$\dot{\gamma}_0 \cong c_f \Delta \gamma^T \nu_G, \quad (2.39)$$

where c_f is the volume fraction of *fertile* material that can take part in the configurational transformation, either forward or in reverse. When the applied stress σ is much less than $\hat{\tau}$ it is usually possible to give a linearized form of ΔG^* and ΔG_r^* ,

$$\Delta G^* \cong \Delta F_0^* - \frac{\sigma \Delta \gamma^T \Omega_f}{2}, \quad (2.40a)$$

$$\Delta G_r^* \cong \Delta F_0^* + \frac{\sigma \Delta \gamma^T \Omega_f}{2}. \quad (2.40b)$$

With this form, eq. (2.38) can be simplified into a useful form for moderately low stress levels,

$$\dot{\gamma}_{\text{net}} = 2\dot{\gamma}_0 \exp\left(-\frac{\Delta F_0^*}{kT}\right) \sinh\left(\frac{\sigma \Delta \gamma^T \Omega_f}{2kT}\right). \quad (2.41)$$

This gives $\dot{\gamma}_{\text{net}} \rightarrow 0$ when $\sigma \rightarrow 0$. For very low levels of stress, all inelastic processes then have an asymptotically linear form in the stress as follows:

$$\dot{\gamma}_{\text{net}} = \frac{\dot{\gamma}_0 \Delta\gamma^T \Omega_f \sigma}{kT} \exp\left(-\frac{\Delta F_0^*}{kT}\right). \quad (2.42)$$

For high stresses close to the mechanical threshold $\hat{\tau}$ (where ΔG_r^* is much greater than ΔG^*), however, the net strain rate is adequately given by

$$\dot{\gamma}_{\text{net}} = \dot{\gamma} = \dot{\gamma}_0 \exp\left(-\frac{\Delta G^*(\sigma, \hat{\tau})}{kT}\right), \quad (2.43)$$

the familiar Arrhenian expression.

2.5.2 Principal Activation Parameters for Crystal Plasticity

Before considering specific processes of crystal plasticity, we shall develop a number of generally useful and instructive thermal-activation parameters that characterize systems deforming close to their mechanical threshold.

In this range (see Fig. 2.12(c)),

$$\Delta G^* = V \int_{\gamma_s}^{\gamma_u} (\tau(\gamma) - \sigma) d\gamma = V \int_{\sigma}^{\hat{\tau}} \Delta\gamma_a(\tau) d\tau = \int_{\sigma}^{\hat{\tau}} \Delta\gamma_{\text{af}}(\tau) \Omega_f d\tau, \quad (2.44)$$

where $\Delta\gamma_a = \Delta\gamma_0$ is the system activation strain and $\Delta\gamma_{\text{af}} = \Delta\gamma^T$ is the cluster activation strain, with the two being related by eq. (2.32). For many, but not all, thermally activated local deformation processes, the energy barrier is governed by an elastic interaction mechanism which scales with a set of elastic constants, but most prominently with the shear modulus μ alone, which has a known temperature dependence resulting from anharmonic effects. Thus, we can write

$$\Delta G^* = \Delta G_0 \frac{\mu(T)}{\mu_0}, \quad (2.45)$$

where μ_0 and ΔG_0 refer to a reference state which, as a first approximation, can be taken at 0 K. Moreover, we recognize that

$$\Delta G^* = \Delta H^* - T \Delta S^* \quad (2.46)$$

and that

$$\Delta S^* = -\left(\frac{\partial \Delta G^*}{\partial T}\right)_{\sigma, p} \quad \left(\text{because } \frac{\partial \Delta H^*}{\partial T} - T \frac{\partial \Delta S^*}{\partial T} = 0\right), \quad (2.47a)$$

so that

$$\Delta S^* = \Delta G^* \left(\frac{1}{\mu} \frac{d\mu}{dT} \right). \quad (2.47b)$$

Two important and readily measurable properties of an activated process that convey important information about its nature are (a) the shear activation volume Δv_σ^* and (b) the pressure activation volume Δv_p^* , or the activation dilatation. We define these as

$$\Delta v_\sigma^* \equiv - \left(\frac{\partial \Delta G^*}{\partial \sigma} \right)_{p,T} = \Delta \gamma_{\text{af}}^*(\sigma) \Omega_f, \quad (2.48)$$

$$\Delta v_p^* \equiv \left(\frac{\partial \Delta G^*}{\partial p} \right)_{\sigma,T}. \quad (2.49)$$

The determination of these two quantities and other activation parameters, by experimental means, is of great interest for probing the mechanism of activated processes. The additional quantities include ΔH^* , ΔS^* , and ΔG^* .

The kinetic expression for the deformation experiment, eq. (2.43), is the principal vehicle for probing the process, through a number of jump experiments, which involve, in increasing complexity, the imposed strain rate (at constant temperature and pressure T, p), the temperature (at constant stress and pressure σ, p), and the pressure (at constant strain rate and temperature $\dot{\gamma}, T$). From such jump experiments, the following are obtained:

$$\Delta v_\sigma^* = kT \left[\left(\frac{\partial \ln \dot{\gamma}}{\partial \sigma} \right)_{p,T} - \left(\frac{\partial \ln \dot{\gamma}_0}{\partial \sigma} \right)_{p,T} \right] \cong kT \left(\frac{\partial \ln \dot{\gamma}}{\partial \sigma} \right)_{p,T}, \quad (2.50a)$$

$$\Delta H^* = kT^2 \left[\left(\frac{\partial \ln \dot{\gamma}}{\partial T} \right)_{\sigma,p} - \left(\frac{\partial \ln \dot{\gamma}_0}{\partial T} \right)_{\sigma,p} \right] \cong kT^2 \left(\frac{\partial \ln \dot{\gamma}}{\partial T} \right)_{\sigma,p}, \quad (2.50b)$$

$$\begin{aligned} \Delta v_p^* &= kT \left[\left(\frac{\partial \ln \dot{\gamma}}{\partial p} \right)_{\sigma,T} - \left(\frac{\partial \ln \dot{\gamma}_0}{\partial p} \right)_{\sigma,T} \right] \cong -kT \left(\frac{\partial \ln \dot{\gamma}}{\partial p} \right)_{\sigma,T} \\ &= kT \left(\frac{\partial \ln \dot{\gamma}}{\partial \sigma} \right)_{p,T} \left(\frac{\partial \sigma}{\partial p} \right)_{\dot{\gamma},T} = \Delta v_\sigma^* \left(\frac{\partial \sigma}{\partial p} \right)_{\dot{\gamma},T}, \end{aligned} \quad (2.50c)$$

where $(\partial \sigma / \partial p)_{\dot{\gamma},T}$ reflects an important strength differential (SD) effect. To obtain the final form of the expressions, we have noted that $\dot{\gamma}_0$ is given by eq. (2.39), in which the pressure and temperature dependence is negligible, and the stress dependence, which usually arises through the stress dependence of mobilizable configurations $c_f(\sigma)$, is very much lower than that of the principal exponential term when $\sigma < \hat{\tau}$. From the principal temperature dependence of eq. (2.45),

where h is the spacing between slip planes (not shown in Fig. 2.13); ρ_M is the mobile dislocation density ($= N\lambda/V$), where N is the active number of segments of length λ in the given volume V ; Δy is the mean distance of advance of a segment ($= \Delta a/\lambda$), where λ here is the *Friedel sampling length* (to be defined in detail in Chapter 6) of the spacing of obstacles in the plane; Δa^* is the activation area; Δy^* is the activation distance; and c_f is the volume fraction of mobilizable configurations, which transforms into a total slipped area of thickness h per unit volume.

2.6 Selection of Slip Systems in Specific Crystal Structures

While the crystallographic nature of plastic deformation in the form of slip bands had been known for a long time, the more specific identification of the latter with crystal planes dates back to the early experiments on the plasticity of metal single crystals (Schmid and Boas 1935). Two basic and complementary criteria have been used to explain the selection of specific slip systems in crystals (Eshelby 1949).

The energy criterion for the selection of slip planes notes that dislocation line energies are large and that, therefore, the crystal will contain and multiply only those dislocations with the minimum line energy, that is, the ones with the smallest Burgers vectors and the smallest effective interplanar shear stiffness.

The complementary mobility criterion notes, in turn, that the dislocations that will dominate in plastic deformation are the ones that experience the lowest lattice resistance. As will be discussed in detail in Chapter 4, the lattice resistance decreases exponentially with the ratio of the core width of the dislocation to the Burgers displacement, and the former scales with the interplanar spacing. As a first approximation, dislocations on planes with the largest ratio of interplanar spacing to Burgers displacement should have the highest mobility and should, therefore, be dominant in slip.

These criteria correctly predict that in BCC α -Fe the $\{110\}\langle 111 \rangle$ slip system should be the preferred one at low temperatures (Eshelby 1949). Moreover, the same system should be preferred in all other BCC transition metals at low temperatures most of these systems, however, also show slip on the $\{112\}\langle 111 \rangle$ system at intermediate temperatures. In Chapter 4, we shall discuss the reasons for the preference of one or the other system at different temperatures.

In the application of these criteria to other systems such as those with FCC, HCP, and diamond cubic structures, and ionic and covalent compounds, other considerations also play very important roles, as we discuss in Section 2.7. These include (a) reduction of the line energy due to the splitting of perfect dislocations into partial dislocations separated by stacking faults; (b) strong directional bonding in the dislocation core; and (c) charge neutrality. In Table 2.2, we list the prominent slip systems of some of the most common crystalline solids.

TABLE 2.2. Glide elements for metals and some other crystals^a

Crystal structure	Lattice type	Slip plane	Slip direction	Reference
Cu, Au, Ag, Ni, CuAu, α -CuZn, AlCu, AlZn	FCC	{111}	$\langle 110 \rangle$	Seeger (1958)
Al	FCC	{111} {100}	$\langle 110 \rangle$ $\langle 110 \rangle^b$	Seeger (1958)
α -Fe	BCC	{110} {112} {123}	$\langle 111 \rangle$ $\langle 111 \rangle$ $\langle 111 \rangle$	Seeger (1958)
Mo, Nb, Ta, W, Cr, V	BCC	{110} {112}	$\langle 111 \rangle$ $\langle 111 \rangle^c$	Seeger (1958)
Cd, Zn, ZnCd	HCP $c/a > 1.85$	(0001) (00 $\bar{1}$ 0) (11 $\bar{2}$ 2)	$\langle 2\bar{1}\bar{1}0 \rangle$ $[11\bar{2}0]^d$ $[\bar{1}\bar{1}23]^d$	Seeger (1958)
Mg	HCP $c/a = 1.623$	(0001) {10 $\bar{1}$ 1} {10 $\bar{1}$ 0}	$\langle 2\bar{1}\bar{1}0 \rangle^e$ $\langle 2\bar{1}\bar{1}0 \rangle^e$ $\langle 2110 \rangle^f$	Seeger (1958)
Be	HCP $c/a = 1.568$	(0001) {10 $\bar{1}$ 0}	$\langle 2\bar{1}\bar{1}0 \rangle$ $\langle 2\bar{1}\bar{1}0 \rangle$	Seeger (1958)
Ti	HCP $c/a = 1.587$	{10 $\bar{1}$ 0} {10 $\bar{1}$ 1} (0001)	$\langle 2\bar{1}\bar{1}0 \rangle^g$ $\langle 2\bar{1}\bar{1}0 \rangle^g$ $\langle 2\bar{1}\bar{1}0 \rangle^g$	Seeger (1958)
Ge, Si, ZnS	Diamond cubic	{111}	$\langle 101 \rangle$	Seeger (1958)
As, Sb, Bi	Rhombohedral	(111) (11 $\bar{1}$)	$[10\bar{1}]$ $[101]$	Schmid and Boas (1935)
NaCl, KCl, KBr, KI, AgCl, LiF	Rock salt structure	{110} {001}	$\langle 110 \rangle$ $\langle 110 \rangle^h$	Schmid and Boas (1935)
MgTi, LiTi, AuZn, AuCd, NH ₄ Br, NH ₄ Cl, CsI, CsBr, TlCl–TlBr, CsCl	Cesium chloride structure	{110}	$\langle 100 \rangle$	Seeger (1958) Sprackling (1976)

^a From Argon (1996).^b Above 450 C.^c Secondary system at higher temp.^d Above 250 C.^e Above 225 C.^f At RT and below.^g Pyramidal and basal planes less active than prism planes.^h Secondary system at higher temp.

2.7 Dislocations in Close-packed Structures

2.7.1 Dissociation of Perfect Dislocations in FCC

In Section 1.7, we discussed some of the symmetry properties of FCC and HCP structures as variants of a close-packing of spheres on the atomic scale and noted the higher level of symmetry in FCC in comparison with that of the HCP structure. We noted also that since the nearest-neighbor relationships are identical in the two structures and the only small differences in energy that arise were attributable to differences in next-nearest-neighbor interactions, stacking faults are likely to be present in both structures. In FCC, such stacking faults result in a local HCP packing, while in HCP, stacking faults result in a local FCC packing in the fault layer. Moreover, we noted also that intrinsic stacking faults in FCC can arise when a partial shear translation takes the opposing parts of the crystal on a $\{111\}$ plane to the nearest interstitial site C from the starting site B over the lower A layer rather than directly to the identity site B' that would result in restoration of full order in one step, as shown in Fig. 2.14. This translation could then be followed by a further partial translation step from site C to B' , completing the process and removing the intrinsic stacking fault. This introduces the possibility that dislocations on the $\{111\}$ planes in FCC with Burgers vectors of $(a/2)[110]$ type may dissociate into two partial dislocations separated by an intrinsic stacking fault by the following typical dissociation reaction,

$$b = \frac{a}{2}[\bar{1}10] \rightarrow \frac{a}{6}[\bar{2}11] + \frac{a}{6}[\bar{1}2\bar{1}] + SF, \quad (2.54)$$

again depicted in Fig. 2.14. This possibility was recognized first by Heidenreich and Shockley (1948), and the partial dislocations of glide type $(a/6)\langle 211 \rangle$ are referred to as Shockley partial dislocations. Such decomposition reactions should always be energetically favored, since they reduce the line energy by spreading out the previously concentrated line energy of a perfect dislocation of type $(a/2)\langle 110 \rangle$ into two Shockley partial dislocations each with a smaller line energy, albeit at the cost of creating a stacking fault separating them. Since, in isotropic crystals, the line energy of a dislocation is proportional to the square of the Burgers vector,

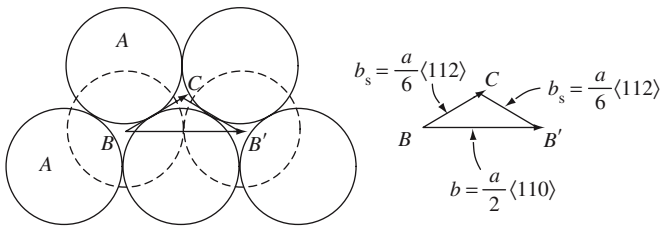


FIG. 2.14. Two partial shear steps on a (111) plane in an FCC crystal.

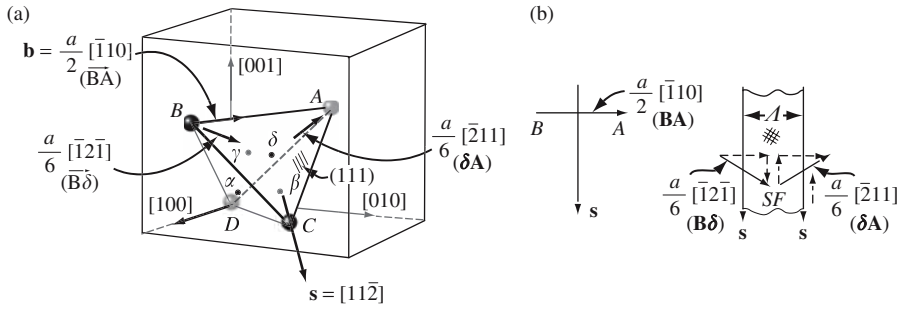


FIG. 2.15. (a) The Thompson tetrahedron in an FCC crystal; (b) the dissociation of a full dislocation $\mathbf{BA}$ into two Shockley partials $\mathbf{B}\delta$ and $\delta\mathbf{A}$, separated by a stacking-fault ribbon of width Λ .

decomposition reactions of this type should always be favored when the product dislocation Burgers vectors make an obtuse angle with each other (Frank's rule, Frank 1949)—not considering the energy of the stacking fault. Since the latter is also an important component of the energy balance, and parallel dislocations near to each other have substantial interaction energies, the overall form of the dissociated dislocation requires a more complete consideration of the energy balance.

Thus, consider for example, an $(a/2)[\bar{1}10]$ edge dislocation on the (111) plane (ABC plane) of the FCC reference cube shown in Fig. 2.15(a) with a line vector s parallel to the $[11\bar{2}]$ direction. Figure 2.15(b) shows schematically this dislocation with Burgers vector $(a/2)[\bar{1}10]$ dissociate into two Shockley partials of Burgers vectors $(a/6)[\bar{1}2\bar{1}]$ and $(a/6)[\bar{2}11]$ with a stacking fault of width Λ separating the two partials. As Fig. 2.15(b) shows, the Burgers vectors of each partial make a 60° angle with the dislocation line vector s of the partials and therefore both have edge and screw components which interact with the corresponding components of the other partial. The total line energy $\mathfrak{F}_1$ of the undissociated edge dislocation is then, per unit length,

$$\mathfrak{F}_1 = \frac{\mu b^2}{4\pi(1-\nu)} \ln\left(\frac{\alpha R}{b}\right), \quad (2.55)$$

while that of the dissociated dislocation is made up of several contributions and is, per unit length,

$$\begin{aligned} \mathfrak{F}_2 = & 2 \frac{\mu b^2}{16\pi(1-\nu)} \ln\left(\frac{\alpha R}{b}\right) + 2 \left(\frac{\mu b^2}{48\pi}\right) \ln\left(\frac{\alpha R}{b}\right) + \frac{\mu b^2}{8\pi(1-\nu)} \ln\left(\frac{R}{\Lambda}\right) \\ & - \frac{\mu b^2}{24\pi} \ln\left(\frac{R}{\Lambda}\right) + \chi_{\text{SF}} \Lambda. \end{aligned} \quad (2.56)$$

In eq. (2.56), the first and second terms represent the self-energies of the edge and screw components, respectively, of the partials. The third and fourth terms represent the interaction energies of the edge components and the screw components, respectively, of the partials (there being no interactions between the opposite edge and screw components), and the last term represents the contribution of the energy of the stacking-fault, of width Λ and specific stacking-fault energy χ_{SF} . The dislocations are considered to lie along the center of a cylinder of radius R , and α is the core cutoff parameter discussed in Chapter 1. The two edge components of the partials are of the same sign, giving a positive interaction energy, while the screw components are of opposite sign, giving a negative interaction energy. Clearly, as the stacking-fault width Λ increases, the interaction energies decrease but the energy of the stacking-fault ribbon increases. A minimum in energy is reached when

$$-\frac{\partial \mathfrak{F}_2}{\partial \Lambda} = 0, \quad (2.57)$$

which gives an equilibrium equation of forces

$$-\frac{\mu b^2}{8\pi(1-\nu)} \frac{1}{\Lambda} + \frac{\mu b^2}{24\pi} \frac{1}{\Lambda} + \chi_{\text{SF}} = 0. \quad (2.58)$$

The solution for the width of the stacking-fault ribbon is

$$\Lambda = b \left(\frac{\mu b}{\chi_{\text{SF}}} \right) \frac{2+\nu}{24\pi(1-\nu)}, \quad (2.59)$$

that is, the smaller the stacking-fault energy, the larger the ribbon width. In passing, we can identify eq. (2.58) to be an equation of force balance. The edge components of the partials, being of the same sign, repel, while the screw components attract, as does the pull of the stacking fault. Substitution of eq. (2.59) into eq. (2.56) gives the total line energy of the dissociated edge dislocation as

$$\mathfrak{F}_e = \frac{\mu b^2}{4\pi(1-\nu)} \ln \left(\frac{\alpha R}{b} \right) - \frac{\mu b^2(2+\nu)}{24\pi(1-\nu)} \left(\ln \left(\frac{\mu b}{\chi_{\text{SF}}} \right) \frac{(2+\nu)\alpha}{24\pi(1-\nu)} - 1 \right). \quad (2.60)$$

The second term in eq. (2.60) gives the energy reduction due to the dissociation, resulting in a fractional energy reduction of

$$\frac{\Delta \mathfrak{F}_e}{\mathfrak{F}_e} = -\frac{(2+\nu)}{6} \left(\ln \frac{(\alpha \Lambda / eb)}{\ln(\alpha R / b)} \right), \quad (2.61)$$

where $e = 2.72$. A very similar procedure can be used to consider the dissociation of an initially perfect screw dislocation.

TABLE 2.3. Equilibrium extensions of edge dislocations on $\{111\}$ planes and fractional line energy reductions in some FCC metals and a solid-solution alloy

Material	χ_{SF} (mJ/m ²)	μ (GPa) ^a	(δ_{eq}/b)	$\Delta\mathfrak{F}/\mathfrak{F}$
Al	200	25.9	1.77	—
Ag	20	26.6	18.59	−0.129
Cu	73	42.1	6.73	−0.059
Cu + 5% Al	5	42.1	98.28	−0.233

^a Effective $\{111\}$ plane shear moduli from Kocks *et al.* (1975).

Table 2.3 gives the fractional energy reduction for edge dissociations for three FCC metals and one solid-solution alloy. For the case of Al, with a very high stacking-fault energy, the dissociation distance is of the order of the core width and therefore the dislocation can be considered as undissociated, while that for the Cu 5% Al alloy, which is characteristic of a number of other similar cases of dilute solid solutions, is quite large. It is this dislocation dissociation which prescribes the slip plane of FCC metals to be the $\{111\}$ plane at low temperatures.

2.7.2 The Thompson Tetrahedron and Other Partial Dislocations

It was noted first by Thompson (1953) that most dislocations of both perfect and partial nature, as well as many dislocation reactions in the FCC structure, can be conveniently visualized with the aid of the tetrahedron shown in Fig. 2.15(a), obtained when the central atom D at the origin is connected to the nearest face center atoms A, B, and C. The four faces of the tetrahedron ABC, BCD, ABD, and ACD represent the four possible close-packed slip planes (111) , $(\bar{1}\bar{1}\bar{1})$, $(\bar{1}\bar{1}1)$, and $(1\bar{1}\bar{1})$, respectively, and the six edges AB, BC, CA, BD, CD, and AD represent the six close-packed directions in space outlining the four close-packed planes. The centers of the triangular faces α , β , γ , and δ facing the corners A, B, C, and D are identified as possible interstitial sites. The perfect edge dislocation and its partials discussed in Section 2.7.1 are shown in Fig. 2.15(b). We note that the arrows connecting corner atom positions to the centers of these faces represent all the possible Shockley partial dislocations, such as the partials $\mathbf{B}\delta$ and $\delta\mathbf{A}$ considered in Section 2.7.1. We note, moreover, that if atom D lies in a close-packed plane then the atoms A, B, and C lie on the adjacent close-packed plane above it and that the magnitude of the arrow connecting D to δ ($\mathbf{D}\delta$) gives the interplanar spacing of close-packed planes. Thus, if a stacking fault of either of the types formed by condensation of vacancies or interstitials in a close-packed plane terminates, it must be surrounded by a partial dislocation of the type ($\mathbf{D}\delta$) along its border. Such partial dislocations are known as Frank partial dislocations (Frank 1950).

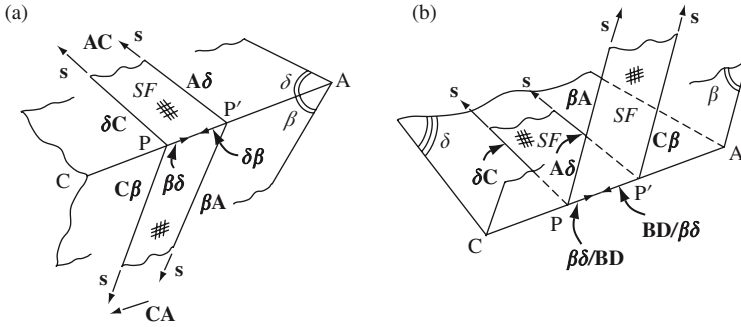


FIG. 2.16. Stair-rod dislocations: (a) of the first kind $\mathbf{B}\delta$; (b) of the second kind $\beta\delta/\mathbf{BD}$.

We now consider two other partial dislocations which play a role in strain hardening. They relate to what happens when a stacking-fault ribbon wraps around a line of intersection between two intersecting $\{111\}$ planes, making either an acute angle or an obtuse angle. Thus, consider first the case of an acute-angle turn in Fig. 2.16(a), where a dissociated $\mathbf{AC}$ dislocation turns from the δ plane into the β plane along their line of intersection $\mathbf{AC}$, in the Thompson tetrahedron notation. This could arise, as depicted in Fig. 2.16(a), if a screw dislocation with Burgers vector $\mathbf{AC}$ turns partly into the δ and partly into the β plane. Considering now the line of intersection as a twofold node on the dislocation, viewing the dislocation on the δ plane away from the line of intersection, the dislocation has a Burgers vector $\mathbf{AC}$ and is dissociated into Shockley partials $\mathbf{A}\delta$ on the right and $\delta\mathbf{C}$ on the left (see Section 2.7.3), according to the dissociation allowed on the δ plane. Correspondingly, viewing the dislocation away from the line of intersection on the β plane, the dislocation has a Burgers vector $\mathbf{CA}$ and is dissociated into Shockley partial dislocations with $\mathbf{C}\beta$ on the right and $\beta\mathbf{A}$ on the left, again according to the allowable dissociation. At the point P' , the Shockley partials $\mathbf{A}\delta$ on the δ plane and $\beta\mathbf{A}$ on the β plane coming together do not have compensating Burgers vectors as required at a twofold node. Therefore we conclude that the two points P' and P must be threefold nodes and that a third partial dislocation lying along the line between P and P' must connect these two points together. It has a Burgers vector $\beta\delta$ viewed from P to P' and $\delta\beta$ viewed from P' to P so that at points P and P' the nodal convention is satisfied, and

$$\delta\mathbf{C} + \mathbf{C}\beta + \beta\delta = 0 \quad (2.62a)$$

and

$$\mathbf{A}\delta + \beta\mathbf{A} + \delta\beta = 0 \quad (2.62b)$$

at points P and P', respectively. This partial dislocation, which has a Burgers vector that connects points δ and β on the Thompson tetrahedron and has a magnitude of $b/3$, and is required to wrap a stacking fault around an acute angle, is referred to as a stair-rod dislocation of the first kind.

By similar considerations, if an **AC** dislocation were to wrap the stacking fault from the δ plane onto the β plane through an obtuse angle, as shown in Fig. 2.16(b), it must be concluded that along the line of intersection of the planes between points P and P' another partial dislocation must be present that has a Burgers vector $\beta\delta/\mathbf{BD}$ viewed from P to P' and $\mathbf{BD}/\beta\delta$ viewed from P' to P, according to the nodal convention

$$\beta\mathbf{A} + \delta\mathbf{C} + \beta\delta/\mathbf{BD} = 0 \quad (2.63a)$$

and

$$\mathbf{C}\beta + \mathbf{A}\delta + \mathbf{BD}/\beta\delta = 0. \quad (2.63b)$$

This new partial dislocation, which has a Burgers vector that connects the center of the line $\beta\delta$ to the center of the line **BD**, or vice versa, and has a magnitude of $\sqrt{2}b/3$, and is required to wrap the stacking fault around an obtuse angle, is referred to as a stair-rod dislocation of the second kind.

2.7.3 The Burgers Vector/Material Displacement Rule

When a dislocation on the {111} plane dissociates as indicated in Fig. 2.15(b), it needs to be determined whether the Shockley partial dislocations are arranged on the right and left side as shown in Fig. 2.15(b) or in the opposite sense. This question can be answered by inspection, considering two close-packed layers such as a {111} layer that contains atom D as shown in Fig. 2.15(a) and the {111} atom layer immediately above it, that is, the ABC layer. If a dislocation between these two layers is displaced parallel to itself, the resulting translation of the lower layer relative to the upper layer identifies the specific shearing direction of the material that moves the lower-layer atoms into the available interstitial sites and thereby fixes the type of the Shockley partial dislocation that accomplishes this. In this manner, a general rule is established readily (Thompson 1953) for the material motion associated with the motion of the dislocation, which states the following.

When a dislocation lying in a close-packed plane, that is, on a face of the tetrahedron, is viewed from the outside of the tetrahedron in the positive line direction and is displaced in a direction 90° clockwise from this positive line direction, the close-packed layer underneath will be displaced in the direction of the Burgers

vector. This indicates that the Shockley partial dislocation on the right side must move the material layer underneath into the only available interstitial site, and must always have a nature such that it has a Roman/Greek letter designation such as **B δ** as shown, for example, in Fig. 2.15(b). It will be followed by the second Shockley partial on the left-hand side, with a Greek/Roman letter designation such as **δ A**, shown in Fig. 2.15(b), which then moves the material back into an identity translation position.

2.7.4 Dislocation Reactions and Sessile Locks

In FCC crystals, with six different kinds of equivalent dislocations that can attract each other strongly, there are a number of important reactions possible that play prominent roles in strain hardening. Such interactions resulting in substantial energy reduction are also present in other crystal structures with a multiplicity of intersecting slip systems, such as the BCC and NaCl structures. Here we consider only the reactions in FCC crystals.

Consider, for example, the ABC plane depicted in Fig. 2.15(a) where two parallel mixed edge and screw dislocations parallel to the $[11\bar{2}]$ direction, one having a Burgers vector **BC** and the other **CA**, encounter each other on the same glide plane. These two dislocations will strongly attract each other, by Frank's rule referred to earlier, to form a single dislocation by the reaction

$$\mathbf{BC} + \mathbf{CA} \rightarrow \mathbf{BA}, \quad (2.64)$$

thereby liberating the line energy of a perfect dislocation. The resulting dislocation **BA**, still lying in the ABC (111) plane, can glide freely in this plane, just as the two dislocations it replaces.

Consider now a more important reaction, where a **DB** mixed dislocation on the $(\bar{1}11)$ γ plane with a line vector parallel to the AB direction encounters a similar mixed dislocation **BC** on the (111) δ plane with its line vector also parallel to the AB direction. These two dislocations, on different glide planes, will nevertheless still strongly attract each other to the line of intersection AB of the two δ and γ glide planes and will undergo a reaction

$$\mathbf{DB} + \mathbf{BC} \rightarrow \mathbf{DC}, \quad (2.65)$$

releasing, again, the energy of an entire perfect dislocation line. The resulting pure edge dislocation parallel to the AB direction will have a Burgers vector **DC** that lies in the (001) plane, which is not a glide plane in FCC crystals. Thus, the product dislocation cannot glide and will be sessile. Such dislocations were considered first by Lomer (1951), as prominent obstacles to glide dislocations in the δ and γ plane that follow on the same plane after the reaction.

Cottrell (1952) noted that in FCC crystals with a relatively low stacking-fault energy, the reacting dislocations would be dissociated as discussed in Section 2.7.1 as

$$\mathbf{DB} \rightarrow \mathbf{D}\gamma + \gamma\mathbf{B}, \quad (2.66a)$$

$$\mathbf{BC} \rightarrow \mathbf{B}\delta + \delta\mathbf{C}, \quad (2.66b)$$

on the γ and δ planes and that the reaction would be more complex. Thus, instead of giving rise to a perfect dislocation of the Lomer type, the two leading partials $\gamma\mathbf{B}$ on the γ plane and $\mathbf{B}\delta$ on the δ plane would react to give a stair-rod dislocation of the first kind,

$$\gamma\mathbf{B} + \mathbf{B}\delta \rightarrow \gamma\delta, \quad (2.67)$$

lying on the line of intersection AB of the γ and δ planes.

The two following Shockley partials $\mathbf{D}\gamma$ on the γ plane and $\delta\mathbf{C}$ on the δ plane remain in their respective planes, separated from the stair-rod dislocation $\gamma\delta$ by stacking faults of width A' , as depicted in Fig. 2.17(a), forming a roof-like 3-D dislocation. This entire extended configuration, still having an overall Burgers vector $\mathbf{DC}$, is sessile similarly to the Lomer dislocation. It is referred to as the Cottrell dislocation. These two sessile dislocations collectively referred to as Lomer–Cottrell (LC) dislocations, can form in a variety of ways, as discussed in Chapter 7, and play a very important role in the strain hardening of FCC crystals as locks, anchoring dislocation clusters to the lattice (Basinski 1964; Prinz and Argon 1980). Of the two reactions, the Cottrell reaction with its 3-D roof-like form results in a substantial energy reduction $\Delta\mathfrak{F}_c$, which can be determined by a procedure analogous to what led to eq. (2.61) for the energy reduction in the dissociation of an edge dislocation in a plane. This determination, involving evaluation of the interaction energies of

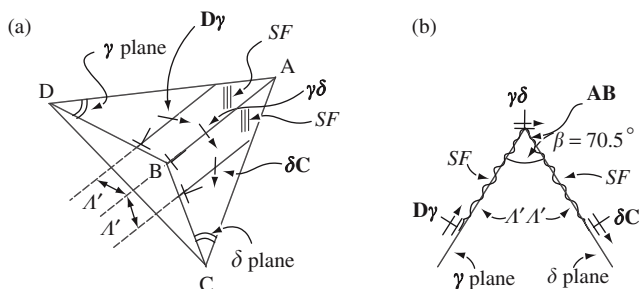


FIG. 2.17. (a) A Cottrell dislocation along the line of intersection AB between the γ and δ planes, showing 3-D dissociation with partials $\mathbf{D}\gamma$, $\gamma\delta$, $\delta\mathbf{C}$; (b) the Cottrell dislocation viewed in the AB direction.

all three partials $\mathbf{D}\boldsymbol{\gamma}$, $\boldsymbol{\gamma}\boldsymbol{\delta}$, and $\boldsymbol{\delta}\mathbf{C}$ with each other, results in

$$\Delta\tilde{\mathcal{E}}_c = -\frac{\mu b^2}{4\pi(1-\nu)} \left\{ \ln\left(\frac{\alpha R}{b}\right) - \frac{(4+3\nu)}{9} \ln\left(\frac{\alpha \Lambda}{b}\right) \right\}. \quad (2.68)$$

Here, the separation

$$\Lambda' = \frac{2}{3(2+\nu)} \Lambda \quad (2.69)$$

between the Shockley partials and the stair-rod dislocation, given in terms of the separation Λ between the partials of the initial unreacted dissociated dislocations given by eq. (2.59), has been incorporated into eq. (2.68). The reduction of energy given by eq. (2.68) is relative to twice the energy of a regular dissociated edge dislocation given by eq. (2.60).

2.8 Plastic Deformation by Shear Transformations

2.8.1 Types of Transformation

Shear transformations in crystals constitute an alternative to slip for inelastic deformation. These transformations include deformation twinning, stress-induced martensitic transformations, and kinking, which we consider briefly below.

2.8.2 Deformation Twinning

Many crystal structures can undergo deformation twinning, in which the lattice inside a specific volume element of characteristic lenticular shape undergoes a crystallographically specific, atomically homogeneous simple shear strain $\gamma^T (= s/h)$ that transforms the interior lattice into a mirror image of the exterior, with respect to a composition plane K_1 , by shear in the η_1 direction through a distance s between planes a distance h apart, as shown in Fig. 2.18(a). Figure 2.19 shows a collection of deformation twins in a Zn crystal. There are a number of important geometrical differences between twinning and slip. If each atomic plane a distance a apart were to undergo slip by the passage of a single dislocation, a homogeneous strain $\gamma (= b/a)$ would result but the slipped volume would remain coherent and indistinguishable from the exterior. In the case of twinning, as shown in Fig. 2.18(b), for the case of a BCC metal, the lattice of the homogeneously sheared volume assumes a mirror relationship with respect to the exterior across the composition plane (112). Twinning shear strains, governed by the specific symmetry properties of the lattice, are unidirectional and always less than the strains the corresponding case of homogeneous slip on every plane. The case of the homogeneous twinning shear shown in Fig. 2.18(b) for the BCC lattice, where a simple shear alone accomplishes the end

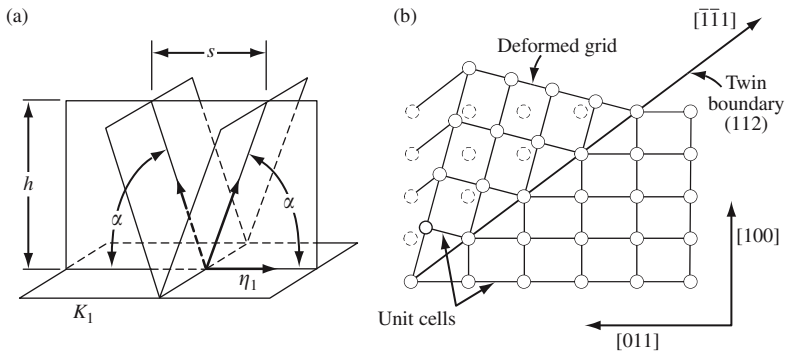


FIG. 2.18. (a) Geometry of an unconstrained twinning process, where K_1 is the composition plane, η is the shear direction, and $\gamma^T = s/h$ is the unconstrained twinning shear strain; (b) geometry of twinning in a BCC crystal.

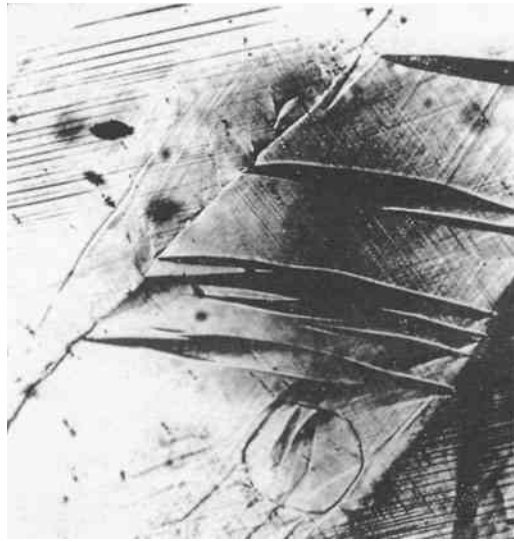


FIG. 2.19. Deformation twins in Zn.

result of relocating all atoms in their proper position, is a special case. In many cases the twinning shear will only transform a sublattice relative to the exterior, and relocating all other atoms properly inside the twinned region will require wholesale internal atom switches, which are referred to as *shuffles*.

The conditions for the nucleation of twins, and for the formation of martensite embryos for that matter, have been discussed in great detail. Elementary considerations of the homogeneous nucleation of twins under stress (Orowan 1954) involve the elastic strain energy of the constrained transformation (Eshelby 1957), the changes in the Helmholtz free energy of the partially stressed lattice, the interface

energy, and the work done by the prevailing stress state. All such analyses show that under the usual levels of applied stresses, the activation free energy for homogeneous twin (or martensite) nucleation is unattainably large. This leads to the conclusion that twin (or martensite) nucleation requires a high local stress concentration (by a factor of 20–50), preexisting embryos, and/or special heterogeneities such as pole sources (Cottrell and Bilby 1951; Venables (1964)) to circumvent the need for nucleation. As a result of these generally acknowledged difficulties, the reported measurements of critical resolved shear stresses for twinning show very large scatter (Venables 1964). Nevertheless, the validity of these conclusions has been well established, in that the occurrence of twinning increases (a) with increasing levels of flow stress through strain hardening, and (b) with decreasing magnitude of the twinning shear strain.

A collection of observed twinning systems in some prominent metals and compounds is presented in Table 2.4. An examination of the magnitudes of the twinning shear strains shows that they are all very substantial. This has important consequences. During the formation and early growth of a twin, the twin assumes a lenticular shape that minimizes the elastic strain energy of transformation (Orowan 1954; Eshelby 1957), which, however, severely concentrates shear at the tip, ahead of the twin. As a result, in crystals with high plastic slip resistance, under uniform levels of stress, twins propagate with a near-sonic velocity (Bunshah 1964; Takeuchi 1966), but under very nonuniform stresses, such as in surface indentations, they can spread quasi-statically, and fully reverse when the stress is removed (Cahn 1954). In crystals with moderate plastic slip resistance, the very high stresses usually initiate local processes of plastic dissipation which level down the high stresses and remove the tendency of the twin to reverse. In the BCC lattice, where the $\{112\}$ composition plane and the $\langle 111 \rangle$ twinning direction are also the slip plane and the slip direction, the plastic dissipation occurs by emission of dislocations in the plane of the twin (emissary dislocations, Sleeswyk 1962) from its tip, thereby terminating the twinning process but propagating and dissipating the concentrated shear. Since twins propagate very fast in materials with even moderate plastic slip resistance, it is not uncommon that stress relief occurs by local microcrack nucleation, particularly in crystals that can undergo cleavage.

In HCP metals and other systems with an insufficient number of independent slip systems for compatible deformation in polycrystals, twinning is an important supplementary form of plastic deformation (Reed-Hill 1964) that enhances formability.

2.8.3 Stress-induced Martensitic Transformations

In many crystal structures, shear transformations exist which transform the sheared region into another crystal structure, such as the well-known *martensite* transformation in carbon steels that converts FCC γ -Fe into a body-centered

TABLE 2.4. Mechanical-twinning elements for metals (Klassen-Neklyudova 1964)

Metal	K_1	η_1	γ^T	Notes
<i>Face-centered cubic</i>				
Cu	(111)	$[11\bar{2}]$	0.707	Only recrystallization twins with these elements occur in Ag, Al, Au, γ -Fe, and Co
<i>Body-centered cubic</i>				
Cr, α -Fe, Mo, Na	$\left. \begin{matrix} (112) \\ (441) \\ (332) \end{matrix} \right\}$	$[11\bar{1}]$	0.707	
W	(112)	$[11\bar{1}]$	0.707	
<i>Diamond cubic</i>				
Ge	(111)	$[11\bar{2}]$	0.707	
<i>Hexagonal close-packed</i>				
Be	$\left. \begin{matrix} \\ \\ \\ \\ \end{matrix} \right\} (10\bar{1}2)$	$[\bar{1}011]$	0.199	$c/a = 1.568$
Cd		$[10\bar{1}\bar{1}]$	0.171	$c/a = 1.886$
Mg		$[\bar{1}011]$	0.129	$c/a = 1.624$
Ti		$[\bar{1}011]$	0.189	$c/a = 1.587$
Zn		$[10\bar{1}\bar{1}]$	0/139	$c/a = 1.856$
Be	$\left. \begin{matrix} (10\bar{1}1) \\ (10\bar{1}3) \end{matrix} \right\}$			Additional forms reported
Mg	(10 $\bar{1}$ 1)	$[10\bar{1}2]$	1.066	Additional forms reported
Ti	$\left. \begin{matrix} (11\bar{2}1) \\ (11\bar{2}2) \\ (11\bar{2}3) \\ (11\bar{2}4) \end{matrix} \right\}$	$[\bar{1}\bar{1}26]$	0.638	$\left. \begin{matrix} \\ \\ \\ \end{matrix} \right\}$ Additional forms reported
		$[\bar{1}\bar{1}\bar{2}3]$	0.957	
		$[\bar{1}\bar{1}22]$	1.194	
		$[\bar{2}\bar{2}43]$	0.468	
	$\{30\bar{3}4\}$	$\langle 20\bar{2}3 \rangle$		
	$\{101\bar{3}\}^*$	$\langle 30\bar{3}2 \rangle$		*150 and 286 °C
Bi	($\bar{1}$ 012)	$[10\bar{1}1]$	0.118	
Hg	($\bar{1}$ 012)	$[10\bar{1}1]$	0.447	
Sb	($\bar{1}$ 012)	$[10\bar{1}1]$	0.146	
	($\bar{1}$ 012)			
α -Zr	$\left. \begin{matrix} (11\bar{2}1) \\ (11\bar{2}2) \\ (11\bar{2}3) \end{matrix} \right\}$			
<i>Tetragonal</i>				
β -Sn	(301)	$[\bar{1}03]$	0.119	$c/a = 0.541$
In	(101)	$[10\bar{1}]$	0.150	$c/a = 1.078$
<i>Orthorhombic</i>				
α -U	(1 $\bar{7}$ 2)	$[312]$	0.228	$x = \text{irrational twin}$
	(112)	$x[372]$	0.228	
	(121)	$x[100]$	0.329	
	(111)	$[12\bar{3}]$	0.24	
	(130)	$[3\bar{1}0]$	0.299	

tetragonal form, after which all such transformations have been named. Other similar and technologically important transformations include an FCC to BCC transformation in Fe–Ni (27–34% Ni) alloys, BCC to hexagonal in Ti, and BCC to face-centered tetragonal in Cu–Zn (40% Zn) alloys, in the case of metals (Klassen-Neklyudova 1964); tetragonal to monoclinic in ZrO_2 , in the case of ceramics; and orthorhombic to monoclinic in polyethylene, with six different possible variants (Bevis and Crellin 1971). While martensitic transformations are basically accomplished by shear, resulting in transformation shear strains of the order of 0.1–0.15, they are usually associated also with a substantial dilatational component of the order of 0.03–0.05. Moreover, the total transformation strain tensor depends on composition. The crystallography of martensitic transformations is usually complex and is made more so by the frequent occurrence of subsequent internal twinning with a number of variants. The early work has been summarized by Klassen-Neklyudova (1964) (including geometrical details of the transformation in many of the above-listed cases), and the more recent work, including assessment of mechanistic models, by Cohen and Wayman (1981) and Olson and Cohen (1986). Many of the problems associated with the energetics of formation and propagation of twins also apply to martensitic transformations. Since stress-induced martensitic transformations can have very substantial and attractive benefits in the toughening of ceramics and high-strength steels, much recent research has been directed toward these effects (see Olson *et al.* 1987).

2.8.4 Kinking

Mineralogists discovered in the nineteenth century (see Mügge 1898) that certain minerals often deform plastically by undergoing organized simple shear by slip inside a narrow band in which the slip planes and the slip directions are perpendicular to the band boundaries, as depicted in Fig. 2.20. The role of this type of deformation in metal crystals with layered slip systems such as Zn was discussed by Orowan (1942). The formation of kink bands in Zn and Cd was investigated further experimentally by Gilman (1954) (see Fig. 2.21), and the conditions for the growth of kink bands by a systematic process of punching-out of dislocation dipoles at the ends of the bands were investigated by Frank and Stroh (1952). Kinking is ubiquitous when crystals with layered slip systems are subjected to compression parallel to the plane of the layers or extended normal to them. As pointed out by Orowan, in well-accommodated kink bands the boundaries of the bands must bisect the lattice misorientation if the bands are to be free of long-range stresses. This necessitates that the planes of the boundaries rotate at half the angular rate of the rate corresponding to increase of shear strain in the interior of the band. This is often not possible, resulting in the formation of cleavage cracks in a variety of ways (Gilman 1954).

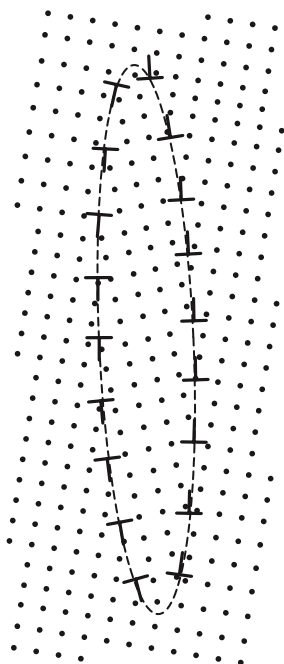


FIG. 2.20. Kinking by organized shearing in a crystal.

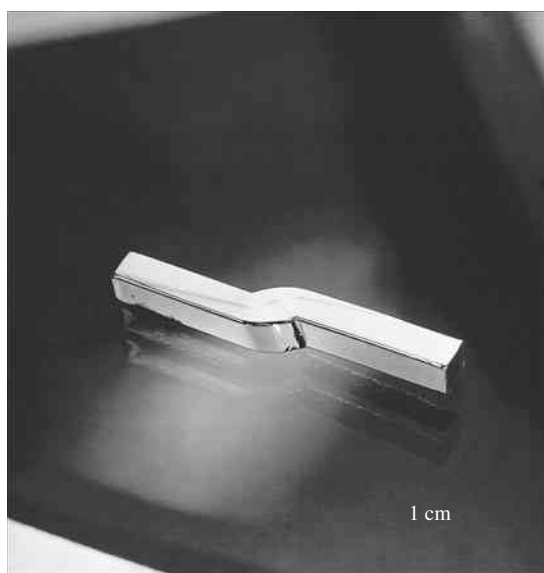


FIG. 2.21. A macroscopic kink in a Cd crystal (courtesy of J. J. Gilman, private communication, 1961).

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OVERVIEW OF STRENGTHENING MECHANISMS

3.1 Introduction

As outlined in Section 1.4, dislocations are the most effective carriers of plasticity in crystalline solids. Under normal circumstances, they are readily generated by purely topological convolution processes as in Frank–Read sources or by double cross-slip of screw dislocations, which can in turn produce new Frank–Read sources readily, all at relatively low levels of stress as discussed in Section 2.3.7. In BCC metals and strongly directionally bonded solids, the glide of dislocations in their slip planes is resisted by a lattice resistance of generally substantial magnitude at low temperatures, giving these materials a high intrinsic plastic resistance to begin with. In comparison, in pure close-packed FCC and HCP metals the lattice resistance in the best slip systems is normally very small, making it necessary to introduce other extrinsic mechanisms to raise the plastic resistance. This is most effectively accomplished by alloying with a second constituent, which, either in the form of a solid solution or as precipitate particles in the host metal, can very effectively elevate the resistance to dislocation motion. In the following section we discuss first why a dislocation mechanics view is essential for properly understanding the remarkable effectiveness of strengthening by small volume fractions of second constituents, either in solution or in the form of nanometer-sized precipitates, which is both qualitatively and quantitatively missed by a heterogeneous continuum plasticity approach. Following this comparison, we introduce briefly the major strengthening mechanisms that will be described in greater detail in Chapters 4–8.

3.2 The Continuum Plasticity Approach to Strengthening Compared with the Dislocation Mechanics Approach

If the plastic shear resistance τ_m of a pure matrix is inadequate, it might be expected that incorporation of a second constituent of much higher plastic resistance would increase the plastic resistance of the pure metal. Composite theory, which deals with such problems, demonstrates that the strengthening that can be achieved depends not only on the plastic resistance of the second constituent but also on its volume fraction c and, above all, on the geometry and form of dispersion of the reinforcing element of this second constituent. Here, to simplify the consideration and make

a direct comparison possible with the actual process of utilizing precipitate particles in the context of crystal plasticity, we consider only spherical or equiaxed reinforcing components and resort to several simplifications. Thus, we consider the strengthening of a matrix of plastic shear resistance τ_m with an equiaxed heterogeneity component of plastic resistance $\tau_h > \tau_m$, at a volume fraction of c , randomly distributed in the matrix.

By an upper-bound consideration assuming that both components, the matrix and the heterogeneity component, deform compatibly by suffering the same plastic shear strain, the overall plastic shear resistance τ_{cc} of the composite, in the continuum model, would be given by composite theory as a rule-of-mixtures sum of the plastic resistances of the two separate components,

$$\tau_{cc} = c\tau_h + (1 - c)\tau_m. \quad (3.1)$$

We note that in reality the plastic resistance of the composite would be considerably less, since the two components would not suffer the same plastic shear strain, and stress concentrations would develop between components (Bao *et al.* 1991). If $\tau_m \ll \tau_h$, as would be the case with pure FCC and HCP metals, and the volume fraction c of the heterogeneity component were not too small, the matrix contribution to the overall plastic resistance could be neglected, giving only

$$\tau_{cc} \cong c\tau_h. \quad (3.2)$$

The implicit assumption of this composite approach is that both the matrix and the particles are able to deform plastically independently from each other, and interact merely across their common borders. In reality, however, if the second constituent is in solid solution in the matrix or is in the form of precipitate particles of nanometer dimensions, the assumptions of composite theory do not hold. In the case of precipitates, the latter would be dislocation-free and would lack a means of deforming independently from the matrix, and the required dislocation multiplication could not be nurtured in them. In the case where the second constituent is in the form of a solid solution, independent deformation of the solute atoms would make little sense. If the second constituent is in the form of precipitate particles that are initially free of dislocations, and are to shear, they need to have lattice coherence with the host matrix and to be shearable only by the passage of matrix dislocations through them. Then, if there is little resistance to dislocation motion in the matrix but there is a high resistance in the precipitate particles, the latter will hold up matrix dislocations, forcing them to bow around them. In this manner, small precipitate particles at relatively low volume concentrations can exercise a large nonlocal effect of impeding the motion of matrix dislocations along their entire length by pinning them at discrete points of contact with the precipitates until the latter are eventually sheared by the force of the line tension of the bowed-out matrix dislocations pressing on them. As we shall discuss in Chapter 6 in detail,

we must take account of important statistical considerations of the sampling of small precipitate obstacles by dislocation lines moving through a field of such obstacles that are to be sheared. These considerations give an overall plastic shear resistance τ_p for the particle field of

$$\tau_p = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{b\langle r \rangle} \right) \left(\frac{3\pi c}{32} \right)^{1/2}, \quad (3.3)$$

where $\langle r \rangle$ is the effective average radius of the spherical precipitates in the slip planes of matrix dislocations and $\hat{K} = \tau_h \pi \langle r \rangle^2$ is the peak shear resistive force of a precipitate particle. If the plastic shear resistance of the matrix is negligible in comparison with that of a precipitate, then eq. (3.3) gives the entire shear resistance of the alloy and is the counterpart of the composite-theory alternative given by eq. (3.2).

Thus, the actual plastic resistance given by eq. (3.3) resulting from the presence of nanometer-sized precipitates is of a very different form from that given by the composite-theory estimate of eq. (3.2). The radically different nature of these two alternatives may be made clearer and more transparent by stating the two results in a similar form. For this, we recognize that the dislocation mechanics resistance τ_p is actually the entire shear resistance that is to be compared with that of the composite model τ_{cc} . The two forms of resistance then become

$$\tau_{cc} = \mu \left(\frac{\tau_h}{\mu} \right) c \quad (3.4)$$

and

$$\tau_c = \tau_p = (0.3)\mu \left(\frac{\tau_h}{\mu} \right)^{3/2} \left(\frac{\pi \langle r \rangle}{b} \right)^2 c^{1/2}, \quad (3.5)$$

where, in order to obtain the form of eq. (3.5), we have introduced the constant-line-tension estimate $\mathcal{E} = \mu b^2/2$ and evaluated the numerical constants.

Comparison of eqs. (3.4) and (3.5) shows that the composite model is independent of the particle size and is linearly dependent on the volume fraction, whereas the model of the actual nonlocal interaction of a dislocation with precipitates depends on the square of the particle size $\langle r \rangle$ and on the volume fraction c to the power $(1/2)$. Thus, for a given set of heterogeneity particles of given size, the rate of change of plastic resistance in the composite model with the particle volume fraction is merely

$$\frac{d\tau_{cc}}{dc} = \mu \left(\frac{\tau_h}{\mu} \right), \quad (3.6)$$

whereas for the nonlocal interaction model it becomes

$$\frac{d\tau_c}{dc} = (0.15)\mu \left(\frac{\pi \langle r \rangle}{b} \right)^2 \left(\frac{\tau_p}{\mu} \right)^{3/2} / c^{1/2}. \quad (3.7)$$

Clearly, while the rate of change of τ_{cc} with c is merely constant, it is unbounded for τ_c as $c \rightarrow 0$, demonstrating the remarkable nonlocal effects of small volume concentrations of precipitates in terms of strengthening; this is a direct consequence of the fact that small precipitates lack a means of deformation independent from the matrix and can be sheared only by the passage of matrix dislocations. In Section 8.12.2 we demonstrate that when precipitates increase in size and become nonshearable but still remain coherent with the host lattice, the nonlocal interaction effect continues to hold and τ_c remains substantially larger than τ_{cc} . The strengthening described by the continuum plasticity theory is only reached when the heterogeneities become large enough to acquire a means of deforming independently from the matrix.

3.3 The Lattice Resistance

In Chapter 4, we shall discuss the intrinsic lattice resistance of pure metals and semiconductors. We demonstrate that in pure FCC metals the lattice resistance is uniformly low for both edge and screw dislocations. In BCC metals, the lattice resistance is high only for screw dislocations, resulting primarily from the nonplanar, spatially spread-out nature of their cores. We develop models for the temperature-assisted motion of screw dislocations and apply these to the strain rate and temperature dependence of the plastic resistance of BCC metals. Comparison with experimental results shows very good agreement.

We consider in some detail the core structure of dislocations in silicon and develop models for the temperature dependence of dislocation velocities for edge and screw dislocations that are in very good agreement with experimental measurements over a wide range of applied stress.

3.4 Solid-Solution Strengthening

When the second constituent introduced for strengthening remains entirely in solution, its interaction with dislocations is diffuse and quite complex. In Chapter 5, we consider only the strengthening caused by substitutional solute atoms and leave out the effect of an interstitial solute. We note that a solute atom can interact with a dislocation either through a size misfit or a modulus misfit. While these two interactions often coexist for a solute, we treat them separately. A solute atom of a

size different from the host atom can interact with the stress field of a dislocation line through a series of random interaction forces that the solute atoms produce. For a spherically symmetrical size misfit, the interaction is primarily with edge dislocations. When a solute atom significantly changes the bonding interaction with its surroundings, it acts as an atomic-size heterogeneity having a different bulk modulus and/or shear modulus, affecting the line energy of a nearby dislocation locally. We show that these two interactions are strong only when the solute resides in planes just outside the core radius of the dislocation. As a set of elemental problems, we calculate first the individual interaction force profiles of solutes with dislocations and evaluate their peak values for prominent solute types, for both edge and screw dislocations in FCC alloys.

We then consider a model, proposed first by Mott (1952) and later by Labusch (1972), of the diffuse interaction of a set of randomly positive and negative interaction forces exerted on a dislocation line under stress, referred to generally as the Mott–Labusch (ML) model of interaction statistics. In the ML model, the dislocation samples a random-error average of solute interaction forces of randomly positive and negative sign along a certain line segment that can advance coherently, independent of other neighboring segments. This provides an estimate of an athermal limit on the plastic resistance of the alloy. The model can be readily modified to account successfully for the temperature-assisted motion of dislocations through a solute field.

We then consider solute strengthening of BCC metals, where the interactions are primarily with screw dislocations and involve the sampling of random solute clusters by wide kinks along the screw dislocations. The model predicts solid-solution softening at low temperatures but hardening at intermediate temperatures, when compared with the behavior of the pure reference BCC metal. Comparisons of predictions with experimental results are generally good.

3.5 Precipitation Strengthening

When the second constituent forms discrete precipitates, they can interact with dislocations through a large variety of mechanisms, which can often coexist in the same precipitate. We consider first each mechanism separately, and then consider how the separate resistances superpose. These separate interactions are systematically discussed in Chapter 6.

The interaction of dislocations with discrete precipitates randomly distributed in space is dealt with in a variety of approximations. For precipitates that interact with glide dislocations at contact, the problem of sampling the precipitates in the glide plane by the dislocation line has been idealized by treating the precipitates as points in the plane that can be overcome when the cusp angle of the contacting dislocation segment wrapped around the precipitate reaches a critical magnitude for shearing

the precipitate. This sampling problem, dealt with by Friedel (1964) and Fleischer and Hibbard (1963) separately, is referred to as the Friedel–Fleischer (FF) statistics of sampling. It results in a plastic shear resistance given by eqs. (3.3) and (3.5) and has wide applicability. For precipitates that have a size and/or modulus misfit, the interaction with the dislocation is more diffuse and extends out of the glide plane of the dislocation. This requires a special summation procedure to account for the out-of-plane interactions of precipitates that do not touch the dislocation in the glide plane, to formally reflect the effect into the glide plane, where the FF statistics of sampling can still be utilized. The interactions become more complex when the precipitates can no longer be idealized as points in the plane, which occurs with higher volume fractions of precipitates. This sampling problem was considered by Schwarz and Labusch (1978) and is referred to as the Schwarz–Labusch (SL) interaction statistics; it introduces important modifications to the FF statistics.

The distinguishing feature of the sampling of precipitates by a dislocation in the FF statistics, compared with the ML statistics for solute atoms, is that in the former, under conditions of flow, the dislocation samples only relatively strong, sparse, discrete opposing interaction forces separately, while in the latter case of diffuse, relatively weak but more plentiful and partly overlapping obstacles, it samples mostly opposing, but also some aiding interaction forces. The transition from the ML form of diffuse sampling to the FF form of discrete sampling occurs when, for relatively weak obstacles with an action range w , normalized peak strength $\hat{K}/2\mathcal{E}$, and concentration c , under a condition of impending flow, the flexed dislocation under stress begins to just sample only opposing interaction forces, while the aiding interaction forces have been relaxed in the geometry of the flexed dislocation. As discussed in Chapter 5, this transition scenario defines a parameter γ , given by

$$\gamma = \frac{\alpha(w/b)^2 c}{\hat{K}/2\mathcal{E}}, \quad (3.8)$$

where $\alpha \approx O(10)$. An approximate analysis indicates that the transition from the FF to the ML form of sampling of obstacles occurs when $\gamma > 1.0$, that is, in cases where the interaction force ($\hat{K}/2\mathcal{E}$) is very weak and the volume concentration c is large. This usually occurs with solutes that have interaction forces typically of the order $\hat{K}/2\mathcal{E} \approx 0.02$ and $w/b \approx 1.7$ and for solute concentrations c exceeding 0.001, when the interactions become diffuse.

In Chapter 6, these forms of sampling are introduced and it is shown that precipitates are in the FF range.

Following our general arguments, specific interaction mechanisms are considered for chemical strengthening (interface shear step formation), stacking-fault strengthening, atomic-order strengthening with ordered superalloy precipitates,

size misfit and modulus misfit strengthening, strengthening by nonshearable particles (the Orowan process), strengthening by spinodal-decomposition microstructures, and strengthening by precipitate-like obstacles in the form of nanovoids and the like. For all cases, specific models are developed, and detailed comparison with experimental results is presented where separate mechanisms have been clearly identified to be present.

3.6 Strengthening by Strain Hardening

Strain hardening, alternatively referred to as work hardening, which has been used for ages to strengthen metals, still remains a process that is at best only semiquantitatively understood, although the basic elements of it have been well characterized and some aspects of it have been well modeled. Strain hardening, first tackled by Taylor (1934), results from the interaction of the strain fields of dislocations as more and more are introduced during plastic flow. The principal complexity of the process arises more from a need to understand how an increasing dislocation density resulting from plastic straining is retained, and less from accounting for the forms of stress field interactions. That is, the principal problem is the need for a mechanistic understanding of the evolution of dislocation microstructures, which can, on the whole, be studied by a variety of forms of microscopy and X-ray diffraction. In Chapter 7, we consider primarily strain hardening in pure FCC metal single crystals and imply that the mechanisms that are involved there will also apply to BCC metals and other crystalline solids.

We start by noting that there are two forms of strain hardening: *interplane hardening*, resulting from interaction of the stress fields of glide dislocations moving on parallel planes, a process considered first by Taylor (1934), and *intraplane hardening*, resulting from dislocations interacting with slip obstacles in their glide plane, studied by many subsequent investigators but first identified separately by Hirsch and Humphreys (1969). It is the intraplane hardening that is of central importance, since it relates directly to the effect of slip obstacles, their forms of increase with strain, and their association with the microstructures that are observed.

After presenting the fundamental experimental manifestations of strain in stress-strain curves and dislocation microstructures, we introduce specific models for the several distinct stages of strain hardening in FCC crystals, where different identifiable forms of interaction are dominant. We note that most stages of hardening in FCC crystals are temperature-independent, except for Stage III, which involves dynamic recovery, that depends on temperature and strain rate but is not affected by diffusion. We provide a specific mechanism for dynamic recovery. We delay consideration of strain hardening in the presence of heterogeneities such as precipitates and grain boundaries to Chapter 8.

3.7 Phenomena Associated with Strengthening mechanisms

A number of separate deformation phenomena, as well as important variants of hardening mechanisms, are considered in Chapter 8. These phenomena include mobile dislocation densities and their dynamics; yield phenomena related to the establishment of mobile dislocation densities; instabilities in mobile dislocation fluxes and jerky flow; dynamic overshoot at low temperatures in underdamped deformation; the Bauschinger effect upon reversal of deformation; interaction between strain hardening and heterogeneities involving geometrically necessary and statistically stored dislocations; plasticity in polycrystals and the evolution of deformation textures; deformation in nanometer-scale grains and the breakdown of the Hall–Petch effect; and, finally, the transition from dislocation mechanics to continuum plasticity.

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THE LATTICE RESISTANCE

4.1 Overview

One of the most fundamental resistances to dislocation motion is that which the discrete lattice offers in a pure crystalline material in a temperature range where diffusion plays no role. We shall consider this resistance from two mechanistically different points of view. First and foremost is the glide resistance, widely referred to as the Peierls–Nabarro (PN) resistance, which results from the pulsing distortions of the dislocation core as it moves through the discrete lattice, often affecting edge and screw dislocations differently, and at very different levels in different crystal structures. We shall examine various models of this resistance, and consider in some detail its temperature and strain rate dependence in BCC metals and to a lesser extent in diamond cubic Si. The second form of resistance, which affects dislocations in all materials quite similarly, is the phonon drag arising from the interaction of a moving dislocation with lattice thermal vibrations, which we have already referred to briefly in Section 2.3.6. We note that the temperature dependence of this resistance will be radically different from that of the lattice resistance.

4.2 Model of a Dislocation in a Discrete Lattice

4.2.1 The Peierls–Nabarro Model of an Edge Dislocation–Updated

Two important questions cannot be answered by considerations of dislocations in linear elastic continua: the removal of the stress singularity at the dislocation core, and the possible variation of the energy of the dislocation with its position in the lattice, which binds it to lattice sites. Both of these problems were addressed first by Peierls (1940) and Nabarro (1947) in an idealized fashion and have been reexamined since by a number of investigators. We shall follow here the original development as modified recently by Joos and Duesbery (1997). The development is based on a minimal concession to atomistic detail in a generally linear elastic formalism for the dislocation field.

Thus, consider the formation of a positive edge dislocation by bringing together two linear elastic half-spaces as shown in Fig. 4.1. The upper half-space contains one atomic layer more than the lower one. When the two half-spaces are brought

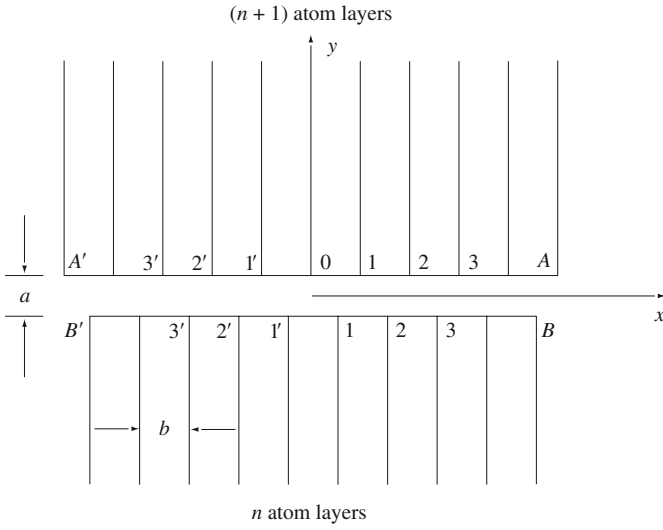


FIG. 4.1. Two elastic half-spaces, before accommodating an edge dislocation between them. The upper half-space has one extra atomic layer.

together to the equilibrium lattice spacing so that the atoms on both sides interact, the two ends at A and B at $x = +\infty$ and A' and B' at $x = -\infty$ will align and spread the pairwise atomic alignment toward the center, thereby forming a positive edge dislocation at the center. The minimal concession to atomic detail is made by introducing a periodic shear resistance to prescribe the relative shear interaction of the two half-spaces across the gap, having the simple sinusoidal form of

$$\tau_{xy} = \tau_{is} \sin \frac{2\pi f(x)}{b}, \quad (4.1)$$

where $f(x)$ is the relative shear displacement between the upper half-space and the lower half-space at any particular position x , and τ_{is} is the maximum value of the resistance, taken as the ideal shear strength. If the interplanar shear resistance is to match the linear elastic shear resistance for small relative displacements across the layer thickness a between the half-spaces, it is required that $\tau_{is} = \mu b / 2\pi a$, where a is the interplanar separation in the y direction. As the two half-spaces interact and the misfit of one atomic layer is concentrated from the two sides at $x = \pm\infty$ toward the center, the resistive shear stresses must increase and cease to be linearly elastic, and reach the ideal shear strength close to the center according to eq. (4.1). Closer in than this distance, the shear resistance will decline to zero at the center of the dislocation by the requirements of point symmetry. This will effectively spread out the misfit around the core of the dislocation and remove the singularity in the stress. Thus, the elastic interactions of the half-spaces become balanced against the

periodic shear resistance everywhere in the x direction. The balance condition is stated by considering the spread-out material misfit at the core as a continuously distributed density of infinitesimal linear elastic edge dislocations, which produce a shear stress at any point x along the junction plane of the half-spaces that must be balanced against the interplanar shear resistance $\tau_{xy}(x)$ through the relative shear displacement $f(x)$ between the half-spaces at that point x (Eshelby 1949). This translates into a singular integral equation

$$\frac{\mu}{2\pi(1-\nu)} \int_{-\infty}^{\infty} \frac{df(x')/dx'}{(x-x')} dx' = \tau_{is} \sin \frac{2\pi f(x)}{b}, \quad (4.2)$$

where the integral represents the shear stress σ_{xy} set up at any point x by packets of distributed infinitesimal Burgers displacements $(df(x')/dx')dx'$ at sites x' integrated over the entire x' axis, subject to the normalization condition

$$\int_{-\infty}^{\infty} \frac{df(x')}{dx'} dx' = b. \quad (4.3)$$

The solution of this integral equation gives the relative equilibrium displacement between the half-spaces in the plane of the dislocation as

$$f(x) = \frac{b}{\pi} \tan^{-1} \frac{x}{\xi} + \frac{b}{2}, \quad (4.4)$$

where

$$\xi = \frac{\mu b}{4\pi(1-\nu)\tau_{is}} \quad (4.5)$$

is the half-width of the spread-out dislocation core. We note that substitution of eq. (4.4) into the shear resistance profile given by eq. (4.1) now gives the distribution of the resistive shear stress $\tau_{xy}(x)$ along the x axis and the new distribution of the shear stress σ_{xy} of the edge dislocation as follows:

$$\sigma_{xy}(x) = -\tau_{xy}(x) = 2\tau_{is} \frac{x\xi}{x^2 + \xi^2}. \quad (4.6)$$

For a sinusoidal resistance with $\tau_{is} = \mu b/2\pi a$ and $\xi = a/[2(1-\nu)]$, the shear stress σ_{xy} along $y = 0$ at $x \gg \xi$ is

$$\sigma_{xy} = \frac{\mu b}{2\pi(1-\nu)x}, \quad (4.7)$$

which is the same as that of the linear elastic dislocation. The stress reaches a maximum value of $\mu b/2\pi a = \tau_{is}$ at $x = \xi$, the half-width of the dislocation,

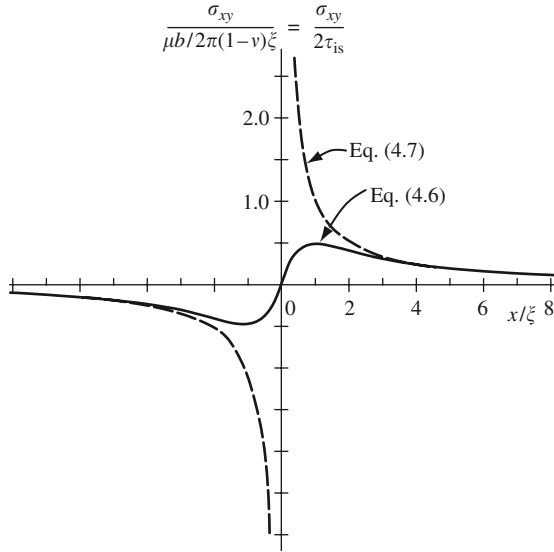


FIG. 4.2. Distribution of shear stress of a positive edge dislocation around the center, in a linear elastic medium (eq. 4.7) and in a medium having an ideal shear strength (eq. 4.6).

which can now be called the core radius, and falls to zero as $x \rightarrow 0$, having the distribution given in Fig. 4.2, where the linear elastic stress distribution that it replaces is shown as a broken curve. It is easy to demonstrate that the introduction of the periodic interplanar shear resistance of eq. (4.1) along the $y = 0$ plane, which removed the singularity in the shear stress σ_{xy} , will, by equilibrium, also remove the singularity in all other stress components around the dislocation core.

The line energy of the dislocation now becomes simply

$$\mathfrak{F} = \frac{b}{2} \int_0^R \sigma_{xy}(x) dx = \frac{\mu b^2}{4\pi(1-\nu)} \ln\left(\frac{R}{\xi}\right), \quad (4.8)$$

which now includes the energy in the entire field, including that of the core.

4.2.2 The Stress to Move the Dislocation

In the PN model of an edge dislocation presented in Section 4.2.1, with its core spread out over a distance 2ξ , there is no variation of the line energy of the dislocation with position in the lattice, since the two half-spaces across the glide plane of the dislocation are still treated as structureless elastic continua. However, when the interactions across the glide plane are considered to arise, to a first approximation,

as a result of the pairwise interactions of atoms on opposite sides of the plane such as between the two atoms 1, etc., shown in Fig. 4.1, the position of the center of the dislocation becomes defined relative to the lattice. Around the center of the dislocation, the pairs of atoms suffer a substantial shear separation due to the concentration of the material misfit there. This shear separation decreases rapidly away from the center along the x axis.

The energy stored in the strip between the half-spaces owing to this shear separation is identified as the *misfit energy*. It is largely concentrated near the center and drops off rapidly away from the center. When the dislocation is displaced from its position of symmetry, the misfit energy varies with a period equal to the interatomic separation distance b , and results in a restoring force which binds the center of the dislocation to its site of symmetry and gives rise to a lattice resistance to motion, which we now determine.

The misfit energy W_m per unit length of dislocation in a strip of atoms of width b at a position x away from the center of the dislocation, due to a relative displacement $f(x)$ between the pairs of atoms across the gap, is

$$W_m(x) = b \int_0^{f(x)} \tau_{xy}(x) df = \frac{b^2}{2\pi} \tau_{is} \left[1 + \cos\left(2 \tan^{-1} \frac{x}{\xi}\right) \right] \quad (4.9)$$

$$= \frac{b^2}{2\pi} \tau_{is} \frac{\chi}{m^2 + \chi^2}. \quad (4.10)$$

Evaluating this misfit energy at atom sites $x = mb$ on the upper layer and summing the contributions from all strips from $m = -\infty$ to $m = +\infty$ gives

$$W_m = \frac{\mu b^2}{4\pi^2 (1 - \nu)} \sum_{m=-\infty}^{m=+\infty} \left(\frac{\chi}{m^2 + \chi^2} \right), \quad (4.11)$$

where we have introduced the dimensionless length parameter $\chi = \xi/b$.

When the center of the dislocation is now displaced by an amount u , the atom layers in the top half-space suffer a total relative displacement in relation to the lower half-space by $f(mx - u)$ at the atom sites. This changes the misfit energy to

$$W_m(y) = \frac{\mu b^2}{4\pi^2 (1 - \nu)} \sum_{m=-\infty}^{m=+\infty} \left(\frac{\chi}{(m - \eta)^2 + \chi^2} \right), \quad (4.12)$$

where we have introduced a new, normalized additional length parameter $\eta = u/b$.

The infinite series in eq. (4.12) has a closed-form representation (Cottrell 1953, p. 98), giving

$$W_m = \frac{\mu b^2}{4\pi (1 - \nu)} \left[\frac{\sinh(2\pi\xi/b)}{\cosh(2\pi\xi/b) - \cos(2\pi u/b)} \right], \quad (4.13)$$

represented again in nonnormalized parameters. We are interested primarily in materials in which $\xi/b > 1.0$ and $2\pi\xi/b \gg 1.0$, permitting a useful expansion of eq. (4.13) to give

$$W_m(u) = \frac{\mu b^2}{4\pi(1-\nu)} \left[1 + 2 \exp\left(-\frac{2\pi\xi}{b}\right) \cos\frac{2\pi u}{b} \right]. \quad (4.14)$$

The lattice resistance to the motion of the dislocation is then

$$\tau_{xy}(u) = -\frac{1}{b} \left(\frac{dW_m(u)}{du} \right) = \frac{\mu}{(1-\nu)} \exp\left(-\frac{2\pi\xi}{b}\right) \sin\left(\frac{2\pi u}{b}\right), \quad (4.15)$$

which has a maximum value at $u = b/2$ as might be expected and is referred to as the Peierls–Nabarro resistance, equal to

$$\tau_{PN} = \frac{\mu}{(1-\nu)} \exp\left(-\frac{2\pi\xi}{b}\right). \quad (4.16)$$

We note from eq. (4.5) that ξ is inversely proportional to the ideal shear strength of the material, as first noted by Foreman *et al.* (1951), and that a low ideal shear strength increases the half-width of the dislocation and drastically reduces the PN resistance. In the case of a sinusoidal resistance profile, the ideal shear strength is $\tau_{iss} = \mu b/2\pi a$ and the dislocation half-width is $\xi_s = a/2(1-\nu)$. Thus, we incorporate this important dependence into the expression in eq. (4.16) to obtain

$$\tau_{PN} = \frac{\mu}{(1-\nu)} \exp\left(-\frac{2\pi\xi_s}{b}n\right), \quad (4.17)$$

where

$$n = \frac{\tau_{iss}}{\tau_{is}} = \frac{\mu b}{2\pi a \tau_{is}} \quad (4.18)$$

is the ratio of the ideal shear strength for the unrealistic sinusoidal resistance profile to the ideal shear strength for the more realistic shear resistance profiles of most metals, which are significantly skewed, as depicted in Fig. 4.3 by curve 2. The ratio n is often a factor of 2 or larger.

The PN resistances¹ stated in eqs. (4.15–4.17) are to be interpreted as the deformation resistance of the material at $T=0$ K without any thermal assistance.

¹ In the literature, the PN resistance is widely referred to as the Peierls stress. This terminology is unfortunate and unsuitable for our purposes, where we distinguish a resistance as a material property, not to be confused with an applied stress. This differentiation is vital when thermal activation is considered, as we shall see repeatedly in later sections.

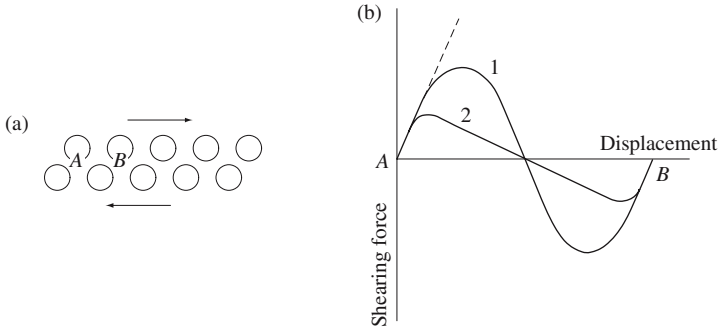


FIG. 4.3. Interatomic shear resistance profiles. Curve 1, for an idealized sinusoidal resistance. Curve 2, for a more realistic skewed form, characteristic of many metals.

Wang (1996a) has effectively taken $n = 2$ and has compared the calculated PN resistances with the motion of edge dislocations for a large variety of materials, including FCC metals (on $\{111\}$ planes), BCC metals (on $\{110\}$ planes), HCP metals on the basal plane, and some other crystals on their appropriate slip planes. He has compared these predictions with the best experimental results, extrapolated back to 0 K, and found good agreement, as shown in Fig. 4.4.²

Analyses similar to what was presented in Sections 4.1 and 4.2 have also been carried out for screw dislocations by Leibfried and Dietze (1949). Since screw dislocations often have important complications, as we discuss in Sections 4.3 and 4.4 below, we shall not present a simple PN-type argument for them.

It is essential to note here that the PN theory discussed above and the comparison of its predictions with experimental measurements that was done by Wang (1996a) should be approached with caution, particularly in relation to BCC metals, in which all existing evidence indicates that the high PN resistance as extrapolated to 0 K is governed by screw dislocations and not by edges, as we discuss further below. The PN resistance for rigid edge dislocations is difficult to verify, since no experiments can be carried out at 0 K, and the kinematics of thermally assisted motion of edge dislocations is still contentious (see Duesbery and Xu 1998). Thus, the above arguments are best viewed from a historical perspective and the experimental results as pertaining to screw dislocations, albeit with a necessary modification by removal of the characteristic plane strain factor $(1 - \nu)$.

² Actually, Wang's theoretical model differs from the one presented here and has the same form as the original PN model, albeit with a very important correction (Wang 1996b). However, he did not consider the important modification due to Foreman *et al.* (1951), which amounts to the introduction of the factor n in eq. (4.17). Taking $n = 2$, as is quite realistic and as we have done, the findings of Wang (1996a) become immediately relevant.

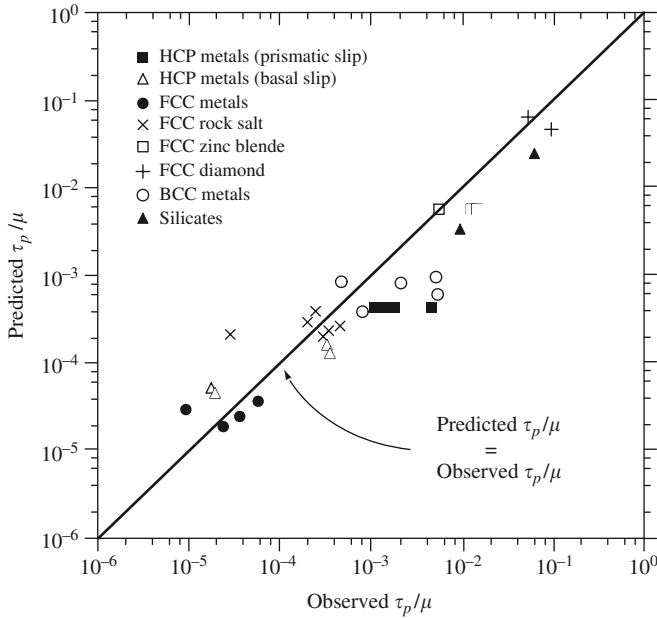


FIG. 4.4. Peak lattice resistances of a variety of materials, predicted by the modified Peierls–Nabarro theory compared with estimates derived from experiments (Wang 1996b; courtesy of Elsevier).

4.3 Inception of Plastic Deformation

4.3.1 HCP and FCC Metals

In Section 2.6 we stated two complementary criteria involved in the selection of specific crystallographic slip systems, where one of these was based on the ease of dislocation mobility, that is, the systems that have the lowest resistance to dislocation motion prescribe the slip system. Equations (4.17) and (4.18) indicate that the planes of lowest lattice resistance are those on which the dislocation core width is largest, or, stated alternatively, those for which the ratio of the interplanar spacing a to the Burgers displacement b is largest.

If plasticity involves the motion of dislocations, then in pure substances having no other impediments to dislocation motion, it must be initiated when the shear stress applied on a slip system can overcome the lattice shear resistance. Schmid and Boas (1935), who carried out some of the pioneering studies of the kinematical and kinetic aspects of plasticity in metal single crystals, stated that *plasticity caused by crystallographic slip is initiated when the resolved shear stress σ_{nd} on the slip plane, in the slip direction, reaches a critical value*, which we interpret to be the lattice shear resistance τ_l . This criterion, which is referred to as the Schmid law

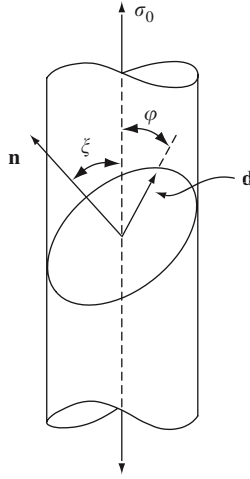


FIG. 4.5. Geometrical relation between a crystallographic slip plane, its slip direction' and the tensile stretch direction in a single crystal.

(Schmid 1924), is satisfied in a tension experiment, as shown in Fig. 4.5, when

$$\sigma_{nd} = \sigma_0 \cos \varphi \cos \xi = \tau_1, \quad (4.19)$$

where σ_0 is the applied tensile stress at yield, ξ is the angle between the tensile axis and the slip plane normal $\mathbf{n}$, and φ is the angle between the tensile axis and the slip direction $\mathbf{d}$ (parallel to the Burgers vector).

The geometrical factor

$$m = \frac{\sigma_{nd}}{\sigma_0} = \cos \varphi \cos \xi \quad (4.20)$$

that relates the applied stress to the resolved shear stress on the slip plane is referred to as the *Schmid factor*. Schmid and Boas verified their criterion in an extensive set of experiments on HCP single crystals of Zn, Cd, and Mg, where slip occurs preferentially on the basal plane. Figure 4.6 shows the results in a plot of σ_0/τ_1 versus the associated Schmid factor (Duesbery 1989). The curve is the predicted dependence given in eq. (4.20) and agrees very well with the data. Schmid and Boas stated, moreover, that the critical shear stress was uninfluenced by either pressure or any other component of the stress tensor not directly involved in establishing the resolved shear stress, at least as far as HCP metals were concerned. The verification of the Schmid law for FCC metals, where noncoplanar slip systems can be active, was also good but not as extensively demonstrated. For a more comprehensive discussion of the early crystal plasticity experiments, see Duesbery (1989).

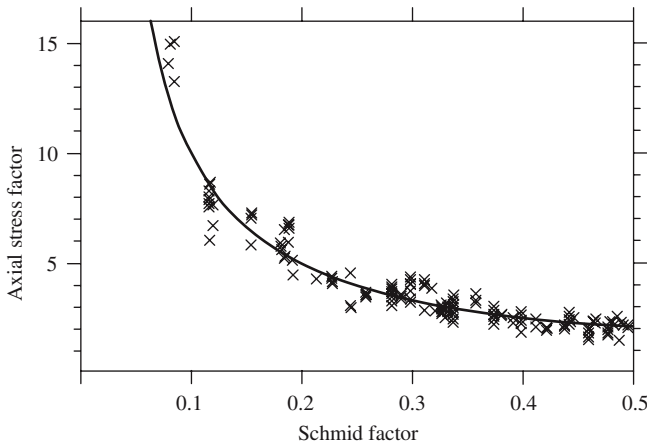


FIG. 4.6. Demonstration of the validity of the Schmid law for the inception of plastic deformation by slip on the most preferred slip system in the hexagonal crystals Zn, Cd, and Mg, deforming by basal glide, showing a plot of σ_0/τ_l against the Schmidt factor (Duesbery 1989; courtesy of Elsevier).

4.3.2 BCC Metals

For BCC metals, significant departures from the Schmid law were found by Taylor and Elam (1923) on Fe at the time when Schmid stated his law for HCP metals, and such departures were found later by Taylor (1928) on β -brass, also with a BCC structure. These departures were of several different types and involved complexities that varied with temperature. First, on the basis of slip band observations on deformed crystals, it was noted that while both the $\{110\}$ and $\{112\}$ planes were good slip planes when the resolved shear stress was maximized on them, in other instances, when the maximum-resolved-shear-stress plane (*mrssp*), was removed from the $\{110\}$ or $\{112\}$ planes, slip was either on the *mrssp* or approximately on a neighboring plane, with the slip bands showing wavy features when viewed in the slip direction. Moreover, the critical resolved shear stress (*crss*), on the *mrssp* at the inception of slip varied considerably, dependent on the position P of the tensile axis in the standard triangle of the stereographic projection (see Fig. 4.7), as listed in Table 4.1 for a variety of BCC transition metals deformed at 77 K. Second, when the *mrssp* coincided with the $\{112\}$ plane, a clear asymmetry was noted, dependent on whether or not the direction of slip coincided with the twinning shear direction. Then, the *crss* was significantly lower than in the opposite case, when the slip direction was opposite to the twinning shear direction.

The departures from the Schmid law are best represented with a terminology introduced by Taylor and adopted by all subsequent investigators in the presentation of slip response. This is shown in Fig. 4.7 for a tensile axis P in the center region

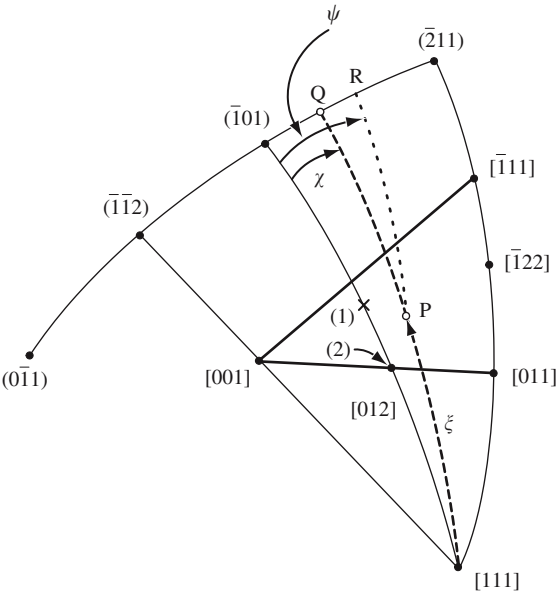


FIG. 4.7. Stereographic representation of expected and unexpected slip planes of a BCC crystal with a center orientation P and a slip direction [111].

TABLE 4.1. Orientation dependence of the *crss* (resolved on the *mrssp*) of BCC metals at 77 K, for a strain rate of circa 10^{-4} s^{-1} , in MPa^a

Orientation schmid factor	Tension				Compression				Reference
	011	001	111	uvw	011	001	111	uvw	
	0.472	0.472	0.315	0.500					
W	730	353	416	485					Rose <i>et al.</i> (1962)
W	584	283	255						Argon and Maloof (1966)
Ta	323	179			172	273			Sherwood <i>et al.</i> (1967)
Ta	353	283			264	221			Hull <i>et al.</i> (1967)
Ta	406	279	337	338					Ferris <i>et al.</i> (1962)
Ta	400	278	293	275	259	221	309	300	Byron (1968)
Fe	270		200						Keh and Nakada (1967)
Mo	471	151			393	137			Sherwood <i>et al.</i> (1967)
Mo					577	214	647	320	Stein (1967)
Mo	500	170		250					Guiu and Pratt (1966)
Nb	254	165			108	214			Sherwood <i>et al.</i> (1967)
Li-0.65Mg					48.4	57.6		61.0	Saka and Taylor (1981)
K ^b	2.94	3.13	2.93	4.52					Basinski <i>et al.</i> (1981)

^a Data from Duesbery (1989).

^b Extrapolated to 0 K.

of the standard stereographic triangle, for which the best slip direction is $[111]$ at an angle ξ , and three candidates for the slip plane are $(\bar{1}\bar{1}2)$, $(\bar{1}01)$, and $(\bar{2}11)$, with the $(\bar{1}01)$ plane being the best positioned. The plane of maximum resolved shear stress, however, is Q, lying on the same great circle as P and the slip direction. The angle of this plane with the $(\bar{1}01)$ plane is designated as χ . Experiments, however, often indicate that the actual slip may be on a different plane R at an angle ψ from the $(\bar{1}01)$ plane. Since, with a relatively wide margin, the angle ξ of the tensile axis P is of less importance, as long as the latter remains in the center of the triangle (away from the corners or edges), the response of slip has been characterized by the two angles χ and ψ , for χ ranging from -30° (at $(\bar{1}\bar{1}2)$) to $+30^\circ$ (at $(\bar{2}11)$). The comprehensive experiments of Šesták and Zarubová (1965) and Šesták *et al.* (1967) on an Fe–3%Si alloy give the best demonstration of these anomalies. Their results are shown in Fig. 4.8, where the dependence of the angle ψ of the actual slip plane on the angle χ of the *mrssp* is shown in tension and compression at 77 K (Figs. 4.8 (a, b)) and, again in tension and compression, at room temperature (Figs. 4.8 (c, d)). At 77 K, a clear preference for slip on the $(\bar{1}01)$ plane is observable, as the chosen tensile axis P shifts the *mrssp* to between $(\bar{1}\bar{1}2)$ to $(\bar{2}11)$. At room temperature, however, apart from a slight tendency for preference for the $\{112\}$ planes, there is on the whole a prevalence of slip on the *mrssp*. In both cases there is a very clear demonstration of a tension–compression asymmetry.

Finally, there are also cases of quite anomalous forms of slip behavior exhibited by many BCC transition metals at cryogenic temperatures when the tensile axis P moves closer to the $[001]$ axis of high symmetry in the triangle. Here slip, instead of appearing on the preferred $(\bar{1}01)$ plane, appears on the $(0\bar{1}1)$ plane (Seeger 2001).

These slip phenomena, unique to BCC transition metals, are all consequences of the changing characteristics of the three-dimensionally extended nature of the core configurations of screw dislocations, which have much lower mobility than edge dislocations and govern the slip character, as well as the plastic resistance. These special characteristics of the cores of screw dislocations and their possible role in explaining the departures from the Schmid law and related anomalies were first pointed out by Hirsch (1960) and require a more detailed study of the characteristics of screw dislocation cores, which we consider next in Section 4.4.

For a more expanded discussion of slip anomalies and departures from the Schmid law, see Duesbery (1989).

4.4 Structure of the Cores of Screw Dislocations in BCC Metals

The accumulating evidence of the breakdown of Schmid's law and the presence of a variety of anomalous slip behavior in BCC metals, partly related to the lack of mirror symmetry about the $\{111\}$ plane, resulted in a suggestion by Hirsch (1960) that the explanation must lie in some unique threefold symmetry in the structure

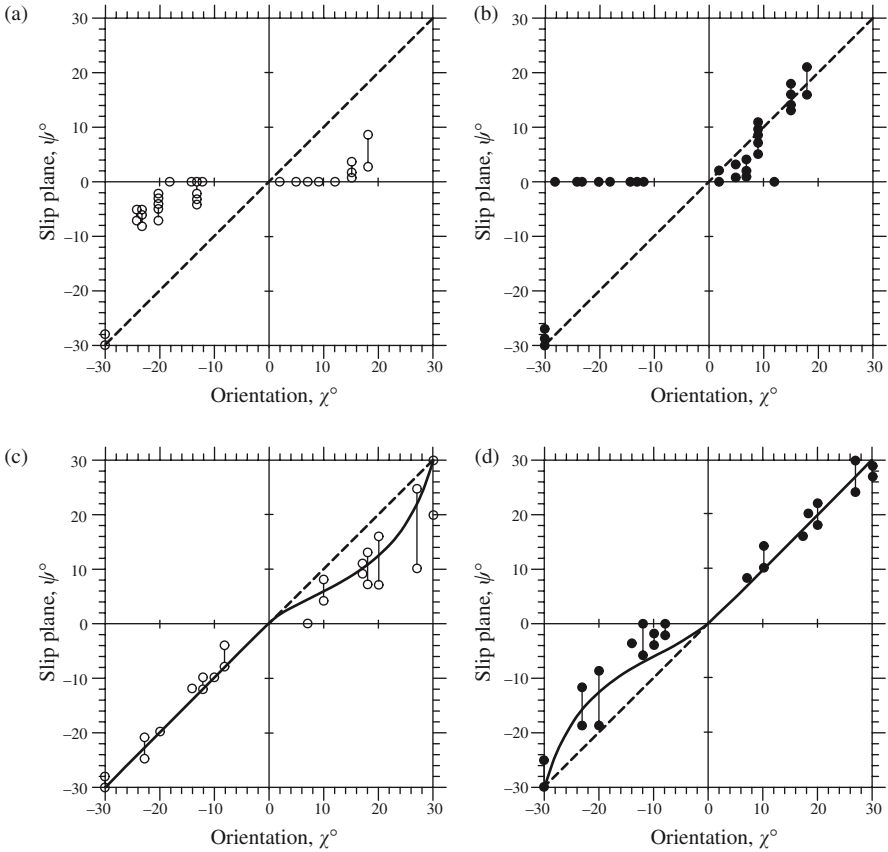


FIG. 4.8. Experimentally determined relationships between the observed angles ψ and χ for slip in Fe – 3%Si alloy crystals with a center orientation: (a) and (b), for tension and compression, respectively, at 77 K; (c) and (d), for tension and compression, respectively, at 296 K. These results demonstrate clearly the geometrical asymmetries exhibited by BCC metals (Sěsták and Zarubová 1965; Sěsták *et al.* 1967).

of the cores of screw dislocations. Studies of the preyield plasticity (Solomon and McMahon 1968), which indicated that in BCC metals the mobility of edge dislocations far exceeds that of screw dislocations and that transmission electron microscopy (TEM) of deformed Fe (see Fig. 4.9, which shows a similar case in W, another BCC metal) showed primarily quasi-straight screw dislocations in $\{110\}$ plane sections, are all in support of this proposition and have established the key role of screws in controlling the rate process of slip in BCC metals. The early suggestions were based on threefold spatial dissociations of the $(a/2)\langle 111 \rangle$ screw

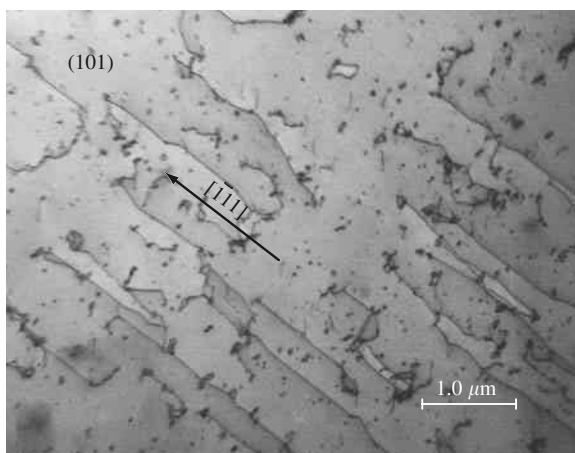


FIG. 4.9. TEM micrograph of the (101) slip plane of a W single crystal deformed at 295 K, showing primarily screw dislocations that are interacting with previously deposited glide debris (Argon, unpublished).

dislocations on either the $\{110\}$ or the $\{112\}$ planes as

$$\frac{a}{2}\langle 111 \rangle \rightarrow 3 \left[\frac{a}{6}\langle 111 \rangle \right], \quad (4.21)$$

with a threefold star-shaped stacking fault attaching them to the center of the screw dislocation. It was thought that motion of such spatially dissociated screw dislocations would require collapse of their extended structure back into a $\{110\}$ plane (or a $\{112\}$ plane, as the case may be).

It was thought that this would explain not only their low mobility but also some of the anomalous slip behavior. Such ideas, however, proved to be incorrect through a demonstration by Vitek (1968) that whereas FCC crystals, when sheared on their slip plane, possessed certain metastable directions of lowered ideal shear resistance parallel to the Burgers vectors of Shockley partial dislocations giving support for stable stacking faults, no such parallel response could be expected in BCC metals on the $\{110\}$ planes. This indicated that this structure possessed no corresponding stacking faults and provided no support for the type of screw-dislocation dissociation suggested by eq. (4.21). Thus, if screw dislocations had threefold symmetry, this had to be a core effect, akin to the planar spread of edge dislocation cores as in the PN model. This possibility was studied with a series of atomistic models utilizing pair potentials by a number of investigators, starting with Vitek *et al.* (1970), and by others since (see, for example, Kuramoto *et al.* 1974). Such studies have been of great value, not so much because of their accurate predictions of the threshold PN resistance at 0 K, which nearly all of them overpredicted by roughly an order of magnitude, but because they go a long way toward explaining most of

the anomalous selection of slip systems and the sensitivity of the stress required to move the dislocation to components of the stress tensor other than the resolved shear stress on the slip system, that is, stress components normal to the Burgers vector.

As an example, we present here results of a simulation by Ito and Vitek (2001) for Mo utilizing a variant of a many-body atomic central-force potential due to Finnis and Sinclair (1984). Figure 4.10 shows the crystallographic directions in the (111) plane perpendicular to the principal Burgers vector $(a/2)[111]$ of a screw dislocation in BCC Mo (a), and the traces of the $\{110\}$ and $\{112\}$ planes passing through the $[111]$ direction (b), all of these being perpendicular to the (111) plane.

Figure 4.11 shows the differential displacements of atoms around a screw dislocation line threading through a central interstitial site, where the lengths of the

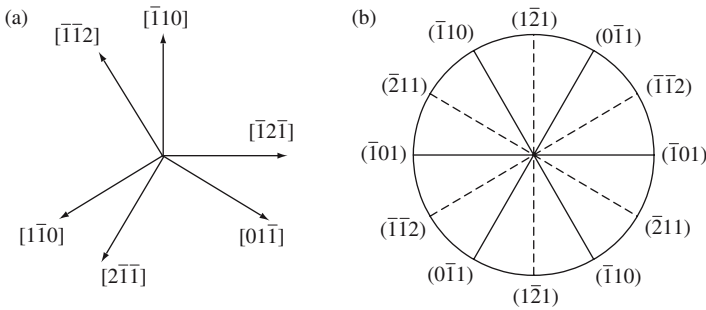


FIG. 4.10. (a) Crystallographic directions in the (111) plane perpendicular to the principal Burgers vector $(a/2)[111]$ of screw dislocations, and (b) the traces of the $\{110\}$ and $\{112\}$ planes through the $[111]$ direction (after Ito and Vitek 2001; courtesy of Taylor and Francis).

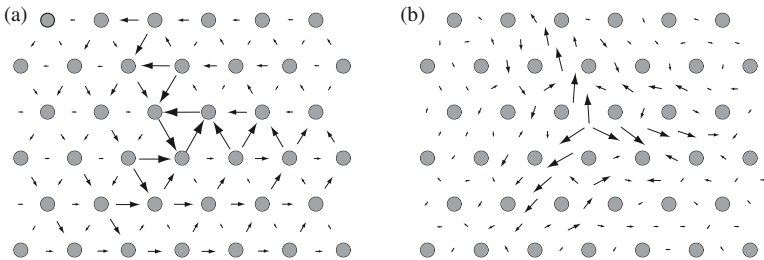


FIG. 4.11. (a) Differential atomic-level displacements of atoms in Mo, around a screw dislocation line normal to the plane of the paper, with the lengths of the arrows in units of $(a/6)[111]$; (b) the distributed small edge components perpendicular to the direction of the screw dislocation, associated with the displacements in (a), that are expected to couple with non-Schmid components of the applied stress tensor (Ito and Vitek 2001; courtesy of Taylor and Francis).

arrows in Fig. 4.11(a) give, in units of $(a/6)[111]$, the displacements between atom rows along the dislocation line and show the nature of the core on the three $\{110\}$ planes passing through the dislocation. Figure 4.11(b) shows the distributed associated small edge components perpendicular to the dissociated, spread core along the same three $\{110\}$ planes. For a simple shear stress applied parallel to the $[111]$ direction on the *mrssp*, making angles χ in the range of -30° from the $(\bar{1}01)$ plane (along the $(\bar{1}\bar{1}2)$ plane) to $+30^\circ$ (along the $(\bar{2}11)$ plane) (see Fig. 4.12(a)), the calculated *crss*, normalized with the elastic constant c_{44} of Mo, is plotted in Fig. 4.12(b), with filled circles for a driving force toward the right with reference to Fig. 4.12(a). The solid curve in Fig. 4.12(b) shows the predicted dependence of the *crss* according to the Schmid law, if it held. We note that the response differs markedly from the Schmid law and that the *crss* on the $(\bar{1}\bar{1}2)$ plane, where shear is in the twinning sense, is considerably lower than that on the $(\bar{2}11)$ plane, where it is in the antitwinning sense.

In a further extension of this study, Ito and Vitek considered the response of Mo to tensile and compressive straining along six special axis orientations of $[001]$, $[012]$, $[011]$, $[\bar{1}22]$, $[\bar{1}11]$, and $[\bar{2}38]$ (orientational (1) in Fig. 4.7) in the standard triangle to probe the critical levels of shear stress on the *mrssp* when it coincides with the $\{112\}$ planes in either the twinning or the antitwinning sense, and with the $(\bar{1}01)$ plane. The results, given in Fig. 4.13, first show a wide spread of the *crss* levels for

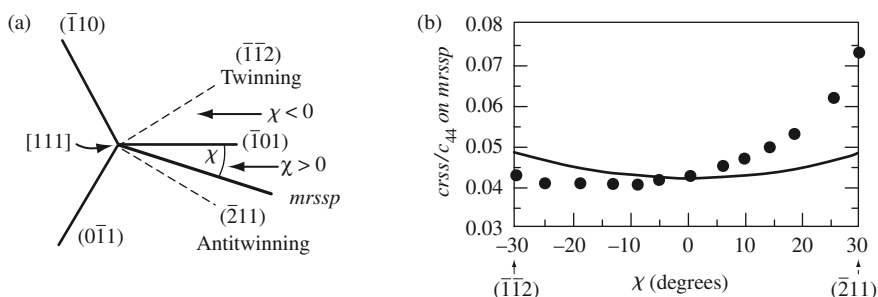


FIG. 4.12. (a) View in the $[111]$ primary slip direction and the mutual orientational relationships of the *mrssp* to the $(\bar{1}01)$ plane and the two unexpected $\{112\}$ -type twinning- and antitwinning-type planes. (b) The solid curve shows the expected dependence of $crss/c_{44}$ on the *mrssp* on the angle ψ if the Schmid law held. The computer-generated results for Mo show that the Schmid law does not hold; for a shear direction that favors the twinning-sense $(\bar{1}\bar{1}2)$ plane for $\chi < 0$, $crss/c_{44}$ is lower than on the *mrssp*. For $\chi > 0$, when $crss/c_{44}$ is lower than the calculated $crss/c_{44}$ on the $\{112\}$ -type planes, the twinning-sense $\{112\}$ planes are still good alternatives to the *mrssp* (Ito and Vitek 2001; courtesy of Taylor and Francis).

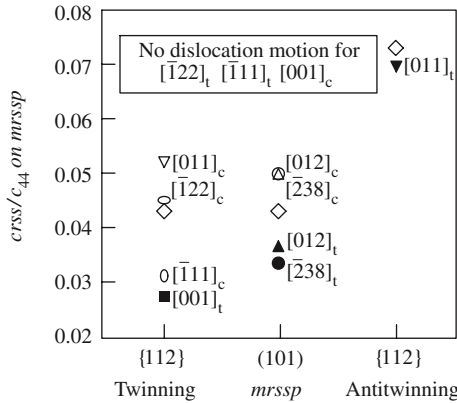


FIG. 4.13. Results of simulation of tensile and compressive straining in Mo along six special axis orientations in the standard triangle, probing the level of stress on the $mrssp$ when it coincides with the $\{112\}$ plane in either the twinning or the antitwining sense, and with the $(\bar{1}01)$ plane. The results demonstrate that the $\{112\}$ -type plane in the twinning sense is always preferred by having the lower $crss$ (Ito and Vitek 2001; courtesy of Taylor and Francis).

tension and compression, with results always being lower for shear on the $\{112\}$ plane when it is in the twinning sense than in the antitwining sense. The study also probed the interaction of the additional fractional edge components of the spread core with other components of the local stress tensor, shown in Fig. 4.11(b) for the six chosen uniaxial stressing orientations. This was investigated also by choosing an appropriate $mrssp$ and applying a shear traction τ normal to the principal Burgers vector, followed by gradually increasing the resolved shear stress on the $mrssp$ in the $[111]$ direction until the dislocation moved. The result is shown in Fig. 4.14 and demonstrates a very direct effect of such shear components normal to $[111]$ on the $crss$ on the $mrssp$. These examples described in Figs. 4.12–4.14 serve to demonstrate the complex nature of the screw dislocation core in BCC metals, which produces a variety of anomalies not present in pure FCC metals, leading to the breakdown of the Schmid law.

4.5 Temperature and Strain Rate Dependence of the Lattice Resistance in BCC Metals

4.5.1 The Nature of Thermal Assistance over a Lattice Energy Barrier

The lattice resistance considered in Section 4.2 is what has to be overcome to move the entire dislocation line rigidly from one lattice valley into a neighboring one over

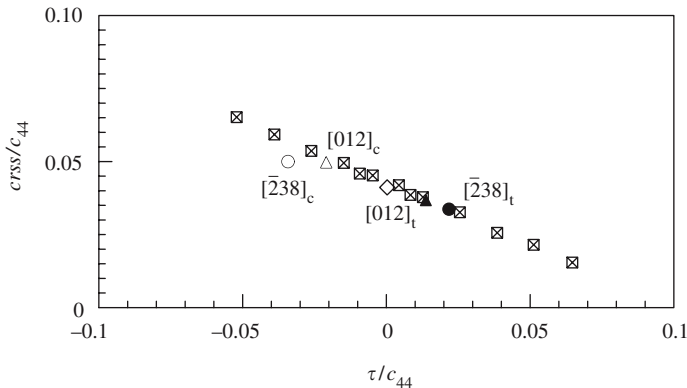


FIG. 4.14. Result of simulation of the effect of a “non-Schmid component” stress, normal to the principal Burgers vector, on $crss/c_{44}$ on the $mrssp$, related directly to the coupling of this stress component with the fractional edge components around the core shown in Fig. 4.11(b), where the same six axis directions as in Fig. 4.13 were considered. These effects demonstrate how the unexpected slip planes can be chosen by subatomic details of the structure of the screw dislocation core (Ito and Vitek 2001; courtesy of Taylor and Francis).

an energy barrier at 0 K, without any help from thermal motion. At higher temperatures, the stress that is required to accomplish the same process will decrease, as the dislocation can be aided by thermal motion. We note that for a PN resistance of the order of 400 MPa for a typical BCC transition metal, with a distance between neighboring lattice valleys of the order of 0.4 nm, and a Burgers displacement of 0.25 nm, the energy barrier separating neighboring lattice valleys is only of the order 3.5×10^{-3} times the line energy of the dislocation. This amounts to only about 0.015 eV per atomic length. Nevertheless, the largest reasonable thermal fluctuation at room temperature is only roughly 1.0 eV, indicating crudely that such a fluctuation can translate at most only a dislocation segment of about 50–60 atomic lengths into the neighboring valley. While the need for maintaining the dislocation unbroken and the effect of a favorable applied driving force will change this estimate, it is clear that the energetically favorable form of advancing the dislocation forward will be through the nucleation of a double-kink configuration over the potential hills as illustrated in Fig. 4.15.

When dislocation segments are pinned between two points not in the same potential valley in the slip plane, they have to straddle over the potential hills bridging these points. Consequently, many preexisting dislocations will possess a density of geometrical kinks of the same sign. Moreover, since kink energies are quite small, it is also quite likely that dislocation lines will contain a certain concentration of kinks of random sign in thermal equilibrium, somewhat like point

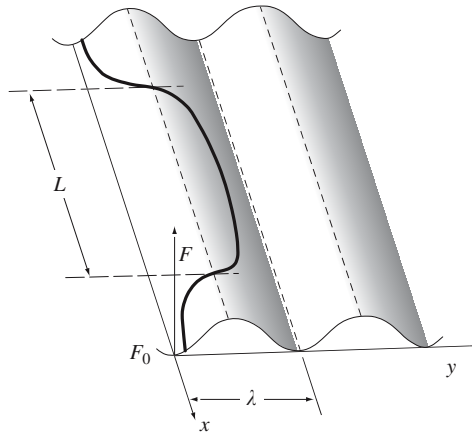


FIG. 4.15. Sketch showing how a screw dislocation in a BCC crystal can advance over the lattice potential by sending out a double-kink segment.

defects strung along on a line. Under an applied stress, the preexisting kinks on such a kinked dislocation line will be driven to the pinning points, resulting in long straight segments lying in a single potential valley. Further motion of such straight segments will require nucleation of kink pairs along the line to advance them from valley to valley over potential hills.

In one model of the above process, two regimes are envisioned to govern dislocation advance by repeated double-kink nucleation. In *Regime I*, under low stress and at higher temperatures, the activation configuration of the kink pair will consist of two fully formed kinks of opposite type separated by a distance L larger than the equilibrium width w of the individual kinks, as shown in Fig. 4.16(a), where the applied stress just holds the two weakly interacting kinks apart. With increasing stress, the distance L in the activation configuration will shrink until at the end of *Regime I*, L decreases to a critical length L_{ce} , roughly equal to the kink width w , and the two kinks begin to touch. At higher stresses, in *Regime II*, the activation configuration will become progressively smaller as the amplitude of the double kink decreases, as depicted in Fig. 4.16(b).

In this model, we deal with *Regime I* through the elastic interaction of fully formed kinks, where the applied stress counteracts only the single activation parameter L . In *Regime II*, we employ a simple parabolic lattice potential, approximating a more likely cosine potential, that has the convenient characteristic of trapping the dislocation in a valley until the applied force reaches the PN resistance. In *Regime II*, we consider the extension of the double kinks to be fixed at L_c and deal with the shrinking activation configuration by employing solutions of the differential equation for a continuous dislocation line. Our model, in a simplified form, will follow

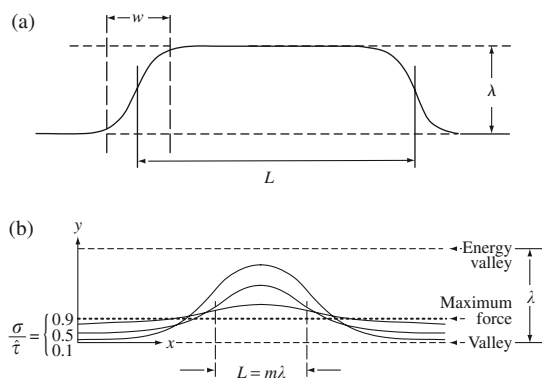


FIG. 4.16. Two limiting forms of idealization of a double kink on a screw dislocation: (a) at low stresses, where the double kink is made up of two well-separated, fully formed right-side and left-side kinks; (b) at high stresses, where the activation configuration is more in the form of a bulge that just places a portion of the line over the peak of the energy ridge.

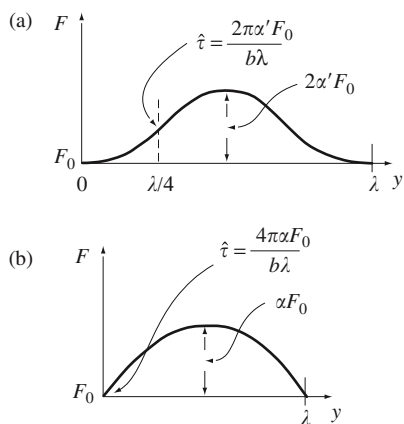


FIG. 4.17. Two empirical forms of the profile of the lattice potential: (a) a “cosine potential”, and (b) a “parabolic potential”.

that of Seeger (1981), which has been most successful in capturing the temperature and strain rate sensitivity of the plastic resistance of BCC transition metals.

In reality, since the core diameter of the dislocation will be of the order of the kink height, envisioning it as a well-defined, sharply delineated entity as depicted in Figs. 4.16 and 4.17 must be recognized to be only a convenient abstraction, where the line notions of a dislocation and linear elastic interactions will be pushed to the limit. However, since at present no successful alternatives have been developed

by a first-principles technique, we shall pursue the line models of double-kink nucleation as outlined above. For a more complete discussion of the various line models of double-kink nucleation, see Kocks *et al.* (1975).

Once the activation configuration of a kink pair is reached, the kinks will propagate apart under the applied stress in opposite directions against phonon drag or a secondary energy barrier, and in the process fold the dislocation line from one potential valley into the next in a smooth and continuous manner.

4.5.2 Lattice Potentials

As remarked above, the actual lattice resistance potential is likely to be complex, as dislocation core structures undergo pulsating atomic-level changes in translating from one low-energy lattice site to the next. Our model of the temperature and strain rate dependence of the lattice resistance will make the idealizations stated in Section 4.5.1 above and is primarily intended to consider the motion of screw dislocations in BCC transition metals. These considerations, however, could also apply to other crystal structures such as the alkali halides NaCl and LiF.

We take as the simplest lattice potential a cosine function having a general description of

$$F_c = F_0 + \alpha' F_0 \left(1 - \cos \frac{2\pi y}{\lambda} \right), \quad (4.22a)$$

where $F_0 = \mathfrak{F}_0$ is the Helmholtz line energy of the dislocation and $2\alpha' F_0$ is the total amplitude of the energy variation over the potential hill between two adjacent potential valleys a distance λ apart, all depicted in Fig. 4.17(a). The cosine potential has its peak resistance $\hat{\tau}$ at $y = \lambda/4$,

$$\hat{\tau} = \left(\frac{1}{b} \frac{dF_c}{dy} \right)_{\max} = \frac{2\pi \alpha' F_0}{\lambda b}. \quad (4.22b)$$

We expect α' to be in the range of a few tenths of a percent for BCC transition metals and $\hat{\tau}$ to be the PN resistance at 0 K. The potential can also be presented in terms of its peak resistance as

$$F_c = F_0 + \frac{\hat{\tau} b \lambda}{2\pi} \left(1 - \cos \frac{2\pi y}{\lambda} \right). \quad (4.22c)$$

For most of our quantitative arguments and comparisons with experimental data, we shall use a simpler parabolic potential, which we define as (Kocks *et al.* 1975)

$$F_p = F_0 + 4\alpha F_0 \left(\frac{y}{\lambda} - \left(\frac{y}{\lambda} \right)^2 \right), \quad (4.23a)$$

with a corresponding peak resistance $\hat{\tau}$ encountered at $y = 0$,

$$\hat{\tau} = \frac{4\alpha F_0}{b\lambda}. \quad (4.23b)$$

Figure 4.17(b) shows the form of this more approximate potential, which has the convenient feature that the dislocation is trapped in the cuspy potential valley until the stress is raised to the peak resistance level $\hat{\tau}$. While this behavior is clearly artificial, it has attractive analytical advantages in treating thermally assisted double-kink formation without serious penalty. Here too the potential can be represented in terms of its peak resistance as

$$F_p = F_0 + \hat{\tau} b\lambda \left(\frac{y}{\lambda} - \left(\frac{y}{\lambda} \right)^2 \right). \quad (4.23c)$$

The parabolic potential can be matched to the cosine potential by requiring that the two have the same peak resistance. This results in a somewhat different energy amplitude coefficient α , where α and α' are related as follows:

$$\alpha = \frac{\pi}{2} \alpha'. \quad (4.24)$$

In the following analysis, we use the constant-line-energy/line-tension approximation and isotropic elasticity.

Other, more complex potentials made up of fourth-order parabolic segments (Seeger 1981) and potentials with intermediate minima at $y = \lambda/2$ (Koizumi *et al.* 1993) have also been used but offer little additional advantage.

4.5.3 Shapes and Energies of Geometrical Kinks

Preparatory to a discussion of double-kink nucleation under stress, we determine the shapes and energies of geometrical kinks by solving the basic differential equation for a dislocation line draping from one lattice potential valley into a neighboring one over a potential hill as depicted in Fig. 4.18.

The shape of a dislocation line containing a kink is obtained from a solution of the dislocation line equation (2.26) for static conditions under no applied stress,

$$\mathcal{E} \frac{d^2 y}{dx^2} - \frac{dF(y)}{dy} = \frac{\mathcal{E}}{2} \frac{d}{dy} \left(\frac{dy}{dx} \right)^2 - \frac{dF(y)}{dy} = 0, \quad (4.25)$$

provided that the line notions are applicable at the core level. The solution has to satisfy the boundary conditions

$$y = 0, \frac{dy}{dx} = 0, \quad \text{and} \quad y = \lambda, \frac{dy}{dx} = 0, \quad (4.26)$$

at both ends of the line, which for the cosine potential should be at $x = -\infty$ and $+\infty$, respectively, for a positive kink, rising with increasing x . For the parabolic

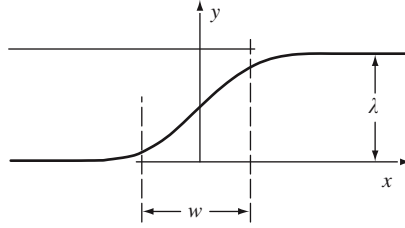


FIG. 4.18. Sketch of the shape of a structural kink of width w in relation to the periodicity spacing λ between valleys of the potential. In realistic potentials the width w is always much larger than λ .

potential, where the dislocation is trapped along most of its length into cusps at $y = 0$ and λ , the same boundary conditions hold, but the kink width becomes sharply defined.

A direct integration of eq. (4.25) gives

$$\left(\frac{dy}{dx}\right)^2 = \frac{2F(y) - F_0}{\mathcal{E}}. \quad (4.27)$$

A further integration of eq. (4.27) subject to the condition that $dy/dx \ll 1.0$ everywhere gives the kink shapes for the two potentials as follows.

For the *cosine potential*,

$$y_c = \frac{2\lambda}{\pi} \tan^{-1} \left\{ \exp\left(\frac{2\pi x}{\lambda} \sqrt{\alpha'}\right) \right\} = \frac{2\lambda}{\pi} \tan^{-1} \left(\frac{1.12\pi x}{w'} \right), \quad (4.28)$$

where the origin is taken at the center of the kink and $w' = 4\xi$ is taken as the kink width, with

$$\frac{w'}{4} = \xi = \frac{\lambda}{2\pi} \frac{1}{\sqrt{\alpha'}} \ln \left(\frac{1}{\tan(\pi/8)} \right) = \frac{0.14\lambda}{\sqrt{\alpha'}} \quad (4.29)$$

defining the kink quarter-width, where, at $x = \xi$ the kink position rises to $\lambda/4$.

For the *parabolic potential*,

$$y_p = \frac{\lambda}{2} \left(1 - \cos \frac{\sqrt{A}x}{\lambda} \right) = \frac{\lambda}{2} \left(1 - \cos \frac{\pi x}{w} \right), \quad (4.30)$$

where $A = (2\hat{\tau}b\lambda/\mathfrak{F})$ and w , the kink width, now sharply defined, is the distance x where the kink has reached the neighboring valley,

$$w = \frac{\pi\lambda}{\sqrt{A}} = \pi\lambda \sqrt{\frac{\mathfrak{F}}{2\hat{\tau}b\lambda}} = \frac{\pi\lambda}{4\sqrt{\alpha'}}. \quad (4.31)$$

The respective shapes of these two different kinks are very similar and are depicted in Fig. 4.18. For α -Fe, the value $\alpha' = 3 \times 10^{-3} (= (2/\pi)\alpha)$ is representative.

The kink energy F_k is then determined in a straightforward manner by integrating the excess energy of line segments draped over the potential contour, over and above the line energy in a reference potential valley, that is,

$$F_k = \int_{-\infty}^{\infty} F(y) ds - F(0), \quad (4.32)$$

where $F(0) = F_0(1 - \alpha')$ for the cosine potential and F_0 for the parabolic potential. Integrations of (4.32) give the two relevant kink energies as

$$F_k^c = \frac{4}{\pi} \lambda F_0 \sqrt{\alpha'} \quad \text{and} \quad F_k^p = \frac{\pi}{8} \left(\hat{\tau} \mu b^3 \lambda^3 \right)^{1/2} = \frac{\pi}{2\sqrt{2}} \lambda F_0 \sqrt{\alpha} \quad (4.32a,b)$$

for the cosine potential and the parabolic potential, respectively. Taking account of eq. (4.29), we note that the energy of the kink in the cosine potential is smaller than in the parabolic potential, which follows from the different profiles of the chosen potentials. Eliminating α' and α between eqs. (4.28) and (4.32a) and between eqs. (4.31) and (4.32b) gives the following for the reference kink widths:

$$w' = \frac{0.713 \lambda^2 F_0}{F_k^c} \quad \text{and} \quad w = \frac{\pi^2}{8\sqrt{2}} \frac{\lambda^2 F_0}{F_k^p}, \quad (4.33a,b)$$

which indicates that $w' = 0.894w$. This follows because the kink in the cosine potential is not sharply delineated.

In the comparison of theoretical predictions with experiments, we rely primarily on the parabolic potential because of the ease in dealing with it analytically.

4.5.4 Double-kink Energy in *Regime I*

At low stresses and higher temperatures, in *Regime I*, we consider the activation configuration of a double kink under stress to consist of two isolated kinks of opposite type a distance L apart as depicted in Fig. 4.16(a), where the kink-kink interaction is just counteracted by the applied stress. Thus, the double-kink activation free energy ΔG_{dk} is made up of the energy $2F_k$ of the two isolated kinks, their interaction energy I_{kk} , and the work done by the applied stress σ , as

$$\Delta G_{dk} = 2F_k + I_{kk} - \sigma b \lambda L. \quad (4.34a)$$

For consistency we represent F_k and I_{kk} by the elastic relations for a kink in an isotropic solid as stated by Hirth and Lothe (1982), and obtain

$$\Delta G_{dk} = \frac{\mu b^2 \lambda}{2\pi(1-\nu)} \beta - \frac{\mu b^2 \lambda^2}{8\pi L} \left(\frac{1+\nu}{1-\nu} \right) - \sigma b \lambda L, \quad (4.34b)$$

where β stands for the particular form of the logarithmic factor involving the inner and outer cutoff dimensions, which become ill-defined in the case of kinks. This lack of clarity will be dealt with when the kink energies are referred to those discussed in Section 4.5.3.

The saddle-point configuration is obtained from

$$\frac{d \Delta G_{\text{dk}}}{dL} = \frac{\mu b^2 \lambda^2}{8\pi L^2} \left(\frac{1+\nu}{1-\nu} \right) - \sigma b \lambda = 0, \quad (4.35)$$

giving for the principal activation coordinate

$$L^* = \left(\frac{1}{8\pi} \left(\frac{1+\nu}{1-\nu} \right) \frac{\mu b \lambda}{\sigma} \right)^{1/2}. \quad (4.36)$$

Substitution of eq. (4.36) into eq. (4.34a) and taking account of the connections to the earlier arguments that gave

$$\frac{\mu b^2 \lambda}{2\pi(1-\nu)} \beta = 2F_k, \quad (4.37)$$

the activation free energy for double-kink nucleation under stress in *Regime I* becomes

$$\Delta G_{\text{dk}}^* = 2F_k \left[1 - \frac{1}{2F_k} \left(\frac{1}{2\pi} \left(\frac{1+\nu}{1-\nu} \right) \sigma \mu b^3 \lambda^3 \right)^{1/2} \right]. \quad (4.38)$$

4.5.5 Double-kink Energy in *Regime II*

The transition from *Regime I* to *Regime II* is considered to occur at a stress level where the activation energies for the two double-kink nucleation models are equal. We shall state this after we develop the expression for double-kink nucleation in *Regime II*, which we deal with through a line tension model based on the solution of the differential equation for the dislocation line.

Under an applied shear stress σ , the differential equation for the dislocation line becomes

$$\mathcal{E} \frac{d^2 y}{dx^2} - \frac{dF(y)}{dy} + \sigma b = \frac{\mathcal{E}}{2} \frac{d}{dy} \left(\frac{dy}{dx} \right)^2 - \frac{dF(y)}{dy} + \sigma b = 0, \quad (4.39)$$

where

$$F(y) = F_0 + \hat{\tau} b \lambda \left(\frac{y}{\lambda} - \left(\frac{y}{\lambda} \right)^2 \right) \quad (4.40)$$

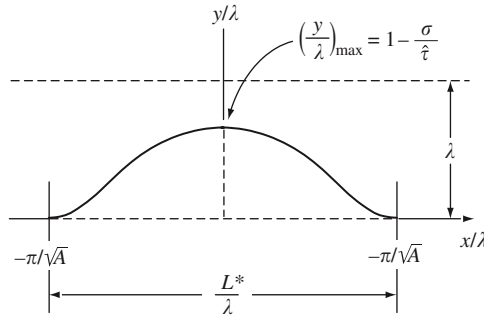


FIG. 4.19. Sketch showing the activated configuration under a stress $\sigma/\hat{\tau}$, in the parabolic potential, where an increase of the stress affects the activation configuration only in the forward direction across the potential contour.

is that for the parabolic potential. Integration twice, under boundary conditions similar to those stated above, gives the double-kink shape as

$$y = \frac{\lambda}{2} \left(1 - \frac{\sigma}{\hat{\tau}} \right) \left(1 + \cos \sqrt{A} \frac{x}{\lambda} \right), \quad (4.41)$$

where A has the same meaning as that introduced in connection with eq. (4.30). The shape of the double kink is shown in Fig. 4.19. Because of the nature of the cuspy parabolic potential, the lateral extent of the double kink remains unaltered with stress, in a somewhat artificial manner. The only effect of increasing stress is then to decrease the amplitude of the double kink. Comparison of the model with experiment in Section 4.6 will show that there is little penalty paid for this constraint.

Finally, the energy of the double kink is obtained similarly to that of a geometrical kink as

$$\Delta G_{\text{dk}}^* = \int_{-\pi\lambda\sqrt{A}}^{\pi\lambda\sqrt{A}} F(y) \, ds - F(0) \, dx - \sigma b y(x) \, dx, \quad (4.42)$$

giving

$$\Delta G_{\text{dk}}^* = 2F_k \left(1 - \frac{\sigma}{\hat{\tau}} \right)^2 \quad (4.43)$$

after making use of eqs. (4.23a), (4.23b), and (4.32b).

The stress level where a transition from *Regime I* to *Regime II* takes place can now be determined by equating the activation energies of the two regimes, that is, eqs. (4.38) and (4.43), together with utilizing the kink energy expressed in terms

of the peak lattice resistance as given in the intermediate form of eq. (4.32b). The transition stress is then obtained from the solution of the third-order equation

$$\left(\frac{\sigma}{\hat{\tau}}\right)^3 - 4\left(\frac{\sigma}{\hat{\tau}}\right)^2 + 4\left(\frac{\sigma}{\hat{\tau}}\right) - \frac{2}{\pi^3} \left(\frac{1+\nu}{1-2}\right) = 0. \quad (4.44)$$

4.6 The Plastic Strain Rate in BCC Metals

4.6.1 The Preexponential Factor and the Net Shear Rate

As outlined in Chapter 2, the plastic strain rate results from the glide motion of dislocations in a series of thermally activated steps under stress, where each step advances the dislocation over an energy barrier, permitting it to sweep out a certain area of the glide plane. Here, account has to be taken also of reverse thermal fluctuations that could possibly undo some of the advance of the dislocation against the applied stress. Prior to considering the kinetic aspects, we consider the kinematics of the production of shear strain when screw dislocations glide by the double-kink nucleation process discussed in Sections 4.5.4 and 4.5.5 above.

Consider the nucleation of a double kink under an applied stress σ in a long dislocation segment, such that it contains the activation length L^* between the kinks. Once the double kink is nucleated as an unstable equilibrium configuration, it expands laterally as depicted in Fig. 4.20, sweeping out a lattice strip of width λ until the entire dislocation line is folded over into the neighboring potential valley. It might be thought that the area $\lambda \Lambda$ swept out by a kink, where Λ is the kink mean free path, is governed by a kinetic balance in which an expanding kink encounters an opposing kink produced by a neighboring nucleation site along the dislocation, and the two kinks annihilate. This would give a condition where the mean time t_a for double-kink nucleation, given by

$$\frac{1}{t_a} = \nu_G \exp\left(-\frac{\Delta G_f^*}{kT}\right), \quad (4.45)$$

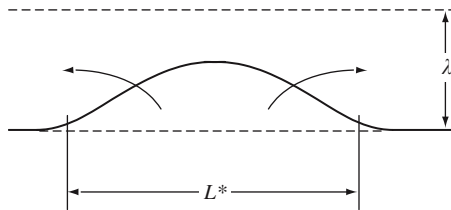


FIG. 4.20. Sketch showing the directions of spreading of an activation configuration, once the free-energy maximum is reached.

is equal to the mean travel time of a released kink moving outward with a velocity v_k over the length Λ where annihilation with a neighboring kink is to occur, that is,

$$\Lambda = v_k \frac{1}{v_G} \exp \left(\frac{\Delta G_f^*}{kT} \right). \quad (4.46)$$

In eqs. (4.45) and (4.46), ΔG_f^* is the activation free energy for nucleation of a kink pair under a stress that will advance the dislocation forward in the direction of the driving force, and v_G is a characteristic normal-mode frequency of the dislocation line segment, roughly of length λ , where the double kink will be nucleated.

In the low-temperature region where the lattice resistance governs, the secondary lattice glide resistance to the motion of kinks along a dislocation line is very low in many types of metal crystals, particularly in the BCC metals of primary interest here. The only resistance present in such cases will be a phonon drag with coefficient $B \cong kT/\Omega v_D$, discussed in Section 2.3.6 and further in Section 4.8 below, giving a kink velocity under stress of

$$v_k = \frac{\sigma b}{B} \quad (4.47)$$

and a mean free kink path length of

$$\Lambda = b \left(\frac{\sigma \Omega}{kT} \right) \left(\frac{v_d}{v_G} \right) \left(\exp \frac{\Delta G_f^*}{kT} \right). \quad (4.48)$$

In eq. (4.48), the terms in parentheses on the right-hand side will typically be of the order of 0.2, 10^2 , and 10^8 , respectively, making Λ/b of the order of 2×10^9 and very much larger than any part dimensions of interest, indicating that the area swept out by kinks upon nucleation will not be governed by a kinetic balance of the kink concentration but rather by a structural length Λ_s such as a distance between pinning points of a dislocation network in the range of $1/\sqrt{\rho}$, the average dislocation mesh length, with ρ being the average dislocation density.

The net plastic shear strain rate $\dot{\gamma}$, combining both forward and reverse events, will then be

$$\dot{\gamma} = \dot{\gamma}_0 \left\{ \exp \left(-\frac{\Delta G_f^*}{kT} \right) - \exp \left(-\frac{\Delta G_r^*}{kT} \right) \right\}, \quad (4.49)$$

where the preexponential factor $\dot{\gamma}_0$ combines several terms:

$$\dot{\gamma}_0 = b \left(\frac{N}{V} \right) (\lambda \Lambda_s) v_G. \quad (4.50)$$

Here N/V is the total density per unit volume of momentary nucleation sites, each of which results in the sweeping-out of a lattice strip $\lambda \Lambda_s$ subsequent to a nucleation

event, and ν_G represents the frequency of oscillation of a dislocation segment of length L^* in its potential well. In eq. (4.49), the second rate factor represents the probability of occurrence of a reverse event that can undo the effect of a forward event. Since such reverse events would have to fold back the entire swept area $\lambda \Lambda_s$ into the initial lattice valley, doing work against the applied stress, ΔG_r^* should be exceedingly high and can be estimated to be typically in the range of (30–50) ΔG_f^* in the plastic-behavior range, making the second factor negligible, giving

$$\dot{\gamma} = \dot{\gamma}_0 \exp\left(-\frac{\Delta G_f^*(\sigma/\hat{\tau})}{kT}\right). \quad (4.51)$$

Finally, we note that

$$\Delta G_f^* = \Delta H_f^* - T \Delta S^* \quad (4.52)$$

and that a certain level of entropy change may also be involved during an activated event. Physically, such activation entropies combine changes in the vibrational frequencies in the activated state and the collection of configurations that can be involved in that activated state. Both contributions are difficult to clarify in an analysis of experimental results and are normally combined into the preexponential factor as a separate factor.

While the preexponential factor $\dot{\gamma}_0$ can be evaluated approximately on the basis of its content, it is operationally more expedient to evaluate the ratio $\ln \dot{\gamma} / \dot{\gamma}_0$. A case in which this has been done successfully was described by Brunner and Diehl (1987) in relation to stress relaxation experiments in α -Fe.

4.6.2 Temperature and Strain Rate Dependence of the Plastic Resistance

On the basis of the arguments in Section 4.6.1, we take the expression for the net shear strain rate, as given by eq. (4.51), as

$$\dot{\gamma} = \dot{\gamma}_0 \exp\left(-\frac{\Delta H^*(\sigma/\hat{\tau})}{kT}\right), \quad (4.53)$$

where ΔH^* is given by eqs. (4.38) and (4.43) for *Regimes I* and *II*, respectively, for the lattice resistance in BCC transition metals.

We start by noting that the plastic shear resistance τ generally has two distinct components,

$$\tau = \tau_\mu + \tau_s(T, \dot{\gamma}), \quad (4.54)$$

that are probed by the applied shear stress σ . The component τ_μ incorporates the effect of long-range internal stresses arising from material misfit and impenetrable slip obstacles such as dense dislocation clusters produced in the course of

strain hardening. It is dependent on temperature only through the mild temperature dependence of the shear modulus, which scales the elastic interactions that govern τ_μ . The second component, $\tau_s(T, \dot{\gamma})$, represents the resistance of the thermally assisted processes discussed in Section 4.6.1 for the lattice resistance. It is often referred to as the effective stress, in the sense that the component σ_s of the applied stress is the effective component that overcomes the rate-dependent resistance τ_s .

With the introduction of eqs. (4.38) and (4.43 into eq. (4.53), the two effective stresses σ_s for *Regimes I* and *II* are obtained. Thus, for *Regime I*, using the intermediate result in eq. (4.32b), one obtains the temperature dependence of σ_s as

$$\frac{\sigma_s}{\hat{\tau}_{sI}} = \frac{\pi^3}{8} \left(\frac{1-\nu}{1+\nu} \right) \left(1 - \frac{T}{T_0} \right)^2, \quad (4.55)$$

where

$$T_0 \equiv \frac{2F_k}{k \ln(\dot{\gamma}_0/\dot{\gamma})} \quad (4.56)$$

represents the temperature where the stress σ_s becomes negligible and the lattice resistance no longer governs. For *Regime II*, one has similarly

$$\frac{\sigma_s}{\hat{\tau}_{sII}} = 1 - \left(\frac{T}{T_0} \right)^{1/2} \quad (4.57)$$

for the temperature dependence of σ_s where T_0 has the same meaning as in eq. (4.56).

To determine the strain rate dependence of the plastic shear resistance at constant temperature, we note that only T_0 is specifically dependent on the shear strain rate. Thus we have

$$\frac{d\sigma_s}{d \ln \dot{\gamma}} = \frac{d\sigma_s}{dT_0} \frac{dT_0}{d \ln \dot{\gamma}}, \quad (4.58)$$

which gives for *Regimes I* and *II*

$$\frac{d\sigma_s}{d \ln \dot{\gamma}} = \hat{\tau}_{sI} \frac{\pi^3}{4} \frac{(1-\nu)}{(1+\nu)} \left(\frac{kT_0}{2F_k} \right) \left(\frac{T}{T_0} \right) \left(\frac{T_0 - T}{T_0} \right), \quad (4.59)$$

$$\frac{d\sigma_s}{d \ln \dot{\gamma}} = \frac{\hat{\tau}_{sII}}{2} \left(\frac{T}{T_0} \right)^{1/2} \left(\frac{kT_0}{2F_k} \right). \quad (4.60)$$

In the identification of the character of the governing mechanism of the lattice resistance, two activation parameters, the shear activation volume Δv_σ^* (eq. 2.47) and the activation enthalpy (the zero-stress limiting form of the activation free energy) ΔH^* are of interest.

For *Regimes I* and *II*, the shear activation volumes are, respectively,

$$\begin{aligned}\Delta v_{\sigma I}^* &= \left(\frac{2F_k}{\hat{\tau}_{sI}} \right) \frac{4}{\pi^3} \left(\frac{1+\nu}{1-\nu} \right) \left(\frac{\hat{\tau}_{sI}}{\sigma_s} \right)^{1/2} \\ &= \frac{1}{\pi^2} \left(\frac{1+\nu}{1-\nu} \right) \left(\frac{\mu}{\hat{\tau}_{sI}} \right)^{1/2} (b\lambda)^{3/2} \left(\frac{\hat{\tau}_{sI}}{\sigma_s} \right)^{1/2},\end{aligned}\quad (4.61)$$

$$\Delta v_{\sigma II}^* = \left(\frac{2F_k}{\hat{\tau}_{sII}} \right) 2 \left(1 - \frac{\sigma_s}{\hat{\tau}_{sII}} \right) = \frac{\pi}{2} \left(\frac{\mu}{\hat{\tau}_{sII}} \right)^{1/2} (b\lambda)^{3/2} \left(1 - \frac{\sigma_s}{\hat{\tau}_{sII}} \right), \quad (4.62)$$

where $\hat{\tau}_s$ is the athermal asymptotic value at $T = 0$.

The activation enthalpies for *Regimes I* and *II*, defined by the equation

$$\Delta H^* = -kT^2 \frac{(\partial \sigma_s / \partial T) \dot{\gamma}}{(\partial \sigma_s / \partial \ln \dot{\gamma})_T}, \quad (4.63)$$

are, in turn,

$$\Delta H_I^* = \frac{T}{T_0} (2F_k)_I = \frac{T}{T_0} \frac{\pi}{4} \left(\hat{\tau}_{sI} \mu b^3 \lambda_I^3 \right)^{1/2}, \quad (4.64a)$$

$$\Delta H_{II}^* = \frac{T}{T_0} (2F_k)_{II} = \frac{T}{T_0} \frac{\pi}{4} \left(\hat{\tau}_{sII} \mu b^3 \lambda_{II}^3 \right)^{1/2}. \quad (4.64b)$$

While eqs. (4.64a) and 4.64b) look identical, the operational values differ, first because of the different $\hat{\tau}_{sI}$ and $\hat{\tau}_{sII}$ and because the factors $(b\lambda)^{3/2}$ in eqs. (4.61) and (4.62) for the kink heights λ in these two regimes can be different, as the dislocations may move on $\{110\}$ planes or $\{112\}$ planes in different stress ranges, as the comparisons of models with experiments discussed below will demonstrate.

We note that in Eqns. (4.55)–(4.64b) the effective athermal resistances $\hat{\tau}_{sI}$ and $\hat{\tau}_{sII}$, are different. The resistance $\hat{\tau}_{sI}$ is the value of the stress σ that makes ΔG_{dk}^* in Eqn. (4.38) vanish, while $\hat{\tau}_{sII}$ is given by Eqn. (4.23b). We note, moreover, that while the T_0 , cut-off temperature, for both Regimes I and II are given by Eqn. (4.56), they may differ somewhat due to small changes in the specific forms of the pre-exponential factors of Eqn. (4.50).

In Section 4.6.3 below, we analyze experimental results for a series of pure BCC transition metals with the aid of the expressions developed in this and previous sections.

4.6.3 Comparison of Theory with Experiments on BCC Transition Metals

The kinetic models outlined in Sections 4.6.1 and 4.6.2 for the rate process of plastic flow in BCC transition metals have been tested in a series of experimental studies on high-purity, single-crystalline materials consisting of α -Fe (Brunner

and Diehl 1987, 1991a,b), Mo (Hollang *et al.* 1997, 2001), Ta (Werner 1987), Nb (Ackermann *et al.* 1983), and W (Brunner 2000). These results have been compared with the theoretical model of Seeger (1981), and found to be broadly consistent with it. We consider these experimental results with the aid of our simplified models outlined in Sections 4.6.1 and 4.6.2 and point out complexities not included in our models, for which the reader will be referred to the sources quoted above. For detailed comparisons we consider the experiments on Mo of Hollang *et al.* (1997, 2001), which present a rather complete range of probing.

As outlined in Chapter 3, there can be a number of coexisting deformation resistance mechanisms active in crystalline media, incorporating both intrinsic lattice resistance and extrinsic types. Here the interest is primarily in the intrinsic mechanisms unaffected by second constituents and grain boundaries. Hence, it is essential to utilize high-purity materials in single-crystalline form and to have a clear means of separation of the dislocation resistance due to strain hardening. To deal with the latter problem, experiments can either be conducted on highly perfect material with a very low dislocation density and consider behavior related only to the initial yield, or develop a state of saturation hardening by a preparatory stage of cyclic plastic flow to achieve a stationary state of strain hardening. Neither of the two techniques is free of problems but the latter, described by Ackermann *et al.* (1983), offers a ready means for repeatability of the experiments on the same sample, and has been used in nearly all the studies referred to above.

With material of such pedigree, there are several experimental techniques available for probing the lattice resistance in a temperature range between 0 K and T_0 . These include (a) probing the initial yield behavior at given imposed plastic shear strain rates $\dot{\gamma}_p (\approx \dot{\gamma})$ of the saturation-hardened state at different well-controlled temperatures, (b) performing strain rate change experiments during steady flow by either increasing or decreasing the strain rate, at given well-controlled temperatures, and (c) performing stress relaxation experiments at a given temperature from a given reference strain rate by stopping the extension rate on the specimens. All these techniques require precise control of the conditions of deformation and have been described in detail in the above references.

As a first step in the analysis of the experimental findings on Mo (Hollang *et al.* 1997, 2001), we recognize that, from the measured plastic shear resistance, the athermal component τ_μ has to be subtracted to obtain the temperature- and strain-rate-dependent “effective” component τ_s ($\hat{\tau}_{sI}$ or $\hat{\tau}_{sII}$). In cyclically conditioned samples, the saturation plastic resistance of the Mo samples on their effective slip systems {112} above the cutoff temperature T_0 where the lattice resistance no longer governs was found to be $\hat{\tau}_\mu = 53$ MPa.

In the studies referred to, two crystal axis orientations in the center of the standard triangle were selected. The first orientation (1) was selected to maximize the Schmid factor m on the ($\bar{1}01$) plane at a level of 0.5. This resulted in Schmid factors of 0.43 on the two alternative slip planes ($\bar{1}\bar{1}2$) and ($\bar{2}11$) (see Fig. 4.7).

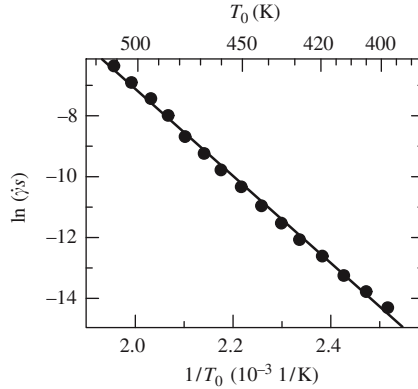


FIG. 4.21. Experimentally determined dependence of the cutoff temperature T_0 on the strain rate in Mo single crystals (Hollang *et al.* 1997; courtesy of Wiley-VCH).

A second orientation (2) was chosen which maximized the Schmid factor on the $(\bar{2}11)$ plane again at 0.43, but the factor on the $(\bar{1}01)$ plane being at 0.40.

In the study of the *Regime I* behavior, which was idealized to involve the nucleation of two well-separated kinks, over a range of plastic strain rate from $5.9 \times 10^{-7} \text{ s}^{-1}$ to $1.7 \times 10^{-3} \text{ s}^{-1}$, the expected monotonic increase of the cutoff temperature T_0 was noted to be as eq. (4.56) predicts, and is shown in Fig. 4.21. The slope of this dependence,

$$-k \frac{d \ln \dot{\gamma}}{d(1/T_0)} = 2F_k, \quad (4.65)$$

gives directly the double-kink energy for two well-defined and well-separated kinks as $1.27 \pm 0.02 \text{ eV}$ ($2.03 \times 10^{-19} \text{ J}$). Comparison of this result with the double-kink energy given by eq. (4.32b),

$$2F_k = \frac{\pi}{4} \left(\hat{\tau}_{sI} \mu b^3 \lambda^3 \right)^{1/2}, \quad (4.66)$$

and using $\mu = c_{44} = 110 \text{ GPa}$ (Hirth and Lothe 1982), $b = 2.72 \times 10^{-10} \text{ m}$, and $\hat{\tau}_{sI} = 690 \text{ MPa}$ (determined from extrapolation back to 0 K of the slope of the temperature-dependent plastic shear resistance in the intermediate temperature range), the kink height λ can be determined as $3.53 \times 10^{-10} \text{ m}$, which is intermediate between the wavelength of the lattice topography on the (110) plane, $\lambda(110) = 2.56 \times 10^{-10} \text{ m}$, and the value on the (112) plane, $\lambda(112) = 4.44 \times 10^{-10} \text{ m}$, but is slightly closer to the value of the latter. The more accurate estimate of Hollang *et al.* (1997) puts the value of λ closer to that on the (112) plane. From this it is concluded that in *Regime I* and even well into *Regime II*, the active slip system is the $\{112\}\langle 110 \rangle$ system for both orientations, even though for orientation (1) the Schmid factor for the $(\bar{1}01)$ plane exceeds that of the two adjacent $\{112\}$ planes.

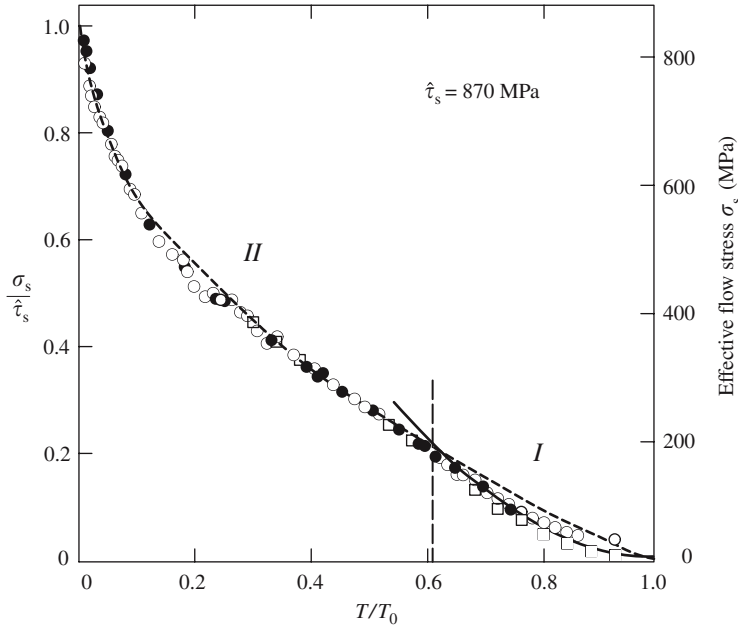


FIG. 4.22. Temperature dependence of the rate-dependent component σ_s of the plastic resistance of Mo, in both *Regime I* and *Regime II*, showing the experimentally determined results compared with the theoretical models of eqs. (4.55) and (4.57) (Hollang *et al.* 2001; courtesy of Elsevier).

The temperature dependence of the flow stress in *Regime I* is given by eq. (4.55). For a Poisson's ratio of $\nu = 0.305$ (Hirth and Lothe 1982), eq. (4.55) becomes

$$\frac{\sigma_s}{\hat{\tau}_{sI}} = 2.06 \left(1 - \frac{T}{T_0} \right)^2. \quad (4.67)$$

This dependence is plotted in Fig. 4.22 as a function of T/T_0 , from 1.0 down to 0.64, where the *Regime I* behavior is superseded by *Regime II* behavior. The agreement with the experimental measurements of Hollang *et al.* (2001) is excellent.

In *Regime II*, Hollang *et al.* state that in an intermediate temperature region between 50 K and 320 K that is, between $T/T_0 = 0.1$ and $T/T_0 = 0.64$, slip remains on the $\{112\}$ systems for both orientations (1) and (2), that the shear resistance extrapolated to 0 K gives a threshold resistance $\hat{\tau}_{sI} = 690$ MPa, as already stated above, and that below 50 K ($T/T_0 = 0.1$) slip is on the $(\bar{1}01)$ plane with an extrapolated threshold resistance of $\hat{\tau}_{sII} = 870$ MPa. We make no such separation and use the simple temperature-dependent plastic-resistance relation of eq. (4.57) for the entire range between $T/T_0 = 0$ and 0.64, but with $\hat{\tau} = 870$ MPa as stated by Hollang *et al.* This dependence is shown in Fig. 4.23 in conjunction with the experimental measurements. Again the agreement of the form of eq. (4.57) with the

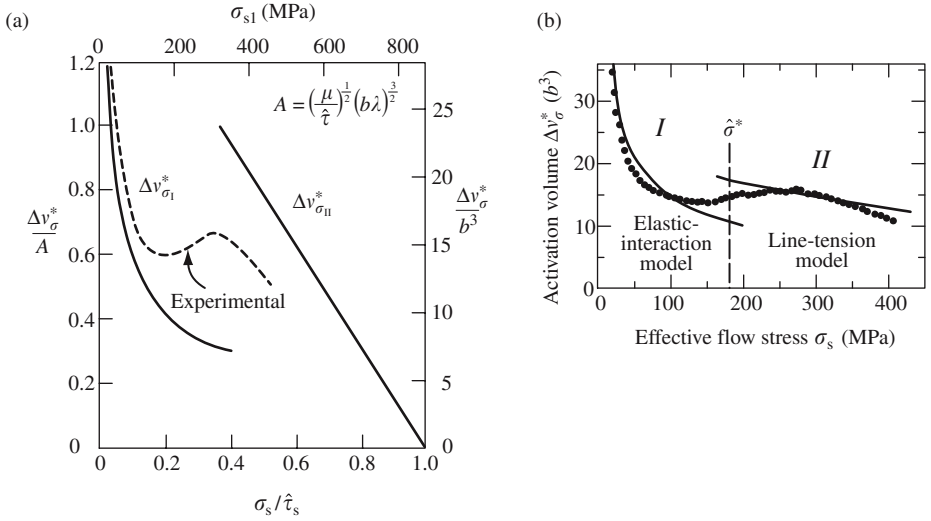


FIG. 4.23. Shear activation volumes and their stress dependence: (a) model results for *Regimes I* and *II*; (b) experimental results for Mo (Hollang *et al.* 2001; courtesy of Elsevier).

experimental results is excellent, without invoking a separation of the resistance process into two parts.

We note here, in passing, a fundamental dependence of considerable importance relating the form of the normalized flow stress $\sigma_s/\hat{\tau}_s$ to the normalized temperature T/T_0 and to the form of the dependence of the normalized activation enthalpy on the normalized stress, $\Delta H^*(\sigma_s/\hat{\tau}_s)/\Delta H_0^*$. This finding follows from eq. (4.55), which we can write as

$$\ln \left(\frac{\dot{\gamma}_0}{\dot{\gamma}} \right) = \frac{\Delta H_0^*}{kT} h \left(\frac{\sigma_s}{\hat{\tau}_s} \right), \quad (4.68)$$

where ΔH_0^* is the energy scale factor of the activation enthalpy and $h(\sigma_s/\hat{\tau}_s)$ is a dimensionless function of the normalized flow stress $\sigma_s/\hat{\tau}_s$. Using eq. (4.56), where we note that $2F_k$ stands for ΔH_0^* , we have

$$\frac{T}{T_0} = \frac{kT \ln (\dot{\gamma}_0/\dot{\gamma})}{\Delta H_0^*} = h \left(\frac{\sigma_s}{\hat{\tau}_s} \right), \quad (4.69)$$

giving, with an operational inversion,

$$\frac{\sigma_s}{\hat{\tau}_s} = h^{-1} \left(\frac{T}{T_0} \right), \quad (4.70)$$

where the right-hand side of eq. (4.70) represents the right-hand sides of eqs. (4.55) and (4.57) for *Regimes I* and *II* in generic form, allowing for differences in $\hat{\tau}_{sI}$ and $\hat{\tau}_{sII}$.

Thus, regardless of mechanism, the form of the dependence on the normalized temperature T/T_0 of the normalized plastic resistance $\sigma_s/\hat{\tau}_s$ reflects directly the dependence on the normalized effective stress of the normalized activation free enthalpy, where T/T_0 transforms into $\Delta H^*/\Delta H_0^*$.

Therefore, on the basis of the above observation, Fig. 4.22 tilted on its side represents the form of the dependence of $\Delta H/\Delta H_0^*$ on $\sigma_s/\hat{\tau}_s$, where $\Delta H_0^* = 2F_K$. Since this figure is a result of using fits of eqs. (4.55) and (4.57) to the experimental data, it also represents an excellent prediction for $\Delta H^* = \Delta H_0^* h(\sigma_s/\hat{\tau}_s)$.

The shear activation volumes, which give a good measure of the size of the activation configuration, as determined from eqs. (4.61) and (4.62) for *Regimes I* and *II*, are plotted in Fig. 4.23(a) as a function of $\sigma_s/\hat{\tau}_s$. They illustrate rather graphically where the mechanistically different processes in these two regimes are dominant. The corresponding experimental values are plotted in Fig. 4.23(b) and are also indicated by a dashed line in Fig. 4.23(a). The basic trends are well represented but the agreement is fair at best, possibly because of the experimental complexities in determining the shear activation volume accurately.

Finally, with decreasing temperature and correspondingly increasing σ_s , the activation enthalpy ΔH^* will decrease monotonically to zero at 0 K as predicted by eqs. (4.64a) and (4.64b) for the two regimes. A direct experimental verification has been given by Brunner (2000) for *W* and is shown in Fig. 4.24. For comparison with theoretical models, we take the results for *Regime I* in the high-temperature region and the measured cutoff temperature of $T_0 = 772$ K for *W* to define the double-kink

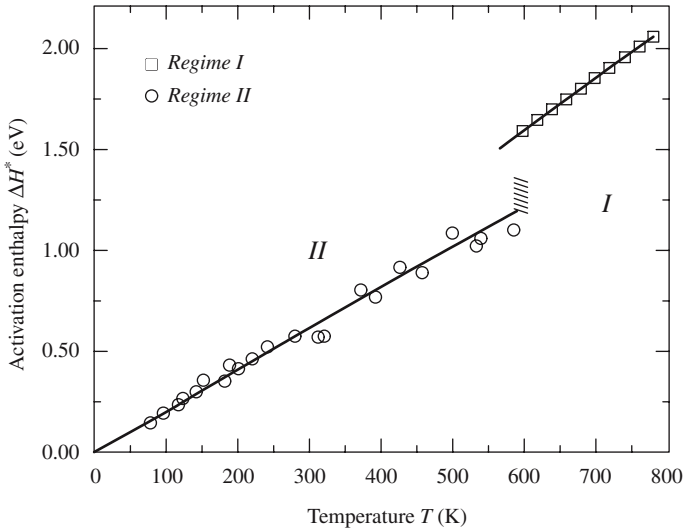


FIG. 4.24. Temperature dependence of the effective activation enthalpy in *Regimes I* and *II* (with changing flow stress) as determined in experiments on *W* single crystals (after Brunner 2000).

energy $2F_k$ as 2.03 eV. We consider this to be based on kinks on the $\{112\}$ planes with a kink height of $\lambda_{112} = 4.47 \times 10^{-10}$ m, characteristic of the W structure. The figure shows, however, that the slope $d \Delta H^*/dT$ is smaller in *Regime II* by a factor of 0.759, resulting in an effective double-kink energy of only 1.54 eV. Since in this regime the active slip planes change from the $\{112\}$ planes to the $(\bar{1}01)$ plane with decreasing temperature, it might be concluded that slip here is on a combination of $\{112\}$ planes and the $(\bar{1}01)$ plane with an average effective kink height of only 3.72×10^{-10} m rather than 4.47×10^{-10} m.

4.7 The Lattice Resistance of Silicon

4.7.1 Dislocations in Silicon

Because of the intense industrial interest in semiconductors for computer devices, the properties of silicon, still the most widely used material for such applications, have received widespread attention ever since its introduction as a transistor material in the early 1950s. While the overwhelming interest in silicon has always been in its electronic properties, it was recognized early that these properties are significantly affected by lattice defects, and particularly dislocations. While first-principles computations (Bigger *et al.* 1992) of the electronic states associated with dislocation cores in Si have shown that the band gap is clear of all states except perhaps for a very shallow band just above the valence band, experiments suggest that some electronic disturbances resulting from the presence of dislocations remain. Moreover, various *n*- and *p*-type doping agents introduced for electronic purposes often interact with dislocations. Furthermore, when dislocations move under stress in Si, they produce electronically active point defects that also affect device performance importantly. For these reasons, dislocations in silicon have been the subject of extensive investigations, and the plasticity of silicon has been studied not only for the purpose of better understanding its effect on electronic properties but also because it constitutes an excellent test bed for fundamental studies of crystal plasticity in general. Here we shall be primarily interested in the nature of the lattice resistance to the motion of dislocations in Si. The interactions of dislocations with electronic properties will be of no interest. For these interactions, the reader is referred to Alexander and Teichler (1991) and the many references cited there.

Silicon is a tetrahedrally bonded solid with a diamond cubic structure (see Fig. 1.4), which is a crystallographic variant of the FCC structure and possesses the same slip systems, that is, $\{111\}\langle 110 \rangle$. However, since it is not a primitive structure and it possesses twice as many atoms per unit cell as the FCC structure (eight instead of four), slip on the $\{111\}$ planes offers two separate possibilities, depicted in Fig. 4.25. This shows a $(1\bar{1}0)$ plane section through the structure, containing two of the four tetrahedral bonds (in the $[111]$ and $[\bar{1}\bar{1}1]$ directions). The (111) plane

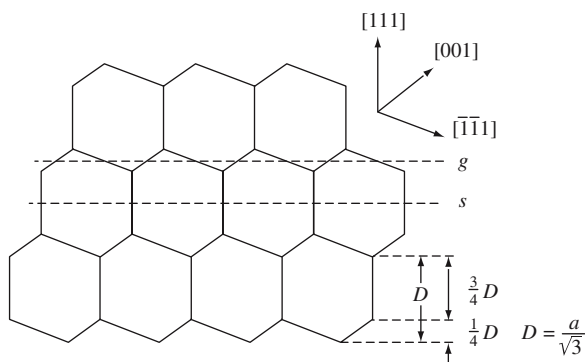


FIG. 4.25. A $(1\bar{1}0)$ plane section across the Si crystal structure, showing two of the four tetrahedral bonds and the outlines of the glide (g) and shuffle (s) planes.

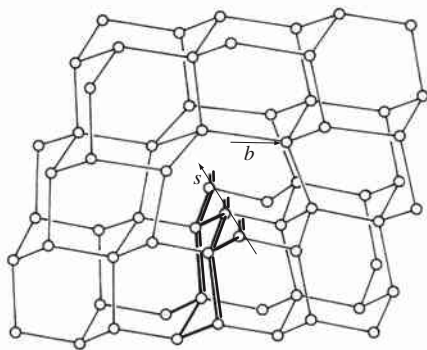


FIG. 4.26. An edge dislocation line on the shuffle plane results in only one broken bond per atomic length along the dislocation line.

is shown by two dashed alternative outlines, marked s and g , standing for *shuffle* and *glide*. Two possible $\langle 110 \rangle$ -type slip directions make angles of $\mp 30^\circ$ with the $(1\bar{1}0)$ plane of the section and are not shown. On the basis of the Peierls–Nabarro analysis of Section 4.2.1, which indicated that the preferred slip planes should be those with the largest interplanar spacing, the logical choice should be slip on the shuffle (s) planes, where, moreover, as shown in Fig. 4.26, only one tetrahedral bond would be broken per atomic length along the dislocation line. However, high-resolution electron microscopy (Ray and Cockayne 1971) of dislocations on the $\{111\}$ glide planes has established that they are dissociated into Shockley partials as in FCC metals. This is only possible on the glide (g) set of planes. Thus, in spite of the fact that the advance of the dislocation along these planes must break three tetrahedral bonds per atomic length, as shown in Fig. 4.27, these must be the preferred slip planes. All the evidence since that discovery has substantiated this finding.

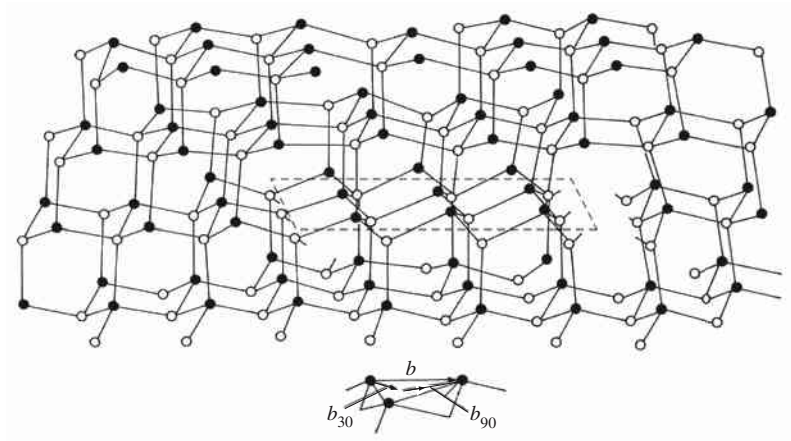


FIG. 4.27. Glide on the glide plane (g) results in the breakage of three tetrahedral bonds across the plane per atomic length without core reconstruction.

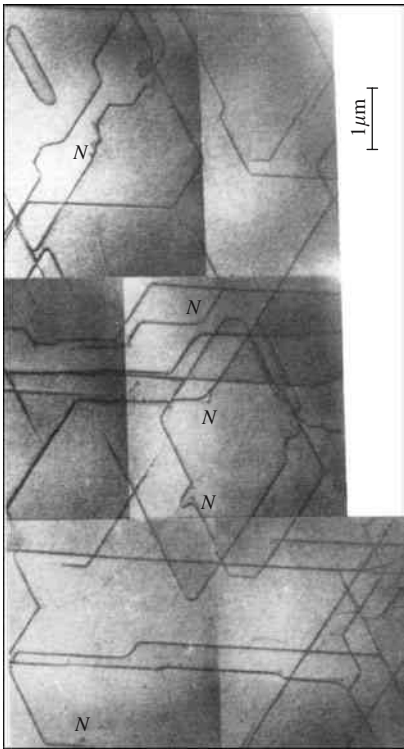


FIG. 4.28. A TEM micrograph of dislocations on a $\{111\}$ glide plane in Si deformed at 750°C (below the temperature T_0) shows only dislocation lines parallel to the $\langle 110 \rangle$ directions (Alexander 1968; courtesy of Wiley-VCH).

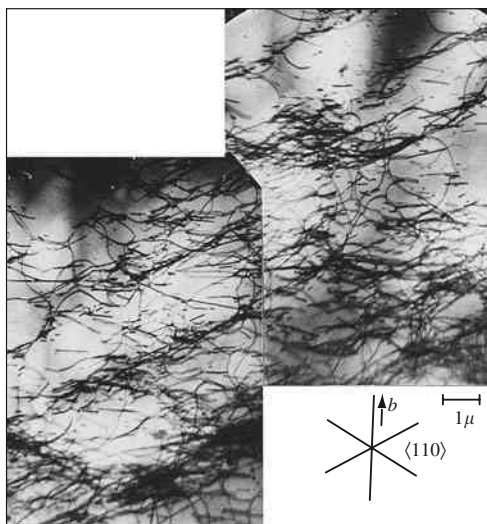


FIG. 4.29. A TEM micrograph of dislocations on a $\{111\}$ plane in Ge, deformed at a temperature well above T_0 where the lattice resistance no longer governs, shows dislocation braids similar to those in Cu but with a high concentration of straight dipolar segments (Alexander 1968; courtesy of Wiley-VCH).

Transmission electron microscopy has established that in Si deformed below a threshold temperature T_0 (as identified below), the retained dislocations lie overwhelmingly parallel to the $\langle 110 \rangle$ directions in the $\{111\}$ glide planes as shown in the typical micrograph of Fig. 4.28 for material deformed at 750°C (Alexander 1968). Clearly, some of these dislocations are of screw nature, while others are of a mixed edge and screw nature with the Burgers vector making a 60° angle with the line direction. This demonstrates that the lattice resistance is very high in Si, with lattice potential valleys lying parallel to the $\langle 110 \rangle$ directions. In Si deformed at temperatures around 420°C , at substantially higher stresses, dislocations perpendicular to $\langle 110 \rangle$ orientations (of edge character) have also been reported but are believed to be made up of closely spaced segments of 60° dislocations (Alexander and Teichler 1991). In Si deformed above T_0 (estimated to be considerably above 1000°C on the basis of a consideration parallel to that of eq. 4.56), where the lattice resistance should no longer be present, dislocation motion should become relatively unrestricted crystallographically, showing forms of aggregation very similar to those in FCC metals with negligible lattice resistance, as shown in Fig. 4.29 for the similar case of Ge (Alexander 1968). Here we shall be interested primarily in the low-temperature behavior ($T \ll T_0$), where the lattice resistance governs. While the majority of our discussion will be related to Si, the behavior of Ge and some semiconductor compounds is very similar.

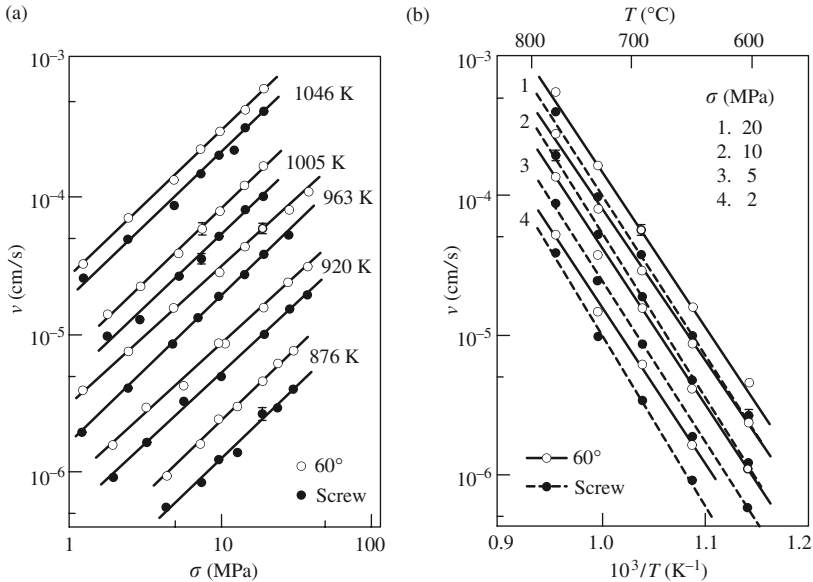


FIG. 4.30. (a) Stress dependence of the velocity of 60° and screw dislocations in normal undoped Si at different temperatures; (b) temperature dependence of the velocities of the same kind of dislocations at different stress levels, all below 40 MPa (Imai and Sumino 1983; courtesy of Taylor and Francis).

4.7.2 Dislocation Mobility in Si

Unambiguous measurements have been made of the dislocation velocity in both intrinsic (pure undoped) and doped Si as a function of the resolved applied shear stress at different temperatures. These measurements were made by determining the initial and final positions of individual dislocations responding to stress pulses of different strengths and of different durations. The initial and final positions of the dislocations were determined either from surface etch pits or from the displacements of the entire dislocation lines in the near-surface region, visualized by means of an X-ray topographic imaging technique, with the latter method being more reliable and less intrusive. Figure 4.30(a) shows the stress dependence of measured velocities at several different constant temperatures in intrinsic Si (Imai and Sumino 1983) obtained using the X-ray topographic imaging technique, and Fig. 4.30(b) shows a kinetic plot of the temperature dependence of these velocities at different constant stress levels, all at relatively low stresses ($\sigma < 10^{-3}\mu$) and temperatures $T < T_0$. The measurements show that under similar conditions, the 60° mixed dislocations move consistently faster than the pure screw dislocations. The observed behavior fits well to the form

$$v = v_0 \sigma \exp\left(\frac{-Q}{kT}\right), \quad (4.71)$$

where for 60° dislocations $v_0 = 1.0 \times 10^4$ m/MPa s, and $Q = 2.20$ eV and for screw dislocations $v_0 = 3.5 \times 10^4$ m/MPa s, $Q = 2.35$ eV represent the magnitudes for the best fits to the data presented in Fig 4.30.

There have been only a few dislocation velocity measurements for stresses above 50 MPa. In intrinsic Si, Alexander *et al.* (1987) noted a definite stress dependence of the activation energy in the range of 30 to 300 MPa but at a temperature of 422°C . Their empirical fit of the data could be interpreted also as showing that the dislocation velocities became nonlinearly dependent on the stress, where the stress exponent m approached 2 for stresses around 400 MPa and became temperature-dependent. While their data was sparse, they also noted an effect of a normal stress acting across the slip plane. In Si-Ge solid solutions, Gillard and Nix (1993) and Gillard *et al.* (1995) have observed similar nonlinearly stress-dependent dislocation velocities in the 100–400 MPa range.

In addition to the larger-than-unity stress exponents for the dislocation velocity at high stresses, there have been reports of stress exponents smaller than unity at small stresses, leveling off at an exponent of unity (George and Champier 1979), in intrinsic Si. While these observations have been attributed to trace impurities, as discussed below, other more intriguing explanations are also possible, as we shall discuss briefly in Section 4.7.5.

Dislocation velocities have also been measured in both n -type (P -doped) and p -type (N -doped) Si at dopant concentrations in the range of $10^{17} - 10^{18}$ atoms/cm³ (George and Champier 1979). Both n -doping and p -doping result in higher velocities than in pure Si for comparable stress and temperature levels, with, however, n -doping being somewhat more effective than p -doping. Parenthetically, comparable experiments on Ge by Patel and Chaudhuri (1966) have shown that while n -doping (with As) results in an increased velocity, p -doping (with Ga) results in a decreased velocity when, in either case, the dopant concentration exceeds 10^{18} /cm³. Moreover, the experiments of Imai and Sumino (1983) have shown that the presence of both oxygen and metallic impurities at relatively low concentrations gives a different than linear dependence of dislocation velocity on stress at low stress levels, as shown in Fig. 4.31. Here we shall focus our attention primarily on the behavior of pure intrinsic Si in the stress range below 50 MPa, where linear behavior holds, but shall consider briefly also the extension of the behavior into the higher-stress range.

4.7.3 Models of the dislocation Core Structure in Si

Reconstruction of Dislocation Cores Since dislocations in Si are dissociated into Shockley partial dislocations, mobility mechanisms must take note of this dissociation. As depicted in Fig. 4.32, the 60° mixed dislocations are composed of two types of Shockley partial dislocation, with Burgers vectors making angles of 30° and 90° with the dislocation line, while the screw dislocations are composed of two 30° partial dislocation; both sets of partials are separated by an intrinsic stacking fault.

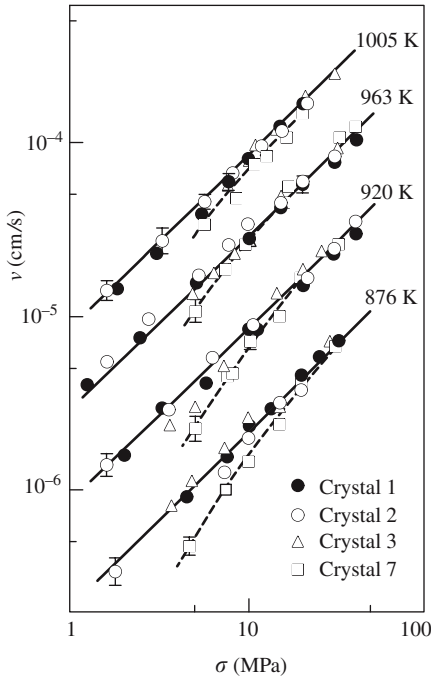


FIG. 4.31. Nonlinear stress dependence of the velocity of dislocations in Si in crystals doped with *O* and some metallic impurities, at different temperatures (Imai and Sumino 1983; courtesy of Taylor and Francis).

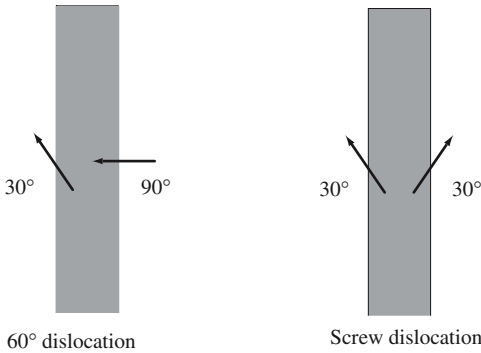


FIG. 4.32. Dissociation of 60° and screw dislocations on the glide planes of Si.

In both cases, structural reconstructions have to take place at the core to eliminate the dangling bonds that would otherwise arise in the core. Figure 4.33(a) shows the structure of an unreconstructed core of a 30° Shockley partial of a screw dislocation as viewed on the {111} glide plane. Here the white atoms in an upper {111} layer

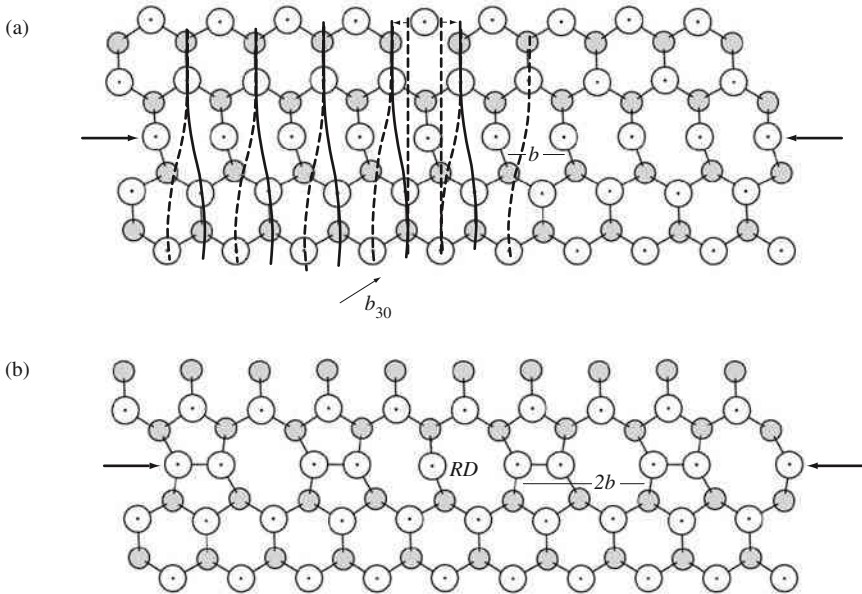


FIG. 4.33. Core of 30° Shockley partial dislocation in Si : (a) the possible unreconstructed core shows a series of dangling bonds at every atomic spacing; (b) reconstruction of such a core results in dimer-type association of neighboring atom pairs but can leave behind reconstruction defects (RD) between reconstructed domains (after Bulatov *et al.* 1995).

are bonded by three tetrahedrally arranged bonds to shaded atoms in a lower $\{111\}$ layer across the glide-set gap. A central vertical bond, depicted by central dots in the circles, connects the upper-layer atoms to a $\{111\}$ plane above, across the larger shuffle-set gap. In the unreconstructed core, the row of atoms along the dislocation core, pointed to by the horizontal arrows, have only three bonds connecting them to their surroundings. While the geometrical features of core structures and their defects are readily visualized, determination of their energies has been done by a variety of computer models, the results of which differ depending on the different interatomic potentials and simulation cells employed. In an early simulation by Bulatov *et al.* (1995) using a Stillinger and Weber (SW) potential, the energy of the unreconstructed core was determined as 2.72 eV per repeat distance b , the Burgers vector length of the full dislocation. In all cases the high-energy structure of the dislocation core is reduced by a reconstruction process such as that shown in Fig. 4.33(b) where the atoms in the upper layer have paired up as a series of distorted dimers to arrange for tetrahedral coordination. This results in a reduction

in energy and an increase in the periodicity spacing by a factor of two. The energy reduction was estimated in the above study to be 0.81 eV for the 30° partial for the same repeat distance. Other estimates range from 0.41 to 1.60 eV (Bulatov *et al.* 2001).

Since the reconstruction into distorted dimers can be initiated independently from different sites along the partial dislocation and can spread along the dislocation, inevitably some atoms will remain unpaired and will form reconstruction defects (*RDs*) between two separate “phases” of reconstruction, as shown in Fig. 4.33(b). The excess energy of such an *RD* has been estimated in the study cited above as 0.81 eV for the 30° partial.

The unreconstructed core structure of the 90° partial and its reconstruction show somewhat greater complexity, as two possible forms of reconstruction, with a single period and a double period, have nearly the same energy, with, however, a considerably lower energy gain in the reconstruction process compared with the 30° partial dislocation, ranging from 0.3 to 1.2 eV, depending on the model used (Bulatov *et al.* 2001). The reconstruction defects in the 90° partial dislocation are also more varied in form and energy. In the case of both the 30° and the 90° partial dislocation, the interest in the *RDs* is because they often serve as convenient nucleation sites for double kinks and also interact with the motion of kinks in the process of strain production. We shall only view these complexities broadly, owing to the sensitivity of their properties to the model. We note here that the structure of kinks in partial dislocations in Si and the resulting effects on the mobility of dislocations are still being actively pursued by first-principles computations (see for example, Valladares *et al.* 1998 and Valladares and Sutton 2006), which will likely lead to further modifications of the picture of dislocation mobility in Si. Our consideration will be limited primarily to the low-stress ($\sigma/\mu < 10^{-3}$) regime, where the stress dependence of the dislocation velocity either is linear as given by eq. (4.71) or becomes slightly nonlinear. As in the BCC case, in this regime the dislocation advances by nucleation of fully formed double kinks that subsequently move apart under stress in a damped viscous manner, overcoming a secondary lattice potential barrier to their motion.

Kinks on Partial Dislocations Figure 4.34 shows the structures of left-hand and right-hand kinks in 30° and 90° partial dislocations, away from *RDs*. In both cases, since the reconstructed cores of partials do not possess structural mirror symmetry about a plane perpendicular to the dislocation, the left-hand and right-hand kinks are not symmetrical and have different energies of formation. The double-kink formation energies in isolation from an *RD* depend on the pair-potential models used for their determination, and range from 1.04 to 1.63 eV for the 30° partial. For the 90° partial dislocation, the corresponding values are roughly around 1.4 eV.

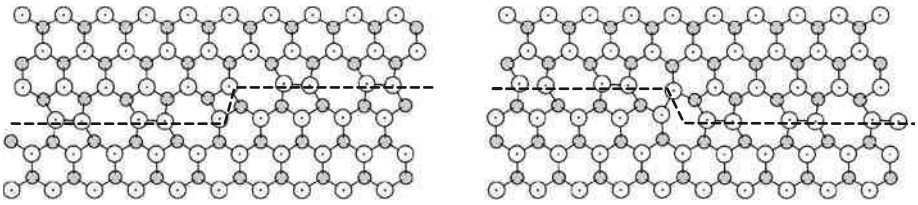


FIG. 4.34. Structures of right-hand (RH) and left-hand (LH) kinks on a 30° Shockley partial. Because of the lack of mirror symmetry along a screw dislocation, the RH and LH kinks have different energies (after Bulatov *et al.* 1995).

Kink Migration Energies As stated above, the kink migration process along a partial dislocation in Si differs fundamentally from that in BCC metals, where the only resistance to kink motion was the phonon drag process of interaction with lattice thermal waves. In Si and other semiconductors, kink migration is impeded by a substantial secondary lattice resistance, which has been studied in detail by Bulatov *et al.* (1995). Predictably, because of the asymmetry along the dislocation line already remarked on above, the kink migration energy of a right-hand kink differs from that of a left-hand kink on a 30° partial. Moreover, because of the period doubling due to reconstruction of the core, an intermediate minimum exists between the stable positions a distance $2b$ apart. The energy barrier for kink motion has been estimated to be between 0.7 and 2.1 eV for the right-hand kink and between 0.7 and 1.5 eV for the left-hand kink, again dependent on the pair-potential model used in the determination.

4.7.4 Model of Dislocation Motion

The Low-stress Regime for $\sigma/\mu < 5 \times 10^{-4}$ As outlined in Section 4.7.3, atomistic models of the structure of the 30° and 90° Shockley partials making up screw and 60° mixed dislocations in Si show considerable detail and range in terms of the energetic aspects of *RDs* for right-hand and left-hand kinks and the energy barriers to their motion. The motion of an extended screw dislocation made up of two partials separated by a stacking fault has been studied by a kinetic Monte Carlo technique without any prejudgment as to the dominant species of core defect (Cai *et al.* 2000, Bulatov *et al.* 2001). Here we shall consider the findings of that study as a guide, but actually refer to an earlier analytical model of Hirth and Lothe (1982) for the diffusive motion of kinks along dislocation lines to provide an overall insight into the mobility of an extended dislocation in Si.

As presented in Chapter 2, the Shockley partial dislocations will have characteristic equilibrium separations Λ_e and Λ_s in extended edge and screw dislocations, given by eq. (2.57) for an edge dislocation and by a corresponding version for the

screw dislocation as follows:

$$\Lambda_e = b \left(\frac{\mu b}{\chi_{SF}} \right) \frac{(2 + \nu)}{24\pi(1 - \nu)}, \quad (4.72a)$$

$$\Lambda_s = b \left(\frac{\mu b}{\chi_{SF}} \right) \frac{(2 - 3\nu)}{24\pi(1 - \nu)}. \quad (4.72b)$$

For the appropriate values in Si, where $\mu = 68.1$ GPa, $b = 0.335$ nm, $\nu = 0.218$, and $\chi_{SF} = 58$ mJ/m², Λ_e and Λ_s are 4.96 nm and 3.02 nm, respectively. Since these stacking-fault widths are only factors of 14.8 and 9.01 larger than the Burgers vector, important effects of correlation should be expected in the motion of the partials as the extended dislocations move. This is indeed the case, as we discuss further below. Nevertheless, for the development of a useful analytic expression for the velocity of an extended dislocation, we first assume that the partials move in an uncorrelated way, but maintaining, on the average, the stacking fault at its equilibrium width. This permits a direct application of the theory of Hirth and Lothe (1982) for dislocation motion over lattice potential contours by the diffusive motion of kinks along the dislocation line, overcoming secondary lattice energy barriers to their motion.

Thus, focusing on the motion of Shockley partials, we note that kinks of a characteristic width w and height λ undergoing Brownian motion along the dislocation line, when no external stress is present, will have a diffusivity D_K , given by (Hirth and Lothe 1982)

$$D_K = \nu_D w^2 \exp\left(-\frac{W_m}{kT}\right), \quad (4.73)$$

where W_m is the magnitude of the energy barrier to kink motion along the dislocation line, and ν_D is the Debye frequency. When a small stress σ ($< 10^{-3}\mu$) is present to exert a force σb_p on a kink, the kink will drift systematically in a direction such that positive work is done, with a velocity v_k , that is,

$$v_k = w\nu_D \left[\exp\left(-\frac{W_m - \sigma b_p \lambda w/2}{kT}\right) - \exp\left(-\frac{W_m + \sigma b_p \lambda w/2}{kT}\right) \right], \quad (4.74a)$$

$$v_k = 2w\nu_D \exp\left(-\frac{W_m}{kT}\right) \sinh\left(\frac{\sigma b_p \lambda w/2}{kT}\right) \approx \frac{\sigma b_p \lambda w^2 \lambda}{kT} \nu_D \exp\left(-\frac{W_m}{kT}\right) \quad (4.74b)$$

for $\sigma b_p \lambda w/2kT \ll 1.0$, giving the well-known expression

$$v_k = \frac{\sigma b_p \lambda}{kT} D_K, \quad (4.74c)$$

where D_K is the diffusion constant of Eqn. (4.73).

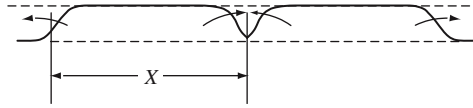


FIG. 4.35. Mean free paths of double-kink separation over characteristic distances X , where kinks formed in other regions encounter and annihilate.

Then, the extended dislocation will move forward through the repeated generation of kink pairs on partials in response to the applied stress, each pair sweeping out a characteristic distance X before they encounter similarly expanding neighboring double kinks, with which they annihilate, and the process can repeat in a systematic manner, as depicted in Fig. 4.35. The forward velocity v_d of the partials, and hence of the extended dislocation, will then be

$$v_d = \lambda X J, \quad (4.75)$$

where J is the production rate of kink pairs per unit length of dislocation line per unit time, and can be viewed as the flux of kink pairs along the line. We view this flux as if it were a classical nucleation process involving the product of the formation probability of double-kink embryos and the flux of kinks crossing the border of the embryos, that is, across the saddle-point configurations of double kinks, as follows:

$$J = c_{dk} v_k. \quad (4.76)$$

In eq. (4.76), c_{dk} is the thermal-equilibrium concentration of double kinks along a straight dislocation line, and v_k is, of course, the kink velocity aided by the applied stress σ .

The thermal-equilibrium concentration of double kinks, c_{dk} , when no stress is applied, is determined by a classical procedure analogous to obtaining the thermal-equilibrium concentration of point defects in a lattice. In this procedure, the free energy of the system, consisting of the energy of formation of the defects and the configurational entropy of placement of the defects among the available lattice sites, is minimized with respect to the defect concentration. In this case the “defects” are kink pairs and the “lattice” sites consist of the discrete available sites along the dislocation line. The result, as explained by Hirth and Lothe (1982) in more detail (in their Chapter 14), is

$$c_{dk} = \frac{1}{w^2} \exp\left(-\frac{2F_k}{kT}\right), \quad (4.77)$$

where $2F_k$ is the double-kink formation energy, which must be taken as the sum of the energies of right-hand and left-hand kinks when the two have different energies.

When an applied shear stress σ is present, aiding the formation of double kinks, their equilibrium concentration becomes enriched. Following considerations similar to the treatment of double-kink nucleation under stress in BCC metals in Section 4.5.4 (eq. 4.38), one has

$$c_{dk} = \frac{1}{w^2} \exp\left(-\frac{2F_k - (\mu\sigma b_p^3 \lambda^3 / 2\pi)^{1/2}}{kT}\right). \quad (4.78)$$

However, at the stress levels under consideration of $\sigma < 10^{-3}\mu$ and for typical kink pair formation energies of the order of 1.5 eV, the stress-dependent term modifying the activation energy makes a contribution that is only in the range of 2%, which can be neglected for first-order considerations. This gives back the expression in eqs. (4.76) and (4.77) for the flux of appearance of double kinks as

$$J = \frac{\sigma b_p \lambda}{w^2 kT} D_k \exp\left(-\frac{2F_k}{kT}\right). \quad (4.79)$$

We interpret this flux as the stress-assisted nucleation rate of double kinks per unit length per unit time.

Finally, the dislocation velocity v_d is determined from eq. (4.75) by noting that the characteristic distance X swept out by a double kink during its lifetime between nucleation and annihilation is

$$X = 2v_k \tau_k, \quad (4.80)$$

where τ_k is the lifetime of a kink pair, which in steady-state motion of the dislocation must also be the time for reappearance of a new kink pair inside the swept-out zone of length X , giving

$$J = \frac{1}{X \tau_k}. \quad (4.81)$$

Eliminating τ_k between eqs. (4.80 and (4.81) then gives

$$X = \left(\frac{2v_k}{J}\right)^{1/2} = \sqrt{2}w \exp\left(\frac{F_k}{kT}\right), \quad (4.82)$$

which gives, finally, for the dislocation velocity,

$$v_d = \sqrt{2} \frac{\sigma b_p \lambda^2}{wkT} D_k \exp\left(-\frac{F_k}{kT}\right) = \sqrt{2} \frac{\sigma b_p w \lambda^2}{kT} v_D \exp\left(-\frac{F_k + W_m}{kT}\right). \quad (4.83)$$

In the form obtained here, the expression above should be taken to be for screw dislocations. For 60° dislocations, with different partials (90° and 30°) in the

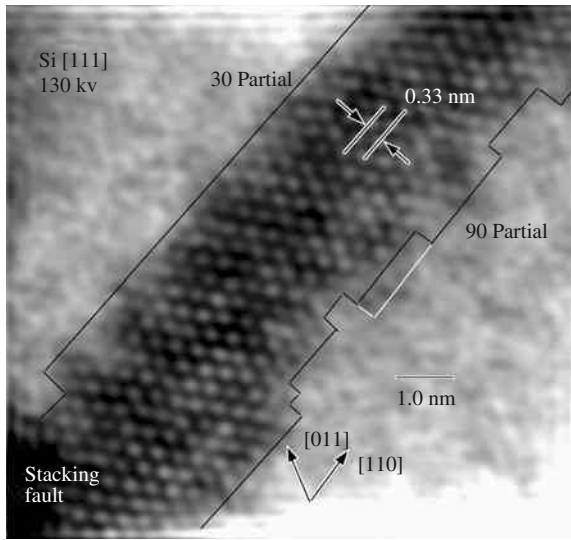


FIG. 4.36. An HRFREM (high-resolution forbidden-reflection electron micrograph) with lattice resolution showing large concentrations of kinks on 90° partials in comparison with 30° partials along an extended 60° mixed dislocation in Si (Kolar *et al.* 1996; courtesy of APS).

extended dislocation with different kink structures and energies, the assumption of uncorrelated advance of the partials would not be a good one.

As the measurements shown in Fig. 4.30(a) indicate, 60° dislocations always move significantly faster than screw dislocations. While the reason for this is not entirely clear, it is consistent with the lower double-kink formation energies estimated for these dislocations from the atomistic models discussed in Section 4.7.3. This has been verified also by direct high-resolution microscopy of extended dislocations that showed double-kink concentrations on 90° partials higher by a factor of three than those on 30° partials, as shown in the direct lattice-imaging micrograph of Fig. 4.36 (Kolar *et al.* 1996).

The High-stress Regime for $\sigma/\mu > 10^{-3}$ At stresses well above σ/μ of 10^{-3} , dislocation velocities become markedly stress-dependent (Alexander *et al.* 1987; Gillard *et al.* 1995), requiring an extension of the Hirth and Lothe (1982) framework. This is readily possible by incorporating the contributions to nonlinear behavior briefly discussed above into the stress dependence of the kink drift expression in eq. (4.74b) and the kink–kink interaction effect in the equilibrium concentration of double kinks along a dislocation line given by eq. (4.78). Following the arguments leading to eqs. (4.75)–(4.83), we obtain a more complete expression for the dislocation

velocity as a function of stress,

$$v_d = 2\sqrt{2}\lambda v_D \exp\left(-\frac{F_k + W_m}{kT}\right) (\sinh s) \left(\exp As^{1/2}\right), \quad (4.84)$$

where

$$s = \left(\frac{\sigma\theta_2}{\mu T}\right) \quad (4.85a)$$

$$\text{and} \quad A = \frac{1}{2}\sqrt{\frac{\theta_1^2}{\theta_2 T}}, \quad \theta_1 \equiv \left(\frac{\mu^2 b_p^3 \lambda^3}{2\pi}\right)^{1/2} \frac{1}{k} \quad \theta_2 \equiv \frac{\mu b_p \lambda w}{k}, \quad (4.85b-d)$$

and where the quantities θ_1 and θ_2 represent some characteristic temperature parameters. Equation (4.84) gives a gradual transition from a linear dependence of the dislocation velocity to an increasingly nonlinear one with increasing stress, which is best represented by the local slope m of the expression in eq. (4.84), giving

$$m = \frac{d \ln v_d}{d \ln s} = s \coth s \left[1 + \left(\frac{A}{2s^{1/2}} \right) \tanh s \right] \quad (4.86)$$

The dependence of m on s is plotted in Fig. 4.37 for the following values: $\mu = 68.1$ GPa; $b_p = 0.433b$, where $b = 0.331$ nm is the Burgers vector of the complete dislocation; $\lambda = 0.33$ nm; $w = 0.385$ nm; and $A = 3.234$ (for $T = 875$ K in the mid range of observations).

In the following section, where models are compared with experiments, it will be seen that this form serves well the nonlinear dependence reported by Alexander *et al.* (1987) and Gillard *et al.* (1995).

4.7.5 Comparison of Models with Experiments

Employing a special high-resolution forbidden-reflection transmission electron microscopy (HRFREM) technique (Kolar *et al.* 1996) capable of resolving individual kinks along 90° and 30° partial dislocations on a 60° mixed dislocation in Si, Alexander *et al.* (1999) were able to make a number of important direct measurements on the mobility of individual kinks. In a preparatory experiment, at 420°C under an applied stress that forced the partials apart, the extended dislocation was first expanded, and then the actual observation with the HRFREM technique was carried out at a higher temperature under no stress, where the stacking fault shrank back to its equilibrium width by systematic formation and motion of double kinks on the partials. Through this technique, the investigators were able to determine the activation energy for kink motion to be 1.24 eV and the individual kink energy to be 0.73 eV on the 90° partial dislocation, giving a total activation energy for

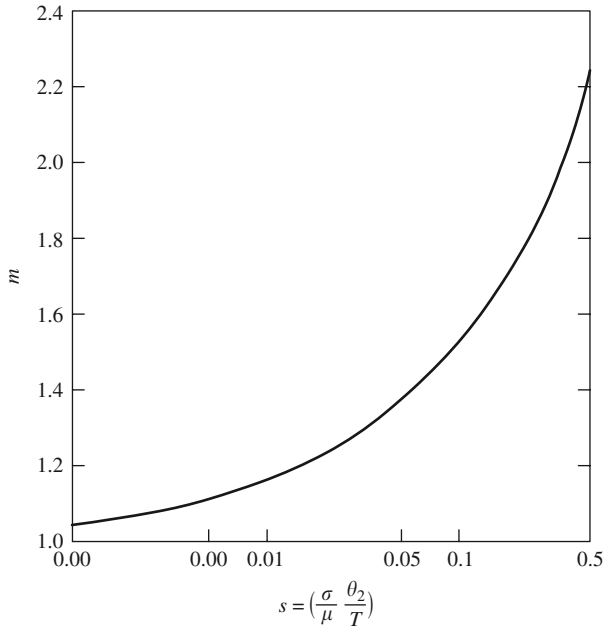


FIG. 4.37. The parameter m representing the stress dependence of the dislocation velocity in Si for stresses above 50 MPa; this dependence is due to important nonlinear effects and is given in eqs. (4.84) and (4.85a–d).

dislocation motion of 1.97 eV. These measurements compare well with the experiments of Nikitenko *et al.* (1987), who, using a special stress-pulsing technique, found for the activation energy of kink mobility and the energy of formation of individual kinks values of 1.6 eV and 0.53 eV, respectively, giving a total activation energy for dislocation motion of 2.13 eV. Other investigators, using other techniques, have reported overall activation energies for dislocation motion ranging up to 2.56 eV (Bulatov *et al.* 2001). These experimental results, while somewhat different, are also in broad agreement with those determined from the computer models of partial-dislocation core structures presented in Section 4.7.3.

For a more direct comparison of models with experiments, we consider the measurements of Imai and Sumino (1983) on screw dislocations with a pair of 30° partials, for which the dislocation velocity is linearly dependent on stress for stresses less than 40 MPa. For example, for a reported activation energy $Q = F_k + W_m = 2.38$ eV at a stress of 10 MPa and temperature of 876 K, a velocity of 1.2×10^{-6} cm/s was reported, with an experimental preexponential factor of $v_{0e} = 5.3 \times 10^7$ cm/s. For these conditions, the theoretical model of eq. (4.83) gives a velocity of 2.0×10^{-10} cm/s for the structural parameters $b_p = 0.143$ nm, $\lambda = 0.331$ nm, and $w = 0.385$ nm, together with $v_D = 1.36 \times 10^{13} \text{ s}^{-1}$, resulting in

a theoretical preexponential factor $v_{0t} = 9.60 \times 10^3$. The theoretical model underpredicts the experimental measurements by a factor of 5.52×10^3 for this case. Comparing other velocities at different temperatures and stresses, Imai and Sumino concluded similarly that the Hirth and Lothe model underpredicts the experimental results by close to a factor of 10^4 . A similar conclusion was reached also for the high-stress results of Alexander *et al.* (1987). This discrepancy is easily resolved. In the model, it is understood that the total activation energy Q under consideration must be the free energy of activation ΔG^* , while in the experimental analysis the investigators determined an activation enthalpy ΔH^* , with the missing activation entropy contribution $T \Delta S^*$ being reflected as a factor $\exp(\Delta S/k)$ into the preexponential factor. An estimate of the missing activation entropy contribution $T \Delta S^*$ can readily be made by use of eq. (2.45b) with the argument that both F_k and W_m will involve elastic interactions and will be subject to the temperature dependence of the elastic properties.

On that basis, the expected activation entropy contribution should be

$$\frac{\Delta S^*}{k} = \left(\frac{\Delta H^*}{kT} \right) \left(-\frac{d \ln \mu}{d \ln T} \right) / \left[1 + \left(-\frac{d \ln \mu}{d \ln T} \right) \right] = \frac{v_{0e}}{v_{0t}} = 8.62. \quad (4.87)$$

With the reported value $\Delta H^*/kT = 31.5$, the expected temperature dependence of the shear modulus needed to explain the discrepancy must be

$$-\frac{d \ln \mu}{d \ln T} = 0.376. \quad (4.88)$$

This, upon integration, gives a temperature-dependent shear modulus $\mu(T)^3$

$$\mu(T) = \mu_R \left(\frac{T_R}{T} \right)^{0.376}, \quad (4.89)$$

where $\mu_R = 68.1$ GPa at room temperature and $T_R = 296$ K, taken as room temperature, is used to evaluate the integration constant. This required temperature dependence predicts a shear modulus at the melting point of $\mu_m = 0.49\mu_R = 33.35$ GPa, all of which is perfectly consistent with the known temperature dependence of the shear modulus of Si (McSkimin and Andreatch 1964). Thus, with the exception of the effects of dopants and trace impurities on dislocation mobility, the model of Hirth and Lothe (1982), with its high-stress extension, furnishes an excellent quantitative framework for the mobility of dislocations in intrinsic Si.

³ The expression given in eq. (4.88) is inapplicable at $T = 0$ since the temperature dependence of μ near $T = 0$ would be subject to restrictions from the third law of thermodynamics.

Finally, the nonlinear low-stress behavior reported by George and Champier (1979) in intrinsic Si has been largely explained by the kinetic Monte Carlo model of Cai *et al.* (2000) referred to in Section 4.7.4. The motion of a dissociated screw dislocation bounded by two 30° partials was simulated there, including all effects of nucleation and motion of all types of kinks, and their mutual interactions with each other on both partials and with the applied stress. Cai *et al.* noted that when the stacking-fault width Λ_s is an integer multiple of the kink height λ , that is, the motion of the partials is “commensurate”, the processes along the two partials must be highly correlated at low stress levels. This results in dislocation velocities lower than in cases where the stacking-fault width is not an integer multiple, where the processes along the partials are not correlated, that is, the motion is “incommensurate”, as assumed in the Hirth and Lothe model presented in the previous section. This close interaction between partials and their correlated motion becomes less strong at higher stresses, and above a stress of 25 MPa the behavior of the commensurate configuration and the incommensurate one give the same result of a linear dependence of dislocation velocity on stress, as shown in Fig. 4.38. Thus, the kinetic Monte Carlo model furnishes a smooth bridging between the George and Champier (1979) measurements and those of Imai and Sumino (1983). This demonstrates the sensitive dependence of the dislocation mobility on the stacking-fault energy governing the width of the extended dislocation.

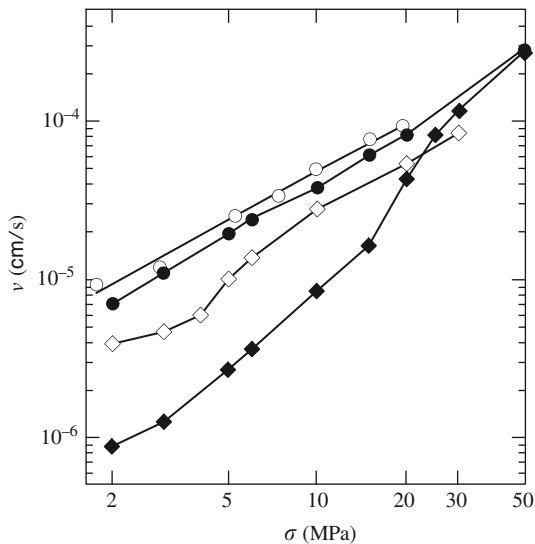


FIG. 4.38. Commensurate and incommensurate advance of extended dislocations relative to the spacing of the Shockley partials on an extended dislocation can explain the departures from linear behavior at low stress levels (after Cai *et al.* 2000).

4.8 The Phonon Drag

When dislocations move in a perfect crystal or in a crystal with a threshold resistance $\hat{\tau}$, under a stress σ considerably in excess of $\hat{\tau}$, the velocity of the dislocation will be governed by its interaction with phonons, that is, the thermal waves in the lattice. Provided the dislocation velocity is well below the velocity of sound c , the phonon interaction will give rise to a drag force that will balance the force applied by the stress in accordance with a relation

$$Bv_d = b\sigma, \quad (4.90)$$

where B is termed the drag coefficient. When the applied stress is near the threshold stress, a more complex interaction will govern the situation, (Frost and Ashby 1971).

Several different specific mechanisms have been considered for dislocation–phonon interactions and have been analyzed in detail. These have been assessed by Kocks *et al.* (1975). The mechanisms that have been considered include (a) thermal shear stress fluctuations that set dislocation segments into “fluttering” vibrations, radiating elastic energy; (b) scattering of phonons from the elastic inhomogeneity of the dislocation core; and (c) the dissipation of energy when volume elements around a moving dislocation are alternately stressed and unstressed. It was suggested by Granato (1968) that these mechanisms are most likely different interpretations or idealizations of the same physical process and one should not attempt to superpose them. In all cases of these interactions with phonons, a stationary dislocation will experience no net effect, since it will be subjected to phonon fluxes isotropically from all sides, while a moving dislocation will experience larger fluxes against its direction of motion and less from behind, resulting in a drag linear in velocity as given by eq. (4.90).

Provided that the net velocity resulting from the interactions of the dislocation with phonons is small in comparison with the sound velocity c , the drag coefficient B will be directly proportional to the thermal energy density per atom in the solid and is given by

$$B = \frac{kT}{\Omega v_D}, \quad (4.91)$$

where Ω is the atomic volume and v_D is the Debye frequency.

The best experimental determinations of the drag coefficient have been performed by measuring the displacements of individual dislocations in nearly perfect crystals due to well-controlled stress pulses at different temperatures. Such measurements have shown that, like the specific heat capacity, which reaches a constant value in units of thermal energy density per atomic volume kT/Ω above the Debye

temperature, so does the drag coefficient. Below the Debye temperature, there is a similar characteristic decrease.

In the approach to absolute zero temperature, when the phonon drag decreases sharply, it is replaced by a complementary process of electron drag in metals, effective in the low cryogenic region, which we shall not consider further.

For further discussions of the drag coefficient and an overview of the mechanisms, the reader is referred to Kocks *et al.* (1975) and to Hirth and Lothe (1982).

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References for Further Study in Depth on the Lattice Resistance

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SOLID-SOLUTION STRENGTHENING

5.1 Overview

Incorporation of a second constituent into a pure metal offers a very flexible means of strengthening. As was briefly discussed in Chapter 3 and as will be described further in Section 5.2, the second-constituent atoms can aggregate in a variety of forms of increasing size, from randomly dispersed individual solute atoms, through solute atom clusters, to second-phase precipitate particles, which may be shearable or nonshearable by dislocations. These different forms of dispersion result in interactions with dislocations, having different strengths and ranges.

In this chapter we consider the interactions of individual solute atoms with dislocations. First we consider the effects of moderate concentrations of individual substitutional solute atoms in FCC and HCP metals with a normally negligible lattice resistance in the low-temperature range, where the solute atoms are immobile in the lattice. We then consider the much more complex interactions of solute atoms with dislocations in BCC metals, again only in the low-temperature range, where, however, a substantial lattice resistance would now have been present in the pure reference metal. These two very different but complementary cases provide a detailed insight into the mechanisms governing the temperature and strain rate dependence of the plastic resistance, not only for these specific cases but also for some other cases which we shall not discuss in the chapter. The latter cases include interactions of dislocations with interstitial solute atoms, and cases at moderately elevated temperature where the solute becomes mobile in the lattice on the same timescale as the motion of dislocations, resulting in instabilities in the form of jerky glide phenomena. We shall also not consider the complex interactions of strain hardening with solute strengthening. Some of these effects will be considered in Chapter 8.

5.2 Forms of Interaction of Solute Atoms with Dislocations in FCC Metals

5.2.1 Overview

Several forms of interaction of solute atoms with both edge and screw dislocations have been considered in detail by a number of investigators (see Neuhauser and

Schwink 1993 for a review). Of these, the most important are the ones that result in elastic interactions with dislocations that cause either a local attraction or a repulsion. They fall broadly into two categories: (a) the size misfit interaction, where a solute atom, substitutionally replacing a host atom, is either somewhat larger or somewhat smaller than the atom it replaces; and (b) the modulus misfit interaction, where the solute atom creates a local environment of different elastic properties. While other variations of these two interactions exist and are important, in this chapter we consider in some detail only these two main forms. Other more complex cases, such as interstitial solute atoms, and divalent ions substitutionally replacing monovalent ions in alkali halides, which produce nonspherically symmetrical, tetragonal distortions will be excluded. Still other so-called chemical interactions, where solute atoms segregate to stacking faults of extended dislocations, will also be left out. For these and some other less important cases, the reader is referred to the comprehensive reviews of Neuhauser and Schwink (1993) and Haasen (1979, 1996).

The interaction of solute atoms with dislocations in BCC metals in the temperature range where the lattice resistance is present is delayed to Section 5.5.

5.2.2 The Size Misfit Interaction

The Edge Dislocation Consider, as shown in Fig. 5.1(a), a straight positive edge dislocation parallel to the x axis, interacting with a solute atom at a position y, z (or r, θ) possessing a positive size misfit, that is, it has an atomic radius r larger by Δr than the radius r_0 of the host atom it replaces substitutionally. We refer to this size misfit by the parameter $\epsilon_s = \Delta r/r_0$. In this definition, Δr is considered to be the effective rigid-equivalent increase in r , combining all higher-order considerations of the elastic volumetric compliance of the solute atom relative to the compliance of the cavity in the host that it fills.

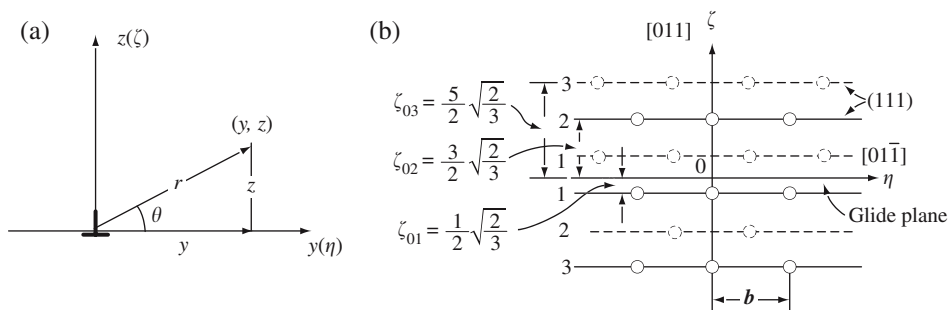


FIG. 5.1. (a) Coordinate axes for a dislocation line; (b) spacing of $\{111\}$ planes parallel to a glide plane in the FCC lattice.

For the edge dislocation depicted in Fig. 5.1(a), the interaction energy I_{es} of the solute with the dislocation arises simply through the mean normal stress σ_m set up by the elastic field of the dislocation at the site y, z of the solute atom, opposing (or aiding) the material misfit of the solute. The interaction energy, as stated first by Cottrell (1948), is then

$$I_{es} = 3\sigma_m\epsilon_s\Omega = -\frac{(1+\nu)\mu b\epsilon_s\Omega z}{\pi(1-\nu)r^2}, \quad (5.1)$$

where, in the “distant-field” solution, we have taken the mean normal stress $\sigma_m (= (2/3)(1+\nu)(\sigma_{rr} + \sigma_{\theta\theta}))$ of the dislocation to be uniform over the atomic site, interacting with the net local dilatation $3\epsilon_s\Omega$ of the solute linearly, where Ω is the atomic volume of the host lattice. The elastic interaction of a solute atom with a dislocation is of such short range that it will be sufficient to consider only atoms nearest to the glide plane of the dislocation at a distance, z_0 , which we fix later. Introducing normalized coordinates $\eta = y/b$ and $\zeta = z/b$, and considering ζ to be fixed at $\zeta_0 = z_0/b$, the normalized size misfit interaction energy of a positive edge dislocation with a solute atom having a positive misfit becomes

$$I_{es}(\eta) = -\frac{(1+\nu)\mu b^3\epsilon_s(\Omega/b^3)\zeta_0}{\pi(1-\nu)}f_{es}(\eta), \quad (5.2a)$$

where

$$f_{es}(\eta) = \frac{1}{\eta^2 + \zeta_0^2}. \quad (5.2b)$$

The interaction force $F_{\eta es}$ of the solute with an edge dislocation moving in the $z = 0$ plane is

$$F_{\eta es} = -\frac{1}{b}\frac{\partial I_{es}}{\partial \eta} = -\frac{2}{\pi}\left(\frac{1+\nu}{1-\nu}\right)\mu b^2\epsilon_s\left(\Omega/b^3\right)\zeta_0 f'_{es}(\eta), \quad (5.3a)$$

where

$$f'_{es}(\eta) = \frac{\eta}{(\eta^2 + \zeta_0^2)^2}. \quad (5.3b)$$

This indicates that a solute with a positive misfit at $\eta > 0$ and $\zeta > 0$ repels a positive edge dislocation.

The Screw Dislocation The linear elastic field around a screw dislocation does not possess a mean normal stress. However, as assessed by Stehle and Seeger (1956) and considered as a source of interaction by Fleischer (1963), the nonlinear elastic behavior of the material at the core of a screw dislocation sets up a short-range

dilatation field which produces a corresponding local short-range mean normal stress σ_m , which, by linear elastic considerations, is assessed to be

$$\sigma_m = \frac{2(1+\nu)\mu}{3(1-2\nu)} \frac{B}{4\pi^2} \left(\frac{b}{r}\right)^2, \quad (5.4)$$

where B is estimated to be between 0.3 and 1.0. A consideration similar to that for the edge dislocation then gives the size misfit interaction energy I_{ss} of a solute with a screw dislocation as

$$I_{ss} = 3\sigma_m \epsilon_s \Omega = \frac{(1+\nu)\mu b^2}{(1-2\nu)2\pi^2} B \epsilon_s \frac{\Omega}{r^2}, \quad (5.5)$$

or, in normalized coordinates,

$$I_{ss} = \frac{(1+\nu)\mu b^3}{(1-2\nu)2\pi^2} \epsilon_s \left(\frac{\Omega}{b^3}\right) B f_{ss}(\eta), \quad (5.6a)$$

where

$$f_{ss}(\eta) = \frac{1}{\eta^2 + \zeta_0^2}. \quad (5.6b)$$

This gives a corresponding interaction force between the solute and the screw dislocation

$$F_{\eta ss} = -\frac{1}{b} \frac{\partial I_{ss}}{\partial \eta} = \frac{(1+\nu)}{(1-2\nu)} \frac{\mu b^2}{\pi^2} \epsilon_s \left(\frac{\Omega}{b^3}\right) B f'_{ss}(\eta), \quad (5.7a)$$

where

$$f'_{ss}(\eta) = \frac{\eta}{(\eta^2 + \zeta_0^2)^2}. \quad (5.7b)$$

Thus, apart from the different scale factor, the form of the size misfit interaction of a solute atom with a screw dislocation is identical to that for an edge dislocation.

5.2.3 The Modulus Misfit Interaction

The Edge Dislocation Solute atoms introduced substitutionally into a host lattice affect both the overall bulk modulus and the overall shear modulus by altering the local atomic interactions. The change is generally larger in the shear modulus and in most instances results in a global reduction of the shear modulus of the alloy. While this will result in a corresponding change in the dislocation line energy by a global effect, a more important effect will be a local change of the line energy due to the change in the elastic strain energy density in a volume Ω at a point y, z

occupied by the solute atom. Thus, considering the arrangement in Fig. 5.1(a), a solute atom producing an effective local change of the shear modulus $\Delta\mu$ over the volume Ω will result in an interaction energy with an edge dislocation on the $z = 0$ plane of

$$I_{e\mu} = \left(\frac{\sigma_m^2}{2K} + \frac{\sigma_{r\theta}^2}{2\mu} \right) \epsilon_\mu \Omega, \quad (5.8)$$

where K is the bulk modulus and $\epsilon_\mu = \Delta\mu/\mu$ is the modulus misfit parameter. The two terms in the parentheses represent the volumetric and shear elastic energy densities, respectively. When given as a function of the site coordinates r and θ of the solute atom relative to the center of the positive edge dislocation, they are given by

$$\frac{\sigma_m^2}{2K} = \frac{(1+\nu)(1-2\nu)}{48\pi^2(1-\nu)^2} \frac{\mu b^2}{r^2} \sin^2 \theta, \quad (5.9a)$$

$$\frac{\sigma_{r\theta}^2}{2\mu} = \frac{\mu b^2}{8\pi^2(1-\nu)^2 r^2} \cos^2 \theta. \quad (5.9b)$$

Examination of the two contributions to the local elastic energy density given by eqs. (5.9a) and (5.9b) shows that, apart from the angular factors, the energy density contributed by the volumetric term is only about 8% of that of the second term. Therefore, we shall consider only the second contribution, coming from the shear field. Then, in normalized distance coordinates η and $\zeta = \zeta_0$,

$$I_{e\mu} = \frac{\mu b^3 \epsilon_\mu (\Omega/b^3)}{8\pi^2(1-\nu)^2} f_{e\mu}(\eta), \quad (5.10a)$$

where

$$f_{e\mu}(\eta) = \frac{\eta^2}{(\eta^2 + \zeta_0^2)^2}. \quad (5.10b)$$

The associated interaction force with the dislocation on the glide plane $z = 0$ is

$$F_{\eta e\mu} = \frac{\mu b^2 \epsilon_\mu (\Omega/b^3)}{4\pi^2(1-\nu)^2} f'_{e\mu}(\eta), \quad (5.11a)$$

where

$$f'_{e\mu}(\eta) = \frac{\eta(\eta^2 - \zeta_0^2)}{(\eta^2 + \zeta_0^2)^3}. \quad (5.11b)$$

The Screw Dislocation Ignoring second-order effects, since a screw dislocation possesses no dilatational strain field, the modulus misfit interaction becomes simply

$$I_{s\mu} = \frac{\sigma_{\theta z}^2}{2\mu} \epsilon_\mu \Omega, \quad (5.12)$$

which gives, in normalized distance coordinates,

$$I_{s\mu}(\eta) = \frac{\mu b^3 \epsilon_\mu (\Omega/b^3)}{8\pi^2} f_{s\mu}(\eta), \quad (5.13a)$$

where

$$f_{s\mu}(\eta) = \frac{1}{(\eta^2 + \zeta_0^2)}. \quad (5.13b)$$

The interaction force on the glide plane is, in turn,

$$F_{\eta s\mu} = \frac{\mu b^2 \epsilon_\mu (\Omega/b^3)}{4\pi^2} f'_{s\mu}(\eta), \quad (5.14a)$$

where

$$f'_{s\mu}(\eta) = \frac{\eta}{(\eta^2 + \zeta_0^2)^2}. \quad (5.14b)$$

5.2.4 Combined Size and Modulus Misfit Interactions

Since solutes in general will possess both a size and a modulus misfit, the total interaction energy and interaction force must combine both effects.

For the *edge dislocation*, the combined interaction gives

$$I_e = I_{es} + I_{e\mu}. \quad (5.15)$$

Because of functional differences between the two contributions of the size and modulus misfit effects, we deal with these separately at first, in the form

$$I_{es} = \frac{\mu b^3 (\Omega/b^3)}{8\pi^2 (1-\nu)^2} \epsilon_s f_{es}(\eta), \quad I_{e\mu} = \frac{\mu b^3 (\Omega/b^3)}{8\pi^2 (1-\nu)^2} \epsilon_\mu f_{e\mu}(\eta) \quad (5.16a,b)$$

where

$$f_{es}(\eta) = \frac{-8\pi(1-\nu^2)\zeta_0}{(\eta^2 + \zeta_0^2)}, \quad f_{e\mu}(\eta) = \frac{\eta^2}{(\eta^2 + \zeta_0^2)^2}. \quad (5.17a,b)$$

Similarly, we consider the interaction forces separately, in the form

$$F_{\eta es}(\eta) = \frac{\mu b^2 (\Omega/b^3)}{4\pi^2 (1-\nu)^2} \epsilon_s f'_{es}(\eta), \quad F_{\eta e\mu}(\eta) = \frac{\mu b^2 (\Omega/b^3)}{4\pi^2 (1-\nu)^2} \epsilon_\mu f'_{e\mu}(\eta), \quad (5.18a,b)$$

where

$$f'_{es}(\eta) = \frac{(-8\pi(1-\nu^2)\zeta_0)\eta}{(\eta^2 + \zeta_0^2)^2}, \quad f'_{e\mu}(\eta) = \frac{\eta(\eta^2 - \zeta_0^2)}{(\eta^2 + \zeta_0^2)}. \quad (5.19a,b)$$

We delay combining these two contributions to a later point, where we introduce specific numerical evaluations that will then permit clarification of the levels of the individual contributions to the total interaction.

For the *screw dislocation* the combined interactions give

$$I_s = I_{ss} + I_{s\mu} = \frac{\mu b^3 (\Omega/b^3)}{8\pi^2(1-\nu)^2} \epsilon'_s f_s(\eta), \quad (5.20)$$

where

$$f_s(\eta) = \frac{1}{(\eta^2 + \zeta_0^2)} \quad (5.21)$$

and

$$\epsilon'_s = \frac{4(1+\nu)(1-\nu)^2 B}{(1-2\nu)} \epsilon_s + (1-\nu)^2 \epsilon_\mu, \quad (5.22)$$

which can be considered as the effective total misfit parameter for screw dislocations.

The net interaction force for screw dislocations is then

$$F_{\eta s}(\eta) = \frac{\mu b^2 (\Omega/b^3)}{4\pi(1-\nu)^2} \epsilon'_s f'_s(\eta), \quad (5.23)$$

where

$$f'_s(\eta) = \frac{\eta}{(\eta^2 + \zeta_0^2)^2}. \quad (5.24)$$

In presenting the net interactions of solutes with edge and screw dislocations, we have chosen common scale factors for all interaction energies and interaction forces and absorbed the different numerical factors into the normalized functions $f(\eta)$ and $f'(\eta)$. This will make comparisons of the individual contributions to the total interaction easier. To permit such comparisons and to gain better insight, it is necessary to evaluate the range-dependent factors specifically.

As shown in Fig. 5.1(b), the $\{111\}$ planes in FCC metals are separated by a distance $h = (\sqrt{2/3})b$, where the atoms of plane (1) nearest to the glide plane are only a distance $z = (1/\sqrt{6})b$ away. Since the core radius of dislocations in FCC metals can be taken as $r \approx b$, the interaction of a solute in these nearest planes would fall inside the core, in which the linear elastic theory presented above would

be inapplicable, and the interaction should be substantially weaker by virtue of the much lower stresses near the center of the dislocation (see also Olmsted *et al.* 2006). Therefore, we take for $\zeta_0 = \sqrt{3}/2$ the distance from the glide plane to the next-nearest atom planes (2), which are now just outside the core radius. For this choice, we plot the range-dependent functions given in eqs. (5.17a,b), (5.19a,b), (5.21), and (5.24) in Fig. 5.2.

Examination of Fig. 5.2 shows that for edge dislocations, even when we consider that the modulus misfit parameter ϵ_μ is, on average, a factor of 10 larger than the size misfit parameter ϵ_s , the modulus misfit interactions are at a level of only roughly 10% of the size misfit interaction. Thus, taking account of primarily the size misfit interaction of the edge dislocation in Fig. 5.2(a) and the total interaction of a screw dislocation with the size and modulus misfit effects in Fig. 5.2(c), we note that the shapes of the curves have an identical range as a function of η . The peak resistive forces occur at $\eta = 0.7$ and fall to half of their peak value at $\eta = 1.7$, which we take as a parameter characterizing the range of the solute interaction w/b in FCC metals.

Considering the peak values of the resistive forces, we obtain the combined total normalized resistive forces $\hat{K}/2\mathcal{E}$ for the edge and screw dislocations,

$$\begin{aligned} \left(\frac{\hat{K}}{2\mathcal{E}} \right)_e &= \frac{(\Omega/b^3)}{4\pi^2(1-\nu)^2} (\hat{F}_{es}\epsilon_s + \hat{F}_{e\mu}\epsilon_\mu) \\ &= -(0.181\epsilon_s + 4.31 \times 10^{-3}\epsilon_\mu), \end{aligned} \quad (5.25a)$$

$$\begin{aligned} \left(\frac{\hat{K}}{2\mathcal{E}} \right)_s &= \frac{(\Omega/b^3)}{4\pi^2(1-\nu)^2} F_s(6.37B\epsilon_s + 0.49\epsilon_\mu) \\ &= (0.041B\epsilon_s + 0.0867\epsilon_\mu). \end{aligned} \quad (5.25b)$$

The sign of the maximum resistive force in eq. (5.25a) indicates that for both positive size misfit and modulus misfit parameters of the generally encountered magnitudes, an edge dislocation will feel a net repulsive force. When considering the overall effect of the interaction of either type of dislocation with solute atoms, we shall take the absolute values of these total interaction forces, as we discuss below. Such total interaction effects will then be used in comparing the solute-strengthening models with experimental results.

In Figs. 5.2(a) and (c), we show also the distribution of the size misfit interaction force $F_{es}(\eta)$ and total screw interaction force $F_s(\eta)$ for solutes in an atom plane (3) away from the glide plane by a distance $\zeta_0 = (5/\sqrt{6})$. Here the levels of the peak interaction forces are only 21.5% of those on the atom planes (2). Moreover, since the interaction forces are odd functions of η , these forces will take either plus or minus signs depending on the position of the solute relative to the center of the dislocation. Thus, these more distant interactions, appearing randomly, will make

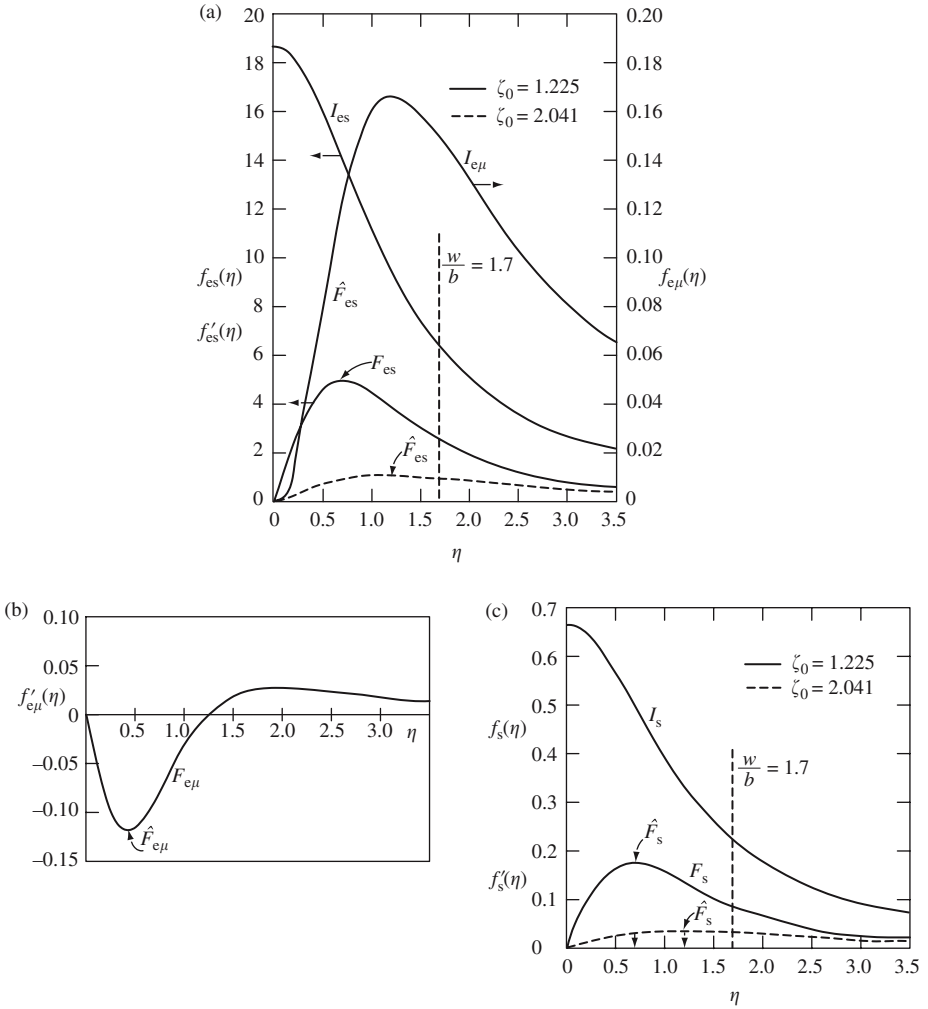


FIG. 5.2. (a) Interaction energies I_{es} and $I_{e\mu}$ of an edge dislocation in an FCC metal with solute atoms having a size (s) and modulus (μ) misfit, as a function of the normalized coordinate η of separation between the dislocation and the solute atom, and interaction force F_{es} between a solute and an edge dislocation, all for solute atoms in the atom planes (2) shown in Fig. 5.1(b). The dashed curve gives the interaction force F_{es} with a solute with a size misfit in the atom planes (3). (b) Interaction force between a solute with a modulus misfit and an edge dislocation. (c) Interaction energy I_s and interaction force F_s between a screw dislocation and a solute with a combined size and modulus misfit in the atom planes (2). The dashed curve gives again the total interaction force between a screw dislocation and a solute with a combined size and modulus misfit located in the atom planes (3).

only very small net contributions to the main interaction of the dislocation with solute atoms on the atom plane (2) and will be neglected.

We note that while there are individual differences in the misfit parameters ϵ_s and ϵ_μ for different alloy additions, in general the levels of ϵ_μ are about a factor of 10 larger than those of ϵ_s . With this recognition, we conclude that for edge dislocations the size misfit interaction will usually dominate over the modulus misfit interaction, while the opposite will hold for screw dislocations. Moreover, evaluation of eqs. (5.25a) and (5.25b) for specific cases shows that in most cases $(\hat{K}/2\mathcal{E})_s$ is a factor of 2 or more larger than $(\hat{K}/2\mathcal{E})_e$, indicating that the mobilities of screws should be less than those of edges. However, since TEM observations of dislocation microstructures in FCC alloys do not show an overwhelming anisotropy in the line direction, this suggests that the line connectivity of edges and screws provides a certain averaging of the individual mobilities during plastic flow. Therefore, we combine the misfit parameters into a simple arithmetic mean value of

$$\left(\frac{\hat{K}}{2\mathcal{E}}\right) = \frac{1}{2} \left(\left(\frac{\hat{K}}{2\mathcal{E}}\right)_e + \left(\frac{\hat{K}}{2\mathcal{E}}\right)_s \right). \quad (5.26)$$

Comparison with experimental results demonstrates that this average gives considerably better correlation with experiments. Furthermore, in evaluating the individual factors $(\hat{K}/2\mathcal{E})_e$ and $(\hat{K}/2\mathcal{E})_s$, we have picked the absolute values of ϵ_s and ϵ_μ , disregarding their sign, again on the argument that the specific functions $F_{es}(\eta)$ and $F_s(\eta)$ are odd functions of η and that in nearly every case the contributions of either ϵ_s or ϵ_μ to the total peak interaction force will dominate. Other authors such as Fleischer (1963) and Haasen (1979) have introduced arbitrary multipliers to the ϵ_s and ϵ_μ terms to obtain better correlation with experiments or have even resorted additionally to geometrical means, which automatically pick only positive contributions (Labusch 1972; see also Haasen 1996). In the arguments above, we have refrained from adopting such arbitrary multipliers.

5.3 Forms of Sampling of the Solute Field by a Dislocation in an FCC Metal

As was considered briefly earlier, there are two rather different limiting forms of sampling, either discrete or collective, of a field of pointlike obstacles consisting of solute atoms or larger nanosized particles or precipitates by a moving dislocation. Here our primary perspective will be substitutional solute atoms located in the two nearest planes of atoms outside the core, on both sides of the plane of motion of the dislocation line. As was demonstrated in Section 5.2, solute atoms in atom planes just outside the core radius of a dislocation, that is, in the planes (2) in Fig. 5.1(b), interact most strongly with a dislocation, while those in the atom planes (3) and

beyond interact too weakly to be of much importance. The solute atoms in the nearest (1) planes are in a “dead” zone. Thus, ignoring the solute atoms in the planes (1), the mean spacing l of solute atoms in the planes (2), projected onto the glide plane of the dislocation, is related to the atomic concentration c of the solute in the alloy by

$$c = \frac{\Omega}{2(\sqrt{2}/3)bl^2}, \quad (5.27)$$

where the denominator represents the volume in the combined thickness of the two atom planes (2) allocated to a solute atom. Noting that in the FCC lattice $\Omega = (\sqrt{2}/2)b^3$, we have

$$\left(\frac{l}{b}\right)^2 = \frac{\sqrt{3}}{4c}. \quad (5.28)$$

The mean spacing l_1 along the dislocation line of the vertical projections of the sites of solutes on the glide plane, given by a simple search process, is given by

$$2wl_1 = l^2, \quad (5.29)$$

which gives

$$l_1 = \frac{\sqrt{3}b^2}{8wc}, \quad (5.30)$$

where w is the effective range of interaction of a solute on the glide plane, discussed in Section 5.2 and shown in Figs. 5.2(a) and (c).

Consider now a reasonable arrangement of solute sites A, A', and B on the glide plane $z = 0$ as shown in Fig. 5.3. Somewhat arbitrarily but consistent with the

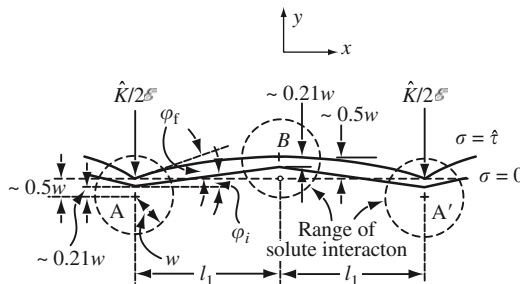


FIG. 5.3. Sketch of a flexible dislocation interacting with three centers of solute atoms in atom planes (2) showing the initial zigzag shape of the dislocation under no applied stress and the critical configuration under a threshold stress permitting the dislocation to go through all three interaction fields of the solutes.

sampling condition of eq. (5.29), the straight dislocation cuts across the interaction fields of the solutes at points roughly halfway between the centers of their sites and their borders at w . Considering the shapes of the interaction force curves in Fig. 5.2, this would make the straight dislocation pass close to the points of the maxima of the interaction forces, acting on the dislocation with alternating signs. In response, under no applied stress, the straight dislocation would move in directions such as to lower these forces, even at the cost of some increase in line length. Clearly, the dislocation would reach an equilibrium shape that reduced the interaction forces on it. For the purpose of our perspective, we assume that the resulting shape would place segments of the line somewhere between the points of the force maxima and the centers of the sites, as shown in Fig. 5.3. Then, under no applied external stress, on the basis of examination of the force profiles $f'(\eta)$, we conclude that the dislocation line would be subjected to a series of alternating positive and negative point forces of interaction of roughly $(3/4)\hat{K}$, where we have introduced the terminology of referring to the resistive forces F of obstacles as K and their peak values as $\hat{K}$. Thus, under this initial condition of no applied stress, the dislocation line will have a zigzag shape as it accommodates itself to the random forces of interaction of the solute atoms. The dislocation segments then make angles $\phi_i \cong 0.21w/(l_1/2)$, on average, relative to the reference straight form of the line. According to the conditions assumed, the components of the line tension forces $2\mathcal{E}\phi_i$ must be equilibrated at the three contact points by the reaction forces $(3/4)\hat{K}$, or, in initial equilibrium under no applied stress,

$$\phi_i = \frac{3}{4} \left(\frac{\hat{K}}{2\mathcal{E}} \right) \cong 0.42 \left(\frac{w}{l_1} \right). \quad (5.31)$$

When a shear stress is applied that displaces the dislocation line in the positive y direction, the resistive forces at points A and A' will increase, while that at point B will decrease. Following the scenario developed by Kocks *et al.* (1975), we consider a critical state where, under the applied stress, the resistive forces on A and A' just reach the peak resistance $\hat{K}$ of the interaction while that at point B drops to zero. In this critical condition, the forces of the applied stress are then resisted entirely by contacts that oppose the motion, and all interaction forces that have previously aided the motion have been relieved. Thus, the dislocation can move ahead by overcoming only a set of discrete opposing forces, even though in the initial unstressed state it was subjected to both opposing and aiding interaction forces relative to the chosen direction of advance. This critical flow state of the dislocation segment is sketched in Fig. 5.3 by a circular arc connecting points A and A'; the line at point B experiences no force and the line tension angle there is zero. In this critical flow state, the dislocation line segments at the contact points

A and A' make angles

$$\phi_f = \frac{w}{l_1} \quad (5.32)$$

with the x axis, resulting in a change of angle

$$\Delta\phi = \phi_f - \phi_i \cong 0.58 \frac{w}{l_1}, \quad (5.33)$$

which evokes a change in resistive force at these points of

$$\Delta K = 2\mathcal{E} \Delta\phi \cong \frac{1}{4} \hat{K}. \quad (5.34)$$

Upon substitution of eqs. (5.30) and (5.33), eq. (5.34) defines a parameter γ (Nabarro 1985),

$$\gamma \cong (0.58) \frac{32}{\sqrt{3}} \left(\frac{w}{b}\right)^2 \frac{c}{(\hat{K}/2\mathcal{E})}. \quad (5.35)$$

If in the critical flow state, $\gamma = 1.0$, the condition is reached where the dislocation advances while probing only opposing interaction forces. Alternatively, and more insightfully, the condition $\gamma = 1.0$ defines a critical solute concentration

$$c_c = 0.93 \left(\frac{b}{w}\right)^2 \left(\frac{\hat{K}}{2\mathcal{E}}\right). \quad (5.36)$$

When $\gamma < 1.0$ and $c < c_c$ where the dislocation encounters in the flow state only discrete opposing obstacles, the situation is the same as what occurs when the second constituent consists of larger precipitate particles that are sampled as discrete opposing obstacles. This latter form of sampling of pointlike obstacles was referred to as Friedel statistics (Friedel 1964) in Chapter 3 and pertains to precipitation strengthening. We shall discuss precipitation strengthening in detail in Chapter 6. In the present chapter we consider only the other limiting form of obstacle sampling, which corresponds to the conditions $\gamma > 1.0$ and $c > c_c$, where, in the flow state, the dislocation is subjected to an array of pointlike obstacles where some of them oppose its motion while others aid it, and the dislocation advances in a complex manner with a wiggly shape. In Chapter 3, this form of statistical sampling of obstacles was referred as the Mott or Labusch statistics (Mott 1952; Labusch 1972).

According to eq. (5.36), the critical concentration c_c above which solid-solution strengthening applies is roughly around 10^{-3} for the solute interaction range $w/b = 1.7$ given in Fig. 5.2(a) and a typical value of the normalized peak resistance $\hat{K}/2\mathcal{E} \approx 0.015$, which we shall establish in Section 5.4.3 when models are compared with experimental results.

5.4 The Solid-solution Resistance of FCC Alloys

5.4.1 The Athermal Resistance

We consider first the plastic resistance of a moderately concentrated alloy for which $c > c_c$ at a very low temperature, where the latter plays no role in reducing the resistance. Under no applied stress, an otherwise straight dislocation in a randomly dispersed field of solute would be subjected to interaction forces from the nearest effective solute atoms located in the two neighboring atom planes (2) of the glide plane shown in Fig. 5.1(b). Under the random attractive or repulsive forces of interaction of the solute, the dislocation will accommodate itself by assuming the randomly zigzagging shape discussed in Section 5.3, which is sketched in Fig. 5.4(a). Under these random interaction forces, estimated in Section 5.3, to be roughly $0.75(\hat{K}/2\mathcal{E})$, the neighboring segments l_1 of the dislocation between solutes suffer random relative rotations of $2\phi_i$ given by eq. (5.31), as depicted in Fig. 5.4(b). In this initial configuration, no net force is applied to the dislocation over any length larger than several times l_1 . When a stress is applied that tends to displace the dislocation, the interaction forces along the dislocation line will change, with some increasing and others decreasing. Following the pioneering considerations of Mott and Nabarro (1948) (see also Kocks *et al.* 1975 and Nabarro 1985), we envision that coherent lengths Λ of the dislocation contacting n solute atom interactions advance and begin to sample more retarding than aiding obstacle forces, until the characteristic length Λ acquires an average forward curvature and advances at its center by a characteristic obstacle interaction distance $2w$. Upon reaching this state of flow, the dislocation samples a net number $\sqrt{n}$ of retarding forces at their peak levels $\hat{K}$ in a random-error sense.

This critical flow configuration is repeated in a hunting manner in the form of randomly alternating wiggles along the entire length of the dislocation as it moves ahead. Figure 5.4(c) depicts this critical flow configuration, where the coherent length Λ with an average forward curvature $1/R$ subtends along its length an average bow-out half-angle of

$$\Phi = \phi_i \sqrt{n}, \quad (5.37)$$

obtained as a random-error sum of the average plus and minus half-angles ϕ_i of the relative segment rotations among the n solute contact interactions. From the geometry of the slight bow-out,

$$R\Phi^2 = 4w, \quad 2\Phi R = \Lambda = nl_1, \quad (5.38a,b)$$

Eqs. (5.37) (5.38a,b) give

$$n^{3/2} = \frac{8w}{l_1\phi_i}. \quad (5.39)$$

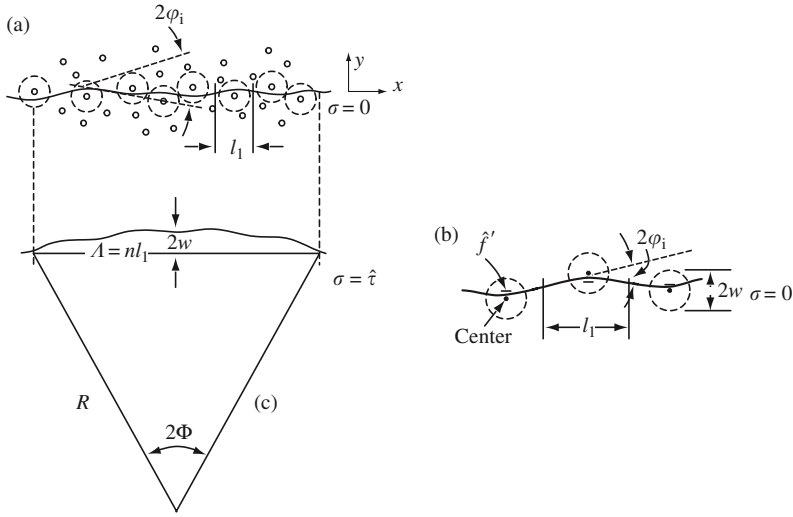


FIG. 5.4. (a) and (b) A dislocation sampling a solute atom field under no applied stress; (c) an independent length Λ of dislocation under a threshold stress $\hat{\tau}$ advancing at its center by a characteristic distance $2w$.

Using eqs. (5.30) and (5.31) in eq. (5.39),

$$n = 13.44 \left(\left(\frac{w}{b} \right)^2 \frac{c}{\hat{K}/2\mathcal{E}} \right)^{2/3} \quad (5.40)$$

and

$$\Lambda = nl_1 = (2.91)b \left(\left(\frac{w}{b} \right) \frac{1}{c} \frac{1}{(\hat{K}/2\mathcal{E})^2} \right)^{1/3} \quad (5.41)$$

is obtained.

Finally, on the assumption that in this critical flow configuration the force of the applied shear stress σ along the coherent length Λ is resisted by only a random-error sum of peak retarding forces $\hat{K}$, that is,

$$\sigma b \Lambda = \sqrt{n\hat{K}} \quad (5.42)$$

the overall threshold plastic shear resistance is obtained as

$$\sigma = \hat{\tau} = 1.26 \mu \left(\frac{w}{b} \right)^{1/3} c^{2/3} \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{4/3}. \quad (5.43)$$

We identify this as the terminal plastic shear resistance of the field of solute atoms and, as it is a threshold material property, refer to it by the symbol $\hat{\tau}$.

5.4.2 Thermally Assisted Advance of a Dislocation in a Field of Solute Atoms in an FCC Metal

To employ the procedures of thermal-activation analysis developed in Chapter 2, we consider a reasonable form for the dependence of the glide resistance $\tau (< \hat{\tau})$ on the forward displacement y , averaged over the coherent length Λ , to be a sinusoidal function, namely

$$\tau = \hat{\tau} \sin \frac{\pi y}{2\bar{y}}, \quad (5.44)$$

where $\bar{y} = (4/3)w$ is where the averaged forward displacement y reaches the peak resistance $\hat{\tau}$.

Using the thermal-activation analysis described in Chapter 2 (Kocks *et al.* 1975), the free-energy barrier ΔG^* for this form of advance of the coherent length Λ , acting independently from the rest of the line, under an applied shear stress $\sigma < \hat{\tau}$ is

$$\Delta G^*(\sigma, \hat{\tau}) = \int_{y_1(\sigma)}^{y_2(\sigma)} (\tau \Lambda b) dy - (\sigma \Lambda b)(y_2 - y_1), \quad (5.45)$$

where $\tau \Lambda b$ represents the resistive force on the segment of length Λ , and $y_1(\sigma)$ and $y_2(\sigma)$ represent the coordinates of the stable and unstable (saddle-point) positions of equilibrium of the bowed-out segment under the applied shear stress σ . Integration gives the dependence of ΔG^* on the stress as

$$\Delta G^* = \Delta G_0^* \left[(1 - \beta^2)^{1/2} - \frac{\pi}{2} \beta \left(1 - \left(\frac{2}{\pi} \right) \sin^{-1} \beta \right) \right], \quad (5.46a)$$

where

$$\Delta G_0^* = \frac{16}{3\pi} (b \Lambda w) \hat{\tau}, \quad \text{and} \quad \beta = \frac{\sigma}{\hat{\tau}}. \quad (5.46b,c)$$

The associated (shear) activation volume Δv^* is

$$\Delta v^* = - \left(\frac{\partial \Delta G^*}{\partial \sigma} \right)_T = \frac{8}{3} (b \Lambda w) \left(1 - \frac{2}{\pi} \sin^{-1} \beta \right). \quad (5.47)$$

The stress-dependent functions in eqs. (5.46a) and (5.47) can be replaced quite accurately with simpler forms to give

$$\Delta G^* = \Delta G_0^* g(\beta), \quad (5.48a)$$

$$\Delta v^* = \Delta v_0^* (\beta), \quad (5.48b)$$

where

$$\Delta G_0^* = (5.55)(\mu b^3) \left(\frac{w}{b}\right)^{3/2} \left(\frac{\hat{\tau}}{\mu}\right)^{1/2}, \quad g(\beta) = (1 - \beta)^{3/2}, \quad (5.49a,b)$$

and

$$\Delta v_0^* = (8.71)b^3 \left(\frac{w}{b}\right)^{3/2} \left(\frac{\mu}{\hat{\tau}}\right)^{1/2}, \quad v(\beta) = (1 - \beta)^{1/2}; \quad (5.50a,b)$$

the dependence of these functions on c and $\hat{K}/2\mathcal{E}$ has been incorporated into $\hat{\tau}/\mu$, which will make more insightful observations possible.¹

Under these considerations, the velocity of a dislocation subjected only to the applied stress is

$$v = \bar{y}v_G \exp \left\{ -\frac{\Delta G_0^*}{kT} \left(1 - \frac{\sigma}{\hat{\tau}}\right)^{3/2} \right\}, \quad (5.51)$$

resulting in a shear strain rate of

$$\dot{\gamma} = b\bar{y}\rho_m v_G \exp \left\{ -\frac{\Delta G_0^*}{kT} \left(1 - \frac{\sigma}{\hat{\tau}}\right)^{3/2} \right\}, \quad (5.52)$$

where v_G , the frequency of oscillation of a dislocation line in a quadratic potential well of extent $2w$, is roughly $5 \times 10^{11} \text{ s}^{-1}$, and ρ_m is the mobile dislocation density, which we shall consider in more detail in Chapter 8. For the present purpose, we consider that the entire preexponential factor

$$\dot{\gamma}_0 = b\rho_m\bar{y}v_G \quad (5.53)$$

is matched to the experimental results by embedding it into a characteristic temperature T_0 , defined as

$$T_0 \equiv \frac{\Delta G_0^*}{k \ln(\dot{\gamma}_0/\dot{\gamma})}, \quad (5.54)$$

which becomes part of the temperature dependence of the plastic shear resistance obtained by inverting eq. (5.52) as

$$\sigma(T, \dot{\gamma}) = \hat{\tau} \left(1 - \left(\frac{T}{T_0} \right)^{2/3} \right). \quad (5.55)$$

¹ The numerical constants in eqs. (5.49a) and (5.50a) differ somewhat from what were reported by Argon (2004). There, slightly more approximate arguments were used.

The temperature T_0 is the temperature at which dislocations can overcome the solid-solution resistance with an infinitesimally small applied stress. Actually, as discussed in Chapter 2, at very small stresses the form of eq. (5.52) developed for stresses in the range of the mechanical threshold needs modification to consider thermal fluctuations that can reverse the motion of the dislocation, leading to a hyperbolic-sine dependence of the stress on the strain rate.

5.5 Comparison of Solid-solution-strengthening Models for FCC Metals with Experiments

5.5.1 Overview of Experimental Information

The entire phenomenology of solid-solution strengthening is considerably more complex than the models of it presented in the previous sections. Therefore, before the model results are compared with experiments, we present first an overview of the experimental facts related to the level of the plastic shear resistance and its dependence on temperature and solute concentration, and identify certain phenomena that cannot be fully explained quantitatively with satisfactory models and for which only reasonable qualitative explanations can be given.

Figure 5.5 shows the temperature dependence of the critical resolved shear stress (CRSS) of Ag–Al alloy single crystals for several atomic concentrations of the Al solute, ranging up to $c = 0.06$, as well as the reference curve for pure Ag (Hendrickson and Fine 1961). The curves show that while the plastic shear resistance of pure Ag is only about 1.0 MPa and has the very small temperature dependence characteristic of all pure FCC metals, alloying with Al at relatively modest concentrations raises the shear resistance substantially and introduces also a substantial temperature dependence at low temperatures. However, this temperature dependence levels off sharply around room temperature (300 K) in a plateau region, where, nevertheless, it remains dependent on the alloy concentration. Other FCC metal alloys such as Cu and Au alloys show very similar effects. Figure 5.6 shows the temperature dependence of the CRSS of several Cu alloys reported by Basinski *et al.* (1972), of which we shall consider two, Cu–0.5% Al and Cu–1.0% Al, for more detailed comparison with experimental results. Here also, the plastic shear resistance shows a leveling off at a plateau at around 300 K. There is, however, a more remarkable effect observable at very low temperatures, where the flow stress shows a more severe perturbation, which has been referred to as a “dynamic overshoot” effect and will be discussed more fully in Chapter 8. At cryogenic temperatures, where the phonon resistance becomes very small, the motion of the dislocations becomes underdamped. In this case, a dislocation released from

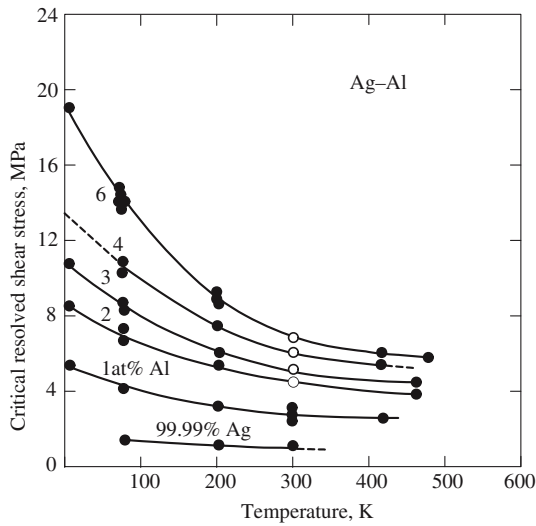


FIG. 5.5. Dependence on temperature of the CRSS in Ag–Al solid-solution alloys of different Al concentration (Hendrickson and Fine 1961; courtesy of AIME).

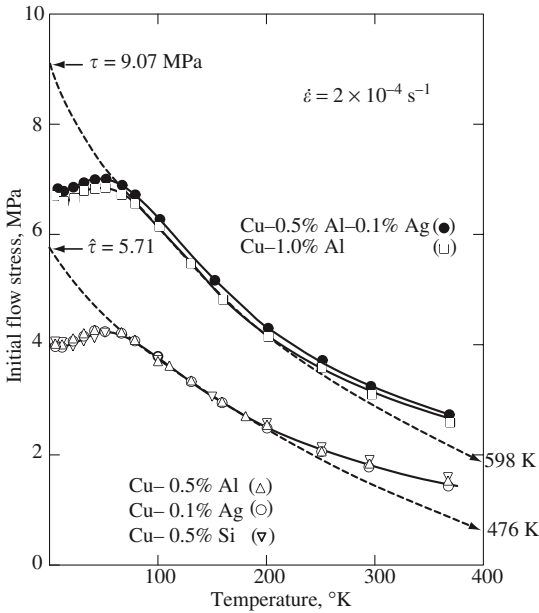


FIG. 5.6. Dependence on temperature of the CRSS for flow of several Cu solid-solution alloys with Al, Ag, and Si. At low temperatures, a dynamic overshoot effect results in a major modification of the behavior, and above room temperature the flow stress levels off in a plateau region (after Basinski *et al.* 1972).

a set of obstacles can acquire considerable kinetic energy, which permits it to overcome several further barriers at a reduced stress. The dotted extensions of the two curves of interest in this range, toward 0 K, show what the result would have been if the dynamic overshoot were not present.

5.5.2 Peak Solute Interaction Forces

Fleischer (1963) was the first to consider quantitatively the size misfit and modulus misfit effects of solute atoms with both edge and screw dislocations, and compared the results with published solid-solution-strengthening effects in the literature. In Section 5.2, the analysis of Fleischer was developed in considerable detail, leading to the two combined peak interaction forces $(\hat{K}/2\mathcal{E})_e$ and $(\hat{K}/2\mathcal{E})_s$ for edge and screw dislocations given by eqs. (5.25a) and (5.25b). On the basis that the terminal resistance $\hat{\tau}$ at 0 K depends on the atomic concentration according to $c^{2/3}$, as is demonstrated below, the experimental values of the peak normalized interaction forces of solutes with a dislocation are obtained (with a simple modification of the Fleischer results) as

$$\left(\frac{\hat{K}}{2\mathcal{E}}\right)_{\text{exp}} = \left[\frac{d\sigma}{dc^{2/3}} / 1.26 \left(\frac{w}{b}\right)^{1/3} \mu \left(1 - \left(\frac{T}{T_0}\right)^{2/3}\right) \right]^{3/4}. \quad (5.56)$$

Through eq. (5.56), the reported solute concentration dependence of the flow stress at room temperature can be projected to 0 K by making use of eqs. (5.43) and (5.54).

In Fig. 5.7, the experimental peak normalized resistances $(\hat{K}/2\mathcal{E})$ for both edge and screw dislocations are compared with the theoretical model results of

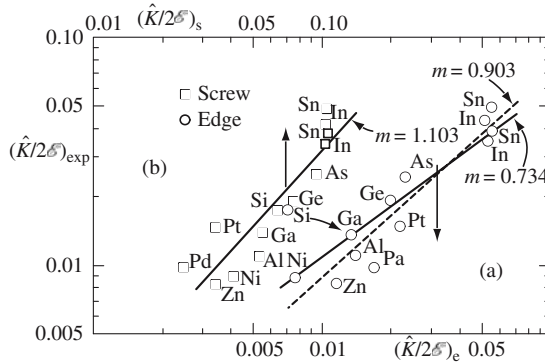


FIG. 5.7. Relationship between experimentally determined peak resistive forces of obstacles and those determined from models for Cu alloys: (a) edge dislocations; (b) screw dislocations (data from Fleischer 1963).

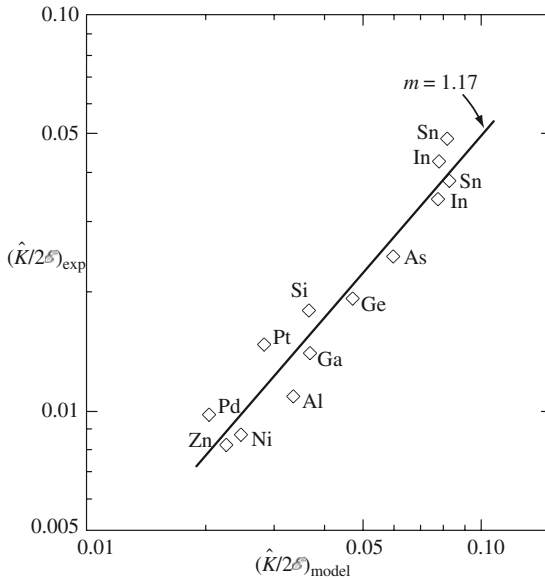


FIG. 5.8. Relationship between experimentally determined peak resistive forces of obstacles for alloy components in Cu and an arithmetic average of the forces determined from models (data from Fleischer 1963).

eqs. (5.25a,b) for 11 Cu solid-solution alloys. The figure shows that, as already stated above, the strengthening effect is stronger by more than a factor of two for edge dislocations than for screw dislocations. For screw dislocations, the best-fit linear dependence gives a slope of $m = 1.10$, reasonably close to the expected value of unity. For edge dislocations, because of the anomalously strong interaction of Si, the slope is only $m = 0.734$. If the Si point is excluded, however, the slope becomes 0.903, well within range of the expected slope of unity. If the two separate edge and screw interactions are combined on the basis of the arguments in Section 5.2.4, the results, as shown in Fig. 5.8, follow a tighter pattern, albeit with a slope somewhat further removed from unity. We emphasize here again that, in the determination of the theoretical peak interaction results, no arbitrary multipliers were introduced in the model, unlike what was done by both Fleischer (1963) and Haasen and coworkers (Haasen 1979; Haasen 1996) to improve the agreement between the theoretical model and experiments. The results presented in Fig. 5.8 will be used in further evaluations.

5.5.3 Dependence of Flow Stress on Solute Concentration

Several investigators (Kan and Haasen (1970) and Jax *et al.* (1970) for Au alloys; Trojanova and Haasen (1975) for Ag alloys; and Kratochvil and Neradova (1971)

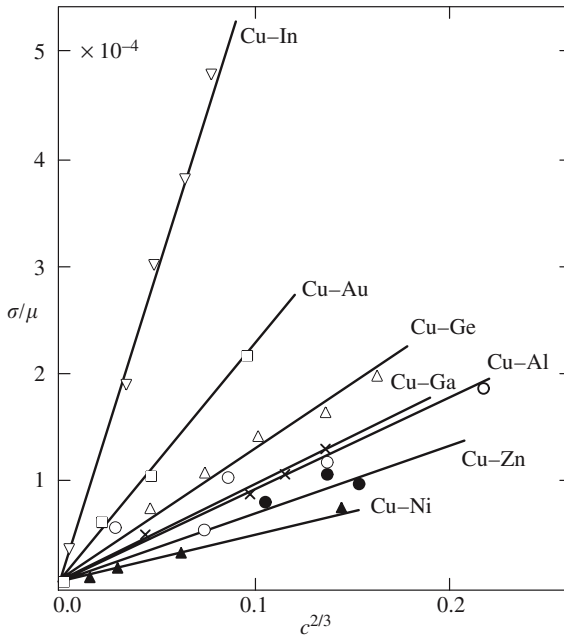


FIG. 5.9. Dependence of normalized CRSS for flow on solute concentration $c^{2/3}$ for seven Cu alloys (after Kratochvil and Neradova 1971).

for Cu alloys) have explored critically and in considerable detail the dependence of the shear resistance on solute concentration, either at room temperature or in the plateau region. Figure 5.9 shows the results of Kratochvil and Neradova (1971) on Cu alloys, giving convincing evidence for a dependence on $c^{2/3}$ rather than $c^{1/2}$ or even c .

5.5.4 Comparison of Temperature Dependence of CRSS between Experiments and Theoretical Model

Considering the temperature dependence of the CRSS results of Basinski *et al.* (1972) given in Fig. 5.6, we note that if the curves for Cu–0.5% Al and Cu–1.0% Al are extrapolated to 0 K, through the dynamic overshoot effects, terminal shear resistances $\hat{\tau}$ of 5.71 and 9.07 MPa are obtained. We note first that the ratio of these resistances is precisely equal to the ratio of the atomic concentrations of Al raised to the 2/3 power.

Ignoring the slight dependence of the global shear modulus on solute concentration by taking $\mu = 55.4$ GPa, picking the ratio $w/b = 1.7$ as suggested in Figs. 5.2(a) and (c), and taking $\hat{K}/2\mathcal{E} = 0.026$ as interpolated from Fig. 5.8, we obtain from eq. (5.43) terminal shear resistances of 19.23 and 30.53 MPa, respectively, for the

two alloys, suggesting that the theoretical coefficient in eq. (5.43) should have been 0.37 rather than 1.26. Considering the rather broad assumptions made in developing the model for the athermal resistance $\hat{\tau}$, a numerical discrepancy of this magnitude is acceptable. Proceeding further, if the curves for the alloys between 0 K and 250 K (before reaching the plateau) are extrapolated to zero shear resistance, cutoff temperatures T_0 of 476 K and 598 K, respectively, are obtained for the two alloys. From eqs. (5.55) and (5.49a), we obtain for the imposed strain rate $\dot{\gamma} = 2 \times 10^{-4} \text{ s}^{-1}$ a preexponential strain rate factor of $\dot{\gamma}_0 = 10^4 \text{ s}^{-1}$ for the alloy with $c = 0.01$. Considering that $\dot{\gamma}_0 = b\rho_m v_0$ where ρ_m is the mobile dislocation density and $v_0 = \bar{\gamma}v_G = (4/3)wv_G$, where v_G is about 10^{11} s^{-1} as an estimate of the base frequency of oscillation of the coherent length Λ (Section 5.4) in its potential well, a mobile dislocation density of roughly 10^{12} m^{-2} would be estimated. Within the relatively simple framework of the model of Section 5.4.2 for the thermally assisted motion of dislocations, these estimates are reasonable and consistent with the rough estimates of about $1\text{--}3 \times 10^{10}/\text{m}^2$ of Ney *et al.* (1977), obtained in connection with measurements of velocities of individual dislocations in dilute Cu–Al alloy single crystals.

Finally, considering the determinations of $\hat{\tau}$ and T_0 for the Cu–1.0% Al alloy, the form of the temperature dependence of the CRSS in the region below the plateau resistance given by eq. (5.54) is plotted in Fig. 5.10 as the lower curve, demonstrating a rather good representation. This leaves the nature of the plateau

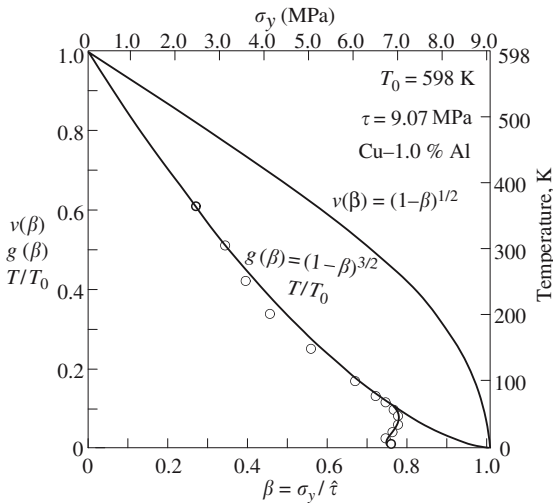


FIG. 5.10. Dependence of $\sigma_y/\hat{\tau}$ for flow on reduced temperature, and dependence of normalized activation volume on $\sigma/\hat{\tau}$ for a Cu–1 atom % Al alloy (data from Basinski *et al.* 1972).

region, above room temperature, to be explained. We consider some views about this in Section 5.5.7 below.

5.5.5 Summary of Solid-solution Strengthening of FCC Alloys

Haasen and coworkers have performed the most thorough investigation of solid-solution-strengthening effects in FCC alloys of Cu, Ag, and Au (Jax *et al.* 1970; Trojanova and Haasen 1975) in the plateau region where the shear resistance is relatively temperature-independent, for reasons that are still not too clear. It is unfortunate that all of these experiments were performed in this region rather than at lower temperatures, where the interaction of solute atoms and dislocations is far better understood. These findings could then have been projected to 0 K as was done in eq. (5.56). Nevertheless, these experiments in the plateau region have shown the basic dependences obtained in Section 5.4, albeit lacking overall quantitative agreement with theoretical models. The responses of the experimental results for these three alloy systems are presented in Figs. 5.11(a), (b), and (c), respectively, as plots of $[(1/\mu)(d\sigma/dc^{2/3})]^{3/4} (= \alpha(\hat{K}/2\mathcal{E})_{\text{exp}})$ versus the best estimates of the theoretical combined peak interaction parameters $(\hat{K}/2\mathcal{E})$ given in eq. (5.26) where, to emphasize again, no arbitrary multipliers were used (unlike in the original presentation in the works cited above). For the plots in Fig. 5.11, all the theoretical $(\hat{K}/2\mathcal{E})$ factors were also re-calculated from the Labusch parameter (see Haasen 1979, 1999), which used a geometrical mean of the experimentally determined size and modulus misfit parameters, incorporating an arbitrary multiplier. If the experiments had been carried out below the plateau region and projected to 0 K, the slope of the dependences given by the coefficient α would have been $\alpha = 0.37(w/b)^{1/3} = 0.44$, where the modified coefficient has been used in place of 1.26, discussed in Section 5.5.4. The values are instead 0.156, 0.555, and 0.536 respectively, for these alloy systems. Clearly, however, a temperature correction factor such as that used in eq. (5.56) could have brought these factors well into agreement with the expectation of the model, to $\alpha = 0.44$. Nevertheless, the overall agreement in form is pleasing and supports the arguments of Section 5.4, to within a necessary numerical adjustment.

5.5.6 The “Stress Equivalence” of the Solid-solution Resistance of FCC Alloys

Basinski *et al.* (1972) have drawn attention to a remarkably universal feature of the solid-solution strengthening phenomenon and its temperature dependence that is obeyed by most, if not all, FCC and HCP substitutional alloys with a moderately high solute concentration, which they have labeled “stress equivalence”. The “universal” observation is that in all solid-solution alloys of Cu, Ag, and Mg, and by implication

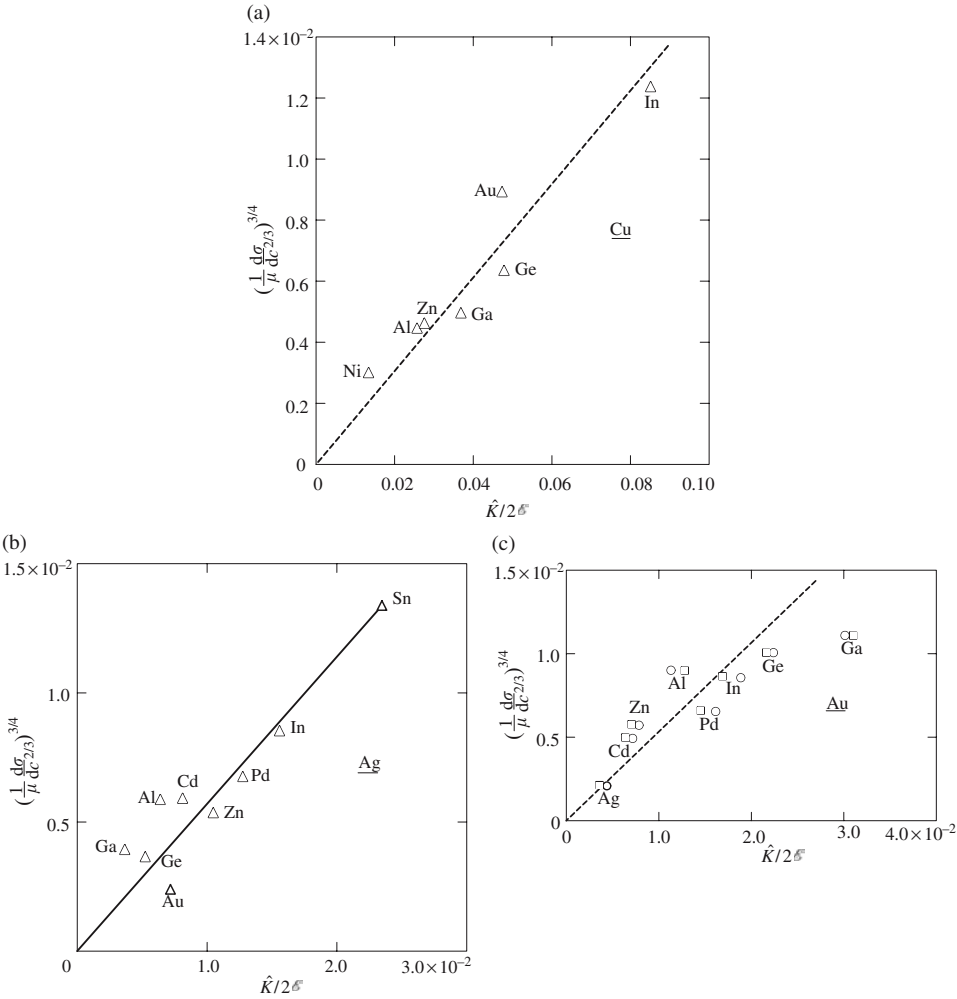


FIG. 5.11. (a) Dependence of CRSS/ μ on $c^{2/3}$ plotted against peak obstacle resistive force $\hat{K}/2\mathcal{E}$, for eight Cu solid-solution alloys (from Kratochvil and Neradova 1971); (b) the same dependence for nine Ag solid-solution alloys (Trojanova and Haasen 1975; courtesy of Carl Hanser, GmbH); (c) the same dependence for eight Au solid-solution alloys (after Haasen and Kratochvil, 1973).

all other FCC and HCP metals, the activation volume at the CRSS depends only on a reference plastic resistance. While this observation should in principle be based on the athermal resistance $\hat{\tau}$, this quantity is not readily accessible, first because it requires experiments near 0 K, which are difficult to perform, and second because

such experiments are seriously affected by the dynamic overshoot effect referred to above. Therefore, Basinski *et al.* (1972) based the correlations on the initial yield stress (the CRSS) at 78 K and 298 K.

To demonstrate this universal concept of solid-solution strengthening of FCC and HCP alloys, it is necessary that eq. (5.52) for the plastic strain rate, with the preexponential term given in eq. (5.53), be abbreviated in the form

$$mkT = \Delta G_0^* (1 - \beta)^{3/2}, \quad (5.57)$$

where $m = \ln(\dot{\gamma}_0/\dot{\gamma})$ and $\beta = \sigma/\hat{\tau}$. This needs to be reformulated, together with eqs. (5.50a,b), into

$$\left(\frac{mkT}{\mu b^3}\right) \frac{1}{B} \left(\frac{\mu}{\sigma_y}\right)^{1/2} = \frac{(1 - \beta)^{3/2}}{\beta^{1/2}} \quad (5.58)$$

and

$$\left(\frac{\Delta v^*}{b^3}\right) \frac{1}{B} \left(\frac{2}{3}\right) \left(\frac{\sigma_y}{\mu}\right)^{1/2} = \beta^{1/2} (1 - \beta)^{1/2}, \quad (5.59)$$

where $B = 5.55(w/b)^{3/2}$ and σ_y is the CRSS at either 78 K or 298 K.

Elimination of β between eqs. (5.58) and (5.59) and introduction of the dimensionless variables

$$V = \frac{\Delta v^*}{b^3}, \quad (5.60a)$$

$$C = \frac{mkT}{\mu b^3}, \quad (5.60b)$$

and

$$s = \frac{\sigma_y}{\mu} \quad (5.60c)$$

results in a quadratic equation in the variable s in terms of the normalized activation volume V and the material constants B and C (Argon 2004),

$$\left(\frac{4}{9}V^2\right)s^2 + \left(\frac{4}{3}VC\right)s - \left(\frac{3}{2}\frac{B^2C}{V} - C^2\right) = 0, \quad (5.61)$$

which gives the sought-after universal dependence of the activation volume on only the yield stress (CRSS) σ_y at either 78 or 298 K. This applies regardless of whether the alloys are Cu-, Ag-, or Au-based, and regardless of what the alloy content is. Considering the Cu–1.0% Al alloy examined before, for which $\hat{\tau} = 9.07$ MPa and $T_0 = 598$ K, giving $m = (\ln(\dot{\gamma}_0/\dot{\gamma})) = 17.52$ as discussed above, and taking

$w/b = 1.7$ as suggested by Figs. 5.2(a) and (c) with a shear modulus of 55.4 GPa, the solution of eq. (5.61) gives the lower curve in Fig 5.12 for 78 K. The alloy data of Basinski *et al.* (1972) for 78 K and 298 K are shown by open circles and open diamond symbols, respectively, and represent 14 Cu–Al alloys, three Cu–Ag alloys, two Cu–Si alloys, one Cu–Ni alloy, and two ternary Cu–Al–Ag alloys. (For identification of individual alloys, Basinski *et al.* (1972) should be consulted.) Clearly, the model given by eq. (5.61) parallels the data for 78 K quite well for stress levels from 1.0 to 20 MPa. Below 1.0 MPa, the inevitable grown-in forest dislocations become more important obstacles than solute atoms, even in high-perfection single crystals. Using the same material constants for 298 K, however, gives rather poor agreement (not shown), as was noted earlier by Argon (2004). While the reason for this is not too clear, it is most likely that at this temperature, which is already in the plateau region, the solute atoms interact with dislocations differently, as we discuss in Section 5.5.7 below. However, a choice of $w/b = 2.8$, with all other material parameters remaining the same, gives a tolerably better agreement with this data, as Fig. 5.12 shows. The reason why w/b should have a different value at 298 K is not clear but may be associated with the plateau properties discussed below. In connection with this, see also Section 8.4 on the Portevin–Le Chatelier effect.

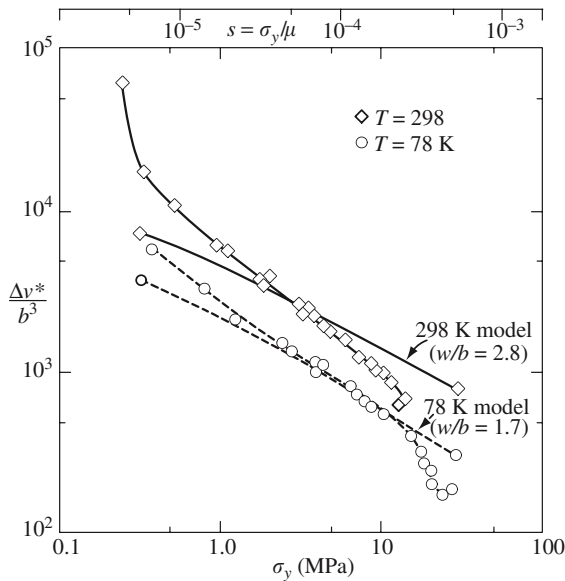


FIG. 5.12. Comparison of the predictions of the model of “stress equivalence” with experimental results for solid-solution-strengthened Cu alloys with a variety of solutes of different concentrations, assembled by Basinski *et al.* (1972) (Argon 2004; courtesy of Carl Hanser, GmbH).

The most striking feature of this observation of “stress equivalence” is that it demonstrates that substitutionally added solute atoms interact with dislocations in a remarkably similar manner, where all differences arising from alloy type and solute concentration are reflected in one single parameter of the CRSS of the alloy system.

5.5.7 The Plateau Resistance

While the plateau resistance of solid-solution strengthening in FCC metals in the form of a leveling-off effect of the temperature dependence of the CRSS at around room temperature is well documented for Cu, Ag, and Au alloys, there is no convincing model that can explain it quantitatively. The fact that the resistance scales with solute concentration as a $c^{2/3}$ relation in the same way as the resistance below the plateau region indicates that it is based on a solute–dislocation interaction very similar to the lower-temperature mechanism and that it is not a separate athermal process. From the considerable difference between the stress equivalence patterns at 298 K and 78 K and the fact that part of this difference may come from a different range w of solute interaction and from the steeper dependence of the activation volume on the CRSS, it is clear that while qualitative similarities remain in the sampling of solute atoms by dislocations, important differences begin to occur in the plateau region. Since no deformation instabilities such as the serrated yielding effects characteristic of higher temperatures are found, it can be concluded that large-scale solute mobility is largely absent. Nevertheless, some-short range drag of solute by dislocations is likely, particularly in the core region, where solute in the nearest atom planes on both sides of the glide plane may be involved (for a somewhat more expanded discussion, consult Haasen (1996)).

5.6 Solid-solution Strengthening of BCC Metals by Substitutional Solute Atoms

5.6.1 Overview of Phenomena

While solute atom interactions with dislocations in BCC metals have some common features with the corresponding cases in FCC metals, the presence of a strong lattice resistance at low temperatures makes superposition of the two mechanisms of lattice resistance and solute resistance quite complex. This is because the lattice resistance at low temperatures in BCC metals is predominantly governed by screw dislocations, while the resistance to edge dislocation motion is generally of secondary importance. The quite different mobilities of screw and edge dislocations result in different forms of sampling of solute atoms by them, which affect both the plastic resistance and its strain rate dependence quite differently in the temperature

ranges where lattice resistance is present. Thus, substitutional additions of solute raise the shear resistance strongly in the room temperature range, whereas increasing the solute concentration for many solute species actually produces a reduction in the lattice resistance at low temperatures and results in *solid-solution softening* rather than strengthening, indicating a distinct interference between the two mechanisms. Iron, the most investigated BCC alloy system, exhibits a plateau in shear resistance around room temperature and somewhat above, similar to that of FCC metals, shown in Fig. 5.13 for single crystals with orientations near the center of the standard triangle. The plateau resistance has a strain rate dependence that features an activation volume with a stress dependence characteristic of that of FCC alloys, while the strain rate dependence of the flow stress at low temperature exhibits the features of kink mobility involving double-kink nucleation characteristic of pure BCC metals. These conflicting observations are indicative of the complexities of the behavior of BCC solid-solution alloys.

In comparison with the abundance of studies of FCC alloy systems, there are comparatively fewer experimental studies of BCC solid-solution alloys. Similarly, theoretical studies of BCC alloys are also quite few. We consider in depth only Fe alloys, where we shall rely primarily on the work of Takeuchi and coworkers

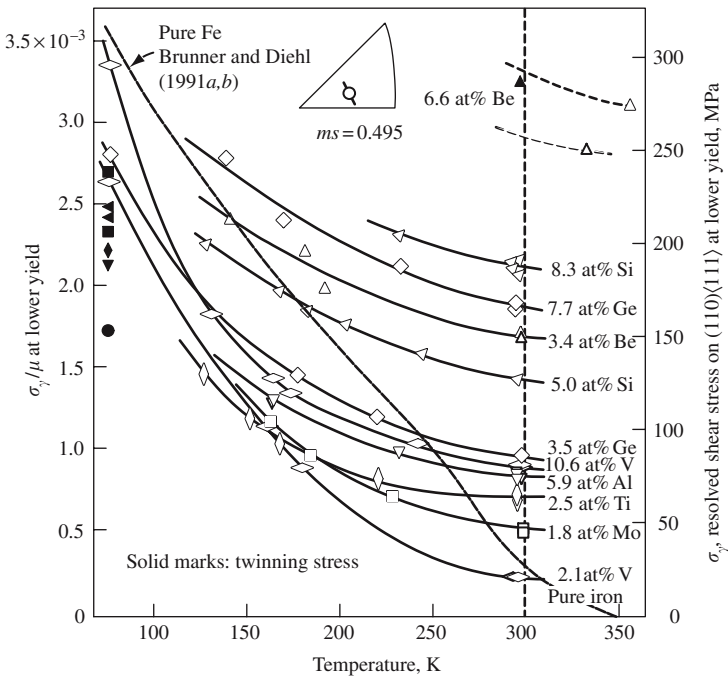


FIG. 5.13. Dependence of CRSS on temperature in Fe alloy single crystals with six different solutes Al, Be, Ge, Mo, Si, and V (after Takeuchi *et al.* 1968).

(Takeuchi *et al.* 1968; Takeuchi 1969). For theoretical considerations, we shall find the work of Suzuki (1979) most helpful. As with FCC metals, only substitutional alloy systems will be considered.

To develop the necessary familiarity with the different features of Fe alloys, we start by presenting in the following section the main experimental findings of Takeuchi and coworkers.

5.6.2 Experimental Manifestations of BCC Solid-solution Alloy Systems

Figure 5.13 shows the temperature dependence of the CRSS (σ_y , at the lower yield point resolved on the $\{110\}\langle 111 \rangle$ slip system) of six Fe alloy single crystals containing solutes of Al, Be, Ge, Mo, Si, and V, at strain rates in the low 10^{-4} /s range (for full numerical details, Takeuchi (1969) should be consulted). For reference, the temperature dependence of the CRSS of pure Fe (Brunner and Diehl 1991a,b) is also given in the figure. It is clear that at room temperature, alloy additions produce substantial rises of the CRSS. However, it is also striking that the temperature dependence of the alloy systems is lower in an intermediate-to-low temperature range, making the shear resistances of the alloy, lower than that of pure Fe, resulting in alloy softening. At 77 K, most crystals yield by twinning rather than by slip. Figure 5.14 shows that the dependence of σ_y/μ in the plateau region at room temperature on alloy concentration is quite linear, with the possible exception of the Fe–V system, which shows also other anomalies. Figure 5.15 shows that with the

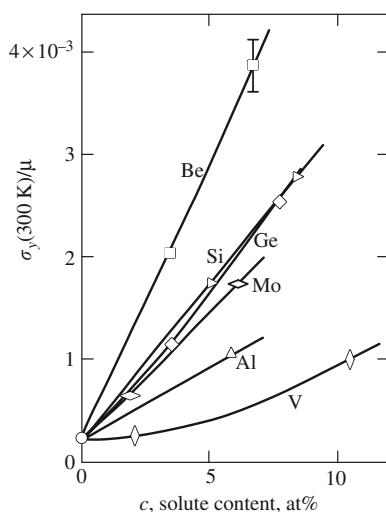


FIG. 5.14. Dependence of reduced CRSS at 300 K on alloy content for the same six Fe alloy single crystals as in Fig. 5.13 (Takeuchi 1969; courtesy of the Physical Society of Japan).

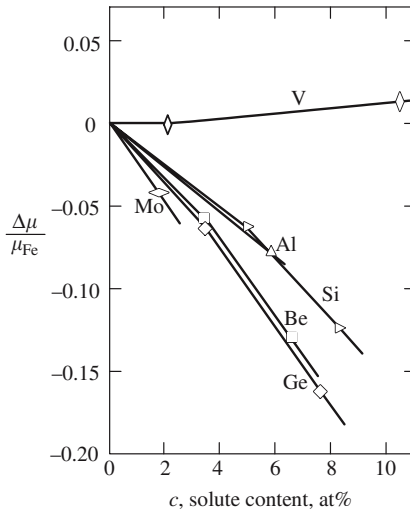


FIG. 5.15. Fractional change of shear modulus with alloy content for the same six Fe alloy single crystals as in Fig. 5.13 (Takeuchi 1969; courtesy of the Physical Society of Japan).

exception of V, which produces a slight increase in stiffness with increasing alloy concentration, all the other five alloy systems result in a substantial and nearly linear decrease in the shear modulus of the alloy. Of some importance in the discussion of the kinetics of plastic flow behavior at the CRSS is a significant dependence of the initial dislocation density on solute content, which has been observed by transmission electron microscopy. This is shown in Fig. 5.16. Moreover, strain rate change experiments at 300 K have given activation volumes $\Delta v^*/b^3$ in the range of 50–200, decreasing with alloy content as shown in Fig. 5.17.

5.7 Interactions of Solute Atoms with Screw Dislocations in BCC Metals

5.7.1 Overview of Model of Interaction of Solute Atoms with Screw Dislocation Cores

As discussed in detail in Chapter 4, the lattice offers a resistance primarily to only screw dislocations in BCC metals by virtue of the complex spatially extended nature of their cores, while the resistance to edge dislocations is at a much lower level. Consequently, at low temperatures, where the lattice resistance is present in the reference pure metal, the interactions of solute atoms of primary importance are with screw dislocations in BCC metals. Here the modulus misfit and size misfit effects of solute atoms continue in a separate specific form into the interactions with

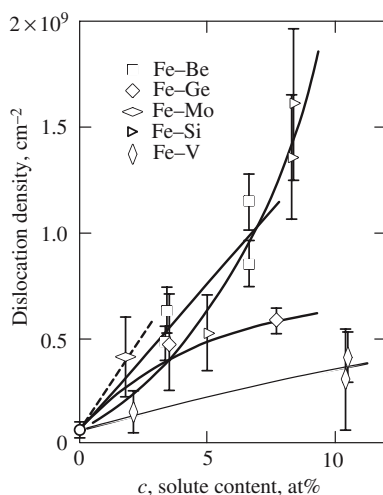


FIG. 5.16. Dependence of the initial dislocation density at the CRSS for the same Fe alloy single crystals as in Fig. 5.13 (Takeuchi 1969; courtesy of the Physical Society of Japan).

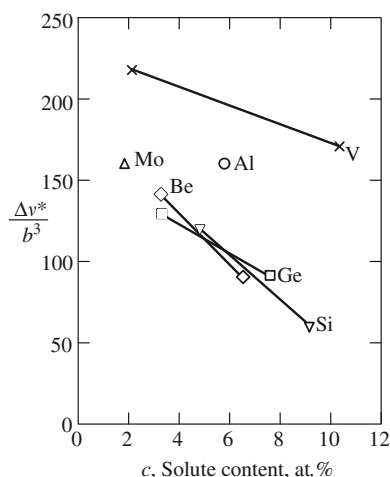


FIG. 5.17. Dependence of activation volume on alloy content as determined from strain rate change experiments at 300 K for the same Fe alloy single crystals as in Fig. 5.13 (Takeuchi 1969; courtesy of the Physical Society of Japan).

screw dislocation cores. However, the mode of thermally assisted advance of screw dislocations remains the same in the presence of solute atoms as in the reference pure metal, that is, advance occurs by nucleation of double kinks under stress and the motion of these kinks along the dislocation line. However, the solute affects the

kinetics of both double-kink nucleation and kink motion significantly. In the latter category while the kink motion in the pure metal is impeded only by the phonon resistance, which is nearly always negligible, in the solid-solution alloy the kinks encounter solute atoms and atom clusters, which produce stronger impediments to their motion. Here, the kink, which has a substantial width in the range of 10 or more atomic distances, has little difficulty in overcoming the impediments of individual solute atoms but becomes subject to significant resistance when larger random groupings of solute atoms parallel to the dislocation line are sampled.

In the plateau, where in the usual Regime I region of the reference pure metal (see Chapter 4) the resistance to double-kink nucleation would have been very small, a screw dislocation, nevertheless, still remains straight and parallel to the $\langle 111 \rangle$ direction. It then samples the field of solute atom interactions essentially as a straight line, athermally. At temperatures below the plateau, where, in the Regime I region of the reference pure metal, the double-kink nucleation resistance begins to rise and the dislocation now advances by motion of kinks along the dislocation line, the overall resistance to the motion of the screw dislocation becomes governed by random large groupings of solute atoms that have ranges comparable to the kink width. Finally, at still lower temperatures, the motion of screws become governed by the difficulty of double-kink nucleation, albeit at levels of double-kink energies that are considerably reduced owing to the generally negative nature of the modulus misfit interactions of solute atoms with a screw dislocation. In comparison, the interactions of solute atoms with edge dislocations, which in the reference pure metal are relatively unimpeded by the lattice resistance, are considered to remain largely the same as in FCC metal alloys.

5.7.2 Interaction of Solute Atoms with Screw Dislocation Cores

Recognizing that the essence of solid-solution strengthening in BCC metals is the interaction of solute atoms with screw dislocations, Suzuki (1979) considered first in an approximate manner the structure of screw dislocation cores as being made up of six atom rows of high energy, parallel to the $\langle 111 \rangle$ direction. The nonlinear elastic interactions between the atom rows were based on the relative axial displacements between the rows and were calculated through a pair of cosine potentials that had the periodicity of the interatomic separation. Figure 5.18 shows the resulting levels of elastic strain energy stored in the individual rows per unit length, in units of $A\mu b^2/4\sqrt{3}\pi^2$ (where $A \approx 1.06$). Clearly, the largest strain energies reside in six atom rows immediately around the center of the dislocation. These we identify as making up the dislocation core. The elastic strain energies associated with atom rows outside this core are substantially lower and decrease rapidly on moving away from the core, and are predictable by linear elastic models of a screw dislocation. On the basis of the size misfit and modulus misfit parameters ϵ_s and ϵ_μ of solute atoms in BCC metals determined by Takeuchi (1969), Suzuki estimated the size

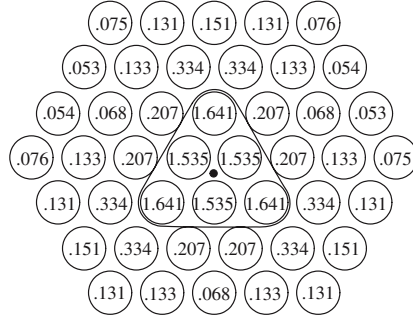


FIG. 5.18. Elastic strain energy per unit length in atom rows parallel to the $\langle 111 \rangle$ direction as determined from the Suzuki model of a screw dislocation in pure Fe (Suzuki 1979; courtesy of Elsevier).

misfit and modulus misfit interaction energies I_s and I_μ of solute atoms with the dislocation core when the latter are incorporated into any one of the atom rows of the core, to obtain

$$I_s = 3\Omega\epsilon_s \Delta P = 3\alpha \left(\left(\frac{\Omega}{b^3} \right) (\mu b^3) \right) \epsilon_s \quad (5.62)$$

and

$$I_\mu = 0.54 \left(\frac{\mu b^3}{2\sqrt{3}\pi^2} \right) \epsilon'_\mu. \quad (5.63)$$

The considerations resulting in these determinations are largely similar to those discussed in Section 5.2 for FCC metals, where the size misfit interaction results from the dilatation inside the core due to the nonlinear elastic interactions of core atoms with each other, which is suppressed by the linear elastic surroundings, creating a pressure ΔP inside the core of magnitude $\Delta P = \alpha\mu$, with the best estimate for α being 0.1. The effective modulus misfit parameter ϵ'_μ differs from $\epsilon_\mu (= 1/\mu d\mu/dc)$ through the incorporation of the usual dilatation effect of a solute in its immediate surroundings, producing a definite change in stiffness and resulting in a dependence of ϵ_μ on ϵ_s also. While the details of these considerations are beyond our scope here, we list the magnitudes of misfit parameters ϵ_s , ϵ_μ and ϵ'_μ for six solute species in Fe in Table 5.1 (Suzuki 1979).

The combined effect of the separate size and modulus misfit effects on the total interaction energy I of a solute atom with the core is then

$$I = \frac{0.54}{\sqrt{3}\pi^2} \left(\frac{\mu b^3}{2} \right) \epsilon'_\mu + 6\alpha \left(\frac{\Omega}{b^3} \right) \left(\frac{\mu b^3}{2} \right) \epsilon_s. \quad (5.64)$$

TABLE 5.1. Misfit parameters of six solute species in Fe alloys (from Takeuchi 1969)

Solute element	ϵ_s	ϵ_μ	ϵ'_μ	$\epsilon'_\mu + 1.521\epsilon_s$
Al	+0.063	-1.32	-0.69	-0.594
Be	-0.077	-1.82	-1.04	-1.157
Ge	+0.066	-2.00	-0.93	-0.829
Mo	+0.103	-2.27	-0.96	-0.803
Si	-0.023	-1.39	-0.85	-0.815
V	+0.039	+0.12	+0.34	+0.399

We consider this total interaction energy as the binding energy E_b of a solute atom to a screw dislocation core and state it in final form as

$$E_b = \frac{0.54}{\sqrt{3}\pi^2} \left(\frac{\mu b^3}{2} \right) [\epsilon'_\mu + 1.52\epsilon_s], \quad (5.65)$$

where the term in square brackets is the combined misfit parameter, the magnitude of which is also given in Table 5.1 for the six solute species in Fe.

5.7.3 Binding Potential of Solute to Screw Dislocation Cores

For detailed considerations, it is useful to associate the binding energy of eq. (5.65) with a potential contour, which can define a resistive force $K(y)$ as a function of the separation distance y of a solute from a dislocation, resembling the considerations represented in Fig. 5.2 in a more compact, albeit approximate, manner.

Thus, consider the free energy $F(y)$ of the binding interaction of a solute given as a function of the separation distance y by the symmetrical function

$$F(y) = E_b \exp \left\{ - \left(\frac{y}{w} \right)^2 \right\}, \quad (5.66)$$

shown in Fig. 5.19(a), where w is the interaction range. The resistive force $K(y)$ between the dislocation and the solute is then

$$K(y) = - \frac{dF(y)}{dy} = \frac{2E_b y}{w^2} \exp \left\{ - \left(\frac{y}{w} \right)^2 \right\}, \quad (5.67)$$

with a maximum resistance $\hat{K}$ occurring where $dK/dy = 0$, at $y = w/\sqrt{2}$, giving

$$\hat{K} = \alpha \left(\frac{E_b}{w} \right) \quad (\alpha = 0.858). \quad (5.68)$$

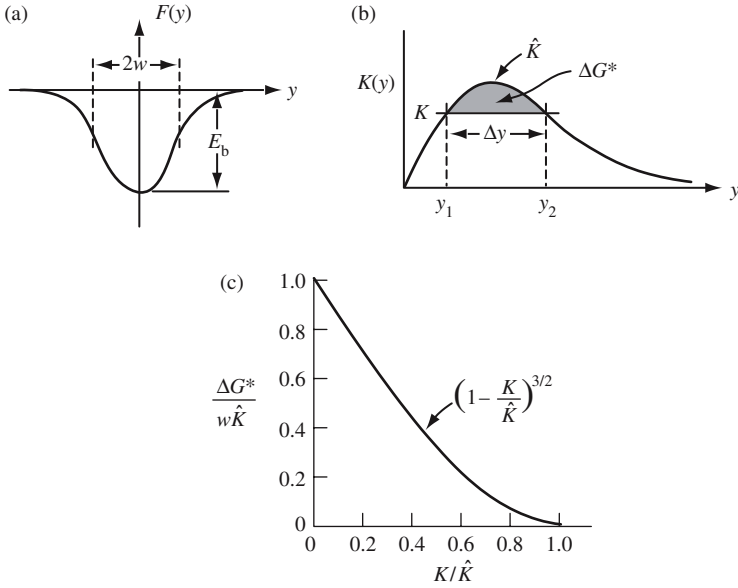


FIG. 5.19. (a) Sketch of the binding potential chosen to represent the interaction of a solute with a kink in a screw dislocation; (b) the force–distance curve of a solute acting on a kink in a screw dislocation; (c) dependence of the activation free energy on $\sigma/\hat{\tau}$ for penetration of a kink in a screw dislocation into the force field of a solute.

If the interaction is between a kink of height λ and a solute atom, the resistance τ_{km} to the motion of the kink along the dislocation is

$$\tau_{\text{km}} = \frac{1}{b\lambda} \frac{dF(y)}{dy} = \frac{K(y)}{b\lambda}, \quad (5.69)$$

with a peak value of

$$\hat{\tau}_{\text{km}} = \frac{\hat{K}}{b\lambda} = \alpha \left(\frac{E_b}{b\lambda w} \right). \quad (5.70)$$

In instances where the thermally assisted overcoming of a solute obstacle by a kink under a stress $\sigma < \hat{\tau}_{\text{km}}$ is of interest, the stress dependence of the activation free

energy ΔG^* is, with reference to Fig. 5.19(b),

$$\begin{aligned}\Delta G^* &= \Delta F^* - \Delta W^* = b\lambda \int_{y_1}^{y_2} (\tau(y) - \sigma) dy \\ &= w\hat{K} \int_{K/\hat{K}}^{1.0} \frac{\Delta y}{w} \left(\frac{K}{\hat{K}} \right) d\left(\frac{K}{\hat{K}} \right).\end{aligned}\quad (5.71)$$

Numerical integration of the final form of eq. (5.71) gives, to a high degree of accuracy (Fig. 5.19(c))

$$\frac{\Delta G^*}{w\hat{K}} = \left(1 - \frac{K}{\hat{K}} \right)^{3/2}. \quad (5.72)$$

5.8 The Shear Resistance

5.8.1 The Athermal Resistance at the Plateau

As noted by Takeuchi (1969) in his experiments on Fe solid-solution alloys, the CRSS shows negligible temperature dependence above 300 K and is a linear function of the solute concentration c , as shown in Fig. 5.14. We note further from Fig. 5.13 that at this temperature the lattice resistance is still present in the reference pure Fe, albeit at a much reduced level. From this we must conclude that at the CRSS for double-kink nucleation, the Regime I behavior is accomplished at a stress below the flow stress but the screw dislocations are still straight and parallel to the $\langle 111 \rangle$ directions and are sampling the field of the solute as a “rigid” line. Under these conditions the lattice resistance is estimated to be 21 MPa, from Figs. 5.13 and 5.14. The remainder of the flow stress is then attributed to a solid-solution resistance due to a collection of solute resistive forces $\hat{K}$ acting on the line at an average spacing of d_s per atom row in the core.

Considering that the effective range of occupation of a solute in a row of atoms is $2w$, the mean spacing d_s of solute atoms in a row of core atoms is

$$d_s = \frac{2w}{c}, \quad (5.73)$$

and then the contribution of solute atoms to the resistance in a row of core atoms is $(\hat{K}/b)(c/2w)$. But, since there are six atom rows in the core of the screw, the

athermal resistance τ_a in the plateau should be

$$\tau_a = 3 \left(\frac{\hat{K}}{b} \right) \left(\frac{c}{w} \right) = 3\alpha \left(\frac{E_b}{b^3} \right) c \left(\frac{b}{w} \right)^2, \quad (5.74)$$

where we recall that $\alpha = 0.858$ as given in eq. (5.65), with

$$\frac{E_b}{b^3} = \frac{0.54}{2\sqrt{3}\pi^2} (\epsilon'_\mu + 1.52\epsilon_s) \mu (1 + \epsilon_\mu c) \quad (5.75)$$

being slightly altered; the last factor accounts for the global change of the shear modulus with solute concentration in Fe alloys, and the modulus misfit parameters ϵ_μ are as given in Table 5.1.

5.8.2 Resistance Governed by Kink Mobility

In the temperature range immediately below the plateau, where double-kink nucleation by either the *Regime I* or *Regime II* processes is relatively readily accomplished, the plastic resistance is governed by kink mobility, involving the interaction of kinks with solute atoms. However, this resistance does not derive from the interaction of kinks with individual solute atoms but rather from interaction of broad kinks of width $w_k \gg b$ with clusters of solute atoms, for subtle reasons. This needs to be demonstrated before more complex processes are considered. Thus, from eq. (5.70), which gives the peak resistance $\hat{\tau}_{km}$ offered by an individual solute atom to the motion of a kink, using the parameters given by eq. (5.75) together with the b/w ratio obtained in the preceding section and $\lambda/b = 1.44$ for the kink height on the $\{110\}$ planes, we estimate the peak resistance $\hat{\tau}_{km}$ to be 570 MPa for an Fe–Si alloy of Si concentration 0.05 as an example. While this is a very substantial resistance, the spatial extent $w = 1.05b$ of the binding potential trough is extremely small, resulting in a total work of separation of only the order of $3.16kT$ at the low edge of the plateau. This renders the individual solute atom obstacles readily penetrable with little thermal help. Consequently, if the solute is to govern kink mobility, this will have to come from larger solute clusters of similar peak strength but of considerably longer range.

In providing a model for the factors controlling kink mobility over solute clusters, Suzuki (1979) noted first that the typical kink on a screw dislocation is not sharp but is spread out over a characteristic width w_k , which is given, for example, for the cosine potential model of the lattice resistance topography, by (eq. 4.21a)

$$\frac{w_k}{b} = \frac{\lambda}{2b} \frac{1}{\sqrt{a'}}, \quad (5.76)$$

where in BCC metals the core energy scale factor is $\alpha' = 3.45 \times 10^{-3}$, and $\lambda/b = 1.44$ on the $\{110\}$ planes (1.63 on the $\{112\}$ planes). The kink width

$w_k = 12.4b$ makes it possible for large solute atom clusters to impede kink motion. While quite rare along a dislocation, these larger clusters act as deep obstacles in the kink path, requiring substantially more work to overcome them, and thus are more resistant to thermally assisted penetration.

An important insight into the arrangement of such large solute clusters along a dislocation came from Suzuki, who noted that there must be a certain concentration of impenetrable obstacles along the dislocation line with spacing Λ that divides it into independent compartments. The mobility of these segments Λ is governed by the mobility of kinks of width w_k at the single largest solute cluster, anywhere within the compartment, where the kink spends the longest time in overcoming the obstacle with thermal assistance under stress. All other individual solute atoms and lesser clusters should then require a comparatively negligible time to be overcome. The nature of these impenetrable obstacles themselves was left unclear by Suzuki beyond stating that they gave rise to sessile “super jogs” that were dragged along by emission of point defects. A clue to the nature of these impenetrable obstacles came from the TEM studies of Hattendorf and Büchner (1990) of Fe–P alloys, where screw dislocations were found to have a certain concentration of well-defined deep cusps such as those shown in Fig. 5.20, marked A and B. Since the density of such cusps in the plane of observation is close to the likely density of forest screw dislocations and since encounters of glide screw dislocations with forest screw dislocations result in sessile contacts that can be displaced only by emission of point defects, we attribute these cusps to such screw–screw encounters. From the measured dislocation densities at the CRSS in Fe alloys reported by Takeuchi (1969), as given in Fig. 5.16, we estimate that for an Fe–Si alloy with a Si solute concentration of 0.05, for example, $\Lambda = 1/\sqrt{\rho} = 1.2 \times 10^{-4}$ cm, which is close to the density of the cusps in Fig. 5.20.

Proceeding further and following Suzuki (1979), we state that the probability $P(m_0, n)$ that there exists a cluster of m_0 excess solute atoms among the average number n of solute atoms contained in a screw dislocation segment of length equal to the kink width w_k is given by a Poisson distribution,

$$P(m_0, n) = 1 - \frac{1}{\sqrt{2\pi}} \int_{m_0}^{\infty} \exp \left\{ -\frac{1}{2} \left(\frac{m}{4n} \right)^2 \right\} dm, \quad (5.77)$$

on the assumption that clustering of solute occurs randomly, and where $n = 3(w_k/b)c$ for a cross section of three atom rows across the direction of advance of the core during glide. A transformation of variables to $k = m_0/\sqrt{2n}$ then permits a restatement of eq. (5.77) as a normal-distribution integral

$$P(m_0, n) = \frac{1}{2} \left(1 - \frac{1}{\sqrt{2\pi}} \int_{-k}^k \exp \left(-\frac{x^2}{2} \right) dx \right), \quad (5.78)$$

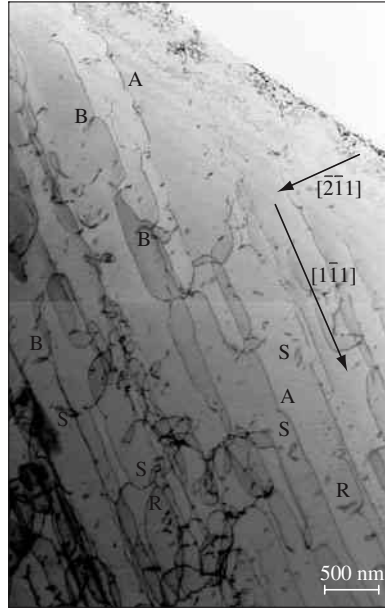


FIG. 5.20. TEM micrograph of parallel screw dislocations on $\{110\}$ planes in an Fe-0.45 at % P alloy crystal, showing deep cusps at sites marked A and B, thought to be impenetrable sites where screw glide dislocations encounter forest screw dislocations (Hattendorf and Büchner 1990; courtesy of Carl Hanser, GmbH).

which, for large arguments x (larger than 4), gives asymptotically, as the first term of a series expansion,

$$P(m_0, n) = \frac{1}{\sqrt{4\pi}} \frac{\exp(-m_0^2/4n)}{(m_0^2/4n)^{1/2}}. \quad (5.79)$$

The number m_0 of excess solute atoms in a cluster that renders the probability unity of finding such a cluster among the total number of segments Λ/w_k along a dislocation line of length Λ in a given compartment is then given by

$$P(m_0, n) \left(\frac{\Lambda}{w_k} \right) = 1. \quad (5.80)$$

This permits determination from eqs. (5.78) and (5.79) of the excess number m_0 of solute atoms in the single largest cluster in a compartment of length Λ where a kink of width w_k spends the longest activation time t_a in overcoming the cluster and moves the length Λ ahead by the kink height λ . Considering, for example, as specific cases, the Fe-Si alloys of Takeuchi (1969) for solute concentrations of 0.05 and 0.083, we find $m_0 = 5.7$ and 7.34 respectively, on average.

These values then make it possible, in turn, to determine the temperature dependence of the plastic shear resistance in the range where it is controlled by the kink mobility.

We view the cluster of m_0 excess solute atoms as residing in a potential trough of half-width w_m and choose this to be the kink half-width $w_k/2$. Moreover, viewing the core, as depicted in Fig. 5.18, to have a cross section of three atom rows of high energy in the glide path, we consider the cluster of m_0 solute atoms to be distributed among these rows in a variety of ways, but all within a dimension $2w_m \approx w_k$. This gives the terminal resistance $\hat{\tau}_{km}$ due to the kink-mobility-governed process as

$$\hat{\tau}_{km} = \frac{\alpha E_b}{b^3} \frac{(m_0/3)b}{(w_k/2)} \left(\frac{b}{\lambda} \right) = \frac{2}{3} \left(\frac{\alpha E_b}{b^3} \right) m_0 \left(\frac{b}{\lambda} \right) \left(\frac{b}{w_k} \right) \quad (5.81)$$

a variant of the expression in eq. (5.70).

Correspondingly, the plastic shear strain rate $\dot{\gamma}$ is given by the usual kinematic-kinetic expression of Orowan (1934),

$$\dot{\gamma} = b \rho_m \bar{v}_k, \quad (5.82)$$

where

$$\bar{v}_k = \frac{\lambda}{t_{ac}} = \lambda \nu_G \exp \left\{ -\frac{\Delta G_{0m}^*}{kT} \left(1 - \frac{K}{\hat{K}} \right)^{3/2} \right\} \quad (5.83)$$

is the average kink velocity in one compartment Λ , t_{ac} is the thermal activation time across the cluster of excess m_0 solute atoms for the kink, ρ_m is the mobile dislocation density, and $\nu_G = O(5 \times 10^{11} \text{ s}^{-1})$ is the “attack frequency” of the kink for overcoming the cluster. Then, since $\Delta G_{0m}^* = m_0 E_b$ and $K/\hat{K} = \sigma_m/\hat{\tau}_{km}$ where σ_m is the component flow resistance at a given temperature T , we have

$$\dot{\gamma} = \dot{\gamma}_0 \exp \left\{ -\frac{m_0 E_b}{kT} \left(1 - \left(\frac{\sigma_m}{\hat{\tau}_{km}} \right) \right)^{3/2} \right\} \quad (5.84)$$

and, by the usual inversion,

$$\sigma_m(T) = \hat{\tau}_{km} \left(1 - \left(\frac{T}{T_{0m}} \right)^{2/3} \right), \quad (5.85)$$

where the required double-kink nucleation rate is readily provided for under the component stress σ_m , and does not govern. In eqs. (5.84) and (5.85),

$$\dot{\gamma}_0 = b^2 \left(\frac{\lambda}{b} \right) \rho_m \nu_G \quad (5.86a)$$

and

$$T_{0m} = \frac{\Delta G_{0m}^*}{k \ln \dot{\gamma}_0 / \dot{\gamma}}, \quad (5.86b)$$

and $\dot{\gamma}$ is the strain rate imposed by the machine.

Comparison of the results of Sections 5.8.1 and 5.8.2 with experiments leads to the conclusion that the athermal resistance and the kink-mobility-controlled resistance superpose directly. Since the two effects act on the same dislocations simultaneously, the interpretation poses a challenge. We conclude first that above a certain temperature, where the stress to nucleate a double-kink is lower than what is required for kink mobility, double kink nucleation is not a controlling factor. The reverse holds below this temperature. In the athermal plateau range, the resistance of the excess solute clusters discussed above can be readily overcome if the temperature is above T_{0m} . Then, a screw dislocation still remains straight if the temperature is below T_0 , where the lattice resistance of the reference pure metal vanishes. However, in the temperature range $T_{0m} < T < T_0$, where kink mobility does not govern but the lattice resistance is still present, the straight dislocation needs to negotiate the field of randomly dispersed solute atoms, whether clustered or not. Under these conditions, no intermediate line form is available to take the straight dislocation through the field in small independent increments of line length, but only a monolithic advance of the line is possible, which cannot be thermally assisted. But, when the temperature is below T_{0m} , the large clusters begin to offer additional resistance, requiring the extra flow stress component σ_m to be supplied as well, and kink mobility governs.

5.8.3 Double-kink-nucleation-controlled resistance

With decreasing temperature, a point is reached where the entire applied stress is insufficient to supply the double-kink nucleation rate required to replenish mobile dislocations and keep them mobile over the solute barriers. Then, consideration of the kinetics of both double-kink nucleation and kink mobility becomes necessary. Experimental evidence shows that in this low-temperature region the double-kink nucleation is generally aided by the presence of the solute, resulting in solid-solution softening. Thus, we consider how this becomes possible within the framework of the structure of the core and its interactions with a solute discussed in Section 5.7.

As discussed in that section, the strain energy in the core is primarily associated with the six atom rows parallel to the $\langle 111 \rangle$ direction in the Suzuki (1979) model of the screw dislocation core. Thus, the elastic strain energy of the core, as estimated by Suzuki (Fig. 5.18) is, per unit length,

$$F_{\text{core}} = \left(\frac{\mu b^2}{4\sqrt{3}\pi^2} \right) \sum_{i=1}^6 I_i, \quad (5.87)$$

where

$$\sum_{i=1}^6 I_i = 9.528, \quad (5.88)$$

giving

$$F_{\text{core}} = 0.139\mu b^2. \quad (5.89)$$

The elastic strain energy concentrated in shells of atom rows outside the core drops off rapidly, as given by the linear elastic model of the screw dislocation. We estimate the core radius r_c of the screw dislocation in the Suzuki model from

$$\pi r_c^2 = \sqrt{3}\lambda^2, \quad (5.90)$$

which, for a dislocation on a $\{110\}$ plane, where $\lambda/b = 1.44$, gives $r_c = 1.07b$. Then, in a crystal of substantial size with, say, $R = 10^4b$, we determine the strain energy F_{lin} outside the core to be

$$F_{\text{lin}} = \frac{\mu b^2}{4\pi} \ln \frac{R}{r_c} = 0.728\mu b^2, \quad (5.91)$$

and the fraction C that the core energy makes up of the total is

$$\frac{F_{\text{core}}}{F_0} = C = 0.160. \quad (5.92)$$

As discussed in Chapter 4, experiments indicate that in the unmodified metal the variation of the core energy due to displacement of the dislocation in the lattice-potential topography results in a variation of the line energy with position of

$$\Delta F = \alpha' F_0 \left(1 - \cos \left(\frac{2\pi y}{\lambda} \right) \right) \quad (5.93)$$

in the cosine potential model, giving

$$\Delta F_{\text{max}} = 2\alpha' F_0 = 2\alpha' \left(\frac{F_{\text{core}}}{C} \right), \quad (5.94)$$

where comparisons with experiment indicates that $\alpha' \approx 3.45 \times 10^{-3}$ in the pure metal.

When the metal is modified with a solute species, we assume that the principal modifications of the elastic strain energy of relevance are in the core and that the fraction C remains largely the same. Naturally, the changes in the distant strain

field must also be considered by rescaling the shear modulus. Thus, considering the solute-modified alloy in the same framework as the unmodified alloy, we take

$$\Delta F(c) = 2\alpha'(c)F_0(c) = 2\alpha'(c)\frac{F_{\text{core}}(c)}{C}, \quad (5.95a)$$

that is,

$$\frac{\alpha'(c)}{\alpha'} \approx \frac{F_{\text{core}}(c)}{F_{\text{core}}}. \quad (5.95b)$$

This gives

$$F_{\text{core}}(c) = \left[\left(\sum_{i=1}^6 I_i \right) (1 - c) + 6cE_{s0} \right] \mu(c)b^2, \quad (5.96)$$

where

$$E_{s0} = \frac{0.54}{2\sqrt{3}\pi^2} (\epsilon'_\mu + 1.52\epsilon_s), \quad (5.97)$$

with $\sum_{i=1}^6 I_i = 0.139$ remaining the same by the assumed similitude.

$$\text{This gives } \alpha'_0(c) = \alpha'_0 \left[(1 - c) + c \frac{(6)(0.54)}{(0.139)2\sqrt{3}\pi^2} (\epsilon'_\mu + 1.52\epsilon_s) \right], \quad (5.98a)$$

where α'_0 is that for the pure metal. Upon evaluation of the numerical factors,

$$\alpha'_0(c) = \alpha'_0 [1 + (0.682 (\epsilon'_\mu + 1.52\epsilon_s) - 1) c]. \quad (5.98b)$$

Recognizing from Table 5.1 that the effective misfit factor is negative for most alloys (except for the alloying element V), we see that modification with a solute lowers the core energy variation significantly and therefore results in solid-solution softening in most alloys.

The terminal lattice resistance $\hat{\tau}_l$ due to double-kink nucleation then becomes

$$\hat{\tau}_l = \pi \alpha'(c) \mu(c) \left(\frac{b}{\lambda} \right), \quad (5.99a)$$

where $\mu(c) = \mu (1 + c\epsilon_\mu)$ and $\alpha'(c)$ is given by eq. (5.98b).

Finally, in compact form, we have

$$\hat{\tau}_l(c) = \pi \alpha'_0 \mu (1 + c\epsilon_\mu) \left(\frac{b}{\lambda} \right) g(c), \quad (5.99b)$$

where

$$g(c) = [1 + (0.682 (\epsilon'_\mu + 1.52\epsilon_s) - 1) c]. \quad (5.100)$$

We note that when $c \rightarrow 0$ eq. (5.98b) reverts back to the form for the unmodified alloy.

Since in the low-temperature region the double-kink nucleation in the reference pure metal will be given by considerations of the *Regime II* process, where the double-kink energy governs the kinetics of nucleation, we have for the modified alloy

$$2F_k(c) = \frac{\pi}{2} \left(\frac{\mu b^3}{2} \right) \left(\frac{\hat{\tau}_1(c)}{\mu(c)} \right)^{1/2} \left(\frac{\lambda}{b} \right)^{3/2} (1 + c\epsilon_\mu), \quad (5.101)$$

where

$$\frac{\hat{\tau}_1(c)}{\mu(c)} = \pi \alpha' \left(\frac{b}{\lambda} \right) g(c). \quad (5.102)$$

This gives for the stress dependence of the activation free energy

$$\Delta G_{dk}^*(c) = 2F_k(c) \left(1 - \frac{\sigma}{\hat{\tau}_1(c)} \right)^2, \quad (5.103a)$$

and a strain rate expression of

$$\dot{\gamma} = \dot{\gamma}_0 \exp \left\{ -\frac{2F_k(c)}{kT} \left(1 - \frac{\sigma}{\hat{\tau}_1(c)} \right)^2 \right\} \quad (5.103b)$$

in the range where double-kink nucleation is the controlling process.

The determination of the overall temperature dependence of the plastic shear resistance σ requires an interactive consideration of three processes, the athermal resistance, the resistance to kink motion, and the resistance to double-kink nucleation, as discussed above. We consider this in the next section.

5.8.4 Combination of Resistances

The three resistance mechanisms, namely athermal at the plateau, kink-mobility-controlled, and double-kink-nucleation-controlled, superpose in a definite way. As already discussed above, at temperatures immediately below the plateau resistance the kink-mobility-controlled process begins to add to the overall resistance, while double-kink nucleation continues to be easy until, at an even lower temperature, the stress required for double-kink nucleation becomes comparable to that for kink mobility.

Thus, the overall strain rate $\dot{\gamma}$ is, by the kinematic-kinetic expression of Orowan (1934),

$$\dot{\gamma} = b^2 \rho_m(c) \left(\frac{\lambda}{b} \right) \frac{1}{t_{ac} + t_{adk}}, \quad (5.104)$$

where t_{ac} is the thermal activation time for the overcoming of an m_0 -fold solute cluster by a kink, which is taken also as a first approximation to the time for the

traversal of a kink over the length of a compartment Λ , and t_{adk} is the thermal activation time for double-kink nucleation. We then obtain

$$\frac{1}{t_{\text{ac}}} = \nu_{\text{G}} \exp \left\{ -\frac{m_0 E_{\text{b}}}{kT} \left(1 - \left(\frac{\sigma - \tau_{\text{a}}}{\hat{\tau}_{\text{km}}(c)} \right) \right)^{3/2} \right\}, \quad (5.105)$$

$$\frac{1}{t_{\text{adk}}} = \nu_{\text{G}} \exp \left\{ -\frac{2F_{\text{k}}(c)}{kT} \left(1 - \left(\frac{\sigma}{\hat{\tau}_{\text{l}}(c)} \right) \right)^2 \right\}, \quad (5.106)$$

and in combined form,

$$\dot{\gamma} = \dot{\gamma}_0 \left[\exp \left\{ \frac{2F_{\text{k}}(c)}{kT} \left(1 - \frac{\sigma}{\hat{\tau}_{\text{l}}(c)} \right)^2 \right\} + \exp \left\{ \frac{m_0 E_{\text{b}}}{kT} \left(1 - \left(\frac{\sigma - \tau_{\text{a}}}{\hat{\tau}_{\text{km}}(c)} \right) \right)^{3/2} \right\} \right]^{-1}, \quad (5.107)$$

where $\dot{\gamma}_0$ is given by eq. (5.86a) and the athermal resistance τ_{a} is given by eq. (5.74). While at temperatures below roughly $T_{0\text{m}}$ the first exponential term in square brackets should dominate, the opposite should be true above $T_{0\text{m}}$, where the second exponential should be dominant; nevertheless, a simple separation of the processes is not advisable, in view of the form of eqs. (5.104), (5.105), and (5.106). When comparing the combined model expression in eq. (5.107) with experiment in Section 5.9, we shall solve eq. (5.107) for the stress σ at different temperatures between 0 K and above the temperature $T_{0\text{l}}$ where the lattice resistance in the pure metal becomes negligible.

5.8.5 The Strain Rate Dependence of the Flow Stress in the Plateau Range

As discussed in Section 5.6.2, the strain rate change experiments of Takeuchi (1969) on six solid-solution alloys of Fe at room temperature in the plateau range resulted in a well-defined set of activation volumes Δv^* that decreased with increasing alloy content, as summarized in Fig. 5.17. Since the temperature dependence of the flow stress in the plateau, however, was negligible, the pronounced strain rate dependence may appear inconsistent and requires explanation. Such behavior is possible if, whereas the principal plastic resistance is governed by the athermal motion of the sluggish screw segments of expanding dislocation loops, the strain rate dependence is governed by the expanding attached edge components. The latter experience negligible lattice resistance but nevertheless need to overcome the field of individual solute atoms, while also being subjected to the long-range internal back stress of the expanding, largely parallel, screw segments. This is depicted in Fig. 5.21(a), where a typical edge dislocation segment EE' attached

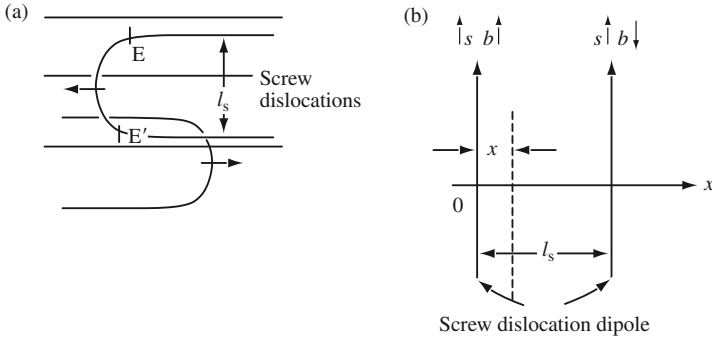


FIG. 5.21. (a) Sketch showing bowed edge segments (EE') squeezing between straight, parallel, sluggishly moving screw dislocation segments setting up an internal back stress; (b) method of computation of the area-averaged internal back stress.

to screw dislocations must advance in the back-stress field of the parallel screw dislocation segments of alternately opposite sign, which are on average a distance l_s apart. We estimate this average internal back stress $\bar{\sigma}_i$ as the area average of the local stresses of a typical screw dislocation dipole of separation l_s , as shown in Fig. 5.21(b). Defining the local back stress σ_i at a site a distance x away from one of the screws in the dipole, we have

$$\sigma_i = \frac{\mu b}{2\pi x} + \frac{\mu b}{2\pi (l_s - x)}. \quad (5.108)$$

The internal back stress $\bar{\sigma}_i$, averaged over the range l_s , experienced by a typical edge segment is then

$$\bar{\sigma}_i = \frac{1}{l_s} \frac{\mu b}{2\pi} \int_b^{l_s-b} \left(\frac{1}{x} + \frac{1}{l_s - x} \right) dx,$$

which gives

$$\bar{\sigma}_i = \frac{\mu b}{2\pi l_s} \ln \left(\frac{l_s - b}{b} \right).$$

Upon taking $l_s \approx (\rho_m)^{-1/2}$, where ρ_m is the mobile screw dislocation density, we obtain

$$\bar{\sigma}_i = \frac{\mu}{2\pi} \left(b^2 \rho_m \right)^{1/2} \ln \left(\frac{1}{b^2 \rho_m} \right). \quad (5.109)$$

Then, finally, the activation volume given by earlier arguments is

$$\frac{\Delta v^*}{b^3} = (8.71) \left(\frac{w}{b}\right)^{3/2} \left(\frac{\mu}{\hat{\tau}_e}\right)^{1/2} (1 - \beta)^{1/2}, \quad (5.110)$$

where $\hat{\tau}_e$ is the terminal resistance of the solute field to edge dislocations as derived in eq. (5.43), where $\beta = ((\sigma - \bar{\sigma}_i)/\hat{\tau}_e)$. Since $\sigma = \tau_a$, the shear resistance in the plateau region, $\beta = (\tau_a - \bar{\sigma}_i)/\hat{\tau}_e$.

In considering the thermally activated motion of edge dislocations, we return to the arguments of Section 5.4 and use a variant of eq. (5.25a) adapted to the BCC lattice spacing for dislocations moving on $\{110\}$ planes. Thus, referring to eq. (5.3a), we consider the principal resistance to the edge dislocations to come from a size misfit interaction and only secondarily from a modulus misfit interaction, as stated in eq. (5.25a), where the size-misfit-related interaction force is

$$F_{es\eta} = -\frac{2}{\pi} \left(\frac{1+\nu}{1-\nu}\right) \mu b^2 \epsilon_s \left(\frac{\Omega}{b^3}\right) \zeta_0 f'_{es}(\eta), \quad (5.111)$$

and where

$$f'_{es} = \frac{\eta}{(\eta^2 + f_0^2)^2}. \quad (5.112)$$

Taking $\zeta_0 = \sqrt{2/3} = 0.816$ as the normalized nearest glide plane spacing in the BCC lattice, we find the maximum resistance $\hat{f}'_{es} = 0.596$ at a separation of the dislocation from the solute of $\eta = 0.471$, which gives the following for the contribution to $(\hat{K}/2\mathcal{E})_s$ from the misfitting-solute interaction given by eq. (5.111) for $\Omega/b^3 = 0.770$ (the value for the BCC lattice):

$$\left(\frac{\hat{K}}{2\mathcal{E}}\right)_s = 0.434 \epsilon_s \quad (5.113)$$

Considering the secondary modulus misfit interaction of the solute with the edge dislocation to be the same as that given in eq. (5.25a), we finally have for the combined peak normalized interaction force

$$\left(\frac{\hat{K}}{2\mathcal{E}}\right) = 0.434 \left(\epsilon_s + 9.92 \times 10^{-2} \epsilon'_\mu\right). \quad (5.114)$$

Finally, $\hat{\tau}_e$ is still given by eq. (5.43), where however, the ratio w/b requires some modification for BCC metals, which we shall state when comparing the model findings with experimental results.

5.9 Comparison of Model Results with Experiments

5.9.1 The Athermal Resistance at the Plateau

Taking the athermal plastic resistance as that given in eqs. (5.74) and (5.75), we consider first a comparison of the model predictions with experimental results for Fe–Si alloys, at $T = 300$ K for solute concentrations $c = 0.050$ and 0.083 . For these alloys, the CRSS values obtained from Fig. 5.14 are 107.2 MPa and 169.9 MPa, respectively. Meanwhile, the resistances calculated from eq. (5.74) are $119(b/w)^2$ MPa and $184(b/w)^2$ MPa, respectively. To obtain agreement of the model results with the experiments, we conclude that for Fe–Si alloys b/w must be 0.968 on average. Similar consideration of other Fe alloys has established that this ratio is different for them and depends systematically on the effective misfit parameter itself. This is demonstrated in Fig. 5.22.

The form of eqs. (5.74) and (5.75) shows that the plastic shear resistance scales with the shear modulus, which depends also on the solute content. It is linearly dependent directly on the solute content and depends on the effective misfit parameter directly and again to the same degree through the ratio $(b/w)^2$, as shown in Fig. 5.23. Thus,

$$\frac{d\tau_a/\mu(c)}{dc} = \frac{3\alpha}{2} \frac{(0.54)}{\sqrt{3}\pi^2} \left(\frac{b}{w} (\epsilon' + 1.52\epsilon_s) \right)^2 (\epsilon'_\mu + 1.52\epsilon_s). \quad (5.115)$$

The experimentally established rate of change of the normalized shear resistance with solute concentration as a function of the effective modulus misfit parameter is shown in Fig. 5.23 by the filled symbols. The slope of the curve is quite close

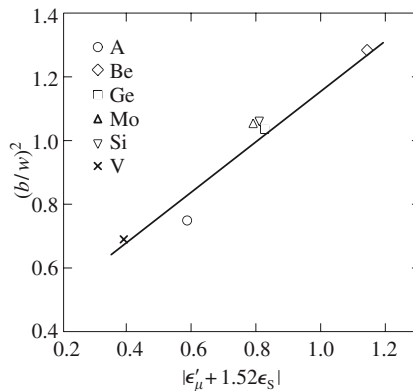


FIG. 5.22. Dependence of $(b/w)^2$ on the effective misfit parameter of solutes interacting with screw dislocations in six Fe alloys, as determined from a match of the model and experiments for the athermal resistance.

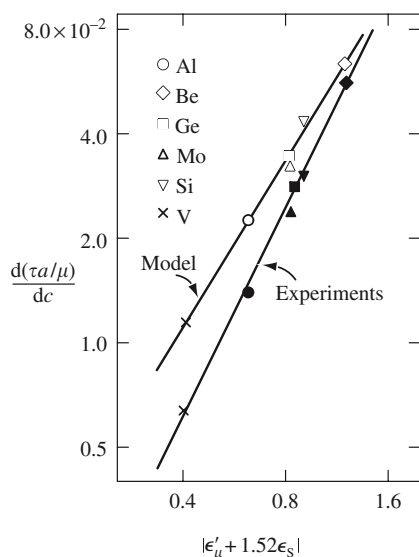


FIG. 5.23. Dependence on alloy concentration of the normalized CRSS at the plateau on the effective misfit parameter for six Fe alloy single crystals; experimental results (solid symbols) from Takeuchi (1969).

to 2, indicating that the dependence is quadratic, which is in keeping with the form of eq. (5.115) and Fig. 5.22. The model predictions of eq. (5.115) are also plotted in Fig. 5.23, by open symbols. The departure of the model predictions from the experimental results is due to the fact that the determined linear dependence of $(b/w)^2$ on the effective misfit parameter involves also a slight shift. Nevertheless, the agreement is quite satisfactory. The source of the dependence of b/w on the effective misfit parameter, however, remains unclear.

5.9.2 Kink-mobility-controlled Plastic Resistance

As discussed in Section 5.8.2, below the athermal resistance range (below T_{0m}), the plastic resistance is governed by the rate at which independent segments of length Λ between impenetrable obstacles overcome solute clusters containing a definite number, m_0 of excess solute atoms. These clusters are overcome by thermal assistance under stress as a kink of width w_k penetrates through them. Here, in a comparison with the experiments of Takeuchi *et al.* (1968), we consider in some detail only the case of the two Fe–Si alloys with solute concentrations of 0.050 and 0.083, as a typical example of other substitutional Fe alloys. We recall also from Section 5.8.2 that the impenetrable obstacles, which are thought to be encounters of gliding screw dislocations with forest screw dislocations, are spaced a distance

of $\Lambda = 1.2 \times 10^{-4}$ cm apart, on the basis of experimentally measured dislocation densities at yield of the alloy with a Si concentration of 0.050. Moreover, we recall that the typical kink width w_k is $12.6b$ in Fe on the $\{110\}$ glide planes and that the number m_0 of excess solute atoms in the largest cluster that can be sampled by a kink within the distance Λ is 5.7 and 7.34, respectively, for the two alloys.

The comparison is started by obtaining from eq. (5.81) the terminal plastic resistance $\hat{\tau}_{km}$ for glide on the $\{110\}$ planes, using $\lambda/b = 1.44$ and appropriate values of the misfit parameter and shear modulus, as 151.5 MPa and 182.2 MPa, respectively, for the two selected Fe–Si alloys.

The temperature dependence of the resistance component $\sigma_{km}(T)$ governed by kink mobility can then be determined next in the usual manner through the use of eqs. (5.69), (5.70), and (5.72), which gives

$$\sigma_{km}(T) = \hat{\tau}_{km} \left(1 - \frac{T}{T_{0m}} \right), \quad (5.116)$$

where

$$T_{0m} = \frac{\Delta G_{0m}^*}{k \ln(\dot{\gamma}_0/\dot{\gamma})} \quad \text{and} \quad \frac{\Delta G_{0m}^*}{k} = \frac{m_0 E_b}{k}. \quad (5.117a,b)$$

For the alloy with $c = 0.050$, using appropriate values $|\epsilon'_\mu + 1.52\epsilon_s| = 0.815$ (Table 5.1), $\mu(c) = 7.17 \times 10^4$ MPa (Fig. 5.15), $m_0 = 5.7$, $\rho_m = 5 \times 10^8/\text{cm}^2$ (Fig. 5.16), $\nu_G = 5 \times 10^{11}/\text{s}$ we determine that

$$\frac{\Delta G_{0m}^*}{k} = 5822 \text{ K} \quad (5.118a)$$

and

$$\ln \left(\frac{\dot{\gamma}_0}{\dot{\gamma}} \right) = \ln \left(b^2 \left(\frac{\lambda}{b} \right) \rho_m \frac{\nu_G}{\dot{\gamma}} \right) = 20.21, \quad (5.118b)$$

giving, for the imposed machine strain rate of $\dot{\gamma} = 3.7 \times 10^{-4}/\text{s}$ used in the experiments of Takeuchi (1969), $T_{0m} = 288 \text{ K}$.

Similar considerations for the second Fe–Si alloy, with $c = 0.083$, give $T_{0m} = 328 \text{ K}$.

In summary, we have the following temperature-dependent plastic resistances for the two selected alloys: for $c = 0.050$,

$$\sigma_{km}(T) = 151.5 \left(1 - \left(\frac{T}{288} \right)^{2/3} \right), \quad (5.119a)$$

and for $c = 0.083$

$$\sigma_{km}(T) = 182.2 \left(1 - \left(\frac{T}{328} \right)^{2/3} \right), \quad (5.119b)$$

for the machine strain rate of $\dot{\gamma} = 3.7 \times 10^{-4}$ /s. Relations for other strain rates require determination of other appropriate T_{0m} values from eqs. (5.117a,b) and (5.118a,b).

As explained in Section 5.8.2, these resistances determined for kink-velocity-governed processes, primarily the process of overcoming large solute clusters by kinks, are superimposed on the athermal resistance in the entire temperature range below T_{0m} , down to temperatures where double-kink nucleation becomes the controlling process.

5.9.3 Double-kink-nucleation-controlled Resistance

As discussed in Section 5.8.4, below a certain temperature the activation times for double-kink nucleation begin to exceed the activation times for kinks overcoming solute clusters and the former begin to control the plastic resistance by the considerations developed in some detail in Section 5.8.3. To deal with this transition to double-kink nucleation control and finally to develop forms for the temperature dependence of the plastic resistance (CRSS) from 0 K to the athermal resistance range, we evaluate the expression in eq. (5.107) to provide a comparison with experiment. For this purpose, as a representative case, we again select the two Fe–Si alloys with solute concentrations of $c = 0.050$ and 0.083 .

First we state eq. (5.107) in a more transparent form,

$$\exp \left\{ \frac{2F_k(c)}{kT} \left(1 - \frac{\sigma}{\hat{\tau}_1} \right)^2 \right\} + \exp \left\{ \frac{m_0 E_b}{kT} \left(1 - \frac{\sigma - \tau_a}{\hat{\tau}_{km}} \right)^{3/2} \right\} = \frac{\dot{\gamma}_0}{\dot{\gamma}}, \quad (5.107a)$$

where

$$\dot{\gamma}_0 = b^2 \rho_m(c) \left(\frac{\lambda}{b} \right) v_G. \quad (5.120)$$

Thus, at intermediate temperatures where double-kink nucleation is readily accomplished, the first exponential becomes relatively negligible and the expression reverts to eq. (5.84), while at low temperatures, where double-kink nucleation is the difficult step, the second exponential is relatively negligible and the expression reverts to eq. (5.103b). For the case of $c = 0.050$, we take the following values: $\rho_m = 5 \times 10^8/\text{cm}^2$ (Fig. 5.16), $m_0 = 5.7$, $\mu(c) = 7.17 \times 10^4$ MPa (Fig. 5.15), and $|\epsilon'_\mu + 1.52\epsilon_s| = 0.815$. These values give for $c = 0.050$ $\dot{\gamma}_0/\dot{\gamma} = 5.98 \times 10^8$, $\hat{\tau}_1(c') = 487$ MPa, $2F_k/k = 8870$ K, $m_0 E_b/k = 5822$ K, $\tau_a = 107.2$ MPa (Section 5.9.1), and $\hat{\tau}_{km} = 151.5$ MPa (Section 5.9.2). Correspondingly, for $c = 0.083$, we take $\rho_m = 1.5 \times 10^9/\text{cm}^2$ (Fig. 5.16), $m_0 = 7.34$, and $\mu(c) = 6.69 \times 10^4$ MPa (Fig. 5.15). These values give, in turn, for $c = 0.083$, $\dot{\gamma}_0/\dot{\gamma} = 1.79 \times 10^9$, $\hat{\tau}_1(c') = 439$ MPa, $2F_k/k = 8130$ K, $m_0 E_b/k = 6998$ K, $\tau_a = 169.9$ MPa (Section 5.9.1), and $\hat{\tau}_{km} = 182.1$ MPa.

With these values, the solution of eq. (5.107a) gives the overall temperature dependence of the plastic resistance for the entire range of processes from double-kink nucleation control at the lowest temperatures to the combination of processes of kink mobility-control and athermal resistance. This is plotted in Fig. 5.24 for the two alloys. The available data points from Takeuchi *et al.* (1968) are also plotted on the figure. We note first that according to the model, for the alloy with $c = 0.050$, the transition from kink mobility control (dotted curve) to kink nucleation control (solid curve) occurs roughly at $T = 225$ K, delineated by the dashed vertical line, while the available data points tend to follow the kink-mobility-controlled resistance model down to nearly 125 K. For the alloy with $c = 0.083$, the transition from kink mobility control (dashed curve) to double-kink nucleation control (dash-dotted curve) is suppressed down to roughly 20 K, delineated by the second dashed vertical line; that is primarily due to the more substantial effect of solute softening on the lattice resistance in this alloy of higher solute concentration. The few available

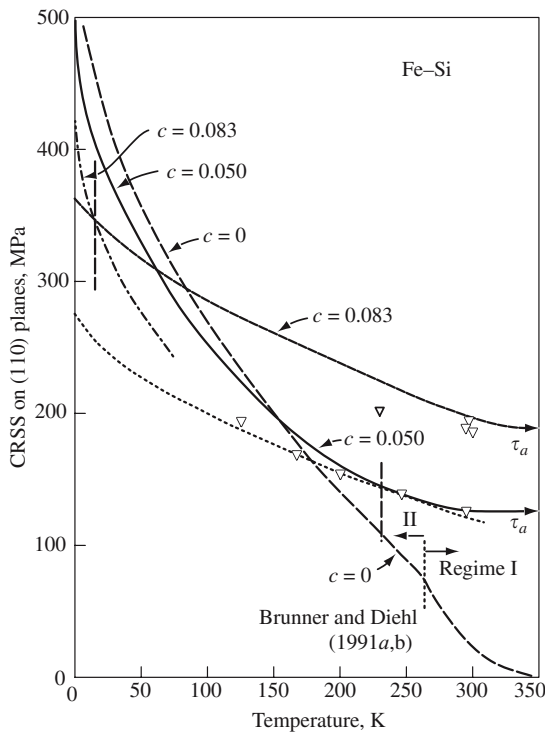


FIG. 5.24. Dependence of the calculated CRSS on temperature between 0 K and the edge of the plateau region, for two Fe-Si alloys with concentrations of $c = 0.05$ and 0.083 , as determined from the model. Also shown are the temperature dependence of the CRSS of pure Fe and data pertaining to the alloys.

data points lie somewhat under the model predictions but tend to parallel again the kink-mobility-controlled resistance. As Fig. 5.13 shows, in nearly all alloys twinning limits the rise of the plastic resistance at around 75 K.

Finally, the model shows that the temperature-dependent curves of the combined resistances for the two alloys cross at a temperature of 65 K, below which the alloy with the higher solute concentration exhibits a lower plastic resistance; this is principally due to the lowered resistance to double-kink nucleation, demonstrating a clear solid-solution-softening effect.

5.9.4 Strain Rate Dependence of the Flow Stress in the Plateau Region, and Activation Volumes

Here our principal comparison will be of the activation volumes measured in the plateau range by Takeuchi (1969) and the model predictions of Section 5.8.5 that attribute the strain rate dependence of the flow stress to motion of edge segments through the solute field between sluggishly moving screw dislocation segments. The model estimates of the activation volume are based on eq. (5.110). The key element in the determination of the activation volumes from the model is an estimate of the internal stress set up by the parallel mobile screw dislocation segments based on eq. (5.109), where the required mobile dislocation density is taken from the TEM measurements of Takeuchi (1969) of the screw dislocation density at the CRSS. The dislocations observed here have been interpreted to be all mobile—clearly a debatable assumption, the lack of reliability of which is reflected in the starred (*) entries under $(1 - \beta)^{1/2}$ in Table 5.2, where the use of eqs. (5.109) and (5.111) gave unacceptable values. These values have been rejected. The remaining values of this quantity based on the magnitudes of parameters given in Table 5.2 are considered to be generally reliable, since they represent a consistent trend as a function of solute concentration. This trend is shown in the upper left inset of Fig. 5.25. For the alloys that gave unreliable determinations of $(1 - \beta)^{1/2}$, the trend of the inset was used to obtain a reasonable value for this quantity based on the solute content. With this proviso, the values of the relevant parameters for the ten Fe alloys listed in Table 5.2 were used to calculate activation volumes at the CRSS from eqs. (5.110), (5.111), and (5.114). These calculated activation volumes, given in Table 5.2, together with the experimentally determined values, shown in Fig. 5.17 and also listed in Table 5.2, have been plotted against each other in Fig. 5.25. While there is considerable scatter, no doubt due to experimental errors as well as minor inadequacies of the model, the relationship between the model and the experiments (with the exception of the dilute Fe–V alloy) indicates that the rate dependence of the flow stress in the plateau must indeed come from the kinetics of motion of edge dislocation segments between sluggishly moving parallel screw dislocation segments.

TABLE 5.2. Parameters required to determine activation volumes for edge dislocation motion in Fe alloys

Alloy	c	μ (GPa)	w/b^a	$ \epsilon_s + 0.992 \times 10^{-3} \epsilon'_\mu $	ρ_m (cm^{-2}) ^b	$\bar{\sigma}_i$ (MPa) ^c	τ_a (MPa) ^d	$\hat{\tau}_e$ (MPa) ^e	$(1 - \beta)^{1/2}$	$\Delta v^*/b^3$ (model)	$\Delta v^*/b^3$ (experiment)
Fe–Al	0.0585	70.5	1.163	0.0700	3.33×10^8	78.23	73.68	133.8	*	127	160
Fe–Be	0.0343	72.1	0.882	0.0873	4.163×10^8	88.16	148.6	117.3	0.696	123.4	140
Fe–Be	0.0662	66.6	0.882	0.0873	1.181×10^9	126.6	268.4	168.1	0.396	69.35	90
Fe–Ge	0.0346	71.7	1.005	0.0753	4.69×10^8	93.9	83.16	100.7	*	152.2	130
Fe–Ge	0.0767	64.1	1.005	0.0753	5.96×10^8	91.56	163.2	152.9	0.729	75.46	90
Fe–Mo	0.0179	73.3	0.971	0.1126	3.084×10^8	78.67	45.47	112.1	*	153.4	160
Fe–Si	0.050	71.7	0.968	0.0315	5.730×10^8	100.7	126.3	39.79	0.597	193.7	120
Fe–Si	0.083	66.9	0.968	0.0315	1.500×10^9	142.2	185.5	52.01	0.4139	119.0	60
Fe–V	0.0208	76.4	1.210	0.042	1.502×10^8	59.88	18.95	35.76	*	380.5	220
Fe–V	0.1043	77.5	1.210	0.042	2.44×10^8	75.1	83.37	110.9	0.962	88.87	170

^a Determined from Fig. 5.22.

^b From Fig. 5.16,

^c Eq. (5.109).

^d From Fig. 5.14.

^e Eq. (5.43).

* Unreliable estimates due to unreliable determinations of $\bar{\sigma}_i$.

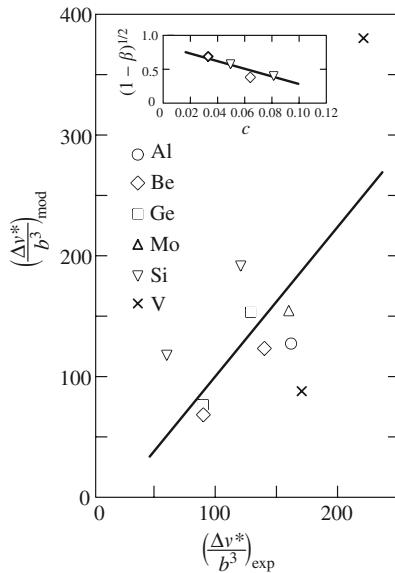


FIG. 5.25. Calculated versus measured activation volumes determined by strain rate change experiments at room temperature for six Fe alloy single crystals with different alloy contents. The upper left inset shows the dependence of the factor $(1 - \beta)^{1/2}$ on solute content for Be and Si alloys, used to establish the trend for all alloys.

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References for Further Study in Depth on Solid Solution Resistance

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PRECIPITATION STRENGTHENING

6.1 Overview

Discrete obstacles, particularly in the form of precipitates of second phases with coherent or incoherent interfaces, and dispersed crystalline or glassy nanoscale particles, but also aggregates of point defects of various types and even nanoscale cavities, can all interact with dislocations to act as slip obstacles that can quite significantly increase the plastic shear resistance of a metal or alloy. Since the shear strength of many precipitates can increase when their size is increased by aging the alloy at an elevated temperature, precipitate strengthening has proved to be a very flexible and powerful industrial process. In other instances, in certain alloys which possess binodals in their solid state enclosing a spinodal region that can undergo decomposition into different spatially interconnected nanoscale domains of different elastic properties, dislocation motion can also be significantly impeded, resulting in strengthening.

Much attention was given to the thermodynamics and kinetics of the precipitation of second-phase particles and the factors controlling their sizes and shapes, well before any mechanistic understanding was developed of their effect on plastic resistance. For a more recent general discussion of precipitation, dealing with the thermodynamic and kinetic aspects of the formation of second phases either by nucleation or growth of precipitates or by spinodal decomposition, the reader is referred to Martin and Doherty (1996) and also to Wagner and Kampmann (1991), who also provide an extensive guide to key references in this field.

We shall briefly touch on the most important aspects of this subject of precipitation in Section 6.2 to develop an essential insight into the factors controlling discrete precipitate formation, as well as the decomposition of alloys in unstable regions of equilibrium into different domains by spinodal decomposition. This will provide some necessary guidance on the factors that control the strengthening mechanisms associated with precipitate coarsening and the like.

In the remainder of this chapter, the main emphasis will be on a number of mechanisms by which precipitates of various different types and other discrete nanoscale heterogeneities affect the plastic resistance of alloys. This subject saw much pioneering attention by a number of earlier investigators, who laid down the foundations and assessed the early developments. These assessments include the reviews of Kelly and Nicholson (1963), Brown and Ham (1971), and Gerold (1979)

and the more recent reviews of Ardell (1985, 1994), Nembach and Neite (1985), and Reppich (1993). While this chapter will make much use of these assessments, consulting these works directly is also highly advisable.

In Section 6.3, a brief catalog will be presented of the principal forms of interaction between discrete particles and dislocations, as well as the interactions of other heterogeneities acting as particles, including an assessment of the possible extent of thermal activation in overcoming such obstacles.

In Section 6.4, the various ways in which glide dislocations sample discrete obstacles, ranging from the more embryonic forms to fully mature dispersoids, will be presented. Here, we present the relevant theoretical details of specific, individual interaction mechanisms of precipitates with dislocations to develop models for plastic resistance and compare these with corresponding experimental results.

6.2 Formation of Second Phases in the Form of Precipitate Particles, Heterogeneous Domains, or other Lattice Defect Clusters

6.2.1 Discrete Precipitates

When a solid solution is cooled below its solvus temperature, a new solid phase will begin to have a lower chemical free energy than that of the subcooled, supersaturated solid solution and will become metastable to composition fluctuations. As the chemical free-energy difference increases with decreasing temperature below the solvus, an increasing thermodynamic driving force develops for precipitation of a new phase. However, the formation of a precipitate with a separate structure will also involve the introduction of an interface between the two component phases that opposes the formation of a nucleus of the new phase by raising the free energy. This keeps the subcooled solid solution metastable to small compositional fluctuations. For sufficiently large compositional fluctuations possessing the structure of the new phase locally and thereby also possessing the appropriate interface, the free-energy barrier can be overcome to form a nucleus of the new phase. If this nucleus, in its simplest form, involves no significant dilatational or shape misfit and if its elastic properties can be assumed to be roughly the same as those of the solid solution and to be isotropic, the precipitate of the new phase will have a spherical shape.

Under the above rather idealized conditions, the total Gibbs free-energy change ΔG of the system of the supersaturated-solution matrix and the precipitate involves a chemical free-energy reduction ΔG^v per unit volume and an increase of the interface energy χ_i per unit interface area of the possible nucleus. The chemical free-energy difference will increase with decreasing temperature below the solvus, depending on the concentration contour of the solvus, while the interface energy will be relatively independent of temperature. The free-energy difference ΔG associated with the formation of a nucleus of radius r of the new phase described by

the classical “droplet model” of homogeneous nucleation will then be

$$\Delta G = 4\pi r^2 \chi_i - \frac{4\pi r^3}{3} \Delta G^v(T, c), \quad (6.1)$$

where the first term on the right-hand side describes the increase in interface energy and the second term describes the chemical free-energy difference between the precipitate phase and the supersaturated solid-solution matrix. Figure 6.1 shows schematically the change of the overall free energy of the system with increasing nucleus radius r at constant pressure p , temperature T , and solute concentration c . Clearly, concentration fluctuations that result in precipitate nuclei smaller than r^* will not be stable and will dissolve again to decrease the free energy. Only fluctuations that result in nuclei with radii equal to or larger than r^* have the ability to grow and reduce the free energy ΔG . The critical-size nucleus that renders the free energy maximum and establishes the free-energy barrier ΔG^* to nucleation requires that

$$\left(\frac{\partial \Delta G}{\partial r} \right)_{p,T,c} = 0, \quad (6.2)$$

giving

$$r^* = \frac{2\chi_i}{\Delta G^v(T, p, c)}, \quad (6.3)$$

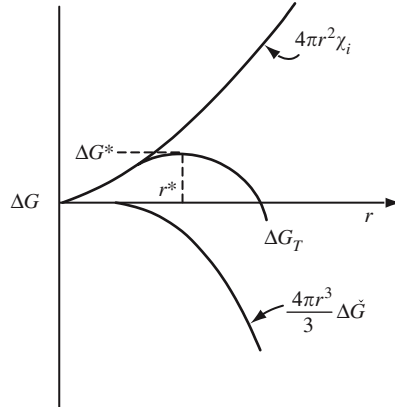


FIG. 6.1. The conventional “droplet model” for nucleation of a new phase from a supersaturated mixture, involving a critical balance of interface energy and volume chemical free energy when the overall free energy reaches a maximum with increasing radius of the nucleus.

which, upon substitution of eq. (6.3) into eq. (6.1), gives

$$\Delta G^* = \frac{16\pi}{3} \frac{\chi_i^3}{(\Delta G^v(T, p, c))^2}. \quad (6.4)$$

Equation (6.3) demonstrates that the larger the interface free energy χ_i , the larger will be the critical nucleus and the more substantial will be the activation energy ΔG^* required to form this critical-size nucleus.

To appreciate the probability of formation of such a nucleus, it is instructive to estimate r^* and ΔG^* for a typical case. For this we consider formation of Al_2Cu precipitates in an Al-Cu solid solution α with undercooling below the solvus. The aluminum-rich part of the Al-Cu equilibrium diagram is shown in Fig. 6.2. At the eutectic temperature of 548°C the solubility of Cu is somewhat in excess of 2 atom %. Consider a specific alloy with $c = 0.016$ of Cu (4 wt% Cu). In the usual precipitation reaction, the α solid solution is quenched to a temperature of about 150°C and would be held there for isothermal precipitation. The solvus temperature for this alloy composition is close to 500°C . Thus the precipitation reaction considered is at an undercooling of 350°C . The chemical free-energy

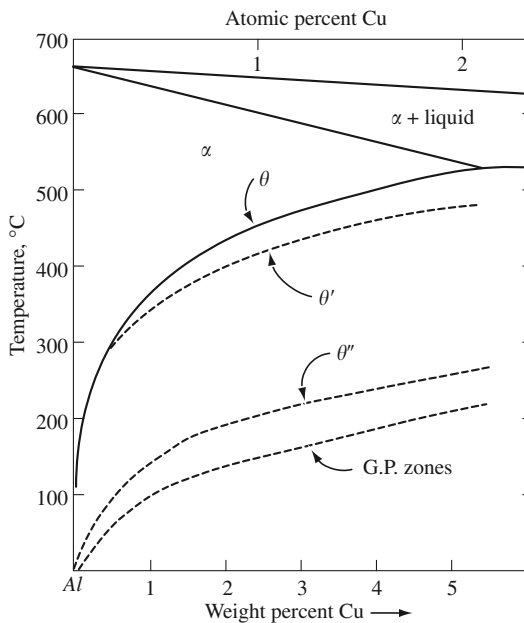


FIG. 6.2. The Al-Cu equilibrium diagram at its Al-rich end, showing various intermediate forms of $\text{Al}_x\text{-Cu}$ compounds θ , θ' , θ'' , and G. P. zones.

change $\Delta G_{\alpha\theta}^v$ between the α solid solution and the θ phase of Al_2Cu expected to precipitate is given by (Doherty 1996) as

$$\Delta G_{\alpha\theta}^v = \frac{c_\theta - c_\alpha}{1 - c_\alpha} \frac{RT}{V^m} \ln \left(\frac{c_\alpha}{c_0} \right), \quad (6.5)$$

where c_θ is the atomic concentration of Cu in the phase Al_2Cu , that is, $c_\theta = 0.33$, c_α is the atomic concentration of Cu at the solubility limit of the α solid solution at the precipitation temperature of 150°C , that is, $c_\alpha = 4 \times 10^{-4}$, R is the universal gas constant, T is the precipitation temperature of 423 K, and V^m is the molar volume of the Al_2Cu θ phase, estimated to be equal to $M_{w\theta}/\rho_\theta = 32.16 \text{ cm}^3/\text{mol}$ for $M_{w\theta} \cong 117.5 \text{ g/mol}$ and $\rho_\theta \cong 3.65 \text{ g/cm}^3$. In eq. (6.5), $c_0 = 1.6 \times 10^{-2}$ is the atomic concentration of Cu in the alloy chosen. With these values, $\Delta G_{\alpha\theta}^v = -1.50 \times 10^2 \text{ J/cm}^3$. Little is known about the interface energy χ_i between α and the Al_2Cu phase. On the assumption that the interface is likely to be incoherent, one might take as a crude estimate for the energy about one-third of the surface free energy of molten Al, that is, $\chi_i = 3 \times 10^{-5} \text{ J/cm}^2$. These estimates for $\Delta G_{\alpha\theta}^v$ and χ_i give $r^* = 40 \text{ \AA}$ and $\Delta G^* = 2.03 \times 10^{-17} \text{ J}$, or 125.7 eV.

This rather crude argument establishes quite clearly that the critical nucleus under a condition of homogeneous nucleation that involves equilibrium estimates of the interface energy will be exceedingly large and will involve an unattainably large activation energy. Interestingly, this conclusion is quite general and applies to many other cases of homogeneous nucleation as well, leading to a general conclusion that the critical nucleus must be made up of nonequilibrium transitional forms of matter such as diffuse interfaces or embryos of much lower energy. Thus, for example, if the critical nucleus were of a shape that permitted a coherent interface of much lower energy, near the energy of a twin boundary or stacking fault, the energy of the nucleus could be dramatically reduced. Consider, for example, an energy χ_i of the order of an intermediate-range stacking-fault energy, that is, about 50 mJ/m^2 . This would reduce r^* to $6.7 \text{ \AA}$ and ΔG^* to $9.31 \times 10^{-20} \text{ J}$, or only 0.58 eV. This is in fact what happens in the precipitation of a Cu-rich component from a supersaturated Al–Cu solid solution. Guinier (1938) and Preston (1938) found that the first observable form of precipitate consisted of a disk-shaped single layer of Cu atoms roughly $90 \text{ \AA}$ in diameter between $\{001\}$ planes of the Al lattice. These earliest forms of “precipitates” subsequently coarsen into thicker disks, referred to as θ'' and θ' precipitates of Al_2Cu . The solvus contours of these precipitate forms are shown in the partial-equilibrium diagram of Fig. 6.2 (Lorimer 1978).

As stated above, when substantial dilatational and shape misfit is involved in the precipitate and when the precipitate has a crystal structure of different symmetry and different elastic properties from those of the matrix, the precipitate shape of lowest energy will depart significantly from a sphere. Estimates of the effects of the elastic strain energy on the shape of precipitates have suggested that unless the elastic strain energy density becomes a significant fraction of the chemical

free-energy change, in the range of 0.5 or so, the shape of the precipitate is likely to be spherical or at least equiaxed (Aaronson *et al.* 1978).

In the following considerations in this chapter, we shall idealize the shapes of discrete precipitates as spherical. In cases where precipitates involve significant dilatational or shape misfit, precipitation is likely to become heterogeneous along dislocations, grain boundaries, or other incoherent interfaces. Such cases have been discussed by Doherty (1996), and will not be considered here.

6.2.2 Spinodal-decomposition Domains

A different process of second-phase formation occurs in some binary and ternary alloys, where a binodal-type miscibility gap is present in either the liquid region or the solid state, such as in the Au–Ni, Au–Pt, and Al–Zn systems in the case of the solid state. Consider such a system with a prominent binodal in a binary equilibrium diagram, such as in the case sketched in Fig. 6.3(a), which represents the state either in a liquid or in a solid solution. At a temperature T_0 where a continuous liquid or a continuous solid solution exists, the chemical free energy as a function of alloy composition is concave upward everywhere, as shown in Fig. 6.3(b), with a minimum in the midrange because of the higher configurational (mixing) entropy occurring there. As the temperature decreases below the critical temperature T_c , and a miscibility gap develops, either in the liquid or in the solid solution, as shown, for example at a temperature T_1 , alloys with compositions c_a and c_b at their solvus reach borderline metastability. Thus, for example, for $c'_0 > c_a$, the chemical free energy begins to exceed that at $c_0 = c_a$ as shown in Fig. 6.3(b), and a thermodynamic driving force develops toward precipitation in relation to the reference state at T'_1

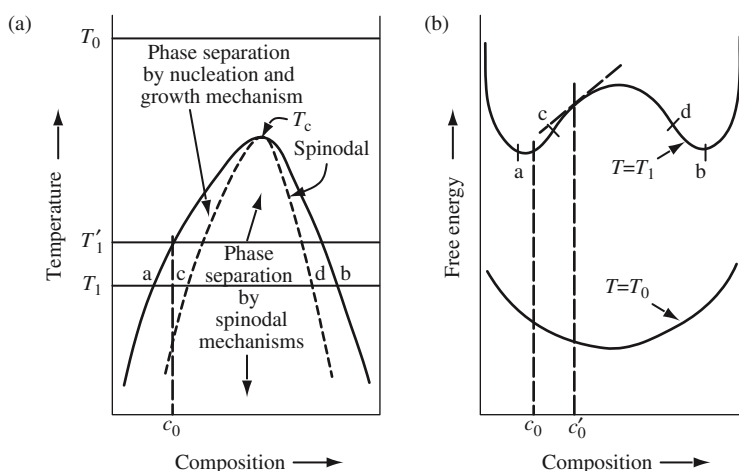


FIG. 6.3. A pair of associated equilibrium phase diagrams (a), and free-energy diagrams (b) in a typical system exhibiting a bimodal (after Cahn 1968).

on the solvus for this composition, creating conditions for a regular precipitation reaction through the formation of a critical nucleus as described above. However, at $T = T_1$ for solute compositions $c'_0 > c_c$ where an inflexion develops in the free-energy/composition profile, a new and different condition sets in. Thus, for $c_c < c'_0 < c_d$, where the curvature of the free-energy/composition profile becomes negative, the alloys in this range lose metastability and become unstable. This inner region of the binodal is referred to as a *spinodal*. For compositions inside the spinodal region, any local virtual variation away from the average concentration c'_0 would result in a decrease in chemical free energy, as indicated by the tangent line constructed on the free-energy diagram. Thus, in this region any compositional perturbation would grow and result in a diffusion-controlled, progressive phase separation into two domains with compositions c_c and c_d , developing an interface gradually and not through nucleation. The theory of spinodal decomposition was pioneered by Cahn and Hilliard, who have summarized these developments separately in a lucid manner (Cahn 1968; Hilliard 1970).

A principal departure of the spinodal-decomposition reaction is that the interdiffusion of species is no longer driven by a random walk down a concentration gradient as in a dilute environment, but rather up a concentration gradient through the dominant attractive interactions of like species toward each other, increasing the concentration gradient further. This is reflected in the effective diffusion constant $\tilde{D}$, which becomes

$$\tilde{D} = D_0 \left(\frac{\partial^2 f}{\partial c^2} \right) \exp \left(\frac{-Q}{RT} \right), \quad (6.6)$$

where $f(c)$ is the free energy of a molecular cluster of composition c . Thus, while the usual diffusional mobility factor remains positive, in the spinodal region ($\partial^2 f / \partial c^2$) is negative, making the effective diffusion constant also negative (Cahn 1968). The final spinodal-decomposition reactions, affected by a variety of factors including coherency strains and elastic anisotropy of the evolving components, results in an interpenetrating domain morphology with a structural wavelength typically in the range of 100 Å, as shown in Fig. 6.4 for the case of a Cu–5 wt% Ti alloy aged at 723 K for 100 minutes.

The development of interpenetrating domain morphologies that is a characteristic feature of spinodal decomposition is not found only in crystalline metallic alloys but also in certain amorphous metal alloys, covalent glasses, and glassy polymers. In this chapter, however, we shall consider only the strengthening effects in crystalline metallic alloys in Section 6.4.8.

6.2.3 Defect Clusters and Nanovoids

Entities similar to discrete precipitates also commonly occur through clustering of point defects, particularly vacancies, into a variety of defect clusters. These can

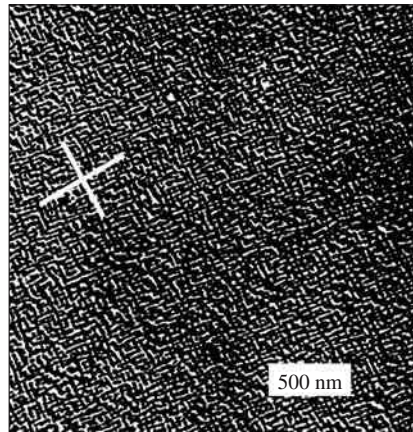


FIG. 6.4. Electron micrograph of a spinodal microstructure of a Cu–Ni–Fe alloy, viewed on a $\{001\}$ plane face. The arms of the white cross are parallel to other $\{001\}$ plane faces (micrograph courtesy of D. Laughlin).

develop through rapid quenching from elevated temperatures where the thermal-equilibrium concentration of vacancies is high, but are also found widely in alloys irradiated by high-energy particles, which produce point defects copiously by *knock-on* events, and these defects subsequently collapse into defect clusters of various forms. In some other cases, particularly when Ni alloys are subjected to high fluences of energetic neutrons, which result in transmutation products that often include gaseous He, irradiation results in He-filled nanovoids, which act as very effective obstacles to dislocation motion and cause severe radiation-induced loss of ductility as well as swelling.

We shall consider some examples of such defect clusters and nanovoids, as they cause significant elevation of plastic resistance.

6.3 Sampling of Precipitates by Dislocations

6.3.1 Precipitate Shapes and Sizes

As stated in Section 6.1, we idealize all precipitates to be of spherical shape. Even with this idealization, it must be recognized that since precipitates tend to grow at a constant volume fraction, this will be at the expense of some that will shrink, establishing a characteristic size distribution at all times. Moreover, since precipitates can be probed by dislocations in slip planes at various levels away from their center, they will present circular cross sections of different sizes on the slip plane even if all precipitates are of the same size. For a first consideration, where the growth-related size distribution will be strongly skewed toward the large particle

size range, we shall consider all precipitates to have sizes close to the maximum of the distribution and take the actual radius of the precipitate to be r_0 . Then, the most probable radius $\langle r \rangle$ of the circular area of a random section of such a precipitate by any slip plane should be

$$\langle r \rangle = \int_0^{r_0} r_0 \left(1 - \left(\frac{z}{r_0} \right)^2 \right)^{1/2} \frac{dz}{r_0} = \frac{\pi}{4} r_0, \quad (6.7)$$

where z is the distance of the section from the precipitate center. Similarly, the most probable area $\langle a \rangle$ exposed by an intersecting slip plane will be

$$\langle a \rangle = \pi \langle r^2 \rangle = \pi \int_0^{r_0} r_0^2 \left(1 - \left(\frac{z}{r_0} \right)^2 \right) \frac{dz}{r_0} = \frac{2\pi}{3} r_0^2 = \frac{32}{3\pi} \langle r \rangle^2. \quad (6.8)$$

This establishes immediately the average distance l between precipitate centers on the slip plane for a precipitate volume fraction of c , as

$$l = \left(\frac{\langle a \rangle}{c} \right)^{1/2} = \langle r \rangle \left(\frac{32}{3\pi c} \right)^{1/2} = r_0 \left(\frac{2\pi}{3c} \right)^{1/2}. \quad (6.9)$$

6.3.2 Two Forms of Interaction of Precipitates with Dislocations

There are two fundamentally different forms of interaction of dislocations with precipitates. In the first category, which we label *hard-contact interactions*, the dislocation senses the precipitate only on contact. Such interactions include interactions with precipitates that have elastic properties closely similar to those of the matrix and have incoherent interfaces that prevent a dislocation from smoothly entering the precipitate. This category also includes interactions with precipitates that have coherent interfaces but upon being sheared offer resistance due to a substantial internal lattice resistance, or require the formation of some internal fault such as a stacking fault or antiphase boundary and also require the formation of an interface ledge resulting from shearing.

In the second category are those precipitates and other defect clusters, including nanovoids, that interact with the strain field of a dislocation through their size misfit or modulus misfit, owing to significantly different elastic properties compared with those of the matrix. These interactions, which we label *soft-contact interactions*, have effective interaction cross sections larger than the actual size of the precipitate.

In the following section, we deal first with hard-contact interactions, which for a dilute precipitate volume fraction c can be idealized as interactions with points, that is, $\langle r \rangle / l \ll 1.0$. This idealization will permit the development of some generally useful statistical concepts.

6.3.3 Statistics of Sampling Random Point Obstacles in a Plane

Consider, as depicted in Fig. 6.5(a), a set of shearable precipitates of relatively small volume fraction c , all of the same kind, dispersed randomly at an average distance $l (\gg \langle r \rangle)$ from each other in a slip plane, interacting with the dislocation only on contact. When idealized as point obstacles, the precipitates will make no

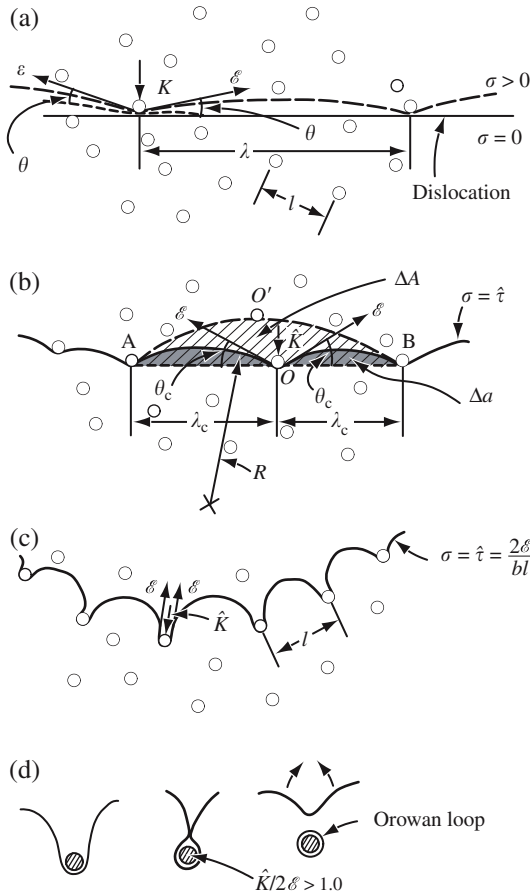


FIG. 6.5. Schematic of a dislocation advancing in a field of weak pointlike slip obstacles, probed by the dislocation on contact: (a) just below the flow condition, the dislocation probes obstacles a distance λ apart; (b) at flow, the dislocation penetrates through the central obstacle at O when $K = \hat{K}$; (c) for very strong obstacles, the mean distance between contacted obstacles is equal to the mean obstacle spacing in the plane; (d) impenetrable obstacles are circumvented when dislocations squeeze between obstacles, leaving "Orowan" loops around them.

contact with a straight dislocation under no applied stress (solid line in Fig. 6.5(a)). When a shear stress is applied, exerting a force on the dislocation, the latter will advance, make contact with some precipitates, and begin to bow between them, as depicted in Fig. 6.5(a) by the slightly cuspy dashed curve. This dislocation will stop when the force on it evokes resistive forces K from the precipitates that have been contacted. At a typical contact point, force equilibrium is reached when the force f exerted by the line tension of the bowed segments of the dislocation, making angles θ relative to the line of first tangency at contact with the precipitate, is balanced by the resistive force K of the precipitate, that is, when

$$f = 2\mathcal{E} \sin \theta = K, \quad (6.10)$$

as shown in Fig. 6.5(a) on the left. The average distance λ between the contacted precipitates along the dislocation line will initially be much larger than the average distance l between precipitate centers in the plane. With increasing stress, the dislocation will bow out further into the field and make contact with more precipitates, and the average distance λ between contacts will decrease. Eventually, the forces exerted on the precipitates will reach their peak resistive force $\hat{K}$ and the dislocation will shear through them as the shear resistance $\hat{\tau}$ of the obstacle field is overcome. As Friedel (1956) has noted, under this condition when flow is reached and a typical dislocation can sweep through the obstacle field unhindered, the advancing segments of the dislocation that have lost one contact with a sheared obstacle must on average gain a new contact upon bowing out, so that steady advance is possible. This establishes a definite relationship between the critical mean distance λ_c between obstacles along the dislocation, the mean obstacle spacing l in the plane, the peak obstacle resistance $\hat{K}$, and the shear strength $\hat{\tau}$ of the obstacle field.

Consider now a typical local precipitate-shearing configuration, depicted in Fig. 6.5(b), where the obstacle at the center point O is sheared when the bow-out angles θ of the neighboring segments have reached θ_c ; the average segment lengths between neighboring obstacles along the dislocation line will have become λ_c , as the applied stress σ reaches the peak shear resistance $\hat{\tau}$ of the obstacle field. Then, the total force $\hat{\tau}b\lambda_c$ on the segments of length λ_c is transferred to the central obstacle, where it overcomes the peak obstacle resistive force $\hat{K}$, that is,

$$\hat{\tau}b\lambda_c = 2\mathcal{E} \sin \theta_c = \hat{K}. \quad (6.11)$$

Under this flow condition, the released segments AO and OB, having sheared through the obstacle O, will bow out into an arc AB, and must just touch another obstacle at, say, O', depicted by the dashed curve. At steady flow, the cuspy, kite-shaped area AOBO' must be on average equal to l^2 , the area allocated to a typical obstacle. If the areas swept out by the bowed-out segments AO and OB, just before shearing of the obstacle O, are denoted by Δa , the area AOBO' must be given by

$$\text{Area AOBO}' = l^2 = \Delta A - 2 \Delta a, \quad (6.12)$$

where ΔA is the area ABO' after the shearing of the obstacle O . If the peak resistive force $\hat{K}$ of the obstacle is weak, that is, θ_c is small, a simple geometrical consideration gives

$$\Delta a = \frac{(2R\theta_c)^3}{12R} = \frac{\lambda_c^3}{12R} \quad (6.13)$$

and

$$\Delta A = \frac{8\lambda_c^3}{12R}, \quad (6.14)$$

where

$$\frac{1}{R} = \frac{\hat{\tau}b}{\mathcal{E}} \quad (6.15)$$

is the curvature of the bowed-out segments AO , OB , and AB after shearing of the obstacle at O , all in equilibrium with the stress $\hat{\tau}$. Then, through the flow condition of eq. (6.12), we obtain directly the relations for the critical segment length λ_c :

$$\frac{\lambda_c}{l} = \left(\frac{2\mathcal{E}}{\hat{\tau}bl} \right)^{1/3} = \frac{1}{(\hat{K}/2\mathcal{E})^{1/2}}. \quad (6.16a)$$

The critical segment length λ_c is referred to as the *Friedel sampling length*. While the condition given in eq. (6.16a) is strictly applicable to flow for weak obstacles, we shall also use it below but close to the flow condition under a stress $\sigma < \hat{\tau}$, in the form

$$\frac{\lambda}{l} = \left(\frac{2\mathcal{E}}{\sigma bl} \right)^{1/3}, \quad (6.16b)$$

for certain considerations of thermally assisted overcoming of precipitate obstacles when $\sigma < \hat{\tau}$.

Finally, from eqs. (6.11) and (6.16a), the principal expression for the shear resistance $\hat{\tau}$ of the precipitate field is obtained as

$$\hat{\tau} = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bl} \right) \quad (6.17)$$

in terms of the mean obstacle spacing l in the plane, or

$$\hat{\tau} = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{br_0} \right) \left(\frac{3c}{2\pi} \right)^{1/2} \quad (6.18)$$

in terms of the actual particle radius r_0 and the atomic concentration c of the precipitates.

We note that the generalized interpretation of the sampling length λ given by eq. (6.16b) shows that as the stress is increased, the dislocation does indeed contact more obstacles and the length λ decreases. Moreover, as given by eq. (6.16a), when the strength of the obstacles increases, the critical sampling length λ_c decreases, and when $\hat{K}/2\mathcal{E} \rightarrow 1.0$, $\lambda_c \rightarrow \ell$. At that stage the peak resistance $\hat{\tau}$ of the obstacle field becomes $2\mathcal{E}/bl$, which amounts to the condition that the dislocation line shape becomes a semicircle with a diameter equal to l . Then, the two segments wrap around the precipitate as depicted in Fig. 6.5(c), and the dislocation exerts the maximum force $f = 2\mathcal{E}$ on the precipitate. Precipitates with resistive forces $\hat{K} > 2\mathcal{E}$ can no longer be sheared but will be bypassed as the dislocation bows around such nonshearable precipitates, pinches off, and leaves behind loops around the precipitates, as depicted by the sequence in Fig. 6.5(d). This terminal mode of interaction of a dislocation with nonshearable precipitates was recognized first by Orowan (1948) and is referred to as the Orowan mechanism, with the quantity

$$\hat{\tau} = \left(\frac{2\mathcal{E}}{bl} \right) \quad (6.19)$$

being referred to as the Orowan stress (or resistance). This simple relationship is central in most alloy development practice. We shall discuss this range of interaction of dislocations with nonshearable precipitates or particles in more detail in Section 6.4.7 and give more accurate expressions for it. The interpretation of eq. (6.17), which results directly from the considerations of Friedel (1956) and, independently, Fleischer and Hibbard (1963), referred to as the Friedel–Fleischer (FF) relationship, is pleasing in that it leads to a seemingly correct limiting behavior when $\hat{K}/2\mathcal{E} \rightarrow 1.0$. However, it departs from reality when $\hat{K}/2\mathcal{E}$ reaches values of about 0.3, when the bow-out angles θ become too large for the small-angle approximations that lead to eq. (6.16a) to be valid.

To explore precipitate–dislocation interactions in the range of large obstacle resistances and to test the validity of the Friedel–Fleischer assumptions in their range of applicability, a number of searching computer simulations were carried out by Kocks (1966), Foreman and Makin (1966), and Hanson and Morris (1975a,b), using different methods of analysis. With small variations between them, these simulations have all shown that as $\hat{K}/2\mathcal{E}$ increases from about 0.3 to 1.0 in a field of randomly dispersed point obstacles, the dislocation advances by finding channels of easier passage where the obstacle spacing is larger than the mean value l , and the plastic resistance of the field of obstacles falls below the prediction of the FF relationship of eq. (6.17). Figure 6.6 shows the simulation results of Foreman and Makin (1966) and those of Hanson and Morris (1975a) for the dependence of the peak plastic resistance $\hat{\tau}$, normalized by the nominal Orowan stress, on the normalized peak precipitate shear resistance $\hat{K}/2\mathcal{E}$. The two simulations agree well with each other but both begin to deviate from the FF relation of eq. (6.17) for values of $\hat{K}/2\mathcal{E}$ above 0.3 or bow-out angles θ larger than about 30° .

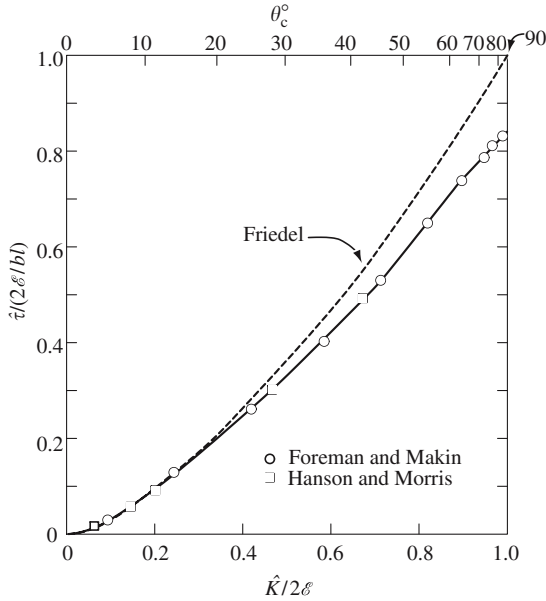


FIG. 6.6. The increase in the terminal shear resistance $\hat{\tau}$ with increasing normalized peak obstacle shear resistive force $\hat{K}/2\mathcal{E}$. The dashed curve is the analytic Friedel relation. The solid curve and data points represent computer simulation results. The angle θ_c is the critical segment bow-out angle at a contacted obstacle during the shear of the obstacle.

All simulations predict a terminal resistance in the range of 0.81–0.84 of the nominal Orowan stress of $2\mathcal{E}/bl$. Ardell (1985) proposed an empirical fit to the simulation results,

$$\hat{\tau} = 0.956 \left(\frac{2\mathcal{E}}{bl} \right) \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(1 - \left(\frac{\hat{K}}{2\mathcal{E}} \right)^2 / 8 \right), \quad (6.20)$$

over the entire range, which leads to a normalized terminal shear resistance $\hat{\tau}/(2\mathcal{E}/bl)$ of 0.84. Hanson and Morris (1975a) have also presented additional statistical information on the distribution of segment lengths between contacting precipitates along the dislocation line as a function of the normalized precipitate resistive force $\hat{K}/2\mathcal{E}$. These dependencies were qualitatively in agreement with the expectations of the analytical FF model.

The simulations referred to above assumed that the dislocation line tension was isotropic and that the precipitates were points. The effects of the anisotropy of the line tension become important when $\hat{K}/2\mathcal{E}$ begins to approach unity and the changes in line curvature affect the sampling process. Some such effects need to

be considered when specific mechanisms in the range of the Orowan resistance are discussed.

When the volume fraction of precipitates becomes large and $\langle r \rangle / l$ is no longer very small, idealization of precipitates as mere points becomes unsatisfactory. Some relevant generalizations were considered by Melander (1977) using simulation techniques similar to those of Hanson and Morris (1975*a,b*) but by considering the precipitates as finite objects.

6.3.4 Sampling Point Obstacles of Different Kinds

Even if precipitates were all of the same kind and could be idealized to be spherical and of the same actual size, they would be sampled by slip planes intersecting them at different levels and would, therefore, possess different strengths on the slip plane. In other instances, several different kinds of precipitates could coexist. In all of these cases, the task is to determine the collective contributions of the precipitates to the total "composite" plastic resistance.

To consider this problem generally, let there be m different types of obstacle, each possessing a peak resistive force $\hat{K}_i$ and having a specific planar point density n_i , or an average spacing $l_i (= n_i^{-1/2})$ in the plane. The total obstacle density of the entire set, N , would then be

$$N = \sum_{i=1}^m n_i. \quad (6.21)$$

If the m different obstacle populations existed by themselves in isolation, they would result in separate plastic resistances of

$$\hat{\tau}_i = \left(\frac{\hat{K}_i}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bl_i} \right) = \beta_i^{3/2} \left(\frac{2\mathcal{E}}{b} \right) n_i^{1/2}, \quad (6.22)$$

where we have used the abbreviation $\beta_i = \hat{K}_i / 2\mathcal{E}$.

The simplest and apparently most direct way of combining these strengthening effects would be to sum them linearly, that is,

$$\hat{\tau}_c = \sum_{i=1}^m \beta_i^{3/2} \left(\frac{2\mathcal{E}}{b} \right) n_i^{1/2} = \sum_{i=1}^m \beta_i^{3/2} \left(\frac{2\mathcal{E}}{b\bar{l}} \right) X_i^{1/2}, \quad (6.23)$$

where $\bar{l} = N^{-1/2}$ is the mean distance between obstacles regardless of type, and $X_i = n_i / N$, the fraction of the obstacle density of type i among the total. We refer to the points as obstacles here rather than as precipitates deliberately, since these arguments will be applicable to irradiation-induced lattice damage and nanovoids and even to forest dislocations threading through the slip plane, as we shall discuss

in Chapter 7. While the estimate of eq. (6.23) appears logical, it leaves it unclear how a glide dislocation samples the field by bowing through it.

Another possibility is to consider that what are really additive are the individual obstacle densities, as stated in eq. (6.21), and that the composite resistance $\hat{\tau}_c$ should be given by a weighted sum of the individual densities, weighted according to the peak resistive forces $\hat{K}_i$ of each kind, taking our cue from the form of the obstacle field resistance $\hat{\tau}_i$ of a particular kind, that is,

$$\hat{\tau}_i^2 = \beta_i^3 \left(\frac{2\mathcal{E}}{b} \right)^2 n_i. \quad (6.24)$$

This leads to a different form for $\hat{\tau}_c$,

$$(\hat{\tau}_c)^2 = \left(\frac{2\mathcal{E}}{b} \right)^2 \sum_{i=1}^m \beta_i^3 n_i = \left(\frac{2\mathcal{E}}{bl} \right)^2 \sum_{i=1}^m \beta_i^3 X_i, \quad (6.25)$$

$$\hat{\tau}_c = \left[\sum_{i=1}^m \left(\beta_i^{3/2} \frac{2\mathcal{E}}{bl} \right)^2 X_i \right]^{1/2} = \left[\sum_{i=1}^m \left(\hat{\tau}_i^{\text{ref}} \right)^2 X_i \right]^{1/2}, \quad (6.26)$$

where we interpret

$$\beta_i^{3/2} \left(\frac{2\mathcal{E}}{bl} \right) = \hat{\tau}_i^{\text{ref}} \quad (6.27)$$

as a reference plastic resistance that would be obtained if all obstacles were only of type i . This form of the flow stress presents no conceptual problem of how the dislocation samples the entire obstacle field. It has been referred to as the Pythagorean sum by Koppenaal and Kuhlmann-Wilsdorf (1964). Computer simulations by Foreman and Makin (1966) and Hanson and Morris (1975*b*) have demonstrated that this Pythagorean sum fits the results best over a wide range of different binary combinations of obstacles, while the direct addition of plastic resistances in eq. (6.23) gives poor results.

6.3.5 Sampling Obstacles of Finite Width

When the precipitate volume fraction becomes large so that $\langle r \rangle/l$ is no longer very small, and in cases where precipitates act on dislocations through strain fields produced by material misfit or when they possess elastic properties markedly different from those of the matrix, several important factors need to be considered that did not enter in the point approximation of obstacles. These arise primarily from the finite interaction cross sections of the precipitates or obstacles, where the dislocation interacts with and overcomes obstacles at a variety of positions on their respective

force–distance curves in the glide plane. This requires a broadened consideration of the problem of the sampling of obstacles by dislocations.

Schwarz and Labusch (1978) (SL) carried out such a simulation of dislocations probing a field of particles with elliptical-type interaction cross sections, having principal normalized width and depth dimensions of $2\xi_0$ and $2\eta_0$ in the x and y directions, respectively, on the slip plane, for dislocations advancing in the y direction, as depicted in Fig. 6.7(a). Of these widths, the one of primary interest is the obstacle depth y_0 , which affects the sampling process most importantly and is related directly to the form of the force–distance curve of the interaction of the dislocation with the precipitate when that interaction is a soft-contact interaction. In their simulation, following the procedures of Hanson and Morris (1975a), SL considered two generic forms of force–distance curve, depicted in the upper and lower insets of Fig. 6.8, which have been labeled as *energy-storing* and *dissipative* interactions, respectively, for reasons that will become clear when we discuss various interaction mechanisms in Section 6.4 later. An insightful further dimension of the SL simulation is that it also incorporated a phonon drag process affecting the motion of dislocation segments between obstacles when their inertial properties could be important in overcoming obstacles at low temperatures, as we shall discuss in Chapter 8.

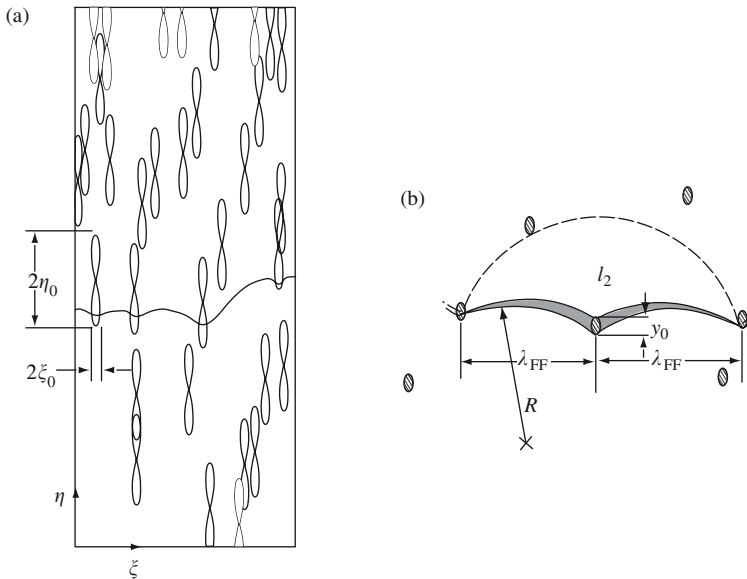


FIG. 6.7. (a) A dislocation advancing through a field of obstacles of finite cross section in the Schwarz and Labusch model (after Schwarz and Labusch 1978); (b) sketch defining the working parameters of the SL model.

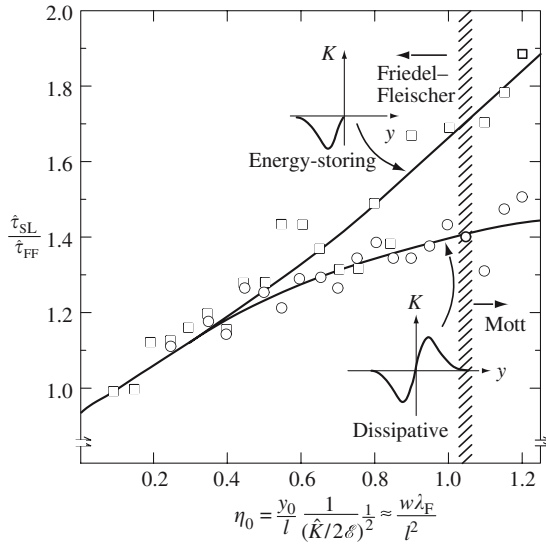


FIG. 6.8. The effect of increasing the SL scaling parameter η on the comparative increase of the terminal shear resistance $\hat{\tau}_{SL}$ in the SL model over $\hat{\tau}_{FF}$ in the Friedel–Fleischer model for both dissipative and energy-storing obstacles (after Schwarz and Labusch 1978).

Here we consider only the principal result of the SL simulation, where the dislocation segments are advancing in a critically damped manner. This is plotted in Fig. 6.8 as the ratio of the shear resistance $\hat{\tau}_{SL}$ of the obstacle field, normalized by the shear resistance $\hat{\tau}_{FF}$ determined by the FF method, plotted as a function of the normalized obstacle depth parameter η_0 given by

$$\eta_0 = \frac{y_0}{l} \frac{1}{(\hat{K}/2\mathcal{E})^{1/2}} = \frac{y_0 \lambda_{FF}}{l^2}. \quad (6.28)$$

The geometrical significance of the normalized obstacle depth η_0 is depicted in Fig. 6.7(b) in the context of the Friedel condition of sampling of the point obstacle field. It is readily demonstrated that

$$\eta_0 = 1.52 \frac{y_0}{b} \frac{c^{1/2}}{(\hat{K}/2\mathcal{E})^{1/2}} \quad (6.29)$$

and that an increase in η_0 amounts to an increase in the volume fraction c of obstacles for a given obstacle depth. For $\eta_0 \leq 0.4$, the Friedel–Fleischer model predicts the obstacle shear resistance of the obstacle field quite well. For larger concentrations c and a given obstacle depth or for larger depths at given concentration, there are significant departures from the FF model. The departures are larger for

obstacles that store energy when being sheared, such as when an internal stacking fault or antiphase boundary is created. The departures are less when the obstacles interact merely by dissipating energy. The differences arise from the changes in the obstacle sampling length λ along the dislocation line. The different forms of the sampling length λ_{SL} can be represented empirically for energy-storing and dissipative interactions as

$$\frac{\lambda_{\text{SL}}^{\text{es}}}{\lambda_{\text{FF}}} = \frac{1.06}{(1 + C\eta_0)}, \quad \text{where } C = 0.67, \quad (6.30)$$

and

$$\frac{\lambda_{\text{SL}}^{\text{dis}}}{\lambda_{\text{FF}}} = \frac{1.06}{(1 + C'\eta_0)^{1/3}}, \quad \text{where } C' = 2.5, \quad (6.31)$$

respectively, and where

$$\lambda_{\text{FF}} = \frac{l}{(\hat{K}/2\mathcal{E})^{1/2}} \quad (6.32)$$

is the previously defined Friedel sampling length. These dependences are plotted in Fig. 6.9.

The best-fit empirical forms for the normalized plastic resistance $\hat{\tau}_{\text{SL}}/\hat{\tau}_{\text{FF}}$ given in Fig. 6.8 are, for the two different obstacle profiles of energy-storing and dissipative type,

$$\frac{\hat{\tau}_{\text{SL}}^{\text{es}}}{\hat{\tau}_{\text{FF}}} = 0.94(1 + C\eta_0) \quad (6.33)$$

and

$$\frac{\hat{\tau}_{\text{SL}}^{\text{dis}}}{\hat{\tau}_{\text{FF}}} = 0.94(1 + C'\eta_0)^{1/3}, \quad (6.34)$$

where the constants C and C' are the same as those given for eqs. (6.30) and (6.31). As Fig. 6.8 shows, the two relationships for the different types of obstacle are identical for $\eta_0 < 0.4$. We note that when η_0 begins to exceed 1.0, that is, when the ratio of the shaded area in Fig. 6.7(b) to the large area swept out after the central obstacle is sheared formally reaches unity, the interaction between the dislocation and the neighboring obstacles becomes complex in the same manner as that encountered for the similar situation in solid-solution strengthening. There, a parameter γ was introduced (eq. 5.35) which separated the phenomena of sampling of dilute obstacles individually by dislocations as characterized by the Friedel–Fleischer process from the sampling of obstacles collectively for higher concentrations by

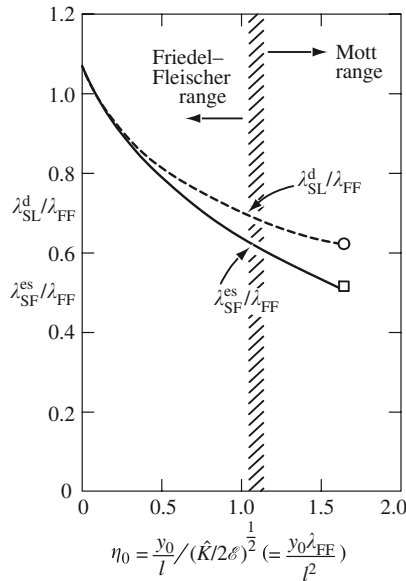


FIG. 6.9. Variation of the obstacle sampling length for dissipative and energy-storing obstacles with the SL scaling parameter η , in units of the corresponding sampling length of the Friedel–Fleischer model.

the Mott–Labusch process, depending upon whether $\gamma \ll 1.0$ for the former case from $\gamma \gg 1.0$ for the latter. Since

$$\gamma = 10.72 \left(\frac{y_0}{b} \right)^2 \frac{c}{\hat{K}/\mathcal{E}} = 4.64\eta_0^2, \quad (6.35)$$

apart from the numerical factors establishing the γ criterion, we can conclude that when $\eta_0 \approx \gamma^{1/2}$ approaches unity the dislocations will begin to sample obstacles collectively in the Mott–Labusch sense, and the motion of a dislocation will exhibit the character of advance associated with that of concentrated solid solutions discussed in Section 5.4. Then, the SL theory will no longer be reliable above a value of η_0 of 1.0. The cross-hatched zones in Figs. 6.8 and 6.9 suggest such a border above which the results of SL are no longer reliable and some form of collective interaction needs to be considered, which is presently not available.

6.3.6 Precipitate Growth, Peak Aging, and Overaging

With few exceptions, the peak resistive forces $\hat{K}$ of precipitates increase with increasing precipitate size. This leads to an attendant increase in plastic resistance at constant temperature, resulting from the diffusion-controlled growth of

precipitates. This is an industrially very important means of controlling the plastic resistance of alloys, referred to as “age hardening”.

Thus, consider a frequently encountered linear dependence of the peak resistive force $\hat{K}$ of an obstacle on r_0 , that is,

$$\hat{K} = r_0 \hat{K}', \quad \text{where } \hat{K}' = \frac{d\hat{K}}{dr_0}. \quad (6.36a,b)$$

Substitution of this into eq. (6.20) and representing l in terms of r_0 , gives

$$\frac{\hat{\tau}}{\mu} = \alpha \left(\frac{r_0}{b} \right)^{1/2} \left(\frac{\hat{K}' b}{2\mathcal{E}} \right)^{3/2} \left(\frac{3c}{2\pi} \right)^{1/2} \left(1 - \left(\frac{\hat{K}' r_0}{2\mathcal{E}} \right)^2 / 8 \right), \quad (6.37)$$

where $\alpha = 0.956$.

Equation (6.37) shows that in the range of shearable precipitates, that is, $\hat{K}/2\mathcal{E} < 1.0$, the term in parentheses in the equation is somewhat under unity and the principal dependence of the shear resistance $\hat{\tau}$ on the precipitate radius is parabolic. This dependence continues until $\hat{K}/2\mathcal{E}$ becomes unity, when the shear resistance $\hat{\tau}$ reaches a peak value of

$$\left(\frac{\hat{\tau}}{\mu} \right)_{\text{peak}} = 0.836 \left(\frac{b}{r_0} \right) \left(\frac{3c}{2\pi} \right)^{1/2}, \quad (6.38)$$

and when the particle radius becomes

$$(r_0)_{\text{peak}} = \left(\frac{2\mathcal{E}}{\hat{K}'} \right). \quad (6.39)$$

Beyond this precipitate size, $\hat{K}/2\mathcal{E} > 1.0$ and the precipitates are no longer shearable. Then, the shear resistance $\hat{\tau}$ begins to decline according to eq. (6.38). This simple model of the dependence of the shear resistance $\hat{\tau}$ on the particle size r_0 is sketched in Fig. 6.10. While it needs to be modified in many respects, as we shall discuss in Section 6.4 for the various different precipitate–dislocation interactions, in broad outline it represents the facts well, at least semiquantitatively.

6.3.7 Thermally Assisted Motion of Dislocations through a Field of Penetrable Obstacles

The Activation Barrier The plastic resistance $\hat{\tau}$ of an obstacle field, with specific emphasis on precipitates, refers to the threshold resistance under athermal conditions, specifically at $T = 0$ K. When the precipitates are shearable, that is, $\hat{K}/2\mathcal{E} < 1.0$, it is in principle possible to overcome the plastic resistance at a stress $\sigma < \hat{\tau}$ by thermal assistance of the obstacle-shearing process. In precipitation

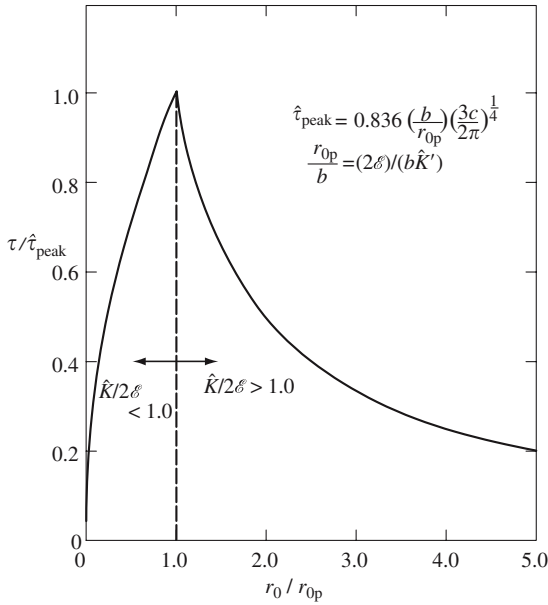


FIG. 6.10. Schematic illustration showing the increase in plastic resistance of an age-hardenable alloy in the underaged behavior range where $\hat{K}/2\mathcal{E} < 1.0$, through the peak strength, to the overaged behavior range where $\hat{K}/2\mathcal{E} > 1.0$, with increasing precipitate radius.

strengthening, the flow stress is often found not to be temperature-dependent, or at most only weakly so (Kelly and Nicholson 1963; Brown and Ham 1971; Ardell 1985, 1994; Reppich 1993). The arguments presented for this finding are usually imprecise and unconvincing. The principal reason for this lack of dependence or weak dependence is more likely that most precipitate-strengthened systems of interest are close to the condition of either peak aging or overaging, where $\hat{K}/2\mathcal{E} \leq 1.0$ and thermal assistance is not possible. There are, however, other considerations, which will become clear when we consider specific cases, that can explain such weak dependence. In the range $\hat{K}/2\mathcal{E} < 1.0$ for precipitates and for other obstacle clusters, under these conditions, the motion of dislocations through some obstacle fields could be thermally assisted. In cases where the obstacle field consists of several different types, some of which have characteristics such that $\hat{K}/2\mathcal{E} > 1.0$, the effect of these must be separated from the analysis when we are dealing only with the obstacles for which $\hat{K}/2\mathcal{E} < 1.0$ holds. This can be done by considering a decomposition of the contributions to the flow stress along the lines presented in Section 6.3.4. Here we shall consider only those cases for which $\hat{K}/2\mathcal{E} < 1.0$.

Thus, consider a typical case, as illustrated in Fig. 2.13(a), of a dilute obstacle field in which the interaction of the dislocation with the obstacles is in the

Friedel–Fleischer range. The dislocation line probes a central shearable obstacle (a precipitate) offering a resistance at contact that is assumed to be derivable from a characteristic resistive force–distance curve $K = K(y)$, where y is measured from the point of contact. The force f_a exerted on the obstacle by the applied stress (considered to be the effective stress) is resisted by K , which is less than but close to the peak resistive level $\hat{K}$, so that the Friedel sampling condition given by eq. (6.16b) is still applicable. Then, under these conditions, the dislocation will penetrate into the obstacle a certain distance y_s , reaching a *stable* equilibrium position, depicted in Fig. 2.13(b) by the point O, or, correspondingly, on the hypothetical force–distance curve in Fig. 6.11 by the position y_s . Under the same applied force $f_a = \sigma b\lambda(\sigma)$, there is another position of equilibrium of the dislocation in the shearing obstacle at y_u , where the equilibrium, however, is *unstable*, representing the saddle-point condition. Between these two equilibrium positions, there is an energy barrier ΔG^* that needs to be overcome to permit a dislocation segment of length 2λ to sweep out an area Δa between the initial shape of the plucked dislocation and the final relaxed shape shown in Fig. 2.13.

The energy barrier to be overcome to reach the saddle-point configuration at O' in Fig. 2.13 is

$$\Delta G^* = \Delta F^* - \Delta W^*. \quad (6.40)$$

As discussed in Chapter 2, the Helmholtz free-energy difference ΔF^* and the work ΔW^* done by the applied stress during activation are, respectively,

$$\Delta F^* = \int_{y_s(\sigma)}^{y_u(\sigma)} \tau(y) b\lambda \, dy = \int_{y_s(\sigma)}^{y_u(\sigma)} K(y) \, dy \quad (6.41)$$

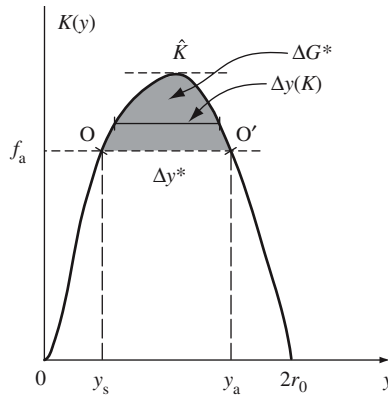


FIG. 6.11. Penetration by thermal assistance through an obstacle at a resistance level $K < \hat{K}$.

and

$$\Delta W^* = f_a \Delta y^* = \sigma b \lambda(\sigma) \Delta y^* = \sigma b \lambda(\sigma) (y_u(\sigma) - y_s(\sigma)), \quad (6.42)$$

giving

$$\Delta G^* = \int_{y_s}^{y_u} (K(y) - f_a) dy = \int_{f_a=K}^{\hat{K}} \Delta y(K) dK = \Delta G^*(f_a, \hat{K}). \quad (6.43)$$

To proceed further, specific forms of $K = K(y)$ need to be considered. We consider three different simple cases, from which some general conclusions can be drawn. More detailed considerations will be given later for a specific case.

(i) Interface step formation upon shearing, for which the force–distance curve is a rectangular box, depicted in Fig. 6.12(a), gives, for shear on the equator,

$$\Delta G^* = 2\hat{K}r_0 \left(1 - \frac{K}{\hat{K}}\right), \quad (6.44a)$$

where

$$\hat{K} = 2b\chi_i, \quad (6.44b)$$

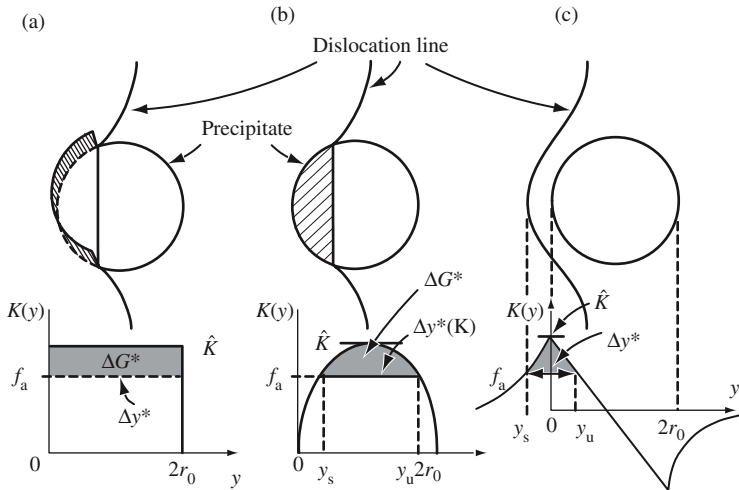


FIG. 6.12. Schematic illustration of three different types of obstacle being overcome by thermal assistance at resistive force levels $K < \hat{K}$: (a) chemical interaction with interface step formation; (b) a hard-contact obstacle with either internal lattice resistance or internal fault production; (c) energy-storing or dissipative soft-contact obstacle.

χ_i being the interface energy of the step formed. Here we have considered the easy case of the shear produced by a screw dislocation.

(ii) High internal lattice resistance, for which the force–distance curve is a semicircle, depicted in Fig. 6.12(b). This gives, for cutting across the equator,

$$K = \hat{K} \sqrt{1 - \left(\frac{\Delta y}{2r_0} \right)^2}, \quad (6.45a)$$

where

$$\hat{K} = 2r_0 b \tau_1. \quad (6.45b)$$

For this, the final activation free energy becomes

$$\Delta G^* = r_0 \hat{K} \left(\frac{\pi}{2} - \sin^{-1} \left(\frac{K}{\hat{K}} \right) - \frac{K}{\hat{K}} \sqrt{1 - \left(\frac{K}{\hat{K}} \right)^2} \right). \quad (6.46)$$

Another specific case of this type is when an extended dislocation in a matrix crystal enters a precipitate in which the stacking-fault energy is substantially lower, resulting in an increased extension of the dislocation inside the particle and a substantial energy reduction, trapping the matrix dislocation in the particle. This case, which is the source of stacking-fault strengthening, involves a number of complexities that we shall develop in some detail in Section 6.4.3 below.

(iii) When the dislocation interacts with an energy-dissipating soft-contact obstacle, such as a particle with size misfit $\epsilon_s = \Delta r/r_0$ or one possessing a modulus misfit $\epsilon_\mu = \Delta\mu/\mu$, having a force–distance curve as shown in Fig. 6.12(c), we have

$$\frac{\Delta y}{r_0} = \left(\frac{\hat{K}}{K} \right)^{1/3} - \frac{K}{\hat{K}}, \quad (6.47)$$

where $\hat{K}$ is

$$\hat{K}_s = \alpha_s (2\mathcal{E}) \epsilon_s (r_0/b) \quad (6.48a)$$

for size misfit or

$$\hat{K}_\mu = \alpha_\mu (2\mathcal{E}) \epsilon_\mu \quad (6.48b)$$

for modulus misfit, with $\alpha_s \approx 3.24$ and $\alpha_\mu \approx 0.12$ (for $r_0/b < 40$), as will be shown below in Sections 6.4.5 and 6.4.6. There, we shall also specifically address the difficult question of how a dislocation interacts collectively with a field of soft-contact obstacles.

These cases give an activation free energy of the form

$$\Delta G^* = r_0 \hat{K} \left[1 - \frac{3}{2} \left(\frac{K}{\hat{K}} \right)^{2/3} + \frac{1}{2} \left(\frac{K}{\hat{K}} \right)^2 \right]. \quad (6.49)$$

In the Friedel–Fleischer range of dilute obstacle concentrations, the ratio $K/\hat{K}$ is

$$\frac{K}{\hat{K}} = \left(\frac{\sigma}{\hat{\tau}} \right)^{2/3} \quad (6.50)$$

by virtue of eq. (6.17), which upon substitution into eqs (6.44a), (6.46), and (6.49) gives expressions for the stress dependence of the activation free energies of the individual mechanisms. These different expressions can be fitted broadly to a general empirical expression that can be stated as follows (Kocks *et al.* 1975):

$$\Delta G^* = \Delta G_0^* \left[1 - \left(\frac{\sigma}{\hat{\tau}} \right)^p \right]^q, \quad (6.51)$$

where the exponents p and q are generally in the range of

$$\begin{aligned} 0.5 &\leq p \leq 1.0, \\ 1.0 &\leq q \leq 2.0. \end{aligned} \quad (6.52)$$

The Strain Rate Expression In the Friedel–Fleischer regime of dilute obstacle concentrations, once the dislocation is activated to the saddle-point configuration of point O' in Fig. 2.13 and sweeps out the area Δa under a stress $\sigma < \hat{\tau}$ but close to $\hat{\tau}$ the probability that the process can be reversed by a reverse thermal fluctuation is negligible, since the energy barrier ΔG_r^* for the reverse fluctuation should be close to a factor $\Delta a/\Delta a^*$ larger than ΔG^* for the forward process. Thus, the plastic strain rate should be simply

$$\dot{\gamma} = \dot{\gamma}_G \exp \left(-\frac{\Delta G^*(\sigma/\hat{\tau})}{kT} \right), \quad (6.53a)$$

where

$$\dot{\gamma}_G \cong b \rho_m \bar{y} \nu_G, \quad (6.53b)$$

and where ρ_m is the mobile dislocation density,

$$\bar{y} \approx \frac{\Delta a}{2\lambda} \approx \frac{l^2}{2\lambda} = \frac{l}{2} \left(\frac{\sigma b l}{2\mathcal{E}} \right)^{1/3} \quad (6.53c)$$

is the average distance of advance of the activated segment of length 2λ , and $\nu_G = O(10^{-2}\nu_D)$ is the attempt frequency of the dislocation segment in the field

of the obstacle resistive force, which is usually in the range of 10^{-2} of the lattice frequency ν_D .

Using the abbreviated form of the activation free energy of eq. (5.51) then gives, together with eqs. (6.53a,b,c),

$$\frac{\sigma}{\hat{\tau}} = \left(1 - \left(\frac{T}{T_0} \right)^{1/q} \right)^{1/p}, \quad (6.54)$$

where

$$T_0 \equiv \frac{\Delta G_0^*}{k \ln(\dot{\gamma}_G / \dot{\gamma})} \quad (6.55)$$

and where

$$\dot{\gamma}_G = \frac{bl\rho_m}{2} \left(\frac{\sigma bl}{2\mathcal{E}} \right)^{1/3} \nu_G \quad (6.56)$$

is weakly stress dependent in a way that can be neglected to a first approximation, making T_0 , the cutoff temperature encountered earlier in Chapters 4 and 5, dependent primarily only on the strain rate.

It is to be recalled that the stress σ in eq. (6.54) is to be considered as the effective stress probing the shearable obstacles and $\hat{\tau}$ as the athermal plastic resistance attributable to only those obstacles.

6.4 Specific Mechanisms of Precipitation Strengthening

6.4.1 Overview

In the following sections, we consider in some detail several mechanisms of particle strengthening and strengthening by other similar discrete obstacles. The mechanisms selected are important and will demonstrate how they can be dealt with in a relatively simple but also uncompromising manner. They will be compared with the most appropriate experimental studies. The coverage is not intended to be complete. The reader is advised to refer to the several comprehensive reviews on the subject that have been listed at the end of the chapter for additional consideration of the mechanisms discussed in this section, as well as others that have been left out.

Most precipitate particles combine more than one mechanism. This makes evaluation of specific cases in general difficult, particularly in relation to the interpretation of experimental results. A combined approach of measurement of the plastic resistance at different temperatures following different aging treatments, and transmission electron microscopy associated with the type of modeling to be

considered, offers the best possibilities. In the cases selected for discussion there will be a dominant mechanism, permitting a reasonable comparison with models. The models of the specific mechanisms presented here will show considerable variation in approach from those of other investigators, even though the detailed physical picture will be the same. This demonstrates that even in this previously well-researched field, different and more precise approaches are still possible. Readers are encouraged to verify the results to their own satisfaction, and are advised to consult some of the original literature, which frequently will give more detailed treatments and information about the experiments.

The principal manifestation of the mechanisms is the level of plastic shear resistance they result in. Since some precipitates trap dislocations, impede their motion, and can affect the dislocation storage rate, which introduces another component into the resistance resulting from dislocation interactions in strain hardening, precipitate mechanisms are best probed by studying the initial yield stress in otherwise undeformed single crystals.

In the discussion of specific mechanisms, we consider examples of the two forms of interaction of dislocations with coherent precipitates that have the same crystal structure as the matrix, presented in Section 6.3.2, labeled *hard-contact interactions* and *soft-contact (or diffuse) interactions*. Among the former form, we consider *stacking-fault strengthening* and *atomic-order strengthening*. Among the latter form, we consider *size misfit strengthening*, often referred to as *coherency strengthening*, and *modulus misfit strengthening*.

All the above forms of strengthening mechanisms can exhibit plastic-resistance maxima, or *peak strengths*, through aging that result from an increase of the precipitate size until a critical size is reached where the precipitates become impenetrable by dislocations, requiring the latter to circumvent them by bowing around the precipitates without penetration, initiating a form of strengthening considered first by Orowan (1948). This form of terminal behavior occurs not only with otherwise coherent precipitates when they exceed a critical (peak) size, but also with all types of incoherently attached precipitates that have a different crystal structure and even with glassy particles, where penetration of dislocations is physically not possible. Therefore, we consider this mechanism separately under the title of the *Orowan resistance*, often referred to also as *dispersion strengthening*.

In all cases of shearing of coherent precipitates by dislocations, a set of new interfaces is produced that offers a definite resistance. We consider this ubiquitous process first.

6.4.2 Chemical Strengthening, or Resistance to Interface Step Production in Shearing

As stated above, one contribution to the plastic resistance comes from the required production of new interfaces when an underaged, coherent particle is sheared by a

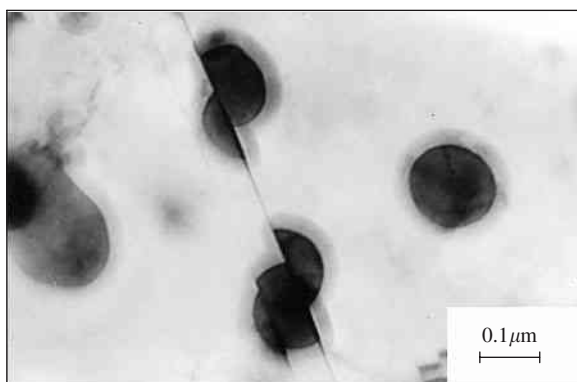


FIG. 6.13. Multiply sheared ordered spherical precipitates in an alloy, showing interface step formation (Gleiter and Hornbogen 1965*b*; courtesy of Wiley-VCH).

TABLE 6.1. Ranges of solid–solid interphase boundary energy χ_i for three types of planar interface^a

Interface type	χ_i (mJ/m ²)
Coherent	5–200
Semicoherent	200–800
Incoherent	800–2,500

^a Howe (1997).

dislocation. A good example of this is shown in Fig. 6.13, which shows a multiply sheared set of internally ordered Ni₃Al particles.

In the simplest case of this process, the precipitate is sheared by a screw dislocation as depicted in Fig. 6.12(a), showing a screw dislocation progressing smoothly through a particle, where the rate of production of a new interface is constant throughout the shearing process and the peak resistive force $\hat{K}$ is simply

$$\hat{K} = 2b\chi_i. \quad (6.57)$$

Here χ_i is the interface energy between the particle, usually an intermetallic compound, and the surrounding matrix. Such energies range from very low to very high values, as indicated in Table 6.1. Since particles with semicoherent and incoherent interfaces are usually not shearable, the typical level of the interface energy can be taken as 100 mJ/m². The peak resistance $\hat{\tau}$ predicted by the Friedel–Fleischer

model for a dilute volume fraction c of particles is then

$$\hat{\tau} = \left(\frac{\hat{K}}{2\mathcal{E}} \right) \frac{2\mathcal{E}}{b\langle r \rangle} \left(\frac{3\pi c}{32} \right)^{1/2}. \quad (6.58)$$

Clearly, since $\hat{K}$ is independent of the particle size, the peak resistance $\hat{\tau}$ decreases monotonically with increasing particle size, and peak aging is not possible for this mechanism alone.

When the particle is approached by an edge dislocation, the entry of the dislocation becomes difficult, since the rate of production of interface energy with displacement of the dislocation at contact becomes unbounded if the dislocation remains straight. Thus, the rate of production of free energy F evokes a resistance

$$K = \frac{dF}{dy} = 2b\chi_i \frac{dx}{dy}, \quad (6.59a)$$

where

$$\frac{dx}{dy} = \frac{1 - y/\langle r \rangle}{\sqrt{2y/\langle r \rangle - \left(\frac{y}{\langle r \rangle} \right)^2}}, \quad (6.59b)$$

which becomes unbounded at both $y = 0$ and $y = 2\langle r \rangle$. However, since the dislocation is not a geometrical line of zero thickness but has finite displacement gradients at its core, the penetration process must be considered as diffuse at the level of the core dimensions over a distance $r = b$. Thus,

$$\hat{K} = 2b\chi_i g \left(\frac{\langle r \rangle}{b} \right), \quad (6.60a)$$

where we take

$$g \left(\frac{\langle r \rangle}{b} \right) = \frac{1 - b/\langle r \rangle}{\sqrt{2b/\langle r \rangle - (b/\langle r \rangle)^2}}, \quad (6.60b)$$

which has a maximum value varying between 2.0 and about 7.0 as $b/\langle r \rangle$ goes from 0.1 to 0.01. Thus, the penetration resistance of particles to edge dislocations should be higher by a factor of roughly 7 or so, in comparison with the case of a screw dislocation. This, however, is not realistic, since the impeded edge dislocation can bow around the particle and begin to enter it at the side as it acquires an increasing screw character. In any case, the normalized peak resistance $\hat{K}/2\mathcal{E} = 2\chi_i/\mu b$ is never likely to exceed 0.05, making it a weak obstacle, albeit always present as an added contribution to the other mechanisms for shearable precipitates.

6.4.3 Stacking-fault Strengthening

Model of a Stacking-fault Strip In FCC metals containing coherent precipitates which are also FCC and where a dislocation in both the matrix and the precipitates can be in extended form, considerable strengthening is possible if the stacking-fault energies in the two media are very different. Thus, when the stacking-fault energy is higher in the precipitate than in the matrix, penetration of a dislocation into a precipitate will raise the line energy locally so as to develop a repulsive interactions at contact with the precipitate, which will reach its maximum value halfway through the precipitate. Alternatively, if the stacking-fault energy is lower in the precipitate, the dislocation will be attracted toward the center of the precipitate, where the line energy will be locally minimized. In this case the maximum resistance to the motion of the dislocation will develop when it is extracted from the precipitate. These interactions were studied first by Hirsch and Kelly (1965) in a general form.

Here we shall consider the second case, of an attractive interaction, for which a good example is provided by the system of Al containing Ag precipitates, where other strengthening mechanisms are substantially absent, since Al and Ag have closely similar lattice parameters and elastic constants. This system was considered in detail by Gerold and Hartmann (1968). We shall expand on their treatment. The geometry of the problem is depicted in Figs. 6.14(a) and (b), showing the lowest-energy position of a dislocation in the center of a precipitate and the most likely configuration near the position of maximum resistive force, occurring on exit of the dislocation from the precipitate.

In the Al matrix, where the stacking-fault energy $\chi_{\text{SF}} = \chi_{\text{m}}$ is 200 mJ/m², the dislocation is only minimally extended. Considering the interaction between an edge dislocation and the precipitate, the dislocation line energy in the matrix, away from a precipitate, is

$$\mathfrak{F} = \frac{\mu b^2}{4\pi(1-\nu)} \ln \left(\frac{\alpha R}{b} \right), \quad (6.61)$$

as previously derived in Chapter 2. The separation w_{m} between the Shockley partials is

$$w_{\text{m}} = \frac{\mu b^2}{24\pi \chi_{\text{m}}} \left(\frac{2+\nu}{1-\nu} \right), \quad (6.62)$$

resulting in a typical stacking-fault width w_{m} of only 0.54 nm, and only a minimal energy reduction of

$$\Delta \mathfrak{F} = -\frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) \ln \left(\frac{\alpha w_{\text{m}}}{eb} \right), \quad (6.63)$$

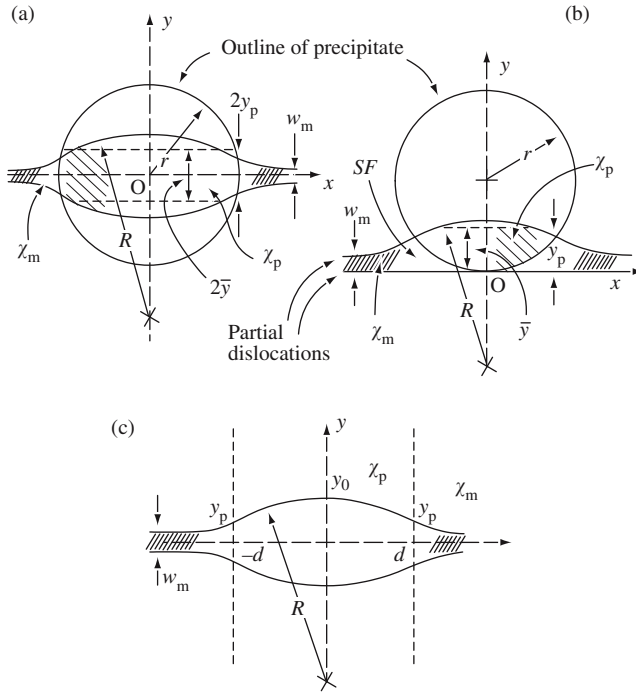


FIG. 6.14. Stacking-fault strengthening involving particles with a much lower stacking-fault energy than that in matrix: (a) a cutting dislocation in the center of a particle; (b) the final "extraction" configuration governing the peak resistance; (c) the configuration in a hypothetical "precipitate strip" model.

which is only roughly 1% of the line energy of an undissociated dislocation and can be neglected. However, when the matrix dislocation enters the particle, as depicted in Figs. 6.14(a) and (b), the extension of the dislocation in the particle is very different from that in the matrix, and important changes in the energy of the dislocation line occur there that govern the interaction of the dislocation with the precipitate. Clearly, the changes in and around the spherical precipitate are complex, primarily because of the curvature of the boundary of the particle.

To simplify the problem, we consider first the interaction of a matrix dislocation with a rectangular strip of width $2d$ of material of lower stacking-fault energy χ_p , as depicted in Fig. 6.14(c). For a large precipitate, this configuration would approximate to the case of the dislocation in the center of the precipitate as sketched in Fig. 6.14(a). The difference between the energy of a dislocation with such a shape and one in the matrix can be obtained by solving for the shape of the extended dislocation and accounting for the various energetic changes. Considering the extended dislocation to be an edge dislocation, we solve first for the shape of the mixed partial dislocation outside the strip in the region $x \geq d$.

Thus, the differential equation for the shape of this partial is given by

$$\mathcal{E} \frac{dy^2}{dx^2} - \chi_m + \frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) \frac{1}{2y} = 0, \quad (6.64)$$

where $y(x)$ is the half-width of the extended dislocation and b is that of the total dislocation. The width w_m of the dislocation outside, well away from the strip, is given by eq. (6.62).

Making the usual substitution

$$\frac{d^2y}{dx^2} = \frac{d}{dy} \left(\frac{1}{2} \left(\frac{dy}{dx} \right)^2 \right) \quad (6.65)$$

and integrating gives

$$\left(\frac{dy}{dx} \right)^2 - \frac{2\chi_m}{\mathcal{E}} \left(y - \frac{w_m}{2} \right) + \frac{\chi_m w_m}{\mathcal{E}} \ln \frac{y}{(w_m/2)} = 0$$

for the boundary conditions

$$\frac{dy}{dx} = 0 \quad \text{at } y = \frac{w_m}{2}, \quad \text{that is, at } x \rightarrow \infty,$$

and

$$y = y_p \quad \text{at } x = \mp d.$$

Considering the difference $y - w_m/2$ to be small in comparison with $w_m/2$ itself permits a second integration, which gives

$$y(x) = \frac{w_m}{2} + \left(y_p - \frac{w_m}{2} \right) \exp \left\{ -C \frac{x-d}{w_m/2} \right\}, \quad (6.66)$$

where $C = 0.294$ for the specific system of an Ag precipitate (or strip) in Al, with $\chi_p = 17 \text{ mJ/m}^2$ for the Ag and $\chi_m = 200 \text{ mJ/m}^2$ for the Al matrix, as already stated above, and $\nu = 0.37$ for Al.

An approximate solution for inside the strip is readily obtained by considering the shape of the Shockley partial inside the strip to be a circular arc of constant curvature, which gives the curvature as

$$\frac{1}{R} = \left(\frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) \frac{1}{2\bar{y}} - \chi_p \right) \frac{1}{\mathcal{E}}, \quad (6.67)$$

where $\bar{y} \approx (4/3)y_p$ is taken to be the average half-distance of separation between the two partial dislocations inside the strip. This gives the shape of the extended

dislocation inside the strip as

$$y(x) = y_0 - \frac{R}{2} \left(\frac{x}{R} \right)^2 \quad \left(\text{for } \frac{d}{R} \ll 1.0 \right), \quad (6.68)$$

where y_0 is the maximum displacement at $x = 0$, which is related to the border displacement y_p at $x = d$ by

$$y_0 = y_p + \frac{R}{2} \left(\frac{d}{R} \right)^2, \quad (6.69)$$

where y_p needs to be determined numerically by matching the ordinates and slopes of the solutions inside and outside the strip, that is, the displacement y and dy/dx at $x = d$. The resulting shapes of the extended dislocation inside and outside the strip are plotted in Fig. 6.15 for three half-widths d of the strip, 1.0, 2.0, and 3.0 nm, which will correspond to the range of precipitate sizes of interest.

The change in energy of the dislocation line relative to a dislocation inside the matrix away from the strip is readily obtainable by accounting for the line energies and the stacking-fault energies. This gives for the dislocation outside the strip

$$\Delta \mathfrak{F}_o = \mathfrak{F}_{of} - \mathfrak{F}_{oi}, \quad (6.70)$$

which becomes

$$\Delta \mathfrak{F}_o = \frac{\mu b^2 w_m}{24\pi} \left(\frac{2 + \nu}{1 - \nu} \right) \left(1.70 \left(\frac{y_p}{(w_m)/2} - 1 \right)^2 \right), \quad (6.71)$$

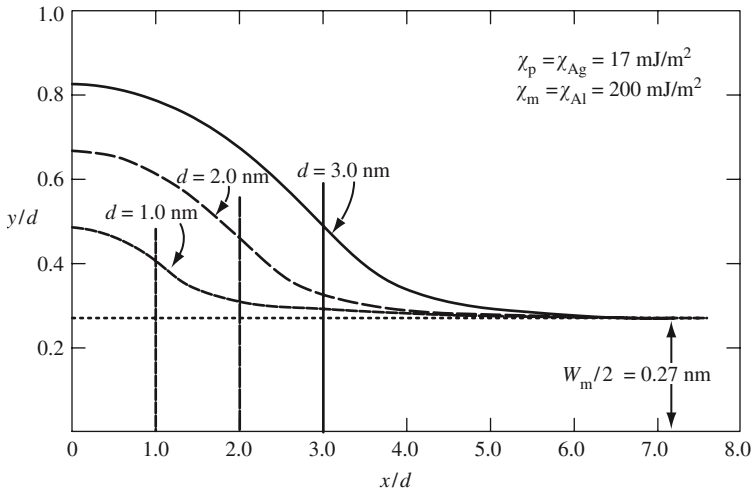


FIG. 6.15. Results for the dislocation extension between partials inside and outside the “precipitate strip” for different strip widths and the Al–Ag pair.

where the numerical constant has again been determined for the properties inside and outside the strip.

Correspondingly, the total change in line energy inside the strip is

$$\Delta\mathfrak{F}_i = \mathfrak{F}_{if} - \mathfrak{F}_{ii}, \quad (6.72)$$

which in turn becomes

$$\Delta\mathfrak{F}_i = \frac{\mu b^2 d}{12\pi} \left(\frac{2+\nu}{1-\nu} \right) f_i \left(\frac{y_p}{(w_m/2)} \right), \quad (6.73)$$

with the function f_i being

$$f_i = \left[-\ln \left(1.333 \frac{y_p}{(w_m/2)} \right) + 0.113 \left(\frac{y_p}{(w_m/2)} - 1 \right) \right]. \quad (6.74)$$

Finally, the total energy change for the strip of width $2d$ is

$$\Delta\mathfrak{F}_{\text{tot}} = \Delta\mathfrak{F}_{io} + \Delta\mathfrak{F}_i, \quad (6.75)$$

giving

$$\Delta\mathfrak{F}_{\text{tot}} = -\frac{\mu b^2 d}{12\pi} \left(\frac{2+\nu}{1-\nu} \right) f_t(d) \quad (6.76a)$$

and

$$\begin{aligned} f_t(d) = & \left\{ \ln \left(1.333 \left(\frac{y_p}{w_m/2} \right) - 0.113 \left(\frac{y_p}{w_m/2} \right) - \frac{(w_m/2)}{d} \right. \right. \\ & \left. \left. \times \left(1.72 \left(\frac{y_p}{w_m/2} \right) - 1 \right)^2 \right) \right\}. \end{aligned} \quad (6.76b)$$

The function $f_t(d)$ is plotted in Fig. 6.16 as a function of d .

Transfer of the Strip Model into a Circular Precipitate The solution of the strip model and the changes in line energy determined above now make it relatively simple to package the effect for a precipitate particle of circular cross section. The strip width $2d$ becomes $2x(y)$, where

$$2x = 2\sqrt{r^2 - y^2} \quad (6.77)$$

for a circular precipitate. The total energy change of the dislocation probing the precipitate as a straight line now becomes, using the strip solution,

$$\Delta\mathfrak{F}(y) = -2r \frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) f_t(d) \sqrt{1 - \left(\frac{y}{r} \right)^2}. \quad (6.78)$$

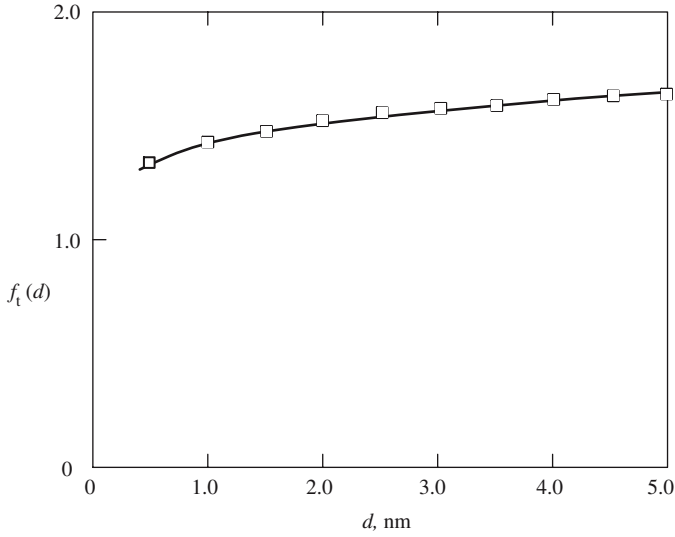


FIG. 6.16. Dependence on the precipitate diameter d of the function $f_t(d)$. See text for explanation.

Since the function $f_t(d)$ given in Fig. 6.16 varies only slowly, as a good approximation we can take $f_t(d) \approx 1.5$, which then gives

$$\Delta \mathfrak{F}(y) = -3r \frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) \sqrt{1 - \left(\frac{y}{r} \right)^2}. \quad (6.79)$$

The resistive force, in turn, is

$$K = \frac{d \Delta \mathfrak{F}(y)}{dy} = \frac{\mu b}{8\pi} \left(\frac{2+\nu}{1-\nu} \right) \frac{y/r}{\sqrt{1 - (y/r)^2}}. \quad (6.80)$$

Considering that the line energy of the dislocation is only minimally affected by splitting, we take the expression given in eq. (6.61) and evaluate it to obtain $\mathcal{E} = 0.90 \mu b^2$, giving a normalized resistive force

$$\frac{K}{2\mathcal{E}} = 0.083 \frac{y/r}{\sqrt{1 - (y/r)^2}}. \quad (6.81)$$

This is the basic force–distance curve for a precipitate in the case of the stacking-fault strengthening mechanism. We note that it is independent of precipitate size. This force–distance curve is plotted in Fig. 6.17. It shows that the resistive force becomes unbounded as $y \rightarrow r$. This indicates that the argument based on the strip model must be modified in favor of a final configuration for the extraction of the dislocation from the precipitate of the kind that is sketched in Fig. 6.14(b).

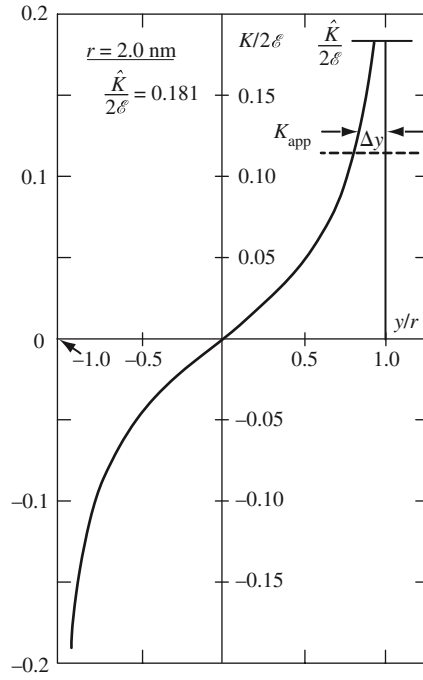


FIG. 6.17. A force–distance curve for a precipitate, offering stacking-fault strengthening.

Thus, considering an antisymmetric configuration as conceived of in Fig. 6.14(b), with the dislocation still maintained straight and the lead partial already outside the precipitate, a variant of the argument given above provides expressions for the changes of energy outside and inside the precipitate as follows:

$$\Delta\mathfrak{F}_o = 1.20 \frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) w_m \left(\frac{y_p}{w_m} - 1 \right)^2, \quad (6.82a)$$

$$\Delta\mathfrak{F}_i = -\frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) 2r f_i \left(\frac{y_p}{w_m} \right). \quad (6.82b)$$

The total energy change relative to a reference dislocation far away in the matrix becomes

$$\Delta\mathfrak{F}_{\text{tot}} \cong -\frac{\mu b^2}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) (2r) f_i \left(\frac{y_p}{w_m} \right), \quad (6.82c)$$

which is very nearly the same as $\Delta\mathfrak{F}_i$, since $\Delta\mathfrak{F}_o$ makes only a negligible contribution for the Al–Ag system. Considering now that a further displacement of the

dislocation by a distance w_m would take it outside the precipitate entirely, we have an expression for the peak resistance,

$$\hat{K} \cong \frac{\Delta \mathfrak{F}_{\text{tot}}}{w_m}. \quad (6.83)$$

In eqs. (6.82b) and (6.82c), the function f_i is

$$f_i = \left[\ln \left(\frac{4}{3} e \frac{y_p}{w_m} \right) - \left(\frac{2}{3} \frac{w_m}{w_p} + \frac{1}{3} \right) \frac{y_p}{w_m} \right] \times \left[2 \left(\frac{w_p}{r} \right) \left(\frac{w_m}{w_p} \right) \left(\frac{y_p}{w_m} \right) - \left(\frac{w_p}{r} \right) \left(\frac{w_m}{w_p} \right) \left(\frac{y_p}{w_m} \right)^2 \right]^{1/2}, \quad (6.84)$$

where $(w_m/w_p) = (\chi_p/\chi_m)$ and $e = 2.72$ (the Naperian number). These considerations give

$$\frac{\hat{K}}{2\mathcal{E}} = 0.055 \left(\frac{r}{w_m} \right) f_i, \quad (6.85)$$

which is precipitate-size-dependent and caps the force–distance curve of Fig. 6.17, as shown there for a precipitate radius of $r = 2.0$ nm, at $\hat{K}/2\mathcal{E} = 0.181$.

Finally, the limiting normalized shear resistance $\hat{\tau}/\mu$ becomes

$$\frac{\hat{\tau}}{\mu} = 1.83 \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{b}{r} \right) \left(\frac{3\pi c}{32} \right)^{1/2} \quad (6.86)$$

for an alloy with a precipitate volume fraction c .

Comparison with Experimental Results Gerold and Hartmann (1968) carried out experiments on Al single crystals containing 1.8 at% Ag, forming precipitates with various different radii obtained by coarsening treatments at 140 °C and 225 °C. The final precipitate sizes were determined by X-ray small-angle scattering. Separate estimates gave the stacking-fault energy of the Ag in the precipitates as 20 mJ/m² for the alloy aged at 140 °C and 60 mJ/m² for that aged at 225 °C. The different aging treatments also resulted in small differences in the Ag concentration, giving some changes in the precipitate volume fraction, for example, $c = 0.02$ for the alloy aged at 140 °C and $c = 0.029$ for the alloy aged at 225 °C. In addition to the room temperature experiments, Gerold and Hartmann also performed experiments at 77 K for both of the differently aged alloys. Some other material parameters used in the comparison of the model with the experiments were for $\mu = 26$ GPa, $b = 0.286$ nm, and $\nu = 0.37$, for Al.

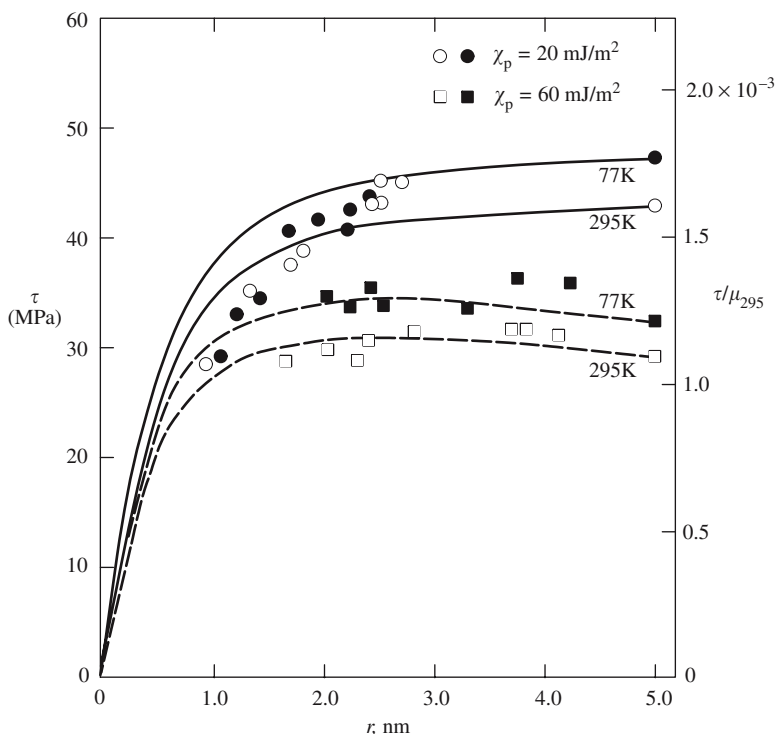


FIG. 6.18. Comparison of stacking-fault strengthening with experimental results of Gerold and Hartmann (1968) for the Al–Ag system with Ag precipitates, showing a dependence on precipitate radius of the alloy shear resistance for two separate aging treatments.

Figure 6.18 shows the computed dependences of the plastic shear resistance τ as a function of precipitate radius for the two Al–Ag alloys, compared with the experimental data. In the determination of the model predictions for 77 K, only changes in stacking-fault energy were considered, and the slight temperature dependence of the shear modulus of Al. The agreement is pleasing.

Thermal Assistance in Overcoming Precipitates Producing Stacking-fault Strengthening Considering the peaked shape of the force–distance curve of Fig. 6.17, the question must arise of the possibility of thermal assistance of dislocations over stacking-fault obstacles and whether the different levels of shear resistance shown for 77 K and 295 K in Fig. 6.18 might have been affected by thermal activation.

For an applied shear stress σ that evokes a resistive force K from a precipitate of radius r_0 , with a volume fraction c , the activation distance Δy , as shown in

Fig. 6.17, is

$$\Delta y = r_0 \left(1 - \frac{K/K_0}{\sqrt{1 + (K/K_0)^2}} \right), \quad (6.87)$$

where

$$K_0 = 2 \frac{\mu b^2}{24\pi} \left(\frac{2 + \nu}{1 - \nu} \right) f_i(r_0) \quad (6.88)$$

and the peak resistance is

$$\hat{K} = \frac{\mu b^2}{24\pi} \left(\frac{2 + \nu}{1 - \nu} \right) \left(\frac{2r_0}{w_m} \right) f_i. \quad (6.89)$$

The activation energy for overcoming the obstacle, by the arguments of Section 6.3.7, is then

$$\Delta G^* = \int_K^{\hat{K}} \Delta y(K) dK = r_0 K_0 \int_K^{\hat{K}} \left(1 - \frac{(K/K_0)}{\sqrt{1 + (K/K_0)^2}} \right) \frac{dK}{K_0}, \quad (6.90)$$

which gives

$$\Delta G^* = \frac{\mu b^3}{24\pi} \left(\frac{2 + \nu}{1 - \nu} \right) \left(\frac{2r_0^2}{bw_m} \right) f_i g \left(\frac{K}{\hat{K}} \right), \quad (6.91)$$

where

$$g \left(\frac{K}{\hat{K}} \right) = \left\{ 1 - \frac{K}{\hat{K}} - \sqrt{\left(\frac{K_0}{\hat{K}} \right)^2 + 1} + \sqrt{\left(\frac{K_0}{\hat{K}} \right)^2 + \left(\frac{K}{\hat{K}} \right)^2} \right\}. \quad (6.92)$$

Evaluation of eqs. (6.91) and (6.92) for the parameters stated above gives for the final activation energy

$$\Delta G^* = \Delta G_0 \left(\frac{r}{w_m} \right)^{0.648} g \left(\frac{K}{\hat{K}} \right), \quad (6.93)$$

where

$$\Delta G_0 = (0.214) \frac{\mu b^3}{24\pi} \left(\frac{2 + \nu}{1 - \nu} \right). \quad (6.94)$$

The dimensionless resistive-force function $g(K/\hat{K})$ is plotted in Fig. 6.19. To determine the temperature dependence of the plastic shear resistance τ from the usual kinetic expression

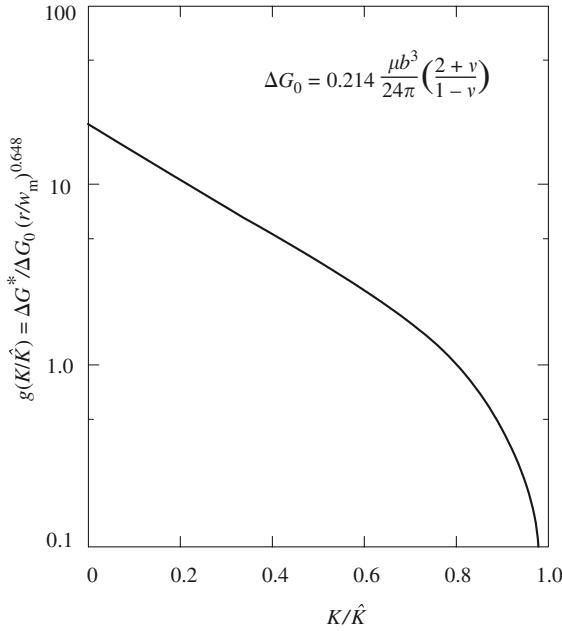


FIG. 6.19. The dependence on $K/\hat{K}$ of the dimensionless resistive-force function $g(K/\hat{K})$. See text for explanation.

$$\dot{\gamma} = \dot{\gamma}_0 \exp \left\{ -\frac{\Delta G^*}{kT} \left(r_0, \left(\frac{K}{\hat{K}} \right) \right) \right\}, \quad (6.95a)$$

the preexponential factor

$$\dot{\gamma}_0 = b \rho_m \lambda \nu_G \quad (6.95b)$$

needs to be estimated first. Taking $\rho_m = 10^9 \text{ cm}^{-2}$, $\lambda = l/(\hat{K}/2\mathcal{E})^{1/2}$, $\nu_G = 10^{11} \text{ s}^{-1}$, $l = r_0(32/3\pi c)^{1/2}$, and $\hat{K}/2\mathcal{E} = 0.25$, together with $r_0 = 3.5 \text{ nm}$ as a representative intermediate size of precipitate, gives for $c = 0.02$

$$\dot{\gamma}_0 = 2.87 \times 10^7 \text{ s}^{-1}. \quad (6.95c)$$

Moreover, considering a machine strain rate of $\dot{\gamma} = 10^{-4} \text{ s}^{-1}$ gives, using eq. (6.95a),

$$kT \ln \left(\frac{\dot{\gamma}_0}{\dot{\gamma}} \right) = \frac{\mu b^3}{24\pi} \left(\frac{2+\nu}{1-\nu} \right) \left(\frac{r_0}{w_m} \right)^{0.648} g \left(\frac{K}{\hat{K}} \right), \quad (6.96)$$

from which $K/\hat{K}$ can be determined as a function of T .

We define, as usual, a cutoff temperature

$$T_0 \equiv \frac{\Delta G_0 g_0(K/\hat{K})}{k \ln(\dot{\gamma}_0/\dot{\gamma})}, \quad (6.97a)$$

where $g_0(K/\hat{K})$ is the value at $K/\hat{K} = 0$, that is, close to 20, as shown in Fig. 6.19. For $r_0 = 3.5$ nm, $w_m = 0.54$ nm for Al, and $\ln(\dot{\gamma}_0/\dot{\gamma}) = 26.4$ on the basis of the above estimates, the temperature T_0 is estimated to be

$$T_0 = 5.58 \times 10^3 \text{ K}. \quad (6.97b)$$

These arguments then give

$$\frac{T}{T_0} = \frac{g(K/\hat{K})}{g_0(K/\hat{K})} = 0.05 g\left(\frac{K}{\hat{K}}\right). \quad (6.98)$$

This functional dependence is plotted in Fig. 6.20(a), together with the associated dependence of $\tau/\hat{\tau} = (K/\hat{K})^{3/2}$. To visualize better any possible temperature dependence of $\tau/\hat{\tau}$ between room temperature and 77 K, the dependence in

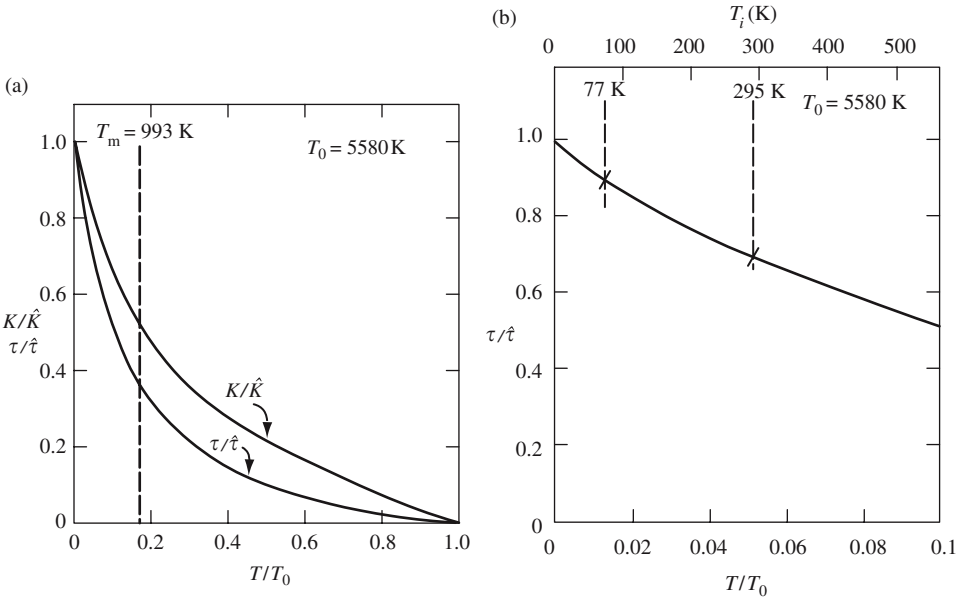


FIG. 6.20. (a) Temperature dependence of the plastic shear resistance (and the obstacle resistive force) in a stacking-fault-strengthened system of Al-Ag, based on information from Gerold and Hartmann (1968); (b) expansion of the low-temperature region of (a).

Fig. 6.20(a) is expanded in Fig. 6.20(b). Comparison of these predictions with the experimental results for the two alloys suggests a somewhat larger temperature dependence beyond what was observed in the experiments. The reason for this small discrepancy is not clear. What is clear, however, is that for the dislocation/precipitate interactions shown by the force–distance curve the interaction depth of the precipitate is quite large, resulting in a very large cutoff temperature. This is a characteristic feature of precipitate obstacles and furnishes an explanation for the generally quite small temperature dependence of the plastic resistance.

6.4.4 Atomic-order Strengthening

Model In alloys containing compositions that can form superlattice ordering, the precipitates can be domains of a superlattice. This occurs in many compositions of Ni and Al, which can give rise to ordered Ni_3Al -type superlattice precipitates. These resist penetration by matrix dislocations, which disorder the precipitates upon shearing and produce antiphase boundaries inside the precipitates.

Figure 6.21 shows a set of Ni_3Al precipitates roughly 12 nm in diameter in a Ni alloy aged at 750 °C, studied in some detail by Gleiter and Hornbogen (1965*a,b*). In the homogeneous supersaturated form of the alloy, dislocations move singly, as shown in Fig. 6.22(a). However, upon formation of ordered precipitates, when these are sheared to form internal antiphase boundaries, dislocations pair up at a characteristic spacing Λ where the second dislocation of the pair, moving in synchronism in the same glide plane, restores internal order in the precipitate. Such a case is shown in Fig. 6.22(b) in the same alloy as in Figs. 6.21 and 6.22(a), after a short two-minute aging at 750 °C that produces ordered precipitates. Figure 6.22(c) shows more pronounced pairing after aging at 750 °C for one hour, resulting in

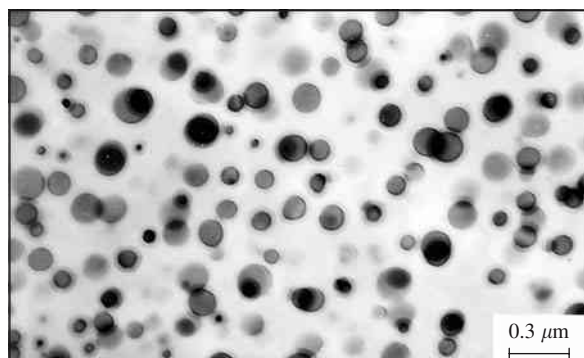


FIG. 6.21. TEM micrograph of ordered Ni_3Al particles in an Al–Ni–Cr alloy (Gleiter and Hornbogen 1965; courtesy of Wiley-VCH).

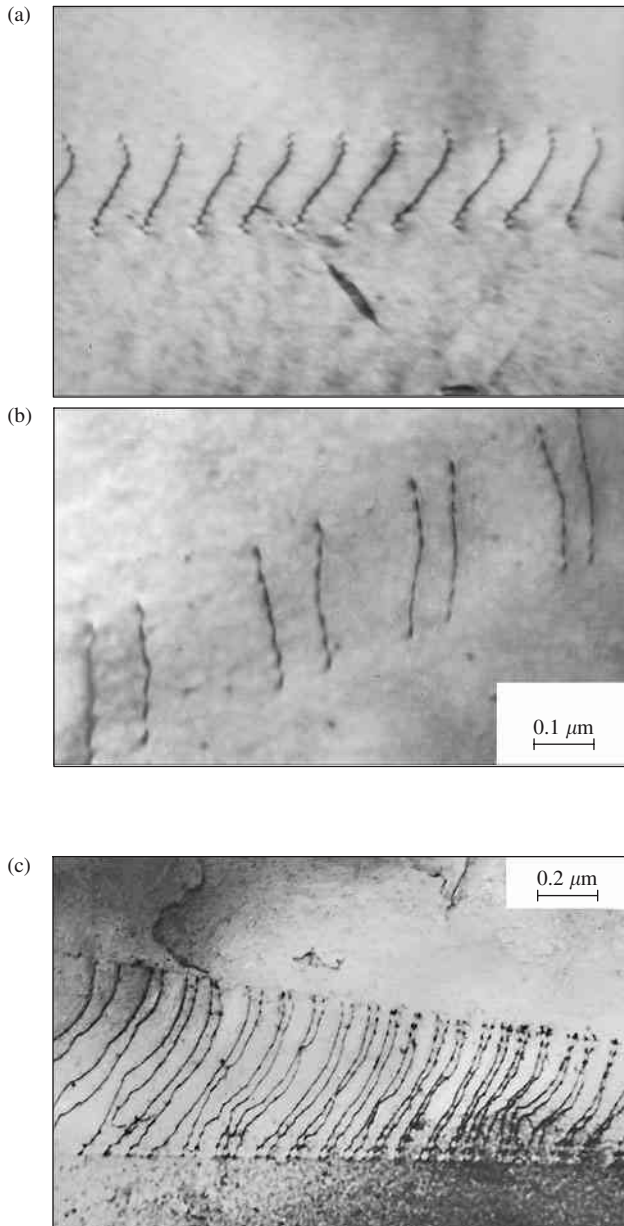


FIG. 6.22. (a) Array of dislocations advancing singly in an initially disordered Al-Ni-Cr alloy; (b) pairing-up of dislocation in the same alloy after 120 s of thermal treatment at 750 °C, resulting in internally ordered Ni_3Al precipitates; (c) array of paired dislocations in the same alloy after precipitate coarsening for 3600 s at 750 °C (Gleiter and Hornbogen 1965*a,b*; courtesy of Wiley-VCH).

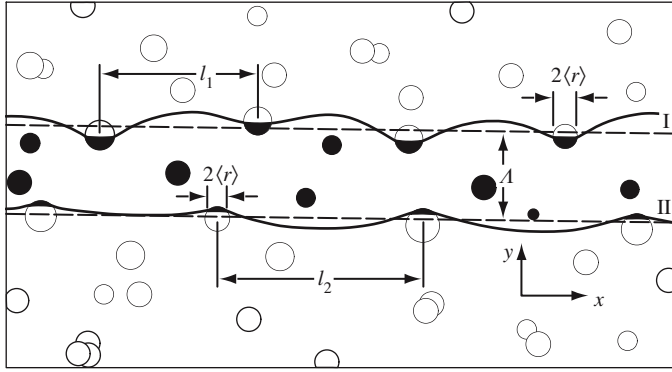


FIG. 6.23. Sketch of the form of probing of ordered Ni_3Al precipitates in an Al–Ni–Cr alloy by a coupled pair of dislocations in the underaged behavior region (after Gleiter and Hornbogen 1967).

substantial coarsening of the precipitates. Figure 6.23 depicts a typical pair of coupled dislocations moving through a field of ordered precipitates in a configuration of steady-state flow. The first dislocation of the pair has sampled precipitates at a spacing l_1 , having sheared them halfway through, where maximum resistance is achieved. The precipitates that have been fully sheared by the first dislocation, now lying inside the strip Λ , all contain antiphase boundaries that attract the second dislocation when contacted by the latter. Since the second dislocation is repelled by the first, it tends to maintain a relatively straight shape. Under an applied shear stress σ , both dislocations are subjected to additional forces resulting from their mutual repulsion, as well as interactions with the antiphase boundaries inside the precipitates. The latter oppose the lead dislocation but aid the second dislocation forward. Force equilibrium for the two dislocations of the pair gives, per unit length,

$$\sigma b + \frac{\mu b^2}{2\pi(1-\nu)\Lambda} - \chi_{\text{APB}} \frac{d_1}{l_1} = 0, \quad (6.99a)$$

$$\sigma b - \frac{\mu b^2}{2\pi(1-\nu)\Lambda} + \chi_{\text{APB}} \frac{d_2}{l_2} = 0, \quad (6.99b)$$

where the force terms represent the Peach–Kochler force of the applied stress, the force of interaction between the two dislocations, and the interaction force of the dislocations with the antiphase boundaries inside the precipitates. In the equations, Λ is the mean distance between the dislocations of the pair and χ_{APB} is the antiphase boundary energy. The mean distance l_1 between the precipitates sampled by the lead dislocation, under the applied stress and the interaction stress with the second dislocation, is taken as the Friedel sampling length $l_1 = \lambda = l/(\hat{K}/2\mathcal{E})^{1/2}$, where

l is the mean center-to-center spacing of precipitates in the glide plane. A reasonable assumption for the ratio d_2/l_2 for the second dislocation, which advances roughly as a straight line, is that it can be taken approximately as the volume fraction c of precipitates.

This strengthening mechanism was considered by a number of investigators, starting with Gleiter and Hornbogen (1965*a,b*), Raynor and Silcock (1970), and Brown and Ham (1971), and more recently by Ardell (1985, 1994) and Reppich (1993), among others. Our treatment follows these earlier considerations with certain modifications and some simplifications.

Adding eqs. (6.99a) and (6.99b) gives the shear resistance experienced by the pair of dislocations in the flow configuration,

$$\sigma = \frac{2\langle r \rangle (\chi_{\text{APB}}/2)}{2\mathcal{E}} \left(\frac{2\mathcal{E}}{bl} \right) \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{1/2} - \frac{\chi_{\text{APB}}}{2b} c, \quad (6.100)$$

where the terms have been arranged to emphasize that the stress concentrates the force on the precipitates through the coupled motion of two dislocations. Thus, the first set of precipitates probed by the lead dislocation act as if they had only half the strength compared with the case of an individual dislocation interacting with precipitates singly. Thus, accordingly, we take

$$\langle r \rangle \chi_{\text{APB}} = \hat{K}, \quad (6.101)$$

defining the peak precipitate resistance. Taking

$$\lambda = \frac{l}{(\hat{K}/2\mathcal{E})^{1/2}} = \frac{l}{(\langle r \rangle \chi_{\text{APB}}/2\mathcal{E})^{1/2}} \quad (6.102)$$

translates the form of eq. (6.100) for the peak plastic resistance $\hat{\tau}$ into

$$\begin{aligned} \sigma = \hat{\tau} &= \left(\frac{\hat{K}}{2\mathcal{E}} \right) \left(\frac{2\mathcal{E}}{bl} \right) \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{1/2} - \frac{\chi_{\text{APB}}}{2b} c \\ &= \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{b\langle r \rangle} \right) \left(\frac{3\pi c}{3z} \right)^{1/2} - \frac{\chi_{\text{APB}}}{2b} c, \end{aligned} \quad (6.103)$$

where c is the volume fraction of precipitates.

The form of eq. (6.103) demonstrates that the paired motion of the two dislocations can best be interpreted as if the lead dislocation, representing the pair, overcomes the resistance of the field (of precipitates of half strength) in the conventional manner, but being aided by the push of the second dislocation through its attractive interaction with previously sheared precipitates.

In the underaged behavior region, where the motion of the two dislocations is only loosely coupled, reformulation of eq. (6.103) gives

$$\hat{\tau} = \left(\frac{\langle r \rangle \chi_{\text{APB}}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{b\langle r \rangle} \right) \left(\frac{3\pi c}{32} \right)^{1/2} - \frac{\chi_{\text{APB}}}{2b} c \quad (6.104a)$$

and

$$\frac{\hat{\tau}}{c^{1/2}} \frac{2b}{\chi_{\text{APB}}} = 1.731 \left(\frac{\langle r \rangle}{L} \right)^{1/2} - c^{1/2}, \quad (6.104b)$$

where we have introduced a new length parameter

$$L = \frac{\mu b^2}{\chi_{\text{APB}}} \quad (6.105)$$

and taken account of the anisotropy of the line energy of edge dislocation segments, which specifies (Ardell 1994; Kocks *et al.* 1975)

$$2\mathcal{E} = 0.393 \mu b^2. \quad (6.106)$$

This expression will hold until the peak alloy strength is reached as the precipitate size increases with continued aging, and the mean spacing Λ between dislocations in the pair steadily decreases.

Equation (6.104b) holds at flow when the lead dislocation repeatedly probes new precipitates by overcoming their peak resistive force $\hat{K}$ halfway across the precipitates. For stress levels less than the alloy peak resistance $\hat{\tau}$ of eq. (6.104b), when a steady state of flow is not possible, the lead dislocation penetrates a precipitate by only a distance $y < \langle r \rangle$ as it evokes a resistive force $K < \hat{K}$ according to a characteristic force–distance curve

$$K = \langle r \rangle \chi_{\text{APB}} \left[2 \frac{\Lambda}{\langle r \rangle} - \left(\frac{\Lambda}{\langle r \rangle} \right)^2 \right]^{1/2}, \quad (6.107)$$

which is plotted in Fig. 6.24 as the continuous semiellipse. Subtraction of eq. (6.99b) from eq. (6.99a) and some simplification gives

$$\begin{aligned} \frac{\Lambda}{\langle r \rangle} &= \frac{1}{\pi(1-\nu)} \frac{L}{\langle r \rangle} \left[2 \left(\frac{3\pi c}{32} \right)^{1/2} \left(\frac{\langle r \rangle \chi_{\text{APB}}}{0.393 \mu b^2} \right)^{1/2} + c \right]^{-1} \\ &= \frac{1}{\pi(1-\nu)} \frac{L}{\langle r \rangle} \left[1.731 c^{1/2} \left(\frac{\langle r \rangle}{L} \right)^{1/2} + c \right]^{-1}, \end{aligned} \quad (6.108)$$

which indicates that as the alloy ages and the precipitates coarsen, $\Lambda/\langle r \rangle$ decreases monotonically.

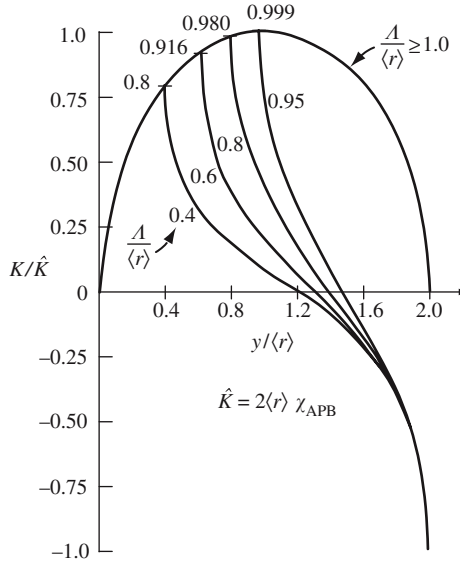


FIG. 6.24. Force–distance curves for shearing internally ordered spherical precipitates for different $\Lambda/\langle r \rangle$ ratios, where $\Lambda/\langle r \rangle < 1.0$ relates to a region of overaging with strongly coupled pairs of dislocations.

When $\Lambda/\langle r \rangle$ becomes unity, the second dislocation just enters a precipitate when the first one is halfway across, at its position of overcoming the peak resistive force $\hat{K}$. When this occurs, the peak alloy shear resistance $\hat{\tau}_p$ is reached. When $\Lambda/\langle r \rangle < 1.0$, the second dislocation, having entered the precipitate, begins to remove the antiphase boundary left by the first dislocation before the first dislocation encounters the peak resistive force. This begins to lower the alloy shear resistance $\hat{\tau}$ by decreasing the rate of production of antiphase boundary. This is reflected in the force–distance curve, which now becomes

$$\frac{K}{\hat{K}} = \frac{K}{\langle r \rangle \chi_{\text{APB}}} = \left[\left(2 \frac{y}{\langle r \rangle} - \left(\frac{y}{\langle r \rangle} \right)^2 \right)^{1/2} - \left(2 \left(\frac{y}{\langle r \rangle} - \frac{\Lambda}{\langle r \rangle} \right) - \left(\frac{y}{\langle r \rangle} - \frac{\Lambda}{\langle r \rangle} \right)^2 \right)^{1/2} \right], \quad (6.109)$$

where y is still the position of the lead dislocation in the precipitate. Since Λ is the distance between the two dislocations, now both in the same precipitate, the remaining terms represent the position of the second dislocation and its effect on the force–distance curve. Figure 6.24 shows four force–distance curves for $\Lambda/\langle r \rangle < 1.0$, for values of 0.95, 0.8, 0.6, and 0.4. These all peak at reduced levels

and have distorted descending parts. It is important to note that the peaking of the force–distance curves for $\Lambda/\langle r \rangle < 1.0$ always occurs just at the entry of the second dislocation into the precipitate, that is, when $y = \Lambda$. Thus, for $\Lambda/\langle r \rangle < 1.0$, the peak precipitate shear resistive force can be given simply as

$$\hat{K} = \langle r \rangle \chi_{\text{APB}} \left(2 \frac{\Lambda}{\langle r \rangle} = \left(\frac{\Lambda}{\langle r \rangle} \right)^2 \right)^{1/2}. \quad (6.110)$$

The condition $\Lambda/\langle r \rangle = 1.0$ represents the peak alloy shear resistance when the precipitate radius reaches a limiting value of $\langle r_p \rangle$, which is obtained from eq. (6.108) for $\Lambda/\langle r \rangle = 1.0$. The dependence of $\langle r_p \rangle$, in units of L , on the volume fraction c of precipitates is plotted in Fig. 6.25. For $\Lambda/\langle r \rangle < 1.0$, the motion of the two dislocations now becomes closely coupled. Figure 6.26(a) shows a micrograph of such a case in a similar alloy aged for a much longer period, where the coarsened precipitates now contain both dislocations at flow.

It is of interest to inquire what the level of the normalized peak resistive force $\hat{K}/2\mathcal{E}$ becomes at and beyond the condition when the peak alloy resistance $\hat{\tau}_p$ is reached. This condition is given by the defining expression

$$\frac{\hat{K}}{2\mathcal{E}} = \frac{\langle r_p \rangle \chi_{\text{APB}}}{2\mathcal{E}} = 2.54 \frac{\langle r_p \rangle}{L}, \quad (6.111)$$

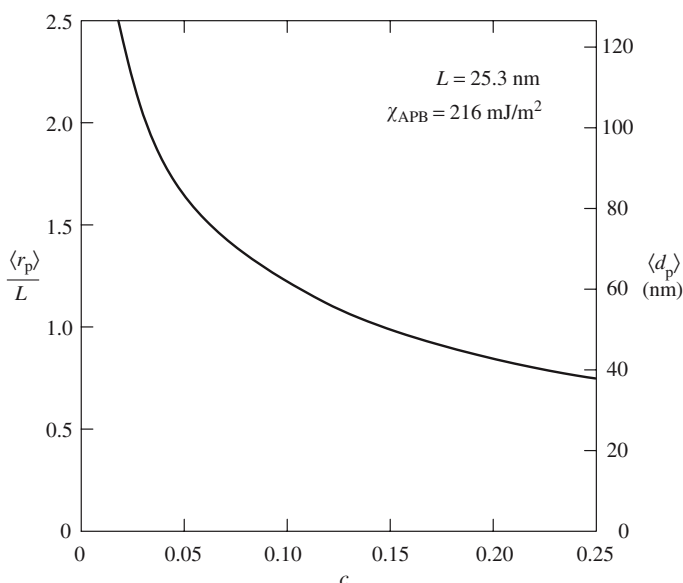
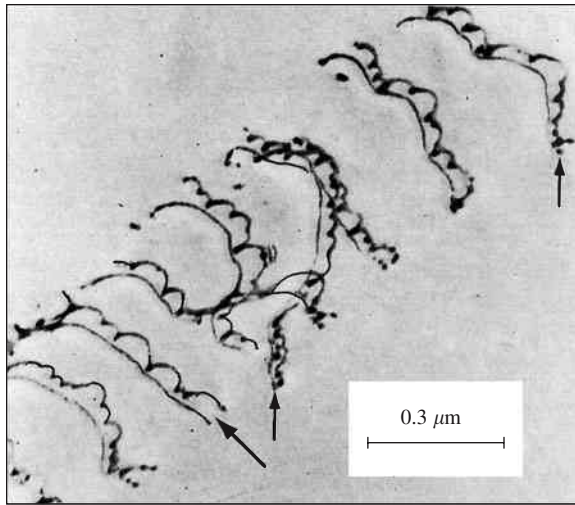


FIG. 6.25. Dependence on precipitate volume fraction of the radius of the precipitates for peak strength.

(a)



(b)

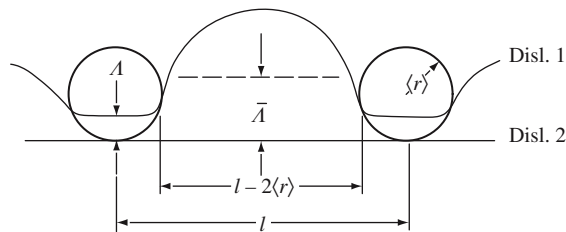


FIG. 6.26. (a) TEM micrograph of strongly coupled motion of dislocation pairs through ordered precipitates in the overaged region; (b) sketch defining the area-averaged dislocation spacing between precipitates (micrograph from Nembach *et al.* 1985; courtesy of Taylor and Francis).

where account has been taken of the line tension anisotropy through eq. (6.106). Examination of Fig. 6.25 shows that for all precipitate volume fractions of interest, $\hat{K}/2\mathcal{E}$ exceeds unity by a substantial factor at the peak alloy shear resistance. However, direct observation of the shapes of dislocations by TEM shows that the Orowan looping process around precipitates does not occur, because the motion of the pair of dislocations remains strongly coupled; both dislocations continue to shear through the precipitates and the second dislocation remains relatively unbowed under the repulsive action of the lead dislocation, largely because, under the condition of paired motion, the precipitates probed by the lead dislocation behave as if of half strength.

Later, below, we address the question of the eventual impenetrability of ordered precipitates.

In the overaged behavior range, when the lead dislocation strongly curves, the approximation of considering both dislocations as quasi-straight as in Fig. 6.23 is no longer correct. As the second dislocation in the lower left corner of Fig. 6.26(a), marked by an arrow, shows clearly, the lead dislocation now samples the precipitates roughly at the mean precipitate spacing, l , in the plane. This demonstrates that the interaction between the two dislocations occurs differently inside the precipitates from outside between them. Therefore, eqs. (6.99a) and (6.99b) need to be modified by introducing an effective interaction distance $\bar{\Lambda}$ for the region between the precipitates based on an area-average spacing, as illustrated in Fig. 6.26(b), which gives

$$\bar{\Lambda} \cong \frac{\langle r \rangle (l - 2\langle r \rangle) + \pi (l - 2\langle r \rangle)^2 / 8}{(l - 2\langle r \rangle)}, \quad (6.112a)$$

$$\frac{\bar{\Lambda}}{\langle r \rangle} = 1 + \frac{\pi}{8} \left(\left(\frac{32}{3\pi c} \right)^{1/2} - 2 \right). \quad (6.112b)$$

With this consideration, the two equations (6.99a) and (6.99b) become, in modified form,

$$\sigma b + \frac{\mu b^2}{2\pi(1-\nu)\Lambda} \left(\frac{2x}{l} \right) + \frac{\mu b^2}{2\pi(1-\nu)\bar{\Lambda}} \left(1 - \frac{2x}{l} \right) - \chi_{\text{APB}} \left(\frac{2x}{l} \right) = 0, \quad (6.113a)$$

$$\sigma b - \frac{\mu b^2}{2\pi(1-\nu)\Lambda} \left(\frac{2x}{l} \right) - \frac{\mu b^2}{2\pi(1-\nu)\bar{\Lambda}} \left(1 - \frac{2x}{l} \right) = 0, \quad (6.113b)$$

where use has been made of the fact that at flow, the second dislocation just touches a precipitate while the first is at a distance $y = \Lambda$ into the precipitate.

As before, adding and subtracting the two equations from each other gives expressions for the plastic shear resistance $\hat{\tau}$ and the separation Λ inside the precipitates as follows:

$$\frac{\hat{\tau}}{c^{1/2}} \frac{2b}{\chi_{\text{APB}}} = 1.085 \left(2 \frac{\Lambda}{\langle r \rangle} - \left(\frac{\Lambda}{\langle r \rangle} \right)^2 \right)^{1/2}, \quad (6.114a)$$

$$\left(\frac{1}{\Lambda} - \frac{1}{\bar{\Lambda}} - \frac{\pi(1-\nu)}{L} \right) \left(\frac{2\langle r \rangle}{l} \right) \left(2 \frac{\Lambda}{\langle r \rangle} - \left(\frac{\Lambda}{\langle r \rangle} \right)^2 \right)^{1/2} + \frac{1}{\bar{\Lambda}} = 0. \quad (6.114b)$$

Utilizing eq. (6.112b) and considering a typical volume fraction $c = 0.075$, eq. (6.114b) is transformed into an equation for the required factor $\Lambda/\langle r \rangle$ as follows:

$$\left(2 \frac{\Lambda}{\langle r \rangle} - \left(\frac{\Lambda}{\langle r \rangle} \right)^2 \right)^{1/2} \left(\frac{\langle r \rangle}{\Lambda} - 0.350 - 2.27 \frac{\langle r \rangle}{L} \right) + 1.18 = 0, \quad (6.114c)$$

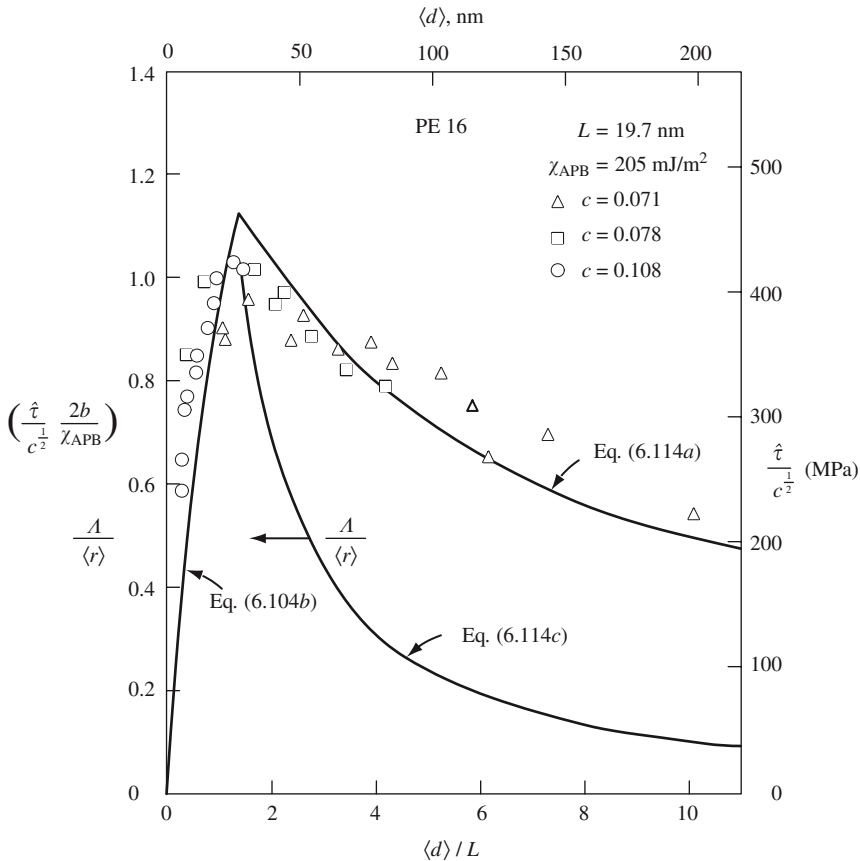


FIG. 6.27. Comparison of the model of strengthening by ordered precipitates with experimental results for a Nimonic PE-16 alloy with three different volume fractions of precipitates (experimental data from Reppich 1993).

which is readily solved numerically as a function of $\langle r \rangle / L$. This dependence is plotted in Fig. 6.27 as a function of $\langle d \rangle / L$.

Comparison with Experimental Results Reppich (1993), Ardell (1994), and Nembach (1997) have all made detailed comparisons of experimental results on Ni-based alloys containing ordered Ni_3Al precipitates with models similar to those presented in this section. Here we shall consider only the results assembled by Reppich (1993) for the special alloy Nimonic PE-16, which has a complex composition consisting of the elements Ni, Fe, Cr, Mo, Al, Ti, and C (for a more detailed breakdown, consult Nembach 1997). The ordered precipitates are of $\text{Ni}_3\text{Al,Ti}$ type, referred to as γ' precipitates, with volume fractions in the range 0.04–0.10, and are

of closely spherical shape. The actual compositions of the matrix and the ordered γ' precipitates vary considerably with heat treatment. Clearly, while the ordered precipitates are of major interest, the alloy matrix is also solid-solution strengthened to some extent. In our comparison, we shall assume that all precipitates, regardless of volume fraction and size, have the same properties, particularly with respect to their antiphase boundary energies. While small differences between the elastic properties of the matrix and the precipitates are most likely present, we shall ignore these, together with the level of possible background solid-solution strengthening in the matrix, and attribute the strengthening of the alloys primarily to the atomic-order strengthening arising from the precipitates.

The dependence on $\langle d \rangle / L$ of the normalized alloy shear resistance ($\hat{\tau} / c^{1/2}$) ($2b / \chi_{\text{APB}}$) for the underaged and overaged regions given by eqs. (6.104b) and (6.114a), respectively, for which the relevant ratios $\Lambda / \langle r \rangle$ are to be obtained from eq. (6.108) and the solution of eq. (6.114c), is plotted in Fig. 6.27. For comparison with experimental results, we take the data for five separate Nimonic PE-16 alloys assembled by Reppich (1993) (his Fig. 7.13); of these, we choose those with volume fractions of 0.071, 0.078, and 0.108. We identify the experimental band of behavior with the model curves of our Fig. 6.27 at the point of peak alloy strength, which occurs at a precipitate diameter $\langle d \rangle_p = 30$ nm, where the alloy peak resistance $\hat{\tau}_p / c^{1/2}$ is close to 425 MPa. The single choice for the antiphase boundary energy that satisfies this match best is $\chi_{\text{APB}} = 205$ mJ/m². This value is between the values of 125 to 180 mJ/m² reported by Reppich (1993) and Ardell (1994) and the value of 250 mJ/m² reported by Nembach (1997). Since most authors have treated χ_{APB} as an adjustable parameter and there are no precise measurements by independent experiments, the value obtained is acceptable. Using the value $\mu = 65$ GPa for the shear modulus of a PE-16 alloy containing a volume fraction $c = 0.075$ reported by Nembach (1997), we obtain for the length parameter $L = 19.7$ nm. This permits the plotting of the individual alloy data for the three chosen PE-16 alloys in Fig. 6.27. Considering the scatter in the experimental data, the agreement between the model and experiments is pleasing.

In Ni–Al alloys containing volume fractions of circa 0.35 of ordered Ni₃Al particles, Haasen and Labusch (1979) found that the model of Schwarz and Labusch (1978) provides good agreement with experimental results. For further comparisons of experimental results with models of varying complexity, the references cited above, namely Reppich (1993), Ardell (1994), and Nembach (1997), should be consulted.

Penetrability of Ordered Precipitates In the arguments given above, no consideration was given to the condition of the actual penetration of a dislocation into an ordered precipitate. Moreover, the possibility of dislocations bowing around impenetrable precipitates was considered to be unlikely only for topological reasons.

Some approximate answers to these questions can now be given under certain assumptions.

Considering a single dislocation pressed against an ordered precipitate of radius $\langle r \rangle$ by an applied shear stress σ , in a field of similar precipitates with a volume fraction c , force equilibrium requires that at entry,

$$\frac{\mathcal{E}}{\langle r \rangle} + \sigma b - \langle \chi_{\text{APB}} \rangle = 0, \quad (6.115a)$$

where the first term represents the line tension force imposed on the precipitate by the arms of the bowing dislocation pressed against it, while the second and third terms represent the effect of the applied stress and the resistance due to the antiphase boundary that would develop during incipient entry of the dislocation. This defines a penetration resistance τ_c of a particle to a forcing dislocation,

$$\sigma = \tau_c = \frac{\chi_{\text{APB}}}{b} - \frac{\mathcal{E}}{b\langle r \rangle}. \quad (6.115b)$$

Here, since the radius of curvature $\langle r \rangle$ is imposed on the dislocation, we take the expression for the isotropic line tension, $\mathcal{E} = (0.5)\mu b^2$.

We note first that the penetration resistance is zero or negative for precipitates of radii $\langle r_c \rangle < 0.5L$, using the length parameters defined by eq. (6.105). Thus, for precipitates in this size range, the considerations developed in the model will apply directly. For precipitates with a size above $\langle r_c \rangle$, a steeply increasing resistance develops. This is shown in Fig. 6.28 in connection with the flow stress of a PE-16 alloy having $\chi_{\text{APB}} = 205 \text{ mJ/m}^2$ as reported above, derived from comparing the model with experimental results. Equation (6.115b) also defines an asymptotic resistance of 867 MPa, which is approached for very large particles. For precipitate diameters $\langle d \rangle > \langle d_c \rangle = 2\langle r_c \rangle$ (shown by the arrow in Fig. 6.28), penetration by the dislocation becomes possible only upon consideration of shear stresses, such as the additional force due to the second dislocation in the paired motion of two dislocations.

Finally, for the question of the possible looping of dislocations around precipitates that are locally impenetrable, we consider the condition of Orowan bowing through looping around impenetrable precipitates, discussed in Chapter 3 and to be expanded on in Section 6.4.6 below. The stress τ_0 under this condition in a field of impenetrable precipitates of mean spacing l in the plane is

$$\tau_0 = \frac{2\mathcal{E}}{b(l - 2\langle r \rangle)} = \frac{2\mathcal{E}(3\pi c/32)^{1/2}}{b\langle r \rangle(1 - 2\langle r \rangle/l)}, \quad (6.116)$$

where the effect of size of the precipitates in affecting the average clear spacing between them is accounted for by the form of the denominator. Taking again an

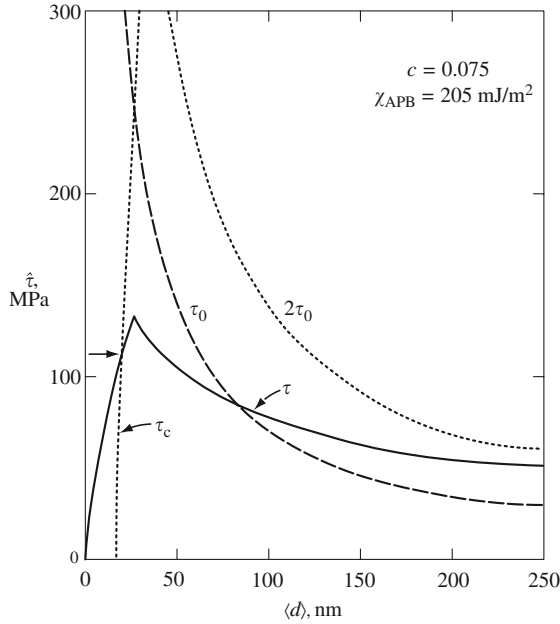


FIG. 6.28. Penetrability of ordered precipitates by paired dislocations and by individual dislocations. See text for discussion.

isotropic line tension and a typical volume fraction $c = 0.075$, we have for the dependence of the Orowan stress on the precipitate diameter

$$\tau_0 = 0.423 \frac{\mu b}{\langle d \rangle}. \quad (6.117)$$

Taking $\mu = 65 \text{ GPa}$ as was done above for PE-16 alloys, the dependence of τ_0 on $\langle d \rangle$ is plotted also in Fig. 6.28. This indicates that for precipitates roughly larger than 85 nm in diameter, the lead dislocation would encounter a lower resistance if it were to loop around the precipitate. However, this is not possible topologically, since the two dislocations in the overaged behavior region are closely coupled and together behave somewhat like a reference dislocation of twice the line tension. This pushes the dependence of the Orowan stress to larger stress levels, shown by the curve labeled $2\tau_0$ in Fig. 6.28. Thus, energetically, the looping of the pair of dislocations becomes possible only for very large precipitates of little interest.

6.4.5 Size Misfit Strengthening (Coherency Strengthening)

Model The size misfit interaction of precipitates with dislocations is the direct counterpart of the size misfit interaction of solutes with dislocations discussed in Chapter 5 and is one of the soft-contact interactions that we consider. The interaction

results from the same phenomenon of a coherently attached precipitate that has the same crystal structure as the matrix but a larger or smaller size than the matrix that it replaces setting up a stress field around it that interacts with the dislocation. As with the corresponding case in solid-solution strengthening, the size misfit interaction of the precipitate is stronger with edge dislocations than screw dislocations, in large part owing to the symmetrical nature of the interaction with edge dislocations.

The interaction of a misfitting spherical precipitate with a straight edge dislocation is particularly simple and has been considered by a number of investigators (for example McClintock and Argon 1966; Gerold and Haberkorn 1966). The more complete problem of the interaction of a misfitting particle with a flexible dislocation of initially edge character has also been considered in detail (Gleiter 1968; Wiedersich 1968). Since the essential features of the interaction for weak obstacles is defined well enough by the idealized case of a straight edge dislocation, we follow here the solutions of Gerold and Haberkorn (1966) but consider later also important changes that arise for strong obstacles.

Consider, as depicted in Fig. 6.29, a spherical precipitate of radius $\langle r_0 \rangle$, having an initial positive size misfit $\delta = \Delta r/r$, where $\Delta r = r_0 - r$, prior to insertion of the precipitate back into the cavity in the matrix of radius r , located at a distance z_0 below the slip plane of an edge dislocation. Because of the different compressibilities of the precipitate and the matrix, the actual effective size misfit parameter ϵ_s upon insertion of the larger precipitate into the matrix cavity that it occupies will be

$$\epsilon_s = \delta / \left(\frac{2\mu(1 - 2\nu_p)}{\mu_p(1 + \nu_p)} \right), \quad (6.118)$$

where μ_p and ν_p are the shear modulus and Poisson's ratio of the precipitate and μ is the shear modulus of the matrix. The shear stress σ_{zy} that can exert a force on

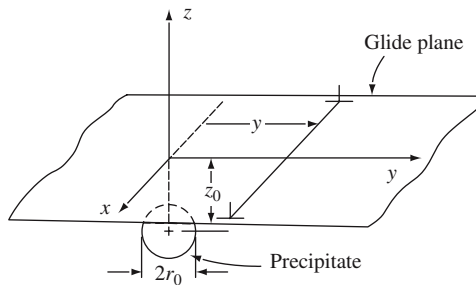


FIG. 6.29. Sketch of an edge dislocation interacting with a precipitate with a size misfit producing a coherency stress field.

the edge dislocation is readily obtained and is

$$\sigma_{zy} = \frac{6\epsilon_s \mu r_0^3 y z_0}{(x^2 + y^2 + z_0^2)^{5/2}} \quad \text{for } (z_0 > r_0), \quad (6.119)$$

for a dislocation a distance y away from the point of shortest distance between the precipitate and the slip plane. The total resistive force K exerted along the dislocation line can be obtained by integration and is

$$K = \int_{-\infty}^{\infty} b\sigma_{zy} dx = 8b\epsilon_s \mu r_0^3 \frac{z_0 y}{(y^2 + z_0^2)^2}. \quad (6.120)$$

Upon introduction of dimensionless lengths

$$\eta = \frac{y}{r_0} \quad \text{and} \quad \zeta_0 = \frac{z_0}{r_0}, \quad (6.121a,b)$$

the total resistive force becomes

$$K(\eta_1 \zeta_0) = K_0 \frac{\zeta_0 \eta}{(\eta^2 + \zeta_0^2)^2} \quad \text{for } \zeta_0 \geq 1.0, \quad (6.122)$$

where

$$K_0 = 8br_0\epsilon_s\mu. \quad (6.123)$$

For $\zeta_0 \geq 1.0$, the distribution of $K(\eta_1 \zeta_0)$ as a function of η is smooth and exhibits a maximum where, for given ζ_0 ,

$$\frac{dK}{d\eta} = 0. \quad (6.124a)$$

This occurs at

$$\eta = \frac{\zeta_0}{\sqrt{3}}, \quad (6.124b)$$

resulting in a maximum resistive force of

$$\hat{K}(\zeta_0) = K_0 \frac{3\sqrt{3}}{16} \frac{1}{\zeta_0^2} \quad (6.125)$$

when $\zeta_0 \leq r_0$ and the dislocation penetrates into the precipitate, where the retarding shear stress vanishes in the interior and the total resistive force exerted on the straight

edge dislocation comes entirely from the stress field external to the precipitate. Then, the resistive force becomes

$$K(\eta, \xi_0, \zeta_0) = K_0 \frac{\zeta_0 \eta}{(\eta^2 + \zeta_0^2)^2} \left(1 - \frac{\xi_0}{2} (2 + \eta^2 + \zeta_0^2) \right), \quad (6.126)$$

where $\xi_0 = x_0/r_0$ gives the position parallel to the x axis where the dislocation penetrates into the precipitate, with ξ_0 , being

$$\xi_0 = (1 - (\zeta_0^2 + \eta^2))^{1/2} \quad \text{for } \zeta_0^2 + \eta^2 \leq 1.0. \quad (6.127)$$

In the range $\zeta_0 \leq 1.0$, the maximum resistive force moves to the surface of the precipitate and for $\zeta_0 \leq \sqrt{3}/2$ it becomes

$$\hat{K} = K_0 \zeta_0 \sqrt{1 - \zeta_0^2}, \quad (6.128)$$

but for $\zeta_0 \geq \sqrt{3}/2$ both the maximum resistive force and its position remain given by eqs. (6.125) and (6.124b). Since the retarding shear stress σ_{zy} reverses sign with ζ_0 , the resistive forces are antisymmetric in ζ_0 . Figure 6.30 shows the dependence

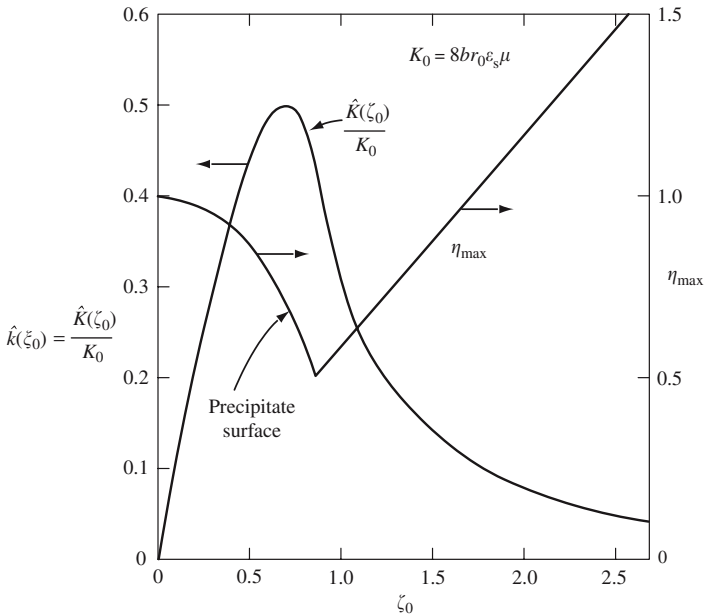


FIG. 6.30. Dependence on the dimensionless separation distance ζ_0 between the center of the precipitate and the plane of the dislocation of the normalized peak resistive force, for a misfitting precipitate with coherency stresses, based on the Gerold and Haberkorn (1966) model.

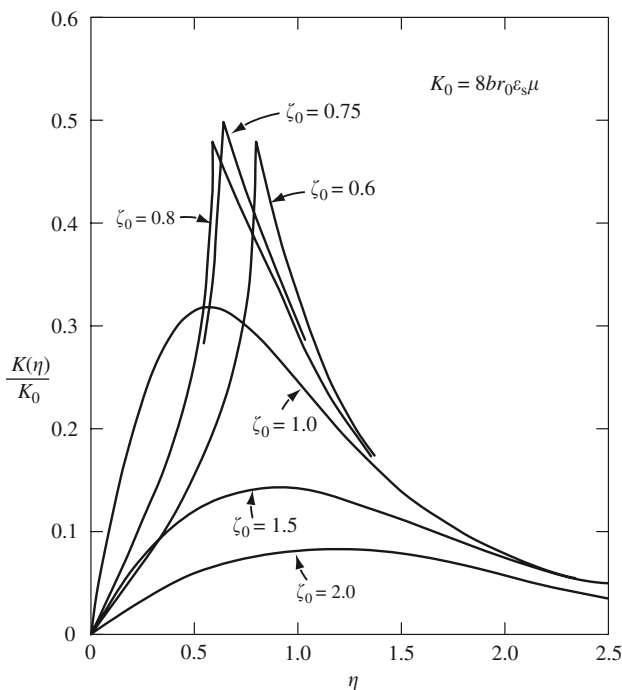


FIG. 6.31. Force–distance curves of precipitates with coherency stresses for various distances of separation ζ_0 of the precipitate center from the glide plane.

of the normalized peak resistive force $\hat{k}(\zeta_0) = \hat{K}(\zeta_0)/K_0$ on ζ_0 , and the position $\eta_{\max}$ where the maximum resistance is reached.

With the information given above, the normalized force–distance curves $K(\eta)/K_0$ for several slip planes at distances ζ_0 away from the precipitate center can be obtained, and are shown in Fig. 6.31 for a straight edge dislocation.

Since the shear stress σ_{xz} is odd in x , the interaction force of a misfitting precipitate on a straight screw dislocation parallel to the x axis has equal and opposite signs for positive and negative x , giving no net force. However, when the dislocation is allowed to flex under the local stress, resulting in a breaking of symmetry, a resistive force can appear. Wiedersich (1968) and Gleiter (1968) obtained solutions of the differential equation for a flexible dislocation line for large bow-out but did not develop these in detail. Such solutions can now be obtained routinely in order to consider complex three-dimensional motion of dislocations (see, for example, Xiang *et al.* 2003, 2004). We shall not pursue these solutions here, but shall note merely the analogy with the corresponding cases of interaction of misfitting solute atoms with both edge and screw dislocations discussed in Chapter 5, where it was demonstrated that this interaction is far stronger with

edges than with screws, making it quite sufficient to consider only straight edge dislocations.

The Plastic Resistance Since all resistive forces scale with K_0 , which depends on precipitate size, the overall plastic resistance also depends on precipitate size and exhibits peak aging behavior when the precipitates become impenetrable. In the range where the dislocation probes the field outside the precipitate, that is, when $\zeta_0 \geq 1.0$, for weak interactions, where the straight-dislocation model is acceptable, the overall plastic resistance is given by eq. (6.18) with $\hat{K}$ being given by eq. (6.125), resulting in an expression dependent on ζ_0 . Since the maximum resistive forces are experienced when the dislocation penetrates the precipitate, that is, when $\zeta_0 < 1.0$, the plastic resistance depends on the effective precipitate-cutting cross section, rather than on r_0 . Moreover, in the development of the overall plastic resistance, any typical edge dislocation interacts with a whole field of precipitates at different distances ζ_0 , requiring a full field averaging. This has been performed by Gerold and Haberkorn (1966) and Brown and Ham (1971), among others. Here we follow a variant of the procedure used by the latter investigators.

Consider a set of precipitates of actual radius r_0 with a volume fraction c , having a size misfit parameter ϵ_s , distributed randomly in space. The number density of such precipitates per unit volume is

$$n = \frac{3c}{6\pi r_0^3}. \quad (6.129)$$

A typical straight edge dislocation sampling all interactions at separate distances z_0 from its glide plane would in effect be sampling their projected effect on the glide plane as a set of real or virtual precipitates with peak strength $\hat{K}(z_0) = K_0 \hat{k}(z_0)$, with $\hat{k}(z_0)$ being given in Fig. 6.30. Since the maximum resistive force of precipitates away from the glide plane drops off sharply beyond $z_0 = 2r_0$, we consider the diffuse effect of precipitates only in strips of thickness $\Lambda = 2r_0$ on both sides of the glide plane. The entire number density N per unit area affecting the glide plane is

$$N = \frac{3c}{4\pi r_0^3} (2\Lambda), \quad (6.130)$$

giving a mean distance l between the projections of virtual precipitates on the glide plane, where

$$l = (N)^{-1/2}. \quad (6.131)$$

For weak obstacles for which the straight-dislocation model is acceptable, the plastic resistance $\hat{\tau}$ due to precipitates, all of the same type with peak resistive

forces $\hat{K}$, is given by eq. (6.17),

$$\hat{\tau} = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bl} \right), \quad (6.132)$$

or, in a form normalized by $2\mathcal{E}/bl$,

$$\hat{\tau}^* = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} = \beta^{3/2}, \quad (6.133)$$

as defined in Section 6.3.4. The plastic resistance $\hat{\tau}_i^*$ on the glide plane due to precipitates in a strip i of thickness dz_0 a distance z_0 away from the glide plane on both sides of the plane would then be, if they represented the entire set of obstacles of density N on the glide plane,

$$\hat{\tau}_i^* = \left(\frac{K_0}{2\mathcal{E}} \right)^{3/2} (\hat{k}_i(z_0))^{3/2}. \quad (6.134)$$

However, the obstacles due to precipitates in the two strips i represent only a fraction

$$dX_i = \frac{2n_i dz_0}{N} = \frac{dz_0}{\Lambda} \quad (6.135)$$

of the overall population. Therefore the total normalized composite plastic resistance $\hat{\tau}_c^*$ must be given by an integral of the contributions of all individual strips i within a range Λ on both sides of the glide plane, that is, by

$$\hat{\tau}_c^* = \left[\int_0^\Lambda \left(\frac{K_0}{2\mathcal{E}} \right)^3 (\hat{k}_i(z_0))^3 \frac{dz_0}{\Lambda} \right]^{1/2}, \quad (6.136)$$

according to the Pythagorean sum of Koppenaal and Kuhlmann-Wilsdorf (1964) and Hanson and Morris (1975*b*). After using the constant-line-tension approximation, giving

$$\frac{K_0}{2\mathcal{E}} = \frac{8\epsilon_s r_0}{b}, \quad (6.137)$$

we obtain

$$\hat{\tau}_c^* = \left[\frac{r_0}{\Lambda} \left(\frac{8\epsilon_s r_0}{b} \right)^3 \int_0^{2.0} (\hat{k}(\zeta_0))^3 d\zeta_0 \right]^{1/2}. \quad (6.138)$$

When this integral is evaluated numerically using the results shown in Fig. 6.30, the composite plastic resistance becomes, in nonnormalized form,

$$\hat{\tau}_c = \frac{2\mathcal{E}}{b} N^{1/2} \hat{\tau}_c^* = 5.813 \left(\frac{\mu b}{r_0} \right) \left(\frac{3c}{2\pi} \right)^{1/2} \left(\frac{\epsilon_s r_0}{b} \right)^{3/2}, \quad (6.139)$$

giving finally

$$\hat{\tau}_c = 4.02\mu \left(\frac{\epsilon_s^3 c r_0}{b} \right)^{1/2}. \quad (6.140)$$

This represents the plastic resistance encountered by an undissociated dislocation in the underaged behavior range in a field of misfitting precipitates.

In most FCC and HCP metals and alloys with a relatively moderate to low stacking-fault energy, dislocations are dissociated. This spreads out the interaction over a larger than atomic range in the misfit stress field of the precipitate and results in significant reduction in the resistive forces. This effect was considered in some detail by Nembach (1984), who established that, to a good degree of approximation, the peak resistive force $\hat{K} = K_0 \hat{k}(\xi_0)$ must be modified by a function Φ that depends on the precipitate particle size and the equilibrium stacking-fault width d_s , having the form

$$\Phi = \beta_1 \left(\frac{r_0}{b} \right)^{\beta_2}, \quad (6.141)$$

where the factors β_1 and β_2 are, in principle, functions of d_s/b . However, apart from relatively large variations for small stacking-fault widths, the dependence of β_1 and β_2 on d_s/b are slight, permitting a simple choice for these as $\beta_1 \approx 0.23$ and $\beta_2 \approx 0.34$ in the range of stacking fault-widths of interest in FCC metals other than Al. Thus, when incorporated into eq. (6.140), the composite plastic-resistance increment for most FCC alloys takes the form

$$\Delta \hat{\tau}_c = 4.02(\beta_1)^{3/2} \mu \epsilon_s^{3/2} c^{1/2} \left(\frac{r_0}{b} \right)^{(1+3\beta_2)/2}, \quad (6.142)$$

and upon introduction of the above values of β_1 and β_2 ,

$$\frac{\Delta \hat{\tau}_c}{\mu c^{1/2}} = 0.44 \epsilon_s^{3/2} \left(\frac{r_0}{b} \right), \quad (6.143)$$

where $\Delta \hat{\tau}_c$ is to be interpreted as the component of the plastic resistance due to the size misfit or coherency stresses of the precipitates, over and above any other contributions that could come from remaining solid-solution resistance and other forms of precipitate-induced resistance.

When the precipitates become larger, they will eventually become impenetrable and the plastic resistance will decline in the post-peak-resistance region. When the plastic resistance is governed by collective interactions as in this case, the criterion $\dot{K}/2\mathcal{E} = 1.0$ is difficult to apply. However, once the precipitates become large enough, they will become nonshearable at nearly any level. In many alloys, this condition is often reached when precipitates lose their coherency stresses and the complex diffuse interactions of dislocations with distant precipitates are no longer present. The flow stress is then governed by the Orowan mechanism and becomes

$$\Delta\hat{\tau}_c = \frac{2\mathcal{E}}{br_0} \left(\frac{3c}{2\pi} \right)^{1/2} = 0.11\mu c^{1/2} \frac{\ln(r_0/b)}{(r_0/b)}, \quad (6.144)$$

where account has been taken of the reduction of the effective line tension when dislocations wrap around small particles, reducing the line tensions of the two arms through their mutual interaction by the logarithmic factor, as we discuss further in Section 6.4.7. Thus, the plastic-resistance increment in the overaged region becomes

$$\frac{\Delta\hat{\tau}_c}{\mu c^{1/2}} = 0.11 \frac{\ln(r_0/b)}{(r_0/b)}. \quad (6.145)$$

Many investigators have reported that in the range of peak resistance, impenetrable precipitates with or without coherency stresses are often overcome by complex maneuvers rather than by a simple Orowan looping process (Gleiter and Hornbogen 1967; Ashby 1969; Hirsch and Humphreys 1969; Brown and Stobbs 1976). These maneuvers, which can involve local cross-slipping steps, result in nonplanar loops roughly parallel to the glide plane or prismatic edge loops normal to the glide plane behind the precipitate, and in some instance in repeated passage of additional dislocations in a series of prismatic loops (Hirsch and Humphreys 1969). Most of these possibilities have been modeled successfully by Xiang *et al.* (2004). Since these maneuvers occur as the dislocation circumvents the precipitate, there is often not much change in the overall flow stress; however, similar subsequent events could result in enhanced strain hardening (Chapter 7).

Comparison with Experimental Results Many investigators have attempted comparison of the coherency strengthening mechanism of size misfit interactions with experimental results. These comparisons have often been frustrating to the point of exasperation of the investigator. Thus, Ardell (1985) examined the experimental findings of eight different investigations on Cu–Co, Cu–Fe, Cu–Mn, and ordered Cu₃Au–Co alloys and found very poor agreement of the model (eq. 6.140 for undissociated dislocations) with the experiments. Ardell's comparisons were reexamined

by Reppich (1993), who found agreement in some cases but still found poor results in most of the others. An important insight has come from the modification to the model by consideration of the extended nature of dislocations in Cu, the matrix metal in most of the alloys. The sources of the poor agreement are now relatively clear and involve other mechanisms at play or even a lack of an appropriate model. We choose for comparison here only the Cu–Co alloys initially studied by Gerold and coworkers (Witt and Gerold 1969; Amin *et al.* 1975). Only in these alloys, Cu–1.4% Co and Cu–2.0% Co, are the conditions favorable for a direct comparison based on observation of coherency stresses by TEM and a relatively close match of the shear moduli of Cu and Co, ruling out an important contribution from modulus misfit strengthening, which is certainly a major factor in the poor agreement in the case of Cu–Fe alloys (Knowles and Kelly 1971), as will be noted in Section 6.4.6. The very low level of strengthening of the Cu₃Au ordered alloys with Co precipitates must be attributed to the possibility that in these ordered alloys, glide is likely to involve superdislocations. Moreover, TEM observations show mostly screw dislocations, making size misfit hardening unlikely. Finally, in the Cu–38% Mn alloy considered by Ardell (1985), the precipitate volume fraction is extremely high, requiring consideration of forms of topological percolation of dislocations between precipitates other than those which the models of Friedel and Fleischer or even Schwarz and Labusch can permit.

Thus, we consider for comparison with the model the Cu–1.4% Co and Cu–2.0% Co alloys of Witt and Gerold (1969) and Amin *et al.* (1975). We have plotted the reported data according to the underaged resistance model of eq. (6.143) and overaged resistance model of eq. (6.145) in Fig. 6.32. The model predictions for the underaged and overaged regions according to eqs. (6.143) and (6.145) are plotted in the figure by the two curves connected by the dashed line. In the overaged region, a misfit parameter of 1.6×10^{-2} was used rather than the reported value of 1.5×10^{-2} . The agreement of the model with the experimental results is reasonably good.

6.4.6 Modulus Misfit Strengthening

Model The modulus misfit interaction of precipitates with dislocations, the second soft-contact interaction that we consider here, is the direct counterpart of the modulus misfit interaction in solid-solution strengthening discussed in Chapter 5. The interaction arises when a precipitate with a shear modulus or bulk modulus different from that of the matrix is sensed by a nearby dislocation, affecting its line energy. As with the size misfit interaction, in the case of the modulus misfit interaction, while the interaction occurs even in the distant field without contact between the dislocation and the precipitate, the interaction is strongest when the dislocation actually penetrates into the precipitate. Two separate solutions have dealt with these two levels of interaction. That of Weeks *et al.* (1969) considered the distant-field interaction in detail, whereas the near-field interaction occurring when the dislocation

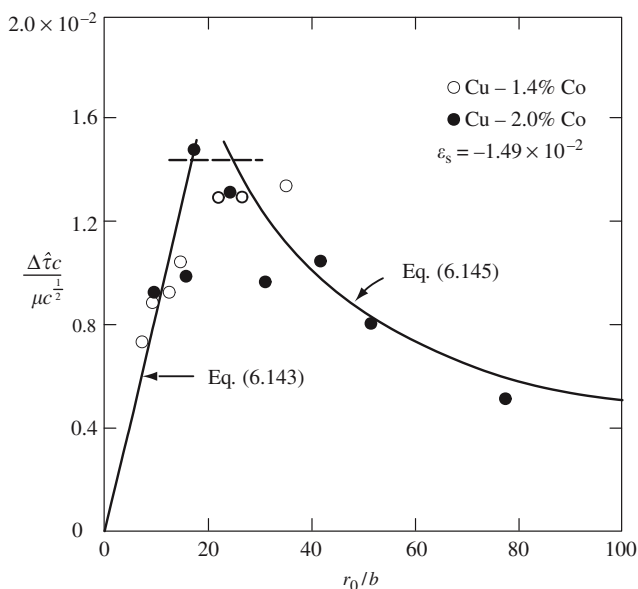


FIG. 6.32. Comparison of the coherency strengthening model with the experimental results for two Cu–Co alloys of Gerold and Haberkorn (1966) and Amin *et al.* (1975).

penetrates into the precipitate was dealt with by Knowles and Kelly (1971). We consider here, in somewhat simplified form, both of these solutions.

Weeks *et al.* considered the elastic interaction energies I_e and I_s of a spherical particle of different shear and bulk moduli with both edge and screw dislocations external to the precipitate particle. We simplify these interaction energies in several ways. First, we consider the fractional changes of the shear modulus, $\Delta\mu/\mu$, and of the bulk modulus, $\Delta K/K$, relative to the matrix as equal, where $\Delta\mu = \mu_p - \mu$ and $\Delta K = K_p - K$, and chose the Poisson's ratios to be $1/3$. This results in the following simplified forms of the interaction energies I_s and I_e of a spherical particle of radius r_0 with a screw and an edge dislocation:

$$I_s = 0.96E_0 \left(\frac{r_0}{r} \right), \quad (6.146)$$

$$I_e = 1.05E_0 \left(\frac{r_0}{r} \right)^2 \left(1 - \frac{2}{3} \sin^2 \theta \right), \quad (6.147)$$

where r is the distance between the center of the precipitate particle and the dislocation line, considered to be straight and parallel to the x axis and on a glide plane with a normal parallel to the z axis; y is the direction of motion of the dislocation;

θ is the angle between the radius vector and the positive y axis; and

$$E_0 = \frac{\mu b^3}{6\pi} \epsilon_\mu \frac{r_0}{b}, \quad (6.148)$$

where $\epsilon_\mu = \Delta\mu/\mu$ is the modulus misfit parameter.

In normalized dimensionless coordinates

$$\eta = \frac{y}{r_0}, \quad \zeta = \frac{z}{r_0}, \quad (6.149a,b)$$

for a precipitate particle with its center a distance $z_0 > r_0$ above the glide plane, these interaction energies become

$$I_s = 0.96E_0 \frac{1}{\eta^2 + \zeta_0^2}, \quad (6.150)$$

$$I_e = 1.05E_0 \frac{1}{\eta^2 + \zeta_0^2} \left(1 - \frac{2}{3} \frac{\zeta_0^2}{\eta^2 + \zeta_0^2} \right). \quad (6.151)$$

The resistive forces K_s and K_e that a precipitate with modulus misfit exerts on straight screw and edge dislocations are then

$$K_s = -\frac{1}{r_0} \frac{\partial I_s}{\partial \eta} = 0.96 \frac{E_0}{r_0} \frac{2\eta}{(\eta^2 + \zeta_0^2)^2}, \quad (6.152)$$

$$K_e = -\frac{1}{r_0} \frac{\partial I_e}{\partial \eta} = 1.05 \frac{E_0}{r_0} \left[\frac{\eta}{(\eta^2 + \zeta_0^2)^2} - \frac{4}{3} \frac{\zeta_0^2 \eta}{(\eta^2 + \zeta_0^2)^3} \right]. \quad (6.153)$$

The maximum values $\hat{K}_s$ and $\hat{K}_e$ are readily obtained from

$$\left(\frac{\partial K_s}{\partial \eta} \right)_{\zeta_0} = 0, \quad (6.154a)$$

$$\left(\frac{\partial \hat{K}_e}{\partial \eta} \right)_{\zeta_0} = 0, \quad (6.154b)$$

which gives

$$\hat{K}_s = 0.673 \frac{E_0}{r_0} \frac{1}{\zeta_0^3} \quad (6.155a)$$

at

$$\eta = \pm \frac{\zeta_0}{\sqrt{3}} \quad (6.155b)$$

and

$$\hat{K}_e = 0.192 \frac{E_0}{r_0} \frac{1}{\zeta_0^3} \quad (6.156a)$$

at

$$\eta = \pm 1.217 \zeta_0. \quad (6.156b)$$

This simple analysis shows that the distant-field interaction of a screw dislocation with a precipitate having a modulus misfit parameter ϵ_μ is stronger by more than a factor of three than that with an edge dislocation, a result quite similar to the effect of misfitting solute atoms with dislocations discussed in Chapter 5.

We shall see, moreover, that when we consider the near-field interactions below, the latter are far stronger than the distant-field interactions and those with screw dislocations are still dominant.

Knowles and Kelly (1971) considered only the interaction of a screw dislocation with misfitting spherical precipitates, in the expectation that this would be dominant, as the distant-field solution has also indicated. In their method of solution, they considered first the elastic interaction per unit length of a straight screw dislocation with a circular cylinder of radius r_0 and modulus $\mu_p > \mu$, parallel to the dislocation line, with its axis lying in the glide plane of the dislocation, that is, at $z_0 = 0$. They then constructed the interaction energy I_s of a screw dislocation with a spherical particle of radius r_0 lying in the glide plane of the dislocation, by assembling interaction energies of infinitesimal parallel discs normal to the x axis of appropriately decreasing radii, to obtain the interaction energy for a sphere by integration. They found

$$I_s = \epsilon_\mu \frac{\mu b^3}{2\pi} \left(\frac{r_0}{b} \right) I'_s(\eta), \quad (6.157)$$

where $I'_s(\eta)$ is a dimensionless function of $\eta = y/r_0$, which also includes a contribution from the core energy when the dislocation has penetrated the particle.

The dimensionless function $I'_s(\eta)$ obtained from the tabulated results of Knowles and Kelly, with certain changes of variables, is plotted in Fig. 6.33 for a precipitate with $\epsilon_\mu > 0$ for several different normalized particle radii r_0/b , as a function of the dimensionless variable η of motion of the dislocation. Clearly, the rate of change of the interaction energy with motion of the dislocation along the equator of the precipitate at $\zeta_0 = 0$ is maximum at the border of the precipitate at $\eta = 1.0$ (if the origin of the y axis is at the center of the precipitate).

The resistive force $K(\eta)$ experienced by the dislocation as it slices through the precipitate is obtained as usual as

$$K(\eta) = -\frac{\partial I_s}{\partial \eta} = -\epsilon_\mu \frac{\mu b^3}{2\pi} \left(\frac{r_0}{b} \right) \frac{1}{r_0} \frac{\partial I_s(\eta)}{\partial \eta} \quad (6.158)$$

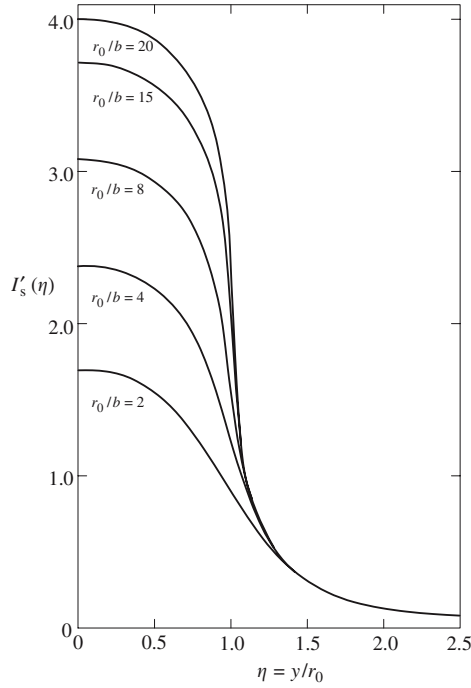


FIG. 6.33. Dependence of the interaction energy of a screw dislocation with a precipitate having a positive modulus misfit, as a function of the normalized distance η in the equatorial plane, based on the Knowles and Kelly (1971) model.

and represents the force–distance curve of the interaction of the screw dislocation with the precipitate. Such force–distance curves for dimensionless radii r_0/b of 4, 8, and 20 are plotted in Fig. 6.34 for positive ϵ_μ . The peak resistive force $\hat{K}$ occurs at $\eta = 1.0$, that is,

$$\hat{K} = -\epsilon_\mu \frac{\mu b^2}{2\pi} \left(\frac{\partial I'_s}{\partial \eta} \right)_{\eta=1.0}, \quad (6.159)$$

and in a form normalized by twice the line tension, in the constant-line-tension model, becomes

$$\frac{\hat{K}}{2\mathcal{E}} = \frac{\epsilon_\mu}{2\pi} f\left(\frac{r_0}{b}\right), \quad (6.160a)$$

where

$$f\left(\frac{r_0}{b}\right) = -\frac{\partial I'_s}{\partial \eta}. \quad (6.160b)$$

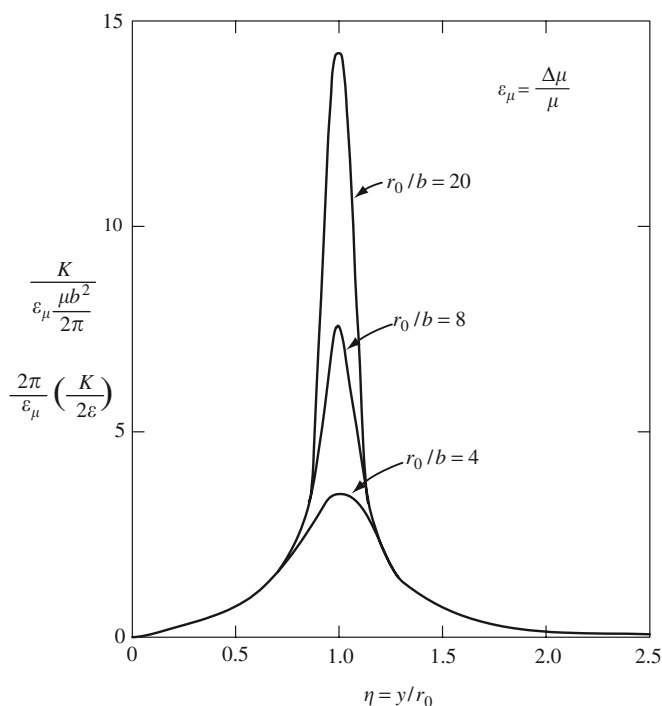


FIG. 6.34. Force–distance curves for precipitates with modulus misfit for three different radii, based on the Knowles and Kelly (1971) model.

The function $f(r_0/b)$, calculated from the tabulated results of Knowles and Kelly (their Table A1), is plotted in Fig. 6.35 over a wide range of precipitate radii, up to $r_0/b = 200$. Clearly, since $f(r_0/b)$ increases linearly, or nearly so, for (r_0/b) values up to roughly 40–50, the modulus misfit interaction should exhibit peak aging behavior with increasing precipitate size.

The Plastic Resistance For relatively dilute precipitate volume fractions where the Friedel–Fleischer model should hold, the plastic-resistance increment in the underaged region should be given by

$$\Delta\hat{\tau} = 0.044\mu\epsilon_\mu^{3/2}c^{1/2}\frac{(f(r_0/b))^{3/2}}{(r_0/b)}. \quad (6.161)$$

The function $(f(r_0/b))^{3/2}/(r_0/b)$ which governs the range over which the underaged behavior holds is plotted in Fig. 6.35. It indeed shows a maximum

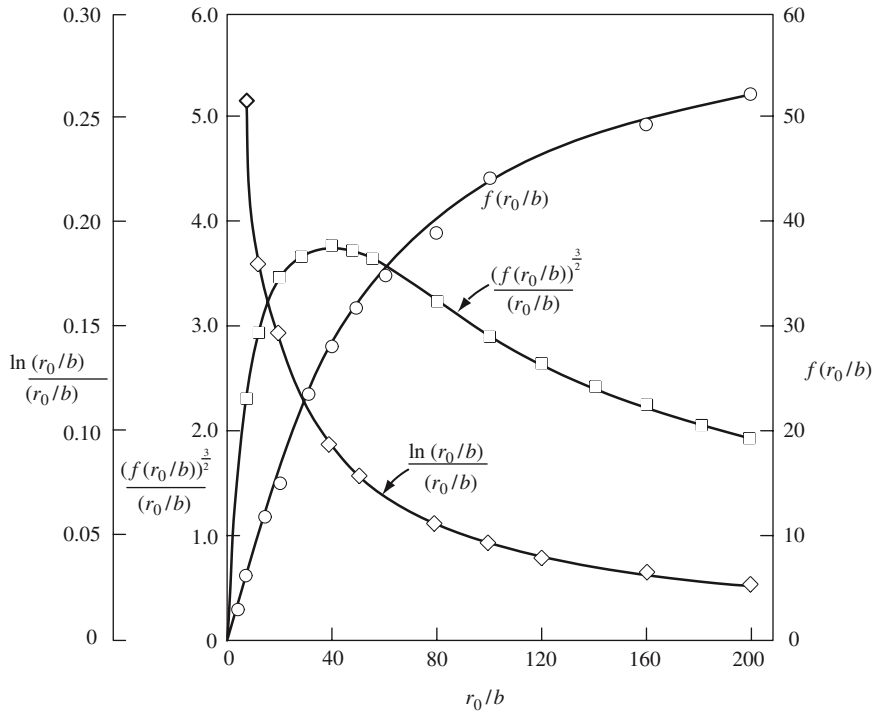


FIG. 6.35. Dependence on particle radius of the functions $f(r_0/b)$ and related functions for precipitates with positive modulus misfit, based on the Knowles and Kelly (1971) model. See text for explanation.

around $r_0/b = 40$, indicating that the peak plastic resistance occurs roughly in this region and that r_0/b values exceeding 40 are in the overaged behavior region.

The theory of Knowles and Kelley (1971) considers a full perfect dislocation interacting with a precipitate particle of different shear modulus. In FCC alloys with extended dislocations, the interaction of an extended dislocation with spherical precipitates has not been addressed in a formal manner as was done by Nembach (1984). As will become clear when we compare the model results with experiments, a correction needs to be made to account for this spread-out interaction of extended dislocations with precipitates that have a modulus misfit. When the stacking-fault width d_s is of the order of the precipitate diameter, an approximate correction could consider the interaction of the precipitate only with a partial dislocation of the total screw dislocation. Thus, since the screw component of a partial dislocation of a total screw dislocation has only half the Burgers vector of the full dislocation, the maximum resistive force $\hat{K}$ as given in eq. (6.159) should be scaled down by a factor of 0.25. When this is incorporated into eq. (6.161), a lower-bound estimate

of the plastic resistance can then be taken as

$$\Delta \hat{\tau} = (5.5 \times 10^{-3}) \mu \epsilon_{\mu}^{3/2} c^{1/2} \frac{(f(r_0/b))^{3/2}}{(r_0/b)} \quad (6.162)$$

for FCC alloy matrices in the underaged behavior region, where $d_s \geq 2r_0$.

Finally, when precipitates become impenetrable, in the overaged behavior region, the Orowan mechanism should apply as given in eq. (6.145).

Comparison with Experimental Results The comparison of model results with experiments for the modulus misfit mechanism is made difficult because precipitate particles with a coherent connection to the matrix often involve both coherency stresses and a modulus misfit. However, there are several cases where the evidence suggests that the modulus misfit mechanism is dominant. Here we select six alloy systems for comparison: (1) Cu–1.36% Fe and (2) Cu–1.15% Fe, investigated by Matsuura *et al.* (1978) and Wendt and Wagner (1980), respectively; and the four alloys (3) Cu–0.95% Fe, (4) Cu–1.45% Fe, (5) Fe–0.9% Cu, and (6) Fe–1.5% Cu, studied by Knowles and Kelly (1971). The first two Cu–Fe alloys were considered as examples for size misfit interactions by both Ardell (1985) and Reppich (1993) with quite unsatisfactory results. We shall see that the rather larger modulus difference in these alloys makes them good candidates for the modulus misfit mechanism. We recognize at the outset that the Cu matrix alloys in the underaged state will involve interactions between precipitates and partial dislocations, while in two Fe matrix alloys the interactions will be between full dislocations and precipitates. In both sets of alloys, since the miscibilities are small, we consider the Fe precipitates in the Cu matrix and the Cu precipitates in the Fe matrix as nearly pure. While the evidence is not fully supportive, we consider the precipitates to be coherent with the matrix, at least in the range of small precipitates.

The following parameters, as given by the various investigators, were assumed: (1) Cu–1.36% Fe, $c = 0.012$, $\epsilon_{\mu} = 0.638$; (2) Cu–1.15% Fe, $c = 0.003$ – 0.011 , $\epsilon_{\mu} = 0.638$; (3) Cu–0.95% Fe, $c = 0.008$, $\epsilon_{\mu} = 0.638$; (4) Cu–1.45% Fe, $c = 0.013$, $\epsilon_{\mu} = 0.689$; (5) Fe–0.9% Cu, $c = 0.008$, $\epsilon_{\mu} = -0.389$; (6) Fe–1.5% Cu, $c = 0.012$, $\epsilon_{\mu} = -0.389$. For other information related to alloy preparation, the original sources should be consulted. Except for the Cu–1.15% Fe alloy, which was studied only in the underaged condition, the alloys were coarsened to give a wide range of precipitate sizes.

The experimental data for the six alloys as reported by the various investigators are given in Fig. 6.36 together with the equations related to the applicable mechanisms. We note that for the alloys (3)–(6) the matrices either did not involve extended dislocations or existed only in coarse form, where the underaged and overaged behavior satisfied eqs. (6.161) and (6.145) quite satisfactorily. For the alloys (1) and (2), which involved either only very small precipitates (alloy (2)) or covered a wide range of precipitate sizes (alloy (1)), the Cu matrix, which contained

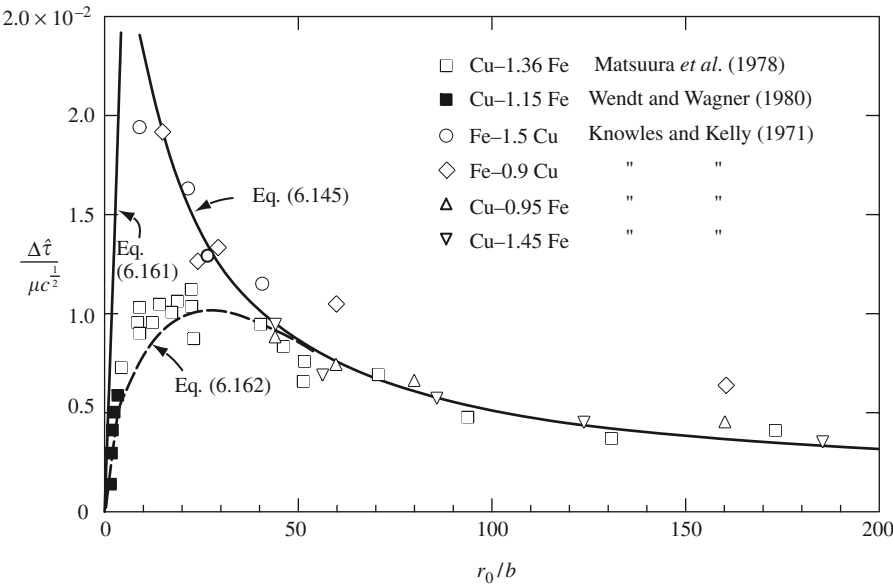


FIG. 6.36. Comparison of the strengthening model for precipitates with modulus misfit with experimental results for six Cu-Fe alloys (data from Knowles and Kelly 1971, Matsuura *et al.* 1978, and Wendt and Wagner, 1980).

extended dislocations, showed clear evidence of a lower level of plastic resistance that was reasonably well related to eq. (6.162) in the underaged region of behavior.

6.4.7 The Orowan Resistance and Dispersion Strengthening

Model In a short discussion paper at a conference on internal stresses, Orowan (1948) presented a model of landmark proportions in the theory of strengthening of alloys by precipitates and impenetrable particles involving dislocations. In this mechanism, which was presented briefly in Chapter 3 and has been invoked in many mechanisms in this chapter to explain the behavior of precipitation-strengthened alloys in the overaged region, it is noted that when particles become nonshearable the alloy shear resistance is derived from the condition of percolation of dislocations through such a field by bowing them around particles. The power and generality of the mechanism come from the fact that the nonshearable particles may be not only very large coherently attached precipitates but also incoherent particles, glassy solids, or even nanosize cavities.

The percolation stress $\hat{\tau}_0$ of a dislocation with line tension $\mathcal{E}$, through a field of impenetrable particles of cutting cross section $2\langle r \rangle$ and mean center-to-center

spacing l , is simply

$$\hat{\tau}_0 = \frac{2\mathcal{E}}{b(l - 2\langle r \rangle)}, \quad (6.163)$$

where the flexible dislocation line has to squeeze through mean gaps of size $(l - 2\langle r \rangle)$ between particles. In the critical condition of extrusion of the dislocation line between particles, where it is forced to have a radius of curvature $\rho = (l - 2\langle r \rangle)/2$, the two opposing arms of the dislocation wrapped around the particle constitute a dipole in which, through the mutual interaction of the opposing arms of the dislocation, the line tension will be substantially reduced through the usual logarithmic dependence, which we take as $\ln(\langle r \rangle/b)$, where the argument of the logarithmic factor is taken as a good measure of the ratio of the mean spacing of the arms to the effective core cutoff distance. We recognize that if the dislocation percolating through the field is on average an edge dislocation, the dipole surrounding the particle will be a screw dipole, and vice versa for a screw dislocation. Thus the resistance to edge and screw dislocations of such a particle field will be (Ashby 1969)

$$\Delta \hat{\tau}_{0e} = \alpha \frac{\mu b}{2\pi l(1 - 2\langle r \rangle/l)} \ln \left(\frac{\langle r \rangle}{b} \right), \quad (6.164a)$$

$$\Delta \hat{\tau}_{0s} = \alpha \frac{\mu b}{2\pi(1 - \nu)l(1 - 2\langle r \rangle/l)} \ln \left(\frac{\langle r \rangle}{b} \right), \quad (6.164b)$$

where $\alpha \approx 0.85$ is the usual statistical sampling factor discussed in Section 6.3.3, and $\Delta \hat{\tau}_{0e}$ or $\Delta \hat{\tau}_{0s}$ indicates that the Orowan resistance is an increment above other possible more diffuse resistances that may also be present.

Considering the two estimates of the resistance to edge and screw dislocations as if they were due to separate mechanisms, employing the Pythagorean sum rule of eq. (6.25) and the relation between l and $\langle r \rangle$ (eq. 6.9), we have finally, for $\nu \approx 1/3$,

$$\frac{\Delta \hat{\tau}_0}{\mu} = 0.132\psi(c) \frac{\ln(\langle r \rangle/b)}{(\langle r \rangle/b)}, \quad (6.165)$$

where

$$\psi(c) = \frac{c^{1/2}}{(1 - 1.085c^{1/2})} \quad (6.166)$$

is plotted in Fig. 6.37.

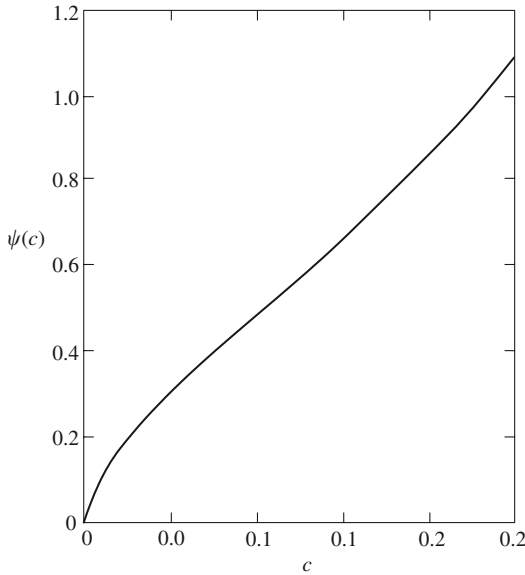


FIG. 6.37. Dependence on particle volume fraction of the function $\psi(c)$ in the Orowan resistance model for impenetrable particles.

For very dilute alloys of small particle volume fraction, where $\psi(c) \rightarrow c^{1/2}$, the form in eq. (6.165) simplifies to the forms already used earlier, for example, eq. (6.145).

Comparison with Experimental Results In earlier sections, we have already demonstrated the connection of our model to experimental results for mechanisms in the overaged behavior regime. Here we consider only two cases, involving glassy silica (SiO_2) particles and BeO particles, both in Cu single crystals, investigated by Ebeling and Ashby (1966) and Jones and Kelly (1968) and comparatively discussed by Ashby (1969). Using the experimental results of these two investigators summarized by Ashby (1969), we have plotted the model estimates of eq. (6.165) against the ratio of resolved shear stress to modulus in the experiments, where additional resistances in the relatively pure Cu single-crystal matrices were negligible. This is shown in Fig. 6.38. While the slopes of the lines connecting the model to the experimental results are somewhat larger than the expected value of unity, the agreement is pleasing and the discrepancy cannot be assigned to other factors such as anisotropy of the line tension and so-called self-interactions of the bowed-out segments across the obstacle particles. Such effects were thoroughly investigated by Bacon *et al.* (1973) and revealed other important sampling differences not included in the usual models that consider dislocations using the line tension approximation

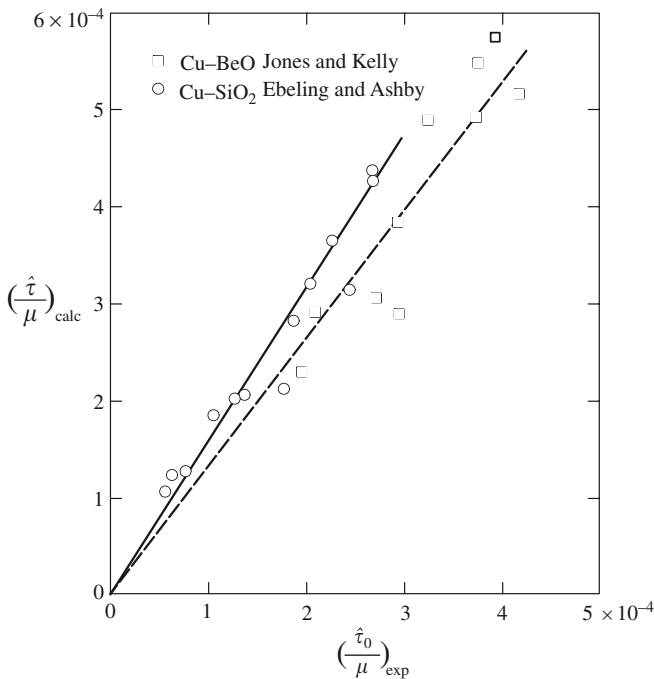


FIG. 6.38. Comparison of model of Orowan resistance with experiments, as arranged by Ashby (1969) (data from Ebeling and Ashby 1966 and Jones and Kelly 1968).

with or without anisotropy, and considerations accounted for in the Schwarz and Labusch (1978) model for finite obstacle dimensions.

6.4.8 Strengthening by Spinodal-Decomposition Microstructures

Model In the microstructures that result from spinodal decomposition, space is tessellated by an organized set of domains of two different compositions, as shown in the typical case of Fig. 6.4 for a Cu–Ni–Fe alloy. Since the different domains have a coherent lattice, the material misfit associated with them will result in corresponding tessellated internal stresses. These interact with dislocations and impede their motion, often producing substantial plastic resistance. Such internal stresses and the associated elevation of plastic resistance were considered first by Cahn (1963), who idealized the domain microstructure as three sets of mutually perpendicular continuous slabs of spinodal domains, each parallel to one of the $\{001\}$ planes. For certain specific slip planes in BCC and FCC crystal structures, Cahn determined the internal stress distribution resulting from the misfitting slabs, obtained equilibrium shapes of both edge and screw dislocation lines snaking through them

under an applied stress in the tessellated internal stress field, and derived the level of the maximum resistive force to their motion. Cahn's procedure was extended further by Kato *et al.* (1980) for dislocation lines in other orientations in the ideally regular domain field to determine different levels of plastic resistance. Noting that the actual spinodal domains are far from ideally regular and are not in the form of continuous slabs as considered by Cahn and Kato *et al.* and that the estimates of the plastic resistance by those authors did not compare at all well with experimental results, Ardell (1985) suggested that the problem fitted better to an earlier model of Mott (1952). In the Mott model, a variant of the Mott and Nabarro (1948) theory of solid-solution strengthening, the randomly wavy equilibrium shape of a dislocation line in a field of attractive heterogeneities with mean diameter w and center-to-center spacing l in the plane, and having binding interactions to dislocations characterized by an energy E_b , is determined by minimizing the line energy of the dislocation to obtain the average bow-out amplitude h and the mean effective spacing L_m of binding contacts along the dislocation line, as depicted in Fig. 6.39. While the possible change in line shape upon application of a stress is unclear in this model to determine the plastic resistance, this uncertainty can be avoided by choosing the maximum resistive force $\hat{K}$ on a typical line segment of length L_m to be that calculated by Cahn (1963) in a fully consistent manner. The plastic resistance increment $\Delta\hat{\tau}$ can then be estimated as

$$\Delta\hat{\tau} = \frac{\hat{K}}{bL_m}, \quad (6.167)$$

where

$$\hat{K} = \frac{0.858E_b}{w/2}, \quad (5.68)$$

as determined in Chapter 5 for tearing the dislocation away from a bell-shaped exponential, binding potential well.

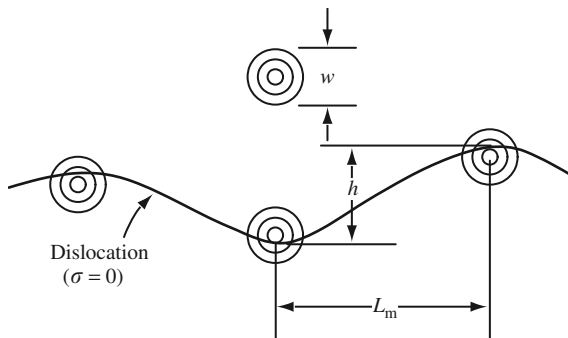


FIG. 6.39. Sketch of interaction of a flexible dislocation with misfitting finite-sized domains in a spinodal microstructure (after Ardell 1985).

Mott's model gives the following for the mean spacing of binding contacts under no stress:

$$L_m = \left(\frac{2\mathcal{E}l}{E_b} \right)^{1/3}. \quad (6.168)$$

Equating l to the main domain wavelength λ of the spinodal slabs, choosing the cross-sectional area $\pi w^2/4$ of a typical binding contact of a discrete spinodal domain to be the average area allocated to such a domain in the $\{111\}$ plane of an FCC crystal, as determined in the Cahn model, that is,

$$\frac{\pi w^2}{4} = \left(\frac{\sqrt{6}\lambda}{2\pi} \right) \left(\frac{\sqrt{2}\lambda}{2\pi} \right), \quad (6.169a)$$

giving

$$w = 0.334\lambda, \quad (6.169b)$$

and taking from Cahn's theory the peak resistance $\hat{K}$ experienced by a typical segment of an edge dislocation, that is,

$$\hat{K} = \sqrt{\frac{2}{3}} A \eta Y \lambda b, \quad (6.170)$$

the plastic resistance increment $\Delta \hat{\tau}$, normalized by the average shear modulus μ , is then obtained as

$$\frac{\Delta \hat{\tau}}{\mu} = 0.204 \left(\frac{\lambda}{b} \right)^{1/3} \left(\frac{A \eta Y}{\mu} \right)^{4/3} \frac{1}{\varphi^{1/3}}. \quad (6.171)$$

In the above expression,

$$A = \frac{(c - c_0)}{3}, \quad (6.172a)$$

$$\eta = \frac{1}{a} \frac{da}{dc} (= \epsilon_s), \quad (6.172b)$$

$$Y = \frac{(c_{11} + 2c_{12})(c_{11} - c_{12})}{c_{11}}, \quad (6.172c)$$

and

$$\varphi = \mathcal{E}/\mu b^2 \quad (6.172d)$$

give the scale factor of the differential compositional amplitude of the spinodal slab, the size misfit parameter, the appropriate anisotropic elastic constant as used in the Cahn model, and the anisotropic line tension factor introduced by Ardell (1985) in the evaluation of the Mott model, respectively.

Comparison with Experimental Results For comparison of the theoretical model results with experiments, we choose four alloys of Cu, Ni, and Fe considered by Ardell (1985). All of these alloys are known to undergo prominent spinodal decomposition. The material parameters A , η , Y , and φ have all been tabulated by Ardell, and will not be presented here. These four alloys had the following compositions by weight:

- Alloy 1 (aged at 625 °C): 51.5 Cu, 33.5 Ni, 15.0 Fe;
- Alloy 2 (aged at 775 °C): same as Alloy 1;
- Alloy 3 (aged at 625 °C): 64.0 Cu, 27.0 Ni, 9.0 Fe;
- Alloy 4 (aged at 625 °C): 32.0 Cu, 46.5 Ni, 22.5 Fe.

The aging of all these alloys was taken to a plateau region in time, beyond which changes in the spinodal morphology we are minimal.

For each alloy composition, four separate evaluations we available. The experimentally determined plastic-resistance ratios $(\Delta\hat{\tau}/\mu)_{\text{exp}}$ and the corresponding $(\Delta\hat{\tau}/\mu)_{\text{th}}$ determined from eq. (6.171) are tabulated in Table 6.2. The pairs of

TABLE 6.2. Comparison of experimental determinations of plastic resistance ratios $(\Delta\hat{\tau}/\mu)_{\text{exp}}$ with theoretical estimates $(\Delta\hat{\tau}/\mu)_{\text{th}}$ for four Cu, Ni, Fe alloys considered by Ardell (1985)

	Alloy 1, aged at 625 °C			
$(\Delta\hat{\tau}/\mu)_{\text{exp}}$	5.05×10^{-4}	8.39×10^{-4}	12.2×10^{-4}	13.4×10^{-4}
$(\Delta\hat{\tau}/\mu)_{\text{th}}$	7.93×10^{-4}	9.65×10^{-4}	11.2×10^{-4}	14.4×10^{-4}
	Alloy 2, aged at 775 °C			
$(\Delta\hat{\tau}/\mu)_{\text{exp}}$	5.39×10^{-4}	7.91×10^{-4}	9.17×10^{-4}	9.45×10^{-4}
$(\Delta\hat{\tau}/\mu)_{\text{th}}$	8.82×10^{-4}	9.59×10^{-4}	10.0×10^{-4}	12.6×10^{-6}
	Alloy 3, aged at 625 °C			
$(\Delta\hat{\tau}/\mu)_{\text{exp}}$	1.58×10^{-4}	3.71×10^{-4}	8.92×10^{-4}	14.3×10^{-4}
$(\Delta\hat{\tau}/\mu)_{\text{th}}$	1.73×10^{-4}	2.40×10^{-4}	3.04×10^{-4}	5.28×10^{-4}
	Alloy 4, aged at 625 °C			
$(\Delta\hat{\tau}/\mu)_{\text{exp}}$	3.51×10^{-4}	8.35×10^{-4}	16.1×10^{-4}	19.0×10^{-4}
$(\Delta\hat{\tau}/\mu)_{\text{th}}$	6.07×10^{-4}	10.8×10^{-4}	13.1×10^{-4}	20.0×10^{-6}

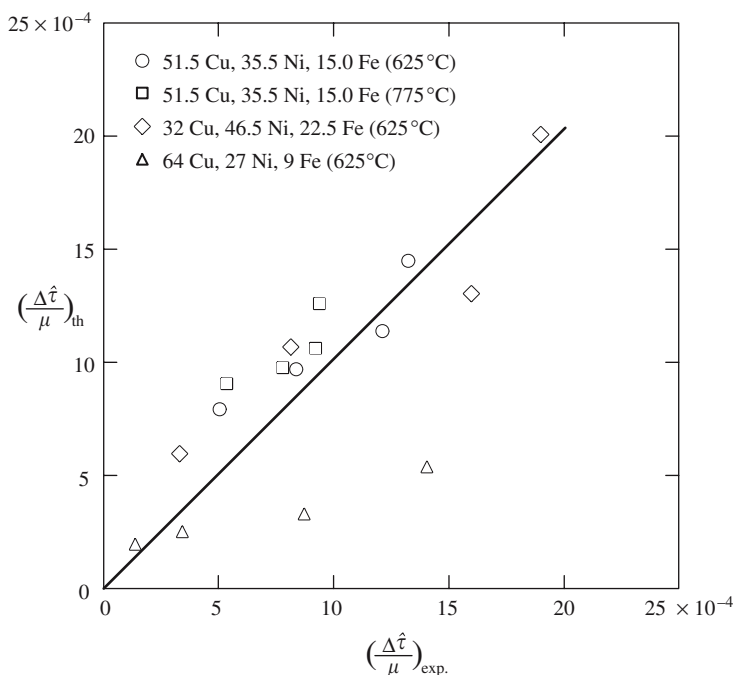


FIG. 6.40. Comparison of the modified Ardell (1985) model of strengthening by spinodal-decomposition microstructure with experimental results (data from Ardell 1985).

values are plotted against each other in Fig. 6.40. The straight line representing the best fit emanating from the origin shows excellent agreement on average.

6.4.9 Precipitate-like Obstacles

Background and Model There are a substantial number of discrete obstacles to dislocation motion that have characteristics similar to precipitates. These range from atomic size to defect clusters of various types, all of which are distinguished from solute atoms by their ability to impede dislocation motion singly as precipitates do, rather than collectively as substitutional solute atoms do. They include vacancy and interstitial clusters, stacking-fault tetrahedra, and sessile dislocation loops, arising from condensation of supersaturations of point defects during quenching and, most prominently, from irradiation damage by high-energy particle fluxes or decay of unstable isotopes. The subject of quench-induced and irradiation-induced lattice defects of precipitate-like character is a broad one, of great importance in the area of embrittlement, particularly of ferrous alloys, but also of FCC metals. Other precipitate-like obstacles are encountered in alkali halides and result from tetragonal distortions produced by individual divalent impurities.

Some of these cases were discussed earlier by Fleischer (1962*a*). Here we select a comprehensive study by Koppenaal and coworkers on the neutron-irradiation-induced hardening of Cu for detailed discussion (Koppenaal and Arsenault 1965; Koppenaal 1968).

The temperature dependence of the plastic shear resistance σ in high-purity (10 ppm total impurity content) Cu single crystals, after a total dose of 8×10^{18} fast neutrons cm^{-2} at an irradiation temperature of around 60–100 °C, is shown in Fig. 6.41. Post-irradiation thermal excursions for various periods at a temperature of 300 °C result in complex reductions of the plastic resistance in the low-temperature range which we shall not consider here. TEM experiments by Makin *et al.* (1962) and Essmann and Wilkens (1964) have established that the primary form of irradiation damage under these conditions is sessile dislocation loops resulting from excess vacancy condensation on {111} slip planes, with a range of diameters up to $50b$; however, most of them are in the smaller size range of $(2-4)b$. Such loops result in substantial local tetragonal distortions out of the {111} planes that produce strong interactions with dislocations and act as precipitate-like potent glide obstacles.

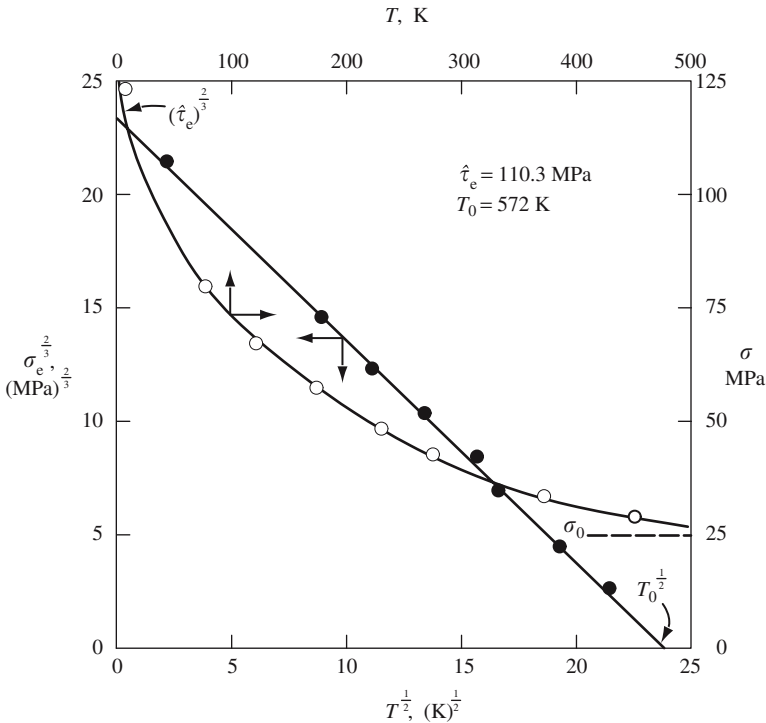


FIG. 6.41. Temperature dependence of effective shear resistance in Cu, irradiated with 8×10^{18} fast neutrons/ cm^2 fluence (data from Koppenaal 1968).

The interaction of such tetragonal-distortion sites with screw dislocations has been considered by Fleischer (1962*a,b*).

For a screw dislocation parallel to a (110) direction moving in a (112) direction on a {111} glide plane, interacting with a tetragonal-distortion site having a principal axis parallel to a (111) direction and situated at the origin, Fleischer (1962*a*) determined an interaction energy I of the form

$$I = \frac{\mu b^4 \Delta \varepsilon}{2\pi \sqrt{6}} \frac{(z + \sqrt{2}y)}{(y^2 + z^2)}. \quad (6.173)$$

In eq. (6.173), $\Delta \varepsilon = \varepsilon_1 - \varepsilon_2$ is the strength of the tetragonal distortion at the defect site in the form the difference between the two principal strain components at the site. Here the probing dislocation line is parallel to the x axis, and is advancing in the y direction on a glide plane with a normal in the z direction. The local resistive force K exerted by the obstacle on the dislocation against its motion in the y direction on a plane a distance z_0 above the obstacle loop, taken as half the loop diameter later, is then

$$K = \left(\frac{\partial I}{\partial y} \right)_z = K_0 g(\eta), \quad (6.174)$$

where

$$K_0 = \frac{\mu b^2 \Delta \varepsilon}{2\pi \sqrt{3}} \left(\frac{b}{z_0} \right)^2, \quad (6.175a)$$

$$g(\eta) = \frac{1 - \sqrt{2}\eta - \eta^2}{(1 + \eta^2)^2} \quad (6.175b)$$

is the normalized, dimensionless force–distance function, and $\eta = y/z_0$ is the dimensionless interaction distance. The function $g(\eta) = K/K_0$ has a complex shape over its entire range of η but is of primary interest only over the range $0.5 \leq \eta \leq 2.0$, where it is large and positive and peaks at a value of 1.15 per atomic volume of the obstacle loop. For a vacancy disk of diameter d in the {111} glide plane, containing a number $n_v = (\pi/2\sqrt{3})(d/b)^2$ atomic volumes, and with its center located a distance $z_0 = (d/2)$ below the glide plane, the peak resistive force becomes

$$\hat{K} = 0.383 \mu b^2 \Delta \varepsilon, \quad (6.176)$$

where in this case $\Delta \varepsilon$ is of order unity. The shape of the force–activation-distance profile $K = K(\Delta \eta)$, required for determination of the activation energy to overcome

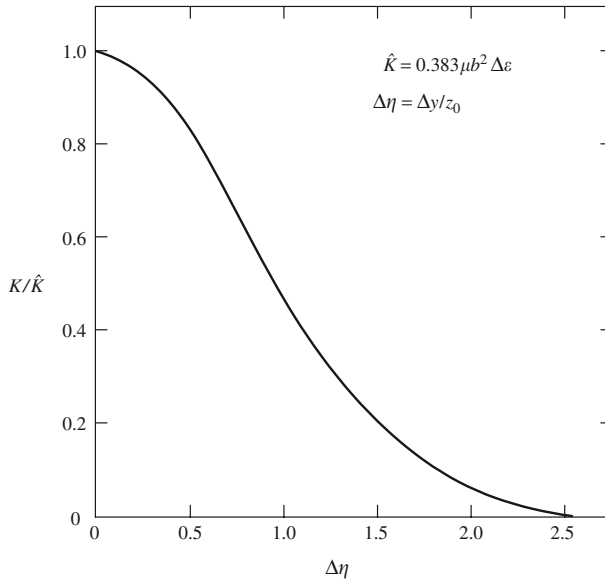


FIG. 6.42. Normalized force–distance curve for a screw dislocation interacting with an irradiation damage site with tetragonal misfit (based on Fleischer 1962a).

the obstacle at a force level K , is shown in Fig. 6.42, where the resistive force is normalized by its peak value $\hat{K}$.

The activation energy ΔG^* for overcoming the obstacle at an applied force $f < \hat{K}$ is the

$$\Delta G^* = \int_{f=K}^{f=\hat{K}} \Delta \eta(f) df = \Delta G_0^* \int_{K/\hat{K}}^{1.0} \Delta \eta \left(\frac{K}{\hat{K}} \right) d \left(\frac{K}{\hat{K}} \right), \quad (6.177)$$

where

$$\Delta G_0^* = z_0 \hat{K}. \quad (6.178)$$

Integration of the force–activation–distance curve of Fig. 6.42 between K and $\hat{K}$ gives the activation energy as a function of $K/\hat{K}$, shown in Fig. 6.43. A very accurate phenomenological fit to the dependence is

$$\Delta G^* = \Delta G_0^* \left(1 - \frac{K}{\hat{K}} \right)^2. \quad (6.179)$$

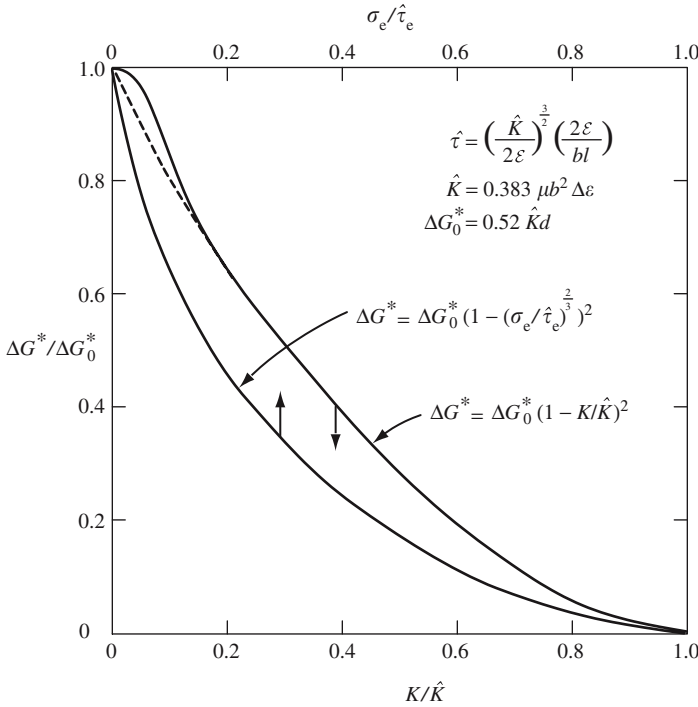


FIG. 6.43. Calculated dependence of the activation free energy on the resistive force due to a tetragonal distortion, based on Fig. 6.42.

While a more accurate form of the expression is shown in Fig. 6.43 by the solid curve, we shall use the truncated dashed curve as an adequate approximation for low values of $K/\hat{K}$.

As Fig. 6.41 shows, the plastic resistance is quite temperature-dependent, making it instructive to determine the shear activation volume Δv_{σ}^* of the rate-controlling process. Assuming crudely that $\mu b^2 \approx 2\mathcal{E}$, we note first from eq. (6.176) that

$$\frac{\hat{K}}{2\mathcal{E}} = 0.383 \Delta\epsilon,$$

where $\Delta\epsilon = O(1.0)$ indicates that the envisioned obstacles in the form of sessile loops are quite strong but still penetrable, and for a mean obstacle spacing l in the glide plane result in a threshold plastic resistance of

$$\hat{\tau} = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bl} \right). \quad (6.180)$$

Moreover, for a resistive force level $K < \hat{K}$ in a thermally assisted plastic shear resistance $\sigma < \hat{\tau}$, we can take

$$\sigma = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bl} \right), \quad (6.181)$$

giving

$$\frac{K}{\hat{K}} = \left(\frac{\sigma}{\hat{\tau}} \right)^{2/3}, \quad (6.182)$$

which, upon substitution into eq. (6.181), gives

$$\Delta G^* = \Delta G_0^* \left(1 - \left(\frac{\sigma}{\hat{\tau}} \right)^{2/3} \right)^2. \quad (6.183)$$

It is important to note that this form of the stress dependence of the activation energy differs fundamentally from that proposed by Fleischer (1962*b*) and used by Koppenaal and Arsenault (1965) and Koppenaal (1968) in analyzing their experimental results. There, for the spacing of obstacles along the dislocation line, the mean spacing l of obstacles in the plane was used rather than the required Friedel sampling length λ_c (eq. 6.16a) for obstacles with $\hat{K}/2\mathcal{E} < 1.0$, as is the case here.

Finally, to determine the shear activation volume, it is necessary to recognize that the stress σ and its threshold value $\hat{\tau}$ must be interpreted as the effective stress σ_e acting on the obstacles, requiring the subtraction of a possible component σ_0 related to other long-range internal stresses or other obstacle resistances; that is,

$$\sigma_e = \sigma - \sigma_0 \quad \text{and} \quad \hat{\tau}_e = \hat{\tau} - \sigma_0 \quad (6.184a,b)$$

should replace σ and $\hat{\tau}$ in eq. (6.183). With these modifications, finally,

$$\Delta G^* = \Delta G_0^* \left(1 - \left(\frac{\sigma_e}{\hat{\tau}_e} \right)^{2/3} \right)^2 \quad (6.185)$$

is obtained, the form of which is also shown in Fig. 6.43. The shear activation volume is then

$$\Delta v_\sigma^* = - \left(\frac{\partial \Delta G^*}{\partial \sigma_e} \right)_{p,T} = \frac{4}{3} \frac{\Delta G_0^*}{\hat{\tau}_e} \left(\left(\frac{\hat{\tau}_e}{\sigma_e} \right)^{1/3} - \left(\frac{\sigma_e}{\hat{\tau}_e} \right)^{1/3} \right). \quad (6.186)$$

The Plastic Resistance From the usual kinetic expression for the shear strain rate

$$\dot{\gamma} = \dot{\gamma}_0 \exp \left[-\frac{\Delta G_0^*}{kT} \left(1 - \left(\frac{\sigma_e}{\hat{\tau}_e} \right)^{2/3} \right)^2 \right], \quad (6.187)$$

the temperature dependence of the effective shear resistance σ_e may be obtained by inversion, and is

$$\frac{\sigma_e}{\hat{\tau}_e} = \left[1 - \left(\frac{T}{T_0} \right)^{1/2} \right]^{3/2}, \quad (6.188)$$

where

$$T_0 = \frac{\Delta G_0^*}{k \ln(\dot{\gamma}_0/\dot{\gamma})} \quad (6.189)$$

represents, as usual, the temperature above which the obstacles under consideration offer no further resistance. Moreover, σ_e must be considered as the increment in resistance derived from this specific obstacle.

Comparison with Experimental Results In view of the difference of our form given in eqs. (6.183) and (6.186) for the activation energy from that used by Koppelaar, we start by establishing its validity by plotting $\sigma_e^{2/3}$ against $T^{1/2}$ in Fig. 6.41. For this, a rate-independent component $\sigma_0 = 25$ MPa has been subtracted from σ , as was suggested by Koppelaar, to obtain σ_e . As Fig. 6.41 shows, this gives a straight-line relationship consistent with the requirement of the form of eq. (6.188). Moreover, from the intercepts of the straight line, we determine that $\hat{\tau}_e = 110.3$ MPa and $T_0 = 572$ K, respectively.

Koppelaar (1968) determined experimentally the activation enthalpy ΔH^* and the shear activation volume $\Delta \nu_\sigma^*$ by a slight variant of the procedure discussed in Chapter 2 and given by eqs. (2.48b) and (2.48a), respectively. Since the temperature dependence of the elastic constants is slight for Cu in the temperature range in which the experimental determinations were carried out, we take $\Delta H^* \approx \Delta G^*$ and compare the measurements directly with our form of the expressions for ΔG^* and $\Delta \nu_\sigma^*$. These comparisons are shown in Figs. 6.44 and 6.45 for all neutron-irradiated and subsequently thermally relaxed samples, indicating that in all these cases of irradiation and post-irradiation thermal excursions the rate-controlling obstacles have remained the same, albeit reduced in density. The agreement of the model with the experimental results looks generally good.

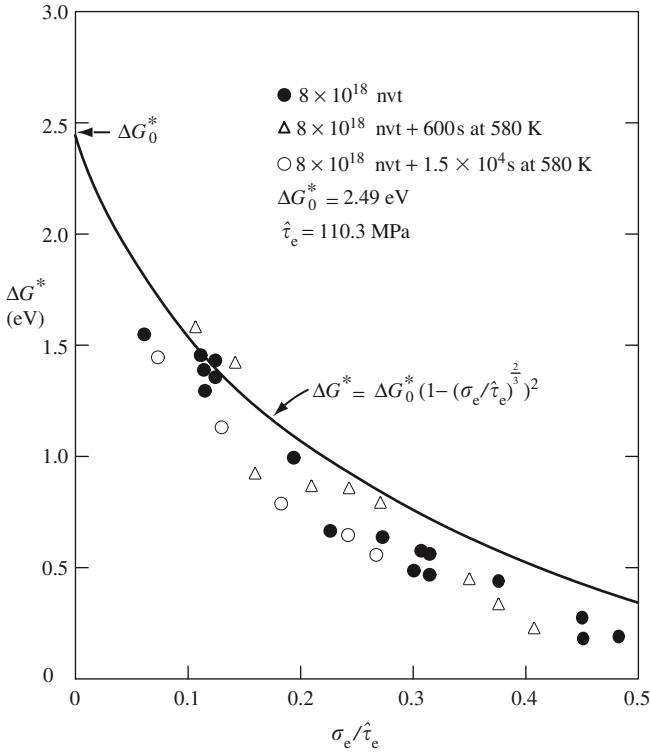


FIG. 6.44. Comparison of model of the dependence on the effective stress of the activation free energy for overcoming a radiation damage site with a tetragonal distortion, with experimental results of Koppenaal (1968).

To interpret the results further, we determine from eq. (6.178) that

$$d = 2z_0 = \frac{2 \Delta G_0^*}{\hat{K}} = \frac{0.58}{\Delta \varepsilon} \text{nm}, \quad (6.190)$$

that is, for $\Delta \varepsilon = 0.5$ – 1.0 , we obtain a diameter of $(2\text{--}4)b$ as conjectured earlier above. Furthermore, we can determine the mean obstacle spacing l in the plane from eq. (6.181), using as a representation for the line tension its anisotropic form for a screw dislocation (Ardell 1985),

$$\mathcal{E} = \frac{\mu b^2}{4\pi} \frac{(1 + \nu(1 - (\hat{K}/2\mathcal{E})^2))}{(1 - \nu)} \ln \left(\frac{l}{b} \right). \quad (6.191)$$

For $\hat{K} = 0.383 \mu b^2 \Delta \varepsilon$, $\Delta \varepsilon \approx 1.0$, a value of

$$\frac{l}{b} = 8.54 \times 10^2 \quad (6.192)$$

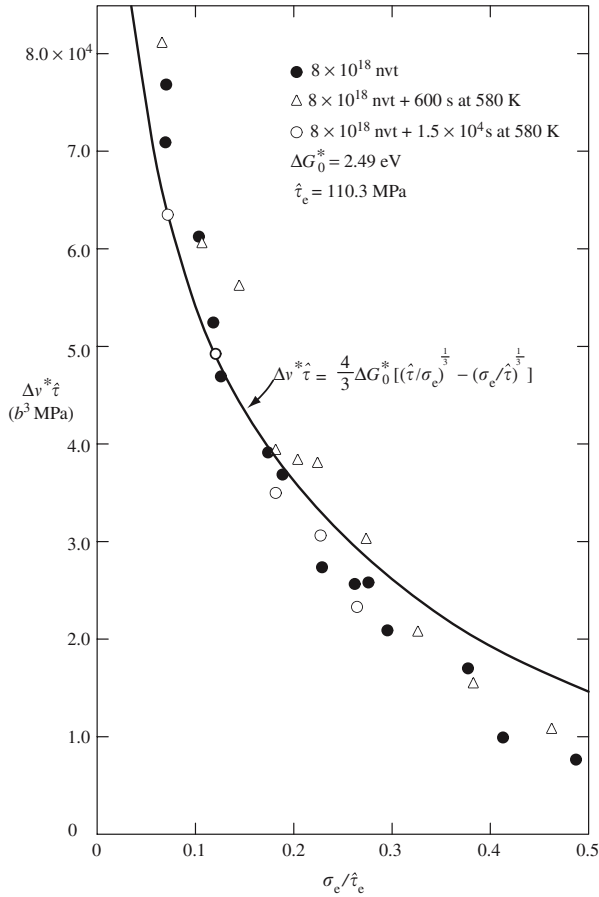


FIG. 6.45. Comparison of model of the dependence on the effective stress of the shear activation volume for overcoming a radiation damage site with tetragonal distortion, with the experimental results of Koppenaal (1968).

obtained using eq. (6.9), and taking $r_0 \approx (d/2)$, we have an effective atomic concentration of $c = 1.15 \times 10^{-5}$ or a mere 10 ppm. Thus, while the assumptions of the model used by Fleischer (1962a,b) are somewhat flexible, this indicates that irradiation-hardening defects are small but quite potent indeed.

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STRAIN HARDENING

7.1 Overview

In this chapter we present the intrinsic mechanisms of “dislocation resistance” resulting from the forms of interaction and intersection of dislocations occurring during quasi-homogeneous plastic flow, and the evolution of this resistance with straining, which is referred to variously as strain hardening or work hardening. This process, by itself and in combination with the solid-solution and precipitation strengthening discussed in earlier chapters, offers a wide range of possibilities for very significantly increasing the plastic resistance of metals and alloys.

Strain hardening has been studied extensively, ever since its mechanism was first discussed in a very limited form by Taylor (1934) in his pioneering publication introducing the concept of a dislocation as a “plasticity carrier”. It is a very complex process and many aspects of it are still intensely debated. In this chapter, we consider only the principal features of the process and present the central mechanisms in single-phase metals and alloys in the low-temperature range where diffusional processes play no role and are unencumbered by other strengthening mechanisms.

In many crystal structures, plastic resistance can be strongly influenced by the lattice resistance. In close-packed pure FCC and HCP metals this resistance is very small, while in other metals where it is high, it becomes negligible above a certain characteristic temperature T_0 , as was discussed in some detail in Chapter 4. When the lattice resistance is negligible as in pure FCC and HCP metals and in other pure crystalline materials above their characteristic temperature T_0 , the processes of strain hardening exhibit common features and are governed by the same mechanisms. Thus, while we consider in this chapter primarily strain hardening in pure FCC metals, which have been the most extensively studied materials, these same processes also govern the behavior of BCC metals, materials with the diamond cubic structure, such as silicon, germanium, alkali halides, and metal oxides, NaCl, LiF, MgO, and the like. However, it must be recognized that the availability of slip systems is governed by crystal symmetry and binding and that additional deformation mechanisms such as twinning, with very different manifestations, can also be present. These can introduce other important degrees of freedom or present restrictions, which will make important changes in the central mechanisms exhibited by the FCC metals that we primarily consider. These differences must be kept in mind.

We recall from Chapter 3 that there are two types of dislocation resistance: the interplane resistance, governed by interactions of mobile glide dislocations in parallel planes, and the intraplane resistance, experienced by glide dislocations in their plane (Hirsch and Humphreys 1970). Strain hardening manifests itself differently in these two resistances. We note here at the outset that while the interplane resistance is largely temperature- and strain-rate-independent, the intraplane resistance has a definite temperature- and rate-dependent component τ_s that results from dislocation intersections. As was demonstrated by Cottrell and Stokes (1955) and Thornton *et al.* (1962), this component is usually a certain fraction of the total intraplane resistance that depends only on the mode of deformation. This Cottrell–Stokes law implies that during much of strain hardening, until the very late stages where dynamic recovery occurs, the rate-controlling forest dislocation obstacles increase as a constant fraction of the total dislocation density with strain.

The early stages of strain hardening can be followed in some detail in single crystals of well-chosen initial orientations to shear strains of perhaps up to 0.25; here, meaningful attribution of strains to individual slip systems and resolution of the stress in such systems is possible. At larger strains, where more than one slip system becomes active, the differences in behavior between single crystals and polycrystals become less as the accumulated dislocation structures reduce in scale and become of very much smaller dimensions than individual grains, and more homogenized views of strain hardening become possible (Kocks and Mecking 2003). Thus, in this chapter we shall consider the early stages of deformation in single crystals and the later stages in polycrystals.

Finally, we note that the considerations of this chapter apply to homogeneous deformation without imposition of lattice curvatures or strong strain gradients, even though some of these develop unavoidably by random statistical processes. Strain hardening in the presence of heterogeneities will be discussed in Chapter 8.

The perspective of strain hardening in this chapter takes account of most of the processes that we consider central. It is necessary to caution the reader, however, that there are many different views about what is central and what is not. Therefore, the reader is encouraged to consult the several detailed reviews that are mentioned at the end of the chapter.

7.2 Features of Deformation

7.2.1 Active Slip Systems in FCC Metals

Since understanding the geometry and crystallography of active slip systems and their interactions is of central importance in the plasticity and strain hardening of FCC single crystals such as Cu, which we consider in detail, it is essential to start with some definitions.

Figure 7.1 introduces the stereographic projection of a cubic crystal and shows the Thompson tetrahedron introduced in Chapter 2, which represents the entire set of four octahedral slip planes and their 12 shared slip directions. For the simplest and least complex conditions of deformation, to achieve slip on a simple slip system, at least at the beginning, the tensile axis is chosen in the central region of the standard stereographic triangle formed by $[100]$, $[110]$, and $[111]$, near the point O shown in Fig. 7.1(a), on a great circle connecting the normal to the $(11\bar{1})$ plane and the $[101]$ direction. This maximizes the resolved shear stress on the $[101](11\bar{1})$ system, which is labeled as the *primary system*. As slip deformation develops on this system, the crystal axis rotates toward the primary slip direction $[101]$. When the axis reaches the $[100]$ – $[111]$ symmetry line, a second slip system, $[110](1\bar{1}\bar{1})$, referred to as the *conjugate system*, becomes equally stressed and, barring certain complications, begins to produce increments of slip equal to those for the primary system. This tends to rotate the crystal axis toward the $[211]$ direction, as shown in Fig. 7.1(a), and tends to increase the resolved shear stress on the *cross-slip system* $[101](1\bar{1}\bar{1})$ (relative to the *primary system*) or $[110](1\bar{1}\bar{1})$ (relative to the *conjugate system*), which previously had not been stressed along the path shown in Fig. 7.1(a). A fourth system, $[10\bar{1}](111)$, referred to as the *critical system*, which remains the third highly stressed system, contributes little to the overall deformation in response

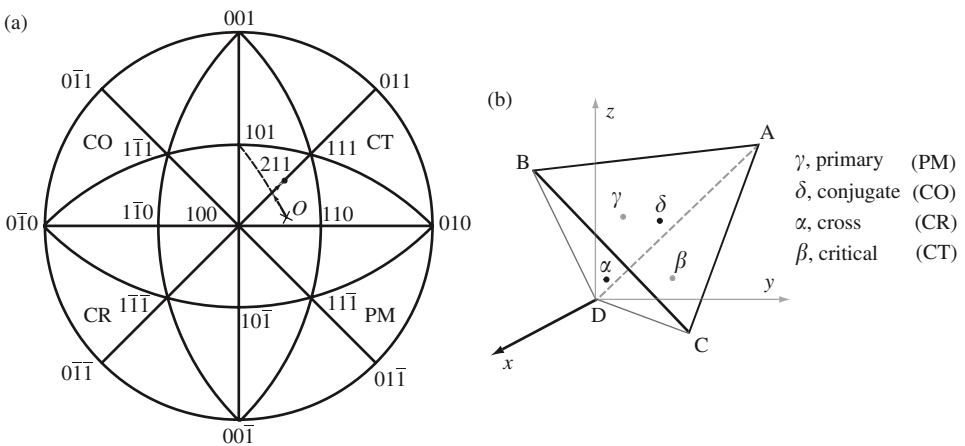


FIG. 7.1. (a) Stereographic projection of the cubic system, (b) the Thompson tetrahedron representing the four $\{111\}$ slip planes and the six slip directions. The four principal slip systems for a crystal with its axis located in the center region of the standard triangle are shown. The primary system (PM) is $(11\bar{1})[101]$, shown as γ on the tetrahedron. The additional slip systems, conjugate (CO), critical (CT), and cross-slip (CR), are also shown in both the projection and the tetrahedron.

to the applied tensile stress but plays an important subsidiary role as will become clear later.

7.2.2 Stress–Strain Curves

Figure 7.2 shows a family of resolved shear-stress–shear-strain curves at room temperature for Cu single crystals in a variety of orientations, identified in the standard triangle on the left (Diehl 1956). A number of important points are apparent. Orientations C14, 18, 21, 25, 26, and 28 all maximize the resolved shear stress initially on only one slip system and therefore show very little initial hardening, in a stage of deformation referred to variously as *Stage I*, *easy glide*, or *laminar slip*. Here the reduced strain-hardening rate (reduced with respect to the shear modulus) $(d\tau/d\gamma)/\mu = \Theta_I/\mu$ is typically in the range $2\text{--}5 \times 10^{-4}$. In comparison, the crystals in orientations C19–22, 23, and 29 are very close to high-symmetry orientations where the resolved shear stress is large on several intersecting slip planes. As a result, crystals in these orientations show only a minimum of Stage I behavior, if any, but exhibit a high strain-hardening behavior almost from the outset, which the crystals of the first set only achieve after the Stage I behavior is exhausted. In this high-hardening regime, labeled *Stage II*, the reduced strain-hardening rates Θ_{II}/μ are typically of the order of 10^{-2} . Polycrystals of high-melting-point pure FCC metals at room temperature exhibit only Stage II behavior from the very outset. This two-mode behavior illustrates a central aspect of all strain hardening by emphasizing that the rate of storage of dislocations, which as the fundamental

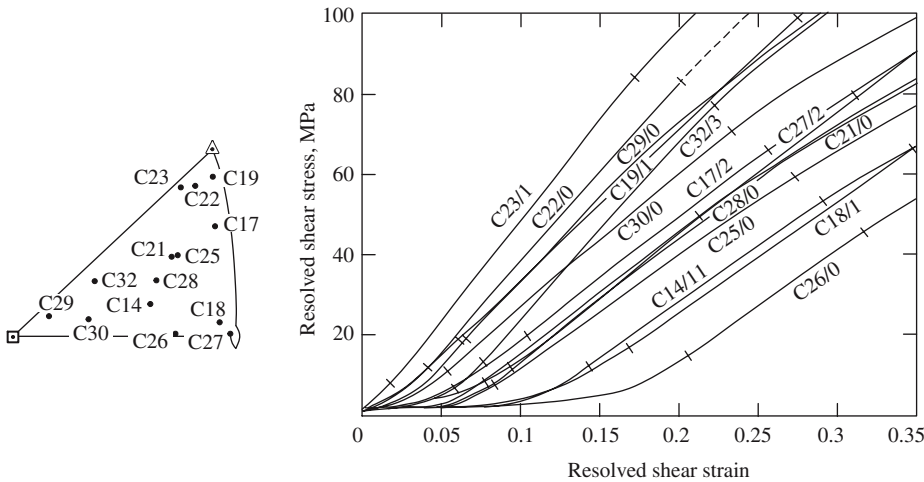


FIG. 7.2. Stress–strain curves of Cu single crystals with different axis orientations, shown in the standard triangle (after Diehl 1956).

basis of strain hardening, is governed by the presence of strong slip obstacles around which such storage can initiate and grow. In heterogeneous alloys with nonshearable particles, these act as very effective sites for clustering of dislocations (Ashby 1966). In pure metals, the obstacles have to be produced by intersecting slip itself. Crystals with an abundance of equivalent slip systems such as those with the FCC, BCC, and NaCl structures fall into this category, while HCP crystals, which deform primarily on the basal plane or on prism planes, do not undergo much intersecting slip until very late in their deformation. Therefore, their hardening behavior is dramatically different from that of the former group. This important fact is demonstrated in Fig. 7.3 using the pioneering work of Schmid and Boas (1935).

The stress-strain curves of Fig. 7.2 show that all hardening curves in Stage II are substantially linear. This behavior continues to a resolved shear strain of roughly 0.25 at room temperature, where the hardening rate begins to decrease as the behavior becomes nonlinear. This new range is labeled *Stage III*. The transition to Stage III is temperature-dependent and occurs at larger strains at lower temperatures. Figure 7.4 shows a resolved-shear-stress-strain curve and the associated strain-hardening rate of a centrally oriented Cu crystal at 77 K (Basinski and

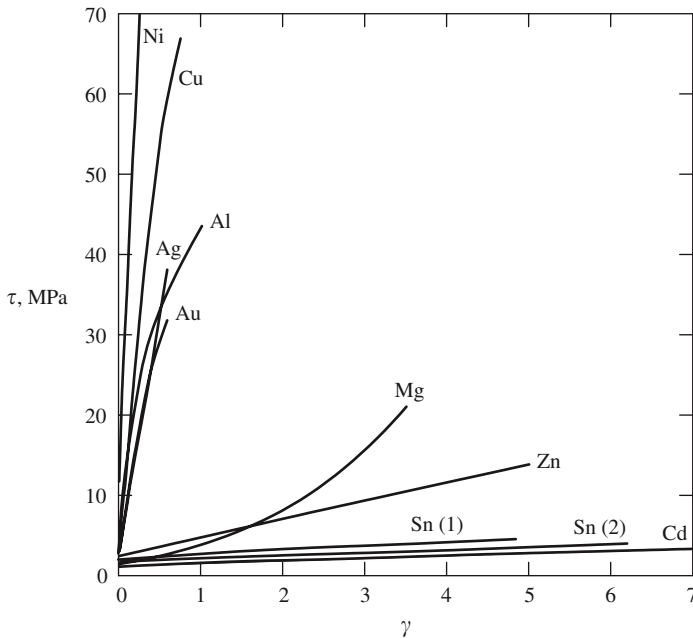


FIG. 7.3. Stress-strain curves of metal single crystals. Those which have intersecting slip systems show substantially higher strain-hardening rates than those which do not (after Schmid and Boas 1935).

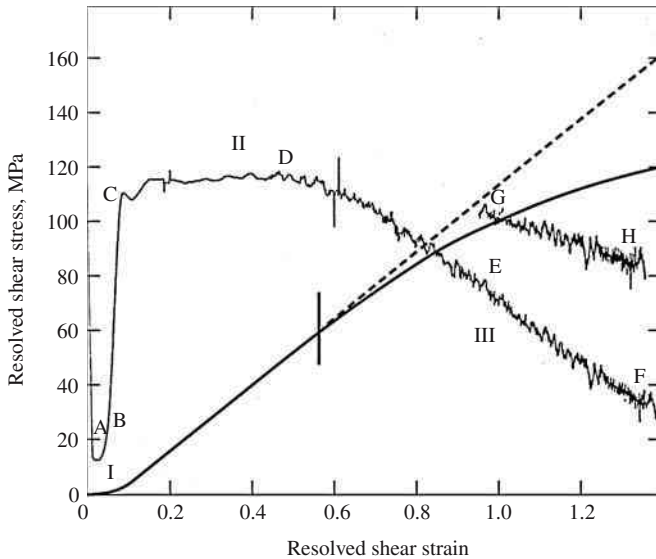


FIG. 7.4. Resolved-shear-stress–shear-strain curve for a Cu single crystal with a central orientation strained at 77 K; the superposed curve shows the strain-hardening rates in Stages I, II, and III. The curve GH was calculated on the assumption that after the axis reaches the $[100]$ – $[111]$ line, deformation will be by symmetrical double slip on the PM and CO systems (after Basinski and Basinski 1979, courtesy of Elsevier).

Basinski 1979). The stretch AB is Stage I. Upon the transition to Stage II, the strain-hardening rate increases by a factor of about 10. The stretch CD represents Stage II. Beyond point D, Stage III sets in. At point E, the crystal axis reaches the $[100]$, $[111]$ symmetry line and the conjugate system also becomes active. The stretch EF was determined on the basis of single slip on the primary system, while GH represents hardening behavior on the assumption of equal activity on the primary and conjugate systems. Clearly, at 77 K the Stage II to III transition has been delayed to a resolved shear strain of about 0.6, from about 0.25 at room temperature.

Upon the transition to Stage III few, if any differences remain between the behaviors of single crystals and polycrystals. Therefore, for the later stages of strain hardening, the stress–strain curves of Al and various lightly solid-solution and precipitation-strengthened alloys of it shown in Fig. 7.5, derived from torsion experiments, can be considered as typical (Rollett 1988). The stage II hardening behavior, which represents only a small range in Al to begin with, is severely contracted close to the stress axis at room temperature on this scale of presentation. The figure shows clearly the parabolic Stage III hardening behavior, terminating roughly at a shear strain of 1.0, beyond which there is a very long stage of low

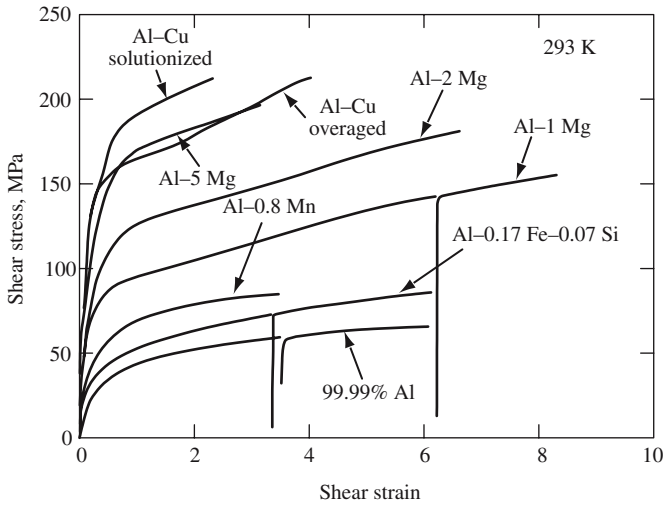


FIG. 7.5. Shear-stress-shear-strain curves of polycrystalline Al and dilute alloys of it at 293 K strained at a rate $\dot{\gamma} = 6.9 \times 10^{-3} \text{ s}^{-1}$ in torsion, showing the large-strain behavior leading into Stage IV (from Rollett 1988).

strain hardening, labeled *Stage IV*, with strains reaching magnitudes of the order of 5–8, during which the reduced flow stress $\hat{\tau}/\mu$ approaches 0.01. The reduced strain-hardening rate in this stage, Θ_{IV}/μ , is of the order of that of Stage I, that is, typically in the range of 5×10^{-4} . At very large strains, Stage IV is terminated by a true saturation of hardening at *Stage V*. (For an expanded discussion of these stages of deformation, see Argon and Haasen 1993.) These stages of strain hardening have counterparts in BCC crystals and in the diamond cubic structure. On the other hand, because of the relative rarity of intersecting slip, Stage I is dominant in HCP crystals of low lattice resistance such as Zn, Cd, and Mg, upto very large strains, where it is called *Stage A*.

Of the five stages of strain hardening, only Stage III and the terminal Stage V are temperature- and strain-rate-dependent. Once reduced with respect to the shear modulus, the hardening rates in Stages I, II, and IV are quite insensitive to temperature and strain rate. This effect is not very apparent in the stress-strain curves, but stands out clearly when the reduced strain-hardening rate Θ/μ is plotted as a function of the reduced slip resistance τ/μ , as shown in Fig. 7.6 for Cu for five different levels of the homologous temperature—all comfortably in the range unaffected by diffusion (Alberdi 1984). The figure shows the strong temperature dependence of hardening in Stage III caused by *dynamic recovery* and the relative temperature independence of the hardening rate in Stage IV. Meanwhile, the basic temperature dependence of the plastic (slip) resistance remains relatively unaltered

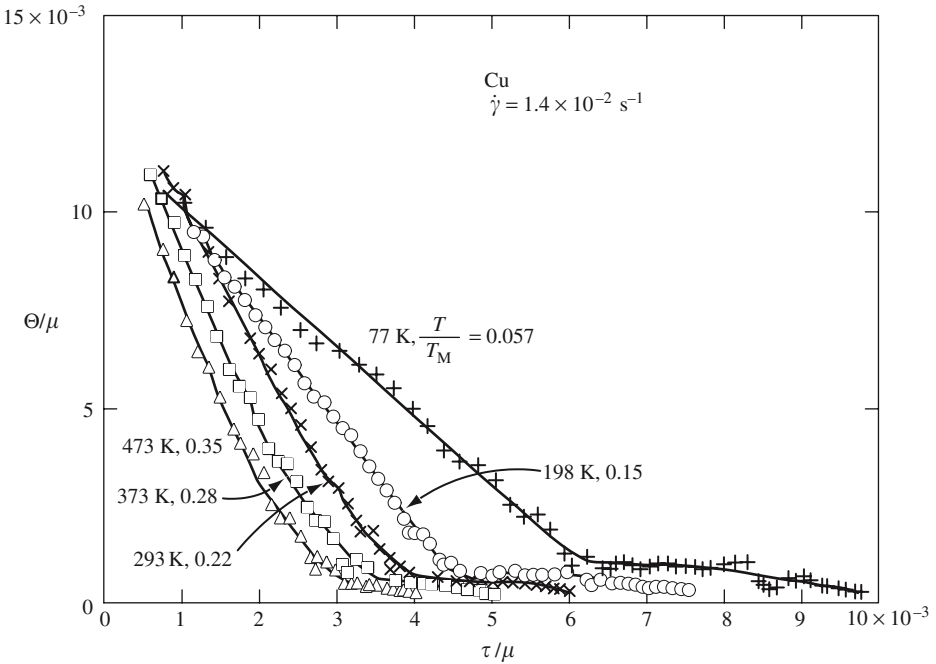


FIG. 7.6. Reduction in strain-hardening rate Θ/μ with increasing plastic shear resistance τ/μ of polycrystalline Cu strained in torsion at a strain rate of $1.4 \times 10^{-2} \text{ s}^{-1}$, at temperatures ranging from 77 K to 473 K (from Alberdi 1984).

over the entire range of deformation, as is demonstrated by the strain rate sensitivity of the reduced flow stress indicated in Fig. 7.7. At low temperatures, there is no important discontinuity in the strain rate sensitivity at the transition between Stage III and Stage IV hardening (marked by an arrow in Fig. 7.7), while at higher temperatures there is a small increase in the dependence at the point of the transition (Alberdi 1984). Similar results have been obtained for Al (Rollett 1988) and Ni and its solid-solution alloys (Hughes and Nix 1989).

Figure 7.8 shows a series of tensile stress–strain curves of pure BCC Mo crystals with a central orientation maximizing the resolved shear stress on a $\langle 111 \rangle \{110\}$ slip system. The curves have not been converted into resolved-shear-stress–shear strain curves, since the investigators could not satisfactorily identify the particular active system (Sestak and Seeger 1978). Nevertheless, it is clear that above a characteristic temperature of circa 413 K, where the lattice resistance has apparently become very small, the stress–strain curves show features very similar to those of FCC Cu shown in Fig. 7.2, that is, Stages I, II, and III are readily discernible. Figure 7.9 shows corresponding resolved-shear-stress–shear-strain curves for slip on the $\langle 110 \rangle \{110\}$ system of high-purity NaCl crystals compressed along the $[001]$ direction at a

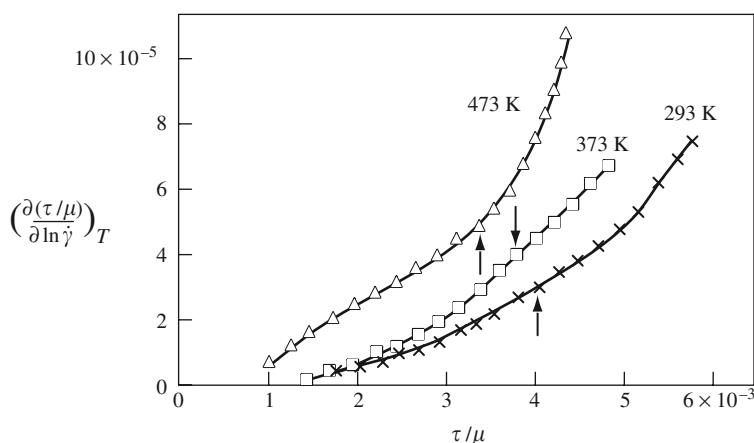


FIG. 7.7. Strain rate sensitivity of the flow stress of Cu polycrystals strained in torsion, versus the plastic shear resistance, for imposed strain rate jumps from $4 \times 10^{-2} \text{ s}^{-1}$ to $4 \times 10^{-3} \text{ s}^{-1}$ at three different temperatures. The transition to Stage IV from Stage III is shown by the vertical arrows (from Alberdi 1984).

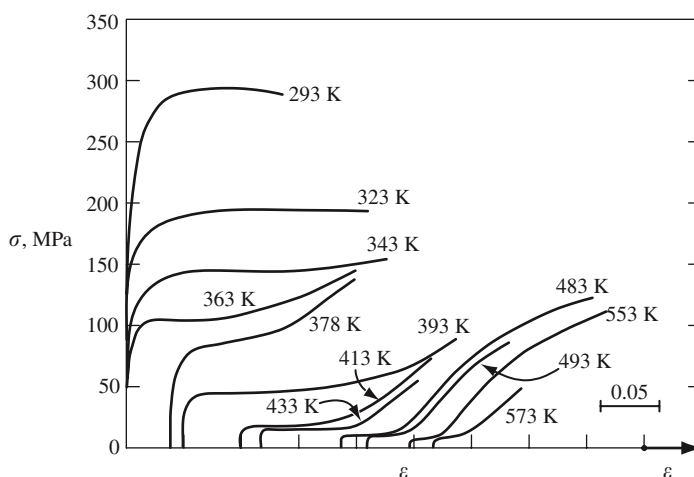


FIG. 7.8. Tensile stress–strain curves of Mo single crystals with a central orientation maximizing the stress on a combination of $\{110\}$ and $\{112\}$ slip planes, strained at a tensile rate of $\dot{\epsilon} = 7 \times 10^{-6} \text{ s}^{-1}$ at temperatures ranging from 293 K to 573 K. At a temperature above 433 K, where the lattice resistance vanishes, the stress–strain curves show three-stage hardening behavior similar to that of FCC crystals (after Sestak and Seeger 1978).

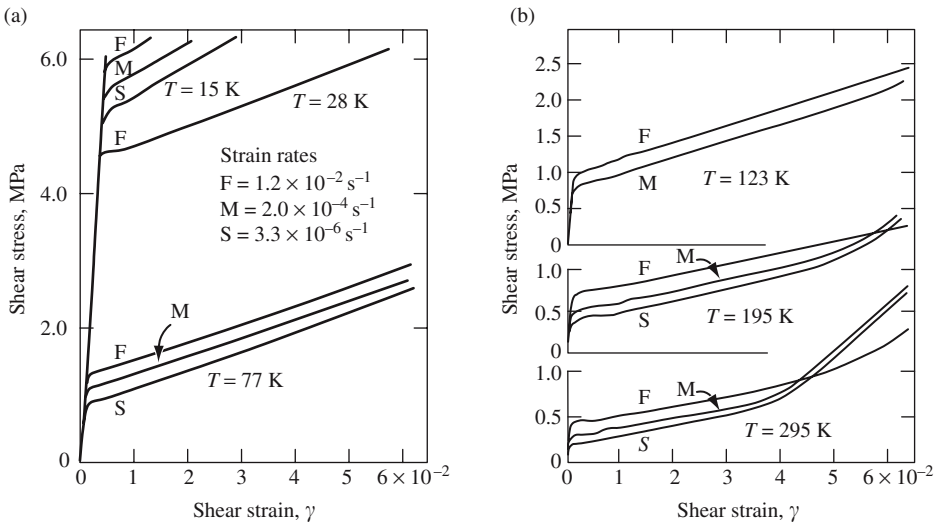


FIG. 7.9. Resolved-shear-stress–shear-strain curves of NaCl single crystals strained at three different rates at temperatures ranging from 15 K to 295 K. At a temperature above 200 K, the curves show hardening ranges I and II similar to those of FCC crystals (Argon *et al.* 1972).

variety of temperatures and three different strain rates (Argon *et al.* 1972). These too show a Stage I and II behavior very similar to that of FCC Cu in spite of the fact that [001] is a multiple-slip orientation, where, however, only one slip system shows activity in a given region of the crystal.

7.2.3 Slip Distributions

Figure 7.10(a) shows the slip line distribution on the surface of a Cu crystal at the end of Stage I at a total shear strain of 0.08 (Seeger 1957). These slip lines are straight, remarkably uniformly distributed, on the average several mm long, and very shallow (circa 2–3 nm deep). In Stage II, the appearance of the slip lines is quite similar to that in Stage I, but the new slip lines added in the course of further straining become progressively shorter, and at the end of Stage II are only about $10\text{ }\mu\text{m}$ long. In Stage III, where the hardening rate continually decreases with strain and becomes temperature-dependent, the incremental additions of slip are in the form of broken-up, coarse, deep slip bands as shown in Fig. 7.10(b) (Seeger 1957). There is ample evidence of widespread activity on the cross-slip system. In Stage IV, the incremental additions of slip bands resemble those of Stage II as shown in Fig. 7.11 for Al (Rollett 1988). The slip line information is important

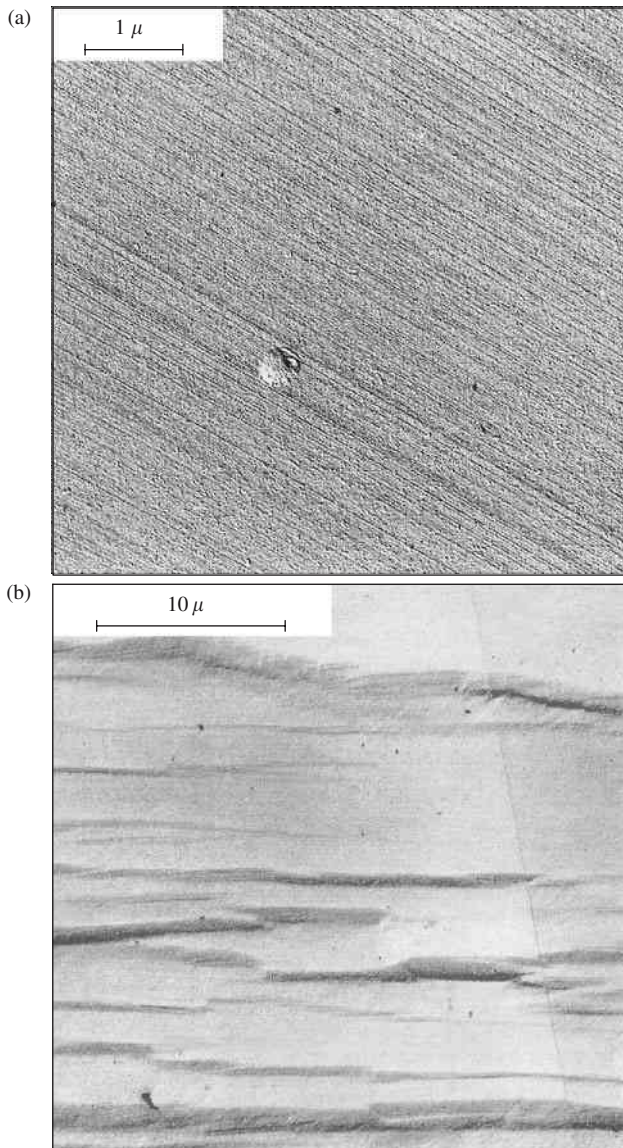


FIG. 7.10. Electron micrographs of (a) slip lines at the end of Stage I ($\gamma = 0.08$) on a Cu single crystal with the orientation C14 of Fig. 7.2, deformed at room temperature; (b) broken-up, coarse slip bands for Stage III of a Cu single crystal of orientation C17 (Seeger *et al.* 1957; Seeger 1957; courtesy of General Electric Co.)

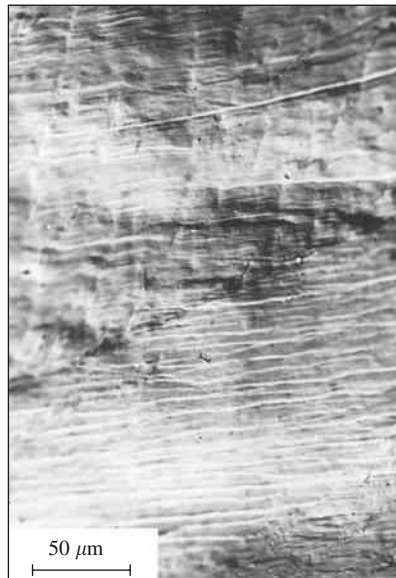


FIG. 7.11. Wavy slip markings resulting from a strain increment of 0.09 in polycrystalline Al deformed in torsion in Stage IV up to a shear strain of 3.58 at room temperature (from Rollett 1988).

for describing the range of elemental slip activity derived from a given active source.

7.2.4 Dislocation Microstructures

Stage I In Stage I the dislocation storage is primarily in the form of well-screened multipolar groups of edge dislocations and the development of some polygonized low-angle walls, shown in the etch pit micrograph of Fig. 7.12 for Cu (Livingston 1962). Transmission electron microscopy confirms that the stored dislocations are predominantly of edge character, indicating that the screw dislocation components have apparently been systematically annihilated by cross-slip in pairwise encounters in parallel slip planes (Mader *et al.* 1963).

Stage II In Stage II, the prevalent form of dislocation storage in the primary slip system is *braids*,¹ roughly perpendicular to the Burgers vector $[101]$ of the primary

¹ The dislocation structures of Figs. 7.13(a),(b) found on primary-plane sections for deformation along the single slip path shown in Fig. 7.1(a) that we have called “braids” represent only an uncomplicated set among other structures found along other loading paths and on other sections, which have been called variously “carpets”, “sheets”, “grids”, etc. For a discussion, see Mughrabi (1975).



FIG. 7.12. Etch pit pattern of multipolar dislocation structures in Stage I for a Cu single crystal (Livingston 1962, courtesy of Elsevier).

system, as shown in the TEM micrographs in Figs. 7.13(a),(b) of a Cu crystal deformed at 77 K to resolved shear stresses of 20 and 40 MPa, respectively (Prinz and Argon 1980). Figure 7.13(a) shows that early in Stage II, the braids contain mainly short edge dislocation dipole segments that have apparently been chopped up by some slip activity on the *conjugate* and *critical* slip systems, and are bunched together. Diffraction contrast analysis of the braids shows a definite concentration of short Lomer–Cottrell (LC)-type straight sessile dislocation segments parallel to the $[1\bar{1}0]$ and $[110]$ directions, which are the lines of intersection between the plane of the primary system and those of the *critical* and *conjugate* systems. These LC segments serve to anchor down the chopped-up edge dipole segments and other monopoles of the primary dislocations. One relatively wide, long braid in Fig. 7.13(a) is parallel to the $[1\bar{1}0]$ direction, where, apparently, the *critical*-plane activity was strong. Figure 7.13(b) shows that at 40 MPa the braids still contain primary dipole segments, but also other straight primary segments that are beginning to have a definite alignment in the $[1\bar{1}0]$ and the $[110]$ directions, where aggregation is clearly catalyzed by activity on the *critical* and *conjugate* systems continuing to furnish LC sessile dislocation segments that pin the primary braids to the lattice. Comparison of the distances of separation D of the braids between Figs. 7.13(a) and (b) shows a clear reduction in the mean spacing of the braids. This spacing

The differences in terminology often result from the difficulty of perceiving 3-D structures from 2-D sections and from the different perceptions of different investigators of possibly the same morphology. Three-dimensional stereoscopic imaging of such microstructures has been rare. For a good example, see Basinski and Basinski (1964). The reader is advised to keep this in mind when consulting other works.

STRAIN HARDENING

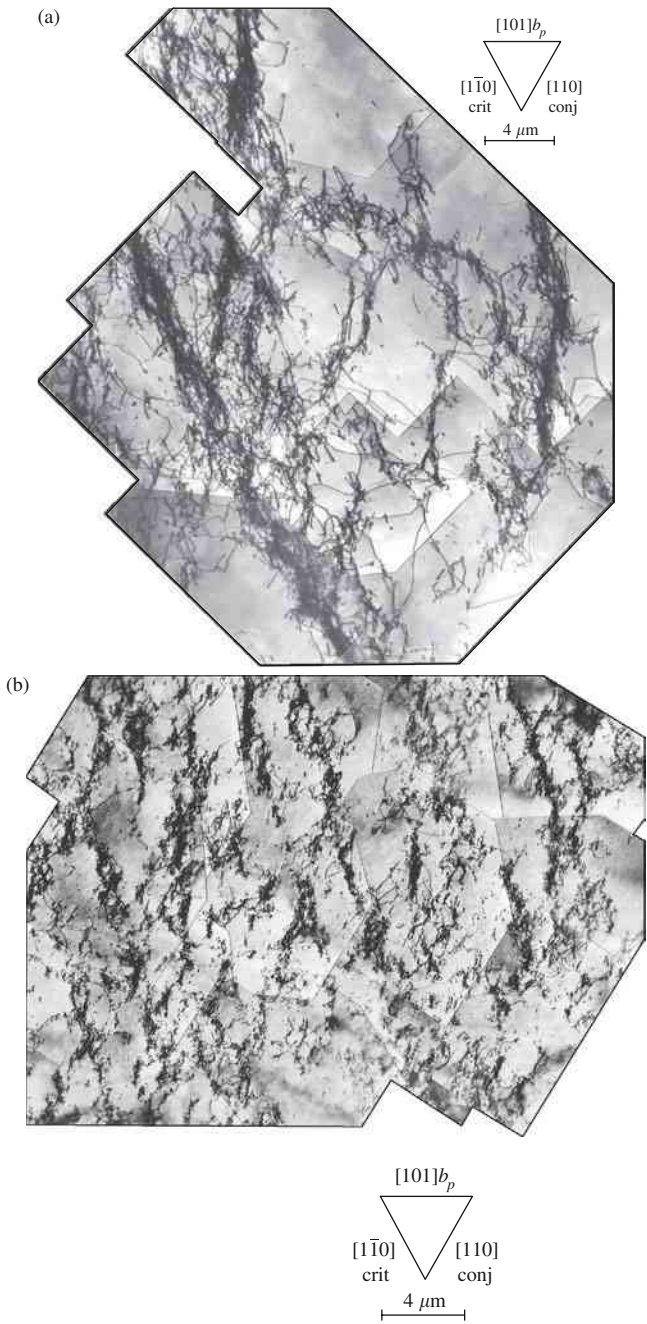


FIG. 7.13. TEM micrographs of dislocation braids in the PM system in Stage II deformation of a Cu single crystal deformed at 77 K to plastic shear resistance levels of (a) 20 MPa and (b) 40 MPa. (c) Dependence of the braid spacing D on the plastic shear resistance τ (Prinz and Argon 1980).

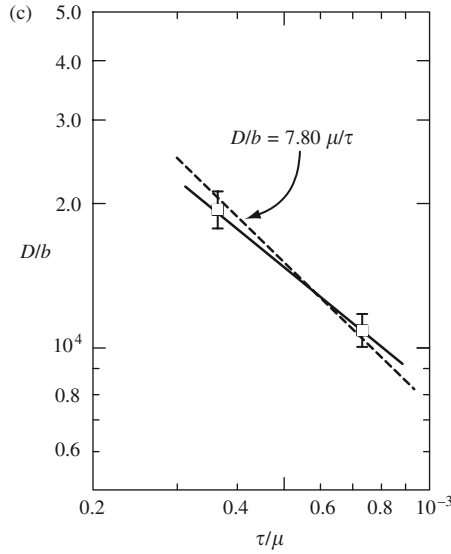


FIG. 7.13. (Contd.)

change is shown in Fig. 7.13(c) and fits closely to a scaling relation of the type

$$\frac{D}{b} = \beta \frac{\mu}{\tau}, \quad (7.1)$$

where the proportionality constant β for this case is 7.80. The form of aggregation of dislocations in the primary system is a particularly well-defined case. There are often less well-defined forms of aggregation in other parts of the crystal. The form of aggregation is even better defined at lower temperatures but becomes more diffuse at room temperature (Prinz and Argon 1980). Nevertheless, the scaling of the spacing of braids or bundles remains as given by eq. (7.1) over nearly all of the range of deformation in Stage II. This form of self-similar aggregation was noted first by Kuhlmann-Wilsdorf (1968), who termed it a *principle of similitude*. It remains a special form of the self-organization of dislocations into quasi-regular groupings that is collectively referred to as *patterning*.

During plastic straining of crystals with single-slip orientations, dislocations of the primary system, which are the main carriers of plastic flow, are stored at slip obstacles of LC type produced by intersections of primary dislocations and forest dislocations threading through the primary plane. Both the primary-plane dislocations and the forest dislocations affect the hardening process directly. Thus, it is important to determine the specific roles of each type and their increase with overall plastic strain. Basinski and Basinski (1979) assembled together the measurements of a number of investigators related to the increase in the density of both primary and forest dislocations with the increase in resolved shear stress in the course of strain

hardening (or, stated in reverse, the rise in the plastic shear resistance of the primary plane associated with the densities of primary and forest dislocations). These measurements were made either by TEM observations or, more reliably (albeit over a more restricted range), by etch pit counts on sections of the primary plane or on the unstressed cross-slip plane through which primary dislocations penetrate. Both forms of dislocation counts give very similar information. The information derived from etch pit counts on Cu and Ag crystals is shown in Fig. 7.14. The data fitting well to a straight line representation of a log-log relationship pertain

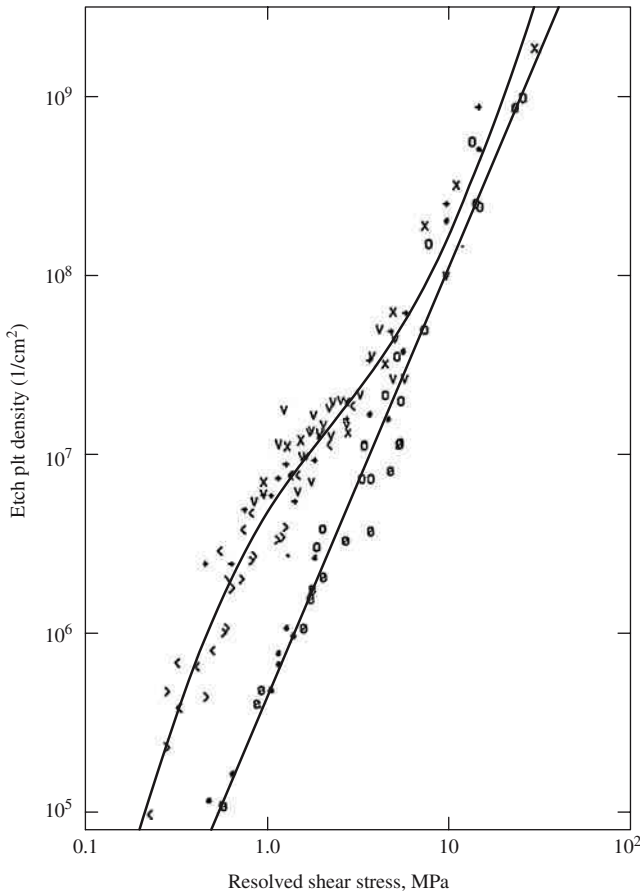


FIG. 7.14. Dependence of dislocation density as determined from surface etch pits, as a function of plastic shear resistance. The sigmoidal curve represents the primary-plane dislocation density, while the straight line represents the forest dislocation density, which more closely obeys eq. 7.2 (Basinski and Basinski 1979, courtesy of Elsevier).

to the forest dislocation density, while the data fitted by a fifth-degree polynomial pertain to the primary dislocation density (for the specific meanings of the symbols, the reader should consult Basinski and Basinski 1979). Figure 7.14 shows clearly that the forest dislocation density, at all levels, except perhaps in Stage I (where other considerations hold), controls the plastic shear resistance of the primary system, even though the primary dislocation density is nearly an order of magnitude larger than the forest density from the beginning of Stage II to half way toward Stage III (at room temperature, 1–10 MPa). The reason for this is clear from Fig. 7.13(a), which shows that a substantial contribution to the primary density comes from well-screened, chopped-up edge dipoles and multipole segments with a short-range interaction. Clearly, in the transition range between Stages I and II when slip activity increases on the conjugate and critical planes, the dislocations of these systems encounter primary dislocations and undergo reactions with them to produce segments of sessile LC obstacles that pin the primary dislocations into braids. This process produces a number of parallel effects. First, the reactions are energetically favored by tending to reduce some line energy in forming the sessile obstacles. Second, they form anchoring points to the primary density, and, finally, equally importantly, they increase the forest density, which basically controls the rate process derived from the more benign primary–forest intersections producing jogs, all of which we discuss in Section 7.3.5. It is important to recognize, however, that glide of the primary dislocations makes the major contribution to the overall strain production, up to 90–95% of the total, while the multiplication and motion of the forest dislocations, stimulated mainly by energetic interactions with the primary dislocations, make only a minor contribution to the total strain, no more than 5–10%, for all secondary systems combined (Basinski and Basinski 1979). Thus, the forest component undergoes rapid multiplication with only a short-range glide activity. The connection between the forest dislocation density and the plastic shear resistance, which holds over five orders of magnitude of change in the average forest dislocation density ρ_f , is one of the central relationships of strain hardening (Nabarro, *et al.* 1964). It fits well to the functional form

$$\tau = \alpha \mu b \sqrt{\rho_f}, \quad (7.2)$$

where $\alpha = 0.45$, as derived from the more complete set of TEM observations. It is important to note that the above relation holds between the plastic shear resistance and the average forest dislocation density, whereas while the areal distribution of the latter on the slip plane may not be random or quasi-uniform at all.

The form of aggregation of stored dislocations shown in Figs. 7.13(a) and (b) relates to the unstressed state of the material after deformation and gives no information on how dislocations move between sources to the eventual storage sites, individually or in groups. To determine this, Mughrabi (1971) subjected samples in the stressed state to neutron irradiation at 4.2 K, prior to unloading followed by

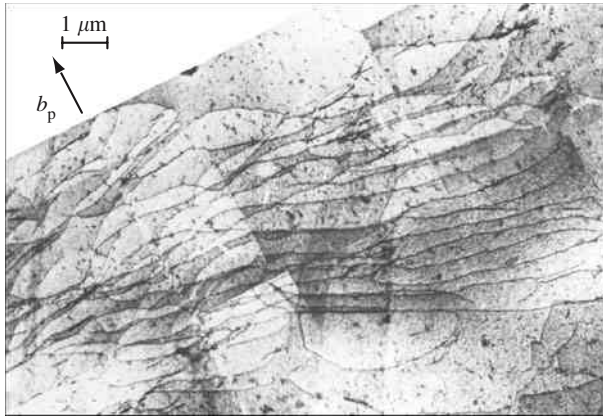


FIG. 7.15. TEM micrograph of active glide dislocations on the PM plane, pinned in place by neutron irradiation, representing groups of dislocations in motion at 77 K at a plastic shear resistance of 12 MPa in Stage II (Mughrabi 1971, courtesy of Taylor and Francis).

sectioning to examine the dislocation structures by TEM. The irradiation damage, in the form of nanosize sessile loops (Section 6.4.9), captured the mobile dislocations in transit. Figure 7.15 shows a set of such mobile dislocations, numbering between 10 and 15, pinned in transit by the visible damage sites. Since the irradiation periods were relatively long, the arrangement of the dislocations does not necessarily relate to any specific short increment of time during actual deformation. Nevertheless, the observations indicate that some group-like motion of glide dislocations occurs and that the number captured in a group, such as that in Fig. 7.15, correlates well with the observed slip line depth.

In Stage II, as noted above, with each increment of strain, dislocations in the primary system are stored after moving characteristic distances related to the spacing of storage sites such as braids, where sessile obstacles formed by reactions between primary and secondary dislocations serve to pin them. As we discuss later in Section 7.3, all these processes of reaction and aggregation scale linearly with strain and produce linear strain hardening, relatively independent of temperature and rate of deformation.

Stages III and IV During the evolutionary development of braids and the refinement of their scale occurring in Stage II, dislocation structures constantly adjust their form by line-tension-driven rearrangements to lower their energy by processes termed “punctuated glide” by Cottrell (2002), wherever possible and within the prevailing constraints. In pure FCC metals and other pure crystals, above a characteristic

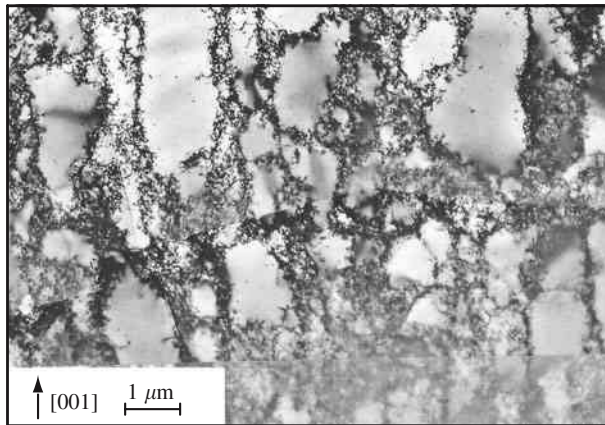


FIG. 7.16. TEM micrograph of closed dislocation cells formed in a [100]-oriented Cu single crystal deformed at room temperature into Stage III (Mughrabi *et al.* 1986, courtesy of Taylor and Francis).

temperature where the lattice resistance vanishes, the static dislocation patterns captured in TEM images constitute constrained *low-energy dislocation structures*, termed LEDS for short by Kuhlmann-Wilsdorf (see Kuhlmann-Wilsdorf 2002 for a historical perspective on these ideas). As a consequence of such ongoing internal rearrangements of dislocations in braids and their increasingly denser packing, toward the end of Stage II a new form of dislocation structure in the form of closed cells develops, which prevails, in various forms of refinement, through Stage III and Stage IV of hardening. Figure 7.16 gives an example of a cellular dislocation structure in a Cu crystal strained in the [001] direction at room temperature to a resolved shear stress of 75.6 MPa on the active slip system, constituting an early Stage III form (Mughrabi *et al.* 1986). Figure 7.17, in turn, represents the cell structure in an Al polycrystal strained in torsion to an equivalent shear strain of $\gamma = 3.46$, or well into Stage IV. Comparison of Figs. 7.16 and 7.17 shows that in Stage III the cells are readily discernible, but have relatively thick and fuzzy walls, while in Stage IV the cells are very clearly defined and have quite narrow walls with a very high dislocation density. In both cases, nearly all visible dislocations are in cell walls and the cell interiors are nearly empty.

Mughrabi *et al.* (1986) have shown that with the evolution of closed cells, starting in late Stage II and continuing in later stages, long-range lattice elastic strains with principal axes parallel and perpendicular to the principal extension direction develop and increase in intensity throughout Stages III and IV. We return to the significance of these in Section 7.3.

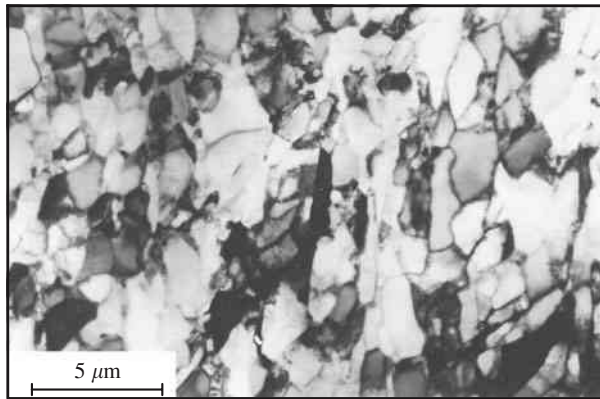


FIG. 7.17. TEM micrograph of a typical cellular dislocation microstructure in a pure Al polycrystal deformed to a shear strain of 3.46 into Stage IV at room temperature (from Rollett 1988).

Hughes and various collaborators (Hughes *et al.* 1997; Hughes and Hansen 2000, 2001; Hughes 2002) have made detailed studies of the statistical features of the prevalent boundaries of cells and larger groupings in polycrystalline Cu and Al and have identified two major types, which they have called *geometrically necessary boundaries* (GNBs) and *incidental dislocation boundaries* (IDBs). Figure 7.18 shows a sketch of an arrangement as obtained from a pure Al sample cold-rolled to an equivalent strain of 0.1 (roughly a shear strain of 0.17), which for polycrystalline Al puts the behavior well into Stage III. The dense GNBs divide regions with random relative lattice rotations, indicated in the figure. The narrow boundaries delineating equiaxed cells in between the GNBs are the IDBs. These are the cells presented in Figs. 7.16 and 7.17. The random lattice rotations across the IDBs are in the range of 0.5° and 2.0° for the above level of strain. For both types of boundaries, the lattice rotations increase with strain less than linearly, as shown in Fig. 7.19. Hughes (2002) has noted that the distributions of the distances D^{GNB} and D^{IDB} between boundaries and the distribution of the misorientation angles across them, when normalized by the average spacing D_{av} and average misorientation angle θ_{av} , obey relatively well-defined forms, shown in Fig. 7.20, which signify the presence of definite statistical processes of self-organization. Modeling of such boundary formation by Dawson *et al.* (2002) strongly suggests that the more prominent GNBs are present only in polycrystalline material and arise from intercrystalline constraints. They are generally absent in single crystals deformed quasi-homogeneously. Therefore, we consider them to be of less importance in the overall picture of the dislocation structures and their relation to strain-hardening levels.

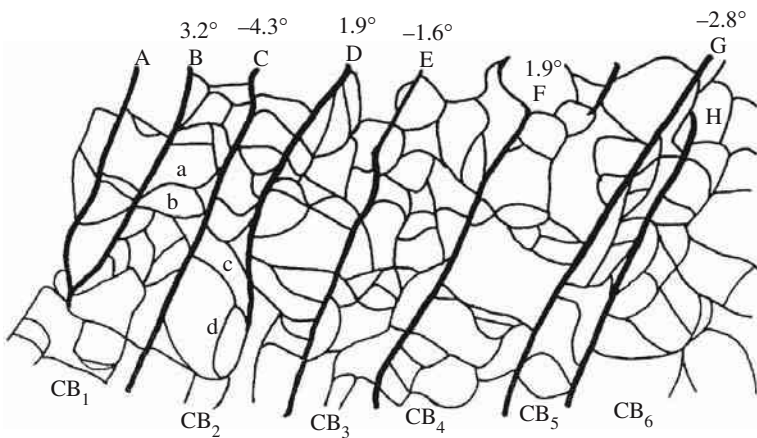
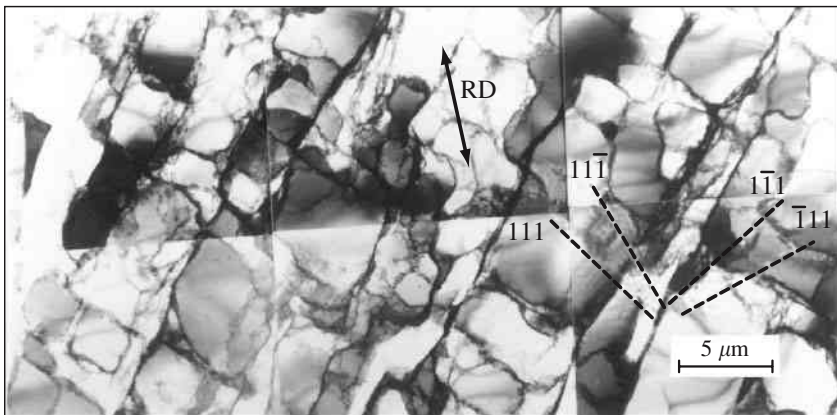


FIG. 7.18. Dislocation microstructure of polycrystalline Al cold-rolled to a thickness reduction of 0.1: (a) TEM image showing geometrically necessary boundaries and incidental boundaries (the cells referred to primarily in this chapter); (b) a sketch better delineating the GNBs and IDBs visible in (a) (Hughes *et al.* 1997, courtesy of Elsevier).

Noting that the braids in Stage II (Figs. 7.13(a) and (b)) contain a high concentration of sessile LC segments, Prinz and Argon (1980) suggested that they could not be readily displaced during the refinement of scale but that such refinement must occur by cell division, where larger cells in the distribution offer additional trapping processes that produce new braids in between, in which secondary systems of the critical and the conjugate system can develop more readily before they interact with the primary system. This purely rate-independent kinematical process is also in keeping with the linear nature of hardening in Stage II. However, once closed cells are formed in Stage III and the hardening rate decreases and becomes

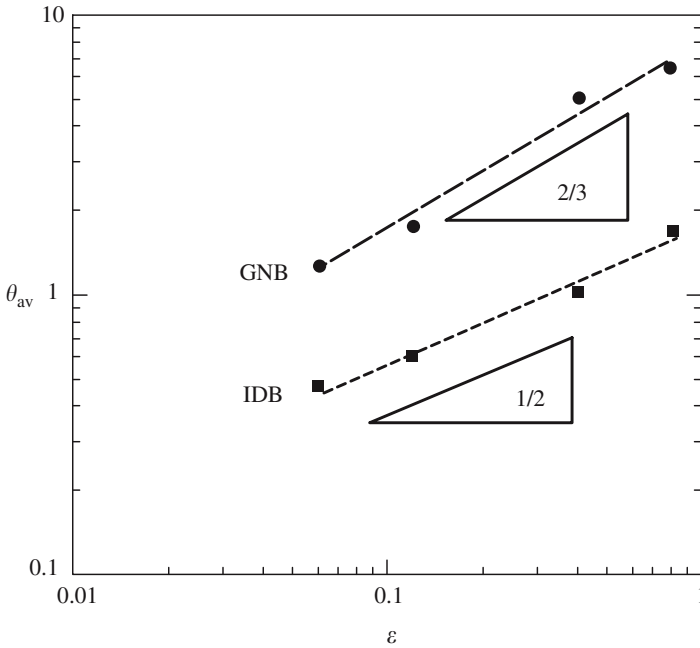


FIG. 7.19. Shear-strain dependence of random lattice misorientation across GNBs and IDBs. Note that the lattice misorientation of the IDBs increases with the square root of the shear strain, suggesting a random-error form of relation (Argon and Haasen, 1993; Hughes *et al.* 1997; courtesy of Elsevier).

rate-dependent, there is evidence that uninterrupted, rate-independent addition of sessile obstacles ceases and other rate-dependent processes begin to dismantle previously formed sessile obstacles. This permits the cell walls to translate and undergo a new form of self-organization directly associated with dynamic recovery. This self-organization has been considered by Haehner and Zaiser (1999) and Zaiser and Haehner (1999) as a noise-induced transition of dislocation structures into cellular form, occurring at high levels of flow stress in plastic flow of a material with very low strain rate sensitivity of the flow stress, as in the case of FCC metals and BCC metals above a critical temperature where the lattice resistance is no longer present. In this process, the full description of which is outside the scope of this chapter, the acoustic noise arising from large local fluctuations in the mobile dislocation fluxes produced by their repeated arrest and release by cell walls creates structural fluctuations that are instrumental in the alteration of cell patterns. Such alterations, in turn, are facilitated directly by dynamic recovery processes occurring in cell walls associated with the removal of pinning LC obstacles that have hitherto prevented

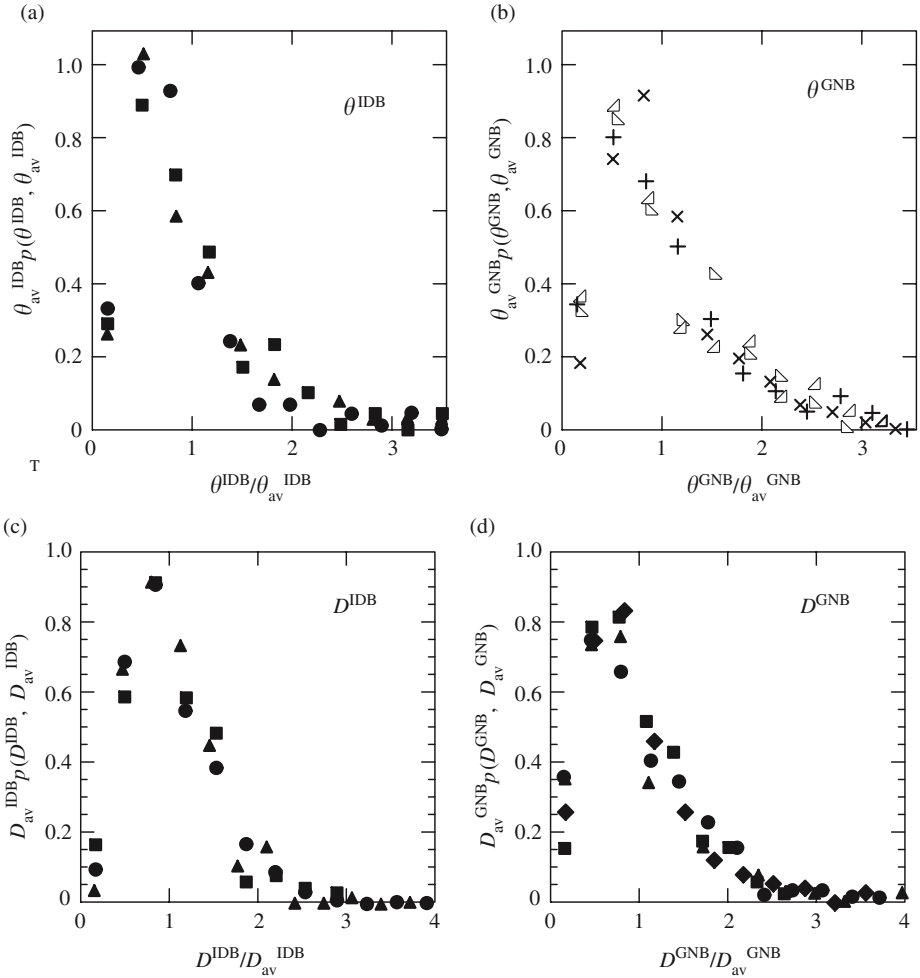


FIG. 7.20. Normalized frequency distributions of lattice misorientation angles, normalized by the population average of the angles: (a) for rolled Ni, IDBs; (b) for rolled Al, GNBs. Normalized frequency distributions of spacings D between boundaries: (c) for rolled Ni, IDBs; (d) for rolled Ni GNBs (Hughes 2002, courtesy of Elsevier).

boundary mobility. The signature of such stochastic self-organization is the formation of cell patterns having a definite fractal character shown, for example, in Fig. 7.21 (Haehner and Zaiser 1999). In Section 7.3.7, we present a model of such dynamic recovery.

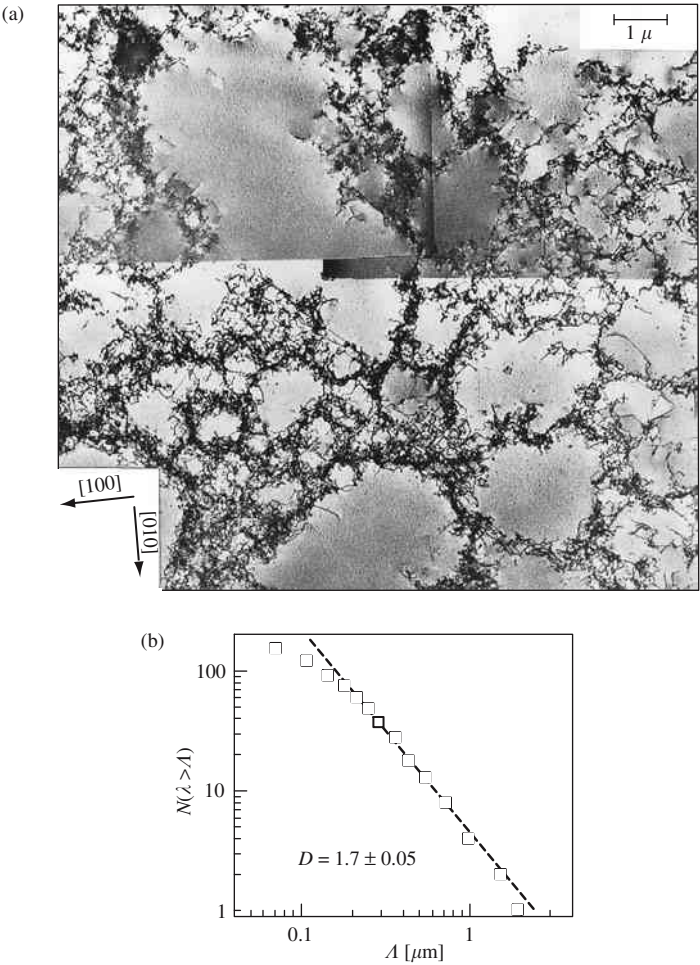


FIG. 7.21. (a) TEM micrograph of self-similar dislocation cells in a [100]-oriented Cu single crystal deformed at room temperature to a resolved shear stress of 37.3 MPa; (b) the corresponding cell size distribution shows clear fractal characteristics (Haehner and Zaiser 1999 courtesy of Elsevier).

7.3 Strain-hardening Models

7.3.1 Overview

The plastic resistance is influenced by strain rate and temperature and evolves with plastic strain as a consequence of strain hardening. In pure FCC metals with negligible lattice resistance and in other structures above a critical temperature, the

strain rate dependence of the plastic resistance is governed by the thermally assisted intersections of primary and forest dislocations. These intersections are basically of two types. The first type, which are referred to as *attractive* and *repulsive junctions* (Saada 1960), are those which, regardless of strength, can be accomplished at the cost of only forming jogs on the intersecting dislocations. These intersections can be thermally assisted. The second type is the intersections between screw dislocations that require continued point defect production for every atomic increment of separation, and cannot be thermally assisted. In Section 7.3.2, we discuss first the rate-controlling dislocation intersections of the first type. In the following Sections 7.3.3–7.3.6, we discuss the evolution of the plastic resistance in models for strain hardening for Stages I–IV. We state at the outset that strain-hardening models have been advanced by many investigators, and generally incorporate very detailed and specialized scenarios that have been passionately defended by their adherents. Some references to these are given at the end of the chapter. In the models that we present in the following sections, we develop only what can arguably be considered as the central processes, consistent with the detailed features of hardening behavior presented in Section 7.2, and compare these models with experimental results.

7.3.2 Dislocation Intersections

In crystal structures of high symmetry as in FCC and BCC metals, the intersection of dislocations of active slip systems plays an important role. First, many intersections between primary and forest dislocations result in *sessile dislocation segments* that act as strong pinning points, which anchor dislocation braids and tangles and become the main source of strain hardening. Second, other types of intersection, where the separation of locally entrapped pairs of dislocations can be more readily accomplished with thermal assistance, become the main source of the rate-controlling process in the glide of primary dislocations.

Here we take up the simplest scenario, follow the process of dislocation intersection in an FCC crystal oriented for single slip, and consider the intersections of the primary dislocations with those of secondary forest systems in the geometry outlined in Section 7.2.1. These processes were considered first by Saada (1960) and in more detail by Baird and Gale (1965). We follow the arguments of the latter.

Consider a generic intersection configuration as depicted in Fig. 7.22, where the primary glide dislocation $A_2-A'_2$ on the plane (2) is undergoing an intersection at point F with the forest dislocation $A_1-A'_1$ on the inclined plane (1). The intersections that can occur under these conditions are of several types. There are *attractive junctions*, resulting from the encounter of a pair in which the two dislocations are attracted to the line of intersection of the two planes by longer-range elastic interactions which result in line energy reduction and, as depicted in the figure, in a reaction product dislocation P_0Q_0 that is formed along the line of intersection. The primary dislocation in the plane (2) is reduced to segments A_2P_0 and $Q_0A'_2$, while

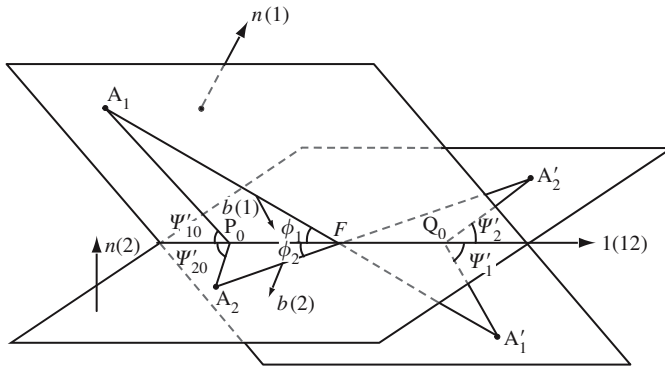


FIG. 7.22. Sketch showing a junction reaction in the making, when two glide dislocations of inclined slip systems intersect, forming a variety of attractive junctions, in particular, LC junctions (after Baird and Gale 1965).

the forest dislocation is correspondingly reduced to segments A_1P_0 and $Q_0A'_1$, with the points A_1, A'_1, A_2, A'_2 being considered fixed for the purpose of simplifying the analysis. The reacted configuration, with the P_0 and Q_0 nodes corresponds to a state with no applied stress. The formation of the reaction product dislocation P_0Q_0 results in significant lowering of energy when compared with the energies of the unreacted segments it replaces. The segment either can be sessile, that is, with a Burgers vector that does not lie in a good slip plane, or could possibly be glissile in either of the two intersecting planes. In either case, the successful passage of the two dislocations through the intersection point F eventually requires the formation of jogs on the two dislocations, which, however, is not the main energy barrier to the intersection. The more energy-consuming process requires pushing back points P_0 and Q_0 to the point F under stress to reverse the initial reaction that formed the product dislocation P_0Q_0 . If the slip plane and slip direction of the primary dislocation can have all 24 possible values (counting positive and negative directions of motion separately), there will be 216 possible reactions in a particular crystal, of which 24 will be sessile, 43 glissile, and 144 not attractive, that is, in which the intersection requires only production of jogs (Baird and Gale 1965). Thus, in addition to the attractive junctions, there are other intersections, where the approaching dislocations repel each other, forcing the intersection to occur with the pair approaching at right angles, with a minimum of long-range interaction. These make up the 144 possibilities listed above and are termed *repulsive junctions*.

Still another type of intersection of dislocations is that between pure screw dislocations, which, as remarked in Chapter 2, is associated with some topological barriers through a need to produce point defect trails of either vacancy or interstitial type. These cannot be assisted by thermal fluctuations.

The intersections of principal interest here are those related to attractive junctions. As can be appreciated from Fig. 7.22, the many degrees of freedom of the intersection process, involving the location of the pinning points A_1 , A'_1 , A_2 , A'_2 , the approach angles (ϕ_1 and ϕ_2), and the multiplicity of secondary-system planes, make a complete analysis of all possibilities quite formidable. Baird and Gale considered only the most prominent possibilities for junctions subjected to stress, and related their findings to a framework of intersections of primary dislocations with a random field of forest dislocations acting as point obstacles in the manner of glide dislocations going through a field of pointlike precipitates. Considering both FCC crystals, for which the product dislocations are either sessile, consisting of Lomer–Cottrell-type segments, or of glissile type, parallel to $\langle 110 \rangle$ directions, and BCC crystals, in which the product dislocations are all sessile, of Cottrell type, parallel to $\langle 100 \rangle$ directions with $\langle 001 \rangle$ -type Burgers vectors, Baird and Gale (1965) determined the average peak strengths $\hat{K}/2\mathcal{E}$ of the attractive junctions in both structures. Their results are given in Table 7.1.

Hirth (1961) considered a broader set of possible sessile dislocation junctions and suggested other types, referred to as *Hirth locks*. These, however, have rarely been observed and do not appear to play a prominent role in strain hardening. In comparison, Lomer–Cottrell-type sessile dislocations have been widely observed and are quite prominently encountered at cryogenic temperatures (Essmann 1965). They can form in a variety of pairwise interactions of different slip systems when the tensile-axis direction is in various parts of the standard triangle. The prevalence of the different sets of LC locks is depicted in Fig. 7.23 (Nabarro *et al.* 1964).

TABLE 7.1. Average effective strengths $\hat{K}/2\mathcal{E}$ of attractive junctions in FCC and BCC metals (Baird and Gale 1965)

FCC crystals
Sessile (Lomer–Cottrell-type junctions); $\hat{K}/2\mathcal{E} = 0.440$ (with a range of 0.397–0.506).
All attractive junctions, including glissile types; $\hat{K}/2\mathcal{E} = 0.375$ (with a range of 0.330–0.428).
BCC crystals
Sessile (Cottrell-type junctions); $\hat{K}/2\mathcal{E} = 0.407$ (with a range of 0.353–0.468).
All attractive junctions, including glissile types; $\hat{K}/2\mathcal{E} = 0.282$ (with a range of 0.243–0.307).

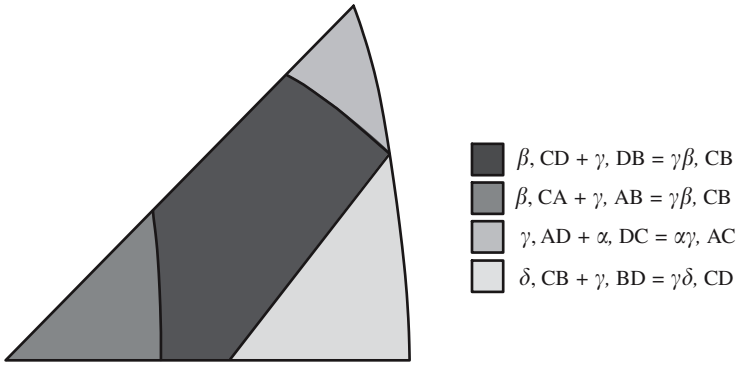


FIG. 7.23. Prevalence of the formation of LC junctions when dislocations of the four slip systems intersect, for different crystal axis orientations in the standard triangle (after Nabarro *et al.* 1964).

For comparison of the strengths of attractive junctions with experiment, we consider the force–activation-distance curves $K(\Delta y^*)$ determined for Cu by thermal analysis of forest dislocation intersections in Stage I deformation by Argon and East (1979). In stress jump experiments conducted at different temperatures on highly perfect Cu single crystals, prestressed to two reference critical resolved shear stress levels of 0.3 and 0.6 MPa, the curves of cutting force K versus activation distance Δy^* were determined by the standard forms of thermal-activation analysis presented in Chapter 2 from the relations

$$\Delta y^* = \frac{\Delta a^*}{\lambda} = \frac{kT}{b\lambda} \left(\frac{\partial \ln \dot{\gamma}}{\partial \sigma} \right), \quad (7.3a)$$

$$K = \sigma b\lambda = 2\mathcal{E} \left(\frac{\sigma bl}{2\mathcal{E}} \right)^{2/3}. \quad (7.3b)$$

In eqs. (7.3a) and (7.3b), $\lambda = l(2\mathcal{E}/\sigma bl)^{1/3}$ is the Friedel sampling length discussed in Chapter 6 and $l = 1/\sqrt{\rho_f}$ is the mean forest dislocation spacing (5.13×10^{-5} m and 2.04×10^{-5} m for the two different reference flow stress levels, revealed on primary-plane sections by etch pits). The two force–distance curves $K = K(\Delta y^*)$ shown in Fig. 7.24, most probably resulting from intersections between primary-plane and critical-plane dislocations, have a common peak resistance $\hat{K} = 3.5 \times 10^{-10}$ N but different activation distance curves. This difference, which is of little interest here, is most likely a result of the rather special conditions created by the very large forest dislocation spacing (Argon and East 1979). Considering an anisotropic line tension $\mathcal{E} = 6.13 \times 10^{-10}$ N appropriate to the conditions of the experiments, we can determine $\hat{K}/2\mathcal{E} = 0.285$, but taking

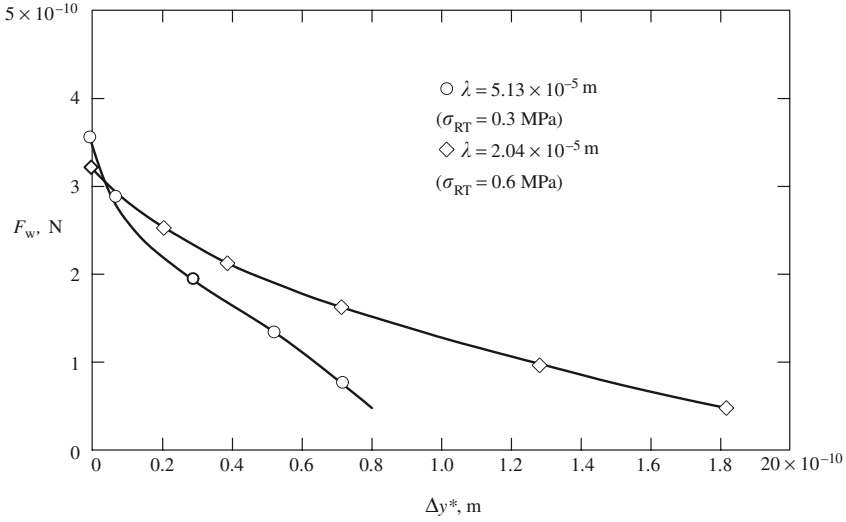


FIG. 7.24. Force–distance curves measured for primary dislocations intersecting forest dislocations, probably forming attractive junctions, in Stage I in Cu for two different levels of primary-plane plastic shear resistance, assembled from strain rate jump experiments at different temperatures (Argon and East 1979).

$\mathcal{E} = 3.11 \times 10^{-10} \text{ N}$ instead, which would be more appropriate for the Stage II flow stress levels to which the Baird and Gale analysis relates, we obtain $\hat{K}/2\mathcal{E} = 0.564$. These two estimates are to be compared with the average value of 0.375 shown in Table 7.1 for glissile attractive junctions.

From these considerations, we conclude that both the sessile junctions and the glissile attractive junctions have finite peak strength and that all attractive junctions can be dissolved under certain conditions during flow by thermal assistance.

Finally, we note that the analysis of Baird and Gale relates to unextended dislocations. When dislocations are extended, made up of partials separated by stacking faults, there are significant differences in the form and strength of the sessile junctions. Thus, in particular, the strengths of LC junctions in Al and Ag, which have a very high (0.104 J/m^2) and a very low (0.016 J/m^2) stacking-fault energy, respectively, have been computationally studied by Shenoy *et al.* (2000). From these simulations, for an appropriate ratio of 0.75 of the stress on the critical plane to that on the primary plane, we determine an LC junction dissolution stress of $\sigma/\mu = 0.014$ for the critical resolved shear stress on the primary plane. This exceedingly high stress comes about from the constrained choice of very short distances between pinning points, of the order of only the reaction product dislocation line length, about $40 \text{ \AA}$, giving an l/b ratio of only about 16. From these conditions, we estimate a normalized LC junction strength of $\hat{K}/2\mathcal{E} = (\sigma l/\mu b)^{2/3} = 0.363$,

which compares tolerably well with the value of 0.44 determined by Baird and Gale. Of greater interest, however, is the inference from Shenoy *et al.*'s study that the more extended Cottrell junction should be significantly stronger than the unextended Lomer junction, and that the dependence of the strength on the stacking-fault energy could be important. We return to this observation later in discussing dynamic recovery in Stage III deformation. We must note in passing that all these analyses are rather idealized, since they consider the forest dislocations to be like immovable precipitate-type obstacles, while there is much evidence that in the intersection process the primary dislocation, in forcing the forest dislocation, can significantly pluck and displace the latter, resulting in some enrichment of the forest dislocations in front of the primary dislocations (East 1971).

7.3.3 Stage I Strain Hardening

As outlined in Section 7.2, in Stage I the hardening is a result of aggregation of edge dislocations into multipoles by mutual trapping while the screw dislocation segments of expanding loops are eliminated by cross-glide of opposite pairs on neighboring glide planes, giving rise to pairs of superjogs on the edge dislocation as observed by Essmann (1965). This is depicted in Fig. 7.25 which shows also that such superjog pairs can act as dislocation sources. Unlike the strain hardening in other stages of deformation, where it is based on dislocation interaction,

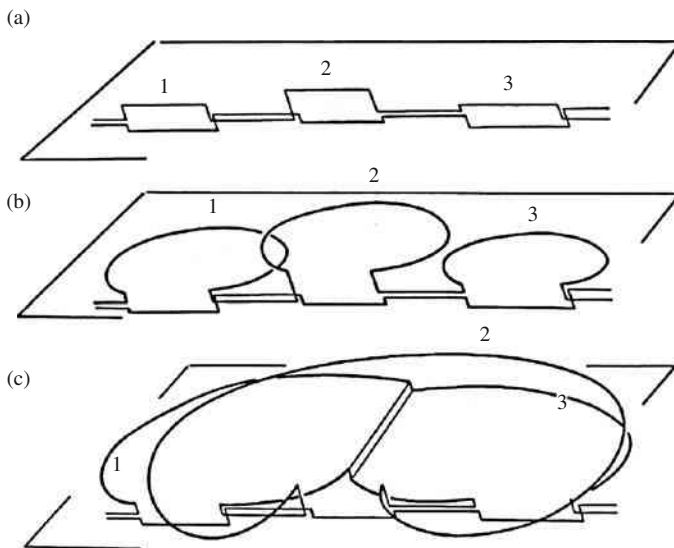


FIG. 7.25. Sketch of the formation of dislocation sources from a partially captured segmented edge dislocation. The stages (a), (b), and (c) show the expected forms at different stages of development (Argon 1969).

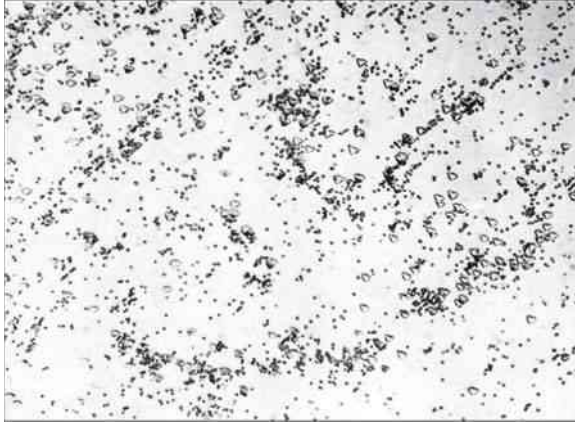


FIG. 7.26. Preferential formation of initial multipolar clusters by trapping of edge dislocations (small pits), occurring in the neighborhood of grown-in sessile dislocations (large pits), revealed by a two-step etching experiment, before and after a small imposed strain increment (Argon and Brydges 1968).

hardening in Stage I is due to the rise in applied stress required to maintain a constant dislocation flux through gradually thickening multipoles.

Since edge dislocation multipoles can be readily displaced, their initial “seed-ing” is accomplished by the grown-in dislocation network, as is shown in Fig. 7.26, in which the grown-in dislocations appear as the larger etch pits that start the aggregation process of multipoles (Argon and Brydges 1968). As the multipoles grow in cross section, they impede the flux of the primary glide dislocations. Experiments indicate that forest dislocations also increase slowly in response to stresses acting on the secondary slip systems (Argon and East 1979), but sessile-junction formation is rare, and the flux is controlled by thermally assisted glissile-type attractive-junction decomposition. Dislocation sources form by superjog formation when screw dislocations are annihilated by cross-glide, and active sources are eventually inactivated when new edge dislocations are trapped in the fringes of the thickening multipoles where the active sources existed. The hardening results from the increase of stress required to maintain the needed dislocation flux (Argon and East 1968; Argon 1969). Here we present only a skeletal outline of a detailed statistical theory (Argon 1969; see also Nabarro 1985).

The primary dislocation flux ϕ that needs to be maintained constant,

$$\phi = \frac{\dot{\gamma}}{b} = \rho_m \bar{v}, \quad (7.4)$$

is a result of a kinetic balance of mobile-dislocation generation and inactivation, where the average dislocation velocity is governed by a combination of phonon

drag and thermally assisted forest dislocation intersections that can given as

$$\bar{v} = \frac{\sigma b}{B} \left[1 + \frac{\sigma b}{l_f B \nu_G} \exp \left(\frac{\Delta H_{fc}^*}{kT} \right) \right]^{-1} = \frac{\sigma b}{B_c}. \quad (7.5)$$

In eq. (7.5), l_f is the mean forest dislocation spacing, B is the phonon drag coefficient, ν_G is a frequency factor for forest cutting, and ΔH_{fc}^* is the stress-dependent activation free energy for forest cutting. For the purpose of simplifying the analysis, an effective stress-dependent drag coefficient is introduced, as follows:

$$B_c = B \left[1 + \frac{\sigma b}{l_f B \nu_G} \exp \frac{\Delta H_{fc}(\sigma)}{kT} \right]. \quad (7.6)$$

Considering the conditions in an established flow process, we take the half-width of the capture cross section of a multipole as w and, in a 2-D representation of edge dislocation flow, take the number density of multipoles as N_0 given their mean spacing $h \cong 1/\sqrt{N_0}$.

Flux maintenance in channels of width h requires a balance between the rate of mobile-dislocation-density production and the rate of inactivation, where the production is at the border of multipoles where active sources exist, and inactivation is accomplished by the entire multipole cross section.

Emission of a mobile dislocation occurs when a previously emitted dislocation has moved away from a source by the mean distance l_m between mobile dislocations, that is,

$$\left(\frac{d\rho_m}{dt} \right)^+ = \frac{2N_0}{\Delta t_n} \quad (7.7a)$$

where the time step Δt_n for nucleation is

$$\Delta t_n = \frac{l_m}{\bar{v}} = \left(\frac{b}{\dot{\gamma} \bar{v}} \right)^{1/2}, \quad (7.7b)$$

giving

$$\left(\frac{d\rho_m}{dt} \right)^+ = 2N_0 \left(\frac{\dot{\gamma} \bar{v}}{b} \right)^{1/2}. \quad (7.7c)$$

Correspondingly, the rate of reduction of the mobile density is based on the fraction of the mobile dislocation flux intercepted by multipoles having a capture cross section of $2(w + y_c) \approx 2w$, where

$$y_c = \frac{\mu b}{8\pi(1-\nu)\sigma} = \frac{\alpha \mu b}{\sigma}, \quad (7.8)$$

and where y_c is the capture range for edge dislocations at a typical multipole fringe and $\alpha = 1/[8\pi(1-\nu)]$. The rate of decrease of the mobile dislocation density is

then

$$\left(\frac{d\rho_m}{dt}\right)^- = -2\left(\frac{\dot{\gamma}}{b}\right)(w + y_c)N_0. \quad (7.9)$$

Constancy of the primary dislocation flux requires that

$$\left(\frac{d\rho_m}{dt}\right)^+ + \left(\frac{d\rho_m}{dt}\right)^- = 0, \quad (7.10)$$

or that

$$\left(\frac{\dot{\gamma}\bar{v}}{b}\right)^{1/2} = \frac{\bar{v}}{w + y_c}, \quad (7.11)$$

giving the current multipole capture half-cross-section

$$w \simeq \left(\frac{b\bar{v}}{\dot{\gamma}}\right)^{1/2} = \left(\frac{b}{\dot{\gamma}}\right)^{1/2} \left(\frac{\sigma}{\mu}\right)^{1/2} \left(\frac{\mu b}{B_c}\right)^{1/2}, \quad (7.12)$$

where y_c has been neglected in comparison with w and use has been made of eq. (7.5).

Meanwhile, the rate of increase of the multipole cross section w by acquisition of dislocations in a characteristic capture channel y_c is

$$\dot{w} = \frac{\eta y_c}{\Delta t_a}, \quad (7.13)$$

where η is a capture efficiency, considered to be quite low since capture occurs only at short superjog pairs on the margins, and where $\Delta t_a = 1/y_c\phi = b/\dot{\gamma}y_c$ is the arrival time of a dislocation in a capture channel of width y_c at the border of a multipole. This gives

$$\dot{w} = \mu \left(\frac{\dot{\gamma}}{b}\right) \left(\frac{\alpha\mu b}{\sigma}\right)^2. \quad (7.14)$$

Differentiation of eq. (7.12) with respect to time and equating with eq. (7.14),

$$\left(\frac{\sigma}{\mu}\right)^{3/2} \frac{d}{dt} \left(\frac{\sigma}{\mu}\right) = 2\eta\alpha^2 \left(\frac{\dot{\gamma}B_c}{\mu}\right)^{1/2} \dot{\gamma} = 2\eta\alpha^2 \left(\frac{\dot{\gamma}B_c}{\mu}\right)^{1/2} \frac{d\gamma}{dt}, \quad (7.15)$$

gives

$$\left(\frac{\sigma}{\mu}\right)^{3/2} d\left(\frac{\sigma}{\mu}\right) = \eta C d\gamma, \quad (7.16)$$

where

$$C = 2\alpha^2 \left(\frac{\dot{\gamma} B_c}{\mu} \right)^{1/2} \quad (7.17)$$

and the normalized strain-hardening rate in Stage I is given crudely by

$$\frac{\Theta_I}{\mu} = \eta C \frac{1}{(\sigma/\mu)^{3/2}}, \quad (7.18)$$

indicating that the slope of the stress-strain curve is not constant but decreases with increasing stress. While this is exaggerated by the simple model, experiments show a somewhat milder form of the effect.

To compare the estimated hardening rate with experiment, we recall that the effective drag coefficient B_c will far exceed the phonon drag coefficient $B = kT/\Omega v_D$. We estimate this effective drag coefficient as

$$B_c = \frac{\sigma b}{\dot{\gamma}} = \left(\frac{\sigma}{\mu} \right) \left(\frac{\mu b^2 \rho_m}{\dot{\gamma}} \right), \quad (7.19)$$

where we have used $\rho_m \approx 10^6 \text{cm}^{-2}$, estimated by Argon and Brydges (1968) from etch pit measurements at a strain of $\gamma = 0.01$ and a flow stress of 0.6 MPa. This modifies eq. (7.18) to

$$\frac{\Theta_I}{\mu} = \frac{2\eta\alpha^2(b^2\rho_m)^{1/2}}{(\sigma/\mu)}, \quad (7.18a)$$

giving

$$\left(\frac{\Theta_I}{\mu} \right)_{\text{model}} = 1.52 \times 10^{-2} \eta. \quad (7.20a)$$

The actual hardening rates measured by Argon and Brydges (1968) were

$$\left(\frac{\Theta_I}{\mu} \right)_{\text{exp}} = 3.64 \times 10^{-4}. \quad (7.20b)$$

This suggests that the capture efficiency of dislocations by the fringes of the multi-poles is only 2.4% in the simple model. In the detailed statistical theory (Argon 1969), these effects are correctly accounted for, requiring no ex-post-facto adjustment. The basic outline of the simple model, however, correctly reflects the physical processes.

7.3.4 Stage II Strain Hardening

There have been many models of hardening for Stage II deformation. The early models were reviewed by Nabarro *et al.* (1964), and a more recent review of special aspects of this stage of hardening can be found in Nabarro and Duesbery (2003). The early models introduced a number of important concepts related to the generation of slip obstacles and discussed the role of long-range stress due to possible dynamic groupings of glide dislocations in pileups. However, the latter were not found experimentally by Mughrabi (1975) in the stylized forms advocated by the theoretical models, but nevertheless those experiments showed some group-like motion of primary dislocations.

The essence of Stage II hardening can be captured by relatively uncomplicated dimensional analysis based on the deformation features enumerated in Section 7.2 and arguments put forth by Kuhlmann-Wilsdorf (1962) and Kocks (1966). Thus, here, the primary emphasis will be to develop a mechanistic rationale for the rate-independent linear hardening of Stage II. Other aspects related to transients due to changing of the deformation direction, including reversal of deformation and the Bauschinger effect, as well as the development of long-range internal stresses, will be presented in the context of Stage III and Stage IV deformation or will be delayed to Chapter 8.

In Stage II, the principal plastic resistance is considered to arise from the interaction of glide dislocations of the primary slip systems with a combination of individual penetrable forest obstacles and impenetrable braids or clusters pinned in place by *LC locks*, as shown in Figs. 7.13(a) and (b). The plastic resistance resulting from the intersection of attractive or repulsive penetrable forest obstacles can be represented as if they were shearable precipitates of peak resistive force $\hat{K}_p$:

$$\hat{\tau}_p = \alpha \left(\frac{2\mathcal{E}}{bl_p} \right) \left(\frac{\hat{K}_p}{2\mathcal{E}} \right)^{3/2}. \quad (7.21)$$

The nonshearable braids, containing denser collections of LC locks that have definite large cross sections on the glide plane at an average spacing D , could then be treated as if they were dispersoids. However, this would create a difficulty in that such packaging of clustered dislocations would leave unclear the actual dislocation content of those pseudodispersoids. Instead, relying on the well-established empirical observations presented in Fig. 7.14 that all forest dislocations influence, albeit in different ways, the plastic shear resistance as if they were separate, unclustered, and randomly distributed, the dislocation content of the braids can also be considered in that manner, with the proviso that they are impenetrable in Stage II, giving

$$\hat{\tau}_i = \alpha_i \left(\frac{2\mathcal{E}}{bl_i} \right), \quad (7.22)$$

where $\alpha_i = 0.85$. Combining the penetrable and impenetrable components of the forest resistance by a Pythagorean superposition gives the overall resistance,

$$\hat{\tau} = \left(\hat{\tau}_i^2 + \hat{\tau}_p^2 \right)^{1/2}, \quad (7.23)$$

where $\hat{\tau}_p$ would then be expected to be attenuated by thermal assistance. This implies for the overall resistance

$$\hat{\tau} = \alpha' \left(\frac{2\mathcal{E}}{bl} \right) = \alpha' \mu b \sqrt{\rho}, \quad (7.24)$$

where ρ is the total forest density, clustered or unclustered; this is related to the other components by

$$\rho = \left(\frac{1}{l} \right)^2 = \rho_p + \rho_i = \left(\frac{1}{l_p} \right)^2 + \left(\frac{1}{l_i} \right)^2, \quad (7.25)$$

where l is the total average forest spacing regardless of type.

Defining the ratio of the impenetrable component of the plastic resistance to the total shear resistance as

$$R = \frac{\hat{\tau}_i}{\hat{\tau}} = \frac{l}{l_i} \quad (7.26)$$

gives

$$\alpha' = \alpha_i \left[R^2 + \left(\frac{\hat{K}}{2\mathcal{E}} \right)^3 (1 - R^2) \right]^{1/2}, \quad (7.27)$$

where $\alpha' = \alpha = 0.45$ as defined earlier through eq. (7.2) and $\alpha_i = 0.85$ as stated above on the basis of the statistics of sampling of pointlike obstacles (Section 6.3.2).

As eq. (7.24) indicates, a change in the total shear resistance $\hat{\tau}$ must result from a strain-induced change in the overall forest obstacle density ρ ;

$$d\hat{\tau} = \frac{\alpha' \mu b}{2} \frac{d\rho}{\sqrt{\rho}}. \quad (7.28)$$

This change in the obstacle density results from an increment of shear strain

$$d\gamma = \frac{b}{V} d \left(\sum_i \Delta a_i \right) = \frac{b}{V} d(L\Lambda), \quad (7.29)$$

where the shear on all of Δa_i represents the entire area swept out by primary glide dislocations during the strain increment in a volume V , L is the mean free path

length of glide dislocations, and Λ is the total primary-dislocation line length that has moved in the volume V of the crystal from sources to storage sites at braids or clusters.

Thus, Λ/V is the total density of moved and subsequently stored primary dislocations during the increment of strain. Considering that the increment in the obstacle density in eq. (7.28) must come directly from the increment in the density of blocked glide dislocations stored during the increment of glide strain, it should follow that

$$d\gamma = bLd\rho, \quad (7.30)$$

where it has been assumed that L varies more slowly and has been considered constant during the increment. The similitude represented by eq. (7.1) suggests that L should be a multiple of the current mean overall spacing l of forest dislocations, that is,

$$L = C_l l = \frac{C_l}{\sqrt{\rho}}, \quad (7.31)$$

and closely related to the mean spacing D of impenetrable braids, that is, $L \approx D$. This gives, through eqs. (7.1), (7.2), (7.28), (7.30), and (7.31), the Stage II strain-hardening rate

$$\frac{d\hat{\tau}}{d\gamma} = \frac{\Theta_{II}}{\mu} = \frac{\alpha'^2}{2\beta} = 1.30 \times 10^{-2} \quad (7.32)$$

for

$$\alpha' = 0.45 \quad \text{and} \quad \beta = 7.8,$$

as stated earlier in connection with eqs. (7.1) and (7.2)

This hardening rate is roughly a factor of 2 too high from the experimentally reported hardening rates and indicates that the mean free path length is roughly a factor of 2–3 larger than the mean braid spacing observable in the micrographs. This conclusion is also in keeping with the slip line lengths in Stage II measured by Seeger *et al.* (1957), which for a flow stress of $\hat{\tau} = 40$ MPa give a half slip line length of 8.3×10^{-4} cm, while for the same stress twice the braid spacing is 5.5×10^{-4} cm from Fig. 7.13(c).

As the shear resistance of the primary plane increases as described above, slip activity also continues in a coupled manner on the secondary systems in response to the resolved shear stress on them and stimulated also by stress concentrations that build up in the primary system. The main effect of this activity is to continue to form sessile junction locks by reactions with primary dislocations that impede the latter at braids. However, the locks and the primary dislocations that continue to be

blocked there also impede activity on the secondary systems. Experiments show that while the densities of the secondary dislocations increase as fast as the stored primary density, their mean free paths of glide are very short and the consequent contribution they make to the overall system strain does not exceed 5–7% of the total (Basinski and Basinski 1979). This has been attributed to the fact that multiplication of secondary dislocations, very locally, is a response to relax the short-range stress concentrations produced by the primary dislocation groups. The net effect of the stifled activity on the secondary systems, resulting in a rapid increase of the glide resistance on them, is termed *latent hardening* and the ratio of the current resistance of the secondary system to that of the primary system is termed the *latent hardening ratio*. This ratio is around 2 for Cu and its dilute solid-solution alloys at a primary flow stress of 1 MPa at the start of Stage II, and continuously decreases slowly to about 1.25 toward the end of Stage II at a primary flow stress close to 80 MPa at room temperature (Basinski and Basinski 1979).

Returning to eq. (7.27), for $\alpha' = 0.45$ and $\alpha_i = 0.85$ the calculated ratio R is 0.49. Experiments by Cottrell and Stokes (1955) have established that during Stage II deformation the ratio of the temperature-sensitive component $\hat{\tau}_\rho$ to the insensitive component $\hat{\tau}_i$ remains a constant, that is, also R . Our estimate of 0.49 is close to the *Cottrell–Stokes ratio*, when extrapolated back to 0 K.

All the above considerations and estimates, while falling short of a comprehensive theoretical model, are in keeping with the assumption of similitude in the model, and represent the physical facts well.

7.3.5 Ingredients of Stage III Hardening

Dislocation Cell Formation and Long-range Internal Stresses As discussed in Section 7.2.4, toward the end of Stage II and persisting into Stages III and IV, a cellular dislocation microstructure forms when the braids of Stage II close up and are rearranged. Hughes and collaborators Hughes *et al.* 1997; Hughes and Hansen 2000, 2001 viewed these microstructures primarily as boundaries, namely *geometrically necessary boundaries* and *incidental dislocation boundaries*. As discussed in Section 7.2.4, the GNBs, which result from dispersal of intergranular constraints in polycrystals, will be considered here as of lesser importance than the IDBs' which are associated with a statistical process of aggregation of the dislocations evolving from the braids of Stage II. They are the only prevalent feature in single crystals in Stages III and IV. Their formation, their refinement in scale with increasing flow stress, and the gradually increasing lattice misorientations between them are all integrally linked to fluctuations in the primary dislocation fluxes that, in turn, result from interaction of the latter with the cell walls through which they must pass. How these processes were viewed collectively as *flow-noise-induced* transformations by Haehner and Zaiser (1999), resulting in a unique fractal distribution of cell sizes, was presented in Section 7.2.4. Here we shall be mainly interested in how cells are

associated with the buildup of long-range internal stresses that directly or indirectly relate to the strain-hardening rate in Stages III and IV of deformation, and how they relate to the dynamic recovery processes that directly affect the monotonic decrease of the strain-hardening rate from the Stage II levels and make it temperature-dependent.

The idea that the cellular dislocation microstructure can be viewed as a composite material with components of different hardness for the cell interiors and cell walls has been discussed by a number of authors (Prinz and Argon 1984; Nix *et al.* 1985; Haasen 1989; and, most prominently, Mughrabi *et al.* 1986 and Mughrabi and Ungar (2003)). It was Mughrabi *et al.* who very directly pointed out the role of the cellular microstructure in the development of long-range lattice elastic strains in distinction to what have often been referred to as long-range stresses, which very often have been related not to stressed but to impenetrable dislocation clusters or braids, and manifest themselves as resistances rather than adverse stresses. Nevertheless, the presence of long-range stresses, even in Stage II before the cellular dislocation microstructure has formed, has also been well established in a variety of ways, particularly by magnetization measurements in Ni by Kronmüller (1967). The source of these stresses is not clear, and is certainly different from the mechanism to be described below.

To clarify how a cellular microstructure with relatively lower shear resistance $\hat{\tau}_c$ inside cells with a very low dislocation content and with cell walls with a much higher dislocation content and a correspondingly higher shear resistance $\hat{\tau}_w$ deforms plastically in a compatible manner, we resort to a definitive form of elastic-inclusion analysis pioneered by Eshelby (1957) (Argon and Haasen 1993). For this purpose we consider, in a thought experiment, an initial reference state of the structure where neither the cell interior nor the walls are stressed when no external stress is applied. We consider a periodic pattern of cells and concentrate our attention on a generic equiaxed element of volume V having a fraction f of cell wall and $1 - f$ of cell interior, made up of an elastically homogeneous and isotropic material where, however, the cell interiors have a shear resistance $\hat{\tau}_c$ and the cell walls $\hat{\tau}_w$. For simplicity, we consider only a simple shear response in which eventually, under fully developed flow, both cell interiors and cell walls must shear. The present concern is to understand only the momentary shear flow and not the hardening behavior, which we deal with in the following sections. Since $\hat{\tau}_c < \hat{\tau}_w$, upon application of an external shear stress σ plastic deformation will start inside the cell. In the stepwise process of bringing the cell to flow, we first cut out the cell interior of volume $V(1 - f)$ and impose on it internally, outside the body, a forward (in the same direction as the eventually applied shear stress σ) transformation shear strain γ^T (a stress-free plastic strain). Then we shear this transformed material elastically back into its initial shape, and fit it perfectly into the hole from which it was cut out. Naturally, in the periodic setting, the same process is repeated simultaneously in all neighboring cells. To keep the elastically back-strained inclusion element

undistorted, a set of tractions are applied to the sheared inclusion. The tractions can be viewed as a shear stress $\sigma_c (= -\mu\gamma^T)$ applied to the inclusion. Under this initial condition, the elastic strain energy inside the whole volume element is present only in the inclusion of volume $V(1-f)$ and is

$$E_{ci} = \left(\gamma^T\right)^2 \frac{\mu V}{2} (1-f). \quad (7.33)$$

Next, the externally applied traction σ_c is slowly removed. The elastically back-strained cell block relaxes and sheds some of its energy to the previously unstressed cell wall material, still all by a set of elastic processes. In the final relaxed state, the elastic strain energy in the cell block has now been reduced to

$$E_{cf} = \gamma_c^2 \frac{\mu V}{2} (1-f), \quad (7.34a)$$

with the elastic strain energy imparted to the cell wall being

$$E_{wf} = \gamma_w^2 \frac{\mu V}{2} f, \quad (7.34b)$$

where γ_c and γ_w are the final elastic shear strains in the cell block and the cell wall, respectively, to be determined below, when the entire assembly is brought to plastic flow. Meanwhile, the removal of the constraining traction will amount to work done by a reverse shear stress $\sigma_a = -\gamma^T \mu (1-f)$ now applied to the cell block and cell wall together, which will elastically strain the cell block from an initial strain γ^T to a final strain γ_c and the cell wall from an initial strain of zero to a final strain γ_w . This work increment ΔW is derived from the partial unloading of the cell block, involving on the whole no exchange of work with external agencies. The work increment is

$$\Delta W = \Delta W_c + \Delta W_w = \sigma_a \left(\gamma_c - \gamma^T \right) \frac{V}{2} (1-f) + \sigma_a \frac{\gamma_w V f}{2}, \quad (7.35a)$$

$$\Delta W = -\frac{\gamma^T \mu V}{2} \left[\left(\gamma_c - \gamma^T \right) (1-f)^2 + f (1-f) \gamma_w \right]. \quad (7.35b)$$

Since the load shedding from the cell block to the cell wall is not accompanied by any external effect, the final energies and the work exchange must be equal to the initial elastic strain energy in the cell block, that is,

$$\begin{aligned} & \left(\frac{\mu V}{2} \right) \left\{ \gamma_c^2 (1-f) + \gamma_w^2 f - \gamma^T \left[\left(\gamma_c - \gamma^T \right) (1-f)^2 + f (1-f) \gamma_w \right] \right\} \\ &= \left(\frac{\mu V}{2} \right) \left(\gamma^T \right)^2 (1-f). \end{aligned} \quad (7.36)$$

Moreover, since there is no external stress applied in the final state, the average stress in the entire volume element must be zero, that is,

$$\gamma_c (1 - f) + \gamma_w f = 0. \quad (7.37)$$

Simultaneous solution of eqs. (7.36) and (7.72) gives

$$\gamma_c = \gamma^T f, \quad \gamma_w = -\gamma^T (1 - f). \quad (7.38a,b)$$

In this final state, the residual stress in the cell block and cell wall will be

$$\sigma_{\text{cres}} = -\mu \left| \gamma^T \right| f, \quad (7.39a)$$

$$\sigma_{\text{wres}} = \mu \left| \gamma^T \right| (1 - f), \quad (7.39b)$$

where $\left| \gamma^T \right|$ is the absolute magnitude of the transformation shear strain. The overall stress in the composite is zero.

Finally, to bring the assembly to plastic flow, an external shear stress σ is applied in the direction of the transformation shear strain previously imparted to the inclusion prior to its reinsertion in its elastically back-sheared form into the initial cavity. This final shear stress is to bring the cell interior to its flow stress $\hat{\tau}_c$ and the wall to its flow stress $\hat{\tau}_w$ simultaneously, by selecting the proper level of the transformation shear strain γ^T . Thus,

$$\sigma - \mu \left| \gamma^T \right| f = \hat{\tau}_c, \quad (7.40a)$$

$$\sigma + \mu \left| \gamma^T \right| (1 - f) = \hat{\tau}_w. \quad (7.40b)$$

This gives

$$\left| \gamma^T \right| = \frac{\hat{\tau}_w - \hat{\tau}_c}{\mu}, \quad (7.41)$$

and upon substitution of this value into either (7.40a) or (7.40b), the overall composite plastic resistance $\hat{\tau}_{\text{cm}} (= \sigma)$ is obtained in a pleasing rule-of-mixtures form,

$$\hat{\tau}_{\text{cm}} = \hat{\tau}_w f + \hat{\tau}_c (1 - f). \quad (7.42)$$

Then, in the unloaded state, long-range residual lattice elastic shear strains will be present in the cell interiors and the cell walls,

$$\gamma_{\text{cres}} = \frac{\sigma_{\text{cres}}}{\mu} = -\frac{(\hat{\tau}_w - \hat{\tau}_c) f}{\mu}, \quad (7.43a)$$

$$\gamma_{\text{wres}} = \frac{\sigma_{\text{wres}}}{\mu} = -\frac{(\hat{\tau}_w - \hat{\tau}_c) (1 - f)}{\mu}. \quad (7.43b)$$

If, at a finite temperature, the plastic shear resistance of the cell walls is attenuated by dynamic recovery inside them, then $\hat{\tau}_w(T) < \hat{\tau}_w$ will result, and the overall

composite plastic resistance $\hat{\tau}_{\text{cm}}(T) < \hat{\tau}_{\text{cm}}$ will be temperature-dependent, as will be the case in Stage III.

Such residual lattice elastic strains were first measured by Mughrabi *et al.* (1986) by an elegant X-ray diffraction method that showed clear displacement of diffraction peaks, as well as peak broadening, permitting separate measurement of the lattice elastic strains and, simultaneously, the dislocation content of both the cell interiors and the cell walls in Cu single crystals of [001] orientation, deformed in tension. The modeling presented there, while more complicated, is completely equivalent to that presented above. Clearly, the long-range lattice elastic strains produced in this manner will interact with the glide dislocations of the primary plane and will impede their motion, or on stress removal will result in some reverse deformation typical of the Bauschinger effect. Below, in discussing strain hardening in Stage IV, a different character of long-range internal stress will be presented which, even though it does not interact with glide dislocations, nevertheless directly affects the strain-hardening rate.

Strain Hardening and Recovery in a Cellular Dislocation Microstructure As discussed in some detail in Section 7.2.4, the dislocation microstructure of Stages III and IV must be viewed as a set of topologically continuous, fully interconnected cell walls of high dislocation density, encapsulating cell interiors of such a low dislocation density that it can be neglected in comparison with that in the cell walls. This view is different from many previous idealizations and is at first sight not supported by a cursory observation of microstructures viewed without the benefit of stereo imaging. If indeed the assumed picture is correct, then in a cellular microstructure of micron-size cells, routine viewing of TEM foils of circa $0.1\mu\text{m}$ thickness should result in a significant probability of capturing bridging boundaries parallel to the plane of the foil. Correspondingly, a significant fraction of cells should appear to contain no dislocations. In instances when stereo pairs of cell wall micrographs have been presented, as by Basinski and Basinski (1979), not only has this view been reinforced, but also many rather thick and fuzzy-appearing cell walls could actually be recognized to be considerably thinner when their inclination with respect to the plane of the foil could be visualized. This idealization of the cellular microstructure remains intact throughout Stages III and IV and, if anything, becomes more accentuated. It is meant to apply to only those cases where the lattice resistance to dislocation motion is negligible, that is, either to pure FCC metals or to BCC metals (Lan *et al.* 1991a,b) and diamond cubic structures (Brion and Haasen 1985; Haasen 1989; Siethoff *et al.* 1984, 1986) at elevated temperatures where the lattice resistance becomes negligible, as was discussed in Chapter 4.

We propose that cell interiors resist shear by the development of a back stress as discussed under “Dislocation Cell Formation and Long-range Internal Stresses” above and not because of a resistance associated with an operable dislocation density. On the other hand, the shear resistance of the cell walls is clearly a result of the

high dislocation density in them. This decomposition of the material has immediate important consequences. First, we note that the cell interiors can only harden by a special free-energy-storing mechanism that we present below. There can also be no rate-controlling process of deformation inside the cells, barring a ubiquitous phonon drag, since there are no thermally penetrable obstacles.

In our model, all rate-controlling processes for the flow stress, such as forest cutting and all evolutionary processes of hardening and recovery, reside in the cell walls—including those that result in the maintenance of the long-range stresses in the cell interiors, as was pointed out above. The scenario developed here was outlined earlier by Argon and Haasen (1993).

7.3.6 Components of Strain Hardening in Stage III

Strain Hardening in Cell Walls Comparison of slip line lengths Λ and cell sizes D indicates that the strain-producing primary dislocation flux ϕ ($\text{m}/\text{m}^2\text{s}$) traverses both cell interiors and cell walls during fully developed plastic flow but undergoes a filtering process in the cell walls. For some applications we divide this bidirectional flux of cross-flowing mobile dislocations into two unidirectional fluxes $\phi/2$ flowing through each other.

When the bidirectional flux traverses the cell interiors it will suffer no important attenuation. The residence time of glide dislocations inside the cell is too short to result in any meaningful encounters with other dislocations. On the other hand, when the flux traverses cell walls a certain fraction c ($= D/\Lambda$) will be extracted and stored in one of several different identifiable components of the cell wall dislocation density and will form dislocation junctions. Most often, the storage will be in the form of redundant dislocations held up by the various cell wall obstacles. Considering that the boundary will adjust its thickness relatively slowly, it can be considered constant for small incremental processes. The rate of increase of the cell wall dislocation density is then

$$\frac{d\rho_w^+}{dt} = \frac{2c\phi}{t_w} \quad (7.44)$$

and, since $\phi = \dot{\gamma}/b$,

$$\frac{d\rho_w^+}{d\gamma} = \frac{c}{bt_w}, \quad (7.45)$$

where t_w is the cell wall thickness, which is considered to continue to be a fraction f of the cell diameter D .

Moreover, since the cell wall plastic resistance τ_w is expected to be related to the cell wall dislocation density by eq. (7.2),

$$\tau_w = \alpha \mu b \sqrt{\rho_w}, \quad (7.46)$$

$$\frac{d\tau_w^+}{d\gamma} = \frac{d\tau_w}{d\rho_w} \frac{d\rho_w^+}{d\gamma} = \frac{\alpha^2 \mu^2 b c}{2t_w \tau_w}. \quad (7.47)$$

For a columnar cell model, we then obtain

$$\frac{d\tau_w}{d\gamma} = \frac{\alpha^2 \mu^2 b c}{\tau_w f D}. \quad (7.48)$$

Here it is assumed that τ_w will gradually decrease with recovery-related processes, which we discuss below, in step with a decrease in D , maintaining a proportionality relation to the cell wall plastic resistance as in eq. (7.1).

Strain Hardening in Cell Interiors To understand the strain-hardening process inside cells, we start by noting that the lattice misorientation across cell walls increases slowly with increasing plastic strain, as already stated in Section 7.2.4. This can be explained by noting that for a multitude of reasons, there are likely to be fluctuations in the unidirectional dislocation fluxes, resulting in different dislocation capture rates by a cell wall from the two cross-flowing flux components. The excess of one component over the other will gradually and randomly build up a lattice misorientation θ across the walls that begin with the inception of deformation. The lattice misorientation is associated with a geometrically necessary wall dislocation content.

Consider first a set of such neighboring dislocation layers in cell walls producing a random change of orientation θ , with the cell walls normal to the principal glide system, as shown in Fig. 7.27(a), and consider an elementary encounter of one of these cell boundaries with two opposite glide dislocations of the main flux. As shown in Fig. 7.27(b), because the planes of the two approaching dislocations make an angle θ with each other, when they meet at the boundary, they will not exactly cancel, but will leave behind a fractional sessile dislocation with Burgers vector θb in an elastically unstrained background. The rate of deposition of such dislocations producing material misfit at the boundary will be

$$\frac{dN}{dt} = \frac{d}{dt} \left(\frac{1}{h} \right) = \phi, \quad (7.49)$$

that is, equal to the dislocation flux. Thus, if the lattice were to remain stress-free, this effect would lay down misfit dislocations of Burgers vector θb at vertical spacings h per unit time, equal to the dislocation flux, and would result in alternating layers of sessile misfit dislocations in the neighboring boundaries which would

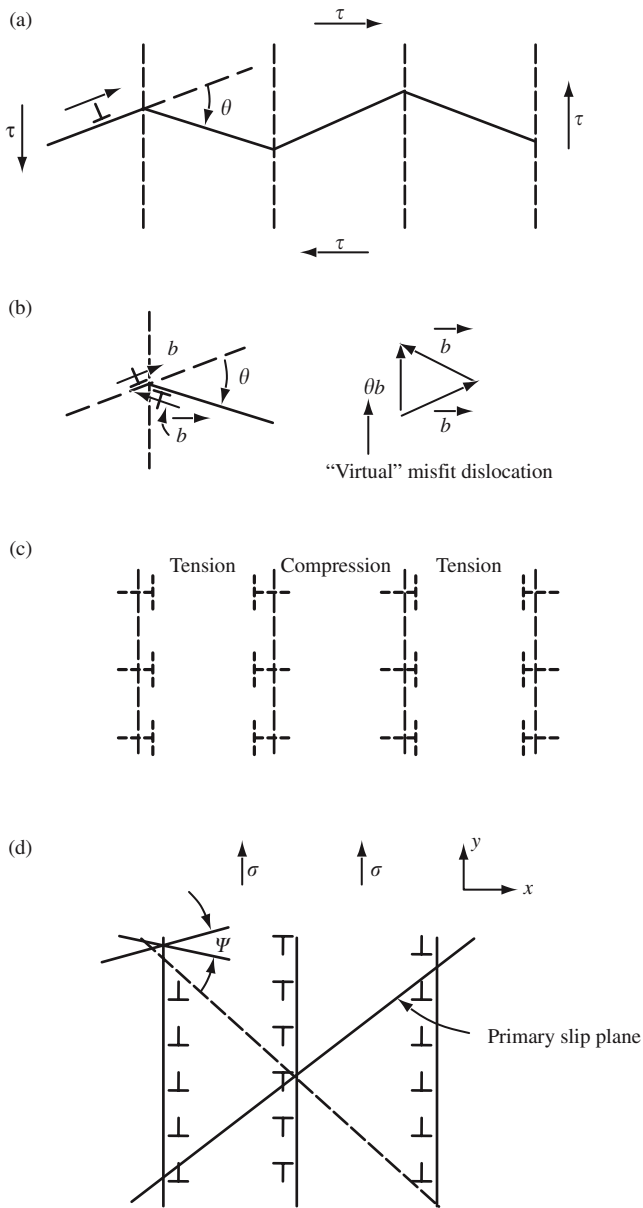


FIG. 7.27. Production of a large misfit stress when mobile dislocations cut through cell walls with a lattice misorientation angle θ in a simple shear model: (a) cell walls and glide planes; (b) virtual-dislocation production upon cutting; (c) alternating tensile and compressive misfit stresses if no virtual dislocations are left behind; (d) a more realistic tensile-extension mode of straining with one active glide system (solid line) (Argon and Haasen 1993).

make randomly alternating elongated or shortened columns of cell material. If, on the other hand, the neighboring columns were to remain coherent, they would be subjected to randomly alternating tension or compression rather than the buildup of dislocation walls, as depicted in Fig. 7.27(c). Then the time rate of buildup of lattice elastic strains ϵ_c^e in the alternating columnar cells would be

$$\left| \frac{d\epsilon_c^e}{dt} \right| = \theta b \frac{d}{dt} \left(\frac{1}{h} \right) = b\theta\phi \quad (7.50)$$

or, in terms of the plastic shear strain γ ,

$$\left| \frac{d\epsilon_c^e}{d\gamma} \right| = \frac{1}{\dot{\gamma}t} \left| \frac{d\epsilon_c^e}{dt} \right| = \theta. \quad (7.51)$$

The stored elastic strain energy per unit volume due to these lattice elastic strains is

$$\Delta F = \frac{E(\epsilon_c^e)^2}{2} \quad (7.52)$$

in this simple shear model of the primary system.

The eventual comparison of the predictions of the model with experimental results necessitates consideration of a more realistic geometrical framework, shown in Fig. 7.27(d), where the cell walls are now considered to be parallel to a direction of principal tensile extension and the principal glide plane is taken to be at a 45° inclination with respect to the principal tension direction. We still consider, however, a single primary glide system rather than a multiplicity of systems arranged symmetrically with respect to each other. This assumption is realistic, since much evidence indicates that even if a multiplicity of slip systems are present in a high-symmetry orientation, usually one slip system dominates locally (Basinski and Basinski 1979). In this new framework, the effect of the cutting of the geometrically necessary cell wall dislocations, producing a lattice misorientation θ with respect to the glide dislocations of the slanted primary slip system, is still the production of a normal stress acting across the primary slip plane. The resulting lattice elastic strain in the coordinates of the slanted slip system is still given by eq. (7.51), but because of the constraint of the surroundings, other stresses and strains also appear, which for a plane strain model result in a slightly modified expression for the elastic strain energy of misfit (Argon and Haasen 1993),

$$\Delta F = \frac{\mu(\epsilon_c^e)^2}{1 - \nu}. \quad (7.53)$$

Experiments (Rollett 1988; Hughes and Hansen 2000, 2001) show that the lattice misorientation angle increases with strain. Since the lattice misorientation arises from random fluctuations in the components of the bidirectional dislocation

flux as it traverses the cell walls, the increase of the lattice misorientation angle θ should depend on the square root of the plastic shear strain γ in a manner akin to a random error accumulation process (Argon and Haasen 1993; Pantleon 1996; Pantleon and Hansen 2001):

$$\theta = B\gamma^{1/2}, \quad (7.54)$$

where B is a proportionality constant of order 1.75×10^{-2} (Rollett 1988; Hughes 2002). This gives a specific dependence of the absolute value of the lattice elastic strains ϵ_c^e ,

$$|\epsilon_c^e| = \frac{2}{3}b\gamma^{3/2}, \quad (7.55)$$

obtained as an integral of an equation of the form of eq. (7.51), and a more precise description of the elastic strain energy of misfit,

$$\Delta F = \frac{4}{9} \frac{B^2 \gamma^3 \mu}{(1 - \nu)}. \quad (7.56)$$

We note that this elastic strain energy is stored as a result of shear deformation, but not in a mode that can directly retard the shear rate in the primary system by a back stress. The tensile and compressive stresses in the columnar cells as sketched out in Fig. 7.27(c) have no shear components in the primary slip system. In a real material where processes are less ideal, some random positive and negative shear stress components could also appear on the active slip plane, which would retard mobile dislocations further by the well-known consequences of internal stress variations, considered in detail by Li (1968). We shall neglect this effect. Regardless of the mode mismatch in the stress, since the buildup of the elastic strain energy is kinematically coupled to the shear strain in the primary system, an organized shear resistance will result from this rate of storage of free energy, which we attribute to the cell interiors. This unidirectional resistance τ_c can be defined as

$$\tau_c = \frac{\partial \Delta F}{\partial \gamma} = \frac{4}{3} \frac{B^2 \gamma^2 \mu}{(1 - \nu)}. \quad (7.57)$$

The resulting hardening rate attributable to the cell interiors is then

$$\frac{d\tau_c}{d\gamma} = \frac{8}{3} \frac{B^2 \gamma \mu}{(1 - \nu)}. \quad (7.58)$$

This athermal shear resistance τ_c of the cells is quite considerable. What is, however, more revealing is that the kinematically linked normal stresses σ_c acting across

the primary slip system are very substantial (Argon and Haasen 1993) and are of the order of

$$\sigma_c = \frac{2\mu|\epsilon_c^e|}{(1 - \nu)}. \quad (7.59)$$

They are of randomly positive or negative sign in different cells. While the monotonic increase of lattice misorientation with strain can continue unabated, the elastic strain ϵ_c^e in cells cannot increase indefinitely. We expect that when these normal strains reach a critical value they will begin to trigger stress relaxation events of a threshold nature to limit their buildup, by nucleating accommodating forms of slip inside the cell columns. The dislocations arising from such critical slip activity inside the cells would be deposited on the cell walls, where they would act as misfit relief dislocations. Such stress relaxation should be nucleation-controlled and should have a rate dependence different from that of cross-slip or climb. This will usher in Stage V of steady-state flow with negligible strain hardening (Argon and Haasen 1993).

7.3.7 Recovery Processes in Stage III

Resistance versus Internal Stress We distinguish between processes of recovery in which there is a reduction in deformation resistance attributable to a reduction in dislocation density, and processes of stress relaxation associated with dislocation nucleation, triggered by large internal stresses that reduce those stresses. The first type is associated with cell walls and constitutes dynamic recovery, while the second type is associated with cell interiors and is stimulated by the stresses σ_c of the cell interiors given by eq. (7.59).

Dynamic Recovery in Stage III In stage III, since the hardening of the cell interior results from the development of intercellular misfit as described above and the dislocation content is negligible in the cells, no important recovery process is attributed to the cell interiors. Thus, the important process of dynamic recovery associated with Stage III that results in systematic and monotonic reduction of the strain-hardening rate from the levels in Stage II is attributed entirely to processes occurring in cell walls.

Perhaps the most controversial aspect of the cellular patterning is this mechanism of the dynamic recovery associated with it. Ever since the proposition of Seeger (1957) that dynamic recovery is controlled by massive cross-slip of large groups of screw dislocations past long linear LC barriers, it has been generally accepted that dynamic recovery is governed by cross-slip. There is little firm evidence for this proposition, which in various modifications has persisted up to the present (see, for example, Zaiser and Haehner, 1998). Indirect support for the proposition is taken to be provided by the temperature dependence of τ_{III} , where

a departure from Stage II linear hardening is observed. This suggests a correlation of τ_{III} with the stacking-fault energy (Mitchell 1964), which might follow from a constriction model (Schoeck and Seeger 1955) of the stacking-fault ribbon between partials during cross-slip. However, other facts are generally not supportive. First, it is well known that widespread cross-slip is already prevalent as early as in Stage I, where screw dislocations are largely absent, presumably owing to their cross-slip-controlled pairwise annihilation (Mader *et al.* 1963). There is scant TEM evidence for significant group-like screw dislocation motion out of the primary plane in any stage of deformation. Thus, it is far more likely that dynamic recovery is not a consequence of thermally assisted *overcoming* of slip obstacles by cross slip but, rather, is due to thermally assisted *removal* of LC segments pinning down cell walls. The mechanism of this is suggested to be a variant of the static thermal-recovery mechanism studied in some detail by Prinz *et al.* (1982), where recovery was observed by TEM to consist of a progression of core-diffusion-controlled elimination of pinning points (LC junction segments), after each of which very substantial cancellation of surrounding redundant dislocation line length by spontaneous glide was observed. Thus, in dynamic recovery, a similar set of processes are envisioned to occur in a thermally assisted manner under the applied stress during flow, without any diffusion, where the dislocations of the primary flux assist in the decomposition of LC junctions in the cell walls, with possible help from the substantial flow noise resulting from repeated arrest and release of the flux by the wall network. We now consider this process, which was outlined earlier (Argon 2002).

We consider a planar cell wall of thickness t_w normal to the main dislocation flux, where $t_w = fD$, with D being the mean cell size. The dislocation density of the wall is ρ_w , which is largely of a redundant nature, containing only a minor component of a geometrically necessary set responsible for the developing lattice misorientation θ between cells. The redundant density is prevented from spontaneous cancellation by a substantial concentration of attractive network junctions, discussed in Section 7.3.2. Of these attractive junctions, a significant fraction is of Lomer–Cottrell type, which are the most resistant to decomposition. They are accumulated during Stage II monotonically in direct proportion to the plastic strain, by reactions between primary dislocations and a variety of secondary dislocations. While these junctions have a high resistance to decomposition, they are not impenetrable obstacles. As indicated in Table 7.1, on average, their normalized strength $\hat{K}_{LC}/2\mathcal{E}$ is only 0.44. Nevertheless, during Stage II, the applied stress is insufficient to decompose them. At a given temperature T of deformation, however, above a stress τ_{III} , while more LC junctions continue to form as before, more and more of the accumulated LC junctions are also beginning to be decomposed by the impinging primary glide dislocations, with some thermal assistance. This results in a systematically decreasing strain-hardening rate and, in the process, permits the dislocation structure to transform from the open-braid type of Stage II shown in Figs. 7.13(a), and (b) to the closed cellular form shown in Fig. 7.21(a).

Evidently this transformation is aided by flow noise, as envisioned by Haehner and Zaiser (1999). An accurate mechanistic description of this recovery process remains difficult. However, it is possible to construct a satisfactory scenario in which the dislocations of the primary flux, resisted inside the cells only by phonon drag, impinge on the cell wall network, not only driven by the applied stress but also possessing an increasingly substantial component of kinetic energy. While the latter effect must reverberate through the segments of the wall network and aid the decomposition of LC segments, much like random thermal fluctuations, this effect is difficult to quantify and will not be directly considered. In the recovery scenario it is assumed only that the impinging dislocations repeatedly strike and pluck the network segments and have a certain probability of striking near an LC junction and forcing it to decompose, much like a forest-dislocation-cutting process.

Considering the cell wall dislocation network to be idealized as a cubical one with segments of length l_w , tied together by junctions at corners, the density N_j of such network junctions per unit volume of cell wall should be approximately

$$N_j = \frac{1}{l_w^3} = \rho_w^{3/2}. \quad (7.60)$$

According to Baird and Gale (1965), the fraction of attractive junctions in the total should be approximately $6/13 = 0.46$, of which perhaps $2/3$ are expected to be of LC type giving the volume concentration N_{LC} of such junctions to be approximately

$$N_{LC} = 0.3(\rho_w)^{3/2}. \quad (7.61)$$

The density of LC junctions projected onto the cell wall should then be roughly $N_{LC} \cdot t_w$, giving the mean distance l_{LC} between them in the the cell wall as roughly

$$l_{LC} = (N_{LC} \cdot t_w)^{-1/2} = ((0.31)^{1/2} \rho_w^{3/4} t_w^{1/2})^{-1}. \quad (7.62)$$

When a mobile dislocation of the primary flux impinges on a cell wall and decomposes an LC junction, the dislocation density in a volume element that is a small multiple of l_{LC}^3 will be cancelled by “glide collapse” of the redundant cell wall density. Thus, the fraction of the total line length in the through-thickness wall volume $l_{LC}^2 t_w$ that would be cancelled prescribes the fractional recovery process per successful impingement as

$$\frac{d\rho}{\rho} = \left(\frac{d\rho_w}{\rho_w} \right)^- = - \left(\frac{l_{LC}^3}{l_{LC}^2 t_w} \right) = - \frac{l_{LC}}{t_w}. \quad (7.63)$$

Such a cancellation step should occur for every time increment Δt , giving the cell wall dislocation density reduction rate

$$\left(\frac{d\rho_w}{dt} \right)^- = -\rho_w \frac{l_{LC}}{t_w \Delta t}. \quad (7.64)$$

In steady-state flow, Δt must be equal to the mean activation time τ_a of thermally assisted decomposition of an LC junction by an impinging mobile dislocation at an activation rate of

$$\frac{1}{t_a} = \nu_G \exp \left\{ -\frac{\Delta \mathfrak{F}_{LC}}{kT} \left(1 - \left(\frac{\sigma_e}{\hat{\tau}_w} \right)^{2/3} \right)^2 \right\} \quad (7.65)$$

where $\Delta \mathfrak{F}_{LC} = \mathfrak{F}_C w_{LC}$ is the binding energy of an LC junction to its lattice site, and where $\mathfrak{F}_C$ is the line energy reduction per unit length of a Cottrell reaction given by eq. (2.68) relative to two associated edge dislocations that enter the reaction, and w_{LC} is the average length of the junction segment shown in Fig. 7.22 as P_0Q_0 , and σ_e is the effective component of the applied stress. The stress-dependent form of the activation energy in eq. (7.65) results from taking the force–distance curve of junction decomposition as a linear expression similar to the nearly linear form determined experimentally for forest dislocation cutting (Fig. 7.24); $\hat{\tau}_w$ is the athermal threshold resistance of the cell wall, assumed to be related to the wall dislocation density ρ_w by eq. (7.2). Evaluation of the expression in eq. (2.68) for Cu, with a stacking-fault energy of $\chi_{SF} = 45 \text{ mJ/m}^2$ and other appropriate elastic constants (Hirth and Lothe 1982) gives the binding energy per unit length of the Cottrell lock

$$\mathfrak{F}_C = -0.767 \mathfrak{F}_e \quad (7.66)$$

in terms of the line energy $\mathfrak{F}_e$ of a dissociated edge dislocation. When we relate the cell wall thickness to the average cell diameter and then to the plastic resistance through eq. (7.1), eq. (7.64) gives, finally, the dynamic recovery rate of the cell wall resistance as

$$\frac{d(\tau_w/\mu)^-}{d\gamma} = -1.3 \left(\frac{\tau_w}{\mu} \right) \frac{\nu_k}{\dot{\gamma}} \exp \left\{ -\frac{\Delta \mathfrak{F}_{LC}}{kT} \left(1 - \left(\frac{\sigma_e}{\hat{\tau}_w} \right)^{2/3} \right)^2 \right\}. \quad (7.67)$$

Since no recovery takes place inside the cells, this constitutes the entire recovery rate in Stage III. This results in a monotonic reduction of the Stage III strain-hardening rate with increasing plastic resistance τ_w .

Before proceeding further to obtain the overall strain-hardening rate in Stage III and its temperature dependence, we use eq. (7.67) to probe the temperature dependence of τ_{III} , the plastic shear resistance where a departure from the linear strain-hardening rate of Stage II is noted. Experiments show that this relation is of the form (Mitchell 1964).

$$\frac{\mu(0)}{\mu(T)} \frac{\tau_{III}(T)}{\tau_{III}(0)} = \exp \left(-\frac{T}{T_0} \right), \quad (7.68)$$

where $\tau_{\text{III}}(0)$ is the back-extrapolated value at $T = 0$. Considering eq. (7.66) for Cu, we define a characteristic temperature

$$T_0 \equiv \frac{\Delta \mathfrak{F}_{\text{LC}}}{k} = \frac{\mathfrak{F}_{\text{c}} w_{\text{LC}}}{k} \approx 5 \times 10^4 \text{ K} \quad (7.69)$$

taking for w_{LC} a value of about $2b$. Thus, if the detectability threshold of a departure from linear hardening is taken as a recovery fraction ϵ , defined as

$$\epsilon = \frac{1}{(\tau_{\text{w}}/\mu)} \left(\frac{d(\tau_{\text{w}}/\mu)}{d\gamma} \right)^{-}, \quad (7.70)$$

we have from eq. (7.67)

$$\Pi = \ln \left(\frac{(1.3) v_{\text{G}}}{\epsilon \dot{\gamma}} \right) = \frac{T_0}{T} \left(1 - \left(\frac{\tau_{\text{III}}/\mu}{\tau_{\text{III}}(0)/\mu} \right)^{2/3} \right)^2, \quad (7.71)$$

where $\sigma_{\text{c}}/\hat{\tau}_{\text{w}}$ has been interpreted, as a special case, as the ratio of $\tau_{\text{III}}(T)$ to $\tau_{\text{III}}(0)$, normalized appropriately by the shear modulus. Using the information given by Mitchell (1964), which states that $\tau_{\text{III}}(0)/\mu = 3 \times 10^{-3}$ and $\tau_{\text{III}}/\mu = 2.6 \times 10^{-4}$ for Cu at $T = 500 \text{ K}$, eq. (7.71) can be used to evaluate the factor Π as 37.35, which for $v_{\text{G}} = O(10^{10}) \text{ s}^{-1}$ and $\dot{\gamma} = 10^{-4} \text{ s}^{-1}$ gives $\epsilon = 8 \times 10^{-3}$, which represents a very reasonable level of a detectability threshold for an observable departure from the constant Stage II strain-hardening rate. With this, the entire temperature dependence of τ_{III}/μ for Cu between 0 K and 500 K can be determined and is plotted in Fig. 7.28 as the solid curve, together with the experimentally determined dependence given by Mitchell (1964), represented by the broken line. The fit is quite good; any small departure of the model relation from the actual data can be attributed to a departure of the actual force-distance relation from the straight-line form assumed for junction decomposition.

7.3.8 Total Strain-hardening Rate in Stage III

The overall strain-hardening rate in Stage III for stresses larger than τ_{III} is, for eq. (7.42),

$$\frac{\Theta_{\text{III}}}{\mu} = f \left\{ \left(\frac{d(\tau_{\text{w}}/\mu)}{d\gamma} \right)^{+} + \left(\frac{d(\tau_{\text{w}}/\mu)}{d\gamma} \right)^{-} \right\} + (1-f) \left(\frac{d(\tau_{\text{c}}/\mu)}{d\gamma} \right)^{+}, \quad (7.72)$$

where $f = t_{\text{w}}/D$ is taken as the area fraction of the cell wall component under the condition that $f \ll 1.0$. Since in much of Stage III, until the transition to Stage IV is reached, the strain-hardening rate of the cell interiors, represented by the last term in eq. (7.72), is an order of magnitude smaller than the cell wall contribution, the

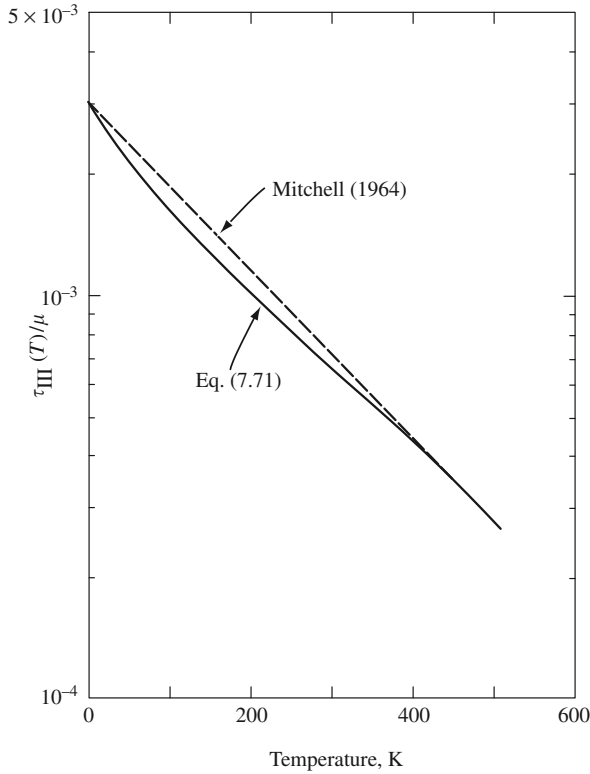


FIG. 7.28. Comparison of the model result for the dependence of the threshold stress τ_{III} on the absolute temperature given by eq. (7.1) with the actual measurements of Mitchell (1964).

overall strain-hardening rate becomes

$$\frac{\Theta_{III}}{\mu} = \frac{\alpha^2 c}{2\beta} - (1.3)f\left(\frac{\tau_w}{\mu}\right) \frac{\nu_G}{\dot{\gamma}} \exp \left\{ -\frac{T_0}{T} \left(1 - \left(\frac{\sigma_e}{\hat{\tau}_w} \right)^{2/3} \right)^2 \right\}, \quad (7.73)$$

where σ_e is the effective shear stress applied on the primary system, and in the first term on the right-hand side we have made use of eq. (7.1).

While the temperature dependence of the strain-hardening rate is consistent with the dynamic recovery rate, the relation of eq. (7.67) is awkward to evaluate, since little information is available on the ratio $\sigma_e/\hat{\tau}_w$ for Stage III, where the convenient Cottrell–Stokes condition of proportionality between the rate-dependent and rate-independent components no longer holds. Therefore, to explore this dependence, a more global empirical condition will be introduced. In eq. (7.73), the exponential factor is noted to be nearly identical to the factor giving the temperature and stress dependence of the shear strain rate, where, however, the reference temperature T_0

is no longer related to the energy to decompose LC junctions but instead is related to the energy to overcome an overall average junction strength, which, according to Baird and Gale (1965), will be roughly 15% lower, as indicated in Table 7.1. Thus, within a constant factor A , which can be treated as adjustable, the exponential factor in eq. (7.73) can be replaced with $\dot{\gamma}(\sigma_e, T)/\dot{\gamma}_0$, where the numerator represents the stress and temperature dependence of the shear strain rate, to be distinguished from the $\dot{\gamma}$ appearing in eq. (7.73), which represents merely the imposed strain rate in the experiment. This permits the evaluation of the temperature dependence of the reduction of the normalized strain-hardening rate Θ_{III}/μ with increasing τ/μ that is shown in Fig. 7.6 as the principal manifestation of dynamic recovery in Stage III.

A direct differentiation of eq. (7.73) gives

$$\frac{d(\Theta_{III}/\mu)}{d(\tau/\mu)} \cong -(1.3) \frac{f A v_G}{\dot{\gamma}_0} \left(\frac{\tau}{\mu} \left(\frac{\partial \ln \dot{\gamma}}{\partial (\tau/\mu)} \right) \right). \quad (7.74)$$

As stated earlier, the factor $A v_G/\dot{\gamma}_0$ can be treated as adjustable. Since specific information on the factor $(\ln \dot{\gamma})/\partial(\tau/\mu)$ is given in Fig. 7.7 for some of the information represented in Fig. 7.6, the systematic decrease of the slope of the normalized strain-hardening rate with increasing normalized overall plastic resistance can be evaluated for the temperatures 293, 373, and 473 K. When these three sets of information are integrated starting from a strain-hardening rate of Θ_{II}/μ at the end of Stage II as a function of τ/μ , the form of the decrease of Θ_{III}/μ with increasing τ/μ can be determined readily. The results are plotted in Fig. 7.29 and are compared with the three experimental curves of Fig. 7.6. The agreement is very good down to very low levels of hardening where the hardening contribution of the cell interiors begins to become observable, as we discuss below.

We note that in the above analysis of dynamic recovery, the stacking-fault energy has not entered explicitly. It does enter indirectly through the dependence of the binding energy of the Cottrell dislocation to its lattice site given by eq. (7.33).

7.3.9 Strain Hardening in Stage IV

As the model expression in eq. (7.73) indicated and as the experimental results of Fig. 7.6 show, continued dynamic recovery with increasing plastic shear resistance results in an asymptotic decrease of the net strain-hardening rate of Stage III to a very low level, in the range of $2 - 3 \times 10^{-4} \mu$ in general (for crystalline Cu, about 5×10^{-4} (Alberdi 1984)). A linear extrapolation of the hardening curves leads to an apparent zero hardening rate at a plastic shear resistance τ_{III_s} . We interpret this as a consequence of a saturation of the plastic resistance of the cell walls at a limiting dislocation density ρ_{ws} where spontaneous cancellation of neighboring edge dislocations in parallel glide planes of very close spacing becomes prevalent, as postulated by Essmann and Mughrabi (1979). However, as the strain hardening

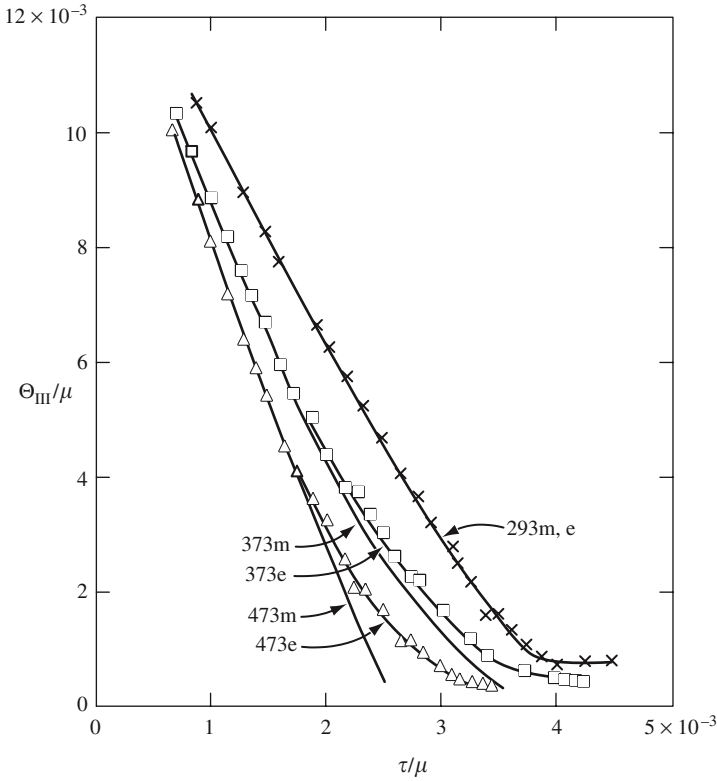


FIG. 7.29. Comparison of the temperature-dependent decrease of the normalized strain-hardening rate Θ_{III}/μ with increasing τ/μ predicted by the dynamic-recovery model, with three of the experimental results reported by Alberdi (1984) shown in Fig. 7.6, for which information on $\alpha \ln \dot{\gamma}/\partial(\tau/\mu)$ was available.

of the cell walls approaches saturation with increasing τ/μ , the rate-independent hardening process of the cell interiors discussed above in Section 7.3.6 that has been continuing all along in Stage III now comes to fore.

Thus, we attribute the Stage IV hardening, with its relative independence from the temperature and strain rate, to the continued evolution of the internal stresses in the cell interiors in the range where the alternating positive and negative normal stresses σ_c in the cell columns remain below a critical threshold σ_{cth} so that stress relaxation inside the cells is largely absent, that is,

$$\Theta_{IV} = (1 - f) \frac{d\tau_c}{d\gamma}, \quad (7.75)$$

Then, through eq. (7.58), the Stage IV strain-hardening rate is

$$\frac{1}{\mu} \Theta_{IV} = \frac{8}{3} \frac{(1-f)}{(1-\nu)} B^2 \gamma. \quad (7.76)$$

On the other hand, the composite plastic resistance given by eq. (7.42) rises much less rapidly beyond $\tau = \tau_{III_s}$, where the Θ versus τ curve extrapolates to zero hardening. When τ_w reaches a limiting saturation resistance τ_{ws} in the cell walls, beyond this point the composite resistance becomes simply

$$\tau = f \tau_{ws} + (1-f) \tau_c \quad (\tau \geq \tau_{III_s}), \quad (7.77)$$

where, of course, τ_c continues to be given by eq. (7.57). To compare our theory with experimental results, we evaluate the Stage IV hardening rate of eq. (7.76) and the composite plastic resistance of eq. (7.77) at around $\tau = \tau_{III_s}$, where Stage III ends and Stage IV begins.

At the beginning of Stage IV, where $\gamma \approx 1.0$, and $\theta = 1.75 \times 10^{-2} = B$ (that is, about 1.0° based on the misorientation measurements of Rollett (1988) on pure Al), for $\nu = 0.3$ and $f = 0.1$ we have from eqs. (7.57) and (7.76)

$$\frac{\tau_c}{\mu} = 5.8 \times 10^{-4}, \quad (7.78a)$$

$$\frac{1}{\mu} \Theta_{IV} = 1.05 \times 10^{-3}. \quad (7.78b)$$

Moreover, taking for $\tau_{III_s} = 4 \times 10^{-3} \mu$ the value from Fig. 7.6 for Cu at room temperature, we calculate from eq. (7.77) an estimate of the terminal saturation resistance τ_{ws} of the cell walls,

$$\frac{\tau_{ws}}{\mu} = 3.48 \times 10^{-2}, \quad (7.78c)$$

and an associated saturation wall dislocation density ρ_{ws} ,

$$\rho_{ws} = 2.15 \times 10^{17} \text{m}^{-2}, \quad (7.78d)$$

which, as mentioned above, is now at the level of spontaneous dislocation cancellation determined by Essmann and Mughrabi (1979) from much experimental evidence.

This Stage IV hardening rate is somewhat high in comparison with published values in the range of $1 - 5 \times 10^{-3}$. Clearly, the processes of free-energy storage by misfit-induced long-range stress production are less efficient in practice than is given in eqs. (7.53)–(7.58).

Returning to the expression for the Stage IV strain-hardening rate in eq. (7.76), we note that it increases somewhat with increasing shear strain. By using eq. (7.42)

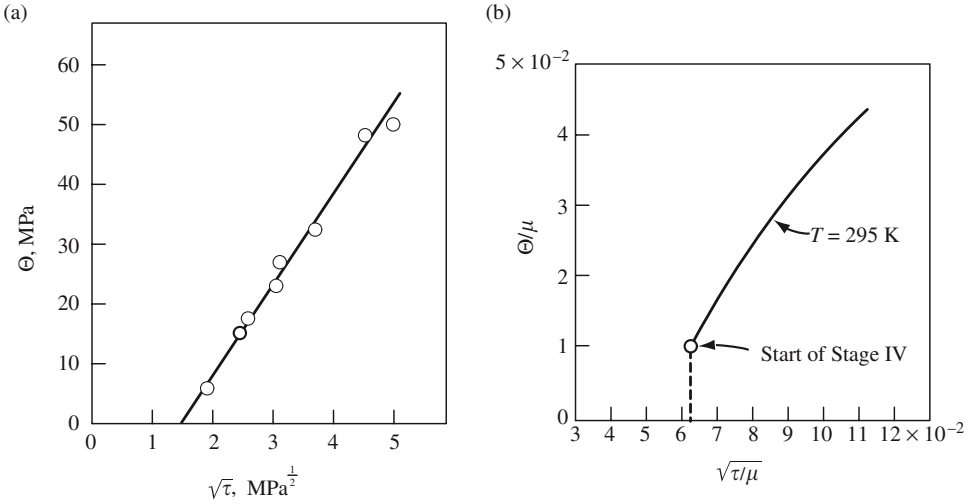


FIG. 7.30. (a) Dependence of the slight increase of the Stage IV strain-hardening rate with the square root of the plastic shear resistance τ as measured by Anongba *et al.* (1988); (b) the same result derived by the Stage IV hardening model given by eqs. (7.79) and (7.80).

for the cell shear resistance and eq. (7.76) for the overall shear resistance, it can be presented in the more interesting form of

$$\frac{\Theta}{\mu} = \frac{4\sqrt{3}}{3} B \sqrt{\frac{1-f}{1-\nu}} g\left(\frac{\tau_{ws}}{\tau}\right) \sqrt{\frac{\tau}{\mu}}, \quad (7.79)$$

where

$$g\left(\frac{\tau_{ws}}{\tau}\right) = \sqrt{1 - f\left(\frac{\tau_{ws}}{\tau}\right)} \quad (7.80)$$

is a slowly varying function of τ ranging from 0.36 at the beginning of Stage IV to somewhat higher values with increasing plastic resistance, for the values of B, f , and τ_{ws} for Cu given above. This expression is of the type shown in Fig. 7.30(a). Some experimental results are plotted in Fig. 7.30(a) for Cu. The results of the model are plotted in Fig. 7.30(b) for $\tau > \tau_{III}$, for Cu at room temperature. The parallelism to the form shown in Fig. 7.30(a) is good, albeit in a range of much higher flow stresses than the high-temperature values given in Fig. 7.30(a).

For more extensive comparisons of the Stage IV hardening processes with experiments, the reader is referred to Argon and Haasen (1993).

7.3.10 Stage V Deformation with No Strain Hardening

As stated above, in the hardening of the cell interiors through the monotonic increase in misfit-related lattice elastic strains, the normal stresses σ_c across the primary system given by eq. (7.59) will continually increase and must eventually reach unsustainable levels. This will usher in a terminal recovery process inside cell interiors based on spontaneous dislocation emission from cell walls that will prevent a further increase of σ_c . The exact kinematic nature of this nucleation process is unclear. A scenario for it was sketched out by Argon and Haasen (1993). It can be expected that such a process can then lead to ideal plastic flow at a normalized plastic shear resistance τ/μ in the range $4 - 10 \times 10^{-3}$, as the information on Cu shown in Fig. 7.6 suggests.

7.4 Strain Hardening in Other Crystal Structures

In Section 7.3, the various stages of strain hardening were presented in detail for FCC crystals. This choice was made because the best quantitative information on this complex evolution process has been experimentally explored in the greatest detail in FCC crystals and also extensively modeled. It is expected that processes similar in both kinematical and energetic character will also be present in other crystal structures at temperatures above the point where the lattice resistance vanishes. Some support for this was presented in Section 7.2.2, where some stress-strain curves for BCC Mo and for NaCl were shown. In these cases, as well as in the diamond cubic Si and Ge (Argon and Haasen 1993) very similar processes of sessile-obstacle production by intersecting slip, leading, for example, in the BCC structure to sessile [100] Cottrell dislocations or other more complex fourfold nodes explored by Bulatov *et al.* (2006), are expected to prevail. These are expected to play roles very similar to the Lomer-Cottrell locks in FCC.

In a very comprehensive experimental investigation of the strain-hardening behavior of HCP Mg single crystals with a tensile-axis orientation that maximized the resolved shear stress on the basal planes, Hirsch and Lally (1965) established that in this crystal structure with an almost complete absence of intersecting slip, the only form of hardening that is present is what corresponds to Stage I in FCC, which they termed *Stage A*.

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DEFORMATION INSTABILITIES, POLYCRYSTALS, FLOW IN METALS WITH NANOSTRUCTURE, SUPERPOSITION OF STRENGTHENING MECHANISMS, AND TRANSITION TO CONTINUUM PLASTICITY

8.1 Overview

In previous chapters, separate strengthening mechanisms were presented by concentrating in each case on how they offered resistance to dislocation fluxes at given strain rates. Effects of resistances with increasing characteristic structural length scales, starting with the *lattice resistance* to fields of substitutional *solute atoms* and *precipitate particles*, were considered, where the resistances, represented fixed defect states with no structural evolution involving either thermal effects or alterations through plastic strain. In distinction to these static resistances, the dislocation resistance that was considered arises from interactions and intersections of dislocations with each other and involves a complex evolution of defect structure during plastic flow.

In the present chapter, we start by considering directly the processes that establish and control dynamic dislocation fluxes and their creation where they were initially not present, often requiring overstress, leading to prominent *yield phenomena* and factors that govern the establishment of mobile dislocation densities. We then consider dynamic flux instabilities referred to as *Portevin–LeChatelier effects*, which lead to jerky flow when solute atoms acquire short-range mobility and are able to interact with mobile dislocations in a quasi-periodic manner involving arrest and release processes, when the strain rate sensitivity of the flow stress becomes negative. We consider *dynamic overshoot* instabilities at very low temperatures, when fluctuations in fluxes arising from the motion of dislocations through thermally penetrable discrete obstacles such as solute atoms or forest dislocations become underdamped in the near absence of phonon damping.

We consider the elevation of the plastic resistance in polycrystals above that of unconstrained single crystals, when sets of crystallographic slip processes in randomly misoriented individual grains are required to partition in such a way as to achieve compatible flow in grain assemblies. We then consider, more specifically,

grain-size-dependent changes in the plastic resistance due to compartmentalization of the strain hardening in grains of decreasing size, starting from the inception of flow to large plastic strains; these changes are collectively referred to as the *Hall–Petch effect*. We pursue this effect to polycrystals of nanometer scale and consider fundamental changes in the mechanism of plastic flow when the confinement of crystal plasticity becomes too severe for its development.

Finally, we consider a number of interactions of individual mechanisms of strengthening with each other and demonstrate that such interactions are generally complex and that their superposition can rarely be described by simple additivity. Some of these interactions also explain effects that could not be dealt with adequately in earlier chapters. One such interaction is that which occurs between precipitate particles and dislocation resistance in strain hardening, where concepts of *geometrically necessary dislocations* arise, leading to consequences referred to as *strain gradient plasticity*.

8.2 Yield Phenomena

For plastic flow to develop at an imposed strain rate, a definite dislocation flux needs to be present. When special initial conditions exist in a nearly perfect crystal or when radical perturbations are imposed on the deformation, the establishment of the appropriate dislocation flux often cannot immediately follow. Very exaggerated conditions of this type occur when, in its initial state, the crystal forced to deform is highly perfect, containing no or only very few dislocations. Cases where this can arise have been encountered in metal whiskers of submicron diameter and in nanowires. A stress–strain curve of an initially nearly perfect submicron-size Cu whisker deformed by Brenner (1958), shown in Fig. 8.1(a), exhibits an exceedingly large yield phenomenon, where an overstress of about 370 MPa, nearing a substantial fraction of the ideal shear strength, was required to initiate flow by dislocation emission from atomic-size surface steps. Once an appropriate level of dislocation flux could be established locally, a plastic front spread along the length of the whisker in the form of a Lueders band as shown in Fig. 8.1(b), at a much reduced flow stress of about 10 MPa. When Au nanowires were stretched by a special nanomanipulator, deformation was observed to occur in discrete steps, where apparently a series of Shockley partial dislocations were inserted at a critical resolved shear stress, estimated to be 1.8 GPa (Marszalek *et al.* 2000). Very similar but less exaggerated yield phenomena were noted by Johnston and Gilman (1959) in LiF, by Alexander and Haasen (1968) in Ge, by Yonenaga *et al.* (1987) in GaAs, and by Yonenaga and Sumino (1989) in GaP, to cite just a few prominent examples. Some industrially more relevant cases include cases where the preexisting sets of dislocations have been locked in to the lattice by segregation of misfitting

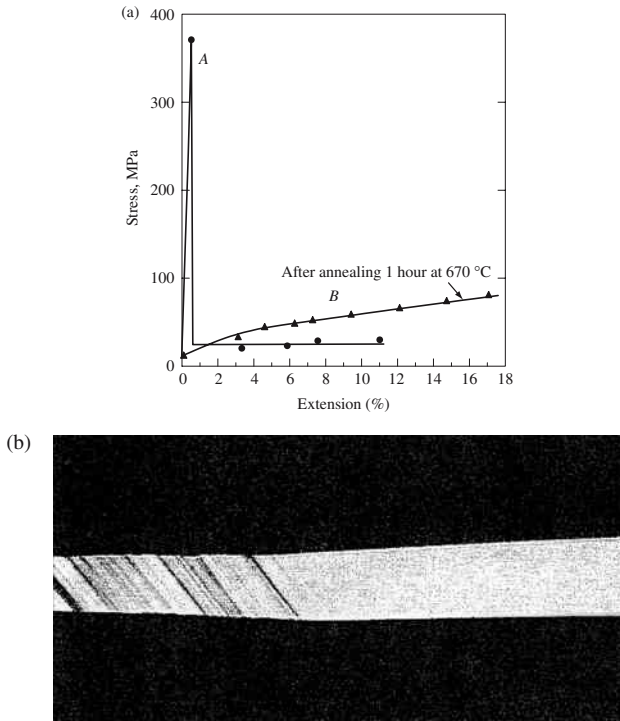


FIG. 8.1. (a) Stress–strain curve of an initially virgin whisker undergoing a very exaggerated yield phenomenon (A), followed by deformation at a much lower flow stress. Annealing at 670 °C has left enough dislocation density in place, causing no new yield phenomenon upon restraining (B). (b) The appearance of a Luders band spreading along the length of a whisker during postyield behavior in a whisker of the type shown in part (a) (Brenner 1958, courtesy of Wiley).

impurities, as in the case of α -Fe. There, misfitting interstitial C atoms diffuse to the dislocation cores and significantly lower their energy or even result in the formation of strings of nanoscale carbide precipitates along the dislocations (Low and Turkalo 1962), pinning the latter down and rendering them initially immobile. In such cases, initiating plastic flow also requires the overcoming of large yield phenomena. Figure 8.2 shows such a case encountered in normalized 1020 steel (Paxton and Bear 1955). Here, path A represents the effect of unloading and immediate reloading, where the previously mobilized dislocations are available, while path B represents the effect of unloading followed by a 10 minute excursion to 100 °C, sufficient for the interstitial C atoms to relock the dislocations, resulting in a new yield phenomenon.

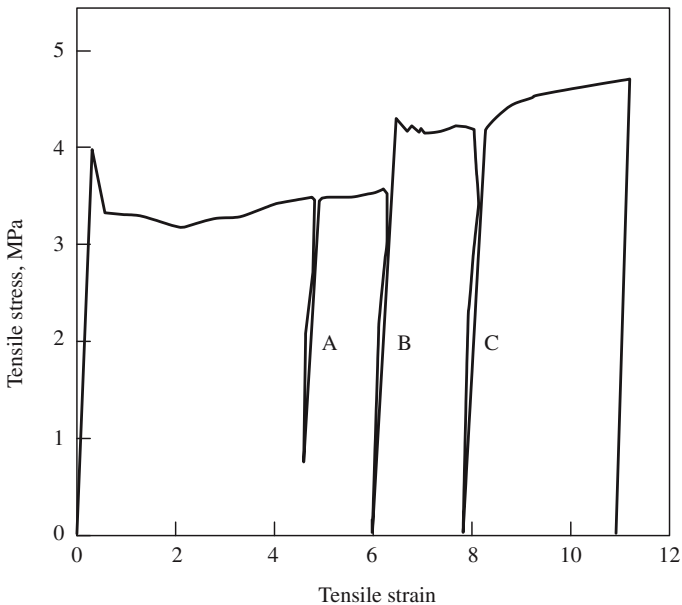


FIG. 8.2. A yield phenomenon in normalized 1020 steel strained at a rate of 10^{-5} s^{-1} . Path A represents unloading and immediate reloading, with no new yield phenomenon. Path B represents reloading after a thermal treatment at 100°C for 10 minutes, sufficient to lock the previously mobile dislocations and requiring a new yield phenomenon upon restretching. Even a 1 minute thermal treatment at 100°C brings back some sign of a yield phenomenon. (after Paxton and Bear 1955).

The theory of intrinsic yield phenomena was developed by Johnston (1962); it was intended to explain the behavior of nearly perfect LiF crystals, but it also represents more broadly the behavior of all such initial glide transients. The principal ingredients of Johnston's theory of the yield phenomenon are a strong stress dependence of the dislocation velocity, a kinematic expression for the plastic strain rate, and the phenomenological observation of Johnston and Gilman (1959) that the mobile dislocation density in such initially dislocation-starved crystals increases in direct linear proportion to the total plastic shear strain by kinematical processes of multiplication. The relevant expressions, coupled with the statement that the deformation occurs at constant total strain rate (that is, the sum of the elastic stretching rate and the plastic strain rate is constant), permits calculation by numerical methods of the response of an initially dislocation-deficient crystal. Johnston's calculated response of LiF crystals with different densities of initial mobile dislocations is shown in Fig. 8.3(a). These curves compare quite well with

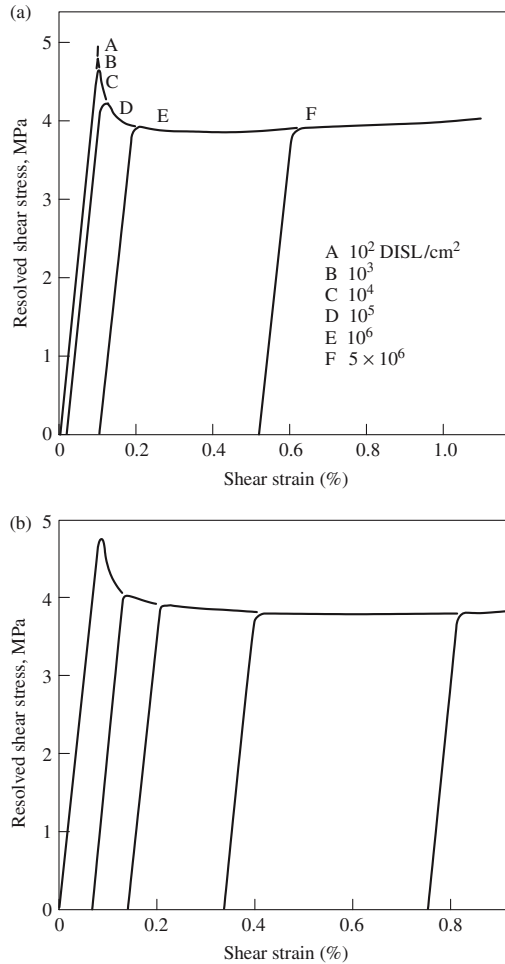


FIG. 8.3. (a) Calculated yield drops in stress–strain curves of LiF for several different assumed initial mobile dislocation densities. (b) Experimental stress–strain curves with unloading and reloading at mobile dislocation density levels corresponding to the curves shown in part (a) (after Johnston 1962).

the experimental stress–strain curves of such crystals shown in Fig. 8.3(b), giving the effect of increasing amounts of prestrain, which is to be construed as increasing densities of mobile dislocations, obtained by repeatedly straining through the upper yield point, unloading quickly, and straining again.

Yield phenomena manifest themselves also in unloading and reloading when a sizable fraction of the mobile dislocation density can be demobilized in various ways. Such behavior has been studied by Haasen and Kelly (1957) in Cu.

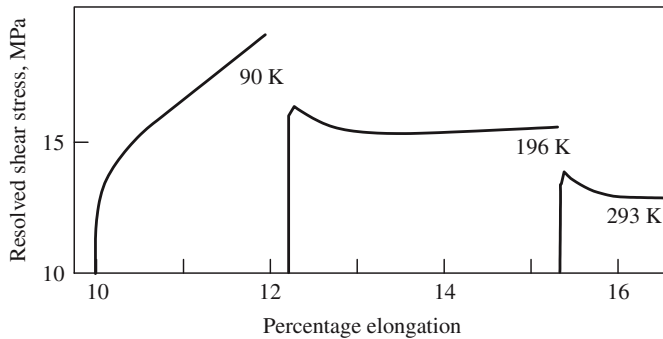


FIG. 8.4. Yield drops in Al observed at 196 K and 293 K after Stage II hardening at 90 K, resulting from massive dynamic recovery (after Cottrell and Stokes 1955).

There exist other yield phenomena of a fundamentally different nature, due to massive strain softening. These occur in a partially strain-hardened crystal when either the conditions or the direction of deformation is changed by increasing the temperature or going, for example, from tension to torsion. A pronounced form of this occurs in the deformation of Al crystals reported by Cottrell and Stokes (1955), who deformed Al crystals in tension at 90 K, into the high Stage II hardening regime and then continued the deformation at 196 K and 293 K. These authors found not only an expected reduction of the plastic resistance but also a substantial strain-softening effect, as shown in Fig. 8.4, resulting from the instability of the low-temperature Stage II dislocation structure to perturbations at more elevated temperatures due to the onset of massive dynamic recovery.

8.3 Balance between the Interplane and the Intraplane Resistances and the Mobile Dislocation Density

In Chapters 3 and 7, the notion was introduced that the dislocation resistance should be considered to be made up of an interplane resistance τ_1 and an intraplane resistance τ_2 . The former resistance is a result of the encounters of mobile dislocations with each other on parallel planes an average distance H apart, with an average distance CH between dislocations in their own plane giving, on average, a mobile dislocation in a cell of CH^2 in area, and giving a mobile dislocation density $\rho_m = 1/CH^2$. We consider the usual kinematic strain rate equation

$$\dot{\gamma} = b\rho_m\bar{v} \quad (8.1)$$

and a simple phenomenological kinetic expression for the dislocation velocity,

$$\bar{v} = v_0 \left(\frac{\tau_2}{\hat{\tau}_2} \right)^n \exp \left(\frac{-\Delta F_0}{kT} \right), \quad (8.2)$$

where $\hat{\tau}_2$ represents the athermal threshold level of τ_2 , v_0 represents a geometrical preexponential velocity factor, ΔF_0 is the activation energy of the intraplanar resistance mechanism, and n is a phenomenological stress exponent. Then, we consider the interplanar resistance τ_1 as that which would pertain to a critical stress for a pair of screw dislocations to pass by each other,

$$\tau_1 = \left(\frac{\mu b}{4\pi} \right) \sqrt{C \rho_m}, \quad (8.3)$$

and take

$$\sigma = \tau = \tau_1 + \tau_2 \quad (8.4)$$

at flow, with the abbreviations

$$B = bv_0 \exp \left(-\frac{\Delta F_0}{kT} \right) \quad \text{and} \quad D = \frac{\mu b}{4\pi} \sqrt{C} \quad (8.5a,b)$$

substituted into eq. (8.1). This gives

$$\dot{\gamma} = B \rho_m \left(\frac{\sigma - D \rho_m^{1/2}}{\hat{\tau}_2} \right)^n. \quad (8.6)$$

We note that $\dot{\gamma}$ is maximized when ρ_m reaches

$$\rho_{\text{mss}} = \left(\frac{2}{2+n} \frac{\sigma}{D} \right)^2 = \frac{1}{C} \left(\frac{8\pi}{2+n} \right)^2 \left(\frac{\sigma}{\mu b} \right)^2. \quad (8.7)$$

We take this as a steady-state mobile dislocation density, since a perturbation away from it in either direction would decrease the strain rate and the plastic dissipation rate (Alexander and Haasen 1968; Argon 1970). Considering that in low-temperature plasticity $n \approx 30$ –40, and taking $C \approx 10$ as reasonable, for

a flow stress of $\sigma/\mu \cong 10^{-3}$, as is typical of the mid-range between Stages II or III,

$$\rho_{\text{mss}} \cong O(6 \times 10^7 \text{ cm}^{-2}) \quad (8.8)$$

is obtained. This gives, as an estimate of a typical ratio of the interplane resistance to the flow stress,

$$\frac{\tau_1}{\sigma} = \frac{2}{2+n}, \quad (8.9)$$

which for low-temperature plasticity is typically 0.06 or, generally, negligible in comparison with the intraplane resistance, as long as n is large. In more strain-rate-sensitive materials or under conditions with smaller n , the effect of the interplane resistance, or the dependence of the behavior on the changes in mobile dislocation density, can be more significant.

8.4 The Portevin–Le Chatelier Effect and Jerky Flow

In many FCC and BCC solid-solution alloys and in some low-carbon steels, in certain intermediate-temperature ranges where long-range lattice diffusion of the solute should be absent, plastic flow can exhibit a jerky character. This generally occurs upon yielding with a yield phenomenon, followed by a small prestrain, and is associated with aging effects where some elevation of the plastic resistance is observed following a short rest period upon unloading. All these are indicative of pinning of mobile dislocations by short-range rearrangement of a solute around the dislocations. This phenomenon, known as the Portevin–Le Chatelier effect, is considered a nuisance in deformation processing. Since clear aging effects and dislocation pinning are present but the temperatures are too low for lattice diffusion of the solute, Cottrell (1953) proposed that such diffusion must nevertheless still be facilitated by a supersaturation of vacancies produced during plastic flow itself. While such an effect has never been fully established or discounted, there has been much evidence against it, suggesting that the diffusion that is clearly present must be of a short-range nature and along dislocation cores, which act as conduits for short-range redistribution of the solute along the dislocation line as it contacts the solute atoms (see also Section 5.5.7). The phenomenon suggests that slow-moving dislocations are more strongly impeded by rearrangement of solute around their core, while fast-moving dislocations feel less of an impediment, indicating that the phenomenon is associated with a negative strain rate dependence of the flow stress. This is found indeed to be the case. Here, we consider the findings of Kocks and coworkers, who have had a major role in clarifying the mechanism of the jerky-flow phenomenon in Ni–C and Al–Mg alloys (Mulford and Kocks 1979; Wycliffe *et al.* 1980; Kocks *et al.* 1985).

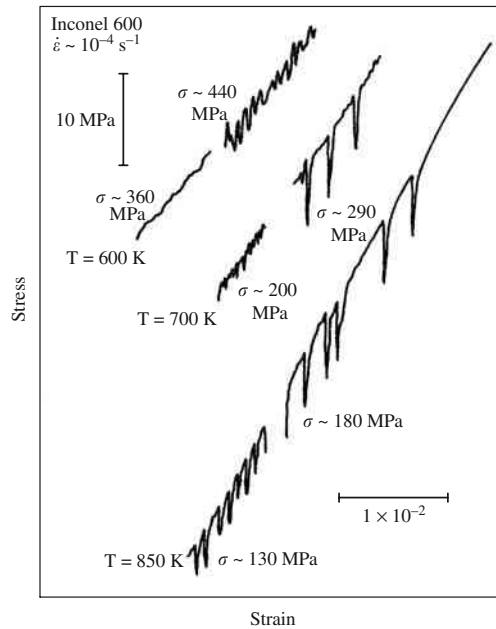


FIG. 8.5. Examples of different types of jerky flow observed in Inconel 600 tested in compression, including disappearance of jerky flow at 850 K. The temperatures and nominal stress levels of each curve are indicated (Mulford and Kocks 1979, courtesy of Elsevier).

Figure 8.5 shows typical portions of serrated flow stress curves in Inconel 600, a solid-solution Ni–C alloy, deformed at a uniaxial strain rate of 10^{-4} s^{-1} in compression at 600 K, 700 K, and 850 K. The curve at 850 K shows that after some jerky straining, the flow stress curve becomes smooth again at this strain rate. Figure 8.6 shows the regions in the strain-rate–temperature domain where jerky extension is prominent (Region II) in Inconel 600; it is absent at low temperatures (Region I), since the solute, C, is immobile there, whereas at high temperatures in Region III it is initiated at the start but disappears at large strains. The key information about the dependence of the strain rate sensitivity of the flow stress on stress is shown in Fig. 8.7(a) for temperatures of 200–600 K and in Fig. 8.7(b) for 700–960 K. Figure 8.7(a) shows that the behavior between 200 and 350 K is entirely stable, with a positive slope of $\Delta\sigma/\Delta \ln \dot{\epsilon}$ versus σ . Above 350 K, the slope becomes negative but $\Delta\sigma/\Delta \ln \dot{\epsilon}$ still remains positive over a significant stress range, giving stable behavior. However, for 500 and 600 K, $\Delta\sigma/\Delta \ln \dot{\epsilon}$ becomes negative at lower stress levels with increasing temperature, as demonstrated in Fig. 8.7(b), but returns to positive territory at 900 K or is on the way to doing so at 700 K.

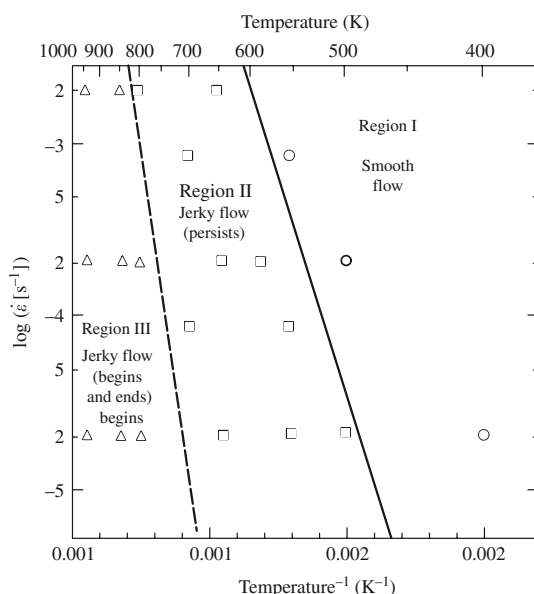


FIG. 8.6. The temperature and strain rate regime of jerky flow in Inconel 600 (Mulford and Kocks 1979, courtesy of Elsevier).

In each case, jerky flow sets in at or very soon after $\Delta\sigma/\Delta \ln \dot{\epsilon}$ becomes negative. The change in the dependence of $\Delta\sigma/\Delta \ln \dot{\epsilon}$ with increasing temperature shown in Fig. 8.7 at constant strain rate is paralleled by changes in the strain rate at constant temperature, as shown in Fig. 8.8 for Inconel 600 at 400 K. This demonstrates clearly that the jerky flow is governed by a kinetic process governing the rise in the solid-solution resistance through aging by a core-level redistribution of solute. Very similar results were obtained also for Al-1.0 at % Mg by Mulford and Kocks (1979). The general behavior of the dependence of $\Delta\sigma/\Delta \ln \dot{\epsilon}$ on stress at different strain rates at a constant strain rate or at a constant temperature for different strain rates was modeled by Schwarz (1982). He assumed that the overall plastic resistance results from a linear superposition of the solid-solution resistance and the dislocation resistance, and that changes due to aging affect only the solid-solution resistance. In the model, the sampling of the solute by the dislocation is assumed to remain unaltered in form but undergoes a change in the strength of the interaction with short-range diffusional rearrangement, leading to local saturation at a characteristic time. This apparent deepening of the “binding potential” for the solute-dislocation interaction, at constant dispersal of the solute but possibly associated with some form of clustering, is reminiscent of the implied change of behavior at 298 K in the discussion of the “stress equivalence” in Section 5.5. The

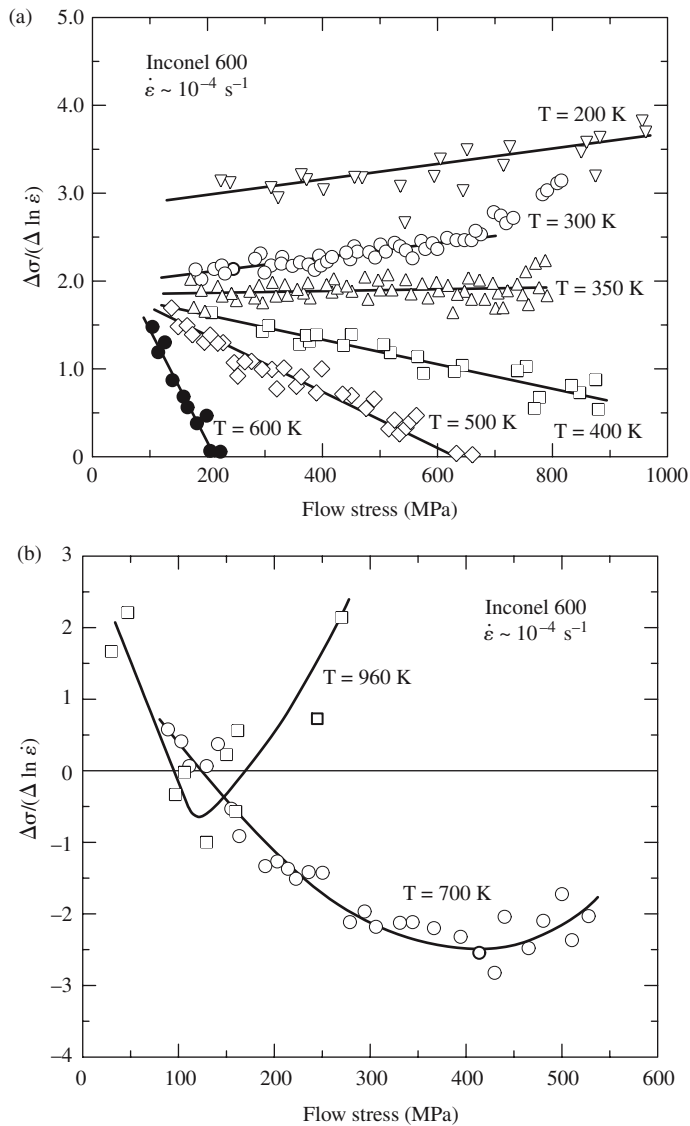


FIG. 8.7. (a) The changing strain rate sensitivity of the flow stress, $\partial\sigma/(\partial \ln \dot{\epsilon})$ with flow stress level in Inconel 600 at different temperatures. Jerky flow sets in when $\partial\sigma/(\partial \ln \dot{\epsilon})$ becomes negative (Mulford 1979). (b) The more complex behavior of Inconel 600 at higher temperatures, where jerky flow disappears at higher strain levels at 960 K (Mulford and Kocks 1979, courtesy of Elsevier).

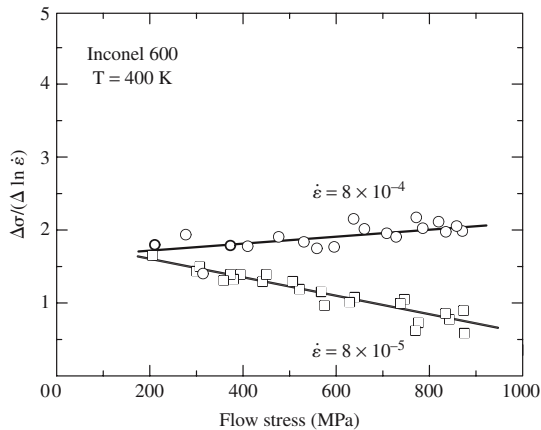


FIG. 8.8. Corresponding tendency toward jerky flow in Inconel 600 deformed at 400 K at different strain rates (Mulford and Kocks 1979, courtesy of Elsevier).

Schwarz model, using appropriate kinetic parameters for core diffusion obtained from internal-friction experiments, predicts a minimum in $\Delta\sigma/\Delta \ln \dot{\epsilon}$ with stress similar to what is shown in Fig. 8.7(b). The minimum increases with increasing temperature owing to an entropy contribution to the “binding energy” of the solute in the solute–dislocation interaction, making $\Delta\sigma/\Delta \ln \dot{\epsilon}$ eventually positive and leading to regular behavior at high stress levels. The minimum also shifts to higher stresses at higher temperatures for a constant strain rate and to lower stresses at higher strain rates for a constant temperature, all in keeping with the experimentally observed behavior.

Kok *et al.* (2002, 2003) have demonstrated with finite-element models the spatiotemporal distribution of the localization of banding phenomena in tensile plastic flow of polycrystals using such a kinetic model of the Portevin–Le Chatelier effect, with a very good correlation with experimental findings.

8.5 Dynamic Overshoot at Low Temperatures

In many solid-solution alloys and, to a lesser extent, in FCC metals with negligible lattice resistance and a relatively large temperature dependence of the flow stress, a prominent perturbation is found to occur in the flow stress at cryogenic temperatures. The effect consists of a peak in the flow stress at a certain low temperature and a reversal of the temperature dependence of the flow stress below that temperature as $T \rightarrow 0$ K. A clear example of this reversal was encountered in Section 5.5 in the temperature dependence of dilute Cu–Al solid-solution

alloys around 70 K, shown in Fig. 5.6. This phenomenon, which results from the overcoming of thermally penetrable localized solute obstacles or, possibly, forest dislocations by the inertial effects of glide dislocations, is referred to as a *dynamic overshoot effect*. Under conditions of a vanishingly low lattice resistance and an ever-decreasing phonon drag with decreasing temperature, when dislocation segments waiting at localized obstacles are released by thermal activation, they can achieve conditions of free flight and can overcome subsequent obstacles in their path by effectively trading increased kinetic energy for potential energy of interaction with obstacles, provided the flow stress has approached the threshold resistance and free-flight conditions can be reached between overcoming an initial array of obstacles and new arrays.

A simple estimate demonstrates that the explanation of the dynamic overshoot is more complex than described above. The kinetic energy of a dislocation line in free flight, subject only to a linear drag B , is

$$\mathfrak{F}_{\text{KE}} = \left(\frac{\mathfrak{F}_0}{2} \right) \left(\frac{\sigma b}{\nu_D b B} \right)^2 = \left(\frac{\mathfrak{F}_0}{2} \right) \left(\frac{\sigma}{\mu} \right)^2 \left(\frac{\mu \Omega}{kT} \right)^2, \quad (8.10)$$

where $\mathfrak{F}_0$ is the line energy per unit length at rest, and certain well-known substitutions have been used for the shear velocity of sound to reach the final form. Introducing a temperature parameter $T^* = (\mu \Omega / \sqrt{2} k)$, which is roughly 3800 K for Cu, and considering the Cu–1.0% Al alloy of Fig. 5.6, where the peak resistance in the flow stress $\sigma/\mu = 1.25 \times 10^{-4}$ is reached at $T = 70$ K, the kinetic energy of a dislocation segment equal to the mean spacing l of solute obstacles in the plane, that is, the value at $l \cong 10b$ for a solute concentration $c = 0.01$, gives $\mathfrak{F}_{\text{KE}}/\mathfrak{F}_0 = 0.5 \times 10^{-3}$. This gives $\mathfrak{F}_{\text{KE}}$ for such a segment, which is an energy of only 0.25 of kT at 70 K. While this is quite small, it is not inconsequential for flow stresses close to the mechanical threshold at very low temperature. However, upon release from a line of obstacles, the dislocation will reach near-free-flight conditions only after moving a distance $\bar{y}$ that can be estimated similarly as

$$\bar{y} = b \left(\frac{\sigma}{\mu} \right) \left(\frac{T^*}{T} \right)^2. \quad (8.11)$$

Under the above conditions, at $T = 70$ K this is roughly $4l$, or much larger than the mean spacing of obstacle arrays in the plane, indicating that the released dislocation would encounter another line of obstacles well before it reached free-flight conditions.

Moreover, considering the complex “hunting” form of overcoming solute obstacles by the dislocation line described in Chapter 5, it is clear that the dynamic overshoot effect requires more precise consideration. This was provided by Schwarz and Labusch (1978) in simulations that were discussed in Chapter 6. In these simulations, the motion of a dislocation through a field of randomly placed discrete obstacles of different normalized cutting depths $\eta_0 = y_0/\lambda$, defined in Fig. 6.7(a), is considered under the action of different levels of the phonon damping coefficient $B (= kT/\nu_G\Omega)$, represented by a normalized damping parameter γ defined by

$$\gamma = \frac{Bl}{(4\mathcal{E}m\beta)^{1/2}} = \frac{(kT/\mu\Omega) (\mu/(2\mathcal{E}/bl))}{(\hat{K}/2\mathcal{E})^{1/2}}, \quad (8.12)$$

where $m = (\mathfrak{F}_0/c^2)$ is the effective mass per unit length of the dislocation line, introduced in Chapter 2. The findings of Schwarz and Labusch are shown in Fig. 8.9, where S_c is the normalized stress $\sigma/\hat{\tau}$ required to initiate dislocation motion through the solute field and S_L is the normalized stress to keep it moving through the field. The figure shows that at temperatures where phonon damping is critical ($\gamma > 1.0$), the stress S_c is the same as S_L . However, with decreasing levels of phonon damping ($\gamma < 1.0$), this damping becomes inadequate and while S_c reaches an asymptotic level, only slightly dependent on γ , S_L continues to fall below S_c . For pointlike obstacles with no depth, that is, $\eta_0 = 0$, below a damping factor

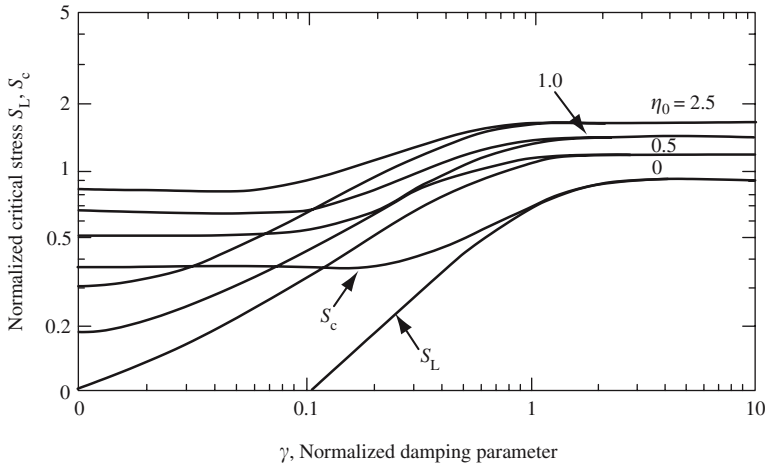


FIG. 8.9. Dependence of the critical stress S_c (crss) to initiate flow and S_L to continue flow, on the normalized damping parameter γ for four different obstacle depths η_0 . For pointlike obstacles ($\eta_0 = 0$), S_L decreases rapidly to very low levels indicating a severe tendency toward dynamic overshoot at low temperatures (after Schwarz and Labusch 1978).

of 0.1, ever-decreasing stresses S_L are sufficient to keep the dislocation moving through the obstacle field. For more realistic cases of obstacles of finite depth, that is, $\eta_0 > 0$, the separation of S_L from S_c continues to set in roughly at $\gamma = 1.0$ but remains finite even at quite low levels of γ (< 0.1). Nevertheless, the dynamic overshoot effect remains. Schwarz *et al.* (1977) and Isaac *et al.* (1978) discussed the more prominent dynamic overshoot effects in alloys that undergo a normal-to-superconducting transition below a critical temperature T_c . Clearly, with this more generalized representation of obstacle depths and the effects of damping, the results of Schwarz and Labusch should also be applicable to some extent to a forest dislocation field, where, however, the phenomenon has not been widely observed, no doubt because of the significantly larger cutting depths of forest dislocations than solute atoms.

8.6 Plastic Deformation in Polycrystals

8.6.1 Plastic Resistance of Polycrystals

While a basic understanding of the mechanisms and kinetics of plastic flow is best obtained from the behavior of single crystals, the crystalline materials of technological relevance are, with very few exceptions, in relatively fine-grained polycrystalline form. Their plastic response has traditionally been dealt with by the phenomenological continuum theory of plasticity. Here we consider the mechanistic and crystallographic aspects of polycrystal plasticity, which not only relates the single-crystal response to the homogenized response of aggregates of larger numbers of grains, but also develops the methodology for predicting, or at least rationalizing, the evolution of the plastic anisotropy and deformation textures that develop as a consequence of the large plastic strains encountered, for example, in forging and plastic forming.

The first consideration was the determination of the initial yield strength of polycrystalline aggregates, given the plastic shear resistances of various slip systems in single crystals. Since the actual solution for the response of a polycrystalline aggregate requires numerical procedures from the beginning, consideration was directed toward obtaining bounds on the plastic resistance of the aggregate.

Thus, Sachs (1928) considered the tensile response of a polycrystal as the orientation average of the least tensile resistances of the individual grains deformed separately. For a grain in which the most favorably oriented slip system has an orientation factor n (the reciprocal of the individual Schmid factor m of eq. (4.19)) in relation to the tensile axis of the polycrystalline aggregate, the tensile resistance Y of that particular grain should then be

$$Y_i = n_i \tau, \quad (8.13)$$

where τ is the plastic shear resistance of the slip plane. The overall tensile resistance Y of the aggregate should then be

$$Y = \int n\tau N(n) \, dn, \quad (8.14)$$

where $N(n)$ is the normalized orientation distribution function of the grains measured by their orientation factors. Thus, the connection between the tensile plastic resistance of the aggregate and the shear resistance of the slip planes becomes simply

$$\left(\frac{Y}{\tau}\right) = \bar{n}_s = \int nN(n) \, dn, \quad (8.15)$$

that is, the population average of the orientation factors of the best slip planes in every grain. For FCC metals, the Sachs average is $\bar{n}_s = 2.238$. Clearly, this approach satisfies neither local equilibrium nor compatibility, but merely a global average equilibrium. For this reason, it is considered to give a lower bound on the estimate of the aggregate tensile yield strength. While it may give some idea on when a local departure from elastic response occurs, it falls quite short of predicting the yield strength of random grain aggregates for the onset of fully plastic response.

Observing that the strains in the individual grains of a polycrystal are often not too different from the overall strain that the aggregate suffers in deformation, and recognizing that compatibility of deformation in the entire aggregate must be satisfied, Taylor (1938) developed another bounding approach based on homogeneous deformation inside grains, with an affine connection to the aggregate whole. Taylor started by recognizing that for a volume-preserving arbitrary plastic deformation, five independent slip systems need to be simultaneously active, in general, to achieve compatibility among the homogeneously deforming, randomly misoriented grains. In FCC metals, where 12 equivalent slip systems are present, there is a substantial redundancy of sets of five systems for each grain. Considering, out of the available sets, only the ones with minimum internal dissipation, that is, the ones for which $\tau \sum_i^5 d\gamma_i$ is minimum, and invoking the principle of virtual work (equivalence of the external work increment and the internal dissipation increment),

$$Y \, d\varepsilon = \tau \left\langle \sum_i^5 d\gamma_i \right\rangle \quad (8.16)$$

must be obeyed, where the brackets indicate an average of the internal dissipation increment over all grains in the aggregate. Thus,

$$\left(\frac{Y}{\tau}\right)_T = \bar{n}_T = \frac{\tau \left\langle \sum_i^5 d\gamma_i \right\rangle}{d\varepsilon} \quad (8.17)$$

gives the Taylor factor $\bar{n}_T = 3.1$ for FCC crystal aggregates (for a somewhat more detailed discussion, see also Cottrell 1953).

The majority of considerations of the initial plastic response of polycrystals have been for FCC metals. It has been argued by Kocks (1970) that in BCC metals, which deform primarily on the $\{110\}\langle 111 \rangle$ slip systems, with additional slip possibilities on the $\{112\}\langle 111 \rangle$ systems, a direct application of the results for FCC metals should be possible—at least for the determination of the Taylor factor for the initial deformation. As pointed out by Groves and Kelly (1963), many crystal structures other than cubic, such as the HCP structure, those with layered structures, and those of many intermetallic compounds and minerals, and even the cubic NaCl structure do not possess five equivalent independent slip systems at low temperatures. The HCP metals, which include many technologically important examples where the second set of slip systems often have much higher slip resistances than the best set, were considered by Hutchinson (1977), who demonstrated that the plastic resistance of a randomly oriented polycrystal then becomes governed by the systems with the highest resistance in the set of systems kinematically necessary for compatible deformation. Many such structures with insufficient slip systems possess in addition other sets of considerably higher resistance, and a multiplicity of twinning systems. The latter, however, are unidirectional and are relatively less effective in aiding the deformation of polycrystals, resulting still in much higher Taylor factors. This difficulty often manifests itself in premature fracture. The deformation of such polycrystalline solids, both in their initial yield behavior and in later stages of deformation, including strain hardening, has received much attention recently (see, for example, Kocks *et al.* 1998). We consider a selection of recent examples of the modeling of deformation textures in the next section.

8.6.2 Evolution of Deformation Textures

As discussed in Section 2.2.3, crystallographic slip results in lattice rotations relative to the external axes of a body. The lattice rotations due to simple glide strain in FCC single crystals were studied in considerable detail by Schmid and Boas (1935). Their basic kinematical nature, resulting from elementary rotations, and their effect on the overall tensile stress–strain curve were already well understood at that time.

In polycrystals, these kinematical ideas find application on the grain scale to determining both the stress–strain curve of the polycrystal and the evolving texture. Texture development has been studied extensively in the context of the Taylor model, which assumes an affine connection between the deformation gradient inside a grain and the global gradient, or the gradient in a local representative volume element (RVE) containing many grains. An important element of the strain hardening of polycrystals is the requirement of compatibility of deformation among grains, with as many as five different slip systems being active in each grain undergoing intersecting slip. Kocks (1970) noted that, therefore, in polycrystals, well before the

textural strengthening becomes significant, the stress–strain curve should exhibit enhanced hardening due to such enforced intersecting slip and that in this range, there should be a good parallel between the stress–strain curves of polycrystals and those of single crystals of high-symmetry orientations, particularly the $\langle 111 \rangle$ orientation in FCC, which is texturally stable. Figure 8.10 shows that this is true not only in FCC (Kocks *et al.* 1968) but also in BCC (Keh 1965), where the agreement is even better.

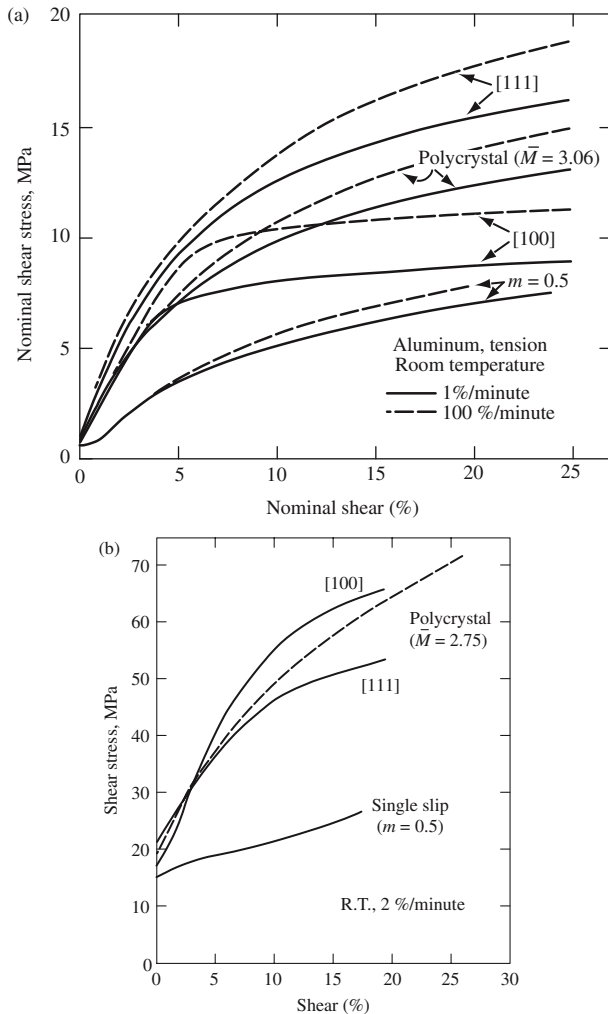


FIG. 8.10. Correspondence between the stress–strain curves of single crystals in high symmetry orientations and polycrystals: (a) in FCC Al (after Kocks *et al.* 1968) and (b) in BCC, Fe (after Keh 1965).

The implementation of the theory of texture development through the Taylor model has been carried out computationally by many investigators for various different deformation histories. These model developments, to which a large number of investigators have contributed, have been discussed in broad outline and also in great detail by Asaro and Needleman (1985), and many new advances have been presented by Kocks *et al.* (1998). An important element here is the introduction of a rate-dependent empirical constitutive kinetic law of the type $\dot{\gamma} = \dot{\gamma}_0(T) (\sigma/\hat{\tau})^m$ for each slip system rather than a critical threshold stress. This has prevented problems of nonuniqueness in the selection of the most appropriate set of slip systems in each grain and their level of activity. The model developments incorporate latent hardening relationships and some interaction between slip systems. Moreover, these model developments, in their most recent applications in finite-element programs, going beyond the Taylor assumption, satisfy not only compatibility but also equilibrium, at least over the RVEs in which they are applied. Extensive application of these developments by Bronkhorst *et al.* (1992) to Cu has given excellent agreement between the predictions and the associated experimental observations, of which a set pertaining to the case of simple shear is shown in Fig. 8.11. Thus, within the presently existing methodology, the development of deformation textures in FCC metals can be followed during relatively complex inhomogeneous deformation by computational techniques. Furthermore, the general level of excellent agreement indicates that the Taylor assumption of an affine connection between the individual grains and the polycrystalline aggregate or the local RVE is often quite satisfactory.

In many crystal structures, such as in FCC metals such as Cu and most HCP metals, such as Mg, Ti, and Zr, deformation twinning together with slip results in important forms of texture evolution that depart considerably from what slip alone produces. Some cases of this type have also been investigated both experimentally and by FEM modeling. Among recent studies is that of Staroselsky and Anand (1998), who have modeled texture development in α -brass under combined slip and twinning; here, twinning systems of the $\{111\}\langle 112 \rangle$ type have been considered in addition to the usual octahedral slip systems $\{111\}\langle 110 \rangle$. In the FEM models, where individual grains were taken as FEM elements, the simulation was carried out both for finite numbers of grains in assemblies, without involving the usual assumption of an affine connection between the macroscopic field and the individual grain field of the Taylor model, and employing the Taylor model itself for comparison. Using appropriate material constitutive parameters, the agreement between the models and experiments was found to be quite good. In some cases of HCP metals such as the Mg alloy AX31B, Staroselsky and Anand (2003) had to resort to considerable internal tuning of parameters of the basal, prismatic, and pyramidal slip systems, with a pyramidal system of type $\{10\bar{1}2\}\langle \bar{1}011 \rangle$, to obtain acceptable agreement between the model and experiments. Once such tuning had been done,

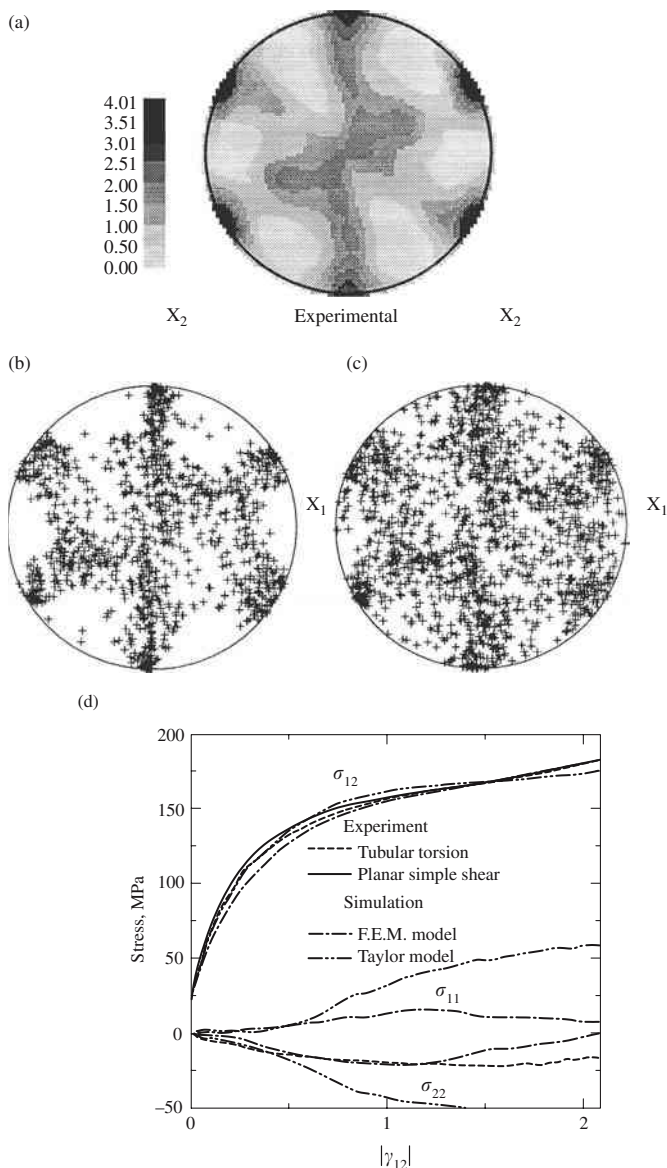


FIG. 8.11. Predictions and experimental measurements of the stress-strain response and texture development in simple shear of polycrystalline Cu showing $\{111\}$ pole figures at a shear strain of $\gamma = 1.4$: (a) experimental; (b) simulation with a Taylor model; (c) results of a finite-element model of a cluster aggregate. (d) Stress-strain curves showing σ_{12} , as well as the spurious stresses σ_{11} and σ_{22} (after Bronkhorst *et al.* 1992, courtesy of the Royal Society).

however, the programs could predict the outcome of more complex deformation problems.

Even more complex crystal structures have been investigated, for example by Schoenfeld *et al.* (1995) and by Parks and Ahzi (1990), where deformation in systems with fewer than the required 5 independent slip systems were available.

8.7 Plastic Deformation in the Presence of Heterogeneities

8.7.1 Geometrically Necessary Dislocations

When heterogeneities of different plastic resistance are introduced into a smooth dislocation flux, as may occur with alloys containing nondeformable precipitates or particles, or when grains of different relative orientation must deform together compatibly as in polycrystals, local “turbulences” occur. Such occurrences are ubiquitous in all cases other than unconstrained single crystals subjected to uniform shear flow. The additional local deformations that are required for compatibility make important changes in the plastic resistance and strain-hardening rate in such heterogeneous materials. Now, the plastic resistance and the hardening rate depend on a local material length scale characterizing the range of the perturbation in the flow, such as the particle size, particle mean spacing, or grain size.

Ashby (1966, 1970, 1971) pioneered the consideration of such problems in crystal plasticity, where the nature of the basic constitutive relation is altered beyond mere considerations of the volume fractions of components of different resistance as would be the case in macroscopic composites. The central concept from which we start in such phenomena is the disruption of smooth dislocation fluxes by local, more or less intense, strain gradients imposed by the heterogeneities, where the displacement incompatibilities must be accommodated locally by *geometrically necessary dislocations*, which, in turn, interact in various ways with the otherwise smooth dislocation flux. It is these GNDs and their effects on the flow stress and strain-hardening rate that we consider next.

8.7.2 Rise in Flow Stress and Enhanced Strain-hardening-rate Effects of Geometrically Necessary Dislocations

In his pioneering work, Ashby (1970, 1971) proposed several equivalent plastic accommodation models of the displacement incompatibilities that develop between a homogeneously shearing continuum and rigid inclusions of a variety of shapes. The basic model, which is particularly useful in understanding the disruptions that such inclusions produce in a homogeneous flux of dislocations, is based on the Eshelby (1957) model in the field of elasticity for considering the effects of general strain transformations in the presence of ellipsoidal inclusions on smooth

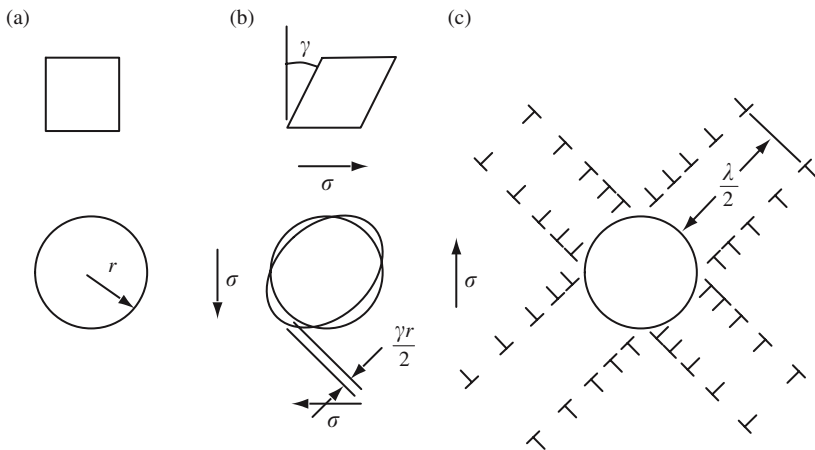


FIG. 8.12. Ashby's model of the production of a secondary plastic zone around a rigid spherical particle by nucleation of a series of geometrically necessary edge dislocation loops to dissipate the concentrated stresses due to deformation-induced incompatibilities during plastic straining of the matrix (Argon *et al.* 1975).

deformation fields, and extends it to plasticity. Thus, consider the thought experiment illustrated in Fig. 8.12, where, as shown in Fig. 8.12(a), a rigid spherical inclusion of radius r is removed and replaced with a sphere of parent material. The continuum is plastically sheared by an amount γ , which distorts the outline of the reference spherical region into an ellipsoid as shown in Fig. 8.12(b). The distorted ellipsoid is now removed and the initially separated rigid sphere is reinserted. This requires the elimination of the displacement incompatibilities, which have a maximum amount of $\gamma r/2$ in four principal directions. For small shear strains in the elastic range, such accommodation can be accomplished in the matrix by local, concentrated elastic deformations. If the inclusion is of nanometer scale and if the surrounding material lacks operable dislocation sources, the local elastic stress concentrations could rise to such a level that the ideal shear strength is reached in the adjoining matrix or dislocation loops are punched out to form plastic accommodation zones, shown in Fig. 8.12(c) (Ashby 1970; Argon *et al.* 1975). Such accommodation zones can have a variety of equivalent forms discussed by Ashby (1970), all of which result generally in quite similar effects. Alternatively, very similar effects of accommodation can also be achieved by the dislocations of the main flux by a series of cross-slip maneuvers, discussed by a variety of investigators, in particular Hirsch and Humphreys (1969), with quite impressive support from TEM observations, an example of which is shown in Fig. 8.13. Very similar forms of maneuvers have also recently been simulated by Srolovitz and coworkers (Xiang *et al.* 2004). In all instances, the principal effects on the flow stress and

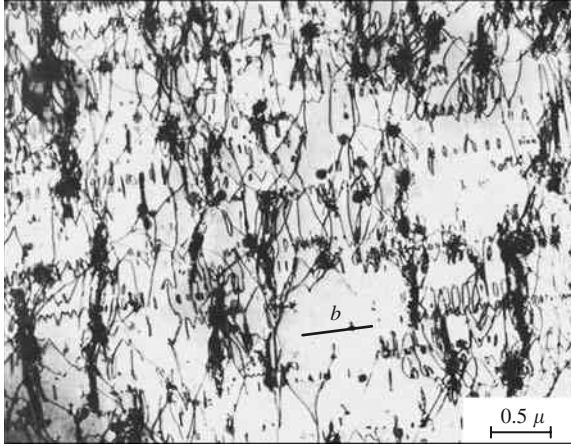


FIG. 8.13. Geometrically necessary dislocations laid up by a series of complex cross-slip maneuvers around rigid inclusions during deformation in Cu (Hirsch and Humphreys 1969, courtesy of MIT Press).

the strain hardening can be understood by use of the equivalent scheme shown in Fig. 8.12 based on the Ashby (1970) model, which in outline proceeds as follows.

If the inclusion is idealized as rigid but the surrounding medium is elastic-plastic in the range where the shear strain γ in the medium is comfortably in excess of its yield strain γ_y , and we consider the form of plastic accommodation to be made up of four sets of punched-out circular dislocation loops, the total number N of loops in the four zones will be

$$N = \frac{2r\gamma}{b}. \quad (8.18)$$

Since the line length Λ of the dislocation in any loop is $2\pi r$, the total increment of line length $d\Lambda$ for an increment of the distant-field plastic shear $d\gamma$ will be

$$d\Lambda = 2\pi r dN = \frac{4\pi r^2}{b} d\gamma. \quad (8.19)$$

This increment of geometrically necessary line length will be dispersed over a local volume V_p encompassing secondary plastic zones of extent $\lambda/2$ or, approximately,

$$V_p \cong \frac{\pi \lambda^2 r}{2}. \quad (8.20)$$

The extent of the secondary plastic zones depends primarily on the glide resistance the loops encounter, which, in turn depends on a variety of other considerations discussed by Argon *et al.* (1975). As shown there, when the plastic shear strain γ in

the matrix exceeds its value at yield by roughly a factor of 10, for volume fractions of several percent of rigid inclusions, the secondary plastic zones of neighboring inclusions begin to overlap and V_p becomes the average volume $V = 4\pi r^3/3c$ allocated to a typical inclusion. Then, the increment $d\rho_G$ of the geometrically necessary dislocation density for an increment $d\gamma$ of the distant field becomes

$$d\rho_G = \left(\frac{c}{r}\right) \left(\frac{3d\gamma}{b}\right) = \left(\frac{1}{\lambda_G}\right) \left(\frac{3d\gamma}{b}\right), \quad (8.21)$$

where $\lambda_G = r/c$ has been introduced as a characteristic length scale of the accommodation process related to spherical inclusions. For inclusions of different shapes and aspect ratios, other characteristic length scales of similar order can be defined, as discussed by Ashby (1971), all of which have relatively similar effects. While the increments in ρ_G develop, the surrounding matrix continues to shear, involving interactions between increments in the GNDs and the dislocations of the surrounding matrix flux. These interactions have been discussed by Ashby (1970, 1971) and by Argon *et al.* (1975). Consideration of such interactions in integrally mixed form involves complexities which will be sidestepped here, by concentrating only on limiting forms of the phenomena of GNDs vis-à-vis statistically stored dislocations in reference problems where the GNDs are absent, or by considering only coarser forms of mixing of the two populations.

In the absence of rigid inclusions, the medium surrounding the inclusions would have undergone strain hardening by statistical processes of dislocation trapping, as discussed in Chapter 7. Thus, considering this statistical storage of glide dislocations in a scenario similar to that of the GNDs, where loops expand to radii r_s where they are stored statistically, the increment of statistically stored dislocation density $d\rho_s$ can be defined kinematically in a similar way

$$d\rho_s = \frac{d\Lambda_s}{V_s} = 2\pi r_s \frac{dN_s}{V_s} = \left(\frac{2}{r_s}\right) \left(\frac{d\gamma}{b}\right) = \left(\frac{1}{\lambda_s}\right) \left(\frac{3d\gamma}{b}\right) \quad (8.22)$$

for an increment of shear strain $d\gamma$, where $d\Lambda_s$ is the increment of statistically stored line length in a volume V_s due to an increment dN_s of loops of radii r_s , and a corresponding length scale $\lambda_s = (3r_s/2)$ has been introduced for comparison purposes.

As pointed out by Ashby (1970), and as should be clear, when $\lambda_G/\lambda_s \gg 1.0$ in a heterogeneous medium with rigid inclusions, the increments in ρ_G are negligible in comparison with the increments in ρ_s , leading to no marked change in the flow stress or hardening rate. However, the opposite holds when $\lambda_G/\lambda_s \ll 1.0$, where increments in ρ_G would be dominant. This is demonstrated in Fig. 8.14, showing the band of increase of the density of statistically stored dislocations with shear strain in single-phase single crystals of FCC metals (Basinski and Basinski 1966). Correspondingly, in similar single crystals of FCC metals containing spherical

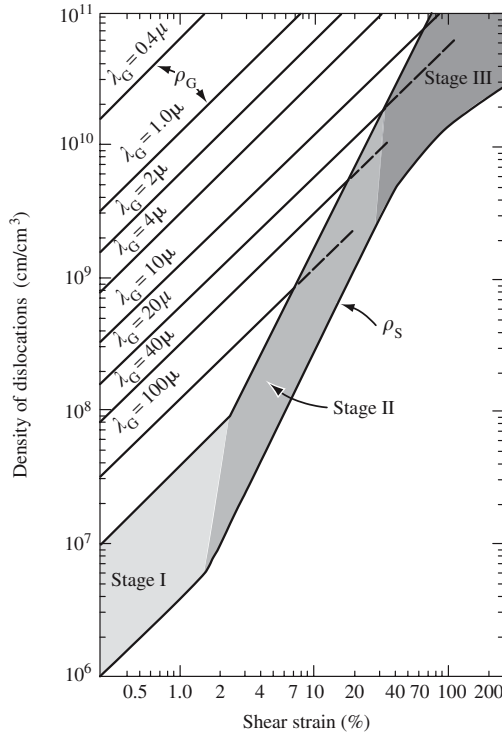


FIG. 8.14. The statistically stored (shaded band) and geometrically necessary dislocation density as a function of shear strain. Note that ρ_G can dominate the total density at small strains, but can be swamped by the statistically stored dislocation density ρ_s at larger strains (Ashby 1970, courtesy of Taylor and Francis).

heterogeneities in the range where $\lambda_G/\lambda_s \ll 1.0$, the increases in ρ_G alone with strain are shown by a series of parallel lines for different lengths λ_G ranging downward from $100 \mu\text{m}$ to $0.4 \mu\text{m}$, where the increases of ρ_G become dominant as the volume fraction c of inclusions increases. The effect of such dominance of ρ_G would be to override the effects of the statistical processes of strain hardening by truncating the earlier portions of the stress-strain curves. This is shown in Fig. 8.15, where the stress-strain curves of Cu single crystals with a center orientation are shown; from left to right, there are for a pure crystal and three crystals with SiO_2 inclusions of 90 nm diameter, having volume fraction c of 0.0033 , 0.0067 , and 0.01 (Ebeling and Ashby 1966). For the cases of curves 2 and 4, λ_G was $20 \mu\text{m}$ and $2 \mu\text{m}$, respectively, indicating from Fig. 8.14 that while in crystal 2, statistical hardening processes were overridden for strains below about $\gamma = 0.2$, for curve 4 the entire stress-strain curve was dominated by the GNDs.

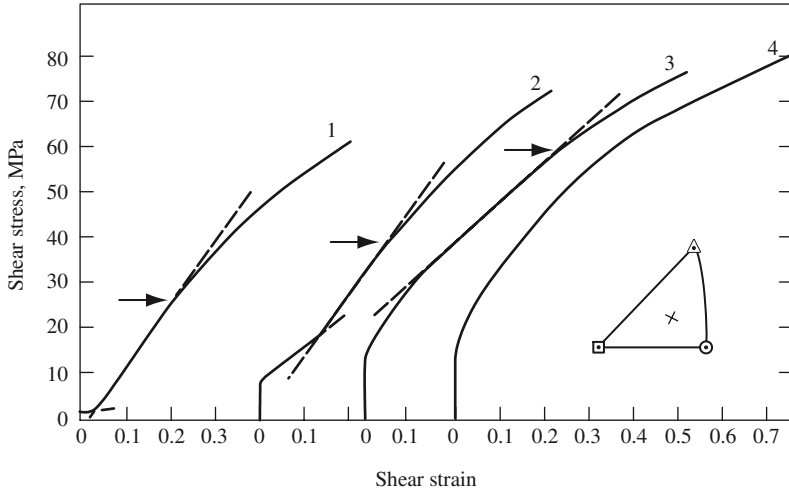


FIG. 8.15. Stress-strain curves of a pure Cu single crystal (1) and identically oriented crystals containing volume fractions $c = 0.0033$ (2), $c = 0.0067$ (3), and $c = 0.01$ (4), of spherical SiO_2 particles of diameter 90 nm (Ebeling and Ashby 1966, courtesy of Taylor and Francis).

Finally, considering that to first order the total dislocation density is given by

$$\rho_T = \rho_s + \rho_G \quad (8.23)$$

and that for purposes of accounting for the flow stress the internal distribution of the different populations may not matter, the total flow stress could be given as

$$\tau_T = \alpha \mu b \sqrt{\rho_T}. \quad (8.24)$$

This would result in a normalized strain-hardening rate of

$$\frac{\Theta}{\mu} = \frac{3}{2} \alpha \frac{(1 + \lambda_G/\lambda_s)}{\lambda_G \sqrt{\rho_T}}, \quad (8.25a)$$

and for ranges where $\lambda_G/\lambda_s \ll 1.0$, in a normalized strain-hardening rate of

$$\frac{\Theta}{\mu} \cong \frac{3}{2} \alpha^2 \left(\frac{\mu}{\tau_T} \right) \frac{1}{\lambda_G/b}, \quad (8.25b)$$

which depends inversely on the length scale of the GNDs. This effect is shown in Fig. 8.16, where the normalized strain-hardening rates in three sets of FCC alloy single crystals are plotted at plastic shear strain levels of 0.05 (Ashby 1970).

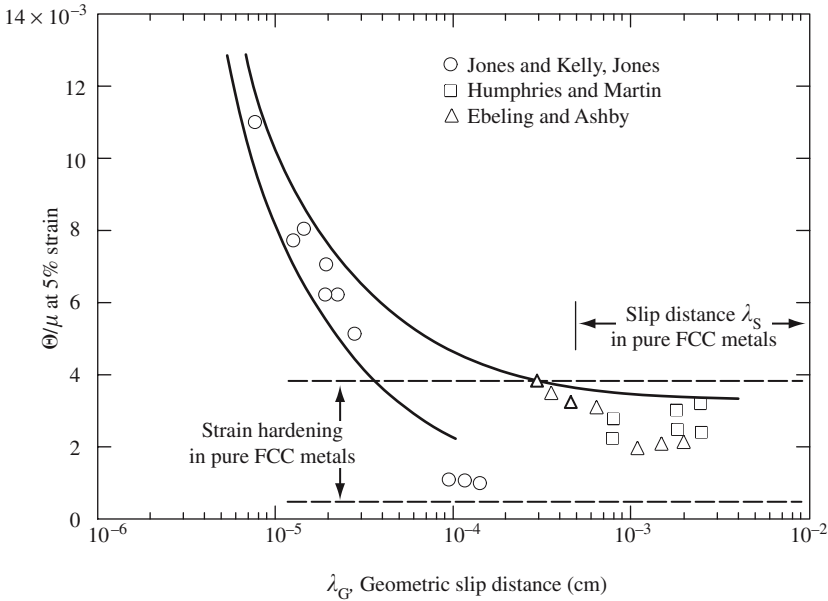


FIG. 8.16. Dependence on the geometrical slip distance λ_G of the rate of strain hardening, Θ , in units of the shear modulus, for cases of dispersion-hardened alloys. The data is related to the work of Jones and Kelly (1968), Jones (1969), Humphreys and Martin (1967), and Ebeling and Ashby (1966) (Ashby 1970, courtesy of Taylor and Francis).

Finally, it must be noted that the integrally mixed nature of the buildup of ρ_G and ρ_s , which was not specifically considered, serves to stabilize the GNDs by a series of inevitable junction reactions between the two sets, as in the case of statistical aggregation of dislocations by themselves. This prevents the GNDs, which are nearly always of the same sign locally, from dispersing or running back when the applied stress is removed.

8.8 Grain Boundary Strengthening

In polycrystals, grain boundaries play two complementary roles. First, as discussed in Section 8.6, since grain boundaries are generally opaque to dislocations passing from one grain into another, compatible deformation of grain assemblies requires selection of different sets of slip systems in different grains. How this results in an elevation of the tensile plastic resistance of polycrystals over the critical resolved

shear stress for pure single crystals by a Taylor factor of 3.1, purely by compartmentalization of slip operations into individual grains, was discussed in Section 8.6.1. Since this analysis was based only on the orientation of slip systems without regard to the mechanism of deformation, the Taylor factor is independent of grain size. Such kinematical considerations also hold for the texture evolution resulting from lattice rotation and are also independent of grain size. However, it is well established that the plastic resistance of polycrystals depends on grain size, with decreasing grain size resulting in increasing plastic resistance, indicating that grain boundaries play a more specific role in strengthening. An explanation of this strengthening effect was advanced by Hall (1951) and Petch (1953) for the initial yield strength of polycrystalline low-carbon steel, based on the observation that in the preyield region some isolated slip activity could be seen in some individual grains but that general yield required percolation of slip processes from grain to grain. It was proposed that this occurs when stress concentrations developed by dislocation pileups initiated in the soft grains and arrested at grain boundaries result in initiation of new slip processes in neighboring harder grains to achieve the required percolation.

Thus, considering a double-ended dislocation pileup on a slip plane in a grain of a polycrystal subjected to an applied tensile stress σ , as a contained shear crack of length d , that is, the grain size, as shown in Fig. 8.17, the concentrated shear stress $\sigma_{xy}(x = \delta)$ at a distance δ away from a grain boundary in an adjoining grain is

$$\sigma_{xy}(x = \delta) = (\sigma - \sigma_0) m \sqrt{\frac{d}{4\delta}} \quad (8.26)$$

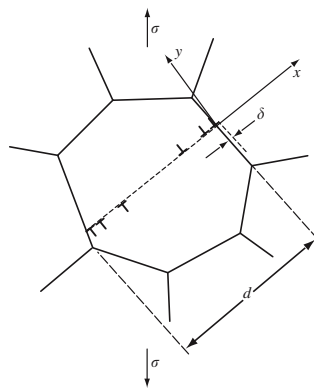


FIG. 8.17. Sketch of a pile-up considered as a Mode II shear crack with flank friction.

(Anderson 1995). In eq. (8.26), σ_0 is a critical tensile stress for initiating relative motion across the faces of the slip plane (the shear crack) in the soft grain and m is the Schmid factor of the slip plane relative to the tensile axis. Then, when σ_{xy} ($x = \delta$) reaches a critical shear stress level σ_{sc} to initiate deformation in neighboring hard grains, percolation of plastic behavior is achieved and $\sigma = \sigma_y$ of the polycrystal is reached. This gives the Hall–Petch relation

$$\sigma_y = \sigma_0 + k_y d^{-1/2}, \quad (8.27)$$

where $k_y = (\sigma_{sc}/m) \sqrt{4\delta}$, and σ_0 is referred to as the friction stress (in tension). While the scenario of Hall and Petch is rational and is supported by experiments for the initial yield stress, a dependence of this type is also found for the flow stress at increasing plastic strains, where there is no evidence for this mechanism or for observable dislocation pileups. Thus, a rationale for such an expression for the flow stress has been sought in other terms. To justify a discussion of alternatives which are based on evidence of different hardening behavior for grains with a different grain size, we consider the experimental findings of Narutani and Takamura (1991). These authors measured the rates of increase of dislocation density in polycrystalline Ni with grain sizes ranging from 22 to 88 μm by measurements of the electrical resistivity as a function of plastic strain. Since the changes in resistivity came from both point defects and dislocations produced in plastic flow, to determine the changes that came from dislocations alone, all deformed samples were given thermal excursions to 623 K, where all point defect contributions were removed.

Figure 8.18 shows the measured change in resistivity as a function of $1/d$, the reciprocal grain size, for experiments carried out at 295 K. Very similar results were obtained also for deformation at 77 K. The resulting expression for the change in resistivity $\Delta\rho_d$ is

$$\Delta\rho_d = \Delta\xi_d(\varepsilon) + \frac{A\varepsilon}{d}, \quad (8.28)$$

where the dependence of $\Delta\xi_d(\varepsilon)$ on strain is monotonic but nonlinear, while the second term shows a linear dependence on strain and reciprocal grain size, with $A = 1.12$ ($\Omega\text{ cm}$) m. Figure 8.19 shows that the flow stress, modified by R , the ratio of the shear modulus at 295 K to that at 77 K, is a unique linear function of the square root of the change in resistivity. Since the resistivity is linearly dependent on the total dislocation density ρ_T , this density too shows the same dependence and is given by

$$\rho_T(\varepsilon) = \frac{(\xi_d(\varepsilon) + A\varepsilon/d)}{\eta}, \quad (8.29)$$

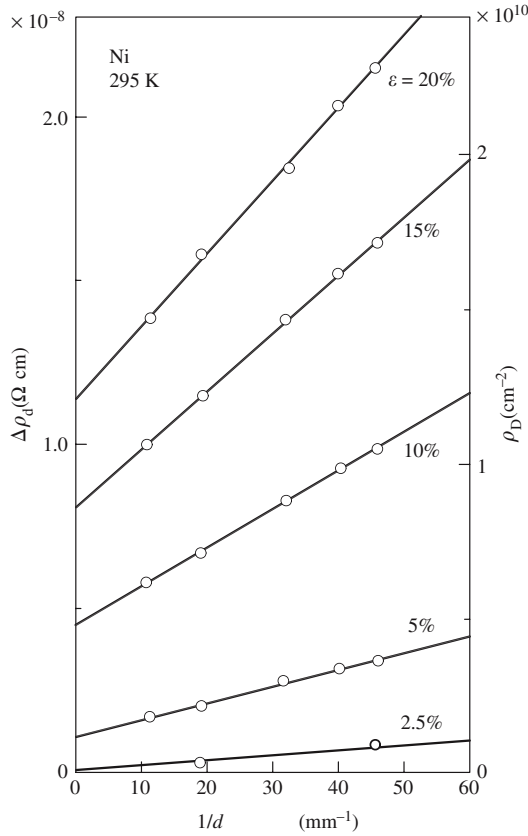


FIG. 8.18. Dependence of the electrical-resistivity change on grain size, resulting from plastic deformation in Ni at 295 K. (Narutani and Takamura 1991, courtesy of Elsevier).

where $\eta = d \Delta\rho_d / d\rho_T = 9.4 \times 10^{-19} \Omega \text{ cm} / (\text{unit dislocation density})$ (Clarebrough *et al.* 1961). It is clear that there are two separate contributions to the total dislocation density, one coming from a grain-size-dependent term and the other being independent of grain size.

Finally, Narutani and Takamura have demonstrated that, as expected, the flow stress depends on the total dislocation density through the well-established relation of eq. (7.2), giving a general, experimentally established relationship for the flow stress

$$\left(\frac{\sigma - \sigma_0}{\alpha \mu b} \right)^2 = \frac{(\xi_d(\epsilon) + A\epsilon/d)}{\eta}, \quad (8.30)$$

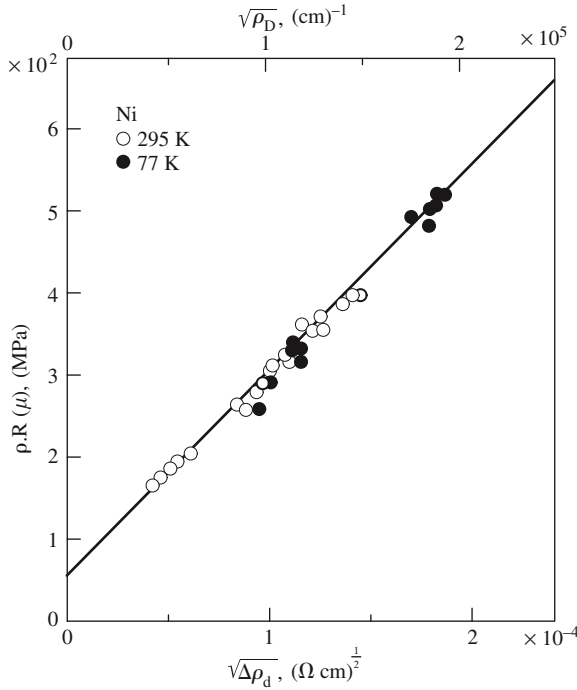


FIG. 8.19. Dependence of “modulus-reduced flow stress” on the change in resistivity ρ_d or on the change in dislocation density ρ_D , resulting from plastic straining at 77 K and 295 K (Narutani and Takamura 1991, courtesy of Elsevier).

where σ_0 represents a friction stress or any component that needs to be overcome to initiate plastic flow. As Fig. 8.18 shows, for small plastic strains less than 0.025, the grain-size-dependent term in eq. (8.30) is dominant. In this case,

$$\sigma \cong \sigma_0 + \frac{\alpha \mu b (A \varepsilon / \eta)^{1/2}}{d^{1/2}}, \quad (8.31)$$

and a flow stress dependence of the Hall–Petch type results. For larger plastic strains, the dependence of the flow stress has both a grain-size-independent and a grain-size-dependent term of Hall–Petch type. Thus, these experiments show that the grain size dependence of the flow stress is an intrinsic feature of the hardening process of polycrystals.

The most satisfactory mechanistic interpretation of this grain size dependence was proposed earlier by Ashby (1970) on the basis of the presence of geometrically necessary dislocations resulting from the additional local deformation gradients that are needed to maintain compatibility among grains. The idea is illustrated in Fig. 8.20, showing in Fig. 8.20(a) a polycrystal that is to undergo plastic flow. As in

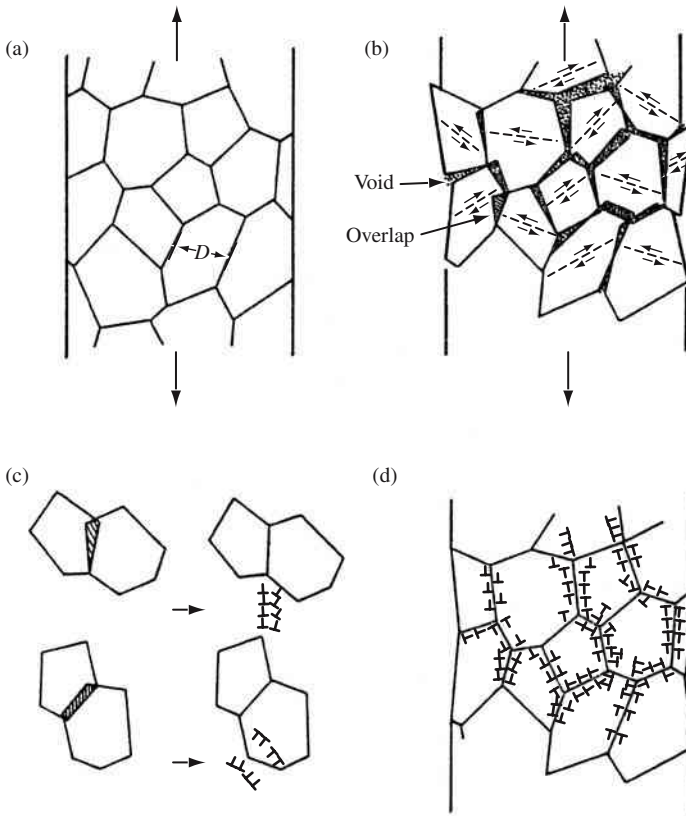


FIG. 8.20. Sketch of Ashby's model of the production of geometrically necessary dislocations in grains of a polycrystal due to dispersal of intergranular deformation-induced incompatibilities resulting from uniaxial straining (Ashby 1970, courtesy of Taylor and Francis).

the Eshelby exercise of Section 8.7.2, if the individual grains are allowed to deform on their best slip systems separately, deformation incompatibilities will arise with dimensions proportional to the grain size, as depicted in Fig. 8.20(b). After fitting the grains together, allowing for an affine contribution equal to the total imposed strain in each grain, there would still be required sets of geometrically necessary dislocations, as depicted in Figs. 8.20(c) and (d). Thus, as in the case of the GNDs associated with rigid inclusions, those that relate to the problem of deformation of polycrystals would have a density with a form of

$$\rho_G = \left(\frac{1}{\lambda_{pc}} \right) \left(\frac{3\gamma}{b} \right), \quad (8.32)$$

where λ_{pc} , the characteristic length of the deforming polycrystal, must be proportional to the grain diameter d . Thus, since the affine component of the deforming grains must be associated with a statistically stored set of dislocations, albeit derived from several interacting sets of slip processes in each grain, the total dislocation density ρ_T for the polycrystal is expected to be of the form given in eq. (8.33) below, in the context of tensile extension,

$$\rho_T = \rho_s(\varepsilon) + \rho_G(\varepsilon) = \rho_s(\varepsilon) + \frac{\beta\varepsilon}{d}, \quad (8.33)$$

where β is a proportionality factor also incorporating $(1/b)$. The introduction of this dislocation density into the flow stress equation gives

$$\left(\frac{\sigma - \sigma_0}{\alpha\mu b} \right)^2 = \rho_s(\varepsilon) + \frac{\beta\varepsilon}{d}, \quad (8.34)$$

which is exactly what the experiments on Ni polycrystals led to. This scenario is also in agreement with the deformation features observed by Hughes *et al.* (1997) and Hughes (2002) in rolled polycrystalline Al and Ni, where dense geometrically necessary boundaries (GNBs) were found. These were absent in plastically deformed unconstrained single crystals, where only open dislocation cells were observed, which Hughes *et al.* (1997) termed incidental boundaries (IDBs). This indicates clearly that the GNBs were due to intercrystalline incompatibilities, as hypothesized in the Ashby model.

8.9 Plasticity in Metals with Nanoscale Microstructure

In homogeneous metals, the flow stress increases with decreasing grain size with a Hall–Petch-type dependence given in eq. (8.34). However, this dependence begins to break down when the grain size decreases into the low nanometer range, where other limiting deformation mechanisms prevail. Some experiments, but mostly computer simulations, have shown that inside grains of size 10–50 nm the development of the required dislocation fluxes for plastic flow begins to become severely curtailed. Instead, simulations (Schiotz *et al.* 1998; Van Swygenhoven *et al.* 1999, 2001; Yamakov *et al.* 2003; Schiotz and Jacobsen 2003) have shown that at such high stress levels a substantial contribution to plastic strain production comes from grain boundary shear, which, however, is associated also with sporadic emission of partial or full dislocations from grain boundary triple junctions or from grain boundary ledges. This grain boundary shear has also been observed to result in considerable restructuring of grain shapes to create continuous shear zones across the polycrystals (Demkowicz *et al.* 2007), often creating a pattern of mutually perpendicular shear zones, resembling the slip line fields of the mathematical theory of

plasticity (Hill 1950). Upon the development of these additional flow mechanisms, the flow stress no longer rises with decreasing grain size but levels off and begins to decrease. Since grain boundary shear does not satisfy compatibility by itself, dislocations are emitted on demand from regions of stress concentration, such as triple grain boundary junctions, to satisfy compatibility. All the indications are that this grain boundary shear is close to a threshold process, occurring near the ideal shear strength of the high-angle grain boundaries. As a consequence, with decreasing grain size, as the grain boundary area per unit volume increases, the flow stress peaks and begins to decrease. This behavior has been termed by Yip (1998, 2004) “the strongest size”. Because of the complexity of the observed phenomena, there have been no models to describe the “turnaround” behavior with decreasing grain size. On the basis of high-temperature simulations, Yamakov *et al.* (2003) have even suggested that the behavior may involve Coble creep-type diffusional flow, which at low temperatures is unlikely to be present because of the extremely high levels of stress, where shear mechanisms almost certainly overtake processes of diffusional matter transport. Here we present a model in keeping with the observations that explains the “turnaround” behavior in the flow stress.

Thus, we consider two competing and also complementary flow mechanisms: grain boundary shear and dislocation plasticity. We assume that the various bridging mechanisms of dislocation injection from stress concentration sites broadly attend to the problem of satisfying compatibility.

We view the grain boundary shear as a flow process in an amorphous metal, albeit constrained into a disordered layer of only a thickness δ , where individual thermally assisted local shear transformations produce a transformation shear strain γ^T (Argon 1979; Argon and Shi 1982). This would give a volume-averaged grain boundary shear strain rate contribution of

$$\dot{\gamma}_{\text{GB}} = \gamma^T \left(\frac{6\delta}{d} \right) \nu_{\text{D}} \exp \left\{ -\frac{Q_{\text{v}}}{kT} \left(1 - \frac{\sigma}{\hat{\tau}_{\text{is}}} \right) \right\}, \quad (8.35)$$

where $6\delta/d$ is the volume of grain boundary material per grain of diameter d ; ν_{D} is the atomic frequency; Q_{v} is the activation energy for viscous flow in an amorphous metal, taken as a terminal level of a narrow distribution of activation energies for shear relaxation (Argon and Kuo 1981); and $\hat{\tau}_{\text{is}}$ is the ideal shear resistance of the disordered grain boundary material, which can be taken to be roughly half the ideal shear strength across an octahedral slip plane in FCC Cu (Ogata *et al.* 2002). The form of eq. (8.35) is rational, in the absence of more precise information about such a shear process.

The corresponding shear rate caused by the process of dislocation plasticity can be taken as

$$\dot{\gamma}_{\text{D}} = \left(\frac{b}{d} \right) \nu_{\text{G}} \exp \left\{ -\frac{\Delta F_{\text{j}}}{kT} \left(1 - \frac{\sigma}{\hat{\tau}} \right) \right\}, \quad (8.36)$$

where it is assumed that plasticity occurs by a succession of thermally assisted releases of dislocations from pinning points, with each traversing the entire grain of diameter d after the release, producing shear strain increments of b/d . In this rate process, we assume that the release is associated with a thermally assisted forest dislocation intersection, where the barrier is that of the jog energy, ΔF_j . We take the threshold shear resistance to be $\beta\mu b\sqrt{\rho}$, where β is an empirical proportionality constant to be determined below; ρ , in the case of nanoscale polycrystals, is considered to be made up of a geometrically necessary set of dislocations, given by $\rho = C\gamma/d$, as postulated by Ashby (1970) and experimentally established by Narutani and Takamura (1991), introduced at the inception of plastic flow, with γ ($= \gamma^p$) taken as the initial plastic shear strain, of the order of the elastic shear strain at yield. The proportionality constant C , with dimensions of reciprocal length, is derived from the dislocation multiplication measurements of Narutani and Takamura, inside grains of polycrystals in the micrometer range, in the early stages of plastic flow. The frequency factor ν_G is taken to be in the range of $10^{-3}\nu_D$, appropriate to dislocation intersections. The existence of such dislocation intersections can be gleaned from the computer simulations of Schiotz and Jacobsen (2003). In an alternative scenario for the dislocation plasticity mechanism, which will not be developed here but can be found elsewhere (Argon and Yip 2006), the rate-controlling process can be conceived of as the thermally assisted emission of dislocation embryos from stress concentration sites such as grain boundary triple junctions.

The rate-dependent forms of $\dot{\gamma}_{GB}$ and $\dot{\gamma}_D$ are clumsy for direct analytical treatment. Noting that the processes will be operating close to their threshold level, we make the convenient phenomenological replacements

$$\exp\left\{-\frac{Q_v}{kT}\left(1 - \frac{\sigma}{\hat{\tau}_{is}}\right)\right\} \rightarrow \left(\frac{\sigma}{\hat{\tau}_{is}}\right)^{m_1}, \quad (8.37a)$$

$$\exp\left\{-\frac{\Delta F_j}{kT}\left(1 - \frac{\sigma}{\hat{\tau}}\right)\right\} \rightarrow \left(\frac{\sigma}{\hat{\tau}}\right)^{m_2}. \quad (8.37b)$$

The best phenomenological fit for the grain boundary shear process for σ approaching $\hat{\tau}_{is}$ gives $m_1 \cong 30$, for Q_v equal to that for grain boundary diffusion, that is, roughly 0.6 eV for Cu. For the dislocation plasticity mechanism, taking $\Delta F_j \approx 0.2\mu b^3$, a similar fit would give $m_2 = 60$. Since both deformation processes are nearly athermal, we take $m_2 = m_1 = m = 30$ to simplify the analysis, at little cost to the accuracy, since at such extreme levels of behavior, changes in the phenomenological stress exponent result in only negligible changes in the strain rate dependence of the plastic resistance. Naturally, if deformation at temperatures other than room temperature were under consideration, m would have to be temperature-dependent, increasing with decreasing temperature below room temperature.

Considering the polycrystal simply as a set of shearing grain boundaries and intervening strips of matrix, representing homogenized grains in series with the grain boundaries, the two mechanisms would both contribute to the overall strain rate, in proportion to the volume fractions of the solid in which they operate, which gives

$$\dot{\gamma} = f \dot{\gamma}_{\text{GB}} + (1 - f) \dot{\gamma}_{\text{D}}, \quad (8.38)$$

where $f = 6(b/d)$ is the volume fraction of the grain boundary material, taking $\delta = b$ as is commonly assumed. We then obtain

$$\dot{\gamma} = \gamma^T f v_{\text{D}} \left(\frac{\sigma}{\hat{\tau}_{\text{is}}} \right)^m + (1 - f) \left(\frac{b}{d} \right) v_{\text{G}} \left(\frac{\sigma}{\hat{\tau}} \right)^m. \quad (8.39)$$

We note that both f and $\hat{\tau}$ depend on the normalized grain size d/b . To proceed, we take the detailed experimental determination of Narutani and Takamura (1991) for the dependence of the (geometrically necessary) dislocation density on grain size at inception of flow to determine the form of the threshold resistance $\hat{\tau}$ for the dislocation plasticity mechanism. Considering an initial plastic shear strain of $\gamma^{\text{p}} = 0.025$ in the evaluation of the factor C , we obtain $C = 6.37 \times 10^7/\text{cm}$ (per unit shear strain). This gives the relevant dislocation density as $\rho = 0.033/(bd)$, and with the value of the empirical constant $\beta = 1.35$ determined by Narutani and Takamura from their experiments on dislocation multiplication, establishes further an expression for the grain size dependence of the threshold plastic resistance,

$$\hat{\tau} = \alpha \mu \sqrt{\frac{b}{d}}, \quad (8.40)$$

with $\alpha = 0.246$. We assume that this expression, obtained in the micrometer range of grain size, projects from the micron to the nanometer range intact.

Introducing the forms of $f = 6(b/d)$ and $\hat{\tau}$ into eq. (8.39) then gives

$$\dot{\gamma} = v_{\text{D}} \left(\frac{\sigma}{\mu} \right)^m \left[\left(6 \gamma^T \frac{b}{d} \right) \left(\frac{\mu}{\hat{\tau}_{\text{is}}} \right)^m + \left(1 - 6 \left(\frac{b}{d} \right) \right) \left(\frac{v_{\text{G}}}{v_{\text{D}}} \right) \left(\frac{1}{\alpha} \right)^m \left(\frac{d}{b} \right)^{m/2} \right]. \quad (8.41)$$

The form of eq. (8.41) suggests that for a constant imposed shear strain rate $\dot{\gamma}$, the flow stress will have a maximum at a certain normalized grain size d/b and that below this grain size, the first term in brackets in eq. (8.41) will dominate through the grain boundary shear process, while above the critical grain size dislocation plasticity will be dominant.

Little definitive information is available for the above dislocation plasticity mechanism in this extreme range beyond what has been stated above. Nevertheless, it is necessary to choose certain material parameters to make it possible for the model

to be compared with computer simulations, such as that of Schiotz and Jacobsen (2003) (SJ). We shall make these choices on the basis of the best available material information or in the most reasonable manner, consistent with the nature of the Hall–Petch phenomenon.

We note first that the peak tensile flow stress reported by SJ is 2.3 GPa, which, for a von Mises factor of $\sqrt{3}$, gives a peak flow stress in shear of 1.33 GPa at a normalized grain diameter d/b of 48, considering the normalization factor b as the Burgers vector of a full dislocation in Cu. Introducing a dimensionless shear flow stress $s = \sigma/\mu$ and a dimensionless grain size $x = d/b$, we obtain from eq. (18.41), through an extremum condition

$$\left(\frac{\partial s}{\partial x}\right)_{\dot{\gamma}} = 0, \quad (8.42)$$

a characteristic equation for the critical grain size x_p for peak resistance

$$x^{(m/2)+1} - \left(\frac{6(m-2)}{m}\right)x^{m/2} - Q = 0, \quad (8.43)$$

where

$$Q = 12\gamma^T \frac{(v_D/v_D)(\alpha/(\hat{\tau}_{is}/\mu))^m}{m}. \quad (8.44)$$

The dimensionless peak flow stress s_p , for a dimensionless grain size x_p or for any other dimensionless grain size x can then be obtained from the flow stress equation derivable from eq. (8.41), that is,

$$s = \left(\frac{\dot{\gamma}}{v_D}\right)^{1/m} \left[\frac{A}{x} + Bx^{m/2} - 6Bx^{(m/2)-1} \right]^{-1/m}, \quad (8.45)$$

where

$$A = \frac{6\gamma^T}{(\hat{\tau}_{is}/\mu)^m}, \quad B = \frac{v_G/v_D}{\alpha^m}. \quad (8.46a,b)$$

To proceed further, we make the following best choices of parameters: $\gamma^T = 0.015$ for the characteristic transformation shear strain, typical of plastic flow of amorphous metals or network solids (Argon and Demkowicz 2006); $\hat{\tau}_{is} = \mu/30$ (Argon 1979); $\alpha = 0.246$ and $m = 30$, both as stated above; $v_D = 10^{13} \text{ s}^{-1}$ and $v_G/v_D = 10^{-3}$, also as stated above; and $\dot{\gamma} = 10^8 \text{ s}^{-1}$, as used by SJ for direct comparison with their results. We note that the latter strain rate is inordinately high in comparison with realistic strain rates and was chosen as such by SJ only for operational purposes. However, as is clear from eq. (8.45), and as already commented on above, its effect on the flow stress is drastically attenuated. This gives

$Q = 6.60 \times 10^{26}$, resulting, from eq. (8.43), in a dimensionless grain size of $x_p = 47.8$ (or $d = 12.2$ nm for Cu), to be compared with the $x_p = 48$ of SJ. Using this dimensionless grain size in eq. (8.45) gives a dimensionless peak stress of $s_p = 0.0276$ (or a peak flow stress of $\sigma_p = 1.52$ GPa), which compares reasonably well with the 1.33 of SJ. Finally, we note that when $d \rightarrow b$ and the material becomes amorphous, the dimensionless shear resistance becomes $s_0 = 0.024$, or 13.4% lower than the dimensionless peak resistance s_p . When $x \gg x_p$, the dimensionless flow stress becomes

$$s = \left(\frac{\dot{\gamma}/v_D}{B} \right)^{1/m} \sqrt{\frac{b}{d}} = 0.1 \sqrt{\frac{b}{d}}, \quad (8.47)$$

which is of the expected Hall–Petch form. We note that this results from the form of eq. (8.40), which was based purely on the Narutani and Takamura experiments on dislocation multiplication in grains of a polycrystal, with no *a priori* use of the Hall–Petch relation.

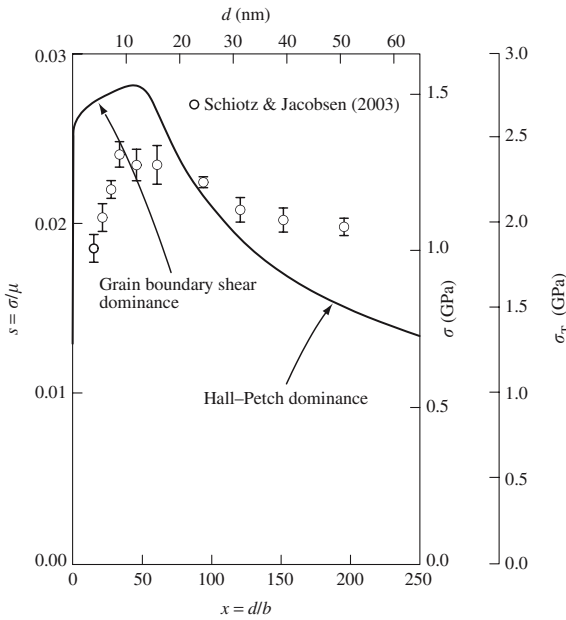


FIG. 8.21. The model behavior for polycrystalline Cu given by eq. (8.45) is shown by the continuous curve using the chosen material parameters presented in the text and is contrasted with the simulation results of Schiotz and Jacobsen (2003), where the latter results have been converted into von Mises shear resistances represented by σ . The far right axis gives the tensile flow stress σ_T , as reported by Schiotz and Jacobsen (Argon and Yip 2006).

The overall model results for the dependence of the shear flow stress on grain size in the peak resistance range are plotted in Fig. 8.21, together with the simulation results of SJ. Clearly, the model results parallel the computer simulation results very well, with neither probably being a fully correct reflection of the actual dependence (Argon and Yip 2006).

8.10 Superposition of Deformation Resistances

In a realistic strengthening process, more than a single resistance mechanism will usually be present. Combining strengthening mechanisms for the purpose of achieving desirable properties in industrial applications is a never-ending goal in alloy development and materials processing, where empiricism still reigns. The combination of intrinsic properties, the effect of alloy composition, and manipulation of microstructure through heat treatment, strain hardening, and the development of deformation-induced texture, and their various interactions are a very broad and complex set, making it difficult to prescribe ideal process paths. Here, we shall not attempt to provide general steps to achieve such goals of property enhancement, but consider only some simple cases where simple superposition of resistances has been applicable.

Considering only the resistances of the lattice, solutes, precipitate particles, and the dislocation resistance, together with their evolution through strain hardening, an assumption widely subscribed to is that, provided the principal length scales of the mechanisms are far apart, the resistances should be linearly superposable for a given state of the microstructure (Kocks *et al.* 1975). However, recalling the findings of the earlier chapters on the interaction of resistances makes it clear that this simple assumption is rarely satisfied.

One of the few instances where the simple linear superposition appears to have support is in FCC metals with a vanishing lattice resistance, where the solid-solution resistance and dislocation resistance are the only two rate-dependent components of the flow stress. This was demonstrated by Mulford (1979) and Mulford and Kocks (1979) in Ni and two Inconel alloys. In this case the basic length scale Λ of the solid-solution resistance, representing the dislocation segment length (eq. 5.41) able to coherently advance through a solute field, is indeed much smaller than the forest dislocation spacing l that governs the flow stress in Stage II of pure FCC metals.

Thus, Mulford and Kocks assumed that the temperature-sensitive components σ_1 and σ_2 of the solid-solution resistance (1) and the dislocation resistance (2) are linearly additive, that is,

$$\sigma = \sigma_1 + \sigma_2. \quad (8.48)$$

Moreover, they noted that in the thermally assisted advance of a dislocation through either a solute field or a field of forest dislocations alone, the works of activation ΔW_i for the two processes separately are

$$\Delta W_1 = \sigma_1 b \Delta a_1, \quad \Delta W_2 = \sigma_2 b \Delta a_2, \quad (8.49a,b)$$

respectively, where the $\Delta a_i = \Delta y_i l_i$ are the activation areas of the two separate mechanisms in isolation, and where the Δy_i are the respective activation distances and the l_i (Δ , and l) are the respective characteristic length scales of the two mechanisms in isolation. When the two mechanisms coexist and a dislocation moves through the combined field, the activation works ΔW_1 and ΔW_2 must necessarily be the same, and will also be the activation work ΔW manifested in the combined set of mechanisms during the activated advance of the dislocation, sampling the two sets of obstacles together as it sweeps across a slip plane over larger distances. Thus,

$$\frac{d(1/\Delta a_i)}{d\sigma_i} = \frac{b}{\Delta W_i} = \frac{b}{\Delta W} = C_i, \quad (8.50)$$

where C_i is a constant for a given state of the microstructure, giving also

$$\frac{1}{\Delta a_i} = \frac{b}{\Delta W} \sigma_i \quad (8.51)$$

and, consequently,

$$\frac{1}{\Delta a} = \frac{1}{\Delta a_1} + \frac{1}{\Delta a_2} \quad (8.52)$$

if the assumption of eq. (8.48) is correct.

If, under this condition, one set of obstacles, that is, either the solute concentration or the forest dislocation density, could be kept constant while the other set was varied, strain rate change experiments that probe the entire set would sense only the set that was varied. Thus, as shown in Fig. 8.22a the measured changes in the reciprocal effective activation area with stress would be parallel to one of two possible straight lines. Experimentally, the alternative that can be readily achieved is to change the forest dislocation density (2) by strain hardening in an alloy with a fixed solute content (1), on the assumption that plastic straining does not alter the solute distribution. Then,

$$d\sigma = d\sigma_2, \quad (d\sigma_1 = 0), \quad d\left(\frac{1}{\Delta a}\right) = d\left(\frac{1}{\Delta a_2}\right). \quad (8.53a,b)$$

However, since

$$\Delta a_2 = \Delta y_2 l_2, \quad (8.54)$$

where the activation distance Δy_2 should not change at constant temperature,

$$\frac{d(1/\Delta a)}{d\sigma} = \frac{d(1/\Delta a_2)}{d\sigma_2} = \frac{(1/y_2) d(1/l_2)}{d\sigma_2} = \frac{b}{\Delta W} = C_2, \quad (8.55)$$

showing that the measured change in $1/\Delta a$ with σ should indeed directly reflect the change in $1/\Delta a_2$ with σ_2 . Since the flow stresses due to the solute and the forest dislocations both obey relationships of the type

$$\sigma_i = \left(\beta_i^{3/2}\right) \left(\frac{2\mathcal{E}}{bl_i}\right), \quad (8.56)$$

where the β_i are the respective normalized peak resistances of the obstacles, then,

$$\frac{d(1/l_2)}{d\sigma_2} = \left(\frac{1}{\Delta y_2}\right) \left(\frac{b}{\Delta W}\right) = \left(\frac{b}{2\mathcal{E}}\right) / \left(\beta_2^{3/2}\right) = C_2 \quad (8.57a)$$

in the above scenario. In the corresponding other alternative,

$$\frac{d(1/l_1)}{d\sigma_1} = \left(\frac{1}{\Delta y_1}\right) \left(\frac{b}{\Delta W}\right) = \frac{b/2\mathcal{E}}{\left(\beta_1^{3/2}\right)} = C_1, \quad (8.57b)$$

and since

$$\beta_1 \ll \beta_2, \quad (8.58a)$$

it follows that

$$C_1 > C_2. \quad (8.58b)$$

The measured changes in $(1/\Delta a)$ with stress for changes in the forest dislocation density should then be along a line with slope C_2 representing a less rate-dependent obstacle, while in the complementary situation, where the solute concentration is varied for a given forest dislocation density, the corresponding straight line would have a steeper slope given by C_1 , representing a more rate-dependent obstacle, as sketched out in Fig. 8.22(a).

Mulford (1979) carried out such an experiment on pure Ni and two Ni alloys: a solid-solution-strengthened Inconel 600 and an age-hardened Inconel X750 containing impenetrable γ' (Ni_3Ti , Al) precipitates. His results are shown in Fig. 8.22(b). The changes in $1/\Delta a$ with σ for pure Ni follow a straight line going through the origin as required by the Cottrell–Stokes law, while the changes in the solid-solution-strengthened Inconel 600 follow a line with slope C_2 as expected, proving indeed that the assumption of eq. (8.48) was correct. For the case of the Inconel X750 alloy with impenetrable γ' precipitates, where the latter impose a

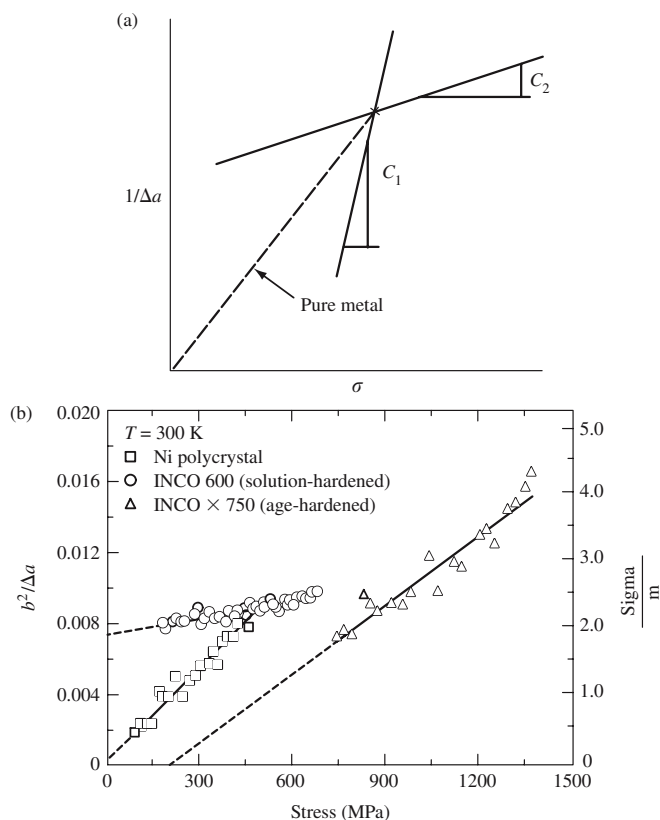


FIG. 8.22. (a) In a solid-solution alloy of an FCC metal, the dependence of the reciprocal activation area of the temperature-sensitive component of the flow stress on flow stress. Starting with a pure metal obeying the Cottrell–Stokes law, if a variation of either the forest density (2) or the alloy concentration (1) is produced as a thought experiment, $1/\Delta a$ will change along a line with a slope C_2 or C_1 , respectively. (b) Experiments in Inconel 600 and in Inconel X750 plotted in comparison with results for pure Ni. See text for explanation (Mulford 1979, courtesy of Elsevier).

rate-independent threshold resistance σ_0 , the dependence of $1/\Delta a$ will follow a line

$$\frac{1}{\Delta a} = \frac{1}{\Delta a_2} = \left(\frac{b}{\Delta W} \right) (\sigma - \sigma_0), \quad (8.59)$$

with a slope nearly parallel to that for pure Ni, as appears to be nearly the case.

Apart from this well-documented case a linear superposition should also be possible between solid-solution strengthening and precipitation strengthening in

FCC metals. However, in most other cases, the superposition of resistances is more complex.

In BCC metals, where the lattice resistance affects primarily screw dislocations as discussed in Chapter 5, solute atoms interact with kinks on screw dislocations, not individually but in clusters, and such solute atoms can either result in elevation of the flow stress or produce solution softening if the effect of the solute on the elastic properties is a reduction of the shear modulus, as is most often the case. There are also interactions of solute atoms with edge dislocations, much like those in FCC metals, but these effects are shadowed by the interaction with screw dislocations, as discussed also in Chapter 5.

In, the case of precipitates, where several individual mechanisms can coexist for the same precipitate, the superposition of mechanisms occurs in the force–distance curves. On the other hand, the superposition of the effects of different precipitate particle types on the flow stress occurs by the Pythagorean sum, that is, the square root of the sum of the squares of the individual flow stresses due to different particle types, as discussed in Chapter 6.

Solute atoms, in their interaction with dislocations in FCC metals, also exhibit instabilities at both cryogenic temperatures, where the very low levels of phonon damping result in dynamic-overshoot softening as discussed in this chapter, and at temperatures in the plateau region, where short-range solute ordering can occur inside dislocation cores, in which case Portevin–Le Chatelier effects such as jerky glide phenomena follow, again as discussed in this chapter.

8.11 The Bauschinger Effect

When monotonic plastic deformation is interrupted and reversed, the initial form of response is reverse elastic deformation as the concurrently elastically strained lattice unloads. This effect, however, is soon followed by a departure from a pure elastic response in the form of reverse plastic straining, which often begins well before the elastic unloading is completed, as depicted in Fig. 8.23. Continued reverse deformation will then show a very soft initial response on the way to large reverse plastic flow in the form of a rounded stress–strain curve, as depicted in Fig. 8.23. This gradual and softened onset of reverse plastic flow is referred to as the *Bauschinger effect*, after its discoverer (Bauschinger 1886).

The general characteristics of the Bauschinger effect are that (a) the Bauschinger strain β_n at any point in the reverse-deformation half-cycle increases with the initial flow stress level, and (b) upon reaching the condition of reverse flow, the material shows a permanent softening, as illustrated in Fig. 8.23. Moreover, these features of the reverse behavior are relatively resistant to removal by mild forms of heat treatment that do not produce major alterations in the total dislocation density and the microstructure.

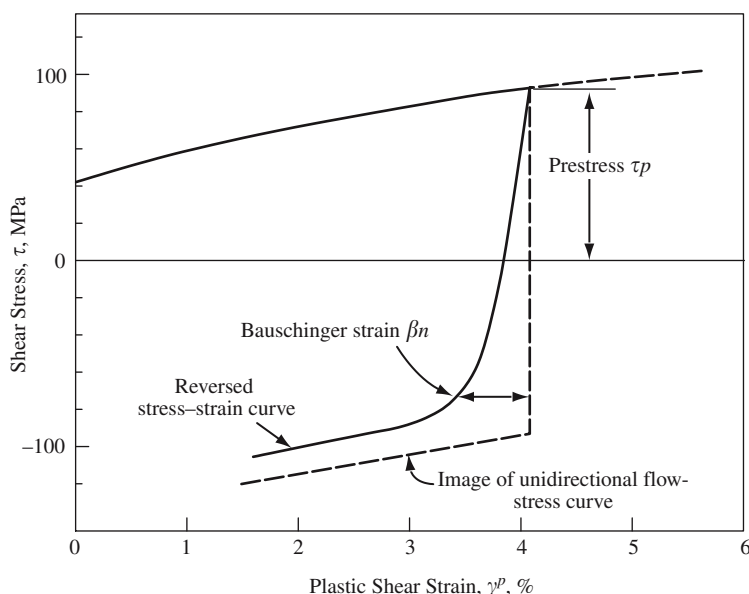


FIG. 8.23. The Bauschinger effect in torsion in decarburized polycrystalline steel (after Deak 1961).

The Bauschinger effect manifests itself most strongly in polycrystalline materials. It is, however, present also in single slip. This was investigated in the case of basal slip in Zn single crystals by Edwards and Washburn (1954), who probed the deformation resistance not only in the reverse direction but also in other directions in the basal plane, between the forward and the fully reversed directions. They found that while there was early inelastic response in all directions, the departure from elastic unloading was maximum in the fully reversed direction. A very prominent form of Bauschinger effect was reported by Brown and Stobbs (1975) and Asaro (1975) in lightly dispersion-strengthened Cu single crystals, where the Orowan loops of dislocations shown in Fig. 6.5d, and other polarized dislocation structures left around impenetrable dispersoids, set up image stresses in the matrix which resulted in the observed soft initial reverse response. These loops, however, are also responsible for the permanent softening effect when long-range reverse deformation begins and the glide dislocations moving in the reverse direction remove some of the Orowan loops.

The fundamental cause of the Bauschinger effect is the inhomogeneous nature of plastic flow, in which plastic strain increments, whether by slip or by shear transformations, occur in more intense form in separate volume elements, resulting in the continued development of residual back stresses in the elastic or less intensely deformed background, as was discussed in detail in Section 7.3.5 for the case of the

development of dislocation microstructures. Thus, upon reversal of the strain, the residual stresses around the plastic inhomogeneities will hasten reverse deformation in the regions that have contributed to the deformation most recently. This process intensifies throughout strain hardening. In a polycrystalline material, where grain boundaries are natural barriers to long-range slip, the development of grain-range residual stresses accentuates the Bauschinger effect, owing in part to the buildup of a geometrically necessary dislocation density as was discussed in Section 8.7.

8.12 Phenomenological Continuum Plasticity

8.12.1 Conditions of Plastic Flow in the Mathematical Theory of Plasticity

As in the parallel cases of the kinetic theory of gases, dealing with the motions of individual molecules, and the continuum mechanics of compressible fluids, dealing with the flow of dense gases, macroscopic problems of plastic flow on the technological level are best handled by the continuum theory of plasticity and not by dislocation mechanics. Here we merely describe the motivation for expecting a connection between crystal plasticity mechanisms and this formal theory of plasticity.

In its rate-independent idealization, the theory of plasticity takes its cue from crystal plasticity by starting with the assumptions that (a) a plastic continuum is incompressible, and (b) deformation cannot be produced by a pure mean normal stress (pressure) but requires a critical deviation away from a state of pure pressure. Thus, it is taken at the outset that plastic work is produced by the stress deviators s_{ij} , which are related to the full stresses σ_{ij} by subtraction of the mean normal stress:

$$s_{ij} = \sigma_{ij} - \delta_{ij}\sigma_m, \quad (8.60)$$

where $\delta_{ij} = 1$ for $i = j$ and 0 for $i \neq j$, and σ_m is the mean normal stress,

$$\sigma_m = \frac{1}{3} (\sigma_{11} + \sigma_{22} + \sigma_{33}), \quad (8.61)$$

where σ_{11} etc. are the three normal stresses at a point. Then, in an initially isotropic solid, plastic flow in three dimensions is initiated when the equivalent stress in shear σ_e , defined by the deviatoric yield condition or the von Mises condition, stated as

$$\sigma_e \equiv \sum_{i,j}^3 \sqrt{\frac{1}{2}s_{ij}s_{ij}} \equiv k = \beta\tau, \quad (8.62)$$

is satisfied. In eq. (8.62), τ is the individual critical slip plane shear resistance, and $\beta(n_T/\sqrt{3})$ is the corresponding Taylor factor connecting the critical shear resistance τ of the slip plane to the macroscopic plastic shear resistance k (where $k = Y/\sqrt{3}$,

with Y being the polycrystalline plastic resistance in tension) (McClintock and Argon 1966).

In the rate-dependent form of the yield condition, τ is replaced by the temperature- and strain-rate-dependent shear strength related to $\hat{\tau}$ by eq. (6.54). This makes $k = k(T, \dot{\gamma})$ temperature- and strain-rate-dependent.

One way in which the mathematical theory of plasticity deals with strain hardening is by assuming that the fundamental polycrystalline stress/strain relation derived from a tension experiment giving $Y = Y(\varepsilon^p)$, where ε^p is the tensile plastic strain (or its shear counterpart $k = k(\gamma^p)$ where γ^p is the plastic shear strain) applies also in three-dimensional deformation. For this, the tensile plastic strain must be generalized to a so-called work-equivalent tensile plastic strain ε_e^p (or its shear counterpart γ_e^p) defined in incremental form by

$$d\varepsilon_e^p \equiv \sum_{ij}^3 \sqrt{\frac{2}{3}} de_{ij} de_{ij} \equiv \frac{d\gamma_e^p}{\sqrt{3}}, \quad (8.63)$$

where the strain deviators de_{ij} are obtained by formally subtracting the dilatational increment $d\theta (= \sum_i d\varepsilon_{ii})$ from the strain increment, that is,

$$de_{ij} = d\varepsilon_{ij} - \delta_{ij}d\theta, \quad (8.64)$$

In this equivalent description of three-dimensional behavior, the kinetic shear rate expressions of eqs. (2.43) and (6.51) for the slip plane translate to the three-dimensional polycrystalline form to result in an expression of the type

$$\dot{\gamma}_e^p = \dot{\gamma}_0 \exp \left\{ -\frac{\Delta F_0}{kT} \left(1 - \left(\frac{\sigma_e}{\beta \hat{\tau}} \right)^p \right)^q \right\} = \sqrt{3} \dot{\varepsilon}^p, \quad (8.65)$$

where $\dot{\gamma}_0$, ΔF_0 , $\hat{\tau}$, and the exponents p and q still relate to the slip plane properties.

How the solid strains plastically in three dimensions, once the generalized yield condition of eq. (8.62) is satisfied, and how the individual plastic strain increments are distributed according to the so-called associated flow rule, is beyond our scope here and can be found in many references on continuum plasticity (see, for example, McClintock and Argon 1966).

8.12.2 Transition from Dislocation Mechanics to Continuum Mechanics

In the analogous problem of gas flow, a transition from molecular flow (Knudsen flow) to compressible continuum fluid mechanics is made when the mean free path length of the gas molecules becomes very small on the scale of the system. Unfortunately, in the plastic deformation of solids, the conditions for the transition from crystal plasticity to continuum plasticity are not so simple. The effects of

crystal plasticity can persist to quite a large scale. We shall demonstrate this here in the special case of the precipitate-particle resistance mechanism, where we consider initially shearable precipitates and progressively coarsen them until they become nonshearable and quite large, to investigate at what scale the problem reverts to the well-known rule of mixtures of continuum mechanics.

Consider a volume fraction c of initially shearable particles of diameter d having an internal plastic shear resistance τ_p . The overall (composite) shear resistance (according to Fig. 6.5) is then given by

$$\tau_c = (1 - c) \tau_m + \frac{\hat{K}}{b\lambda}, \quad (8.66)$$

where $\hat{K} = \tau_p b d$ is the peak precipitate shear force and λ is the Friedel sampling length, as defined by eq. (6.16a). Then, using eq. (6.16a), we obtain eq. (6.18) in modified form,

$$\tau_c = \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{3/2} \left(\frac{2\mathcal{E}}{bd} \right) \left(\frac{3c}{2\pi} \right)^{1/2} + (1 - c) \tau_m. \quad (8.67)$$

The corresponding shear resistance τ_{cc} in the continuum model, by the rule of mixtures, based on an upper-bound consideration of uniform strain in all components, is given by

$$\tau_{cc} = c\tau_p + (1 - c) \tau_m. \quad (8.68)$$

We note that eq. (8.67) can also be written as

$$\tau_c = \tau_p \left(\frac{\hat{K}}{2\mathcal{E}} \right)^{1/2} \left(\frac{3c}{2\pi} \right)^{1/2} + (1 - c) \tau_m, \quad (8.69)$$

where $\tau_c > \tau_{cc}$, because, as already noted in Chapter 3, $(\hat{K}/2\mathcal{E})^{1/2}(3c/2\pi)^{1/2} > c$. This is a direct demonstration of the effectiveness of the nonlocal nature of precipitate strengthening in comparison with the continuum model. The difference between the nonlocal dislocation model and the continuum model increases further as $(\hat{K}/2\mathcal{E}) \rightarrow 1.0$ and the precipitates become nonshearable. When $(\hat{K}/2\mathcal{E}) > 1.0$, however, and the precipitate resistance decreases by overaging, a new resistance emerges in the form of a long-range image stress τ_i , produced by the Orowan loops left around the precipitates of the type shown in Fig. 6.5(d). To a first approximation, this image stress is

$$\tau_i = n \left(\frac{2\mathcal{E}}{bd} \right) c, \quad (8.70)$$

where n is the number of Orowan loops left around particles. When n becomes, n_s , the steady-state loop number, the particle is sheared under the pinching stress of

the loops, step by step, since presumably

$$n_s \frac{2\mathcal{E}}{bd} = \tau_p. \quad (8.71)$$

Then, in this regime of $(\hat{K}/2\mathcal{E}) > 1.0$, the composite resistance becomes

$$\tau_c = \tau_p c + \frac{2\mathcal{E}}{bd} \left(\frac{3c}{2\pi} \right)^{1/2} + (1 - c) \tau_m, \quad (8.72)$$

that is, it begins to approach the continuum form of eq. (8.68) but still contains the additional overaging component of resistance, which decays only slowly with $1/d$. Thus, the continuum limit is approached only slowly when

$$d \gg \frac{2b}{\sqrt{\pi c}} \left(\frac{\mu}{\tau_p} \right). \quad (8.73)$$

For a 95% approach to the continuum limit for typical values $c = 0.1$ and $\tau_p/\mu \approx 10^{-2}$, it is necessary that $d > 10^4 b$ or about $2.5 \mu\text{m}$.

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